UNIFORMITY, PRIME SUPPORT, AND REACHABLE LATTICES IN APPROACHES TO THE *abc* CONJECTURE

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ABSTRACT. We develop proof reductions, obstruction theorems, and formally checked finite certificates for several routes toward the standard *abc* conjecture. The conjecture is neither proved nor disproved here. Our common method is to separate a global uniformity statement from its local algebraic and combinatorial inputs, prove the latter before formalization, and test every proposed strengthening against all of its hypotheses. Exact prime-support identities and radical formulas isolate sufficient height and counting gates; explicit Pell, affine, Mersenne, and local-field examples delimit claims that those gates alone can support.

The project sharpens several active interfaces. In the IUT-facing model, primitive local factors give the exact ef coefficient, total-degree product-Haar normalization, and a pointed same-pilot reduction, while the source-level Ind3 comparison remains open. At the Mersenne endpoint, an exact Farey prefix–tail inequality shows that linear energy would force a polynomial swarm of distinct depth-three rows; the critical arithmetic counting exponent remains open. In the affine route, divisibility-maximal intersection tops admit ownership-preserving Cauchy aggregation and a strict non-arm energy reduction, leaving a concrete catalogue-sparsity estimate. In the Pell route, all-index polynomial norm and derivative identities and the full Taylor–Hensel digit law isolate the fixed-coefficient squarefull exclusion.

An earlier gain/packet/trace checkpoint adds five exact boundaries. For primitive nonunit triples, the canonical product gives $1/3 < A_{\text{can}} < 1/2$ and the correlated identity $q = 1+X-Y$; a uniform defect-flag budget is proved equivalent to standard *abc*, without being assumed. An infinite primitive family refutes the uncompensated synchronized-packet bound at exponent $4/3$, while the exact *ab*-compensated implication remains a live route. Finally, adjacent signed Pell traces encode simultaneous squarefullness as fourth-power support. Two complete local collisions show that depth two need not promote to depth five, but do not decide the simultaneous two-channel gate.

A new labelled valuation-incidence complex retains the arm, prime, exponent, defect filtration, and local congruence of every selected vertex. Its top face gives an exact rational-power half-space for *abc*. A dyadic infinite family refutes its fixed absolute-budget selector, and a balanced two-prime family refutes the first coefficient-one binary-scale repair. Allowing all three arms still leaves a raw weighted-defect gate false: an infinite primitive Pythagorean-square family is a complete-premise obstruction. An ordered fractional successor gives an exact height reduction, but the odd prime family $(1, p^2 - 1, p^2)$ forces an inaccessible source of mass $\log p$ and refutes its uniform gate at $\varepsilon = 1/4$. The same family refutes the ordered endpoint-flow gate, whose conditional implication to *abc* remains valid. Merely deleting the order reduces optimal unmatched mass to the positive scalar height defect. The first costed bidirectional repair is also false: prime-hypotenuse Pythagorean squares force energy at least $\frac{1}{4}\log p$ while their conductor is at most $3\log p$. We then isolate an indivisible prime-packet boundary gate which implies *abc* and does not collapse pointwise to the scalar relaxation. Linnik's least-prime theorem then supplies a prime-neighbour family that refutes its exclusive-ownership rule while satisfying $h \le r$. A shared CRT-incidence successor then lets a saturated hyperedge certify several endpoint powers while charging each sink capacity once. It gives a new height bridge and is strictly non-scalar at $(343, 625, 968)$. Arbitrary partial sharing collapses back to the scalar defect, while a saturated-only child is refuted by $(2, 15^n - 2, 15^n)$ for $n \equiv 284 \pmod{310}$. The exact shared-incidence uniform estimate, weighted multi-face, and genuine homological successors remain open.

We also introduce a divisor-valued contact surface. Its standard Bloch five-term boundary gives an exact quadratic Veronese peeling and a valuation-layer expansion of logarithmic contact area. Primitive Pythagorean squares furnish full-premise infinite counterexamples to two single-cell gates and to several fixed one-move policies. These examples do not refute arbitrary multi-cell fillings or the *abc* conjecture. Every finite scan is recorded without asymptotic extrapolation, and every formal claim is separated from its paper-only analytic or arithmetic bridge. The remaining pointed tensor comparison, critical Farey counting, affine catalogue sparsity, fixed Pell squarefull exclusion, compensated packet compression, canonical excess/slack control, and calibrated five-term filling estimates are explicit open obligations.

1. THE TARGET AND THE ROLE OF THE RESULTS

Throughout, an *abc triple* means positive integers a, b, c such that $a + b = c$ and a, b, c are pairwise coprime. Write

$$r = \mathrm{rad}(abc), \qquad R = \log r, \qquad h = \log c.$$

Here $\mathrm{rad}(n)$ is the product of the distinct prime divisors of n. Pairwise coprimality is equivalent to $\gcd(a, b, c) = 1$ under the additive relation. The target is the standard assertion

$$(1) \qquad \forall\varepsilon > 0 \quad \exists C_\varepsilon \in \mathbb{R} \quad \forall(a, b, c) : \qquad h \leq (1 + \varepsilon)R + C_\varepsilon.$$

Exponentiation gives the usual multiplicative constant. The repository's `ABCConjecture` uses $\log \max(a, b, c)$, which equals h here. Its definition is not changed in this work.

The quantifier order in (1) is essential. Its negation requires one fixed positive ε and violations for every additive constant, hence at unbounded height. A finite set of high-quality triples cannot supply that negation: it is absorbed into C_ε. Likewise, a sparse exceptional-set bound does not imply that the exceptional set is finite.

Our results concern several specific interfaces. Section 2 uses finite prime support to test a proposed analytic construction. Section 3 bounds distinct outputs of two amplification constructions under explicit sufficient radical certificates. Section 5 records the cancellation that an endpoint approach must preserve. Section 6 supplies a local generator certificate relevant to possible-image constructions. Sections 8 and 9 examine effective arithmetic geometry and recent literature. In particular, Corollary 8.5 proves effective finiteness of a specified moving-index family. These components are partial theorems and applicability tests; none constructs the missing uniform bound in (1). Section 4 extends the amplification test from sufficient certificates to actual radicals. Section 7 uses the original admissible-arrow definitions and identifies a precise Ind2 hull equality. Section 10 supplies constructions for the full family of abc triples, together with their exact remaining endpoint and denominator costs. Sections 11 and 12 construct full local Galois arrows and distinguish their specified point, pre-ideal and whole-product sources. Sections 13, 14 and 15 supply explicit arithmetic realizations, simultaneous local formulas and global initial-data constructions. Section 16 proves a uniform existence theorem at unbounded height. Section 17 proves the two-prime bound and the rational-return and odd-part fibre statements. Section 18 determines an entire rational isogeny class and its intrinsic bounds. Section 19 calculates its exact minimum Weil height, retaining the contribution of the reduced denominator. Section 20 computes the point hull for every independent finite torsion representative of the standard theta values. Section 21 proves canonical local membership in one synchronized base branch, keeping the predecessor carrier and the transfer direction explicit. Section 22 constructs the arithmetic bundles and calculates their degrees and reference changes. Section 23 gives the exact three-prime reduction, the mixed-full radical theorem, and the newest Campana counting boundary. Section 24 proves the failure of every coefficient below six on one Frey family and the finite selected-place product-hull theorem. It does not prove that six is an attainable optimal coefficient. Section 25 gives the exact positive uniformity equivalence, the counting threshold, the residual-kernel and polynomial counterexample obstructions, and the conditional Pell disproof gate. Section 28 gives the corrected local-volume holonomy ledger and the latest affine, Pell, and Danilov continuations. Section 29 gives a literal asymptotic counterfamily to the remaining printed global-ω hypothesis in a claimed 2026 disproof, while retaining and broadening its valid prime-power radical-neighbour target. Section 30 lifts the same-pilot comparison to a faithful labelled prime/unit signature before taking volume, and gives complete counterexamples to three weaker coordinate interfaces. Section 31 splits a generic fixed-stage packet into primitive field factors, restores zero to the signature, and states the exact source-level covariance still needed for IUT. Section 32 audits the newest LANA conditional implication, proves the empty-set obstruction to its current local-volume interface, and isolates the missing abstract-to-integer abc bridge. Section 33 gives an inhabited p-adic ball repair, proves its exact orbit constraints, and supplies the uniform comparison package that would carry abstract Statement I to the integer target. Section 34 realizes that package with zero error on the actual open rational tripod, proves that Statement I for this shadow is exactly equivalent to integer abc, and uses full-premise models to separate the three semantic inputs still missing for an intended theory. Section 35 then matches the genuine degree-one rational points, log-canonical height, different, and conductor to their source definitions and isolates the still-missing intended arithmetic height-theory instance and upstream Statement I proof. Section 39 constructs a simultaneous two-arm CRT packet and tests the two necessary marginal excess gates against a complete canonical counterexample. Section 40 proves cubic-product separation for a full fixed three-arm certificate, closes its large-modulus boundary, and derives the quantitative certificate-entropy obstruction while retaining all adaptive and correlated affine routes. Section 41 combines all three affine congruences in the determinant, forces canonical collinearity, and derives the stronger repeated-kernel entropy requirement with its exact boundary counterexamples. Section 42 proves the sharp determinant bound for three point-adaptive certificates, improves the canonical support constants, and retains the unresolved common-kernel selection problem. Section 43 proves the line cap, large-label fibre and multiplicity bounds, and the third-gcd energy identity; an actual box-wide collinear example isolates the remaining overlap gate. Section 44 extracts the exact arithmetic period on a ray, proves cube-root and weighted cubic-energy bounds away from constant-arm directions, and treats the vertical direction under an individual arm cap. Section 45 computes the actual Euler-totient catalogue mass and proves that coincident divisor labels can only increase its cubic energy, while preserving the separate singleton- novelty and signed-line gates. Section 46 treats arbitrary signed primitive directions, proves the strict capture squeeze and exact linear caps on all three canonical arm levels, and reduces the remaining global energy to one non-arm inverse-period sum and a singleton baseline. Section 58 separates base order blocks from the LTE lifting remainder and proves unconditional radical bounds at prime Mersenne layers. Section 59 identifies the exact divisor-average endpoint and proves why the available coarse order estimates do not control the moving weighted fibre. Sections 61– 64 resolve the super-Wieferich contribution into finite depth layers, give pointwise and divisor-average sufficient gates, and certify the odd-order counterexample at 3511 without closing the surviving route. Section 67 proves the exact divisor moment, its $O(\log\log(3m))$ moving localization, the actual cyclotomic cap, and the resulting near-diagonal exceptional-mass equivalence. Section 68 replaces that moving range by the exact family of fixed-polylogarithmic co-divisor gates, with its atomic and diffuse failure alternatives. Section 69 then reduces every surviving linear mass to deep lifts, transition clustering, or extreme small order. Section 70 removes the auxiliary exception threshold, localizes the actual block mass, and raises the three-arm coefficient to every value below the sharp algebraic ceiling $1/3$. Section 71 uses the injective exact-order multiplier to remove the localized low arm and raises the remaining two-arm coefficient to every value below $1/2$. Section 72 moves the one-copy cutoff to every fixed positive log-log slack and removes the growing low-multiplier part of the deep arm using Yamada's pointwise valuation bound. Section 73 replaces every fixed positive power by an arbitrarily slowly divergent factor, adds the exact multiplier character table, and records the precise arithmetic and abstract endpoint counterexample boundaries. Section 49 adds the exact modulo-eight and quadratic-character restrictions in the Pell packet, while Section 50 compresses a hypothetical squarefull packet into two odd-exponent kernels and retains a third quotient digit. Section 51 splices both Pell channels into a norm-one Lucas sequence, proves the exact support-unit staircase at every multiplication order, and recovers the opposite channel signs from a paired companion correction. Section 52 constructs the companion staircase in closed form, proves its coefficientwise differential correlation, and combines coherent every-order splitters with vertexwise character incidence and certified finite exclusion through prime index 271. Section 36 audits the current public IUT source, compiles an admissible-domain quantifier repair, and proves the exact finite index of a congruence order with surjective component projections. Section 37 realizes the repaired local domain with actual Haar measure, derives the local-degree packet coefficient, and isolates the still-unconstructed tensor and same-pilot steps. Section 47 cancels the singleton incidence baseline and gives exact Euler and tail bounds for pairwise inverse-period catalogues. Section 53 joins the quotient ledger to the all-order Lucas system and computes its sharp endpoint curvature. Section 74 closes one critical endpoint arm and identifies the exact-order Farey-energy obstruction for the other. Section 38 derives the primitive-factor degree coefficient and isolates the pointed source comparison still needed by the IUT-facing route. Section 48 aggregates divisibility-maximal affine tops without double-counting their owners. Section 54 proves the polynomial Hensel specialization on paper and records the fixed-parameter squarefull boundary. Section 55 kernel-checks the all-index polynomial identities, the all-support displacement equivalence under explicit scale-unit and transversality hypotheses, finite CRT steering, and the complete moving-parameter witness. Section 56 proves those transversality premises for every actual support prime at the fixed coefficient two and identifies the still-open simultaneous-displacement gate with the prime-index squarefull exclusion. Section 57 rewrites that surviving gate as two adjacent fourth-power support packets, proves the exact projector defect and its Newton lift, and uses complete-premise depth-two collisions to retire only stronger fifth-power promotions. Section 75 converts endpoint energy into a quantitative depth-three row swarm under an explicitly open global counting hypothesis. Section 76 kernel-checks the harmonic prefix budget, frequent-subsequence quantifiers, and finite injection of actual depth-three exact-order rows into the super-Wieferich prime count. Section 77 audits the latest alternative quality metrics, proves their exact packing-coordinate transfer, and gives an abstract full-premise countermodel to divergence transfer without claiming an arithmetic counterexample. Section 78 introduces a three-arm divisor packet spectrum directly on primitive abc triples, proves its exact product, height and orientation laws, and isolates an uncompensated radical-energy candidate that is tested and corrected in the following section. Section 79 proves that the original uncompensated radical-energy gate fails on an infinite primitive family at exponent $4/3$, and retains an exact ab-compensated implication to the required height bound. Section 80 audits the 2026 approximation/power gain proposal, constructs the canonical gain surface and its exact scale groupoid cocycles, and proves that the resulting uniform defect-flag budget is equivalent to the standard conjecture rather than a substitute for it. Section 92 develops the divisor-contact five-term complex, proves the surviving calibrated filling reduction, and uses primitive Pythagorean squares to retire only its single-cell gates. Section 93 constructs the integer exponent-gcd split and the actual finite-support divisor norm, then instantiates the filling inequality under an explicit surface-boundary equality. Section 94 constructs the algebraic and positive generated five-term relations and proves their boundary bridge; arbitrary-target filling existence and the two Gate VF costs remain open. Section 95 analyzes the exact quadratic five-term move, proves its valuation-layer flag, and delimits three fixed one-move policies without rejecting arbitrary fillings. No priority over existing literature is claimed for consequences of the cited classical theorems.

The new statements below were proved mathematically before their corresponding Lean implementations. Separate research agents reviewed the arguments, source hypotheses, and formal scope; this internal review is not external journal peer review. Classical published theorems may be used in the paper, but are not silently represented as proved Lean declarations. The formal scope is specified in Section 96. The literature and public-repository cutoff is September 3, 2026.

Six recent primary sources are useful for locating the boundary. Letendre's short-interval bounds for ω assume a new abc-type Conjecture 1 that the paper presents as stronger than abc [24]; they are therefore test interfaces rather than unconditional inputs here. Zhou's announced effective rational inequalities explicitly invoke the μ_6 version of IUT III, Corollary 3.12 in Proposition 1.9 [31]; this does not remove the source-faithful comparison obligation isolated below. Falk–Harrington–Jones give exact generalized Wieferich/Lucas criteria for monogenic trinomials [23], but no density bound for the fixed-base depth-three packet. Laniewski's 29 August 2026 preprint reduces Mersenne transgression to aggregate ordinary-Wieferich order–defect growth, while explicitly leaving that growth open; it likewise supplies no critical count for the fixed-base depth-three packet [25]. Sankaran's DGM and power-mean quality metrics have an unconditional divergent alternative-quality family, but the packing-efficiency formula $q_{\mathrm{std}} = \eta q_{\mathrm{DGM}}$ shows that this is not a proof or disproof of standard abc; the paper's Theorem 4.15 is an exact reformulation whose missing content is control of η on actual triples [26]. Müller and Taktikos introduce approximation and power gains for abc equations arising from algebraic approximation [27]. Their exact factorization motivates our canonical gain surface, while fixed separate bounds on the two factors yield only their product coefficient and therefore do not by themselves imply the coefficient-$1+\varepsilon$ quantifier of standard abc. Versioned PDFs, extracted text, metadata, and SHA-256 checksums for Letendre, Zhou, and Falk–Harrington–Jones are archived under `research/sources/latest_abc_proposals_2026_09_02/`; the Laniewski source record is archived under `research/sources/mersenne_farey_denominator_entropy_2026_09_02/`, and the Sankaran record under `research/sources/alternative_quality_metrics_2026_09_03/`, and the Müller–Taktikos record under `research/sources/abc_gain_surface_2026_09_03/`.

2. Prime support in short intervals

Let $\omega(n)$ be the number of distinct prime divisors of n, and let $P^+(n)$ be its greatest prime divisor, with $P^+(1) = 1$. For integers $N, Y \geq 1$ and $w \geq 0$, define

$$\mathcal{A}(N, Y, w) = \{1 \leq n \leq N : P^+(n) \leq Y,\ \omega(n) \leq w\}.$$

Theorem 2.1 (Finite prime-power encoding)**.** *With $L = \lfloor \log_2 N \rfloor$, one has*

$$|\mathcal{A}(N, Y, w)| \leq (1 + YL)^w. \tag{2}$$

Proof. Use the alphabet

$$\mathcal{Q} = \{1\} \cup \{q^e : 1 \leq q \leq Y,\ 1 \leq e \leq L\}.$$

It has at most $1+YL$ members; the bases need not be prime. For $n \in \mathcal{A}(N, Y, w)$, write its actual prime factorization as $n = \prod_{j=1}^{t} p_j^{e_j}$, with $p_1 < \cdots < p_t$ and $t \leq w$. Since $2^{e_j} \leq p_j^{e_j} \leq n \leq N$, each $e_j \leq L$. Thus every factor belongs to $\mathcal{Q}$. Append $w - t$ copies of 1 to obtain a word of length w. Its product recovers n, so different integers have different encoded words. Counting all words proves (2). The empty factorization covers $n = 1$ and $w = 0$; $N = 1$ is also included. ☐

Corollary 2.2 (A finite first-moment bound)**.** *Let S be a finite set of distinct positive Y-smooth integers at most N, and put $B = (1 + Y\lfloor \log_2 N \rfloor)^w$. Then*

$$|\{n \in S : \omega(n) \leq w\}| \leq B, \tag{3}$$

$$\sum_{n \in S} \omega(n) \geq (w + 1) \max\{|S| - B, 0\}. \tag{4}$$

Proof. The first set is contained in $\mathcal{A}(N, Y, w)$. Its complement in S has at least $\max\{|S| - B, 0\}$ members, each with at least $w + 1$ distinct prime factors. Summation proves the second assertion. ☐

Suppose now that $y = y(x) \geq 2$ satisfies

$$\log\log x = o(\log y), \qquad \log y = o(\log x), \qquad u = \frac{\log x}{\log y}. \tag{5}$$

For every fixed $\kappa > 0$, Theorem 2.1, with $N = \lfloor 2x \rfloor$, $Y = \lfloor y \rfloor$ and $w = \lfloor \kappa u \rfloor$, gives

$$\#\{n \leq 2x : P^+(n) \leq y,\ \omega(n) \leq \kappa u\} \leq x^{\kappa + o(1)}. \tag{6}$$

Indeed,

$$w \log(1 + YL) \leq \kappa \frac{\log x}{\log y}\Big(\log y + O(\log\log x)\Big) = \kappa \log x + o(\log x).$$

More generally the upper bound is $x^{o(1)}$ whenever $w(\log y + \log\log x) = o(\log x)$.

2.1. The external analytic input.

Write $\Psi(x, y) = \#\{1 \leq n \leq x : P^+(n) \leq y\}$. Younis's unconditional Theorem 1.1 [73], p. 2, states that for each fixed $17/30 < \theta \leq 1$ there is $C_Y(\theta) > 0$ such that

$$\frac{\Psi(x + h, y) - \Psi(x, y)}{h} = \frac{\Psi(x, y)}{x}\left(1 + O_\theta\left(\frac{\log(u + 1)}{\log y}\right)\right) \tag{7}$$

uniformly for

$$x^\theta \leq h \leq x, \qquad \exp\Big(C_Y(\theta)(\log x)^{2/3}(\log\log x)^{4/3}\Big) \leq y \leq 2x.$$

Fix

$$\frac{17}{30} < \theta < 1, \quad C \geq C_Y(\theta), \quad h = x^\theta, \quad y = \exp\Big(C(\log x)^{2/3}(\log\log x)^{4/3}\Big). \tag{8}$$

The lower bound on C is part of the hypotheses, not a disposable constant. Put

$$S_x = \{n \in (x, x+h] : P^+(n) \le y\}, \qquad M_x = |S_x|.$$

The classical long-interval estimate recalled in [51, equations (1.8)–(1.9), p. 415] yields $\Psi(x,y) = x\rho(u)(1+o(1))$ here. Its range $u \le (\log y)^{3/5-\varepsilon}$ is satisfied, for example with $\varepsilon = 1/20$: the power of $\log x$ on the right is $11/30$, larger than the power $1/3$ in u. Also $-\log\rho(u) = u\log u(1+o(1)) = o(\log x)$. Consequently (7) gives

$$M_x = h\rho(u)(1+o(1)) = x^{\theta - o(1)}. \tag{9}$$

Only this mass estimate is needed below; no short-interval moment formula is assumed.

Theorem 2.3 (A lower tail and a first-moment obstruction)**.** *For* (8) *and every fixed* $0 < \kappa < \theta$*,*

$$\frac{\#\{n \in S_x : \omega(n) \le \kappa u\}}{M_x} \le x^{\kappa - \theta + o(1)} \longrightarrow 0. \tag{10}$$

In particular,

$$\liminf_{x\to\infty} \frac{1}{uM_x} \sum_{n\in S_x} \omega(n) \ge \theta. \tag{11}$$

For every fixed $A > 0$*, the proportion of* $n \in S_x$ *with* $\omega(n) \le (\log\log x)^A$ *tends to zero.*

Proof. Every member of S_x is at most $2x$. Divide (6) by (9) to obtain (10). If its numerator is B_x, then

$$\sum_{n\in S_x} \omega(n) \ge \kappa u(M_x - B_x).$$

The lower limit after division by uM_x is at least each fixed $\kappa < \theta$, and hence at least θ. This argument does not require a variable κ or a uniform error term in that variable. Finally,

$$(\log\log x)^A(\log y + \log\log x) = o(\log x).$$

The last observation after (6) bounds the corresponding population by $x^{o(1)}$. Divide again by (9). □

Corollary 2.4 (A false intermediate assertion in a claimed disproof)**.** *Lemma 4.2 of Carella* [14]*, p. 8, equation (4.5), is false in its stated range. The same holds for the second-moment assertion in Lemma 4.4, p. 9, and the relative exceptional estimate (4.24), p. 11.*

Proof. Choose $\theta = 3/5$ and a fixed $C \ge C_Y(3/5)$. Then $h > x^{7/12}$, $y < h < x$, and $y = o(h)$, exactly as required in the first lemma. Its proposed formula would give

$$\frac{1}{M_x}\sum_{n\in S_x} \omega(n) = O(\log\log y).$$

However,

$$\frac{u}{\log\log y} \asymp \frac{(\log x)^{1/3}}{(\log\log x)^{7/3}} \longrightarrow \infty,$$

contradicting (11). Changing the left endpoint of the interval from open to closed adds at most one integer; its contribution is at most $\log_2(2x)$ and is negligible relative to M_x. The proposed second moment is $O(M_x(\log\log y)^2)$, whereas (10) gives a lower bound $\kappa^2u^2M_x(1-o(1))$ for any fixed $0 < \kappa < 3/5$. Finally the threshold in (4.24) is bounded by a fixed power of $\log\log x$. Theorem 2.3 shows that the proportion above it tends to one, although the claimed relative upper bound tends to zero. □

Remark 2.5. This refutes those intermediate assertions, not *abc* and not the possibility of an exceptionally sparse counterexample family. In particular a set of size $x^{o(1)}$ can still be infinite. It is also important to distinguish $o(h)$ from $o(M_x)$: the whole smooth set already has size $o(h)$.

2.2. **Two further transfer tests.** For a fixed integer $k \geq 4$, the intervals

$$(p^k, p^k + (p^k)^{3/5}], \qquad X < p^k \leq 2X,$$

are pairwise disjoint for large X. Indeed, successive distinct integer bases give gaps at least $X^{1-1/k}$, larger than $(2X)^{3/5}$. The finite encoding, using the largest thresholds in this range, shows that at most $X^{o(1)}$ centres can contain a smooth candidate with $\omega(n) \leq 2 \log\log(p^k)$ at the scale (8). The prime number theorem gives $X^{1/k+o(1)}$ centres, so their proportion is at most $X^{-1/k+o(1)}$. This rules out a positive-proportion selection of this particular kind, while leaving a sparse selection unresolved.

Recent work on exceptional sets [7, 12] and on the existence of smooth neighbours [54] addresses different quantifiers. A useful general check is the following. If a counting theorem gives $N(T) \ll T^{\eta+o(1)}$, then a seed of height H contradicts it only when it produces more than $H^{\beta\eta+o(1)}$ *distinct* qualifying outputs of height at most $O(H^\beta)$. Repeated encodings of one output do not count. None of the existence results just cited supplies the required uniform low-radical neighbours or such an amplification theorem.

3. Two explicit amplification families

Let $N_\mu(T)$ count positive primitive triples $A + B = C \leq T$ with $\operatorname{rad}(ABC) < C^\mu$, where $0 < \mu < 1$. The current estimates [7, Proposition 1.1 and Theorems 1.2–1.3] give

$$N_\mu(T) \ll_{\mu,\delta} T^{F(\mu)+\delta}, \qquad F(\mu) = \min\left\{\frac{2\mu}{3}, \frac{23\mu+3}{40}, \frac{3}{5}\right\}, \quad \delta > 0. \tag{12}$$

We test two actual constructions from a primitive seed $a + b = c$. An amplifier contradicting (12) at output height $T = c^K$, for fixed $K > 0$, would need more than $c^{KF(\mu)+o(1)}$ distinct qualifying outputs. Our conclusions concern explicit radical certificates, rather than the entire exceptional set.

3.1. **Inherited congruence moduli.** Fix pairwise coprime positive integers U, V, W, put $M = UVW$, and let $R_M = \operatorname{rad}(M)$. Outputs with $U \mid A$, $V \mid B$, $W \mid C$ satisfy

$$\operatorname{rad}(ABC) \leq \frac{R_M ABC}{M}.$$

Call an output *size-certified* if the right side is at most C^μ.

Proposition 3.1. *For $T \geq 1$, there are at most $1 + \lfloor \log_2 T \rfloor$ size-certified primitive outputs with $C \leq T$, for $0 < \mu \leq 1$. Allowing every inherited template $U \mid a$, $V \mid b$, $W \mid c$ gives at most*

$$\tau(abc)\big(1 + \lfloor \log_2 T \rfloor\big)$$

distinct outputs. For fixed K and $T \leq c^K$ this is $c^{o(1)}$.

Proof. Suppose first $M \geq 2$. Certification implies $R_M AB \leq M$, and $R_M \geq 2$. Consider two outputs in the strip $L \leq A, A' < 2L$. Their determinant $\Delta = AB' - A'B$ is divisible by UV and also by W, since $\Delta = AC' - A'C$. Thus $M \mid \Delta$. The bounds $B' \leq M/(R_M A')$ and $B \leq M/(R_M A)$ give

$$0 < AB', A'B < 2M/R_M \leq M.$$

Consequently $|\Delta| < M$, so $\Delta = 0$. Coprimality implies $A \mid A'$ and $A' \mid A$, hence equality of all coordinates. There is at most one output in each dyadic strip. The strict strip endpoint also handles $R_M = 2$. If $M = 1$, certification forces $AB \leq 1$; only $(1, 1, 2)$ can remain, and only when the exponent permits it.

The number of inherited templates is $\tau(a)\tau(b)\tau(c) = \tau(abc)$. A union bound proves the second claim. For every fixed $\eta > 0$, the elementary estimate $\tau(n) \ll_\eta n^\eta$ follows from $e + 1 \leq 2^e \leq p^{\eta e}$ for sufficiently large primes p and bounded multiplicative costs at the finitely many smaller primes. Since $abc \leq c^3$, this proves the last claim. □

Outputs whose remaining factors have additional powers or smaller actual radicals are not ruled out by this proposition. A count of raw CRT solutions cannot replace the count of distinct certified primitive triples.

3.2. **Square completion on a conic.** Put $P = abc$ and $R_P = \mathrm{rad}(P)$. Positive integer points on

$$ax^2 + by^2 = cz^2 \tag{13}$$

give candidate outputs (ax^2, by^2, cz^2). Retain only primitive outputs. Then $\gcd(x, y, z) = 1$, and the positive coordinates are recovered uniquely from the output. Their radical is at most $R_P xyz$.

Lemma 3.2. *For coprime integers m, n, not both zero, put*

$$Q = an^2 + bm^2, \quad J = an + bm, \quad X = Q - 2nJ, \quad Y = Q - 2mJ,$$

and set

$$D_0 = \gcd(a, m)\gcd(b, n)\gcd(c, m - n), \qquad G = \gcd(X, Y, Q) > 0.$$

Then

$$aX^2 + bY^2 = cQ^2, \qquad \gcd(Q, J) = D_0, \qquad G = \gcd(Q, 2J), \qquad D_0 \mid G \mid 2D_0.$$

Every positive primitive point of (13) is represented by $(X/G, Y/G, Q/G)$ for some such pair. Passing to absolute values only overcounts outputs, and (m, n) and $(-m, -n)$ give the same point.

Proof. Expansion proves the conic identity. Coprimality of m, n gives $\gcd(Q, 2nJ, 2mJ) = \gcd(Q, 2J)$. Here is a valuation check of the remaining gcd, including the prime two. If a prime divides both Q, J, the identities

$$aQ = J^2 - 2bmJ + bcm^2, \qquad bQ = J^2 - 2anJ + acn^2$$

show that it divides abc. At a prime dividing a, the common valuation is $\min(v_p(a), v_p(m))$: compare the terms of $J = an + bm$ and $Q = an^2 + bm^2$, using that b, n are units whenever $p \mid m$. The case $p \mid b$ is identical with the roles exchanged. At $p \mid c$, set $e = v_p(c)$, $r = v_p(n - m)$. If $r = 0$ the common valuation is zero. If $r > 0$, both coordinates are units and $J = a(n - m) + cm$ has valuation $\min(e, r)$ unless $e = r$. In that exceptional case either $v_p(J) = e$, or

$$Q = 2mJ - cm^2 + a(n - m)^2$$

has valuation exactly e. Thus the common valuation is always $\min(e, r)$. This argument does not divide by two. It proves the gcd formula and hence the assertions about G.

In the chart $z = 1$, a line through $(1, 1)$ has equation $(x, y) = (1, 1) + t(n, m)$. Its intersection parameters satisfy $t(2J + tQ) = 0$, giving the displayed second point. Every other rational point determines such a rational direction, which may be chosen primitive. The base point itself comes from the tangent direction $(m, n) = (-a, b)$. Dividing homogeneous coordinates by their gcd gives every primitive point. $\square$

Theorem 3.3. *For every $T > 0$, the number of positive primitive square-completion outputs with $cz^2 \le T$ is at most*

$$\tau(P)\left(1 + 4\pi\sqrt{T/P}\right). \tag{14}$$

The subfamily certified by $R_P xyz \le (cz^2)^\mu$, for $0 < \mu < 1$, has size at most

$$\tau(P)\left(1 + \frac{4\pi}{\sqrt{c}}T^{\mu/2}\right). \tag{15}$$

Proof. Fix the three divisors in D_0 from Lemma 3.2. The conditions

$$d_a \mid m, \qquad d_b \mid n, \qquad d_c \mid m - n$$

define a lattice in $\mathbb{Z}^2$ of index $D_0 = d_a d_b d_c$. Indeed the first two have index $d_a d_b$, and the third residue map is surjective there because these three divisors are pairwise coprime. The height bound and $G \le 2D_0$ imply

$$an^2 + bm^2 = Q \le 2D_0\sqrt{T/c}.$$

This ellipse has area $V = 2\pi D_0\sqrt{T/P}$.

A centred ellipse of area V contains at most $1 + 2V/D_0$ primitive integer directions in a lattice of index D_0, modulo sign. To see this, select representatives in a fixed half-plane and order them by angle. There is at most one primitive vector on each positive ray. Consecutive distinct directions have nonzero determinant divisible by D_0, so their triangles with the origin have area at least $D_0/2$. These triangles have disjoint interiors and lie in the ellipse. The cases of zero or one direction satisfy the bound as well. There are $\tau(P)$ divisor templates, proving (14).

For a certified output write $C = cz^2$. Equation (13) implies $\max(x, y) \geq z$, so $xyz \geq z^2$. Hence

$$R_P C/c \leq R_P xyz \leq C^\mu, \qquad C^{1-\mu} \leq c/R_P.$$

All certified heights are at most $U = \min\{T, (c/R_P)^{1/(1-\mu)}\}$. Moreover, $ab \geq c - 1 \geq c/2$ and $R_P \geq 2$, giving $R_P P \geq c^2$. Therefore

$$\sqrt{U/P} = U^{\mu/2}\sqrt{U^{1-\mu}/P} \leq T^{\mu/2}\sqrt{c/(R_P P)} \leq c^{-1/2}T^{\mu/2}.$$

Apply (14) at height U to obtain (15). ☐

For fixed $K > 0$ and $T = c^K$, (15) is $c^{\max\{0, K\mu/2-1/2\}+o(1)}$. Every term defining $F(\mu)$ is strictly greater than $\mu/2$, so this family cannot exceed the exponent required to contradict (12). The conclusion is limited to the certificate. For example, the seed $(1, 8, 9)$ and point $(7, 2, 3)$ give $(49, 32, 81)$, whose radical is 42, while $R_P xyz = 252$. Since $42^2 < 3^7$, $42 < 81^{9/10}$, although the certificate fails even with exponent one. This example proves that actual exceptional descendants cannot be identified with the certified subfamily; it is not a disproof of abc.

4. Actual radical budgets and a necessary amplification window

Here we count actual prime support, allowing arbitrary new primes and their repeated powers. Fix a positive primitive seed $a + b = c$ and put $P = abc$. *Only in this section, $R = \operatorname{rad}(P)$ denotes the integer radical, rather than its logarithm as in the introductory notation.* Thus $c \geq 2$ and $R \leq P \leq c^3$. All output triples below are ordered.

Theorem 4.1 (A height-independent radical budget)**.** *Let $R \geq 1$ be a squarefree integer and $Y > 0$ a real number. The number $\mathcal{N}(R, Y)$ of all positive primitive triples $A + B = C$ with $R \mid \operatorname{rad}(ABC)$ and $\operatorname{rad}(ABC) \leq Y$ is zero if $Y < R$. If $Y \geq R$, then, with $X = Y/R \geq 1$,*

$$\mathcal{N}(R, Y) \leq 905\, 45^{\omega(R)} X(1 + \log X)^{44}. \tag{16}$$

There is no restriction on the output height C.

Proof. For each output, $D = \operatorname{rad}(ABC)/R$ is a squarefree integer with $D \leq X$ and $\gcd(D, R) = 1$. Fix D and take S to consist of the archimedean place of $\mathbb{Q}$ and the primes dividing RD. The pair $(A/C, B/C)$ solves $u + v = 1$ in S-units. This association is injective: A/C is reduced because $\gcd(A, C) = 1$, so it recovers A, C and then B.

Theorem 1.1 of [52] bounds the number of solutions over a degree-m number field by $(3.1 + 5(3.4)^m)45^{|S|}$. Here $m = 1$ and $|S| = 1 + \omega(R) + \omega(D)$, including the infinite place. The number for this D is consequently at most $904.5\, 45^{\omega(R)+\omega(D)}$, which is smaller than the same expression with coefficient 905.

For an integer $M \geq 1$, let $d_M(n)$ count positive ordered M-tuples of product n. Prime factorization gives

$$M^{\omega(n)} \leq d_M(n) = \prod_{p^e \| n} \binom{e + M - 1}{M - 1}.$$

Fixing the first $M-1$ entries of a tuple, and then dropping their product restriction, yields for every real $X \geq 1$

$$\begin{aligned}\sum_{n\leq X} d_M(n) &\leq X \sum_{n_1\cdots n_{M-1}\leq X} \frac{1}{n_1\cdots n_{M-1}} \\ &\leq X\left(\sum_{1\leq n\leq X}\frac{1}{n}\right)^{M-1} \leq X(1+\log X)^{M-1}.\end{aligned}$$

The empty tuple covers $M=1$, and every sum equals one when $X=1$. Sum the fixed-D estimate, enlarge the sum by discarding coprimality and squarefreeness, and use $M=45$. This proves the bound and finiteness. The case $Y<R$ follows from divisibility of positive integers. □

Corollary 4.2 (Uniformity for a seed)**.** *Fix $K>0$, $0<\mu<1$, and $\varepsilon>0$. The number of outputs with*

$$R \mid \operatorname{rad}(ABC), \qquad \operatorname{rad}(ABC) \leq C^{\mu}, \qquad C \leq T \leq c^K$$

is zero if $T^{\mu}<R$, and otherwise is

$$\ll_{\varepsilon,K,\mu} c^{\varepsilon}\frac{T^{\mu}}{R}. \tag{17}$$

The implied constant is independent of the seed and of T.

Proof. Apply Theorem 4.1 with $Y=T^{\mu}$. For any fixed $\eta>0$, separating the finitely many primes $p<45^{1/\eta}$ gives $45^{\omega(R)} \ll_{\eta} R^{\eta} \leq c^{3\eta}$. Choose $\eta=\varepsilon/6$ before varying the seed. Since $\log(T^{\mu}/R) \leq K\mu\log c$, the remaining logarithmic factor is $\mathcal{O}_{\varepsilon,K,\mu}(c^{\varepsilon/2})$. This proves the assertion. □

To record exactly what repeated primes save, for positive integers P,n set

$$D_P(n)=\prod_{\substack{p\mid n\\ p\nmid P}} p, \qquad E_P(n)=n/D_P(n)\in\mathbb{N}_{>0}.$$

The factor $D_P(n)$ divides n, is squarefree, and is coprime to R. In $E_P(n)$, an old prime contributes its entire exponent, and a new prime of exponent e contributes exponent $e-1$.

Proposition 4.3 (Exact excess accounting)**.** *Let x,y,z be positive integers satisfying $ax^2+by^2=cz^2$, and put $(A,B,C)=(ax^2,by^2,cz^2)$. Then*

$$\operatorname{rad}(ABC)=RD_P(xyz), \qquad \operatorname{rad}(ABC)E_P(xyz)=Rxyz. \tag{18}$$

In particular, $\operatorname{rad}(ABC)\leq C^{\mu}$ *implies*

$$E_P(xyz) \geq (R/c)C^{1-\mu}. \tag{19}$$

These statements do not require the output to be primitive.

Proof. The prime divisors of Pn^2 are exactly those of Pn, so $\operatorname{rad}(Pn^2)=RD_P(n)$. Use $ABC=P(xyz)^2$ to obtain (18). The conic equation makes z^2 a positive weighted average of x^2,y^2. Thus $\max(x,y)\geq z$, and $x,y\geq 1$ give $xyz\geq z^2=C/c$. Consequently $RC/c \leq Rxyz = \operatorname{rad}(ABC)E_P(xyz) \leq C^{\mu}E_P(xyz)$. □

Theorem 4.4 (A necessary amplification window)**.** *Put $\rho=\log P/\log c$ and $\sigma=\log R/\log c$. Fix $K>0$ and $0<\mu<1$. Consider all positive integer square completions whose* unscaled *output (ax^2,by^2,cz^2) is primitive and satisfies $C\leq c^K$ and* $\operatorname{rad}(ABC)\leq C^{\mu}$*. If nonempty, this family has size*

$$\ll_{\varepsilon,K,\mu} c^{\min\{\max(0,(K-\rho)/2),\ K\mu-\sigma\}+\varepsilon} \qquad (\varepsilon>0). \tag{20}$$

For the exponent before ε to exceed $KF(\mu)$ from (12), it is necessary that

$$K>\rho+4\sigma, \qquad \frac{3\sigma}{K}<\mu<\frac{3}{4}\left(1-\frac{\rho}{K}\right)<\frac{3}{4}. \tag{21}$$

All implied constants are uniform in the seed for fixed K, μ, ε.

Proof. Integer square completion retains every prime of P, so Corollary 4.2 gives the exponent $K\mu - \sigma$; nonemptiness makes it nonnegative. The complete fibre bound (14) in Theorem 3.3, together with $\tau(P) \ll_\eta P^\eta$ and $P \le c^3$, gives $\max(0, (K-\rho)/2)$. Taking the smaller bound proves (20). The elementary divisor estimate used here follows by splitting primes at a fixed threshold; for large primes, $e+1 \le p^{\eta e}$, and for each remaining prime the quotient has a finite supremum over $e \ge 0$.

The BBLT exponent is $F(\mu) = \min\{2\mu/3, (23\mu+3)/40, 3/5\}$ [7]. If $\mu \ge 3/4$, all three entries are at least $1/2$, whereas $\max(0, (K-\rho)/2) < K/2$ because $K, \rho > 0$. Thus the desired strict exponent gain is impossible in this range, even for the complete fibre. If $0 < \mu < 3/4$, then $F(\mu) = 2\mu/3$. Both exponents in (20) must exceed $2K\mu/3$, forcing

$$K\mu - \sigma > 2K\mu/3, \qquad (K-\rho)/2 > 2K\mu/3.$$

These inequalities give the two bounds on μ in (21); their interval is nonempty only if $K > \rho + 4\sigma$. □

In particular, along seeds satisfying $K \le \rho + 4\sigma$, this family cannot supply a uniform lower count c^α with a fixed gap $\alpha > KF(\mu)$. Choose ε in (20) smaller than that gap to see this directly. The conclusion is for fixed K, μ; it asserts no uniformity for growing K. An exceptional seed bounds σ from above, and that must not be substituted as a lower bound in the counting estimate. The interval (21) remains open: no sufficient lower bound for actual excesses or distinct outputs has been proved here.

Full support retention also needs its stated domain. The primitive seed $(49, 576, 625)$, of radical 210, has the rational completion $(x, y, z) = (3/7, 1/6, 1/5)$ with primitive output $(9, 16, 25)$ of radical 30. Equivalently, the integer point $(90, 35, 42)$ loses the prime 7 after its output is divided by the common factor 44100. Thus arbitrary projective normalization is not covered by the full-R estimate. If the output itself retains the rational squareclasses ax^2, by^2, cz^2, only primes of odd valuation in P are automatically retained: their corresponding output valuations are nonnegative odd integers.

A separate attempt uses the three quadratic tripod identities

$$((a-b)^2, 4ab, c^2), \quad (a^2, 4bc, (b+c)^2), \quad (b^2, 4ac, (a+c)^2),$$

with $a \ne b$ and each row divided by its actual common divisor. That divisor is 1 or 4: for coprime u, v, the entries $(u-v)^2, 4uv, (u+v)^2$ have no common odd prime, and their gcd is 4 exactly when both u, v are odd. Their primitive radicals are exactly

$$R\,\mathrm{rad}_{\mathrm{odd}}(|a-b|), \qquad R\,\mathrm{rad}_{\mathrm{odd}}(a+2b), \qquad R\,\mathrm{rad}_{\mathrm{odd}}(2a+b),$$

where $\mathrm{rad}_{\mathrm{odd}}$ omits the prime 2. Every new odd prime in the displayed factor is coprime to P, while all seed odd primes survive normalization and every primitive positive triple contains 2. Thus all three branches, and their positive iterates, retain R.

Nevertheless, no branch is guaranteed to preserve quality $q = \log C / \log \mathrm{rad}(ABC)$. The seed $(1, 8, 9)$ gives $(49, 32, 81)$, $(1, 288, 289)$, and $(16, 9, 25)$, with radicals $42, 102, 30$. The first quality is smaller than $\log 9/\log 6$ because $42 > 36$; the last is less than one. For the middle one,

$$\frac{\log 289}{\log 102} < \frac{38}{31} < \frac{\log 9}{\log 6}, \qquad 17^{24} < 6^{38}, \quad 2^{38} < 3^{24},$$

where the integer comparisons prove the two logarithmic inequalities. This refutes only the proposed universal quality-preserving selection among these three branches. Neither finite example is an abc disproof, and more general maps and the remaining window are not excluded.

5. The coupled signed prime-exponent defect

For $n \ge 1$ put

$$E_1(n) = \sum_{\substack{p \mid n \\ v_p(n) = 1}} \log p, \qquad E_3(n) = \sum_{p \mid n} (v_p(n) - 2)_+ \log p.$$

Write $m = \min(a, b)$ and $M = \max(a, b)$ for an abc triple.

Proposition 5.1 (Exact signed identity)**.** *For every positive integer n,*

$$\delta_2(n) := \log n - 2\log \operatorname{rad}(n) = E_3(n) - E_1(n). \tag{22}$$

The coupled defect satisfies

$$\begin{aligned}\Delta &:= \log(Mc) - 2R \\ &= E_3(M) + E_3(c) - E_1(M) - E_1(c) - 2\log \operatorname{rad}(m),\end{aligned} \tag{23}$$

$$2h - 2R - \log 2 \le \Delta \le 2h - 2R. \tag{24}$$

Consequently abc is equivalent to

$$\begin{aligned}\forall \varepsilon > 0 \quad \exists K_\varepsilon \quad \forall (a, b, c) : \qquad E_3(M) + E_3(c) &\le E_1(M) + E_1(c) \\ &\quad + 2\log \operatorname{rad}(m) + 2\varepsilon R + K_\varepsilon.\end{aligned} \tag{25}$$

Proof. For every positive exponent e, $e-2 = (e-2)_+ - \mathbf{1}_{e=1}$. Multiply by $\log p$ and sum over the actual prime factorization to obtain (22). Pairwise coprimality gives $r = \operatorname{rad}(m)\operatorname{rad}(M)\operatorname{rad}(c)$, which proves (23). Since $c/2 \le M \le c$, taking logarithms of $c^2/2 \le Mc \le c^2$ gives (24). An *abc* constant C_ε supplies $K_\varepsilon = 2C_\varepsilon$. Conversely (25) and the lower corridor give $C_\varepsilon = (K_\varepsilon + \log 2)/2$. □

The equivalence does not prove (25); it identifies the entire missing estimate without losing the negative exponent-one terms. Replacing the coupled inequality by separate endpoint bounds fails.

Proposition 5.2 (An unbounded separate-endpoint obstruction)**.** *For every $0 \le \rho < 1$ and $K \in \mathbb{R}$, there is an abc triple with $m = 1$ and*

$$\delta_2(M) > \log \operatorname{rad}(m) + \rho R + K.$$

Proof. Take $(a, b, c) = (1, 2^N, 2^N + 1)$ with $N \ge 1$. Then $m = 1$, $M = 2^N$, and $\delta_2(M) = (N-2)\log 2$. Without making any assumption about the odd neighbour,

$$r \le 2(2^N + 1) \le 2^{N+2}, \qquad R \le (N+2)\log 2.$$

Therefore

$$\delta_2(M) - \rho R \ge ((1-\rho)N - 2 - 2\rho)\log 2 \longrightarrow \infty.$$

This proves the claim. The bound uses only $\operatorname{rad}(n) \le n$ and $\operatorname{rad}(2^N) = 2$. □

This argument neither proves nor assumes that all odd neighbours have large radical. It only rejects a false stronger estimate that discards their possible contribution. The coupled route remains open.

6. Finite certificates for reachable local lattices

Let K be a nonarchimedean local field, $\mathcal{O}$ its valuation ring, π a uniformizer, and $k = \mathcal{O}/\pi\mathcal{O}$ its residue field, with $|k| = q > 1$ and $v(\pi) = 1$. Fix $N \ge 1$, write $L = \mathcal{O}^N$, and normalize additive Haar measure on K^N by $\mu(L) = 1$. For any set $\mathcal{R} \subset L$, let

$$H(\mathcal{R}) = \overline{\operatorname{span}_{\mathcal{O}}(\mathcal{R})}.$$

If nonsingular N-tuples exist in $\mathcal{R}$, define

$$d(\mathcal{R}) = \min_{\substack{r_1,\dots,r_N \in \mathcal{R} \\ \det[r_1 \ \cdots \ r_N] \ne 0}} v(\det[r_1 \ \cdots \ r_N]);$$

otherwise set $d(\mathcal{R}) = \infty$.

Theorem 6.1 (A finite witness for the exact hull measure)**.** *With $q^{-\infty} = 0$, one has*

$$\mu(H(\mathcal{R})) = q^{-d(\mathcal{R})}.$$

If $d(\mathcal{R}) < \infty$, some matrix A whose columns belong to $\mathcal{R}$ satisfies

$$v(\det A) = d(\mathcal{R}), \qquad H(\mathcal{R}) = A\mathcal{O}^N.$$

Proof. In the full-rank case the nonzero determinant valuations are nonnegative integers, so choose A attaining their minimum d. For $r \in \mathcal{R}$, Cramer's rule gives

$$(A^{-1}r)_i = \frac{\det A_i(r)}{\det A},$$

where $A_i(r)$ replaces column i by r. Its columns still belong to $\mathcal{R}$. Every nonzero numerator has valuation at least d; a zero numerator causes no difficulty. Thus $A^{-1}r \in \mathcal{O}^N$ and $\mathcal{R} \subset A\mathcal{O}^N$. Since the columns of A lie in $\mathcal{R}$, their integral spans are equal. The set $A\mathcal{O}^N$ is compact, hence closed.

Integral row and column operations give $UAV = \operatorname{diag}(\pi^{a_1}, \ldots, \pi^{a_N})$, with $U, V \in \mathrm{GL}_N(\mathcal{O})$, $a_i \geq 0$, and $\sum a_i = d$. One may obtain this diagonalization by repeatedly moving an entry of least valuation to the upper left and clearing its row and column by integral operations. The quotient $\mathcal{O}^N / A\mathcal{O}^N$ has q^d elements. Its equal-measure cosets partition L, proving $\mu(A\mathcal{O}^N) = q^{-d}$.

In the rank-deficient case a nonzero K-linear functional annihilates $\mathcal{R}$. Scale its coefficients to lie in $\mathcal{O}$ with at least one unit coefficient. It then defines a surjection $\ell : L \to \mathcal{O}$ whose closed kernel contains $H(\mathcal{R})$. For every $t \geq 1$, $\ell^{-1}(\pi^t \mathcal{O})$ has index q^t in L. Hence $0 \leq \mu(H(\mathcal{R})) \leq q^{-t}$ for all t, proving measure zero. □

Corollary 6.2. *The equality $H(\mathcal{R}) = L$ holds if and only if the reductions of $\mathcal{R}$ span k^N, equivalently if some reachable column matrix has unit determinant. More generally, any measurable $\mathcal{O}$-submodule containing all columns of a reachable matrix A with $v(\det A) = d$ has measure at least q^{-d}.*

Proof. A determinant is a unit exactly when its reduction is nonzero; a spanning set of k^N contains a basis. Apply Theorem 6.1. For the last claim, the target contains $A\mathcal{O}^N$ and measure is monotone. The target need not be contained in L. □

6.1. **Independent and simultaneous actions.** Let $L_j = \mathcal{O}^{d_j}$, with $d_j \geq 1$, and let $T = \bigotimes_{j=1}^{s} L_j$, where $s \geq 1$ and $N = \prod_j d_j$. For nonzero $x_j \in L_j$, write $x_j = \pi^{m_j} u_j$ with u_j primitive.

Proposition 6.3. *Under the independent action of $\prod_j \mathrm{GL}_{d_j}(\mathcal{O})$, the orbit of $\bigotimes_j x_j$ spans $\pi^{\sum_j m_j} T$. The measure of this spanned lattice, normalized by $\mu(T) = 1$, is $q^{-N \sum_j m_j}$.*

Proof. Integral row operations send a primitive vector to any chosen standard basis vector. Choosing these operations independently in each factor gives every standard tensor basis vector multiplied by $\pi^{\sum_j m_j}$. Conversely all orbit vectors lie in this scaled lattice. Its index is $q^{N \sum_j m_j}$, giving the measure. □

Proposition 6.4 (A strict simultaneous-action counterexample)**.** *Over any commutative ring A, identify $A^2 \otimes_A A^2$ with $\mathrm{Mat}_2(A)$. The A-span of $\{v \otimes v : v \in A^2\}$ is exactly the submodule of symmetric matrices. If $A \neq 0$, it omits E_{12}. For $A = \mathcal{O}$ it has Haar measure zero in $\mathrm{Mat}_2(K)$.*

Proof. Every generator has equal off-diagonal entries, a condition preserved by linear combinations. Conversely $e_1 \otimes e_1$, $e_2 \otimes e_2$, and

$$(e_1 + e_2) \otimes (e_1 + e_2) - e_1 \otimes e_1 - e_2 \otimes e_2 = E_{12} + E_{21}$$

span all symmetric matrices. No division by two occurs. The functional $M \mapsto M_{12} - M_{21}$ vanishes on the span and takes value one on E_{12}. In the local-field case, the congruence $M_{12} = M_{21} \pmod{\pi^t}$ occupies a proportion q^{-t} of the integral matrix lattice for every t. Its exact equality locus therefore has measure zero. □

The generating set in Proposition 6.4 is stable under every simultaneous action $g \otimes g$. Thus even the full simultaneous integral linear group does not provide Proposition 6.3. This rejects a specific abstract substitute for independent generation; it does not identify the actual IUT outputs with this diagonal example.

6.2. **A tensor point inside the holomorphic hull.** An important distinction changes the reachability question. The hull in [56, Section 9.10.7, p. 125] is, after the finite-etale decomposition of a tensor algebra, a product of principal local ideals. It is therefore a module over the *whole product of integer rings*. Independent linear actions are not necessary for the following containment.

Let $E_1, \dots, E_m$ be finite extensions of $\mathbb{Q}_p$, and write

$$T = \bigotimes_{i=1}^{m} E_i = \prod_{\alpha} F_\alpha, \qquad B = \prod_{\alpha} \mathcal{O}_{F_\alpha}, \qquad A = \bigotimes_{i=1}^{m} \mathcal{O}_{E_i} \subset T,$$

where the integral tensor product is over $\mathbb{Z}_p$. Tensoring integral bases embeds A as a full lattice. Its elements are integral over $\mathbb{Z}_p$, so $A \subset B$, of finite index $I = [B : A]$.

Proposition 6.5. *For nonzero $\tau_i \in E_i$, put $t = \bigotimes_i \tau_i$. If a B-submodule $H \subset T$ contains a set S with $t \in S$, then*

$$\bigotimes_i \tau_i \mathcal{O}_{E_i} = tA \subset tB \subset H.$$

Both displayed lattices have full rank, and the smallest B-module containing the singleton $\{t\}$ is exactly tB.

Proof. The inverse of t is $\bigotimes_i \tau_i^{-1}$, by multiplication in the tensor algebra. Closure under B gives $tB \subset H$. The inclusion $A \subset B$ and the tensor multiplication rule give the other equality and inclusion. Multiplication by the unit t is a $\mathbb{Q}_p$-linear automorphism and preserves full rank. Finally tB itself is a B-module containing t, proving the singleton assertion. □

The conclusion applies *after* taking the specified holomorphic hull. It does not put tA in the raw tensor-image set or its ordinary $\mathbb{Z}_p$ convex closure. As printed, Section 9.10.7 requires the component generators to be nonzero and integral. An arbitrary bounded nonintegral set is not covered by that definition. With fractional generators allowed, a bounded set containing a unit has a unique such hull: take the least valuation in each nonzero component. This is an explicit extension of the convention, not an assumption that every source output is integral.

Proposition 6.6 (The two closures can differ)**.** *Let $E = \mathbb{Q}_3(\pi)$ with $\pi^2 = 3$. Under*

$$E \otimes_{\mathbb{Q}_3} E \simeq E^2, \qquad a \otimes b \longmapsto (ab, \sigma(a)b), \quad \sigma(\pi) = -\pi,$$

the tensor order satisfies

$$A = \{(x, y) \in \mathcal{O}_E^2 : x - y \in \pi\mathcal{O}_E\}, \qquad [\mathcal{O}_E^2 : A] = 3.$$

It is already a closed $\mathbb{Z}_3$-module, whereas its holomorphic hull is $B = \mathcal{O}_E^2$. The singleton $(1, 1)$ also has holomorphic hull B, while its ordinary convex closure is contained in $\mathbb{Z}_3(1, 1)$.

Proof. Valuations show that $\pi \notin \mathbb{Q}_3$ and $\mathcal{O}_E = \mathbb{Z}_3 + \mathbb{Z}_3\pi$. The images of an integral tensor basis are

$$(1, 1), \quad (\pi, -\pi), \quad (\pi, \pi), \quad (3, -3).$$

Their differences are divisible by π. Conversely write $x = a + b\pi$, $y = c + d\pi$. The asserted congruence is $3 \mid a - c$, and coefficients $(a+c)/2, (b-d)/2, (b+d)/2, (a-c)/6$ express (x, y) in the displayed basis. Reduction of $x - y$ modulo π gives the quotient $\mathbf{F}_3$ and hence the index. Both A and the singleton contain $(1, 1)$; its B-multiples are all of B. The compact module $\mathbb{Z}_3(1, 1)$ is convex and omits $(1, 0)$, proving the last distinction without relying on a choice between affine and absolutely convex conventions. □

This gives a counterexample to the general equality of holomorphic hull and minimum ordinary convex closure in [56, Proposition 9.10.8.1]. It does not refute IUT or abc. In particular the positive containment in Proposition 6.5 must be used on the holomorphic-hull side of this distinction.

6.3. **Native coefficients and compatible volumes.** There is a coherent marking-preserving collation of the relevant local Kummer classes. For a fixed finite extension $E/\mathbb{Q}_p$ and marked copies $i_y : E \simeq E_y$, Kummer theory identifies the integral multiplicative cohomology with

$$\mathcal{K}(E_y) = \varprojlim_n E_y^\times/(E_y^\times)^{p^n}.$$

This is the identification used in [55, Proposition 7.2.2]. At each finite level, the map $i_z i_y^{-1}$ induces an isomorphism of these quotients; their inverse limit gives a topological isomorphism $F_{y\to z}$ with

$$F_{y\to z}\big(\kappa_y(i_y(u))\big) = \kappa_z(i_z(u)), \qquad F_{z\to r}F_{y\to z} = F_{y\to r}.$$

It preserves the unit-completion subgroup and can be chosen coherently every time a label is repeated. This uses one family of the factorwise topological collations in the cited source; it does not assert that every such collation preserves a marked arithmetic element.

For $b_p = p$ when p is odd and $b_2 = 4$, the coefficient in [56, equation (9.7.2.2)] is

$$\lambda_{b_p}(a) = \frac{\log(1 + b_p a)}{b_p}, \qquad a \in \mathcal{O}_E.$$

For $a \neq 0$ it is nonzero and has the same valuation as a. Indeed, with $v(p) = 1$ and $x = b_p a$, every term after the first in the logarithm series has greater valuation:

$$v(x^n/n) - v(x) = (n-1)v(x) - v_p(n) > 0 \quad (n \geq 2).$$

For odd p use $v_p(n) < n - 1$, and for $p = 2$ use $v(x) \geq 2$. The terms tend to zero, so $v(\log(1 + x)) = v(x)$. The field markings commute with this convergent series, yielding the same normalized coefficient in the fixed target. In particular, the background coefficient $\lambda_{b_p}(1)$ is a unit.

Proposition 6.7. *Put $d_i = [E_i : \mathbb{Q}_p]$, $D_T = \prod_i d_i$, and normalize Haar measure on T by $\mu(A) = 1$. For $t = \bigotimes_i \tau_i$,*

$$\mu(tB) = I \prod_i |\tau_i|_{E_i}^{D_T/d_i}, \qquad |x|_{E_i} = |N_{E_i/\mathbb{Q}_p}(x)|_p.$$

For any $c_0 > 0$, the functional $\nu(U) = \mu(U)^{c_0/D_T}$ is monotone, and every measurable $H \supset tB$ satisfies

$$\nu(H) \geq I^{c_0/D_T} \prod_i |\tau_i|_{E_i}^{c_0/d_i} \geq \prod_i |\tau_i|_{E_i}^{c_0/d_i}.$$

A factor prescription $\prod_i \mu_i(V_i)^{\gamma_i}$ for the same integral tensor lattice can be independent of its presentation only if all $d_i\gamma_i$ agree. These equalities are also sufficient on full tensor lattices.

Proof. The I translates of A partition B, so $\mu(B) = I$. Multiplication by τ_i in its tensor factor has determinant $N_{E_i/\mathbb{Q}_p}(\tau_i)^{D_T/d_i}$. Multiplication by t is the composition of these maps, giving the displayed Haar factor. Monotonicity and the positive power give the inequalities. For necessity of balancing, put a scalar p in any one factor of the tensor lattice pA. The proposed values are $p^{-d_i\gamma_i}$ and must agree. If $d_i\gamma_i = c_0$, the normalized functional μ^{c_0/D_T} gives the factor prescription by the same determinant computation, proving sufficiency. ☐

The normalization matters when using an upper estimate from another source. Denote the preceding Haar measure by μ_A, and instead let $\mu_B(B) = 1$. On a measurable region of finite positive measure, put

$$V_A(U) = \frac{\log \mu_A(U)}{D_T}, \qquad V_B(U) = \frac{\log \mu_B(U)}{D_T}.$$

Uniqueness of Haar measure and $\mu_A(B) = I$ give

$$V_A(U) = V_B(U) + \frac{\log I}{D_T}, \qquad V_B(tB) = \sum_i \frac{\log |\tau_i|_{E_i}}{d_i}. \tag{26}$$

The positive power of Haar measure used above need not itself be an additive measure. Only its monotonicity was asserted.

This conversion is required by the original texts: [56, Section 9.10.3.1] assigns value one to the tensor order A, whereas [65, Proposition 1.4(i), author-version p. 13] normalizes the log-volume of the maximal order B to zero. The positive index contribution cannot be carried into a B-normalized upper estimate. The second equality in (26) preserves the native lower bound without that contribution.

For example, if all m factors equal a Galois extension $E/\mathbb{Q}_p$ of degree d, then

$$E^{\otimes m} \simeq E^{d^{m-1}}, \qquad \log_p[B : A] = \frac{(m-1)d^{m-1}\delta_E}{2},$$

where δ_E is the exponent of its discriminant over $\mathbb{Z}_p$. Indeed, the trace Gram matrix on a tensor basis has discriminant exponent $md^{m-1}\delta_E$. On the product maximal order that exponent is $d^{m-1}\delta_E$. Their difference is twice the exponent of the lattice index. Thus

$$V_A(U) - V_B(U) = \frac{(m-1)\delta_E}{2d} \log p.$$

For copies of a fixed selected field $E = L'_w$ over $F = L_{\mathrm{mod,v}}$, the source weight $\gamma = 1/[E : F]$ obeys $[E : \mathbb{Q}_p]\gamma = [F : \mathbb{Q}_p]$. It is thus balanced. A tuple with r theta coefficients $\lambda_{b_p}(a)$ and the remaining coefficients $\lambda_{b_p}(1)$ gives the native lower bound

$$\nu(H) \geq |a|_E^{r/[E:F]} \geq |a|_E^r \quad \text{if } 0 < |a|_E < 1.$$

For a rational Frey curve C, the field of moduli is $\mathbb{Q}$. The underlying punctured curve is defined over $\mathbb{Q}$; its absolute field of moduli, used before adjoining level structure in [56, Section 3.1(4)], is therefore $\mathbb{Q}$. Equivalently the Legendre parameter $-b/a$ gives $\mathbb{Q}(j) = \mathbb{Q}$ in [57, Section 5.3]. The later extensions $L = \mathbb{Q}(i, C[30])$ and $L' = L(C[\ell])$ are Galois over $\mathbb{Q}$ and do not redefine that unlevelled moduli field. Choosing one place of L' above each rational prime gives the place-compatible selected bijection of [56, Section 3.3(14) and Proposition 9.4.2.4(4)]. Thus there is one selected place per prime. The internal field decomposition of $E^{\otimes m}$ still remains.

For a split multiplicative place with $q \in \mathbb{Q}_p$, take a root $a^{2\ell} = q$ in the selected field E. The native module absolute value satisfies $|q|_E = |q|_p^d$, so

$$|\lambda_{b_p}(a)|_E^{1/d} = |q|_p^{1/(2\ell)}.$$

A block with one such entry and background units therefore gives

$$V_B(H) \geq V_B(tB) = -Q_p, \qquad Q_p = -\frac{\log |q|_p}{2\ell} > 0. \tag{27}$$

The chosen marking-induced isomorphisms give an actual tuple in the collation, and convex closure retains it. The tuple-to-tensor map and the module hull then give the stated containment. No independent action on repeated labels is needed.

6.4. **Changing the target of the whole collation.** Fix a family of source groups C_y, a set of configurations of source classes, and a fixed slot set in each block. Require repeated occurrences of a source label to use the same isomorphism. Let Ψ_k be the union of all resulting tuples in a target C_k, where all allowed topological group isomorphisms are used. The same discussion applies to a groupoid of Galois-induced isomorphisms.

Proposition 6.8 (Covariance for a fixed source family)**.** *If $c : C_0 \simeq C_1$ is an allowed target isomorphism, then*

$$\Psi_1 = c^{\mathrm{slots}}(\Psi_0).$$

If c in logarithmic coordinates is induced by a local field marking, the same covariance holds for closed convex closure, tensor image, and the maximal-order module hull. In particular the transported hulls have equal V_A volumes, and equal V_B volumes with the respective normalizations.

Proof. Postcomposition $f \mapsto c \circ f$ is a bijection from $\mathrm{Iso}(C_y, C_0)$ to $\mathrm{Iso}(C_y, C_1)$, with inverse postcomposition by c^{-1}. Apply it to the whole family of maps used by a source configuration. Equal maps on repeated labels remain equal in both directions. Taking the union proves the set equality. For a groupoid the same argument uses closure under composition and inverse.

The marking induces a linear homeomorphism in logarithmic coordinates, so it carries a smallest closed convex region to the corresponding smallest closed convex region. Tensoring the field maps commutes with the map on pure tensors. The resulting algebra isomorphism preserves integrality over $\mathbb{Z}_p$ and thus carries A_0, B_0 to A_1, B_1. Minimality of the coefficient-module hull, applied also to the inverse, proves equality of the transported hulls. Finally uniqueness of Haar measure transports the chosen normalization, and dimensions are unchanged. □

The source family and its classes are hypotheses of this proposition. It justifies a change of target in the all-isomorphism construction of [56, Section 9.7.5.1]; it does not show that every operation intended by a later Frobenius comparison keeps that source family fixed. In particular a favorable native point and covariance for this family cannot be combined with an upper estimate for a different family.

To see the normalization issue directly, write a rescaled untilt norm on the marked finite field as $N_y(x) = |x|_E^{s_y}$, $s_y > 0$. The native Haar contribution is

$$|\lambda_{b_p}(a)|_E^{1/d} = N_y(\lambda_{b_p}(a))^{1/(ds_y)}.$$

Changing s_y alone leaves it unchanged. Also replacing q by q^p rebuilds the principal unit from a^p and gives coefficient $\lambda_{b_p}(a^p)$, whereas raising $1 + b_p a$ to its pth power gives $p\lambda_{b_p}(a)$. They need not agree: for $p = 3, E = \mathbb{Q}_3, a = 3$, their valuations are three and two, respectively. The Frobenius in [55, Corollary 5.6.2] acts on a perfected multiplicative monoid, not by definition on this fixed integral Kummer lattice.

The original IUT IV computation explicitly uses a squared exponent: [65, Theorem 1.10, Step (v), author-version pp. 27–29] takes the jth theta-pilot valuation to be $j^2 v_p(q)/(2\ell)$. Replacing the coefficient above by $\lambda_{b_p}(a^{j^2})$ would give the single-point lower bound $-j^2 Q_p$, rather than $-Q_p$. For $s = (\ell - 1)/2$ the average of these coefficients is

$$\frac{1}{s}\sum_{j=1}^{s} j^2 = \frac{(s+1)(2s+1)}{6} = \frac{\ell(\ell+1)}{12}.$$

Proposition 6.8 can be proved for either fixed family, but it does not identify them. The required favorable lower bound for the correctly powered possible-image family is a further assertion.

6.5. **The source weights for several selected places.** The mixed-place weights can also be read from the original upper source, [65, Remark 1.7.1, author-version p. 17]. Let $F = L_{\mathrm{mod}}$ have degree n, put $h_v = [F_v : \mathbb{Q}_p]$, $d_v = [L'_{w(v)} : \mathbb{Q}_p]$, and $\gamma_v = h_v/d_v$ for $v \mid p$. Then $\sum_v h_v = n$. For a word f of length m in these places put

$$D_f = \prod_i d_{f(i)}, \qquad \rho_f = \frac{\prod_i h_{f(i)}}{n^m}.$$

With B_f the maximal order in the corresponding tensor algebra, the degree-normalized functional on a product of ideal hulls is

$$V_{B,p,m}(H) = \sum_f \frac{\rho_f}{D_f} \log \mu_{B_f}(H_f), \qquad \mu_{B_f}(B_f) = 1.$$

The word weights sum to one and have marginal h_v/n at every slot. Applying the tensor determinant identity therefore gives

$$V_{B,p,m}(tB) = \frac{1}{n}\sum_i \sum_{v|p} \gamma_v \log |\tau_{i,v}|_{L'_{w(v)}}.$$

The corresponding tensor-order normalization differs by exactly $\sum_f (\rho_f/D_f) \log[B_f : A_f]$. This proves the formula on product ideal hulls. It is not a product formula for an arbitrary measurable non-product region. The extra global factor n must be coordinated with the arithmetic-degree convention; it equals one in the rational branch.

These constructions prove the local point-to-lattice implication and fixed-source target covariance with their stated normalizations. They do not yet identify the full Frobenius-shifted source families in [57, Theorem 6.10.1] with those of the later upper estimate. That estimate must concern the same hull and volume functional. Part III already introduces the volume of a containing compact region on p. 121; omission of a hull symbol in a later formula is therefore not, by itself, evidence that a different set was estimated. What is needed is the actual comparison with the region and measure in IUT IV, rather than an identification by notation.

6.6. **What a source-derived application must establish.** The July 2026 LANA report [59] isolates an identification between q-pilot and possible-output volume data in the passage to IUT III, Corollary 3.12 [62]. Joshi's July 30 response [58] refers to distinct arithmeticoids and his Part IV comparison [57, Theorem 6.10.1]. These are source claims to reconstruct, not assumptions used in our local theorems.

Part III [56, Proposition 9.7.5.1] allows specified topological isomorphisms in its collation. Section 6.2 constructs one marking-preserving family and proves containment after the stated holomorphic hull. It does not require unrestricted independent linear actions. The remaining identification must compare the same set of source objects, the same target region, and the same numerical normalization with the later upper estimate.

For a source-faithful finite packet of local sets $\mathcal{R}_v \subset \mathcal{O}_v^{N_v}$ and positive weights λ_v, Theorem 6.1 gives, when all defects are finite,

$$V_H = -\sum_v \frac{\lambda_v}{N_v} d(\mathcal{R}_v) \log q_v.$$

Suppose genuine local target regions Θ_v contain the respective hulls and have finite positive measure. Define, on these same carriers, $V_\Theta = \sum_v (\lambda_v/N_v) \log \mu_v(\Theta_v)$. Let $V_q \in \mathbb{R}$ be the fixed native q-pilot log-volume under the same source normalization. Then

$$\sum_v \frac{\lambda_v}{N_v} d(\mathcal{R}_v) \log q_v \leq -V_q \quad \Longrightarrow \quad V_q \leq V_H \leq V_\Theta.$$

The final inequality follows from local measure monotonicity, followed by the logarithm and the positive weighted sum. A different volume functional on a non-product region requires a separate monotonicity theorem. This is a sufficient numerical interface. The native points and their holomorphic hulls above provide one local construction; identifying them with the region bounded in IUT IV [65], controlling the same normalized determinant sum, and deriving the uniform global comparison remain separate obligations. Archimedean factors and infinite-packet convergence are not covered by the finite local theorem.

7. Admissible unit actions and a cyclotomic trace invariant

This section uses the actual all-open-subgroup condition in [64, Example 1.8(iv), author-version pp. 38–39]. It must be distinguished from allowing every automorphism of one fixed integral lattice. Fix a finite extension $E/\mathbb{Q}_p$, an algebraic closure $\overline{E}$, and $G = \mathrm{Gal}(\overline{E}/E)$. For each finite K/E inside $\overline{E}$, put

$$U_K = \mathcal{O}_K^\times, \qquad L_K = \log U_K \subset K, \qquad I_K = p^{-1} L_K.$$

The logarithm is used only on units. Its kernel on U_K is the finite group of roots of unity in K, and L_K is a full $\mathbb{Z}_p$-lattice in K. These assertions hold also for $p = 2$.

Logarithm induces a G-equivariant additive isomorphism

$$\mathcal{O}_{\overline{E}}^\times/\mu(\overline{E}) \simeq (\overline{E}, +). \tag{28}$$

Indeed the finite-field kernel statement proves injectivity. Given $x \in \overline{E}$, choose n large enough for $\exp(p^n x)$ to converge in a finite extension and choose a p^n-th root y of it. Then y is a unit and $\log y = x$, proving surjectivity. If $H = \mathrm{Gal}(\overline{E}/K)$, the invariants of the quotient in (28) are K, whereas the image of the invariants of the original unit group is precisely L_K.

Under this identification, the group $\mathrm{Ism}(G)$ specified in [64] consists of additive G-equivariant automorphisms F of $\overline{E}$ preserving L_K for *every* finite K/E. The source requires preservation of these lattices, not only of their rational spans. Such a map is $\mathbb{Q}_p$-linear on each K: preservation of $p^n L_K$ implies continuity, which extends integer linearity to $\mathbb{Z}_p$-linearity and then to $\mathbb{Q}_p$-linearity.

7.1. **The all-open condition forces scalar action.**

Lemma 7.1 (Realizing logarithmic sublattices by norms)**.** *For every finite-index $\mathbb{Z}_p$-submodule $J \subset L_E$, there is a finite abelian extension K/E such that*

$$\mathrm{Tr}_{K/E}(L_K) = J.$$

Proof. Let $V = \log^{-1}(J) \subset U_E$, choose a uniformizer π of E, and put $N = \pi^{\mathbb{Z}} V$. This is an open finite-index subgroup of $E^\times$. The local existence theorem of class field theory [67, Chapter I, Theorem 1.4] supplies a finite abelian K/E with $\mathrm{Norm}_{K/E}(K^\times) = N$. For integer-normalized valuations,

$$v_E(\mathrm{Norm}_{K/E}\, z) = f_{K/E} v_K(z).$$

Consequently a norm is a unit exactly when its preimage is a unit, and $\mathrm{Norm}_{K/E}(U_K) = N \cap U_E = V$. The identity $\log \mathrm{Norm}\, z = \mathrm{Tr} \log z$ follows by summing the logarithms of the conjugates. Therefore $\mathrm{Tr}_{K/E}(L_K) = \log V = J$. $\square$

Lemma 7.2 (Sublattice rigidity)**.** *An automorphism of a nonzero finite free $\mathbb{Z}_p$-module preserving every finite-index submodule is multiplication by one element of $\mathbb{Z}_p^\times$.*

Proof. Choose a basis $e_1, \ldots, e_d$. Preservation of $\mathbb{Z}_p e_i + p^n L$ for every n puts the image of e_i in their intersection, which is $\mathbb{Z}_p e_i$: every other coordinate is divisible by every power of p and hence is zero. Write $F(e_i) = \lambda_i e_i$. The same argument for $e_i + e_j$ implies $\lambda_i = \lambda_j$. Thus $F = \lambda\, \mathrm{id}$; surjectivity shows that λ is a unit. The rank-one case is already covered by the first step. $\square$

Theorem 7.3 (Classification of the specified Ism action)**.** *For the group defined in* [64, Example 1.8(iv)]*,*

$$\mathrm{Ism}(G) = \mathbb{Z}_p^\times$$

under (28)*, with the right side acting by scalars.*

Proof. For finite abelian K/E, equivariance gives $F_E\, \mathrm{Tr}_{K/E} = \mathrm{Tr}_{K/E}\, F_K$. Since $F_K(L_K) = L_K$, Lemma 7.1 implies that F_E preserves every finite-index submodule of L_E. Lemma 7.2 gives $F_E = \lambda_E\, \mathrm{id}$ with $\lambda_E \in \mathbb{Z}_p^\times$. Repeat over any finite K/E: the hypothesis includes all finite extensions of K, since their Galois groups are open in G. Thus $F_K = \lambda_K\, \mathrm{id}$. Restriction to the nonzero subfield E gives $\lambda_K = \lambda_E$, proving the assertion on all of $\overline{E}$. Conversely every $\mathbb{Z}_p$-unit scalar preserves all the lattices and commutes with G. It is the image of a unit of $\widehat{\mathbb{Z}}$ under the unit-power action; prime-to-p components act trivially after quotienting by all roots of unity. $\square$

Only preserving L_E would leave the full group $\mathrm{GL}_{[E:\mathbb{Q}_p]}(\mathbb{Z}_p)$; the finite-extension hypothesis is decisive. The full-unit lifting theorem [53, Theorem 7.6] by itself does not classify automorphisms of the torsion quotient. The preceding proof supplies that additional step. For the indeterminacy called Ind2 in [62, Theorem 3.11(i)], the theorem applies to its indicated Ism factors. It does not classify the Galois arrows of Ind1 or the set operations of Ind3.

For example, if $1 < e(E/\mathbb{Q}_p) \le p - 2$, logarithm and exponential on the maximal ideal give $I_E = \pi^{1-e}\mathcal{O}_E$. The unit $u = \log(1+p)/p$ has native valuation zero, and all its Ism images still have valuation zero. None reaches the least valuation $1/e - 1$ of this larger lattice. This statement concerns Ism; it adds no unproved restriction to Ind1.

7.2. **What the procession retains, and an exact Ind2 hull equality.**

The Ind1 source in [62, Theorem 3.11(i)] is the mono-analytic procession of [63, Proposition 6.9(ii)]. At a nonarchimedean place its retained category is $\mathcal{B}(E)^0$, the connected finite etale covers of E, rather than the earlier curve-covering category [63, Examples 3.2(i), 3.3(i), Definition 4.1(iii)–(iv)]. Its self-equivalences, up to natural isomorphism, correspond to $\mathrm{Out}(G_E)$: pullback of finite transitive G_E-sets constructs one direction, and adjoining finite disjoint unions and recovering the fibre functor constructs the inverse.

For fixed capsule labels, the capsule-full poly-isomorphism contains all collections of constituent isomorphisms [63, §0, pp. 33–34; Definition 4.10]. Consequently every local outer Galois automorphism occurs as a representative in an allowed poly-automorphism, with the same representative used at repeated labels if required. To see this directly, identify each labelled category with $\mathcal{B}(E)^0$, conjugate the chosen pullback equivalence to each copy, and use identity maps at the other places. Keep the capsule and stage injections equal to the identity. Each resulting collection belongs to the required full set of constituent isomorphisms, and conjugation preserves the full connecting sets. There is no further requirement to lift back to the curve category. This is a statement about representatives inside poly-morphisms, not a claim that each representative is a distinct object or morphism of the coarse category. It supplies no arbitrary integral linear lift.

Proposition 7.4 (Ind2 does not enlarge a fixed module hull)**.** *On a fixed marked nonarchimedean tensor carrier, let*

$$T = \bigotimes_{i,\mathbb{Q}_p}\Big(\prod_{v\in J_i} E_{i,v}\Big), \qquad A = \bigotimes_{i,\mathbb{Z}_p}\Big(\prod_{v\in J_i} \mathcal{O}_{E_{i,v}}\Big) \subset B,$$

where B is the maximal order of the finite etale algebra T. For any subset $S \subset T$, write $\Gamma_2 S$ for the union of its images under the local Ind2 operators of [62, Theorem 3.11(i)]. *Then*

$$\mathrm{span}_B(\Gamma_2 S) = \mathrm{span}_B(S).$$

The same equality holds for closed B-module hulls.

Proof. Theorem 7.3 makes the action on each local direct summand multiplication by a scalar $\lambda_{i,v} \in \mathbb{Z}_p^\times$. In factor i, their tuple $c_i = (\lambda_{i,v})_v$ is a unit of $\prod_v \mathcal{O}_{E_{i,v}}$. The tensor product operator is multiplication by $c = \bigotimes_i c_i \in A^\times$, with inverse the tensor of the c_i^{-1}. Thus every image of S lies in $\mathrm{span}_B(S)$. The identity Ind2 operator supplies the reverse inclusion of spans. Taking closures gives the closed version. Any extra convention forcing some scalars to coincide leaves the argument unchanged. □

This removes the Ind2 step from a B-module-hull computation once the Ind1 image set and native markings are fixed. It does not remove the distinction between A and B, turn an Ind1 map into a B-linear map, or identify Ind3 with multiplication by a unit.

7.3. **Integral Galois transport preserves trace depth.** A coefficient-compatible Galois arrow consists of a continuous isomorphism $\alpha : G_{E'} \simeq G_E$ and an isomorphism $\beta : \alpha^*\mathbb{Z}_p(1)_E \simeq \mathbb{Z}_p(1)_{E'}$ of integral Tate modules. The cyclotomic character and inertia are group-theoretically determined [68, Propositions 1.1–1.2 and Corollary 1.3]. In particular the coefficient isomorphism can be chosen integrally; one cannot insert multiplication by p as an isomorphism.

Proposition 7.5. *The induced logarithmic map $F_{\log} : E \longrightarrow E'$ satisfies*

$$\mathrm{Tr}_{E'/\mathbb{Q}_p}(F_{\log} x) = c_\alpha\, \mathrm{Tr}_{E/\mathbb{Q}_p}(x) \quad (x \in E) \tag{29}$$

for some $c_\alpha \in \mathbb{Z}_p^\times$. The statement also holds with both logarithmic coordinates divided by p.

Proof. Put $h_E = \log_p \chi_E \in H^1(G_E, \mathbb{Z}_p)$, where the latter coefficient action is trivial. Equivariance of β gives $\chi_E \alpha = \chi_{E'}$, hence $\alpha^* h_E = h_{E'}$. Local duality and the invariant map identify $H^2(G_E, \mathbb{Z}_p(1))$ with $\mathbb{Z}_p$. Integral cohomology transport is an isomorphism, so it acts there by a unit c_α, not by an arbitrary nonzero rational p-adic scalar.

The local reciprocity cup-product formula, for an integral character h and a Kummer class $\kappa(z)$, is

$$\mathrm{inv}(h \cup \kappa(z)) = h(\mathrm{rec}_E\, z).$$

It follows at finite level from the cyclic-algebra description and then by inverse limit; see [67, Chapter III, Proposition 3.6 and §4]. In the arithmetic-Frobenius convention,

$$h_E(\mathrm{rec}_E\, z) = -\log_p \mathrm{Norm}_{E/\mathbb{Q}_p} z = -\mathrm{Tr}_{E/\mathbb{Q}_p} \log_p z \qquad (z \in U_E).$$

The integral unit Kummer submodule is also preserved: it is the annihilator, under this pairing, of the unramified $\mathbb{Z}_p$-character. Characteristic inertia carries a generator of that unramified character module to a unit multiple of a generator, and cup-product naturality preserves its annihilator.

Applying the same naturality with h_E now proves (29) on L_E. Torsion unit classes are killed by logarithm and pair to zero in the torsion-free integral H^2; the argument therefore descends to the actual free unit lattice. Since L_E spans E, $\mathbb{Q}_p$-linearity proves the formula everywhere. Uniform division by p on both sides does not change it. □

Composing with the Ism factors of Theorem 7.3 only changes c_α by a unit. Thus the valuation of a nonzero absolute trace is an invariant of these local Galois/Kummer/Ism arrows. For example, on $E = \mathbb{Q}_7(\pi)$ with $\pi^3 = 7$, the normalized integral log lattice I_E has basis $1, \pi/7, \pi^2/7$. Swapping its first two basis vectors is not an admissible Galois/Kummer arrow: their traces are 3 and 0. No claim that every matrix satisfying (29) is admissible is made.

Theorem 7.6 (A native root-power obstruction)**.** *Let $p \neq \ell$ be odd primes, $b \in \mathbb{Q}_p$ with $N = v_p(b) > 0$ and $\ell \nmid N$, and let $E/\mathbb{Q}_p$ contain μ_ℓ and a with $a^\ell = b$. Write $d = [E : \mathbb{Q}_p]$ and, for $s \geq 1$ with $\ell \nmid s$, put $\tau_s = \log(1 + pa^s)/p$. Then*

$$\operatorname{Tr}_{E/\mathbb{Q}_p} \tau_s = \frac{d}{\ell p} \log(1 + p^\ell b^s), \tag{30}$$

$$v_p(\operatorname{Tr}_{E/\mathbb{Q}_p} \tau_s) = v_p(d/\ell) + \ell - 1 + sN. \tag{31}$$

In particular, distinct such s give distinct orbits under the arrows of Proposition 7.5, even after Ism action.

Proof. The field $F = \mathbb{Q}_p(\mu_\ell)$ is unramified over $\mathbb{Q}_p$. The valuation assumption makes b a non-ℓ-th-power in F, so $K = F(a)$ has degree ℓ over F. Since multiplication by s permutes the exponents modulo ℓ, and ℓ is odd,

$$\operatorname{Norm}_{K/F}(1 + pa^s) = \prod_{\zeta \in \mu_\ell} (1 + p\zeta a^s) = 1 + p^\ell b^s.$$

All factors are principal units. Taking logarithms, dividing by p, and using trace transitivity gives (30): the remaining factor is $[E : K][F : \mathbb{Q}_p] = d/\ell$. For $x = p^\ell b^s$, the term x is the unique least-valuation term in $\log(1 + x)$, since

$$v_p(x^k/k) - v_p(x) = (k-1)(\ell + sN) - v_p(k) > 0 \quad (k \geq 2).$$

This proves (31), including nonvanishing. Its values differ by $(s - t)N$ for distinct labels, which (29) cannot change. □

For a concrete example take $p = 11$, $\ell = 5$, $b = 11$, and $E = \mathbb{Q}_{11}(\pi)$ with $\pi^5 = 11$. The base field contains μ_5; the trace depths of τ_1 and τ_4 are 5 and 8. This is a specified local example, not a construction of all the global initial theta data.

There is a direct local link with a split Tate curve having rational full 2-torsion. If q is its parameter, choose $b^2 = q$. The class of b in $\overline{\mathbb{Q}}_p^\times/q^{\mathbb{Z}}$ is rational, so $g(b)/b \in q^{\mathbb{Z}} \cap \{1, -1\}$. Valuation makes this intersection $\{1\}$; hence $b \in \mathbb{Q}_p$. Rationalizing full ℓ-torsion also rationalizes full 2ℓ-torsion and supplies the required a and μ_ℓ. The theorem consequently applies when $\ell \nmid v_p(q)/2$. It does not assert that every bad prime meets that condition.

For a pure tensor with $m - 1$ factors $u = \log(1 + p)/p$ and one factor τ_s, trace in $E^{\otimes m}$ is the product of the factor traces. Its valuation is

$$(m - 1)v_p(d) + v_p(d/\ell) + \ell - 1 + sN.$$

Local arrows, Ism actions, and factor permutations preserve this number, including when labels require repeated use of one arrow. This separates the exponent-1 input from the exponent-j^2 input for $j > 1$, $j < \ell$. Raising $1 + pa$ to the power j^2, which scales its logarithm, is a different operation from replacing a by a^{j^2}.

7.4. **Why this is not yet a holomorphic-hull comparison.** Multiplying generators by coefficients of the maximal tensor order need not preserve their trace depths. Thus distinct pure orbits do not imply distinct module hulls. The upper-semicompatibility operation Ind3 in [62, Theorem 3.11(ii)] is a further operation, and has not been replaced here by an automorphism of one fixed Kummer module. Nor does changing an untilt's real-valued norm change the native trace in a fixed marked p-adic field.

A converse test is useful: trace preservation alone permits a common least-layer displacement, without establishing an Ind1 lift.

Lemma 7.7 (Common avoidance with fixed trace)**.** *Let k be a field of characteristic different from two, let V be a k-vector space, and let $t, l \in V^*$. Suppose $t(w) = 0$, $l(w) = 1$. For any nonzero $x, y \in V$, there is an automorphism F with $tF = t$ and $l(Fx), l(Fy) \neq 0$.*

Proof. Set $P(v) = v - l(v)w$. Choose a functional h nonzero on each nonzero vector among $P(x), P(y)$. Such a functional exists: separate the first vector; if that functional kills the second, separate the second and use either the new functional or the sum, according to its value on the first. Put $f = hP$, so $f(w) = 0$. The maps $F_z(v) = v + zf(v)w$ have inverses F_{-z} and preserve t. For $v = x, y$, the polynomial $l(v) + zf(v)$ is nonzero: if $l(v) = 0$, then $P(v) = v \neq 0$. Two nonzero affine polynomials have at most two roots altogether. The three distinct scalars $0, 1, -1$ therefore contain a valid choice of z. ☐

For finite Galois $E/\mathbb{Q}_p$ with $2 \leq e \leq p - 2$, one has $I_E = \pi^{1-e}\mathcal{O}_E$, the inverse different. Both $\mathrm{Tr} : I_E \to \mathbb{Z}_p$ and $\mathrm{Tr} : \mathcal{O}_E \to \mathbb{Z}_p$ are onto; the latter follows by tracing first through the tame ramified extension and then through the unramified one. On I_E/pI_E, trace modulo p is independent of any nonzero $\mathbb{F}_p$-coordinate of $I_E/\pi I_E$: the latter vanishes on $\mathcal{O}_E \subset \pi I_E$, whereas trace does not. The lemma gives an automorphism preserving trace modulo p and reaching the least valuation layer for two primitive vectors simultaneously. Choose a basis splitting integral trace. The residue matrix has first row $(1, 0, \ldots, 0)$; lift that row exactly and the others arbitrarily. The determinant is a unit, so this is an integral invertible lift preserving trace exactly. No lift to an absolute Galois automorphism is established by this construction. Determining that image, and then performing all source-prescribed hull operations, remains necessary for a global comparison. No IUT theorem or the abc conjecture has been disproved by these local nonreachability statements.

8. Effective bounds and finiteness in a specified Pell family

We now restrict to positive integers b, A, B, u, v, r_0, s, X and a prime $p \geq 37$ satisfying

$$b = Au^2, \quad b + 1 = Bv^2, \quad b + 2 = 3r_0^2, \quad b + 3 = s^2, \tag{32}$$

$$D = 3AB, \quad Z = b^2 + 3b + 1 = T_p(X), \quad D + 1 \leq X^2, \quad X^p \leq Z. \tag{33}$$

Here T_p is the Chebyshev polynomial of the first kind. These equations define the specified residual packet; no reduction of general abc counterexamples to it is asserted. Set

$$H = \log(b + 2), \quad L = \log(D + 1), \quad C_0 = 12^{331776}, \quad K_0 = \log(2C_0).$$

The packet immediately yields

$$(D + 1)^p \leq X^{2p} \leq Z^2 < (b + 2)^4, \qquad pL < 4H. \tag{34}$$

Theorem 2.2 of Bérczes–Evertse–Győry [6], specialized to $K = \mathbb{Q}$, $S = \{\infty\}$, and exponent two, gives the following published input. If $f \in \mathbb{Z}[t]$ is squarefree of degree $n \geq 3$, with coefficient height $\widehat{h} = \log \max(1, |a_0|, \ldots, |a_n|)$, and $f(x) = y^2$ in integers, then

$$\max\{\mathrm{h}(x), \mathrm{h}(y)\} \leq (4n)^{4096n^4} \exp(50n^4\widehat{h}), \quad \mathrm{h}(t) = \log \max(1, |t|). \tag{35}$$

Proposition 8.1 (Two effective constraints)**.** *Every packet* (32)–(33) *satisfies*

$$H \leq \exp(4300p^5), \qquad H \leq 2C_0 D^{4050}. \tag{36}$$

Proof. First put $f_p(t) = 4T_p(t) + 5$. The relation in (33) gives $f_p(X) = (2b+3)^2$. The identities $T_p' = pU_{p-1}$ and $T_p^2 - 1 = (t^2-1)U_{p-1}^2$ show that a repeated root of f_p would have $T_p = \pm 1$ and $T_p = -5/4$, a contradiction. If M_n is the sum of the absolute values of the coefficients of T_n, then $M_{n+2} \le 2M_{n+1} + M_n$ and $M_0 = M_1 = 1$. Induction gives $M_n \le 3^n$. Thus every coefficient of f_p has absolute value at most $4 \cdot 3^p + 5 \le 3^{p+2}$, and $\widehat{h} \le (p+2)\log 3 \le 4p$. Since $\log(4p) \le p$ for $p \ge 3$, (35) implies

$$H \le \log(2b+3) \le \exp\Big(4096p^4 \log(4p) + 50p^4\widehat{h}\Big) \le \exp(4300p^5).$$

For the second estimate, set $x = b+1$ and $Y = Duvr_0$. The first three equations in (32) give

$$Y^2 = D(x^3 - x). \tag{37}$$

The cubic $D(t^3 - t)$ is squarefree and has coefficient height $\log D$. Equation (35), with $n = 3$, gives

$$\log(b+1) \le 12^{4096\cdot 3^4} \exp(50 \cdot 3^4 \log D) = C_0 D^{4050}.$$

Finally $b + 2 \le (b+1)^2$ for $b \ge 1$, proving the second bound. □

Proposition 8.2 (Descent to the squarefree parameter)**.** *Write $D = d\tau^2$, where d is a positive squarefree integer and $\tau \ge 1$ is an integer. Then every packet satisfies*

$$H \le 2C_0 d^{4050}.$$

Proof. Equation (37) implies $\tau^2 \mid Y^2$. Comparing prime valuations gives $\tau \mid Y$. Write $Y = \tau y$ and cancel the nonzero factor τ^2 to obtain

$$y^2 = d(x^3 - x), \qquad x = b+1.$$

The polynomial $d(t^3 - t)$ has distinct roots and coefficient height $\log d$. Equation (35) gives $\log(b+1) \le C_0 d^{4050}$, and again $H \le 2\log(b+1)$. □

Combining (34) and (36) gives

$$p \ge \left(\frac{\log H}{4300}\right)^{1/5}, \qquad L < 4 \cdot 4300^{1/5} \frac{H}{(\log H)^{1/5}}, \tag{38}$$

$$D \ge \left(\frac{H}{2C_0}\right)^{1/4050}, \qquad p(\log H - K_0) < 16200H. \tag{39}$$

For example, the first upper bound follows by multiplying the lower bound for p by $L > 0$ and applying $pL < 4H$. For the final inequality use $\log H \le K_0 + 4050 \log D$ and $\log D < L$. In a form avoiding fractional powers, the former calculation is

$$(\log H)L^5 \le 4300(4H)^5.$$

These inequalities alone are compatible with unbounded height: they would force both the index and the squarefree kernel to grow, while $\log(D+1)/H$ tends to zero. The following argument supplies the missing absolute index bound and excludes such an unbounded sequence in this specified family. It does not freeze a field or an index in advance.

8.1. **An absolute index bound from normalized logarithmic forms.** The preceding effective height estimate can be combined with a different logarithmic-form argument to prove finiteness, uniformly in the index. For this argument retain only positive integers b, r_0, s, X and an integer $p \ge 3$ satisfying

$$b + 2 = 3r_0^2, \qquad b + 3 = s^2, \qquad Z = b^2 + 3b + 1 = T_p(X), \qquad X > 1. \tag{40}$$

Primality of p, the endpoint kernels A, B, and their congruences are not needed. Put

$$\varepsilon = 2 + \sqrt{3}, \quad \eta = s + r_0\sqrt{3}, \quad \delta = X + \sqrt{X^2 - 1}, \quad W = Z + \sqrt{Z^2 - 1}, \quad H = \log(b+2).$$

The Pell equation gives $\eta = \varepsilon^m$ for an integer $m \ge 1$. The Chebyshev identity gives $W = \delta^p$; δ need not be a fundamental unit. In fact $b \ge 22$: below 22 the fixed Pell equations leave only $b = 1, r_0 = 1, s = 2$, whereas $Z = 5 < T_p(X)$ for $p \ge 3, X \ge 2$. Here $T_p(X) \ge X^p$ follows by induction from its recurrence and $T_{n+1}(X) \ge T_n(X) > 0$.

We use Matveev's original Corollary 2.3 [66], with the normalized coefficient parameter in his equation (1.3). For a real number field of degree d, algebraic numbers $\alpha_i > 1$, and a nonzero form $\Lambda = \sum_{i=1}^{n} b_i \log \alpha_i$ with integral coefficients, take

$$A_i \ge \max\{d\mathrm{h}(\alpha_i), |\log \alpha_i|, 0.16\}, \qquad B_* = \max\left\{1, \max_i \frac{|b_i|A_i}{A_n}\right\}, \qquad \Omega = \prod_i A_i.$$

Then

$$\log|\Lambda| > -C_1(n,1)d^2\Omega\log(ed)\log(eB_*), \qquad C_1(n,1) \le 2^{6n+20}. \tag{41}$$

Unlike Theorem 2.1 of that source, this corollary does not require the logarithms to be independent. It also does not require A_n to be the largest parameter. Both points matter when δ varies.

Lemma 8.3. *For* (40),

$$1 < \frac{\eta^4}{8W} < 1 + \frac{2}{b}, \qquad W > \eta^2, \qquad \log W < 3H.$$

Consequently the real linear form

$$\Lambda = 4m\log\varepsilon - 3\log 2 - p\log\delta = \log\frac{\eta^4}{8W} \tag{42}$$

satisfies $0 < \Lambda < 2/b$ *and* $-\log\Lambda > H/2$.

Proof. For $t > 1$, write $f(t) = t + \sqrt{t^2-1}$, so $2t - 1 < f(t) < 2t$. The Pell identities give

$$\eta^2 = f(2b+5), \quad \eta^4 = f(8b^2+40b+49), \quad W = f(Z).$$

Thus

$$\eta^4 > 16b^2 + 80b + 97 > 16Z > 8W,$$

and

$$\frac{\eta^4}{8W} - 1 < \frac{32b+90}{16b^2+48b+8} < \frac{2}{b}.$$

The final comparison is equivalent to $32b^2 + 90b < 32b^2 + 96b + 16$. Also $W > 2b^2 + 6b + 1 > 4b + 10 > \eta^2$ for $b \ge 22$. Finally $W < 2Z < 2(b+2)^2$ gives $\log W < \log 2 + 2H < 3H$. The strict inequality $\log(1+u) < u$ for $u > 0$ proves the asserted bounds for Λ. Since $(b/2)^2 > b+2$, $-\log\Lambda > \log(b/2) > H/2$. ☐

Theorem 8.4 (An absolute index bound)**.** *Every tuple* (40) *satisfies* $p < 2^{59}$.

Proof. Use the real field $K = \mathbb{Q}(\sqrt{3}, \sqrt{X^2-1})$ of degree $d \le 4$, and in this order choose

$$(\alpha_1, \alpha_2, \alpha_3) = (\varepsilon, 2, \delta), \qquad (b_1, b_2, b_3) = (4m, -3, -p),$$
$$A_1 = 2\log\varepsilon, \qquad A_2 = 4\log 2, \qquad A_3 = 2\log\delta.$$

For integer $X \ge 2$, $X^2 - 1$ lies strictly between two consecutive squares. Hence δ is a quadratic algebraic integer with minimal polynomial $t^2 - 2Xt + 1$ and conjugate δ^{-1}. The absolute heights are

$$\mathrm{h}(\varepsilon) = \tfrac{1}{2}\log\varepsilon, \qquad \mathrm{h}(2) = \log 2, \qquad \mathrm{h}(\delta) = \tfrac{1}{2}\log\delta.$$

These identities, $d \le 4$, and $\delta \ge \varepsilon$ verify all the height and 0.16 conditions in (41). The fields may coincide; no independence assertion is used.

By Lemma 8.3, the normalized coefficients obey

$$\frac{4mA_1}{A_3} = \frac{4p\log\eta}{\log W} < 2p, \qquad \frac{3A_2}{A_3} = \frac{6\log 2}{\log\delta} < 4 \le 2p.$$

For the latter strict inequality use $\delta^2 \ge \varepsilon^2 > 8$. The remaining entries are p and one, so $B_* \le 2p$. Using the same chosen parameters, rather than changing them when estimating B_*, gives

$$\Omega = 16\log\varepsilon\log 2\,\frac{\log W}{p} < 96\frac{H}{p}.$$

Apply (41), with $C_1(3,1) \le 2^{38}$, $d^2 \le 16$, and $\log(ed) < 3$. Comparison with the upper approximation gives

$$\frac{H}{2} < -\log\Lambda < 2^{38}\cdot 16\cdot 3\cdot 96\,\frac{H}{p}\log(2ep).$$

Cancel the positive H and multiply by p to obtain

$$p < 9\cdot 2^{48}\log(2ep) < 2^{52}\log(2ep). \tag{43}$$

With $t = p/2^{52} > 0$, this reads $t < 53\log 2 + 1 + \log t$. The elementary inequality $\log t \le t/2$ gives $t < 2(53\log 2+1) < 108 < 128$. Thus $p < 2^{59}$, independently of the height and all varying fields. □

Corollary 8.5 (Effective finiteness of the specified family)**.** *The tuples* (40) *form a finite set. Every such tuple satisfies*

$$H < \exp(4300\cdot 2^{295}), \qquad b+2 < \exp\big(\exp(4300\cdot 2^{295})\big). \tag{44}$$

In particular the more restrictive residual (32)–(33) *is effectively finite.*

Proof. The first part of the proof of Proposition 8.1 applies to $4T_p(X)+5 = (2b+3)^2$ for every integer $p \ge 3$, giving $H \le \exp(4300p^5)$. Combine it with Theorem 8.4 and exponentiate. The resulting bound leaves finitely many positive integers b. The square equations determine r_0, s, the index is bounded, and $X^p \le Z$ bounds X. If the endpoint data are also retained, then $A \mid b$, $B \mid b+1$, and their positive square multipliers have only finitely many possibilities. □

These bounds are effective but far beyond feasible enumeration. They prove finiteness without listing the exceptional tuples. They improve a fixed-index finiteness statement: the same absolute bound applies while both the index and the quadratic field vary.

Corollary 8.6 (The product Pell unit outside the finite set)**.** *Suppose only that b, r_0, s are positive integers with $b+2 = 3r_0^2$ and $b+3 = s^2$. Put $Z = b^2+3b+1$ and write $Z^2-1 = D_0y^2$ with $D_0 > 1$ squarefree and y a positive integer. Outside the bound in* (44)*, $Z + y\sqrt{D_0}$ is the fundamental positive norm-one unit of the integral Pell group $\mathbb{Z}[\sqrt{D_0}]$.*

Proof. Write $Z + y\sqrt{D_0} = \varepsilon_{D_0}^k$, $k \ge 1$. If $k = 2j$, then $Z = T_2(T_j(X_0))$ for the integer first coordinate $X_0 > 1$ of ε_{D_0}. Thus, with $Y = T_j(X_0)$,

$$2Y^2 = Z+1 = (b+1)(b+2) = 3(b+1)r_0^2.$$

Since $3 \nmid b+1$, the right side has odd 3-adic valuation and the left side has even 3-adic valuation, a contradiction. Therefore k is odd. If $k > 1$, it is at least three and (40) holds with $p = k, X = X_0$. Corollary 8.5 bounds b, proving the assertion. □

This last statement concerns the integral Pell group, without identifying it with the unit group of a larger maximal order. A fundamental moving Pell unit can still have a large regulator relative to its discriminant. Neither finiteness of the nonfundamental cases nor the absolute index bound gives the radical estimate (1), and no reduction of arbitrary abc triples to (40) has been established.

8.2. **A stronger bound from a fixed-norm element.** For the height and kernel estimates in this subsection use only (32), and impose the additional conditions that A, B are squarefree and

$$A \equiv 22 \pmod{24}, \qquad B \equiv 23 \pmod{24}.$$

They specify the squarefree branch under consideration; the more general packet estimates above did not require them. No Chebyshev decomposition is needed except in the displayed consequence for p. In particular $b \ge 22$, $A, B > 1$, $3 \nmid AB$, $\gcd(A,B) = 1$, and $D = 3AB$ is squarefree. Let R_A, R_B denote the regulators of the maximal real quadratic orders.

We use two general tools from the proof of the new result [69, Theorem 2.4 and Lemma 2.5], without applying its irreducible-cubic theorem to a split polynomial. First, in a totally real degree-four field, for a multiplicative group generated modulo torsion by three independent elements g_1, g_2, g_3 and $g \neq 1$ in that group,

$$-\log|1-g| \le C_* \log\max\{2, \mathrm{h}(g)\} \prod_{i=1}^{3} \max\{1, \mathrm{h}(g_i)\}, \tag{45}$$

where $C_* > 0$ is effective and absolute. Second, for an algebraic integer ω of absolute norm $q > 1$ in a real quadratic field K, the group $\mathcal{O}_K^\times\langle\omega\rangle$ has independent elements ξ_1, ξ_2 generating a subgroup modulo torsion of index $1 \le N \le 2$ such that

$$\omega^N = \zeta\xi_1^{e_1}\xi_2^{e_2}, \quad \zeta \in \{\pm 1\}, \qquad \mathrm{h}(\xi_1)\mathrm{h}(\xi_2) \le \frac{3}{8} R_K \log q. \tag{46}$$

Here the set S consists of both real places and no finite place, so the all-or-none condition in Lemma 2.5 is satisfied and $q' = q$. The small-generator estimate follows from the exterior-product theorem of Akhtari–Vaaler [5, Theorem 1.2]; its joint use of the unit and nonunit generators is what saves a regulator factor. Finally Landau's fixed-degree estimate gives an effective absolute C_Q with

$$R_A \le C_Q\sqrt{A}\log(2A), \qquad R_B \le C_Q\sqrt{B}\log(2B). \tag{47}$$

Theorem 8.7. *There is an effective absolute constant C_A such that every packet in this subsection satisfies*

$$H \le C_A\sqrt{A}[\log(2A)]^2.$$

Proof. Put $\varphi = (1+\sqrt{5})/2$ and $\rho = \frac{1}{2}\log\varphi$. Every nonzero algebraic number of degree at most two which is not a root of unity has height at least ρ. Indeed a rational such number has height at least $\log 2$. For a quadratic number, its primitive minimal polynomial has Mahler measure at least two if the leading or constant coefficient has absolute value at least two. Otherwise it is a quadratic unit. For norm minus one a nonzero integral trace gives a root of absolute value at least φ; trace zero is reducible. For norm plus one, traces of absolute value at most two give roots of unity or reducibility, and the other traces give a root of absolute value at least φ^2. The formula $2\mathrm{h} = \log M$ proves the height lower bound.

In $K = \mathbb{Q}(\sqrt{A})$ take

$$\omega = s + u\sqrt{A} = \sqrt{b+3} + \sqrt{b}, \qquad \eta = s + r_0\sqrt{3} = \sqrt{b+3} + \sqrt{b+2}.$$

Their respective quadratic norms are three and one. The latter is a positive power of $\varepsilon_3 = 2+\sqrt{3}$. Apply (46) with $q = 3$. The field $F = \mathbb{Q}(\sqrt{A}, \sqrt{3})$ has degree four, since $A > 1$ is squarefree and $A \neq 3$. Define $\gamma = \omega/\eta$. At the chosen real embedding,

$$0 < \gamma < 1, \qquad 1 - \gamma = \frac{2}{(\sqrt{b+2}+\sqrt{b})(\sqrt{b+3}+\sqrt{b+2})} < \frac{1}{2b}.$$

As $1 \le N \le 2$, it follows that $0 < 1 - \gamma^N < 1/b$.

The group $\langle -1, \xi_1, \xi_2, \varepsilon_3\rangle$ has rank three modulo torsion and contains γ^N. Independence follows because $K \cap \mathbb{Q}(\sqrt{3}) = \mathbb{Q}$, and a rational power of ε_3 must equal its conjugate, forcing that power to be zero. The other two generators are independent by construction. Moreover $\mathrm{h}(\varepsilon_3) < 1$ and $\max(1, \mathrm{h}(\xi_i)) \le \mathrm{h}(\xi_i)/\rho$, giving

$$\prod_i \max(1, \mathrm{h}(g_i)) \le \frac{3\log 3}{8\rho^2} R_A.$$

Both conjugates of ω, η are positive. Their smaller conjugates are $3/\omega < 1$ and $1/\eta < 1$, so their global heights are $\frac{1}{2}\log\omega$ and $\frac{1}{2}\log\eta$. Thus

$$\mathrm{h}(\gamma^N) \le 2\big(\mathrm{h}(\omega) + \mathrm{h}(\eta)\big) < \log(4(b+3)) < H + 2 \le 3H.$$

Applying (45) gives $\log b \ll R_A\log(3H)$, with an effective absolute constant. Since $H \le \log b + \log 3$, we have $H \le M\log(3H)$ for $M = C_1\max(1, R_A) \ge 1$ and an absolute C_1.

For any positive X, M with $X \le M\log(3X)$, one has

$$X \le 2M\log(3M). \tag{48}$$

To prove it, write $t = X/M$ and use $\log t \le t/2$, which follows from $\log(t/2) \le t/2 - 1$ and $\log 2 < 1$. Then $t \le \log(3M) + t/2$, proving (48). Equation (47) bounds M by $C_2\sqrt{A}\log(2A)$ and $\log(3M)$ by $C_3 \log(2A)$. Applying (48) proves the theorem. □

Replacing ω by $s + v\sqrt{B}$, of norm two, gives the analogous bound with $\sqrt{B}\log^2(2B)$. For this endpoint, however, a stronger published input is available and was already identified in the repository. The unit

$$\varepsilon_B = (b+2) + vs\sqrt{B}$$

is the fundamental positive norm-one unit: its first coordinate is $3r_0^2$, so Bennett–Walsh [13, Theorem 1.2] says that its unit index is the first index at which three divides the first Pell coordinate. If the fundamental first coordinate is 1 or -1 modulo three, the Chebyshev recurrence gives only 1 or alternating signs modulo three, respectively. A first coordinate divisible by three therefore forces the fundamental first coordinate itself to be divisible by three, and the index is one. Since $B \equiv 3 \pmod 4$, the order $\mathbb{Z}[\sqrt{B}]$ is maximal and a norm-minus-one unit is impossible modulo four. Consequently

$$H < R_B \le C_Q\sqrt{B}\log(2B). \tag{49}$$

We use this established endpoint result, rather than presenting it as a new consequence of the recent preprint.

Corollary 8.8. *There is an effective absolute $K \ge 1$ such that these packets satisfy*

$$H^4 \le KD[\log(2D)]^6. \tag{50}$$

In particular,

$$D \ge \frac{H^4}{K5^6(\log H)^6}, \qquad \log D \ge 4\log H - 6\log\log H - \log(K5^6). \tag{51}$$

When the latter lower bound is positive,

$$p < \frac{4H}{4\log H - 6\log\log H - \log(K5^6)}.$$

Proof. Square Theorem 8.7 and (49) and multiply. Since $AB = D/3$ and $A, B \le D$, the result is (50). If $D \ge H^4$, the first inequality in (51) is immediate because $H > 2$ and $K \ge 1$. Otherwise $\log(2D) \le 5\log H$, giving that inequality from (50). Taking logarithms proves the second one. Finally use $p\log(D+1) < 4H$. □

The absolute index bound already excludes an unbounded sequence with $p \ge 3$. The kernel estimate (50) remains useful without a Chebyshev decomposition, including cases where the product Pell unit is fundamental. Its polynomial dependence on H does not supply the radical bound required for abc.

Proposition 8.9 (A general split-cubic estimate)**.** *For positive squarefree d, an integral solution of $y^2 = d(x^3 - x)$ with $x \ge 2$ satisfies*

$$\log x \ll d^{2/3}[\log(2d)]^3$$

with an effective absolute implied constant.

Proof. Write $x - 1 = a_0u^2$, $x = b_0v^2$, $x + 1 = c_0w^2$ with positive squarefree coefficients. Adjacent coefficients are coprime and $g = \gcd(a_0, c_0) \in \{1, 2\}$, so $a_0b_0c_0 = g^2d \le 4d$. No two coefficients are equal: equality for adjacent ones would force positive squares to differ by one; equality at the ends forces the common coefficient to divide two and gives the same impossibility or a difference of squares equal to two. The quadratic fields of the three products a_0b_0, b_0c_0, a_0c_0 are therefore distinct.

The three positive norm-one units

$$U = 2x - 1 + 2uv\sqrt{a_0b_0}, \quad V = 2x + 1 + 2vw\sqrt{b_0c_0}, \quad W = x + uw\sqrt{a_0c_0}$$

satisfy $1 < V/U, V/(2W), 2W/U < 1 + 2/x$. Indeed $2t - 1 < t + \sqrt{t^2 - 1} < 2t$ for $t > 1$ gives these inequalities directly at $t = 2x - 1, 2x + 1, x$. Each ratio is generated by two quadratic fundamental units and two; two may also be adjoined for the first ratio. Distinct quadratic fields give independence, and two is an independent nonunit. The global height of each ratio is at most $\log x + O(1)$. Choose the pair whose radicands share $m = \min(a_0, b_0, c_0)$. Landau's bound gives a product of regulators at most

$$C\sqrt{a_0b_0c_0m}\,[\log(2a_0b_0c_0)]^2 \leq C(a_0b_0c_0)^{2/3}[\log(2a_0b_0c_0)]^2.$$

Passing to squarefree radicands can only decrease the discriminant bound. Apply (45) to the selected ratio and then (48), adjusting absolute constants for bounded x. The latter step adds one logarithmic power. Using $a_0b_0c_0 \leq 4d$ proves the claim. □

9. Applicability of recent elliptic-curve estimates

9.1. **A denominator lower bound.** For the Frey curve $E : y^2 = x(x-a)(x+b)$ attached to an abc triple, let $Q = a^2 + ab + b^2$. Its invariant is

$$j_E = \frac{256Q^3}{(abc)^2}.$$

Let n, d be the positive numerator and denominator in lowest terms.

Proposition 9.1. *For every such triple,*

$$c^4 \leq 1024d, \qquad n \leq 256c^6. \tag{52}$$

Proof. Primitivity implies $\gcd(Q, abc) = 1$. Indeed, reduction modulo a prime dividing any one of a, b, c leaves in Q the square of a nonzero other coordinate. Therefore $g = \gcd(256Q^3, (abc)^2)$ divides 256, and $(abc)^2 = gd \leq 256d$. Since $(a-1)(b-1) \geq 0$, we have $ab \geq c-1$; as $c \geq 2$, this gives $2ab \geq c$. Squaring $2abc \geq c^2$ proves $c^4 \leq 4(abc)^2 \leq 1024d$. Finally $Q = c^2 - ab \leq c^2$, so $n \leq 256Q^3 \leq 256c^6$. □

Pasten's 2026 small-denominator theorem [70, Theorem 1.2] applies under a bound $\mathrm{den}(j_E) \leq A_0(\log \mathrm{num}(j_E))^{B_0}$ with fixed positive A_0, B_0. For Frey curves, (52) would imply

$$c^4 \leq 1024A_0(\log 256 + 6\log c)^{B_0}.$$

A positive power of c dominates every fixed power of its logarithm. Thus only bounded c, and consequently finitely many positive triples, can satisfy this hypothesis for fixed A_0, B_0. This is a precise restriction on direct applicability, not a counterexample to the theorem or to other Szpiro strategies.

9.2. **The August 2026 Mordell estimate.** The preprint [69, Theorem 1.3], submitted August 24, 2026, gives an effective bound for $y^2 = x^3 + k$, $k \neq 0$, using

$$\underline{k} = \prod_{q|k} q^{\min(2, v_q(k))}.$$

The stated estimate is

$$\log\max(|x|,|y|) \ll \underline{k}^{1/2}\log^2(2\underline{k})\log(2|k|)\log(\underline{k}\log(3|k|)). \tag{53}$$

We record this new result without treating its publication date as a substitute for checking its hypotheses or proof.

Directly substituting $c_6^2 = c_4^3 - 1728\Delta$ and $\Delta = 16(abc)^2$ gives $k = -27648(abc)^2$. Then $\underline{k}^{1/2} \ll r$, while both $\log(2|k|)$ and $\log\max(|c_4|,|c_6|)$ are comparable with h. The resulting right-hand side has the shape

$$r\log^2(2r)\,h\,\log(2r\log(3c)).$$

The height remains a multiplicative factor on the right; cancellation does not yield an upper bound for c in terms of r. The integral-j result in Theorem 1.6 of the same paper concerns the height of the *curve*, and cannot be substituted for a point-height bound.

There is a more primitive Mordell point, but it preserves the same height factor in this substitution.

Proposition 9.2. *Put $m = abc/2$ and $Y = (a-b)(2a+b)(a+2b)/2$. These are integers, and*

$$Y^2 = Q^3 - 27m^2, \qquad \gcd(Q,m) = 1, \qquad c_4 = 4^2 Q, \quad c_6 = 4^3 Y.$$

There is no integer $z \geq 2$ with $z^2 \mid Q$ and $z^3 \mid Y$. The associated binary cubic is split over $\mathbb{Z}$:

$$U^3 - 3QUV^2 + 2YV^3 = (U - (a-b)V)(U - (a+2b)V)(U + (2a+b)V).$$

It remains reducible after every change of variables in $\mathrm{GL}_2(\mathbb{Z})$.

Proof. At least one of a, b, c is even, so m is integral. If one of a, b is even then one of $2a+b, a+2b$ is even; if both are odd then $a-b$ is even. This proves integrality of Y. Expansion gives

$$4Q^3 - \big((a-b)(2a+b)(a+2b)\big)^2 = 27(abc)^2$$

and the asserted cubic factorization. The standard invariant formulas give the two scalings. Coprimality follows from $\gcd(Q, abc) = 1$. If the proposed z existed, then $z^6 \mid 27m^2$ while $\gcd(z,m) = 1$; thus $z^6 \mid 27$, impossible for $z \geq 2$. An invertible linear change of variables takes each nonzero linear factor to another nonzero linear factor, preserving reducibility. ☐

The normalized parameter is therefore $k = -27m^2$ and its prime support is still that of $3m$. Removing the fixed powers of four does not remove the factor comparable with h on the right of (53). The split cubic also does not meet an irreducible Thue-form hypothesis.

There is also a strict obstruction to one possible transfer from (37). With $U = Dx$ and $V = DY$, it becomes $E_D : V^2 = U^3 - D^2 U$, of j-invariant 1728.

Proposition 9.3. *For $D, k \neq 0$, there is no nonconstant algebraic morphism over $\overline{\mathbb{Q}}$ from E_D to the nonsingular Mordell curve $M_k : v^2 = u^3 + k$.*

Proof. Their geometric endomorphism algebras are respectively $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$: the automorphisms $(x,y) \mapsto (-x, iy)$ and $(u,v) \mapsto (\zeta_3 u, v)$ give these quadratic actions, and in characteristic zero the endomorphism algebra of an elliptic curve has dimension at most two over $\mathbb{Q}$. A nonconstant morphism, translated to send the origin to the origin, is an isogeny. An isogeny induces an isomorphism of rational endomorphism algebras by conjugation. The two imaginary quadratic fields are not isomorphic, a contradiction; see also [71, Chapters III and VI]. ☐

The proposition excludes this morphism strategy, even if k depends on D. It does not exclude other Diophantine encodings or extending the new logarithmic-form methods to a different family.

10. Three auxiliary points and the remaining endpoint cost

The constructions here apply to every positive primitive abc triple. Write its canonical cube decompositions as

$$a = Au^3, \qquad b = Bv^3, \qquad c = Cw^3, \tag{54}$$

where A, B, C are positive cube-free integers. These coefficients, and the three original endpoints, are pairwise coprime within their respective triples. Put

$$R_0 = \mathrm{rad}(ABC), \qquad R_3 = \prod_{\substack{p \mid abc \\ 3 \mid v_p(abc)}} p, \qquad H = \log c.$$

Thus $\mathrm{rad}(abc) = R_0 R_3$ and $\gcd(R_0, R_3) = 1$. The subscript on R_0 denotes the support left after removing cubes, not the primes whose exponents are divisible by three.

10.1. **Actual integral points and their prime costs.** For each omitted endpoint n_i, define the following integers:

i	n_i	d_i	r_i	x_i	y_i
a	a	vw	BC	$4vwBC$	$4BC(a+2b)$
b	b	uw	AC	$4uwAC$	$4AC(2a+b)$
c	c	uv	AB	$-4uvAB$	$4AB(a-b)$.

Set $k_i = 16(n_i r_i)^2$.

Proposition 10.1. *Every row gives an integral point $y_i^2 = x_i^3 + k_i$ with $|x_i|^3 \geq 32c$. If $\rho_i = \operatorname{rad}(2n_i r_i)$, then*

$$\rho_a\rho_b\rho_c = 4^{\mathbf{1}_{3|v_2(abc)}} R_0^3 R_3, \qquad \min_i \rho_i \leq 2^{2/3} R_0 R_3^{1/3}. \tag{55}$$

Proof. For the first row use $(a+2b)^2 - a^2 = 4bc$ and $bc = (vw)^3 BC$; the second is the same identity with a, b interchanged. For the last, $(a-b)^2 - c^2 = -4ab$ gives the negative sign of x_c. The three values of $|x_i|^3$ are respectively $64bc(BC)^2$, $64ac(AC)^2$, and $64ab(AB)^2$. The first two exceed $32c$, and the last is at least $32c$ because $ab \geq c - 1 \geq c/2$.

An odd prime in R_0 occurs in all three $n_i r_i$: once in the omitted endpoint and twice in the coefficient products. A prime in R_3 occurs only in its own omitted endpoint. The prime 2 divides every ρ_i, and exactly one endpoint is even. If its exponent is divisible by three this adds a factor 4 to the preceding product; otherwise no correction is needed. This proves the exact identity; the minimum is at most the geometric mean. □

These are auxiliary curves of j-invariant zero. Their standard 3-isogeny is

$$(x, y) \longmapsto \left(\frac{x^3 + 4k}{x^2}, \frac{y(x^3 - 8k)}{x^3}\right), \qquad E_k \longrightarrow E_{-27k}. \tag{56}$$

On the unextracted omit-b point with $k = 16(abc)^2$, it gives $(4Q, 4T)$, where

$$Q = a^2 + ab + b^2, \qquad T = (a-b)(2a+b)(a+2b).$$

These are $(c_4/4, c_6/8)$ for the original Frey curve. The assertion is an identity involving its invariants; the Frey curve itself has not become isogenous to a $j = 0$ curve. After cube extraction, the same image is generally rational rather than integral.

10.2. **A small field discriminant with an exact growing order index.** Form the monic cubic

$$G_i(Z) = Z^3 - 3x_i Z + 2y_i, \qquad \operatorname{disc} G_i = -108k_i = -1728(n_i r_i)^2. \tag{57}$$

For the omit-b row, put $\alpha = \sqrt[3]{A^2C}$ and $\beta = \sqrt[3]{AC^2} = \alpha^2/A$. Then $\tau_b = -2(u\alpha + w\beta)$ is a root. The omit-a row uses (B, C, v, w) instead, while the omit-c row has root $\tau_c = -2(u\sqrt[3]{A^2B} - v\sqrt[3]{AB^2})$.

If the selected product r_i exceeds one, its pure cubic is irreducible. At a prime of either cube-free coefficient, the valuation of the radicand is not divisible by three. Independence of $1, \alpha, \alpha^2$ then makes τ_i nonrational, so it generates the same degree-three field K_i. In particular the root fields are

$$K_a = \mathbb{Q}(\sqrt[3]{B^2C}), \qquad K_b = \mathbb{Q}(\sqrt[3]{A^2C}), \qquad K_c = \mathbb{Q}(\sqrt[3]{A^2B}),$$

with signature $(1, 1)$. Reducible rows are exactly those with $r_i = 1$.

Proposition 10.2. *In an irreducible row, put $S_i = \operatorname{rad}(r_i)$. There is an order $\mathcal{O}_i \subset K_i$ with*

$$[\mathcal{O}_i : \mathbb{Z}[\tau_i]] = 8n_i, \qquad |\Delta_{K_i}| \mid 27S_i^2. \tag{58}$$

More precisely, there is a larger order $\mathcal{O}_i'$ such that

$$[\mathcal{O}_{K_i} : \mathbb{Z}[\tau_i]] = 8n_i(r_i/S_i)[\mathcal{O}_{K_i} : \mathcal{O}_i']. \tag{59}$$

Proof. Write the two selected coefficients as E, F, their cube bases as h, j, and $r = EF$. The lattice $\mathcal{O} = \mathbb{Z}\{1, \alpha, \beta\}$ is an integral order, because

$$\alpha^2 = E\beta, \quad \beta^2 = F\alpha, \quad \alpha\beta = r.$$

Its trace matrix is $\left(\begin{smallmatrix} 3 & 0 & 0 \\ 0 & 0 & 3r \\ 0 & 3r & 0 \end{smallmatrix}\right)$, and hence its discriminant is $-27r^2$. The coordinate matrices of $1, \tau, \tau^2$ in this basis are

$$M_+ = \begin{pmatrix} 1 & 0 & 8hjr \\ 0 & -2h & 4Fj^2 \\ 0 & -2j & 4Eh^2 \end{pmatrix}, \qquad M_- = \begin{pmatrix} 1 & 0 & -8hjr \\ 0 & -2h & 4Fj^2 \\ 0 & 2j & 4Eh^2 \end{pmatrix}.$$

Their determinants are $8(Fj^3 - Eh^3)$ and $-8(Eh^3 + Fj^3)$, respectively. The absolute value is $8n_i$ in the corresponding row. Since τ is an integral generator of a cubic field, this gives the first equality of (58).

Write $E = E_1E_2^2$, $F = F_1F_2^2$ with the four factors squarefree and pairwise coprime. Put $s = E_2F_1$, $t = E_1F_2$, $\theta = \sqrt[3]{st^2}$, and $\eta = \theta^2/t$. The lattice $\mathcal{O}' = \mathbb{Z}\{1, \theta, \eta\}$ is an order by $\theta^2 = t\eta$, $\eta^2 = s\theta$, $\theta\eta = st$. Its discriminant is $-27(st)^2 = -27S_i^2$. Furthermore $\alpha = E_2\theta$, $\beta = F_2\eta$, so $[\mathcal{O}' : \mathcal{O}] = E_2F_2 = r_i/S_i$. The discriminant-index formula and multiplication of indices prove the remaining claims. □

Thus the small discriminant concerns the field, while the original monogenic order has an index containing the unbounded endpoint n_i. Replacing its polynomial discriminant by the field discriminant in a point-height argument would discard this factor.

10.3. **A full-family relative height estimate.**

Theorem 10.3. *There is an effective absolute constant $C_0 > 0$ such that every positive primitive abc triple, and each of its three rows, satisfies*

$$H \le C_0 S_i[\log(2S_i)]^2 L_i \log(3S_iL_i), \qquad L_i = \log(2n_ir_i), \quad S_i = \mathrm{rad}(r_i). \tag{60}$$

Proof. We use the stated unconditional Theorem 1.7 and the reduction in Section 4 of Pasten's 2026 preprint [69], together with its classical degree-three regulator bound. We keep the actual pure cubic root field, rather than bounding its regulator using $\operatorname{disc} G_i$.

First suppose $r_i > 1$ and put $D = 1728(n_ir_i)^2$. The reduction in that section supplies $\gamma \in \mathrm{GL}_2(\mathbb{Z})$ such that the binary cubic $F = G_i \circ \gamma$ has coefficient height at most $\sqrt{D}$. For $(u_0, v_0) = \gamma^{-1}(1, 0)$ one has $F(u_0, v_0) = 1$ and $\gcd(u_0, v_0) = 1$. The fractional linear change of root leaves the cubic field K_i unchanged. Theorem 1.7, applied with empty prime list, unit right side, and just the two archimedean places of K_i, gives, for $X = \max(3, |u_0|, |v_0|)$,

$$\log X \ll \log(2D) + \mathcal{R}_i \log(2D)\log(2 + 2\mathcal{R}_i\log(2D)), \tag{61}$$

where $\mathcal{R}_i$ is the ordinary regulator. The degrees and the place count are fixed, so the implied constant is absolute and effective. Hessian and Jacobian recovery in the same section gives

$$\log\max(2, |x_i|, |y_i|) \ll \log|k_i| + \log X.$$

These covariants recover the point, not just the auxiliary curve. Proposition 10.1, together with $\log(2D) \le 12L_i$ and $\log|k_i| \le 4L_i$, therefore yields

$$H \ll \mathcal{R}_i^* L_i \log(3\mathcal{R}_i^* L_i), \qquad \mathcal{R}_i^* = \max(1, \mathcal{R}_i).$$

Here all logarithm arguments are bounded away from one because $L_i \ge \log 2$.

The degree-three Landau bound, with class number at least one, and Proposition 10.2 imply

$$\mathcal{R}_i^* \ll S_i[\log(2S_i)]^2.$$

Substitution loses no further logarithmic power. Indeed, for $B = \log(2S_i)$ and $z = S_iL_i \ge \log 2$, one has $B \le 2S_i$ and $L_i \ge 1/2$, hence $S_iB^2L_i \le 16z^3$. Thus $\log(3CS_iB^2L_i) \ll_C \log(3S_iL_i)$ for any fixed positive C. This proves (60) in the irreducible case.

If $r_i = 1$, the selected endpoints are cubes. In a plus row their difference is $n_i = w^3 - h^3$ with $w > h \ge 1$. Thus $n_i \ge w^2$ and $H = 3\log w \le 3L_i/2$. In the minus row $n_i = c$, so $H \le L_i$. As

$S_i = 1$ and $L_i \geq \log 2$, enlarging one absolute constant proves the same bound in all reducible cases. $\square$

Cube-freeness gives $r_i \leq S_i^2$, so L_i may be replaced on the right by $\log(2n_i) + 2\log S_i$. Also

$$S_a S_b S_c = R_0^2, \qquad \min_i S_i \leq R_0^{2/3}. \tag{62}$$

Choosing the omitted endpoint $m = \min(a,b)$ instead gives

$$H \ll R_0[\log(2R_0)]^2 M \log(3R_0 M), \qquad M = \log(2m) + 2\log R_0. \tag{63}$$

It is not legitimate to cancel an unknown height from the right: L_i may be comparable to H. For fixed bounded R_0, (63) does imply $\log m \gg_{R_0} H/\log H$ along an unbounded sequence. A stronger fixed-support statement follows next, with an ineffective threshold.

10.4. **One common curve and its three torsion translates.** Put $N = ABC$ and $t_a = u, t_b = v, t_c = w$. Since $n_i r_i = N t_i^3$, the points

$$Q_i = (X_i, Y_i) = (x_i/t_i^2, y_i/t_i^3)$$

all lie on the same nonsingular curve

$$E_N : \quad Y^2 = X^3 + 16N^2. \tag{64}$$

Primitivity gives $\gcd(t_i, d_i r_i) = 1$, so their reduced X-denominators are exactly

$$\operatorname{den}(X_i) = (t_i/\gcd(t_i,2))^2. \tag{65}$$

Indeed the numerator is $\pm 4 d_i r_i$, and the only cancellation with t_i^2 is $\gcd(t_i^2, 4) = \gcd(t_i, 2)^2$.

Proposition 10.4. *The point $T_0 = (0, 4N)$ has order three on E_N, and*

$$Q_b + T_0 = Q_c, \qquad Q_b - T_0 = -Q_a.$$

Proof. The horizontal tangent at T_0 gives $2T_0 = -T_0$. The slopes from T_0 and $-T_0$ to Q_b are respectively

$$\frac{2Au^2}{vw}, \qquad \frac{2Cw^2}{uv}.$$

Squaring the first and subtracting X_b gives $-4ABuv/w^2 = X_c$; the ordinate from the line through T_0 is $4AB(a-b)/w^3 = Y_c$. The second gives $4BCvw/u^2 = X_a$ and ordinate $-4BC(a+2b)/u^3 = -Y_a$. These computations use only $Au^3 + Bv^3 = Cw^3$ and positive denominators. $\square$

Writing $V = uvw$, the three points $Q_b, Q_c, -Q_a$ have the same image $(4Q/V^2, 4T/V^3)$ under (56). Their changing denominators describe three translates of one rational point; an integral-point bound for a fixed curve does not bound them.

Theorem 10.5 (Asymptotic balance for bounded residual support)**.** *Fix $R_* \geq 1$. For every $\varepsilon > 0$, all sufficiently large positive primitive abc triples with $R_0 \leq R_*$ satisfy*

$$\min(a,b) \geq c^{1-\varepsilon}.$$

Equivalently, along any such sequence with $c \to \infty$, $\log \min(a,b)/\log c \to 1$. The threshold may depend on R_*, ε; the proof does not make it effective.

Proof. For a fixed elliptic curve over $\mathbb{Q}$, Siegel's local approximation theorem, with the even function x, the point at infinity, and the real place, states that

$$\frac{\log \max(1, |x(P)|)}{h_x(P)} \longrightarrow 0 \quad \text{as } h_x(P) \longrightarrow \infty, \quad P \in E(\mathbb{Q}). \tag{66}$$

Here $h_x(P) = \log \max(|A_x|, B_x)$ for $x(P) = A_x/B_x$ in lowest terms, $B_x > 0$. This is [71, Theorem IX.3.1 and Example IX.3.3], not just finiteness of integral points. Since $N = ABC \leq R_0^2 \leq R_*^2$, only finitely many curves (64) occur. Formula (66) is uniform over this finite list.

By symmetry suppose $a = m \leq b$, and set $g = \gcd(u, 2) \in \{1, 2\}$. Equation (65) gives

$$X_a = A_x/B_x, \qquad A_x = 4vwBC/g^2, \quad B_x = u^2/g^2.$$

Moreover $X_a^3 = 64N^2bc/a^2 \geq 64$, so $X_a \geq 4$ and $h_x(Q_a) = \log A_x$. Direct substitution yields

$$\log A_x = \tfrac{1}{3}(\log b + \log c) + \tfrac{2}{3}\log(BC) + \log 4 - 2\log g,$$
$$\log B_x = \tfrac{2}{3}\log a - \tfrac{2}{3}\log A - 2\log g.$$

As $c/2 \leq b < c$, these imply

$$\tfrac{2}{3}H - \tfrac{1}{3}\log 2 \leq h_x(Q_a) \leq \tfrac{2}{3}H + \log 4 + \tfrac{2}{3}\log N, \tag{67}$$
$$0 \leq H - \log a \leq \tfrac{3}{2}\log X_a + \tfrac{1}{2}\log 2. \tag{68}$$

Thus $h_x(Q_a) \to \infty$ and is comparable to H, uniformly for bounded N. Equation (66) makes the right side of (68) equal to $o(H)$. This proves the stated ratio and hence the power inequality. If $b < a$, use Q_b instead. An infinite countersequence to the eventual assertion would contradict the same finite-list application of Siegel, so the quantifier is over all triples with $R_0 \leq R_*$. ☐

10.5. **An infinite fixed-support family and the exact limitation.** Bounded R_0 alone does not imply bounded height.

Proposition 10.6. *There are infinitely many positive primitive abc triples of unbounded height with $R_0 = 3$.*

Proof. On $Y^2 = X^3 - 34992$, let $P = (36, -108)$. Doubling gives

$$2P = (252, 3996), \qquad X(4P) = \left(\frac{882}{37}\right)^2 - 504 = \frac{87948}{1369}.$$

The final numerator has remainder 36 modulo 37, so $4P$ is not integral. The Nagell–Lutz integrality theorem on this integral nonsingular short Weierstrass model implies that P has infinite order; see [72, Theorem 24.21]. The mutually inverse formulas

$$u = \frac{324 + Y}{6X}, \quad v = \frac{324 - Y}{6X}, \qquad X = \frac{108}{u + v}, \quad Y = \frac{324(u - v)}{u + v}$$

identify its rational affine points with $u^3 + v^3 = 9$. No affine rational point has $X = 0$, and $u + v = 0$ is impossible. Nonzero multiples of P therefore supply infinitely many rational solutions. Clear denominators to obtain

$$x^3 + y^3 = 9z^3, \qquad z > 0, \quad \gcd(x, y, z) = 1.$$

Neither x nor y is zero, since 9 is not a rational cube. If $3 \mid x$, reduction modulo three forces $3 \mid y$, and then $27 \mid 9z^3$ forces $3 \mid z$, a contradiction. The same holds for y. A different prime dividing two coordinates divides the third, so x, y, z are pairwise coprime.

If $x, y > 0$, take $(a, b, c) = (x^3, y^3, 9z^3)$. If one is negative, move that term to the other side; for example $x < 0$ gives $((-x)^3, 9z^3, y^3)$. Each resulting triple is positive and primitive. Every prime other than three has exponent divisible by three in its product, while the exponent at three is $2 + 3v_3(z)$. Thus $R_0 = 3$. Only finitely many choices of signs and order give the same positive triple, so infinitely many distinct triples occur. Their heights are unbounded because a bounded height admits only finitely many triples. ☐

This family is consistent with Theorem 10.5. It refutes only a bound depending on the cubic residual support alone: the growing primes of cube-divisible exponent still contribute to $\mathrm{rad}(abc)$. For moving R_0, neither the finite-list threshold in Siegel's theorem nor the field discriminant estimate removes the growing point, denominator, or order-index cost. Controlling that cost uniformly remains a global abc problem.

11. Full local Galois lifts and simultaneous minimum layers

Proposition 7.5 gives a necessary trace constraint for local Galois transport. We now construct actual automorphisms of full local absolute Galois groups and compute their action on specified vectors. The two examples below give a common minimum layer for a background vector and, in the second example, three different square-index vectors. All tensor factors use the same local arrow. The resulting exact maximal-order hulls concern the native families defined here; their identification with a published global collation is a separate question.

Throughout this section, p is odd, $E/\mathbb{Q}_p$ is finite, and the native valuation is normalized by $v_p(p) = 1$. Write

$$G_E = \mathrm{Gal}(\overline{E}/E), \qquad L_E = \log \mathcal{O}_E^\times, \qquad I_E = p^{-1}L_E.$$

The notation for free generators and local coordinate vectors in this section is independent of the global endpoints of an abc triple.

11.1. **A word automorphism and the full relative presentation.** Let $n = [E : \mathbb{Q}_p]$ be even. The original Jannsen–Wingberg construction [74, §1.2, Theorem 2, and §1.4(a), pp. 74–76] uses the full tame group $\mathfrak{T}$, generated by σ, θ, and the relative profinite group

$$V = \widehat{F}_{n+1} * \mathfrak{T}, \qquad Z = \ker(V \longrightarrow \mathfrak{T}), \qquad J = \ker(Z \longrightarrow Z^{(p)}), \qquad \mathcal{F} = V/J. \tag{69}$$

Here $\widehat{F}_{n+1}$ is free profinite on $x_0, \ldots, x_n$, and $Z^{(p)}$ is the maximal pro-p quotient of Z. One then imposes the additional relation

$$\sigma x_0 \sigma^{-1} = W_0(x_0, \theta) x_1^{p^s}[x_1, x_2][x_3, x_4] \cdots [x_{n-1}, x_n], \tag{70}$$

where W_0 belongs to the closed subgroup generated by x_0, θ. The commutator convention is $[x, y] = xyx^{-1}y^{-1}$. The tame relation $\sigma\theta\sigma^{-1} = \theta^{p^f}$ and all internal relations of $\mathfrak{T}$ are retained in (69). In particular, this construction is not the maximal pro-p quotient of the whole absolute Galois group.

Lemma 11.1 (An exact cross-handle word)**.** *In the free group on a, b, c, d_0, put*

$$r = bab^{-1}, \qquad z = d_0 r d_0^{-1}, \qquad \delta = [a, b][c, d_0].$$

The substitution

$$\begin{aligned} F(a) &= a, & F(b) &= d_0 b, \\ F(c) &= zr^{-1}cz^{-1}, & F(d_0) &= zd_0z^{-1} \end{aligned} \tag{71}$$

is an automorphism fixing a and δ exactly. Its inverse is

$$\begin{aligned} G(a) &= a, & G(b) &= r^{-1}d_0^{-1}rb, \\ G(c) &= r^{-1}d_0^{-1}rd_0cr, & G(d_0) &= r^{-1}d_0r. \end{aligned} \tag{72}$$

On the free abelianization it acts as

$$a \longmapsto a, \qquad b \longmapsto b + d_0, \qquad c \longmapsto c - a, \qquad d_0 \longmapsto d_0. \tag{73}$$

Proof. All substitutions act on the rightmost expression first. Set $D = r^{-1}d_0r$ and $s = D^{-1}rD$. From (72),

$$G(b) = D^{-1}b, \qquad G(r) = s, \qquad G(z) = r, \qquad G(c) = sr^{-1}cr, \qquad G(d_0) = D.$$

For example, $G(r) = (D^{-1}b)a(b^{-1}D) = D^{-1}rD$, and $G(z) = D(D^{-1}rD)D^{-1} = r$. Substitution in (71) now gives

$$G(F(a)) = a, \quad G(F(b)) = DD^{-1}b = b, \quad G(F(c)) = rs^{-1}(sr^{-1}cr)r^{-1} = c,$$

and $G(F(d_0)) = rDr^{-1} = d_0$. Conversely, $F(r) = z$, $F(D) = z^{-1}(zd_0z^{-1})z = d_0$, and $F(s) = d_0^{-1}zd_0 = r$. Applying F to the preceding expressions for $G(b), G(c), G(d_0)$ gives b, c, d_0, respectively. Thus both compositions are the identity on all four generators.

The boundary is preserved as a word, not just up to conjugacy:

$$F([a, b]) = ad_0r^{-1}d_0^{-1},$$
$$F([c, d_0]) = zr^{-1}cd_0c^{-1}rd_0^{-1}z^{-1}.$$

Using $z = d_0rd_0^{-1}$, their product reduces to $ar^{-1}cd_0c^{-1}d_0^{-1} = [a, b][c, d_0]$. Finally r and z both have abelian image a, which gives (73) directly. $\square$

Proposition 11.2 (Descent to the full Galois group)**.** *Suppose $n \geq 4$ is even. Apply Lemma 11.1 to $(a, b, c, d_0) = (x_1, x_2, x_3, x_4)$ and fix all other generators, including the tame factor. This defines a continuous automorphism of G_E with the action (73) on abelianization. More generally, a free-word automorphism of $x_1, \ldots, x_n$ descends in this manner if it fixes x_1 and the ordered commutator product in (70) exactly.*

Proof. The substitution and its finite-word inverse extend continuously to the free profinite product V by its universal property. Both leave the projection to $\mathfrak{T}$ unchanged, and hence preserve Z. The kernel J of the canonical maximal pro-p quotient of Z is characteristic in Z. Thus both substitutions preserve J and give inverse automorphisms of $\mathcal{F}$. This verifies the wild-normal-subgroup condition in the original presentation.

They fix σ, θ, x_0, and therefore fix W_0 and all its profinite powers. They fix $x_1^{p^s}$ literally. The first two commutators are fixed as their ordered product, and the later commutators are fixed term by term. Thus the additional relator and its closed normal closure are preserved. The induced maps on the full quotient are inverse; the identification of that quotient with G_E is Jannsen–Wingberg's Theorem 2 with the algebraic closure as extension. The same argument proves the general assertion. □

For clarity, the descended cross-handle automorphism is not inner. The character to $\mathbb{Z}_p$ sending x_4 to 1 and the other generators and tame factor to 0 satisfies the relative presentation. It has value 0 on x_2 and value 1 on $F(x_2) = x_4 x_2$; an inner automorphism would preserve every abelian character. This observation does not require any choice of logarithmic basis.

11.2. **The integral JW basis and the available linear actions.** We keep the same full presentation and its cyclotomic parameters when passing to local reciprocity. Let y_i be the logarithm of the principal-unit image of x_i. Hoshi–Nishio [75, Lemma 1.3 and its proof] prove that $y_1, \ldots, y_n$ are a $\mathbb{Q}_p$-basis of E and that $y_0, \ldots, y_n$ generate L_E over $\mathbb{Z}_p$. The former assertion alone does not give an integral basis.

Lemma 11.3 (Integral upgrade in the small tame range)**.** *Assume $2 \leq e(E/\mathbb{Q}_p) \leq p-2$. Then $L_E = \mathfrak{m}_E$, and*

$$v_i = y_i/p \quad (1 \leq i \leq n) \tag{74}$$

form a $\mathbb{Z}_p$-basis of I_E. For even n, in these same coordinates,

$$I_E \cap \ker \operatorname{Tr}_{E/\mathbb{Q}_p} = \mathbb{Z}_p v_1 \oplus \bigoplus_{i=3}^{n} \mathbb{Z}_p v_i. \tag{75}$$

If additionally $\operatorname{Tr} I_E = \mathbb{Z}_p$, then $\operatorname{Tr}(v_2)$ is a unit.

Proof. The inequality $1/e > 1/(p-1)$ makes logarithm and exponential inverse on $1 + \mathfrak{m}_E$ and $\mathfrak{m}_E$. The prime-to-p torsion units have zero logarithm, proving $L_E = \mathfrak{m}_E$.

Let g, h be the lifts of the tame cyclotomic actions appearing in the original presentation. With the notation of [75, Proposition 1.1], the mixed factor can be written

$$W_0 = \left(x_0^{h^{p-1}} \theta x_0^{h^{p-2}} \theta \cdots x_0^{h} \theta\right)^{\epsilon_p g/(p-1)},$$

where $\epsilon_p \in \widehat{\mathbb{Z}}$ is the projector onto $\mathbb{Z}_p$. The tame relation makes the image of θ torsion in abelianization, so it contributes zero to logarithm. Taking logarithms in (70) gives

$$\left(1 - \frac{g}{p-1} \sum_{r=1}^{p-1} h^r\right) y_0 = p^s y_1. \tag{76}$$

The reduction of h modulo p is not 1. Indeed, trivial inertia action on μ_p would make $E(\mu_p)/E$ unramified, forcing $p-1$ to divide $e(E/\mathbb{Q}_p)$, contrary to the assumed range. Thus the geometric sum in (76) vanishes modulo p. Its parenthesized coefficient is a unit, and $y_0 \in \mathbb{Z}_p y_1$. The remaining y_i therefore generate L_E integrally. Their known rational independence gives an integral basis and proves (74).

For even degree, Kondo's trace-kernel theorem [76, Theorem 1.3] identifies $\ker \mathrm{Tr}$ with the rational span of $v_1, v_3, \dots, v_n$ in this JW basis. Intersecting with the proved integral basis gives (75). Consequently the trace ideal of I_E is generated by $\mathrm{Tr}(v_2)$, which proves the final assertion. □

Write $a = v_1$, $b = v_2$, and $W = \bigoplus_{i=3}^{n} \mathbb{Z}_p v_i$. Let ω be the standard integral alternating form on these remaining handle coordinates, with $\omega(v_3, v_4) = 1$. This form records the handle action; it is not being identified with field multiplication. The covariant logarithmic action of Proposition 11.2 is

$$N_{v_4}(a) = a, \qquad N_{v_4}(b) = b + v_4, \qquad N_{v_4}(x) = x + \omega(v_4, x)a \quad (x \in W).$$

The twist $x_2 \mapsto x_2 x_1$ gives $C_1(b) = b + a$, fixing a, W.

The boundary-preserving surface construction in [76, pp. 19–20 and Theorem 2.17] also gives an individual full Galois lift of every integral symplectic matrix on the remaining handles, fixing $\sigma, \theta, x_0, x_1, x_2$. The construction on p. 19 explicitly includes even degree; the odd-degree restriction later on p. 20 is not used here. In particular, handle exchanges and rotations move v_4 to any displayed basis direction. Conjugating the cross-handle lift therefore realizes N_w for every such direction.

For a general integer coordinate vector $w \in \mathrm{span}_{\mathbb{Z}}\{v_3, \dots, v_n\}$, the linear formulas are

$$\begin{aligned} N_w(Aa + Bb + x) &= (A + \omega(w, x))a + Bb + x + Bw, \\ C_c(Aa + Bb + x) &= (A + cB)a + Bb + x. \end{aligned} \tag{77}$$

The following correction makes their realization precise.

Lemma 11.4 (Ordered central correction)**.** *As identities of the induced linear maps,*

$$N_w N_v = C_{\omega(w,v)} N_{w+v}, \qquad C_c C_d = C_{c+d}, \qquad C_c N_w = N_w C_c. \tag{78}$$

For $w = \sum_i n_i e_i$ in the fixed ordered integer handle basis,

$$N_w = C_{-\sum_{i<j} n_i n_j \omega(e_i, e_j)} N_{e_1}^{n_1} \cdots N_{e_{n-2}}^{n_{n-2}}. \tag{79}$$

Thus every such N_w is the canonical action of an actual full Galois automorphism.

Proof. In coordinates (A, B, x), applying N_v and then N_w gives

$$(A + \omega(v, x) + \omega(w, x + Bv),\ B,\ x + Bv + Bw).$$

Bilinearity gives (78); the other two identities follow directly from (77). Alternation implies $N_w^{-1} = N_{-w}$ and $N_w^m = N_{mw}$ for every integer m. An induction in the displayed left-to-right basis order gives (79). Each basis cross and C_1 already has a full lift. Integer powers, inverses and compositions of those lifts realize the displayed linear image. □

The central identities are not asserted in the full nonabelian automorphism group: lifts may differ by elements acting trivially on abelianization. Nor is a group-homomorphic section of a mapping class group into an outer Galois group required. Individual lifts and the functoriality of their linear action suffice. In particular, any prescribed vector modulo p can be realized by choosing an integer coordinate lift in (79).

11.3. **Covariance and integral Kummer transport.** For $\alpha \in \mathrm{Aut}(G_E)$, let M_α be the canonical covariant action on principal-unit abelianization, extended by logarithm and $\mathbb{Q}_p$-linearity to E. It sends each y_i to the logarithm of the image of $\alpha(x_i)$ and preserves I_E. The norm/trace compatibility of this action is the one in [75, Lemma 2.3].

Proposition 11.5 (The inverse in Kummer transport)**.** *For a continuous automorphism α and an integral equivariant Tate-module isomorphism, let F_α be the induced logarithmic Kummer pullback. Then*

$$F_\alpha = c_\alpha M_\alpha^{-1} \qquad \textit{for some } c_\alpha \in \mathbb{Z}_p^\times. \tag{80}$$

Thus the native valuations attained by M_α are attained by one integral Kummer arrow belonging to α^{-1}.

Proof. The cyclotomic character is determined by the local absolute Galois group, so an equivariant integral Tate-module isomorphism exists; its choices differ by $\mathbb{Z}_p^\times$ [68, Proposition 1.1]. Local duality identifies $H^2(G_E, \mathbb{Z}_p(1))$ with $\mathbb{Z}_p$, and integral transport acts on it by a unit c_α. The cup-product reciprocity pairing gives, for a continuous integral character χ and a Kummer class z,

$$\langle \chi \circ \alpha^{\mathrm{ab}}, F_\alpha z\rangle = c_\alpha \langle \chi, z\rangle. \tag{81}$$

Here the pairing is evaluation of χ on the pro-p completion of $E^\times$ under local reciprocity; see the conventions in Proposition 7.5 and [67, Chapter III, Proposition 3.6 and §4]. The unit submodule is the annihilator of the unramified character, so it is preserved. Torsion classes disappear after logarithm.

On the free unit lattice, the pullback character is $\chi \circ M_\alpha$. These characters separate the lattice after rational extension. Equation (81) therefore says $M_\alpha F_\alpha = c_\alpha \,\mathrm{id}$, proving (80). Applying this to α^{-1} gives $F_{\alpha^{-1}} = c_{\alpha^{-1}} M_\alpha$. A $\mathbb{Z}_p$-unit scalar preserves the native valuation of every vector, simultaneously. Uniform division of logarithmic coordinates by p does not alter the identity. □

11.4. **A degree-ten example.**

Lemma 11.6 (One minimal vector)**.** *Use the integral basis in Lemma 11.3, with even $n \geq 4$, and let π be a uniformizer. Suppose*

$$\operatorname{Tr} I_E = \operatorname{Tr}(\pi I_E) = \mathbb{Z}_p, \qquad u_0 \in \pi I_E, \quad \operatorname{Tr}(u_0) \in \mathbb{Z}_p^\times,$$

and $t_0 \in I_E \setminus \pi I_E$ with $\operatorname{Tr}(t_0) \in p\mathbb{Z}_p$. Then one full Galois automorphism has canonical action M satisfying $Mu_0, Mt_0 \in I_E \setminus \pi I_E$.

Proof. Put $\beta = \operatorname{Tr}(b) \in \mathbb{Z}_p^\times$. The respective b-coordinates are $B_u = \operatorname{Tr}(u_0)/\beta \in \mathbb{Z}_p^\times$ and $B_t = \operatorname{Tr}(t_0)/\beta \in p\mathbb{Z}_p$. If $a \notin \pi I_E$, use C_1. Modulo πI_E, its image of u_0 is $B_u a \neq 0$, and its image of t_0 is unchanged, because $pI_E \subseteq \pi I_E$.

If $a \in \pi I_E$, then $I_E \cap \ker \operatorname{Tr}$ surjects onto $I_E/\pi I_E$: subtract an element of πI_E with the same trace from any representative. Equation (75) implies that some displayed basis vector $w \in W$ is outside πI_E. The actual cross lift N_w changes u_0 by $B_u w + A_u a$ and t_0 by $B_t w + A_t a$, for integral A_u, A_t. Modulo πI_E, the first image is $B_u w \neq 0$, and the second is the original nonzero image of t_0. The same map works for both. □

Theorem 11.7 (A common minimum layer over $\mathbb{Q}_{19}$)**.** *Let*

$$K_0 = \mathbb{Q}_{19}(\mu_5), \qquad \pi^5 = 19, \qquad E = K_0(\pi),$$

and put

$$u = \frac{\log 20}{19}, \qquad \tau = \frac{\log(1+19\pi)}{19}, \qquad t = \tau/19.$$

Then some full Galois automorphism has canonical action M with

$$v_{19}(Mu) = v_{19}(Mt) = -4/5. \tag{82}$$

One integral Kummer arrow attains the same two valuations.

Proof. The order of 19 modulo 5 is 2, so $K_0/\mathbb{Q}_{19}$ is unramified quadratic. The Eisenstein polynomial $X^5 - 19$ and $\mu_5 \subset K_0$ give a Galois extension with $(n, e, f) = (10, 5, 2)$. The logarithm range yields

$$I_E = \pi^{-4}\mathcal{O}_E = \bigoplus_{r=-4}^{0} \mathcal{O}_{K_0}\pi^r.$$

The negative powers in this direct sum have relative trace zero, and $\operatorname{Tr}_{E/K_0}(1) = 5$. Hence $\operatorname{Tr} I_E = \mathbb{Z}_{19}$. Also $1/10 \in \mathcal{O}_E \subset \pi I_E$ has absolute trace 1, proving $\operatorname{Tr}(\pi I_E) = \mathbb{Z}_{19}$.

Leading logarithm terms give $v_{19}(u) = 0$ and $v_{19}(t) = -4/5$. The norm identity

$$\prod_{\zeta \in \mu_5} (1 + 19\zeta\pi) = 1 + 19^6$$

gives the exact traces

$$\mathrm{Tr}(u) = 10u \in \mathbb{Z}_{19}^{\times}, \qquad \mathrm{Tr}(t) = \frac{2}{19^2}\log(1+19^6) \in 19^4\mathbb{Z}_{19}^{\times}.$$

Lemma 11.6 proves (82), and Proposition 11.5 gives the Kummer assertion. □

The covariant action in Theorem 11.7 is not induced by a field automorphism: it changes the native valuation of u from 0 to $-4/5$. This is an assertion about a specified local vector, rather than a classification of all admissible linear maps.

11.5. **A simultaneous finite-family lemma.** The degree-ten argument begins with t already at the minimum layer. The square-index example below does not have this feature. We therefore need a common operation on several vectors initially lying above the minimum layer.

Lemma 11.8 (Simultaneous parabolic reachability)**.** *Let k be a finite field with q elements, and let the finite-dimensional k-vector space*

$$V = ka \oplus kb \oplus W$$

carry the standard nondegenerate alternating form on W, and let B be the b-coordinate. Let $L \subset V$ have codimension two, and suppose $\ker B \to V/L$ *is onto. Assume*

$$U, T_1, \dots, T_r \in L, \qquad B(U) \neq 0, \qquad B(T_j) = 0, \qquad T_j \notin ka,$$

with $r+1 < q$. The linear operations C_c, N_w of (77), together with symplectic operations on W fixing a, b, contain one map sending $U, T_1, \dots, T_r$ all outside L. If $k = \mathbf{F}_p$ and the coordinates come from the preceding integral JW basis, one such map is induced by a full Galois automorphism, using integer coordinate lifts.

Proof. Write $T_j = A_j a + w_j$. The hypotheses imply $w_j \neq 0$, so each functional $w \mapsto \omega(w, w_j)$ is nonzero.

Suppose first that $a \notin L$. Fewer than q proper hyperplanes cannot cover W: in dimension D, their union has fewer than $q \cdot q^{D-1} = q^D$ elements. Choose w with $\omega(w, w_j) \neq 0$ for all j. Then

$$N_w(T_j) + L = \omega(w, w_j)(a + L) \neq 0.$$

If $N_w(U) \notin L$, the construction is complete. Otherwise compose with C_1. It adds the nonzero class $B(U)(a+L)$ to $N_w(U)$ and leaves all the T_j images unchanged because their B-coordinates remain zero.

Suppose now that $a \in L$. Then $\lambda : W \to V/L$ is onto. Also $\lambda(w_j) = 0$ for every j. Avoid $\ker\lambda$ and the r hyperplanes $\ker\omega(-, w_j)$ simultaneously, obtaining $s \in W$ such that

$$\lambda(s) \neq 0, \qquad \omega(s, w_j) \neq 0 \quad \text{for all } j.$$

There are $r + 1 < q$ proper subspaces, so the same union bound applies. The single symplectic transvection

$$S_s(x) = x + \omega(s, x)s \quad (x \in W), \qquad S_s(a) = a, \quad S_s(b) = b$$

has inverse given by changing the plus sign to a minus sign. Expanding its pairing shows it preserves ω; the cross terms cancel and $\omega(s, s) = 0$. Since the original projections vanish,

$$S_s(T_j) + L = \omega(s, w_j)\lambda(s) \neq 0$$

for every j. If $S_s(U) \notin L$, stop. Otherwise choose a displayed basis vector e of W with $\lambda(e) \neq 0$ and apply N_e. Its change on the zero projection of $S_s(U)$ is $B(U)\lambda(e) \neq 0$. It does not change any T_j projection, because $B(T_j) = 0$ and $a \in L$.

For the arithmetic assertion, lift the chosen residue vectors to integer coordinate vectors. Lemma 11.4 realizes each required cross. The lifted transvection is an element of $\mathrm{Sp}(W_{\mathbb{Z}})$ and has an individual full Galois lift by the surface construction cited above. Compose these lifts. The argument uses one map for the whole family, not a different map for each vector. □

11.6. **Three square labels at the prime 139.** We state the next example for a general unit part of the parameter. Let $p = 139$ and choose any $b_0 \in \mathbb{Q}_p$ with $v_p(b_0) = 1$. Put

$$K_0 = \mathbb{Q}_p(\mu_{210}), \qquad \pi^{105} = b_0, \qquad E = K_0(\pi), \qquad q_0 = b_0^2. \tag{83}$$

Thus $\pi^{210} = q_0$ and $\pi^{15} = q_0^{1/14}$ with these choices of roots. The field in (83) is exactly $\mathbb{Q}_p(\mu_{210}, q_0^{1/210})$; no global curve with a prescribed exact parameter is assumed here.

Theorem 11.9 (One full Galois arrow for three square labels)**.** *In* (83)*, set*

$$\begin{gathered} u = \frac{\log(1+p)}{p}, \qquad \tau_s = \frac{\log(1+p\pi^{15s})}{p}, \\ k_s = \left\lfloor \frac{s}{7} + \frac{104}{105} \right\rfloor, \quad T_s = p^{-k_s}\tau_s \qquad (s = 1, 4, 9). \end{gathered} \tag{84}$$

Then $k_1 = k_4 = 1$, $k_9 = 2$. One full Galois automorphism has canonical action M such that

$$v_p(Mu) = v_p(MT_1) = v_p(MT_4) = v_p(MT_9) = -104/105. \tag{85}$$

Consequently

$$v_p(M\tau_1) = 1/105, \qquad v_p(M\tau_4) = 1/105, \qquad v_p(M\tau_9) = 106/105. \tag{86}$$

One integral Kummer arrow attains all these values simultaneously.

Proof. Since $139^2 \equiv 1 \pmod{210}$ and $139 \not\equiv 1 \pmod{210}$, $K_0/\mathbb{Q}_p$ is unramified quadratic. The Eisenstein polynomial $X^{105} - b_0$ and $\mu_{105} \subset K_0$ give a Galois field with $(n, e, f) = (210, 105, 2)$. As $1/105 > 1/138$,

$$I_E = \pi^{-104}\mathcal{O}_E, \qquad pI_E = \pi\mathcal{O}_E, \qquad \pi I_E = \pi^{-103}\mathcal{O}_E. \tag{87}$$

These are equalities of fractional ideals: b_0/p is a rational unit and need not be 1.

The power basis gives $I_E = \bigoplus_{r=-104}^{0} \mathcal{O}_{K_0}\pi^r$. All negative powers in this sum have relative trace zero and $\mathrm{Tr}_{E/K_0}(1) = 105$. Hence $\mathrm{Tr}\, I_E = \mathbb{Z}_p$. Also $1/210 \in \mathcal{O}_E \subset \pi I_E$ has trace 1. Subtracting $\mathrm{Tr}(z)/210$ from $z \in I_E$ proves that $I_E \cap \ker \mathrm{Tr}$ surjects onto $I_E/\pi I_E$. Lemma 11.3 applies.

The integer k_s is exactly the largest k for which $\tau_s \in p^k I_E$. Leading logarithm terms and the exact norm give the following data:

s	k_s	$v_p(T_s)$	$v_p(\mathrm{Tr}\, T_s)$
1	1	$-90/105$	6
4	1	$-45/105$	9
9	2	$-75/105$	13

Indeed, for these s, the seven conjugates of π^{15s} occur fifteen times in the relative norm, and therefore

$$\begin{gathered} \mathrm{Nm}_{E/K_0}(1 + p\pi^{15s}) = (1 + p^7 b_0^s)^{15}, \\ \mathrm{Tr}(T_s) = \frac{30}{p^{k_s+1}} \log(1 + p^7 b_0^s). \end{gathered} \tag{88}$$

The last expression has valuation $s + 6 - k_s$, as displayed. Furthermore $\mathrm{Tr}(u) = 210 \log(1+p)/p$ is a unit.

Pass to the $\mathbf{F}_p$-space

$$V = I_E/pI_E, \qquad L = \pi I_E/pI_E, \qquad \dim V = 210, \quad \operatorname{codim} L = 2.$$

The integral JW basis gives $V = \mathbf{F}_p a \oplus \mathbf{F}_p b \oplus W$, with $\dim W = 208$ and B its b-coordinate. Equation (75) and $\mathrm{Tr}(b) \in \mathbb{Z}_p^\times$ imply

$$B(u) \neq 0, \qquad B(T_s) = 0, \qquad u, T_s \in L, \qquad T_s \neq 0 \text{ in } V.$$

The trace-zero subspace surjects onto V/L by the subtraction argument above. There remains a possible fixed-line obstruction: one or more of the T_s might belong to $\mathbf{F}_p a$.

We remove it by one common field automorphism. The logarithm tail satisfies

$$T_s \equiv \pi^{15s}/p^{k_s} \pmod{pI_E}. \tag{89}$$

For its term of degree $m \geq 2$, use $v_p(m) \leq m-2$ to obtain

$$m(1+s/7) - (k_s+1) - v_p(m) \geq 1 + ms/7 - k_s \geq 1 + 2s/7 - k_s.$$

The last three bounds are $2/7, 8/7, 11/7$, all larger than $1/105$, the minimum valuation of pI_E.

Let $\sigma(\pi) = \zeta_{105}\pi$, fixing K_0. For $h = 0, \ldots, 6$, equation (89) gives

$$\sigma^h(T_s) \equiv \zeta_7^{sh}\pi^{15s}/p^{k_s} \pmod{pI_E}.$$

For each s, these determine seven distinct $\mathbf{F}_{139}$-lines. To see this, the reduction of ζ_7 has order seven and $7 \nmid 138$, so its group meets $\mathbf{F}_{139}^\times$ only in 1. Equality of two lines would force their leading coefficients to have a ratio in that intersection. A nonzero residue difference would be a unit times π^{15s}/p^{k_s}, whose negative valuation precludes membership in pI_E. Since s is prime to seven, the two exponents must be equal modulo seven.

For each of the three labels, at most one of these seven values of h puts $\sigma^h(T_s)$ in the fixed line $\mathbf{F}_p a$. Avoid the at most three excluded values. This one σ^h fixes u, preserves L and trace, and places all three labels outside the fixed line. It is an actual field automorphism, hence has a representative on the full local Galois group.

Apply Lemma 11.8 to these three resulting vectors and u. The numerical condition is $3+1 < 139$. More precisely, in its second case the excluded sets have at most $139^{206} + 3 \cdot 139^{207} < 139^{208}$ elements. The lemma gives one further actual lift sending all four vectors outside L. Composing with the chosen inertia representative proves (85). Linearity and $\tau_s = p^{k_s}T_s$ give (86); Proposition 11.5 supplies the common Kummer arrow. $\square$

The unit b_0/p is fixed by inertia. Thus the proof includes all parameters in (83), rather than only $b_0 = p$. The separate passage from a rational Frey curve to these local data does not enter the simultaneous-reachability argument.

11.7. **Exact native maximal-order hulls.** Let $E/\mathbb{Q}_p$ now be either Galois field above. For $m \geq 1$, write

$$\mathcal{T}_m = E^{\otimes_{\mathbb{Q}_p} m}, \qquad B_m = \prod_{\mathrm{Gal}(E/\mathbb{Q}_p)^{m-1}} \mathcal{O}_E,$$

using the decomposition

$$x_1 \otimes \cdots \otimes x_m \longmapsto \big(x_1\gamma_2(x_2)\cdots\gamma_m(x_m)\big)_{(\gamma_2,\ldots,\gamma_m)}. \tag{90}$$

Thus B_m is the maximal order in this fixed finite etale carrier. In $\pi^r B_m$, the uniformizer is embedded through the first tensor factor, so it has the same valuation in every component. Let Γ_E be the set of integral logarithmic Kummer transports arising from automorphisms of the same G_E, with integral Tate-module identifications. Every $F \in \Gamma_E$ is $\mathbb{Q}_p$-linear and preserves I_E and $p^k I_E$ for all integers k.

Proposition 11.10 (An attained block hull)**.** *Suppose $I_E = \pi^{1-e}\mathcal{O}_E$, and put $\kappa = 1 - 1/e$. Let $u, t \in I_E$, $\tau = p^k t$ with integer k, and suppose one $F_0 \in \Gamma_E$ satisfies $v_p(F_0 u) = v_p(F_0 t) = -\kappa$. Define the coherent repeated-label family*

$$S_m = \{F(\tau) \otimes F(u) \otimes \cdots \otimes F(u) : F \in \Gamma_E\}, \qquad H_m = \mathrm{span}_{B_m}(S_m).$$

Then

$$H_m = \pi^{ek-(e-1)m} B_m. \tag{91}$$

The equality also holds after taking topological closure. More generally, it remains true if the set of source tuples is enlarged to any set lying between the displayed coherent tuples and $p^k I_E \times I_E^{m-1}$.

Proof. For every F, the first factor has valuation at least $k-\kappa$, and each remaining factor at least $-\kappa$. All field embeddings in (90) preserve valuation. Every component of every tensor generator therefore has valuation at least $k-m\kappa$, proving containment in the right-hand side of (91).

The single arrow F_0 gives equality in this valuation in every component. Its tensor is consequently $\pi^{ek-(e-1)m}$ times a unit of the product ring B_m. Its principal B_m-span is the entire right-hand side, giving the reverse containment. This fractional product lattice is closed. The same upper bound holds for every tuple in the stated containing product, while the original attaining tuple remains present; this proves the enlargement assertion. □

The enlargement includes the closed $\mathbb{Z}_p$-convex hull taken in the product of the source coordinate spaces, since that product lattice is closed and $\mathbb{Z}_p$-convex. This does not identify such a convex hull with a B_m-module hull: tensor formation and the final B_m-span remain distinct operations.

Corollary 11.11 (The two exact local calculations)**.** *For Theorem 11.7, the family with one τ factor and $m-1$ background factors has*

$$H_m = \pi^{5-4m} B_m \quad (m \geq 1), \qquad \textit{in particular } H_3 = \pi^{-7} B_3.$$

For Theorem 11.9, let H_j use one τ_{j^2} factor and j background factors, for $j = 1,2,3$. Then one and the same arrow attains all three hulls, and

$$H_1 = \pi^{-103} B_2, \qquad H_2 = \pi^{-207} B_3, \qquad H_3 = \pi^{-206} B_4. \tag{92}$$

Proof. Use Proposition 11.10 with $(e,k) = (5,1)$ in the first case. In the second case $e = 105$, $m = j+1$, and $(k_1, k_4, k_9) = (1,1,2)$. Thus the three exponents are $105k_{j^2} - 104(j+1) = -103, -207, -206$. The common witness is supplied by Theorem 11.9 and the single inverse/unit conversion of Proposition 11.5. □

These nonintegral hulls are exact calculations for stated local source families. They do not establish a comparison between different native markings, a cross-Frobenius identification, or compatible global initial theta data. In particular, identifying the source sets needed to compare a published pilot container with one of these hulls remains a separate requirement.

The free-word identities, the discrete free-group automorphism, the linear central correction, and the finite-family transvection statements have independent Lean proofs. The relative profinite quotient, local class field theory, local duality, integral JW basis, and logarithmic norm calculations are mathematical source arguments, not Lean declarations of local-field theorems in this work. The global abc bound remains an additional obligation.

12. Native pilot points, multiplicative families, and transported ideals

We now distinguish three local inputs: a specified point, the image of multiplication on a log lattice, and a principal maximal-order ideal before transport. There are positive comparisons between these inputs, even for nongeometric Galois arrows. In the attained examples the point and pre-ideal hulls coincide, while saturation to the whole multiplication image produces a strictly larger hull. All the constructions in this section use one native carrier and one normalization; they do not assume the cross-arithmeticoid comparison in [57, Theorem 6.10.1].

12.1. The local objects and the logarithmic scale.

Let $p > 2$ and let $E/\mathbb{Q}_p$ be finite Galois, of degree d and ramification index $2 \leq e \leq p-2$. Normalize $v(p) = 1$, choose a uniformizer π, and write

$$\kappa = \frac{e-1}{e}, \qquad I = p^{-1} \log \mathcal{O}_E^\times = p^{-1}\mathfrak{m}_E = \pi^{1-e}\mathcal{O}_E, \qquad u = p^{-1}\log(1+p). \tag{93}$$

Logarithm and exponential are inverse isometries on $1+\mathfrak{m}_E$ and $\mathfrak{m}_E$, since $1/e > 1/(p-1)$. The tame different is $\mathfrak{D}_{E/\mathbb{Q}_p} = \pi^{e-1}\mathcal{O}_E$, so $I = \mathfrak{D}^{-1}$. The convention $(2p)^{-1}\log \mathcal{O}_E^\times$ differs only by a $\mathbb{Z}_p^\times$-factor, which does not affect the ideals below.

For $m \geq 1$, set

$$T_m = E^{\otimes_{\mathbb{Q}_p} m}, \qquad A_m = \mathcal{O}_E^{\otimes_{\mathbb{Z}_p} m}, \qquad B_m = \text{the normalization of } A_m \text{ in } T_m.$$

The algebra T_m is a product of d^{m-1} copies of E, and B_m is the product of their integer rings. An ideal written $\pi^t B_m$ means the product ideal of valuation t/e in every component, without requiring identical component generators.

Let Γ be a family containing the identity of actual integral Kummer actions on I, or more generally any such family in $\mathrm{Aut}_{\mathbb{Z}_p}(I)$. Each action extends $\mathbb{Q}_p$-linearly to E. We do not suppose that the actual Galois family exhausts this general linear group. With the same arrow in each factor, put

$$\mathcal{H}_\Gamma(S) = \overline{\mathrm{span}_{B_m}\{Fx_1 \otimes \cdots \otimes Fx_m : F \in \Gamma,\ (x_1, \ldots, x_m) \in S\}} \qquad (S \subset E^m). \tag{94}$$

This is an orbit followed by a module hull. Taking a product convex hull before tensoring is a separate operation.

Write Kum for the integral Kummer map on principal units. The standard Bloch–Kato exponential, extended by $p^{-n}\,\mathrm{Kum}(\exp(p^n x))$ when necessary, has inverse $\log_{\mathrm{BK}}^{\mathrm{std}}$ satisfying

$$\log_{\mathrm{BK}}^{\mathrm{std}}(\mathrm{Kum}(1 + pa)) = \log(1 + pa).$$

The following coordinate map is different:

$$\rho = p^{-1} \log_{\mathrm{BK}}^{\mathrm{std}}, \qquad \lambda_p(a) = \rho(\mathrm{Kum}(1 + pa)) = p^{-1}\log(1 + pa). \tag{95}$$

The three displays [56, (9.7.1.1), (9.7.2.1)–(9.7.2.2), PDF p. 110] require this distinction. If the exponential has its standard cohomological scalar meaning and the class is the unscaled class of $1 + pa$ defined on p. 109, its inverse cannot take that class to $\lambda_p(a)$. Indeed, for nonzero integral a,

$$\exp_{\mathrm{BK}}(\lambda_p(a)) = p^{-1}\,\mathrm{Kum}(1 + pa) \neq \mathrm{Kum}(1 + pa).$$

The Kummer class is nonzero because its logarithm has nonzero leading term pa. Formula (95) is a consistent normalized coordinate, but it is not that inverse formula.

Passing from ρ to $\log_{\mathrm{BK}}^{\mathrm{std}}$ multiplies each entry, including each background entry, by p. Since integral Kummer arrows commute with this scalar, every length-m hull satisfies

$$H^{\mathrm{std}} = p^m H^\rho, \qquad V_B(H^{\mathrm{std}}) = V_B(H^\rho) - m\log p, \quad V_B(H) = d^{-m}\log \mu_B(H), \quad \mu_B(B_m) = 1. \tag{96}$$

The determinant of multiplication by p^m on T_m is p^{md^m}, which proves the volume formula. If instead $\mu_A(A_m) = 1$, then $V_A(H) = V_B(H) + d^{-m}\log[B_m : A_m]$. This constant cancels when both sets use the same reference order.

12.2. **A point bridge that survives every integral arrow.**

Proposition 12.1 (Transported root and crystalline point)**.** *For $0 \neq a \in \mathcal{O}_E$, set $r = v(a)$ and $k = \lfloor r + \kappa \rfloor$. Every $F \in \mathrm{Aut}_{\mathbb{Z}_p}(I)$ satisfies*

$$v(F\lambda_p(a)) = v(Fa), \qquad v(Fu) = v(F1). \tag{97}$$

Consequently

$$\mathcal{H}_\Gamma(\{(\lambda_p(a), u, \ldots, u)\}) = \mathcal{H}_\Gamma(\{(a, 1, \ldots, 1)\}). \tag{98}$$

Proof. The choice of k gives $a \in p^k I \setminus p^{k+1} I$. In the convergent expansion

$$\lambda_p(a) - a = \sum_{n \geq 2} (-1)^{n+1} \frac{p^{n-1} a^n}{n},$$

every term has valuation at least $1 + 2r$: for $p > 2$ and $n \geq 2$, $v_p(n) \leq n - 2$. Hence

$$v(\lambda_p(a) - a) \geq 1 + 2r \geq k + 1 - \kappa, \qquad \lambda_p(a) - a \in p^{k+1} I.$$

As F preserves each $p^n I$ in both directions,

$$v(Fa) < k + 1 - \kappa, \qquad v(F(\lambda_p(a) - a)) \geq k + 1 - \kappa.$$

The strict ultrametric inequality proves the first equality in (97); apply it to $a = 1$ for the second.

In every component of T_m, the two nonzero tensors $F\lambda_p(a)\otimes(Fu)^{\otimes(m-1)}$ and $Fa\otimes(F1)^{\otimes(m-1)}$ have the same valuation. Their componentwise ratio is a unit of B_m, so they generate the same principal B_m-ideal. Taking spans and closures proves (98). No E-linearity or multiplicativity of F was used. ☐

The same proof allows different prescribed integral arrows in different factors. Root-of-unity ambiguities can be handled by applying the proposition separately to each permitted ζa and then taking their union; one does not assert $F(\zeta a) = \zeta F(a)$. Also, (98) is a statement about module hulls of raw orbits. The tensor map is not additive in its tuple of inputs, so equality after a product convex hull needs a further argument. An attaining-point argument below supplies it in the explicit example.

12.3. **A whole multiplication image requires a whole unit family.**

Proposition 12.2 (Exact saturated Kummer family)**.** *For $0 \neq a \in \mathcal{O}_E$, the group $U_a = 1 + paI = 1 + a\mathfrak{m}_E$ satisfies, as actual sets,*

$$\rho(\mathrm{Kum}(U_a)) = aI, \qquad \log^{\mathrm{std}}_{\mathrm{BK}}(\mathrm{Kum}(U_a)) = paI. \tag{99}$$

Proof. The ideal $paI = a\mathfrak{m}_E$ lies in the common strict convergence disk of logarithm and exponential. These maps preserve valuations, so $\log(1+paI) = paI$. More explicitly, for every $z \in aI$, $\exp(pz) \in U_a$ and $\rho(\mathrm{Kum}(\exp(pz))) = z$. The inverse series gives the reverse containment. The positive-valuation ideal also makes U_a closed under multiplication and inversion. ☐

In particular, the background family is the whole principal-unit group U_1, not the singleton $1 + p$. Taking products in (99) realizes $aI \times I^{m-1}$ before any arrows or hulls are applied. This equality therefore survives the same transport, product convex closure, tensor map, and final module hull, in any specified order.

Proposition 12.3 (The singleton can be strictly smaller)**.** *Put*

$$P(a) = \mathcal{H}_\Gamma(\{(\lambda_p(a), u, \ldots, u)\}), \qquad S(a) = \mathcal{H}_\Gamma(aI \times I^{m-1}).$$

If $v(a) < \lfloor v(a) + \kappa \rfloor$, then $P(a) \subsetneq S(a)$. The strict inclusion persists if both product inputs first undergo closed $\mathbb{Z}_p$-convex closure.

Proof. Writing $r = v(a)$ and $k = \lfloor r + \kappa \rfloor$, the logarithmic expansion gives $\lambda_p(a)/a \in \mathcal{O}_E^\times$. Also $u \in \mathcal{O}_E \subset I$, so the singleton input lies in the whole input. Every transported singleton lies in the closed product lattice $p^k I \times I^{m-1}$, so every tensor in $P(a)$ has component valuation at least $k - m\kappa$. The whole input at the identity contains

$$(a\pi^{1-e}, \pi^{1-e}, \ldots, \pi^{1-e}).$$

Its tensor has component valuation $r - m\kappa < k - m\kappa$, proving strictness. Convex closure keeps the smaller input in its closed product container and retains the displayed larger input. ☐

Proposition 12.2 supplies a positive cohomological realization of a specified multiplication image. It is an enlargement of a single-class recipe. Proposition 12.3 shows why equality of coefficient norms alone cannot justify omitting this enlargement.

12.4. **One actual arrow for six normalized inputs.** We use the field of (83): $p = 139$, $v(b_0) = 1$, $\pi^{105} = b_0$, and $E = \mathbb{Q}_p(\mu_{210}, \pi)$. Thus $(d, e, f) = (210, 105, 2)$ and $I = \pi^{-104}\mathcal{O}_E$. Put

$$r_0 = \pi^{15}, \quad \alpha_s = r_0^s, \quad \tau_s = \lambda_p(\alpha_s), \quad n_s = \lfloor s/7 \rfloor, \quad k_s = \lfloor s/7 + 104/105 \rfloor \qquad (s = 1, 4, 9),$$

and define two normalized families

$$T_s = \tau_s/p^{k_s}, \qquad X_s = \alpha_s \pi^{-104}/p^{n_s}. \tag{100}$$

Lemma 12.4 (A common arrow for both families)**.** *One actual integral Kummer arrow F satisfies*

$$v(Fu) = v(FT_s) = v(FX_s) = -104/105 \qquad (s = 1, 4, 9).$$

Proof. The triples (n_s) and (k_s) are $(0, 0, 1)$ and $(1, 1, 2)$. The native valuations are

s	1	4	9
$v(T_s)$	$-90/105$	$-45/105$	$-75/105$
$v(X_s)$	$-89/105$	$-44/105$	$-74/105.$

Thus all six vectors are in $\pi I \setminus pI$. The Kummer conjugates show $\mathrm{Tr}(X_s) = 0$, since $105 \nmid 15s - 104$, including for the negative powers occurring here. The direct norm computation used in Theorem 11.9 gives

$$\mathrm{Tr}(T_s) = \frac{30 \log(1 + p^7 b_0^s)}{p^{k_s+1}}, \qquad v(\mathrm{Tr}(T_s)) = s + 6 - k_s \in \{6, 9, 13\}, \qquad \mathrm{Tr}(u) = 210u \in \mathbb{Z}_p^\times.$$

For the norm formula, the seven distinct conjugates of $1 + p\pi^{15s}$ multiply to $1 + p^7 b_0^s$, and there are fifteen repetitions over the unramified quadratic subfield.

Let $V = I/pI$ and $L = \pi I/pI$, of codimension two. Lemma 11.3 gives coordinates $V = \mathbf{F}_p a \oplus \mathbf{F}_p b \oplus W$ in which the b-coordinate is a unit multiple of the trace. The trace is onto on πI: the element $1/210$ lies in $\mathcal{O}_E \subset \pi I$ and has trace one. Consequently $\ker B \to V/L$ is onto. The vector u is in L with nonzero b-coordinate, and the six other vectors are nonzero in $L \cap \ker B$.

There is an actual field automorphism $\sigma(\pi) = \zeta_7 \pi$, fixing $\mathbb{Q}_p(\mu_{210})$. Modulo pI, the leading term of T_s is α_s/p^{k_s}. Proposition 12.1 places its logarithmic error in pI. The characters on T_s and X_s are therefore respectively ζ_7^s and $\zeta_7^{15s-104} = \zeta_7^{s+1}$. All six are nontrivial. Since $\mu_7 \cap \mathbf{F}_{139}^\times = \{1\}$, each of their projective orbits has length seven. Indeed a difference between a nontrivial such scalar and a scalar of $\mathbf{F}_{139}^\times$ is a residue-field unit and cannot annihilate a nonzero vector.

For each vector there is at most one $h \in \{0, \ldots, 6\}$ placing σ^h of that vector on the line $\mathbf{F}_p a$. Avoid all six forbidden choices at once. This common inertia action fixes u and preserves L and the trace. Now the six vectors satisfy all hypotheses of Lemma 11.8, with $r = 6$ and $r + 1 < 139$. Its full-Galois lift sends them and u outside L simultaneously. Finally Proposition 11.5 converts the canonical action M into a Kummer action by taking the inverse Galois arrow: the resulting action is a $\mathbb{Z}_p^\times$-multiple of M. This preserves all seven minimum valuations. ☐

Theorem 12.5 (Exact point and whole-family hulls)**.** *For $s = j^2$, $j = 1, 2, 3$, and $m = j + 1$, define*

$$P_j^\rho = \mathcal{H}_\Gamma(\{(\tau_s, u, \ldots, u)\}), \qquad S_j^\rho = \mathcal{H}_\Gamma(\alpha_s I \times I^{m-1}),$$

where Γ is the full integral Galois–Kummer family. Then

$$P_j^\rho = \pi^{105k_s - 104m} B_m, \qquad S_j^\rho = \pi^{105n_s - 104m} B_m = p^{-1} P_j^\rho. \tag{101}$$

One common arrow attains both bounds in all three blocks. The same equalities hold with intermediate closed product convex hulls.

Proof. The containments $\tau_s \in p^{k_s} I$ and $\alpha_s I \subseteq p^{n_s} I$ give componentwise lower valuation bounds $k_s - m\kappa$ and $n_s - m\kappa$ for the two transported families. Lemma 12.4 attains the first bound on $(\tau_s, u, \ldots, u)$, and the second on $(\alpha_s \pi^{-104}, u, \ldots, u)$, using the same arrow: their first entries are respectively $p^{k_s} T_s$ and $p^{n_s} X_s$. Each attaining tensor has that valuation in every field component. Its principal B_m-span is the entire asserted product ideal, proving the reverse containments and closedness. Since $k_s = n_s + 1$, the two ideals differ by p^{-1}.

The containing product lattices $p^{k_s} I \times I^{m-1}$ and $p^{n_s} I \times I^{m-1}$ are closed and convex. Convex closure cannot violate the upper containments and retains the attaining tuples. ☐

Proposition 12.1 gives the same point hull with $(\alpha_s, 1, \ldots, 1)$; the same argument now also treats its product convex hull. The formulas remain valid for independent integral factor arrows if the diagonal witnesses are retained. For reference, the exponents of π in the two logarithmic scales are

j	m	P_j^ρ	S_j^ρ	P_j^{std}	S_j^{std}
1	2	-103	-208	107	2
2	3	-207	-312	108	3
3	4	-206	-311	214	109

The last two columns add $105m$ to the corresponding ρ columns. Nonintegrality in a ρ column is therefore not a violation of a bound stated in the standard cohomological coordinate.

12.5. **The maximal-order ideal before transport.** For $z = a \otimes 1 \otimes \cdots \otimes 1$ and $\Phi_F = F^{\otimes m}$, put

$$\mathcal{M}_\Gamma(a) = \overline{\operatorname{span}_{B_m}\{\Phi_F(b) : b \in zB_m,\ F \in \Gamma\}}. \tag{102}$$

The ideal zB_m is the input, not an ideal formed after applying Φ_F.

Proposition 12.6 (A trace-dual comparison for one source set)**.** *For $0 \neq a \in \mathcal{O}_E$,*

$$\mathcal{H}_\Gamma(\{(a, 1, \ldots, 1)\}) \subseteq \mathcal{M}_\Gamma(a) \subseteq \mathcal{H}_\Gamma(aI \times I^{m-1}). \tag{103}$$

Proof. The first containment follows from $z \in zB_m$. Take the integral trace dual of A_m in the finite etale algebra T_m. Trace factorizes on pure tensors; choosing an integral basis and its trace-dual basis in each factor therefore gives

$$A_m^\vee = (\mathfrak{D}_{E/\mathbb{Q}_p}^{-1})^{\otimes_{\mathbb{Z}_p} m} = I^{\otimes_{\mathbb{Z}_p} m}.$$

If $b \in B_m$ and $c \in A_m \subset B_m$, the product bc is integral in every component field. Its trace belongs to $\mathbb{Z}_p$, proving $B_m \subseteq A_m^\vee$. Consequently zB_m is contained in the $\mathbb{Z}_p$-span of pure tensors with entries in $aI \times I^{m-1}$. Apply the $\mathbb{Q}_p$-linear map Φ_F, then take the common B_m-span and closure. □

In particular, the preceding native example gives bounds for the same pre-hull input:

$$P_j^\rho \subseteq \mathcal{M}_\Gamma(\alpha_{j^2}) \subseteq S_j^\rho = p^{-1} P_j^\rho, \tag{104}$$

and hence

$$(m\kappa - k_{j^2}) \log p \leq V_B(\mathcal{M}_\Gamma(\alpha_{j^2})) \leq (m\kappa - n_{j^2}) \log p.$$

This is a valid interval bound of width $\log p$. The next result sharpens it and determines the middle ideal exactly in the attained examples.

Proposition 12.7 (The sharper trace-dual bound)**.** *Let $0 \neq a \in \mathcal{O}_E$, $r = v(a)$, and $k = \lfloor r + \kappa \rfloor$. If a $\mathbb{Q}_p$-linear map $\Phi : T_m \to T_m$ satisfies $\Phi(A_m^\vee) \subseteq A_m^\vee$, then*

$$\operatorname{span}_{B_m} \Phi(zB_m) \subseteq \pi^{ek - m(e-1)} B_m, \qquad z = a \otimes 1 \otimes \cdots \otimes 1. \tag{105}$$

In particular this bounds $\mathcal{M}_\Gamma(a)$ for every family of integral factor arrows used above. The upper bound itself does not require Φ to factor as a tensor product.

Proof. Take all trace duals with respect to the absolute algebra trace $\operatorname{Tr}_{T_m/\mathbb{Q}_p}$. Since $E/\mathbb{Q}_p$ is Galois, every component field of T_m is a copy of E. The idempotents of B_m isolate the components of the trace sum, so

$$B_m^\vee = \prod_\alpha I_\alpha \subseteq A_m^\vee = I^{\otimes_{\mathbb{Z}_p} m}, \tag{106}$$

where I_α is the inverse different in the corresponding copy of E. There is no additional degree or component-count factor in this formula. The containment follows from $A_m \subseteq B_m$ by contravariance of the trace dual.

Each component of z has valuation r. Because $r - k \geq -\kappa$, one has

$$p^{-k} zB_m \subseteq B_m^\vee \subseteq A_m^\vee.$$

It follows that $\Phi(zB_m) \subseteq p^k A_m^\vee$. Every pure tensor in $I^{\otimes m}$ has component valuation at least $-m\kappa$, and $(\pi^{1-e})^{\otimes m}$ attains this value in every component. Consequently

$$\operatorname{span}_{B_m} A_m^\vee = \pi^{-m(e-1)} B_m.$$

Multiplying by p^k proves (105). The bounding ideal is closed, so unions and closures preserve this upper bound. Integral factor arrows preserve $I^{\otimes m} = A_m^\vee$, as required. □

Corollary 12.8 (Exact pre-ideal hull)**.** *If the point orbit attains depth $k - m\kappa$ in every component, then*

$$\mathcal{M}_\Gamma(a) = \mathcal{H}_\Gamma(\{(a,1,\ldots,1)\}) = \pi^{ek-m(e-1)}B_m.$$

In the degree-210 three-label example, this gives

$$\begin{gathered}\mathcal{M}_\Gamma(\alpha_{j^2}) = P_j^\rho, \qquad S_j^\rho = p^{-1}\mathcal{M}_\Gamma(\alpha_{j^2}),\\ V_B(\mathcal{M}_\Gamma(\alpha_{j^2})) = (m\kappa - k_{j^2})\log p, \qquad m = j+1, \quad j = 1,2,3.\end{gathered} \tag{107}$$

The same actual full-Galois arrow can witness all three point attainments.

Proof. The point-orbit containment in Proposition 12.6 and the assumed attainment give the reverse inclusion to (105). In the stated example, Proposition 12.1 identifies each transported root-point principal ideal with its normalized crystalline-point ideal, including the background factors. Theorem 12.5 supplies the common attaining arrow and the separate whole-family formula. The measure follows from the depth of the resulting product ideal. □

This comparison has the input type of the containing calculation in [65, Proposition 1.2, PDF pp. 10–11; Theorem 1.10, Step (v), pp. 27–28]. Indeed Φ_F preserves $I^{\otimes m}$, and hence $(\log \mathcal{O}_E^\times)^{\otimes m} = p^m I^{\otimes m}$, as required there. The displayed input is $\phi(p^\lambda B_m)$, with $\lambda = v(\underline{q}^{j^2})$ at a bad place. Here $\underline{q} = q^{1/(2\ell)}$ is the root fixed in [63, Example 3.2(iv), PDF p. 71]. In the example $e = 105, p = 139$, its parameters are

$$d_i = 104/105, \qquad a_i = 1/105, \qquad b_i = -1/105.$$

Thus its displayed containing ideal has depth

$$\lfloor \lambda - d_I - a_I \rfloor - b_I = \lfloor \lambda \rfloor - m\kappa, \qquad \lambda = j^2/7,$$

which is the upper endpoint in (104). By (107) it is a larger containing ideal, equal to p^{-1} times the exact pre-ideal hull in these examples. A loose upper bound is not a contradiction, and its agreement with the whole-family formula does not identify those input sets.

If all tensor coordinates are changed from ρ to the standard logarithm, the pre-ideal input changes from zB_m to $p^m zB_m$. Its hull then becomes $p^m\mathcal{M}_\Gamma(\alpha_{j^2}) = P_j^{\mathrm{std}}$, with measure shift $-m\log p$. This conversion does not insert p^m into a printed $\phi(p^\lambda B_m)$ input without changing its coordinate description.

12.6. **The source distinctions that remain.** Three distinctions are necessary when using these local interfaces in the published constructions.

First, [56, §9.6.2 and Theorem-Definition 9.8.1.1, PDF pp. 109, 114–115] defines a chosen class from $1 + p\,q^{1/(2\ell)}$ and background classes from $1+p$. The optional family in its Remark 9.6.2.1 is printed with exponent $j/(2\ell)$, whereas the squared-exponent pilot in [62, Theorem 3.11; Remark 3.11.1(i)] involves $\underline{q}^{j^2}$. The valuation scaling of [56, (4.2.2.2)] is additional marking data. It does not by itself replace an algebraic root by its squared power. For example, in our fixed native field $\lambda_p(r_0)$ belongs to $pI \setminus p^2I$, while $\lambda_p(r_0^9)$ belongs to $p^2I \setminus p^3I$. No integral log-lattice isomorphism identifies these two specified elements. A change $v \mapsto cv$ changes $v(p)$ as well and does not change this integral content.

Second, the monoid representation in [62, Theorem 3.11(i)(b), PDF pp. 153–154] is a subset equipped with a multiplicative action; the pilot also has the fractional-ideal realization of its Proposition 3.7(v) and Definition 3.8(i). A point, an operator image, and a principal ideal before transport have different quantifiers. Proposition 12.1 supplies an equality for specified point hulls, and Proposition 12.2 supplies a set equality for a specified whole family. Neither permits a B_m-span to be moved across a non-B_m-linear arrow. Proposition 12.7 instead bounds the entire pre-ideal before the arrow is applied; together with point attainment it proves (107) without moving a module span across that arrow.

Third, the standard versus normalized logarithmic map must be the same on both sides of a volume comparison. Equation (96) is its explicit conversion, including background factors. The native fixed-source covariance of an all-isomorphism collation cannot at the same time be interpreted as a change of its source element or source principal ideal. Remark 6.10.2 of [57] itself restricts direct comparison of local quantities at different arithmeticoids.

The degree-210 calculation applies locally to a split rational Frey curve with full rational 2-torsion and $v(q) = 2$. Tate uniformization [71, Theorem C.14.1] makes a chosen square root of q rational: its class is a rational point of order two, so its Galois sign must be a power of q; valuations force that power to be zero. Thus it supplies precisely the b_0 allowed above, without requiring $q = p^2$. This local observation does not establish all the initial data in [63, Definition 3.1].

There is also a small-parameter limitation to using the particular existence proof of [57, §5.8]. Its prime threshold must satisfy $\xi_{\mathrm{prm}} > 10$, because $\theta(10) = \log 210 < 20/3$. With the normalized Q of its Definition 5.4.1 and Theorem 5.7.1, every candidate with $\sqrt{Q} \le \ell = 7$ is therefore included in the proof's small-Q exceptional set, if it is in the chosen bounding domain. The opening display of Section 5.8 omits the degree divisor present in that definition. If this display is instead read literally as a different unnormalized quantity, the normalized value and numerical prime window cannot be reused for it. Neither reading turns a local example into an automatic outside-exception global input. This statement concerns that proof-constructed set, not every possible sufficient exceptional set or a direct verification of initial data.

Finally, the actual global family, its product weights, the Ind3 operation, and the cross-Frobenius comparison are not computed in this section. The trace-dual bound, the attained exact pre-ideal equality, and the two positive Kummer interfaces are local theorems about their explicitly specified sets. Their p-adic analytic and full local-Galois hypotheses have not yet been completely formalized in Lean. They neither prove nor disprove the unconditional abc conjecture.

13. Explicit Frey curves realizing the local fields

The local calculations above use actual finite extensions of $\mathbb{Q}_p$. We now realize their ramification by torsion fields of explicit rational elliptic curves. This separates a local model from a claim that every condition on global initial data has already been supplied. The latter construction is given separately in Section 15.

For positive coprime a, b, put $c = a + b$ and

$$D_{a,b}: \quad y^2 = x(x-a)(x+b), \qquad S = a^2 + ab + b^2.$$

Direct calculation from the Weierstrass coefficients gives

$$\Delta(D_{a,b}) = 16(abc)^2, \qquad c_4(D_{a,b}) = 16S, \qquad j(D_{a,b}) = \frac{256S^3}{(abc)^2}. \tag{108}$$

Reducing S modulo a, b, c gives b^2, a^2, a^2, respectively; thus $\gcd(S, abc) = 1$. At an odd prime $r \mid abc$, the displayed equation is minimal and multiplicative, and its Tate order is $2v_r(abc)$ after a splitting extension. Indeed c_4 is a unit, which both proves minimality and gives the reduction criterion [71, VII.5.1].

Lemma 13.1 (Two transvections and passage to a torsion field)**.** *Let ℓ be a prime and let $G \subseteq \mathrm{GL}_2(\mathbf{F}_\ell)$ be a subgroup. If G contains a nonidentity transvection and an element preserving no $\mathbf{F}_\ell$-line, then G contains $\mathrm{SL}_2(\mathbf{F}_\ell)$. If $H \triangleleft G$ has finite index prime to ℓ, then H also contains this special linear group.*

Proof. Let T be the transvection, with fixed line L, and let g be the second element. The lines L and gL differ. In a basis along them, T and gTg^{-1} have matrices

$$U(s) = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \qquad V(t) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}, \qquad st \neq 0.$$

Their integer powers give $U(x)$ and $V(y)$ for every $x, y \in \mathbf{F}_\ell$, since the field is prime. Elementary row reduction shows that these two root groups generate $\mathrm{SL}_2(\mathbf{F}_\ell)$: for a matrix with nonzero upper-left entry, a lower row operation first clears the lower-left entry, and an upper operation clears the upper-right entry; diagonal matrices are products of elementary matrices. Explicitly,

$W(t) = U(t)V(-t^{-1})U(t)$ has entries $\left(\begin{smallmatrix} 0 & t \\ -t^{-1} & 0 \end{smallmatrix}\right)$, so $W(t)W(-1) = \mathrm{diag}(t, t^{-1})$. If the upper-left entry is zero, a preliminary upper operation makes it nonzero. For the last assertion, the image of T in G/H has order dividing both ℓ and $[G : H]$, hence is trivial. Normality retains gTg^{-1}, so the same argument applies inside H. □

Here and below the arithmetic use of this lemma invokes two classical facts. Good-reduction Frobenius has characteristic polynomial $T^2 - a_rT + r$, with $a_r = r + 1 - \#D(\mathbf{F}_r)$ [71, V.2; VII.4]. At a split multiplicative prime $p \neq \ell$ with $\ell \nmid v_p(q)$, the Tate description $\overline{\mathbb{Q}}_p^\times/q^{\mathbb{Z}}$ supplies a nontrivial inertia transvection on $D[\ell]$: adjoining $q^{1/\ell}$ has ramification index ℓ, while μ_ℓ is unramified. See [77, Theorems 1–2 and Proposition 3]. These statements about arithmetic representations are not consequences of the formal elementary-matrix lemma alone.

Proposition 13.2 (Degree 210 at the prime 139)**.** *Each of the curves*

$$D_{139,279}, \qquad D_{1,2362}$$

has split multiplicative reduction at 139 *with native Tate order two. For either curve D, the mod-7 image over $F = \mathbb{Q}(i, D[30])$ contains* $\mathrm{SL}_2(\mathbf{F}_7)$*. The completion at* 139 *of* $\mathbb{Q}(i, D[210])$ *is*

$$E = \mathbb{Q}_{139}(\mu_{210}, \pi), \qquad \pi^{105} = b_0, \qquad b_0^2 = q, \quad v_{139}(b_0) = 1. \tag{109}$$

It has ramification index 105*, residue degree two, and degree* 210*. The second curve is already a rational Legendre curve, with parameter* -2362*.*

Proof. The factorizations are

$$279 = 3^2 \cdot 31, \quad 418 = 2 \cdot 11 \cdot 19, \qquad 2362 = 2 \cdot 1181, \quad 2363 = 17 \cdot 139.$$

All displayed factors other than 3^2 are prime. For the first curve, reduction at 139 gives $y^2 = x^2(x+1)$; for the second it gives $y^2 = x(x-1)^2$. The respective nodes have two distinct rational tangent directions, of slopes ± 1. Equation (108) gives $v_{139}(\Delta) = 2$ and unit c_4 in both cases.

Counting the actual points modulo five, including infinity, gives 8 and 4, respectively. The two Frobenius polynomials are $T^2 + 2T + 5$ and $T^2 - 2T + 5$; both have discriminant -16, a nonsquare modulo seven. The transvection at 139 and Lemma 13.1 therefore give the special linear subgroup over $\mathbb{Q}$. The field F is Galois and

$$[F : \mathbb{Q}] \mid 2\,|\,\mathrm{GL}_2(\mathbb{Z}/30\mathbb{Z})| = 2 \cdot 6 \cdot 48 \cdot 480 = 276480 = 2^{11}3^3 \cdot 5. \tag{110}$$

Its degree is prime to seven, so the normal-subgroup part of the lemma applies over F.

Full rational two-torsion makes a square root b_0 of q rational over $\mathbb{Q}_{139}$. In fact its Tate class is rational, so for every Galois element σ the quotient $\sigma(b_0)/b_0$ belongs to both $\{\pm 1\}$ and $q^{\mathbb{Z}}$. Its valuation is zero, forcing that quotient to be one. The order of 139 modulo 210 is two, so the cyclotomic field in (109) is unramified quadratic. The polynomial $T^{105} - b_0$ is Eisenstein there. The Tate torsion classes give $\mathbb{Q}_{139}(D[210]) = \mathbb{Q}_{139}(\mu_{210}, q^{1/210}) = E$; fixing their classes forces the root-of-unity multipliers to be one. The unramified quadratic field also contains i. Finally $D_{1,2362}$ is $y^2 = x(x-1)(x+2362)$, as asserted. No equality $q = 139^2$ has been assumed. □

These small examples require care if used with the particular finite-exception construction of [57, §5.8]. Write

$$N_1 = 8105229 = 139 \cdot 279 \cdot 418/2, \qquad N_2 = 2790703 = 2362 \cdot 2363/2.$$

In both cases the normalized Tate quantity is $Q_D = 2\log N_i$. At two, $v_2(j) = 6$; adjoining the full three-torsion gives good reduction there, by the finite-torsion argument below. Summing the odd pole orders of j gives the asserted normalized value. The exact inequalities

$$3^{25} < N_i^2 < 2^{49}, \qquad i = 1, 2,$$

and $2 < \exp(1) < 3$ imply $25 < Q_D < 49$. But the prime threshold in that existence proof must exceed ten, since $\theta(10) = \log 210 < 6 < 20/3$. Thus candidates with $\sqrt{Q_D} \leq 7$ belong to its proof-constructed small-height exceptional set when they lie in the chosen bounding domain. This observation uses the normalized definition of Q_D in Definition 5.4.1 and Theorem 5.7.1 of that source. The opening display of its Section 5.8 omits the degree divisor; a literal unnormalized reading must not reuse the preceding normalized values. This is a limitation of invoking that sufficient existence proof, not a failure of the local calculations or of a direct initial-data construction.

Theorem 13.3 (An exact balanced example at level 43)**.** *Put*

$$p = 1289, \quad \ell = 43, \quad A = p(p^{16} + 428), \qquad (a, b, c) = (A^2, A^2 + 1, 2A^2 + 1), \quad D = D_{a,b}.$$

This is a primitive abc triple, and D is rationally isomorphic to the Legendre curve of parameter $\lambda = -1 - A^{-2}$. For $F = \mathbb{Q}(i, D[30])$, the following assertions hold:

(1) *D is good above two and at 43, and split multiplicative at every odd prime dividing abc.*
(2) *The mod-43 image over both $\mathbb{Q}$ and F contains $\mathrm{SL}_2(\mathbf{F}_{43})$.*
(3) *Every nonzero integral Tate order over F is prime to 43.*
(4) *The normalized Tate quantity satisfies the exact bounds*

$$Q_D = 2\log\left(\frac{abc}{2}\right), \qquad 1224 < Q_D < 1648 < 43^2. \tag{111}$$

(5) *At p, the completion of $K = \mathbb{Q}(i, D[1290])$ has degree 1290, ramification index 645, and residue degree two.*

Proof. The three endpoints are pairwise coprime, and $a + b = c$. The rational substitution $x = A^2 X$, $y = A^3 Y$ gives

$$Y^2 = X(X - 1)(X + 1 + A^{-2}),$$

so no quadratic twist is introduced. The exact residues are $A \equiv 5 \pmod 8$ and $A \equiv 1 \pmod{5 \cdot 43}$. At an odd prime dividing a or b, the reduced equation is $y^2 = x^2(x + 1)$, with tangent slopes ± 1. At a prime dividing c, translation of its node gives $y^2 = X^2(X + A^2)$, with slopes $\pm A$. All these slopes are nonzero. The endpoints are $(1, 2, 3)$ modulo 43, so the discriminant is a unit there.

At two, $v_2(abc) = 1$ and $3A^4 + 3A^2 + 1$ is odd, whence $v_2(j) = 6$. Here is a direct reason that the particular field F gives good reduction. At any completion F_w above two, the group $D[3] \simeq (\mathbb{Z}/3\mathbb{Z})^2$ is rational. If reduction were additive, both the formal subgroup $D_1(F_w)$ and the quotient $D_0(F_w)/D_1(F_w)$ would have no three-torsion. Thus $D[3]$ would inject into $D(F_w)/D_0(F_w)$, whose order is at most four in the additive case, a contradiction. These are [71, VII.2.1, VII.3.1, VII.5.1(c), VII.6.1], valid over finite extensions of $\mathbb{Q}_2$ as well. Multiplicative reduction would require negative j-valuation. The only remaining possibility is good reduction.

At five the reduced model has four points and Frobenius trace two. The discriminant -16 of $T^2 - 2T + 5$ is a nonsquare modulo 43. Also $v_p(A) = 1$, giving native Tate order four at p. Lemma 13.1, followed by (110), proves the image assertion.

For clarity, the assertion about all prime exponents does not use a factorization of the large endpoints. Let $P = \prod_{r \le 511,\ r \text{ prime}} r$ be the actual primorial. The exact integer calculations give

$$a, b, c < 2^{377}, \qquad \gcd(a, P) = 3, \quad \gcd(b, P) = 2, \quad \gcd(c, P) = 17,$$

and $v_3(a) = 2$, $v_2(b) = 1$, $v_{17}(c) = 1$. These identities can be verified by integer Euclidean division; the companion Lean proof also checks the primorial defined as the product over all primes in the indicated range. Every prime $r \ge 512$ has $r^{43} \ge 2^{387} > a, b, c$. Since the endpoints are pairwise coprime, it follows that

$$v_r(abc) < 43 \qquad \text{for every prime } r. \tag{112}$$

At a place $w \mid r$ of F, the nonzero integral Tate order is $2e(w/r)v_r(abc)$. The ramification index divides the Galois degree in (110), which is prime to 43; equation (112) proves the claim.

The normalized sum over F is the sum of the rational odd pole orders, because

$$\sum_{w \mid r} e(w/r) f(w/r) = [F : \mathbb{Q}].$$

This proves the identity in (111). For its strict bounds, put $N = abc/2$. We have $p^{17} < A < 2p^{17}$ and $A^6 < N < 3A^6$. The integer inequalities $3^6 < p < (8/3)^8$ give

$$3^{1224} < N^2 < 9 \cdot 2^{12}(8/3)^{1632} < (8/3)^{1648}.$$

For the last step use $9 \cdot 2^{12} < 3^{12} < (8/3)^{16}$. Together with $8/3 < \exp(1) < 3$, these imply (111) without a floating point approximation.

Finally full rational two-torsion gives $b_0 = \sqrt{q} \in \mathbb{Q}_p$ with $v_p(b_0) = 2$, by the proof of Proposition 13.2. Put $K_0 = \mathbb{Q}_p(\mu_{1290})$ and choose $\pi^{645} = b_0$. The field K_0 is unramified quadratic, since $p \equiv -1 \pmod{1290}$. Writing $\eta = p^2/b_0 \in \mathbb{Z}_p^\times$, the genuine uniformizer is

$$\beta = \pi^{323}/p, \qquad \beta^{645} = p\eta^{-323}, \qquad \pi = \eta\beta^2. \tag{113}$$

The first polynomial in β is Eisenstein. Tate torsion identifies $K_v = K_0(\pi) = K_0(\beta)$, and K_0 contains i. The claimed degrees follow. In particular the native root $q^{1/86} = \pi^{15}$ has valuation $2/43$, not $1/43$. □

The numerical upper bound on the level in [57, Theorem 5.7.1] is also satisfied here: with $\delta = 2^{12}3^3 \cdot 5$ and $d_{\text{mod}} = 1$,

$$\sqrt{Q_D} < 43 < 10\delta\sqrt{Q_D}\log(2\delta \log Q_D).$$

This directly verifies a window for the specified prime. It does not identify that prime with an unspecified existential choice in another theorem. The next sections construct the additional data and an unbounded family independently of that identification.

14. All square labels in a tame valuation-four Tate field

The preceding constructions extend to every square label for an arbitrary prime level. The new step is one inertia choice for both the point and whole-product inputs. We retain the actual full-Galois lifts of Section 11 and the distinct source types of Section 12.

Let $\ell \geq 7$ and p be primes such that

$$p \equiv -1 \pmod{30\ell}, \qquad e = 15\ell, \quad d = 2e, \quad h = (\ell-1)/2.$$

No condition $\ell \equiv 3 \pmod 4$ is needed for this local result. Choose $b_0 \in \mathbb{Q}_p$ with $v_p(b_0) = 2$, and set

$$\begin{aligned} &q = b_0^2, \qquad K_0 = \mathbb{Q}_p(\mu_{2e}), \qquad \pi^e = b_0, \qquad E = K_0(\pi), \\ &\beta = \pi^{(e+1)/2}/p, \qquad r = \pi^{15} = q^{1/(2\ell)}, \qquad \kappa = 1 - 1/e. \end{aligned} \tag{114}$$

All roots are chosen once, before the common arrow below is selected.

Lemma 14.1 (The genuine uniformizer). *The extension $E/\mathbb{Q}_p$ is Galois with ramification index e, residue degree two, and degree d. The element β is a uniformizer; π has valuation $2/e$ and is not one. Writing $\eta = p^2/b_0 \in \mathbb{Z}_p^\times$, one has*

$$\beta^e = p\eta^{-(e+1)/2}, \qquad \pi = \eta\beta^2, \qquad r = \eta^{15}\beta^{30}, \qquad r^\ell = b_0. \tag{115}$$

Moreover

$$I = p^{-1}\log \mathcal{O}_E^\times = \beta^{1-e}\mathcal{O}_E, \qquad \operatorname{Tr} I = \operatorname{Tr}(\beta I) = \mathbb{Z}_p. \tag{116}$$

The trace kernel surjects onto $I/\beta I$.

Proof. The order of p modulo $2e$ is two, so $K_0/\mathbb{Q}_p$ is unramified quadratic. Since e is odd, the exponent $(e+1)/2$ is an integer, and direct calculation gives (115). Its first equation is Eisenstein over K_0, and its second gives $K_0(\beta) = K_0(\pi)$. The roots of unity in K_0 make E a splitting field. These observations prove the field assertions.

One has $p \geq 2e - 1$, hence $e \leq p - 2$. Logarithm and exponential are inverse isometries on $1 + \beta\mathcal{O}_E$ and $\beta\mathcal{O}_E$. Dividing the logarithmic image by p proves the lattice identity; the unit in (115) is not discarded as an element. Tameness gives $I = \mathfrak{D}_{E/\mathbb{Q}_p}^{-1}$, so its trace is integral. Also $p \nmid d$ and $\operatorname{Tr}(1/d) = 1$, with $1/d \in \mathcal{O}_E \subset \beta I$. This proves the trace identities. For $x \in I$, subtracting $\operatorname{Tr}(x)/d \in \mathcal{O}_E$ kills its trace without changing its class modulo βI. □

This field is the full local 30ℓ-torsion field of the split Tate curve with parameter q. Conversely, a split elliptic curve over $\mathbb{Q}_p$ with full rational 2-torsion and Tate valuation four supplies a b_0 as above. Indeed, the rationality of the Tate class of $\sqrt{q}$ makes its Galois ratio a power of q; valuation zero forces that power to be zero. Applying the same argument to both Tate torsion generators proves equality with $\mathbb{Q}_p(\mu_{30\ell}, q^{1/(30\ell)})$, not merely containment. Here we use the Galois-equivariant Tate uniformization theorem [71, Theorem C.14.1].

For $1 \le j \le h$, put $s = j^2$ and define

$$
\begin{gathered}
z_j = r^{j^2}, \qquad n_j = \left\lfloor \frac{2j^2}{\ell} \right\rfloor, \qquad k_j = n_j + 1 = \left\lceil \frac{2j^2}{\ell} \right\rceil, \\
u = p^{-1}\log(1+p), \qquad \tau_j = p^{-1}\log(1+pz_j), \qquad t_j = p^{-k_j}\tau_j, \qquad x_j = p^{-n_j} z_j \beta^{1-e}.
\end{gathered}
\tag{117}
$$

The ratio $2j^2/\ell$ is nonintegral in this range, and $k_j = \lfloor 2j^2/\ell + \kappa \rfloor$.

Lemma 14.2 (Contents and trace depths)**.** *The elements t_j, x_j lie in $\beta I \setminus pI$, while $u \in \beta I$ and $\operatorname{Tr} u$ is a unit. For every label,*

$$
\begin{gathered}
\operatorname{Tr}(t_j) = \frac{30}{p^{k_j+1}} \log(1 + p^{\ell} b_0^{j^2}), \\
v_p(\operatorname{Tr}(t_j)) = \ell - 1 + 2j^2 - k_j \ge \ell, \qquad \operatorname{Tr}(x_j) = 0.
\end{gathered}
\tag{118}
$$

Modulo pI, t_j has the pure leading term z_j/p^{k_j}.

Proof. Write $2j^2 = \ell n_j + a_j$, with $1 \le a_j \le \ell - 1$. Then

$$
e\, v_p(t_j) = 15a_j - e, \qquad e\, v_p(x_j) = 15a_j - e + 1.
$$

Both exponents are strictly greater than $2-e$ and less than 1, which proves the lattice assertions. Also $\operatorname{Tr} u = du \in \mathbb{Z}_p^\times$.

For $R = v_p(z_j) = 2j^2/\ell$, the logarithmic tail has valuation at least $1 + 2R$. Dividing by p^{k_j} leaves valuation at least $1 + R - \kappa \ge 1/e$, so the remainder belongs to pI. The relation $r^\ell = b_0$ and $\gcd(j^2, \ell) = 1$ give

$$
\mathrm{N}_{E/K_0}(1 + pr^{j^2}) = (1 + p^\ell b_0^{j^2})^{15}.
$$

Taking logarithms and then the degree-two trace proves the first formula in (118). Its valuation follows from $v_p(b_0) = 2$, $p \nmid 30$, and $k_j \le j^2$. Finally x_j is a rational unit times $\beta^{30j^2+1-e}/p^{n_j}$. Its exponent is 1 modulo 15, so summing its e conjugates over K_0 gives zero. The geometric sum applies also to a negative exponent. $\square$

Use the integral JW basis of Lemma 11.3, writing it as a, b, W, with W of rank $d-2$. Let B be the b-coordinate. Lemma 14.1 gives $B(v) = \operatorname{Tr}(v)/\operatorname{Tr}(b)$ with unit denominator. Set

$$
V = I/pI, \qquad V_0 = \beta I/pI.
$$

The quotient V/V_0 has dimension two, and $\ker B \to V/V_0$ is onto. Lemma 14.2 places the $2h$ nonzero inputs in $V_0 \cap \ker B$, while $B(u) \ne 0$.

Lemma 14.3 (One inertia exponent for both families)**.** *There is one $\sigma^\nu \in \operatorname{Gal}(E/K_0)$ that puts all t_j, x_j off the line $\mathbf{F}_p a$ in V, without changing u, trace, or membership in V_0. At most*

$$
15h + (h - 1 + \ell) = 9\ell - 9 < 15\ell = e
\tag{119}
$$

of the exponents $0 \le \nu < e$ are forbidden.

Proof. Choose $\sigma(\beta) = \zeta_e\beta$. Since $\gcd(e, p-1) = \gcd(e, 2) = 1$, the roots μ_e have trivial intersection with $\mathbf{F}_p^\times$ after reduction. For a character of order $o \mid e$, its leading projective coefficients therefore run through o distinct $\mathbf{F}_p$-lines. A fixed line is hit at most e/o times as $0 \le \nu < e$ varies. If the valuation of the target vector differs, there is no hit. In every case equality of full vector lines in V requires this leading-line equality, which is all that the following count uses.

The leading character of t_j is $\zeta_e^{30j^2}$, of order ℓ, because $\gcd(e, 30j^2) = 15$. Thus each of the h point inputs forbids at most 15 exponents.

For x_j the exponent modulo e is $30j^2+1$. It is 1 modulo 15, so its gcd with e is either 1 or ℓ. The second possibility requires

$$30j^2+1 \equiv 0 \pmod{\ell}.$$

There is at most one such j in $1,\dots,h$: the two possible roots modulo the prime ℓ are a nonzero pair $j,-j$, with exactly one member in this half range. A nonexceptional character has order e and forbids at most one exponent. The possible exceptional character has order 15 and forbids at most ℓ. Thus these inputs forbid at most $h-1+\ell$ exponents, also a valid bound when no exception occurs. Adding the two bounds gives (119), leaving a common choice of ν. Field inertia preserves valuation and trace and fixes the rational element u, as required. ☐

Theorem 14.4 (A common arrow for every square label)**.** *There is one actual integral Kummer arrow $F \in \Gamma_E$ with*

$$v_p(Fu) = v_p(Ft_j) = v_p(Fx_j) = -\kappa \qquad (1 \le j \le h). \tag{120}$$

The chosen arrow may depend on the field and the chosen roots; within these choices it is the same for all labels and both families.

Proof. Apply Lemma 14.3. Every resulting input has a nonzero W-component, because it lies in $\ker B$ but off $\mathbf{F}_p a$. The other hypotheses of Lemma 11.8 were verified above. Its required inequality is

$$2h+1 = \ell < p.$$

It therefore gives one parabolic composition moving all inputs and u outside V_0. The required residue vectors have integer coordinate lifts; Lemma 11.4 and the full-presentation handle construction realize this composition by one actual Galois-group automorphism. Compose it with the chosen field inertia. Being outside V_0 means valuation $-\kappa$. Finally Proposition 11.5 converts the covariant canonical action to the Kummer action of the inverse automorphism, up to a common $\mathbb{Z}_p^\times$-factor. That factor changes none of the valuations in (120). ☐

For $m=j+1$, use the native algebra T_m and maximal order B_m of Section 12, and define

$$\begin{aligned} P_j^\rho &= \mathcal{H}_{\Gamma_E}(\{(\tau_j,u,\dots,u)\}),\\ S_j^\rho &= \mathcal{H}_{\Gamma_E}(z_jI \times I^{m-1}). \end{aligned}$$

Corollary 14.5 (Two attained endpoint hulls)**.** *For every $1 \le j \le h$,*

$$\begin{aligned} P_j^\rho &= \beta^{ek_j-(e-1)m}B_m,\\ S_j^\rho &= \beta^{en_j-(e-1)m}B_m = p^{-1}P_j^\rho. \end{aligned} \tag{121}$$

The same arrow witnesses every equality, and the spans are closed.

Proof. For the point hull, apply Proposition 11.10 to $\tau_j = p^{k_j}t_j$ and (120). For the whole product, $z_jI \subset p^{n_j}I$ bounds each component valuation below by $n_j - m\kappa$. The first input $z_j\beta^{1-e} = p^{n_j}x_j$ and the repeated remaining input u attain that bound under the same arrow. The attaining tensor is a unit times the asserted power of β in every field component, so its principal B_m-span is the entire product ideal. Closedness follows from that fractional-lattice description. Since $k_j = n_j+1$ and p is a unit times β^e, the ratio is p^{-1}. ☐

The full principal-unit families in Proposition 12.2 realize $z_jI \times I^{m-1}$ cohomologically. Selecting those families is an explicit enlargement of the single-class input; it is not entailed by matching the coefficient norm of one point.

Proposition 14.6 (A trace-dual upper bound for the pre-transport ideal)**.** *Let $A_m = \mathcal{O}_E^{\otimes_{\mathbb{Z}_p} m} \subset T_m$, and take trace duals using the absolute algebra trace on T_m. For any family of $\mathbb{Q}_p$-linear maps $\Phi : T_m \to T_m$ satisfying $\Phi(A_m^\vee) \subseteq A_m^\vee$, one has*

$$\overline{\operatorname{Span}_{B_m}\{\Phi(x) : x \in (z_j \otimes 1 \otimes \cdots \otimes 1)B_m\}} \subseteq \beta^{ek_j-(e-1)m}B_m, \tag{122}$$

where the span includes every map in the specified family. No B_m-linearity or trace preservation of the maps is required.

Proof. Put $Z_j = z_j \otimes 1 \otimes \cdots \otimes 1$. The product decomposition $T_m \simeq \prod_{d^{m-1}} E$ identifies its absolute algebra trace with the sum of the component field traces. Testing against the individual idempotents in B_m gives

$$B_m^\vee = \prod_{d^{m-1}} I \subseteq A_m^\vee = I^{\otimes_{\mathbb{Z}_p} m}.$$

The inclusion reverses $A_m \subseteq B_m$; the last equality follows by tensoring an integral basis and its trace-dual basis. There is no factor equal to the number of product components.

Every component of Z_j has valuation $R_j = 2j^2/\ell$. Since $k_j = \lfloor R_j + \kappa \rfloor$, one has

$$p^{-k_j} Z_j B_m \subseteq B_m^\vee \subseteq A_m^\vee.$$

Thus $Z_j B_m \subseteq p^{k_j} A_m^\vee$. Applying any allowed Φ preserves this containing lattice by $\mathbb{Q}_p$-linearity. Its B_m-span is contained in

$$p^{k_j} \operatorname{Span}_{B_m}(I^{\otimes_{\mathbb{Z}_p} m}) = \beta^{ek_j-(e-1)m} B_m.$$

The last equality follows from factor valuations at least $-\kappa$, attained by $\beta^{1-e} \otimes \cdots \otimes \beta^{1-e}$ in every component. This product ideal is closed, completing the proof. $\square$

Corollary 14.7 (The exact pre-transport pilot and the two scales)**.** *For the pre-transport ideal hull of* (102)*,*

$$\mathcal{M}_{\Gamma_E}(z_j) = P_j^\rho \subsetneq S_j^\rho = p^{-1} P_j^\rho. \tag{123}$$

The equality still holds upon enlarging the allowed tensor arrows to any family of $\mathbb{Q}_p$-linear maps preserving $A_m^\vee$ and containing the already constructed attaining diagonal arrow. Every exponent in (121) *is strictly negative. Consistently multiplying all m coordinates and the corresponding source by p gives*

$$P_j^{\text{std}} = \beta^{ek_j+m} B_m, \qquad S_j^{\text{std}} = \beta^{en_j+m} B_m, \tag{124}$$

whose valuation exponents are strictly positive, and $p^m \mathcal{M}_{\Gamma_E}(z_j) = P_j^{\text{std}}$.

Proof. Proposition 12.1 identifies the point orbit hull with that of $(z_j, 1, \ldots, 1)$, separately after every integral arrow. Since $Z_j \in Z_j B_m$, the attaining diagonal arrow supplies $P_j^\rho \subseteq \mathcal{M}_{\Gamma_E}(z_j)$. Each integral tensor arrow preserves $A_m^\vee = I^{\otimes_{\mathbb{Z}_p} m}$, so Proposition 14.6 proves the opposite inclusion. The same proof works for the larger stated map families. This strengthens Proposition 12.6 for the same pre-transport input; it never moves the B_m-span through a merely $\mathbb{Q}_p$-linear map. The whole-product hull is strictly larger because $p^{-1} P_j^\rho \neq P_j^\rho$.

For $j \le h$, one has $2j^2/\ell < j$, hence $k_j \le j$. Thus

$$ek_j - (e-1)(j+1) \le j - (e-1) < 0.$$

The whole exponent is smaller by e. The scale rule (96) adds em to each exponent, giving $ek_j + m > 0$ and $en_j + m > 0$ and proving (124). $\square$

For additive Haar measure on the entire T_m, normalized by $\mu_B(B_m) = 1$, put $V_B(H) = d^{-m} \log \mu_B(H)$. Then

$$\begin{gathered} V_B(\mathcal{M}_{\Gamma_E}(z_j)) = V_B(P_j^\rho) = (m\kappa - k_j) \log p > 0, \\ V_B(S_j^\rho) = V_B(P_j^\rho) + \log p. \end{gathered} \tag{125}$$

The remaining $\log p$ difference belongs to the larger whole source. After consistent standard scaling, the exact volumes of the pre-transport pilot and the whole source are respectively $-(k_j + m/e) \log p$ and $-(n_j + m/e) \log p$, both strictly negative. Thus the signs of valuation exponents and logarithmic Haar volumes are opposite. This Haar measure is not a probability measure restricted to B_m.

The equality in (123) identifies two native hulls, not their source objects, and it leaves a strict distinction from the whole-product source. This result does not supply global initial data, a compact bounding domain, escape from a finite exceptional set, the additional Ind3 unions, or the cross-Frobenius comparison. The specified native square powers and coordinate scale remain part of its statement.

The finite arithmetic of the half-range exception, the budget $9\ell - 9$, the strict floor/ceiling bracket, and the exponent identities is kernel-checked in the companion module `IUTGeneralTameSquareLabels20260830`. The local-field, logarithmic, full-Galois-lift and pilot arguments above remain paper proofs using the stated mathematical inputs; the companion does not introduce them as Lean axioms. The algebraic trace-dual and transported-span steps are separately checked in `TraceDualPreidealHull20260831` using the actual algebra trace; this does not formalize the local inverse different, tensor-basis identification, or attaining Galois arrow.

15. Constructing the global initial theta data

The local fields of Section 13 arise from a specific rational elliptic curve. We first construct its additional global covering, cusp, and section of places required by the original definition of initial theta data, then extend the construction to every member of Theorem 16.2. The construction uses [63, Definition 3.1(a)–(f)] directly. It does not invoke an existence theorem outside a finite exceptional set, and it keeps the chosen prime and the global base field fixed.

For the first construction put

$$\begin{gathered} \ell = 43, \qquad p = 1289, \qquad A = p(p^{16} + 428), \\ a = A^2, \qquad b = A^2 + 1, \qquad c = 2A^2 + 1, \\ D:\ y^2 = x(x-a)(x+b). \end{gathered} \tag{126}$$

The rational change $x = A^2X$, $y = A^3Y$ identifies D with the Legendre curve of parameter $-1 - A^{-2}$. Define

$$\begin{gathered} F = \mathbb{Q}(i, D[30]), \qquad K = F(D[43]) = \mathbb{Q}(i, D[1290]), \\ X_F = D_F \setminus \{O\}, \qquad C_F = [X_F/\{\pm 1\}], \qquad C_K = C_F \times_F K, \\ S = \{r : r \text{ is an odd prime and } r \mid abc\}. \end{gathered} \tag{127}$$

Fix an algebraic closure $\overline{F}$ containing these fields. Quotients by inversion are stack quotients; in particular, their finite etale covering maps are not claims about unramified maps of coarse curves. The set S is finite and contains 1289.

We first make explicit the two conditions that cannot be replaced by genericity assertions: stable reduction above 2 and the required core.

Lemma 15.1 (The place above 2)**.** *Let k be a finite extension of $\mathbb{Q}_2$ and let E/k be an elliptic curve with rational full 3-torsion and integral j-invariant. Then E has good reduction over k. In particular, D has good reduction at every place of F above 2.*

Proof. Use a minimal Weierstrass equation over the complete valuation ring of k to define $E_0(k)$ and $E_1(k)$. Its residue field is finite and perfect, as required in Chapter VII of [71]. Suppose that reduction is additive. The group $E_1(k)$ has no nonzero 3-torsion, by [71, VII.3.1(a)]; the quotient $E_0(k)/E_1(k)$ is the additive group of the residue field, by [71, VII.2.1 and VII.5.1(c)], and also has no 3-torsion. Hence $E_0(k)[3] = 0$. The rational group $E[3] \cong (\mathbb{Z}/3\mathbb{Z})^2$ would inject into $E(k)/E_0(k)$. The latter group has order at most four in the additive case, by the Kodaira–Neron bound [71, VII.6.1]. This is a contradiction. The cited bound applies in residue characteristic two and over each finite extension k; no unramifiedness assumption is being made. Finally, multiplicative reduction would force $v_k(j(E)) < 0$, whereas the j-invariant was assumed integral. Only good reduction remains.

For (126), A is odd, so $A^2 \equiv 1 \pmod 8$. Writing $S_D = 3A^4 + 3A^2 + 1$, one has

$$j(D) = \frac{256 S_D^3}{(abc)^2}, \qquad v_2(b) = 1, \qquad v_2(S_D) = 0.$$

Thus $v_2(j(D)) = 6$. Every completion of F contains $D[3]$, and the first assertion applies. Notice that the minimal equation over the completion need not be the rational equation in (126). $\square$

The references in this argument have a useful precise location in the archived second edition of [71]: the Chapter VII local-field hypotheses are on printed page 185, and VII.2.1, VII.3.1(a), VII.5.1(c), and VII.6.1 are on printed pages 188, 192, 196, and 200, respectively. In the archived PDF these are pages 201, 204, 208, 212, and 216, respectively.

Proposition 15.2 (The required core)**.** *The hyperbolic orbicurve C_K in (127) is a K-core in the sense of [78, Remark 2.1.1]. The same assertion holds after replacing K by any of its completions.*

Proof. Takeuchi's classification gives exactly four geometric arithmetic once-punctured elliptic curves [80, Theorem 4.1(i)]. Their j-invariants, as computed in [82, Section 3.1, Table 4], are

$$\frac{2^{14}31^3}{5^3}, \qquad \frac{2^2 73^3}{3^4}, \qquad 1728, \qquad 0. \tag{128}$$

All four are integral at 1289. The curve D satisfies $v_{1289}(j(D)) = -4$, by the split multiplicative calculation in Section 13. Thus its once-punctured curve is not among the four arithmetic curves. Arithmeticity is unchanged under finite etale coverings, so the hemi-elliptic quotient is also nonarithmetic over an algebraic closure.

By [78, Proposition 2.7], a nonarithmetic punctured hemi-elliptic orbicurve over an algebraically closed field of characteristic zero is its own core. Proposition 2.3(i)–(ii) of the same paper gives invariance under algebraically closed base extension and descent from a separable closure. Apply these results first over K and then over each completion. They give the asserted cores. ∎

The source locations for this step are author-PDF pages 9–10 and 14–15 of [78], published page 392 (PDF12) of [80], and PDF11 of the pinned arXiv version of [82]. In particular, the argument does not replace the four exclusions by a non-CM hypothesis.

Lemma 15.3 (Decorated quotient transitivity)**.** *Let V be a two-dimensional vector space over a field, equipped with a nondegenerate alternating form. The group $\mathrm{SL}(V)$ acts transitively on pairs $(H, \bar{v})$ consisting of a line $H \subset V$ and a nonzero vector $\bar{v} \in V/H$.*

Proof. Choose a lift v of $\bar{v}$. There is a unique $h \in H$ with $\omega(h, v) = 1$. The pair (h, v) is a symplectic basis. Given a second decorated pair, construct (h', v') in the same way. The linear map $h \mapsto h'$, $v \mapsto v'$ preserves the alternating form, has determinant one, and carries the first decorated quotient to the second. ∎

Theorem 15.4 (Explicit initial theta data)**.** *For every nonempty subset $S_0 \subseteq S$, there exist a hyperbolic orbicurve $\underline{C}_K$, a cusp ϵ of it, and a subset $\mathbb{V} \subseteq V(K)$ mapping bijectively to $V(\mathbb{Q})$, such that*

$$(\overline{F}/F, X_F, 43, \underline{C}_K, \mathbb{V}, S_0, \epsilon) \tag{129}$$

satisfies all of [63, Definition 3.1(a)–(f)]. More precisely, one may fix any line $H \subset D[43](K)$ and any nonzero $\bar{v} \in D[43](K)/H$ in advance, construct a single global cover and cusp from them, and then choose $\mathbb{V}$ with the required local properties. The fields F and K remain exactly those in (127).

Proof. We follow the letters of the source definition, whose full statement is on author-PDF pages 61–63 of [63].

The arithmetic requirements (a)–(c). The field F contains i and all 6-torsion, and is Galois over $\mathbb{Q}$. Section 13 gives

$$[F : \mathbb{Q}] \mid 2|\mathrm{GL}_2(\mathbb{Z}/30\mathbb{Z})| = 276480, \qquad 43 \nmid [F : \mathbb{Q}], \tag{130}$$

and proves that the image of G_F on $D[43]$ contains $\mathrm{SL}_2(\mathbb{F}_{43})$. Its kernel cuts out K. Since the pointed curve has a model over $\mathbb{Q}$, every automorphism of $\overline{\mathbb{Q}}/\mathbb{Q}$ fixes its geometric isomorphism class. Consequently its field of moduli is exactly $F_{\mathrm{mod}} = \mathbb{Q}$.

Every odd bad prime of D belongs to S and is already split multiplicative over $\mathbb{Q}$. All other finite primes except possibly 2 are good. Lemma 15.1 treats 2. These properties persist after the extension to F. The zero section makes the smooth genus-one model stable as a one-pointed curve. At a multiplicative place it gives a stable nodal model, either directly from the minimal cubic or by contracting the unmarked two-valent components of the semistable polygon. Thus X_F has stable reduction at every finite place, in the pointed-curve sense needed in the definition.

For each $r \in S$, Section 13 proves

$$1 \leq v_r(abc) < 43, \qquad \operatorname{ord}_w(q_w) = 2e(w/r)v_r(abc) \quad (w \mid r \text{ in } F).$$

The ramification index divides (130), so 43 is prime to all of these positive Tate orders. It is also different from every $r \in S$, since $(a, b, c) \equiv (1, 2, 3) \pmod{43}$. This verifies all the required order and residue-characteristic conditions for S_0, including the choice $S_0 = S$. There is no need to know the complete factorization of the large endpoints: the finite divisibility certificate and size bound in Section 13 prove the displayed inequalities.

The global covering in (d). Put $V = D[43](K)$ and fix $(H, \bar{v})$ as in the statement. Take the quotient isogeny and its dual

$$\phi : D_K \longrightarrow D_K/H, \qquad \psi : D_K/H \longrightarrow D_K, \qquad \psi\phi = [43].$$

Both are defined over K, since V is rational over K. Their kernels give the exact sequence of constant finite groups

$$0 \longrightarrow H \longrightarrow V \xrightarrow{\phi} Q := \ker\psi \longrightarrow 0. \tag{131}$$

In particular, $Q \cong V/H$ is cyclic of order 43. Set

$$\underline{X}_K = (D_K/H) \setminus Q, \qquad \underline{C}_K = [\underline{X}_K/\{\pm 1\}]. \tag{132}$$

The restriction of ψ is a degree-43 finite etale cyclic cover $\underline{X}_K \to X_K$ with group Q. Its cusps are all K-rational. Since the cover extends etale across the origin of the completed elliptic curve, inertia at that cusp acts trivially on its fiber. The residue-field part of the cusp decomposition group also acts trivially, because every point of Q is rational. The quotient of the fundamental group is therefore trivial on the full cusp decomposition group and surjective on geometric monodromy. As an isogeny cover, it factors through the elliptic abelian quotient. This is precisely the cyclic quotient used in [79, Definition 2.1 and its preceding construction].

Inversion yields the cartesian diagram

$$\begin{array}{ccc} \underline{X}_K & \longrightarrow & X_K \\ \downarrow & & \downarrow \\ \underline{C}_K & \longrightarrow & C_K. \end{array} \tag{133}$$

The lower arrow is finite etale by descent along $X_K \to C_K$. Thus $\underline{C}_K$ has type $(1, 43\text{-tors})^{\pm}$ in the source terminology. It is not necessary, and generally not true, that this lower cover is Galois. By Proposition 15.2, C_K is a K-core. Finite etale commensurable orbicurves have the same category of finite etale correspondences, so this is also the K-core of $\underline{C}_K$.

The nonzero element $\bar{v} \in V/H$ gives a nonzero cusp in Q through (131); its inversion orbit gives a cusp ϵ of $\underline{C}_K$. Both the cover and ϵ are now fixed globally. They will not be changed in the construction at the different primes.

The section and the local cusp conditions in (e)–(f). For each $r \in S_0$, initially choose any place w of K above r. Split Tate uniformization over $F_{w|F}$ and then K_w gives

$$0 \longrightarrow \mu_{43} \longrightarrow D[43](K_w) \longrightarrow \mathbb{Z}/43\mathbb{Z} \longrightarrow 0. \tag{134}$$

All these torsion points come from V. The kernel defines a line $H_w \subset V$, and the class of $q_w^{1/43}$ determines a nonzero vector $\bar{v}_w \in V/H_w$. Changing the root only multiplies by μ_{43}. Reversing the Tate orientation changes the vector's sign, which is permitted for the cusp of the inversion quotient. For the following argument choose one of the two orientations.

The decorated pairs are equivariant under $\operatorname{Gal}(K/F)$. With the convention $(gw)(x) = w(g^{-1}x)$, the map of completions $g : K_w \to K_{gw}$ transports Tate uniformization and its multiplicative subgroup. It also transports the graph generator. Thus

$$(H_{gw}, \bar{v}_{gw}) = g(H_w, \bar{v}_w) \tag{135}$$

for compatible orientation choices; the equality modulo sign does not require those choices. Lemma 15.3, together with the actual $\mathrm{SL}_2(\mathbb{F}_{43})$ in the Galois image, therefore gives $g_r \in \mathrm{Gal}(K/F)$ such that $g_r(H_w, \bar{v}_w) = (H, \bar{v})$. Choose $v_r = g_r w$ as the selected place above r.

The isogeny direction is significant. At this place, write the curve as $E(q)$ and the matched line as $H = \mu_{43}$. Then

$$\begin{aligned} \phi : E(q) &\longrightarrow E(q^{43}), \quad [z] \longmapsto [z^{43}], \\ \psi : E(q^{43}) &\longrightarrow E(q), \qquad [z] \longmapsto [z]. \end{aligned} \tag{136}$$

The kernel of ψ is generated by $[q]$, and $\phi([q^{1/43}]) = [q]$. Accordingly ψ, which defines our cover, is the graph cover obtained by reducing the natural $\Pi_X^{\mathrm{tp}} \twoheadrightarrow \mathbb{Z}$ quotient modulo 43. Its distinguished cusp is the image of the generator, up to sign. This is the graph quotient constructed at the beginning of [79, Section 1]; its profinite completion is the $\widehat{\mathbb{Z}}$ quotient in Definition 2.5(i) of that paper. It follows that $\underline{C}_{v_r}$ has type $(1, \mathbb{Z}/43\mathbb{Z})^{\pm}$, and that ϵ_{v_r} is its specified ± 1 cusp.

At all remaining rational places, including the archimedean place, choose any one place of K above it. Combining these choices with the v_r gives $\mathbb{V}$. The source requires a section of $V(K) \to V(F_{\mathrm{mod}})$, with no equivariance, continuity, or previously fixed section condition. Thus the g_r can be chosen independently at the finitely many bad primes. There is still one global H, one global cover, and one global ϵ; the argument has not substituted different global covers at different places.

The auxiliary theta covers. We finish by checking the auxiliary construction recalled in parts (d)–(f), without extending F. For odd ℓ, inversion determines the canonical geometric subgroups of the theta cover, by [79, Proposition 2.2]. An arithmetic splitting of the indicated cusp extension gives a model over the base field; the choices form a torsor under $H^1(G_k, \mu_{43}) = k^\times/(k^\times)^{43}$. Part (d) of the initial-data definition only invokes the geometric subgroups. Part (e) then invokes their natural local models at each selected bad place; it does not prescribe a single arithmetic splitting over K compatible with all local splittings.

Let $k = K_v$ at a selected bad place. Its residue characteristic is odd and prime to 43, the curve is split multiplicative, and $i \in k$. The nonarithmeticity needed at the beginning of [79, Section 2] follows from Proposition 15.2 and its proof. Moreover rational full 2-torsion implies $q^{1/2} \in k$. Indeed, a square root b_0 represents a rational Tate 2-torsion class. For every $\sigma \in G_k$, the ratio $\sigma(b_0)/b_0$ belongs to $q^{\mathbb{Z}}$ and has valuation zero, so it equals one. Thus $k = k(\mu_2, q^{1/2}) = \ddot{k}$, the field condition of [79, Definition 2.5]. The root-of-unity condition recalled in Remark 1.10.1(ii) of that paper also holds: rational $D[3]$ and the Weil pairing give $\mu_3 \subset F$, while $i \in F$ gives $\mu_4 \subset F$. Hence $\mu_{12} \subset F \subset k$, without any further field extension.

The classical theta series in [79, Proposition 1.4] has value $2i$ plus terms of positive valuation at the point represented by i. This value is a unit, so a k-unit multiple normalizes it to one and gives a class of standard type in Definition 1.9 of that paper. Theorem 1.10(iii) there supplies the compatible $\{\pm 1\}$-structure on the cusp torsor. Passing to the order-43 quotient gives the required splitting class: the two representatives differing by -1 agree in $k^\times/(k^\times)^{43}$ because $(-1)^{43} = -1$. Proposition 2.2 with that splitting gives exactly the local theta cover in Definition 2.5(i). All cusps used here are k-rational; also $\mu_{43} \subset k$ by the Weil pairing on the rational 43-torsion.

These are the local models invoked in [63, Definition 3.1(e)]. They use no global extension obtained by adjoining arbitrary 43rd roots of constants. The oriented covers at the end of part (f) are then determined by the fixed triple $(X_K, \underline{C}_K, \epsilon)$, as in Definition 1.1 and Remark 1.1.2 of [63]. This verifies the final auxiliary construction and completes all parts of the initial-data definition. □

For checking the last part directly, the relevant author-PDF pages of [79] are: the graph quotient on page 11; $\ddot{k} = k(\mu_2, q^{1/2})$ on page 16; Proposition 1.4 on pages 20–21; Definition 1.9 and Theorem 1.10 on pages 27–28; and Definitions 2.1, 2.5 with Proposition 2.2 on pages 32–36. The original PDFs and their hashes are recorded separately, so these locations refer to fixed source objects.

15.1. **A uniform construction along the power-free family.** The covering construction uses no property specific to the number 43 beyond the arithmetic hypotheses already verified. The following criterion states precisely what can vary.

Proposition 15.5 (A rational initial-data criterion)**.** *Let $D/\mathbb{Q}$ be an elliptic curve, let $\ell \geq 7$ be prime, and set*

$$F_D = \mathbb{Q}(i, D[30]), \qquad K_D = F_D(D[\ell]) = \mathbb{Q}(i, D[30\ell]).$$

Suppose that D_{F_D} has good or multiplicative reduction at every finite place, that the image of G_{F_D} on $D[\ell]$ contains $\mathrm{SL}_2(\mathbb{F}_\ell)$, and that $v_{p_0}(j(D)) < 0$ for some rational prime $p_0 > 5$. Let S_0 be a nonempty finite set of odd rational primes, different from ℓ, where $D/\mathbb{Q}$ is split multiplicative. Suppose that every integral Tate order at a place of F_D above S_0 is prime to ℓ.

For any line $H \subset D[\ell](K_D)$ and any nonzero $\bar{v} \in D[\ell](K_D)/H$, the quotient isogeny and its dual construct one global cover $\underline{C}_{K_D}$ and cusp ϵ. A section $\mathbb{V} \subset V(K_D)$ of the rational place map can be chosen such that

$$(\overline{F}_D/F_D, D_{F_D} \setminus \{O\}, \ell, \underline{C}_{K_D}, \mathbb{V}, S_0, \epsilon)$$

satisfies [63, Definition 3.1(a)–(f)]. *No extension of F_D or K_D is required.*

Proof. The universal degree bound $[F_D : \mathbb{Q}] \mid 276480 = 2^{11}3^35$ is prime to ℓ. The field is Galois, contains i and rational full 6-torsion, and the original pointed curve has field of moduli $\mathbb{Q}$. The mod-ℓ kernel cuts out exactly K_D. The reduction hypothesis gives a stable one-pointed model at every finite place, by the same smooth or nodal construction used above. The image and Tate-order hypotheses then give all the other conditions in parts (a)–(c). The orders are imposed over F_D, as in the source, not over K_D.

All four invariants in (128) are integral at every prime greater than 5. The condition at p_0 therefore proves nonarithmeticity by the same classification and excludes no additional curves. The proof of Proposition 15.2 gives a core over K_D and over each completion.

Take the quotient $\phi : D \to D/H$ and its dual $\psi : D/H \to D$ over K_D. Their constant kernel sequence is $0 \to H \to D[\ell] \to \ker \psi \to 0$. Removing $\ker \psi$ and quotienting by inversion constructs the cover exactly as in (132). Its cusps are rational, its geometric monodromy is cyclic of order ℓ, and its full cusp decomposition group acts trivially on the fiber at the origin. It therefore has the type required in part (d), with the same core. The class $\bar{v}$ fixes one nonzero global cusp.

Lemma 15.3 applies over $\mathbb{F}_\ell$. For each $r \in S_0$, choose any place w above it and then an actual $g_r \in \mathrm{Gal}(K_D/F_D)$ carrying its Tate decorated quotient to the fixed $(H, \bar{v})$. Formula (135) is independent of 43, so selecting $g_r w$ gives the required local interpretation of that one global cover and cusp. The dual isogeny is $E(q^\ell) \to E(q)$, $[z] \mapsto [z]$; its graph generator is $[q]$, the image of $[q^{1/\ell}]$. All remaining places may be selected arbitrarily. This verifies the section and generator requirements in parts (e)–(f).

Finally, the auxiliary-cover argument above only uses oddness of ℓ, the original full mod-ℓ torsion, and the explicit local conditions. At a selected place k, rational 2-torsion gives $k = \ddot{k}$; rational 3-torsion and i give $\mu_{12} \subset k$; and the theta value at i is a unit in odd residue characteristic. EtTh's canonical geometric subgroups and natural local theta splittings therefore apply. The sign ambiguity disappears modulo $(k^\times)^\ell$ because $(-1)^\ell = -1$. For the oriented covers, $\ell \geq 7$ is prime to 6, and the original curve's mod-ℓ abelian action is trivial over K_D. There is no extra global arithmetic splitting condition, and no need to adjoin the full ℓ-torsion of D/H. These are exactly the remaining assumptions checked in the preceding proof. ☐

Corollary 15.6 (Initial data throughout the unbounded family)**.** *For every member (ℓ, p, A) of Theorem 16.2, put*

$$F_A = \mathbb{Q}(i, D_A[30]), \qquad K_A = \mathbb{Q}(i, D_A[30\ell]),$$
$$S_A = \{r : r \textit{ is an odd prime}, \ r \mid A^2(A^2+1)(2A^2+1)\}.$$

For every nonempty $S_0 \subseteq S_A$, every line $H \subset D_A[\ell](K_A)$, and every nonzero quotient vector $\bar{v}$, Proposition 15.5 supplies initial theta data with precisely these fields and the chosen prime

ℓ. In particular $S_0 = S_A$ is allowed. The construction introduces no further largeness threshold on ℓ beyond that in the family existence theorem.

Proof. Every odd bad prime is split multiplicative by the family calculation, and all other odd primes are good. Odd A gives $v_2(j(D_A)) = 6$, so Lemma 15.1 gives good reduction over each completion of F_A at 2. Thus the first hypothesis of Proposition 15.5 holds. The family theorem gives the required SL_2 image over F_A. At its prescribed prime $p \geq 30\ell - 1 > 5$, $v_p(A) = 1$ and $v_p(j(D_A)) = -4$, proving the core criterion uniformly. Also $p \in S_A$, so this set is nonempty, while $(a, b, c) \equiv (1, 2, 3) \pmod{\ell}$ gives $\ell \notin S_A$.

For the order condition one must use the precise power-free statement; it is not asserted that A^2 is ℓ-power-free. For $w \mid r$ in F_A, the three mutually exclusive cases give

$$\mathrm{ord}_w(q_w) = \begin{cases} 4e(w/r)v_r(A), & r \mid A, \\ 2e(w/r)v_r(A^2+1), & r \mid A^2+1, \\ 2e(w/r)v_r(2A^2+1), & r \mid 2A^2+1. \end{cases} \tag{137}$$

Each valuation on the right lies between 1 and $\ell - 1$, and $e(w/r)$ divides the prime-to-ℓ degree $[F_A : \mathbb{Q}]$. Since ℓ is odd, all three orders are prime to ℓ. This verifies the last hypothesis of the proposition without the unnecessary stronger inequality $v_r(abc) < \ell$. The proposition now applies to each chosen S_0 and fixed decorated quotient. □

When $p \in S_0$, changing the selected place by g_p leaves the exact local field in the family theorem unchanged up to isomorphism. It still has ramification index 15ℓ, residue degree two, and total degree 30ℓ. In the notation of (149), β remains a uniformizer and

$$v_p(q^{1/(2\ell)}) = 2/\ell.$$

Thus the compatible section does not remove the actual Tate unit or change the valuation to $1/\ell$. The cover and cusp are single fixed global objects for each family member; there is no assertion that distinct curves share one covering or one field.

Remark 15.7 (Scope of the construction). Theorem 15.4 and Corollary 15.6 verify the original global initial data, including one covering and cusp with simultaneous local interpretations for each curve. Neither identifies our prescribed ℓ with an existentially selected prime in [57, Theorem 5.7.1]. For the isolated level-43 example, no exceptional-set avoidance is claimed. For the unbounded family, avoidance of a finite set fixed before the family varies is the separate statement of Corollary 16.4; it uses the original compactly bounded domain specified there.

At the selected place above 1289, the completion is still, up to isomorphism, the exact field of Section 13, with ramification index 645, residue degree two, and total degree 1290. Selecting a Galois-conjugate place changes none of these quantities. The native root $q^{1/86}$ retains valuation 2/43.

Initial data alone do not identify the published pilot family with the explicit native point family, its saturated unit family, or a principal maximal-order ideal formed before transport. The exact local pre-ideal/point-hull equality proved separately applies to its specified native source and integral arrows; this initial-data construction does not identify them with the whole published pilot. The standard Bloch–Kato versus p^{-1} log normalization, the j^2-dependent absolute-value rescaling, the specified order of hull operations, and the remaining Ind3 and global Frobenius comparisons still need their own compatible proofs. Neither the initial-data theorems in this section nor the external core and theta-cover theorems invoked here have been formalized in Lean in this work.

16. An unbounded family with prescribed local level data

The preceding local constructions can be supplied by rational elliptic curves of unbounded height while the relevant local bounding domain remains fixed. We prove this using a least-prime theorem and a finite sieve in one arithmetic progression. The result is an existence family with specified reduction, torsion-image, and level-window properties. No lower bound for its abc quality with an exponent greater than one is asserted.

16.1. **A uniform finite sieve.** An integer is k-power-free if it is not divisible by q^k for any prime q. We shall use the following estimate with k and the progression modulus both growing.

Lemma 16.1 (Power-free values in a progression)**.** *Let $k \geq 5$, let M be a positive even integer, and let $X \geq 1$. Fix a residue class $r \bmod M$, and suppose that, for every integer $A \in [X, 2X]$ in this class, none of*

$$A, \qquad A^2+1, \qquad 2A^2+1$$

is divisible by q^k at a prime $q \mid M$. The number N_{good} of such A for which all three values are k-power-free satisfies

$$N_{\mathrm{good}} \geq \frac{X}{M}\left(1 - 5\sum_{q \text{ prime}} q^{-k}\right) - 1 - 5(9X^2)^{1/k} \geq \frac{3X}{4M} - 1 - 5(9X^2)^{1/k}. \tag{138}$$

Proof. The interval contains at least $X/M - 1$ integers of the prescribed class. If $q \nmid M$, then q is odd. The polynomial T has one root modulo q^k, and each of T^2+1 and $2T^2+1$ has at most two. For the latter assertion, there are at most two roots modulo q, and the derivative is nonzero at every root. A root $x \bmod q^h$ has lifts $x + tq^h$; reduction of

$$f(x+tq^h) \equiv f(x) + tq^h f'(x) \pmod{q^{h+1}}$$

gives exactly one admissible $t \bmod q$. This proves the root bound at every exponent without a uniformity assumption on the prime.

Each of these at most five roots, combined with $r \bmod M$, specifies one class modulo Mq^k. It contributes at most $X/(Mq^k) + 1$ integers. All three polynomial values are positive and at most $9X^2$. Thus only primes

$$q \leq Y := (9X^2)^{1/k}$$

can exclude an integer. Summing these upper bounds over $q \leq Y$ excludes at most

$$\frac{5X}{M}\sum_{q \text{ prime}} q^{-k} + 5Y$$

integers. The sum of the error terms is finite; no error of size one is summed over infinitely many primes. Subtracting gives the first inequality. Finally,

$$\sum_{q \text{ prime}} q^{-k} \leq \sum_{n=2}^{\infty} n^{-k} \leq 2^{-k} + \frac{2^{1-k}}{k-1} \leq \frac{3}{2}2^{-k}.$$

For $k \geq 5$, five times this bound is at most $15/64 < 1/4$. ◻

16.2. **The existence theorem.** The only prime-distribution input needed below is the effective estimate of Xylouris [83, Theorem 1.1, author manuscript p. 9]. With $P(m)$ the maximum, over reduced classes modulo m, of the least prime in the class, it gives an effective absolute constant C such that

$$P(m) \leq Cm^{26/5}. \tag{139}$$

We use the exponent 5.2 of that archived author version throughout; the later improvement is unnecessary. In particular, C is independent of both the modulus and the reduced class.

Theorem 16.2 (An unbounded balanced Frey–Legendre family)**.** *For every sufficiently large prime $\ell \equiv 43 \pmod{60}$, let p be the least prime congruent to $-1 \pmod{30\ell}$, and put*

$$M = 10\ell p^2, \qquad X = \exp(\ell^2/16). \tag{140}$$

There are at least $X/(2M)$ integers $A \in [X, 2X]$ satisfying

$$A \equiv 1 \pmod 2, \qquad A \equiv 1 \pmod{5\ell}, \qquad A \equiv p \pmod{p^2}, \tag{141}$$

for which $A, A^2+1, 2A^2+1$ are all ℓ-power-free. Every such A gives the positive primitive triple and curve

$$(a,b,c) = (A^2, A^2+1, 2A^2+1), \qquad D_A:\ y^2 = x(x-a)(x+b). \tag{142}$$

The curve is rationally isomorphic to the Legendre curve with

$$\lambda_A = -1 - A^{-2}.$$

For $L_A = \mathbb{Q}(i, D_A[30])$, the following assertions hold:

(1) *At p, the curve is split multiplicative with native Tate valuation $v_p(q) = 4$.*
(2) *The mod-ℓ image over $\mathbb{Q}$ and over L_A contains $\mathrm{SL}_2(\mathbf{F}_\ell)$. Reduction at ℓ is good.*
(3) *Every nonzero integral Tate order over L_A is prime to ℓ.*
(4) *The Tate degree normalized by $[L_A : \mathbb{Q}]^{-1}$ is*

$$Q_A = 2\log\frac{A^2(A^2+1)(2A^2+1)}{2}, \qquad \frac{3}{4}\ell^2 < Q_A < \ell^2. \tag{143}$$

The parameters lie in one fixed compactly bounded domain in the original sense specified below, and both $\mathrm{h}(\lambda_A)$ and $\mathrm{h}(j(D_A))$ tend to infinity with ℓ. The threshold is effective in terms of the effective constant in (139); no numerical value of that constant is extracted here.

Existence of the integers. The three moduli in (141) are pairwise coprime. By (139),

$$30\ell - 1 \le p \le C(30\ell)^{26/5}, \qquad M \le C_0\ell^{57/5}, \qquad C_0 = 10C^2 30^{52/5}.$$

The primes dividing M cause no exclusion in Lemma 16.1. At 2, A is odd, so $v_2(A^2+1) = 1$ and $2A^2+1$ is odd. At 5 and ℓ, the three values are $1, 2, 3$. At p, their valuations are $1, 0, 0$. All four primes are distinct.

For $k = \ell$, the error parameter is

$$Y = 9^{1/\ell}\exp(\ell/8).$$

It suffices that $1 + 5Y \le X/(4M)$. Since $1 + 5Y \le 46\exp(\ell/8)$, an explicit sufficient condition is

$$\frac{\ell^2}{16} - \frac{\ell}{8} - \frac{57}{5}\log\ell \ge \log(184C_0). \tag{144}$$

The left side tends to infinity and is increasing for $\ell \ge 43$: its derivative is $\ell/8 - 1/8 - 57/(5\ell) > 0$. Lemma 16.1 now gives the lower bound $X/(2M)$. All constants are independent of the chosen ℓ, p, A.

There are infinitely many eligible primes ℓ. Indeed, if the primes congruent to 43 modulo 60 were a finite list $\ell_1, \dots, \ell_s$, the reduced CRT class congruent to 43 modulo 60 and to 1 modulo each ℓ_i would contain a prime by (139). It would be a new prime in the original progression, a contradiction. □

Arithmetic and height assertions. Pairwise coprimality in (142) follows from $\gcd(A^2, A^2+1) = 1$ and $2(A^2+1) - (2A^2+1) = 1$. The substitution $x = A^2x_1,\ y = A^3y_1$ gives

$$y_1^2 = x_1(x_1-1)(x_1+1+A^{-2}).$$

It is an isomorphism over $\mathbb{Q}$, not a quadratic twist. The general invariants (108) give, with $S_A = 3A^4 + 3A^2 + 1$,

$$\Delta = 16(abc)^2, \qquad c_4 = 16S_A, \qquad j(D_A) = \frac{256S_A^3}{(abc)^2}, \qquad \gcd(S_A, abc) = 1. \tag{145}$$

At an odd prime $r \mid abc$, unit c_4 makes the equation minimal and multiplicative, with Tate order $2v_r(abc)$. Splitness is also uniform in A. If $r \mid a$ or $r \mid b$, the reduced node has tangent slopes ± 1; if $r \mid c$, translation of the node at $x = a$ gives $y^2 = x_1^2(x_1 + a)$, with slopes $\pm A$. In each case they are distinct units. These are the reduction criteria of [71, Chapter VII, Proposition 5.1]; the split Tate description is [77, Theorems 1–2]. At the prescribed p, $v_p(A) = 1$, and hence $v_p(q) = 4$. At ℓ, the endpoints are $1, 2, 3$, so reduction is good.

The good reduction at 5 is the same fixed curve $y^2 = x(x-1)(x+2)$ used in Section 13. It has four $\mathbf{F}_5$-points, giving Frobenius polynomial $T^2 - 2T + 5$. Its discriminant -16 is a nonsquare modulo ℓ, as $\ell \equiv 3 \pmod 4$; thus this image element fixes no line. At p, Tate uniformization supplies a nonidentity inertia transvection of order ℓ, because $p \ne \ell$ and $\ell \nmid v_p(q) = 4$ [77, Proposition 3]. Conjugating its fixed line by the Frobenius element gives a distinct fixed line.

Lemma 13.1 therefore gives the full $\mathrm{SL}_2(\mathbf{F}_\ell)$ subgroup. The same lemma applies over L_A: the latter is Galois, and (110) gives

$$[L_A : \mathbb{Q}] \mid 2|\mathrm{GL}_2(\mathbb{Z}/30\mathbb{Z})| = 276480 = 2^{11}3^35, \tag{146}$$

which is prime to ℓ. The normal image over L_A retains the order-ℓ transvection and its conjugate.

At 2, (145) gives $v_2(j) = 6$, hence potential good reduction [71, Chapter VII, Proposition 5.5]. We check why the specific field L_A already gives good reduction. Potential good reduction makes the inertia image on $T_3(D_A)$ finite. Since all 3-torsion is rational over L_A, that image lies in $1 + 3\,\mathrm{Mat}_2(\mathbb{Z}_3)$, a torsion-free group. To see the last assertion, write a nonidentity element as $1 + 3^hB$, where $h \geq 1$ and $B \not\equiv 0 \pmod 3$. Powers prime to 3 preserve h, and

$$(1 + 3^hB)^3 = 1 + 3^{h+1}\bigl(B + 3^hB^2 + 3^{2h-1}B^3\bigr)$$

increases it by exactly one. Thus no positive finite power is the identity. Inertia is trivial, and the Neron–Ogg–Shafarevich criterion [71, Chapter VII, Theorem 7.1] gives good reduction over every completion of L_A at 2.

For $w \mid r$, where r is odd and divides abc, the integral Tate order over L_A is

$$\mathrm{ord}_w(q_w) = 2e(w/r)v_r(abc).$$

If $r \mid A$, this equals $4e(w/r)v_r(A)$; otherwise it is twice the ramification index times the valuation of one of the two quadratic factors. Each latter valuation, and $v_r(A)$, lies strictly between zero and ℓ. Also $\ell \nmid e(w/r)$, by (146). Since ℓ is odd, every nonzero order is prime to ℓ. We do not need or assert that A^2 itself is ℓ-power-free.

Sum these orders with residue-degree weights and divide by $[L_A : \mathbb{Q}]$. For each r, the identity $\sum_{w|r} e(w/r)f(w/r) = [L_A : \mathbb{Q}]$, good reduction at 2, and $v_2(abc) = 1$ give the exact formula for Q_A in (143). Finally, for $A > 1$,

$$A^6 < \frac{A^2(A^2+1)(2A^2+1)}{2} < 3A^6.$$

Together with $X \leq A \leq 2X$, this yields

$$\frac{3}{4}\ell^2 < Q_A < \frac{3}{4}\ell^2 + 12\log 2 + 2\log 3 < \frac{3}{4}\ell^2 + 16 < \ell^2. \tag{147}$$

The remaining domain and unbounded-height assertions follow from Proposition 16.3 below. □

The prime-to-ℓ order assertion is over L_A, as stated. It is not an assertion over $L_A(D_A[\ell])$, whose ramification at p has an additional factor ℓ. Set $\delta = 2^{12}3^35 = 552960$, the constant for $d_{\mathrm{mod}} = 1$ in [57, Theorem 5.7.1]. Equation (147) also proves both numerical endpoints

$$\sqrt{Q_A} < \ell < 10\delta\sqrt{Q_A}\log(2\delta\log Q_A). \tag{148}$$

For the second inequality, use $\sqrt{Q_A} > \ell/2$ and $\log(2\delta\log Q_A) > 1$. This directly verifies a window for our chosen ℓ; it does not identify it with an existential choice in that source.

16.3. **The full local torsion field.** The local field in this family is determined with its actual Tate unit retained. Put $N = 30\ell$ and $e = 15\ell$. Full rational 2-torsion implies that a square root b_0 of q belongs to $\mathbb{Q}_p$. Indeed, its Tate class is rational, so for every Galois element the ratio of a conjugate to b_0 is a power of q. The ratio has valuation zero, hence is one. Thus $v_p(b_0) = 2$.

Let $K_0 = \mathbb{Q}_p(\mu_N)$, choose $\varpi^e = b_0$, and write $u_0 = b_0/p^2$. Since $p \equiv -1 \pmod N$, $K_0/\mathbb{Q}_p$ is unramified quadratic. The element ϖ has valuation $2/e$; the uniformizer is instead

$$\beta = \frac{\varpi^{(e+1)/2}}{p}, \qquad \beta^e = p\,u_0^{(e+1)/2}, \qquad \varpi = u_0^{-1}\beta^2. \tag{149}$$

The equation for β is Eisenstein over K_0. Consequently

$$E = K_0(\varpi) = K_0(\beta), \qquad [E : \mathbb{Q}_p] = 30\ell, \quad e(E/\mathbb{Q}_p) = 15\ell, \quad f(E/\mathbb{Q}_p) = 2.$$

Tate uniformization identifies

$$\mathbb{Q}_p(D_A[N]) = \mathbb{Q}_p(\mu_N, q^{1/N}) = E :$$

fixing each torsion class forces its root-of-unity multiplier to be one, because a nontrivial power of q has nonzero valuation. The unramified quadratic field K_0 contains i, so E is also the completion at p of $\mathbb{Q}(i, D_A[N])$. It is Galois, being the splitting field obtained from the cyclotomic field by adjoining all roots of $T^e - b_0$. Moreover,

$$e = 15\ell \le p-2, \qquad \gcd(e, p-1) = 1, \tag{150}$$

since $p \ge 30\ell - 1$ and $p-1 \equiv -2 \pmod{15\ell}$. Thus the small tame range of the local arguments occurs at unbounded global height. No substitution $q = p^4$ or $b_0 = p^2$ has been made.

16.4. **A fixed domain and the finite-set quantifier.** Let $U = \mathbf{P}^1 \setminus \{0, 1, \infty\}$. Consider

$$\begin{aligned} \mathcal{K}_\infty &= \{z \in \mathbf{C} : |z+1| \le 1/2\}, \\ \mathcal{K}_2 &= \{z \in \overline{\mathbb{Q}}_2 : |z+2|_2 \le 1/8\}. \end{aligned} \tag{151}$$

For nonarchimedean places, the original compactly bounded condition of [89, Example 1.3(ii), author manuscript pp. 5–6] is formulated on intersection with each finite local extension. It does not require the entire set $\mathcal{K}_2$ to be compact in $\overline{\mathbb{Q}}_2$.

Proposition 16.3 (Fixed local bounds and unbounded rational height)**.** *The sets* (151) *give one compactly bounded domain of support* $\{\infty, 2\}$ *in that original sense. Every member of Theorem* 16.2 *belongs to this domain, and*

$$\mathrm{h}(\lambda_A) = \log(A^2+1) \longrightarrow \infty, \qquad \mathrm{h}(j(D_A)) \ge Q_A \longrightarrow \infty. \tag{152}$$

On all of $\mathcal{K}_2$*, the Legendre invariant has valuation 6.*

Proof. Both sets avoid the three removed points and are stable under the respective Galois actions. The complex set is a compact regular closed disc. For every finite extension $k/\mathbb{Q}_2$,

$$\mathcal{K}_2 \cap k = -2 + 8\mathcal{O}_k$$

is compact and open, hence is the closure of its interior in k. The set $\mathcal{K}_2$ is itself open and closed in $\overline{\mathbb{Q}}_2$, so any separate regular-closed requirement also holds. These are the stated original conditions.

For $A \ge 2$, $|\lambda_A + 1| = A^{-2} \le 1/4$. Since A is odd, A^2 and its inverse are 1 modulo 8, giving $\lambda_A \equiv -2 \pmod 8$. Thus the domain is fixed independently of ℓ, p, A, and the degree bound is one. If $z \in \mathcal{K}_2$, then $v_2(z) = 1$, $v_2(1-z) = 0$, and $v_2(1 - z + z^2) = 0$. The formula

$$j(z) = \frac{256(1-z+z^2)^3}{z^2(1-z)^2}$$

therefore gives $v_2(j(z)) = 6$. On $\mathcal{K}_\infty$, the inequalities $1/2 \le |z| \le 3/2$ and $|1-z| \ge 3/2$ also give the fixed coarse bound

$$|j(z)| \le \frac{256(19/4)^3}{(1/2)^2(3/2)^2} = \frac{438976}{9} < 50000.$$

The numerator and denominator of $\lambda_A = -(A^2+1)/A^2$ are coprime. Its rational height is therefore the first expression in (152). Also $abc/2$ is odd, and (145) reduces to

$$j(D_A) = \frac{64 S_A^3}{(abc/2)^2}$$

in lowest terms. Its denominator alone gives $\mathrm{h}(j(D_A)) \ge 2\log(abc/2) = Q_A$. □

Corollary 16.4 (Avoidance of any fixed finite set)**.** *If a finite set of parameters in* $U(\overline{\mathbb{Q}})$*, or a finite set of geometric elliptic moduli points, is fixed before* ℓ *varies, all sufficiently large members of the family avoid it.*

Proof. Take the maximum of the finitely many parameter heights, or of the heights of their j-invariants, and apply (152). In particular, bounded complex absolute values of these invariants do not bound their rational heights. □

16.5. **The source-interface limits.** The finite-set statement has its exact order of quantifiers: the set must be fixed before the family varies. It cannot be used for an exceptional set chosen separately for each member. There is also a topological difference that prevents silently substituting one definition for another.

Proposition 16.5 (The literal ambient compactness condition)**.** *In the usual valuation topology of $U(\overline{\mathbb{Q}}_2)$, every compact subset has empty interior. Consequently no nonempty compact subset is the closure of its ambient interior.*

Proof. Every nonempty open ball contains, after translation and a sufficiently small nonzero scaling, infinitely many algebraic integral units with distinct residues in $\overline{\mathbf{F}}_2$. For example, one may take lifts in finite unramified extensions. The distances between the scaled points are all the same positive constant. Thus the ball is not totally bounded and cannot be contained in a compact set. A compact set with nonempty interior would contain such a ball, a contradiction. □

Section 5.1 of the archived v2 of [57], p. 50, literally requires nonempty compact subsets in the product of the local algebraic closures and equality with the closure of their ambient interiors. Under the usual valuation topology, the proposition applies to the nonarchimedean factor, and the same obstruction holds for a product containing it. The domain (151) satisfies the original finite-extension-section definition; it must not be described as a compact subset of that full algebraic closure. Using finite-extension sections or a different topology would require stating that interpretation explicitly. This observation does not disprove the original IUT statements.

Finally, the separate existence quantifier in [57, Theorem 5.7.1, p. 53] supplies some prime in a window. Our chosen ℓ satisfies (148) and the arithmetic properties proved above directly, but these facts do not identify it with that existential prime. The covering and marking requirements of [63, Definition 3.1] for these members are supplied separately by Corollary 15.6. They do not identify the complete possible-image source including subsequent indeterminacies. The classical least-prime, elliptic-curve, and local-field inputs in this section remain mathematical source dependencies; they are not asserted as new Lean axioms or as a formal proof of the abc conjecture.

17. Two-prime support and rational-return rigidity

We give a complete elementary bound on one moving-support subclass of abc triples, and then examine a proposed passage from local logarithmic arrows to rational S-unit solutions. These are different conclusions: the first is a special-case height estimate, while the second restricts the number of outputs of a specified construction. Throughout this section, $r = \operatorname{rad}(abc)$ and $R = \log r$, as in (1).

17.1. The complete two-prime subclass.

Theorem 17.1 (A sharp bound on two-prime support)**.** *Let a, b, c be positive integers with $a + b = c$ and $\gcd(a, b) = 1$. If $\omega(abc) \le 2$, then*

$$c \le \frac{3}{2} \operatorname{rad}(abc). \tag{153}$$

Equality holds precisely for $(a, b, c) = (1, 8, 9)$ or $(8, 1, 9)$. Every other triple in this class satisfies $c \le \operatorname{rad}(abc)$.

Proof. The entries are pairwise coprime. Exactly one is even: the addends cannot both be even, and their parities determine that of their sum. If all three entries exceeded 1, each would contribute a different prime divisor, contradicting the support bound. An addend is therefore 1. The case $(1, 1, 2)$ has $c = r = 2$. Otherwise, after exchanging the addends if necessary, write the triple as $(1, N, N+1)$, with $N \ge 2$. Both consecutive integers have a prime factor, and their supports are disjoint. Thus, for an odd prime q and positive integers u, v,

$$\{N, N + 1\} = \{2^u, q^v\}. \tag{154}$$

Suppose first that $2^u = q^v + 1$. If v is even, then $q^v \equiv 1 \pmod 4$, so $2^u \equiv 2 \pmod 4$ and $u = 1$, contrary to $q^v + 1 > 2$. If $v \geq 3$ is odd, the factorization

$$q^v + 1 = (q+1)(q^{v-1} - q^{v-2} + \cdots - q + 1)$$

has an odd second factor greater than 1. Indeed, its v terms are odd modulo 2, and its value is at least $q^2 - q + 1 > 1$. Such a factor cannot divide a power of 2. This ordering therefore requires $v = 1$.

Now suppose that $q^v = 2^u + 1$. An odd $v \geq 3$ is excluded by

$$q^v - 1 = (q-1)(1 + q + \cdots + q^{v-1}),$$

whose second factor is again odd and greater than 1. If v is even, set $Q = q^{v/2} \geq 3$. Both factors in

$$(Q-1)(Q+1) = 2^u$$

are positive powers of 2. Write $Q - 1 = 2^s$ and $Q + 1 = 2^t$, with $1 \leq s < t$. Subtraction gives $2^s(2^{t-s} - 1) = 2$, hence $s = 1$, $t = 2$ and $Q = 3$. Since q is prime, this forces $q = 3$, $v = 2$ and $u = 3$. The resulting pair is $\{8, 9\}$.

In every remaining case $v = 1$, the radical is $r = 2q$, and the larger endpoint is q or $q + 1$, both at most $2q$. For $(1, 8, 9)$ and its exchanged triple, $c = 9$ and $r = 6$. These observations prove both the bound and its equality statement. $\square$

The estimate is uniform as the odd prime varies. For every $\varepsilon > 0$, the entire subclass in Theorem 17.1 satisfies

$$\log c \leq R + \log(3/2) \leq (1+\varepsilon)R + \log(3/2),$$

with no constant depending on the prime support. The proof is elementary and uses no classification theorem for general consecutive perfect powers. No priority claim is made for this classical special case, and no estimate for $\omega(abc) \geq 3$ follows from it.

17.2. **Nonzero rational return under the actual trace constraint.** Let E, E' be finite extensions of $\mathbb{Q}_p$. We retain the coefficient-compatible arrow of Proposition 7.5: a continuous isomorphism $\alpha : G_{E'} \simeq G_E$ together with an integral equivariant coefficient isomorphism

$$\beta : \alpha^* \mathbb{Z}_p(1)_E \simeq \mathbb{Z}_p(1)_{E'}.$$

Its induced logarithmic map $F : E \longrightarrow E'$ is $\mathbb{Q}_p$-linear and satisfies, for some $\kappa \in \mathbb{Z}_p^\times$,

$$\mathrm{Tr}_{E'/\mathbb{Q}_p}(Fz) = \kappa \; \mathrm{Tr}_{E/\mathbb{Q}_p}(z) \qquad (z \in E). \tag{155}$$

This is the previously proved identity (29). Its unit multiplier comes from the integral transport of $H^2(G_E, \mathbb{Z}_p(1)) \cong \mathbb{Z}_p$, together with the reciprocity cup-product formula [67, Chapter III, Proposition 3.6 and §4]. Cyclotomic character, inertia and absolute degree are recovered from the local Galois group [68, Propositions 1.1–1.2 and Corollary 1.3]. In particular the degrees of E and E' agree for these arrows. None of these assertions requires the arrow to be induced by a field isomorphism.

Lemma 17.2 (Rational return and the scalar line)**.** *Suppose $[E : \mathbb{Q}_p] = [E' : \mathbb{Q}_p] = d$ and a $\mathbb{Q}_p$-linear map F satisfies (155). If $z \in \mathbb{Q}_p \subset E$ and $F(z) = z' \in \mathbb{Q}_p \subset E'$, then*

$$z' = \kappa z. \tag{156}$$

For nonzero z, the output is nonzero and $v_p(z') = v_p(z)$. Moreover, one such nonzero return implies

$$F(t) = \kappa t \qquad (t \in \mathbb{Q}_p).$$

Proof. The traces of the two rational elements are dz and dz'. Equation (155) gives $dz' = \kappa dz$. The positive integer d is nonzero in $\mathbb{Q}_p$, so cancellation gives (156), even when $p \mid d$. Its valuation assertion follows from $v_p(\kappa) = 0$. If $z \neq 0$, linearity gives $F(1) = z^{-1}F(z) = \kappa$, and then $F(t) = tF(1) = \kappa t$. $\square$

Equal degree is an explicit hypothesis in this linear-algebra statement. With unequal degrees d, d', the same calculation would instead yield $z' = \kappa(d/d')z$. Uniform division of both logarithmic coordinates by p leaves the displayed trace identity unchanged; independent rescalings by different powers of p do not. The lemma does not assert that every local Galois arrow preserves the rational line. It describes what follows if a specified point returns to that line.

Lemma 17.3 (Depth of an odd-prime principal-unit logarithm)**.** *For an odd prime p and $0 \neq \xi \in p\mathbb{Z}_p$,*

$$v_p\big(\log_p(1-\xi)\big) = v_p(\xi). \tag{157}$$

Proof. Put $e = v_p(\xi) \geq 1$. In the convergent series

$$\log_p(1-\xi) = -\sum_{n\geq 1} \frac{\xi^n}{n},$$

the first term has valuation e. For $n \geq 2$ one has $v_p(n) \leq n-2$: this is immediate for $n = 2$, and for $n \geq 3$ it follows from $v_p(n) \leq \log_3 n \leq n-2$. Every subsequent term therefore has valuation $ne - v_p(n) \geq e+1$, and these valuations tend to infinity. The first term is the unique term of least valuation. The logarithm is nonzero and has valuation e. ☐

For an odd prime $p \mid abc$, choose the following vanishing coordinate and its companion rational S-unit, where S is the full prime support of abc:

$$\begin{array}{c|cc} & \xi_p & u_p = 1-\xi_p \\ \hline p \mid a & a/c & b/c \\ p \mid b & b/c & a/c \\ p \mid c & c/a & -b/a. \end{array} \tag{158}$$

Pairwise coprimality makes every denominator a p-adic unit. Thus $\xi_p \in p\mathbb{Z}_p$, $u_p \in 1 + p\mathbb{Z}_p$, and $v_p(\xi_p)$ is exactly the exponent of p in the indicated endpoint. In the last row $1 - c/a = -b/a$; the negative real sign does not prevent p-adic congruence to 1.

Proposition 17.4 (An endpoint exponent cannot change under rational return)**.** *Let (a, b, c) and (A, B, C) be positive primitive triples, and let an odd prime p divide the same labelled endpoint in both triples. Form u_p, u'_p using that row of (158). Suppose an equal-degree coefficient-compatible logarithmic arrow satisfies the pointwise equality*

$$F_p(\log_p u_p) = \log_p u'_p. \tag{159}$$

Then the exponent of p in the indicated endpoint is the same in the two triples.

Proof. Both logarithms are nonzero rational p-adic elements by Lemma 17.3. Lemma 17.2 equates their valuations, and (157) identifies those valuations with the corresponding endpoint exponents. ☐

The proposition imposes no condition at $p = 2$. The depth identity used here is not valid for every $\xi \in 2\mathbb{Z}_2$. Nor can (159) be replaced by membership of the two logarithms in the same ideal or hull.

17.3. **The exact arithmetic fibre bound.** For $n > 0$, write $n_{\text{odd}} = n/2^{v_2(n)}$. Fix a positive primitive triple (a, b, c) and put $(a_0, b_0, c_0) = (a_{\text{odd}}, b_{\text{odd}}, c_{\text{odd}})$. Let $\mathcal{F}(a, b, c)$ be the set of all positive primitive triples (A, B, C) with

$$(A_{\text{odd}}, B_{\text{odd}}, C_{\text{odd}}) = (a_0, b_0, c_0).$$

These conditions fix all odd-prime exponents in their labelled endpoints, while allowing the even endpoint to move.

Theorem 17.5 (Two possible outputs)**.** *For every positive primitive triple,*

$$|\mathcal{F}(a, b, c)| \leq 2. \tag{160}$$

If the position of the even endpoint is fixed to that of the seed, the fibre contains only the seed. The bound two is attained.

Proof. Every target has exactly one even endpoint, with exponent $k \geq 1$. The three possible positions give

$$\begin{array}{c|c|c} \text{even endpoint} & (A,B,C) & \text{required equation} \\ \hline A & (2^k a_0, b_0, c_0) & 2^k a_0 + b_0 = c_0 \\ B & (a_0, 2^k b_0, c_0) & a_0 + 2^k b_0 = c_0 \\ C & (a_0, b_0, 2^k c_0) & a_0 + b_0 = 2^k c_0. \end{array} \tag{161}$$

Each row admits at most one k, since 2^k multiplies a fixed positive integer and is strictly increasing. Either of the first two rows implies $c_0 > a_0+b_0$, whereas the third implies $c_0 < a_0+b_0$. The third row cannot coexist with either of the first two, so at most two triples occur.

When the even position is fixed, only one row remains, and it already contains the seed. No division by a difference of odd parts was used. In particular, if $a_0 = b_0$, coprimality forces $a_0 = b_0 = 1$ and the same argument applies. When all three odd parts are 1, the only possibility is $(1, 1, 2)$.

Finally, the two distinct triples

$$(4, 3, 7), \qquad (1, 6, 7) \tag{162}$$

are primitive and have the same labelled odd parts $(1, 3, 7)$. Thus the bound is sharp. $\square$

Corollary 17.6 (The specified rational-return construction)**.** *Fix a seed (a, b, c) with full prime support S. Consider positive primitive targets (A, B, C) satisfying all of the following:*

1. *Their exact prime support is S.*
2. *Every odd $p \in S$ belongs to the same labelled endpoint as in the seed.*
3. *For each odd $p \in S$, an equal-degree coefficient-compatible local logarithmic arrow gives (159), with the companion units in (158).*

There are at most two targets. If the even endpoint is prescribed as well, the only target is the seed.

Proof. Proposition 17.4 preserves every odd exponent on the specified support. Primitivity makes its exponents on the other two endpoints zero. Exact support equality excludes any new odd primes, so all three odd parts agree. Apply Theorem 17.5. Identity arrows admit the seed itself. $\square$

The local arrows may depend on the prime and the target; the upper bound requires no common global arrow. Exact support and endpoint-label preservation are hypotheses, not consequences of trace transport. Checking the old primes alone would not exclude new prime factors.

Proposition 17.7 (Sharpness with freely chosen coefficient units)**.** *In the class of pairs (α, β) in which an identity-Galois arrow may be accompanied by any integral Tate-module coefficient unit, the upper bound in Corollary 17.6 is attained by the seed and target in (162).*

Proof. Both triples have support $\{2, 3, 7\}$; the odd prime 3 stays on the second endpoint and 7 on the third. Their companion units are

$$(u_3, u_3') = (4/7, 1/7), \qquad (u_7, u_7') = (-3/4, -6).$$

The vanishing coordinates at 3 are $3/7, 6/7$, and those at 7 are $7/4, 7$. All have valuation one at their indicated prime. Lemma 17.3 therefore shows that

$$\kappa_p = \frac{\log_p u_p'}{\log_p u_p} \in \mathbb{Z}_p^\times \qquad (p = 3, 7).$$

Take $E = E' = \mathbb{Q}_p$ and $\alpha = \mathrm{id}_{G_{\mathbb{Q}_p}}$. Multiplication by κ_p on $\mathbb{Z}_p(1)$ is an integral equivariant coefficient isomorphism. It acts as that same scalar on unit Kummer cohomology and its logarithmic module, so it gives exactly (159). These are principal units at odd primes, where logarithm is injective; no torsion ambiguity affects the stated points. Together with the identity arrows for the seed, this supplies two targets in the specified coefficient-pair class. $\square$

The coefficient convention in this proposition is essential. The purely arithmetic sharpness of Theorem 17.5 has no Galois hypothesis. Its additional realization just proved allows β to be multiplied by a freely chosen element of $\mathbb{Z}_p^\times$. It does not show that every resulting κ_p is induced by a strict automorphism of $G_{\mathbb{Q}_p}$ when a Tate coefficient framing has been fixed. For that possibly smaller class, the upper bound remains valid, but this example alone does not establish sharpness. No arrow condition at $p = 2$ was imposed.

17.4. **Formal scope and the unproved uniform step.** Theorem 17.1, including its equality classification, is formalized in the separate module `ABCTwoPrimeSupport20260831`. Its proof starts with the actual prime-factor support and the repository radical; the prime-power classification is proved rather than assumed. All 22 public theorems in that module were compiled and checked for axiom dependencies. The module `TraceCovariantRationalReturn20260831` proves the algebraic cancellation and scalar-line conclusions using the actual algebra trace and actual dimensions. Trace covariance remains an explicit hypothesis there: this does not formalize its local Galois or Kummer derivation, or the principal-unit logarithm argument. The separate module `ABCOddPartFibre20260831` now proves Theorem 17.5 for the actual natural-number odd part and the unchanged primitive-triple type. It constructs an injection of each entire fibre into a two-element type, proves that the fibre is finite, and establishes cardinality exactly two for the displayed example. This is an arithmetic fibre theorem; the logarithmic endpoint-transfer proposition and the coefficient-pair realization remain the mathematical proofs above, rather than formalized Galois-orbit assertions.

The rational-return obstruction is compatible with the minimum-layer constructions of the preceding sections. Those constructions can move a rational input outside the rational line and then form a hull. Returning to an actual rational S-unit solution is an additional condition. Multiplication by tensor-order coefficients, addition and ideal generation do not automatically preserve a point's trace, so the fibre bound cannot be applied after replacing (159) by hull membership.

No general S-unit method, IUT theorem, complete Ind3 operation or abc conjecture is disproved by this obstruction. Mechanisms introducing new prime support, changing prime labels, or using nonrational intermediate points followed by a proved arithmetic descent remain outside its scope. The uniform estimate (1) for $\omega(abc) \geq 3$, even for a fixed number of primes which themselves vary, is not proved here. Neither finite exceptional sets depending on the support nor density-one bounds supply that missing quantifier.

18. An obstruction in the entire rational isogeny class

For one infinite family of primitive abc triples, we determine the entire rational isogeny class of the actual Frey curve and obtain matching bounds for two intrinsic quantities on that class. The largest minimal discriminant has order c^5, while the smallest complex absolute j-invariant has order c. Thus selecting an arbitrary rationally isogenous representative cannot recover a sixth power of the endpoint from its minimal discriminant.

18.1. **Odd isogenies and prime-degree paths.** We use the following unconditional classification: if an elliptic curve over $\mathbb{Q}$ admits a cyclic $\mathbb{Q}$-isogeny of degree N, then

$$N \in \{1, \dots, 19, 21, 25, 27, 37, 43, 67, 163\}. \tag{163}$$

The prime-degree theorem is due to Mazur [2, Theorem 1]. Its completion by Kenku for composite cyclic degrees is stated in [1, Theorem 2.2, p. 239]. We use this completed statement, including its CM cases. Quotients by Galois-stable finite subgroups, dual isogenies, and descent of uniquely determined factors are the standard characteristic-zero isogeny results of [71, Chapter III].

Proposition 18.1. *Let $E/\mathbb{Q}$ have full rational 2-torsion. If E admits a $\mathbb{Q}$-isogeny of odd prime degree ℓ, then $\ell = 3$.*

Proof. Write the isogeny as $\psi : E \to E'$, and choose distinct nonzero points $T_0, T_1 \in E[2](\mathbb{Q})$. Since ℓ is odd, ψ identifies $E[2]$ with $E'[2]$. Form the rational quotient maps

$$\phi_0 : E \longrightarrow E_0 = E/\langle T_0 \rangle, \qquad \phi_1 : E' \longrightarrow E_1 = E'/\langle \psi(T_1) \rangle.$$

The composite

$$\Phi = \phi_1 \circ \psi \circ \widehat{\phi_0} : E_0 \longrightarrow E_1$$

has degree 4ℓ. We check that its geometric kernel is cyclic; the degree alone is insufficient.

On $E_0[2]$, the dual $\widehat{\phi_0}$ has a kernel of order two. Its image has order two and is contained in $\ker \phi_0$, since $\phi_0 \widehat{\phi_0} = [2]$. Consequently its image is $\langle T_0 \rangle$. The subgroups $\langle \psi(T_0) \rangle$ and $\langle \psi(T_1) \rangle$ are distinct subgroups of order two. Therefore

$$\ker \Phi \cap E_0[2] = \ker \widehat{\phi_0}, \qquad \#(\ker \Phi)[2] = 2.$$

The 2-primary subgroup of $\ker \Phi$ has order four. It cannot be $(\mathbb{Z}/2\mathbb{Z})^2$, so it is cyclic. The ℓ-primary subgroup has prime order and is also cyclic. Their coprime orders imply that $\ker \Phi$ is cyclic of order 4ℓ. The list (163) permits this only for $\ell = 3$. □

The hypothesis here is a rational isogeny, or equivalently a Galois-stable subgroup, and does not require a rational point of order ℓ.

Lemma 18.2. *Any two elliptic curves that are isogenous over $\mathbb{Q}$ are joined by a finite path of rational prime-degree isogenies, up to $\mathbb{Q}$-isomorphism at the endpoint.*

Proof. Let the curves be E, F. By duality there is a rational isogeny $E \to F$. Among all such maps choose θ of least positive degree, and let H be its finite geometric kernel. If H is not cyclic, one of its primary subgroups is not cyclic, and for some prime ℓ the group $H[\ell]$ has dimension two over $\mathbb{F}_\ell$. Here we used $H[\ell] \subseteq E[\ell]$ and $E[\ell] \cong (\mathbb{Z}/\ell\mathbb{Z})^2$. Thus $E[\ell] \subseteq H$. The quotient property gives a factorization

$$\theta = \theta_1 \circ [\ell], \qquad \deg \theta_1 = \deg \theta / \ell^2 < \deg \theta.$$

The factor θ_1 is defined over $\mathbb{Q}$: it is unique, and both θ and $[\ell]$ are defined over $\mathbb{Q}$. This contradicts the choice of θ.

Hence H is cyclic. Its unique subgroup of each divisor order is Galois-stable. Quotienting along a subgroup chain with successive prime-order quotients, then identifying E/H with F, proves the claim. □

This argument replaces the initial map by one of least degree. It does not assert that every given rational isogeny factors through rational prime-degree maps: multiplication by ℓ on a curve with irreducible $E[\ell]$ need not do so.

Lemma 18.3. *Fix an odd prime ℓ. If $E/\mathbb{Q}$ has no rational ℓ-isogeny, neither does any curve rationally isogenous to E by an isogeny of 2-power degree.*

Proof. Take such a curve F and a rational isogeny $\alpha : F \to E$ of 2-power degree, using a dual if necessary. Its restriction to $F[\ell]$ is a Galois-equivariant isomorphism onto $E[\ell]$. A Galois-stable subgroup of order ℓ in $F[\ell]$ would therefore give one in $E[\ell]$, contrary to the hypothesis. □

18.2. **An explicit family and its four models.** For every integer $n \geq 1$, put

$$c = c_n = 1792n + 2, \qquad (a, b, c) = (1, c_n - 1, c_n), \qquad E_c : \ y^2 = x(x-1)(x+c-1). \tag{164}$$

These are positive primitive abc triples, and c_n tends to infinity. Moreover,

$$c \equiv 2 \pmod 8, \qquad c \equiv 2 \pmod 7, \qquad c \geq 1794.$$

The curve E_c is the actual Frey curve of this triple, with full rational 2-torsion.

Lemma 18.4. *The curve E_c in (164) has no rational 3-isogeny.*

Proof. Its displayed discriminant is $16(c-1)^2c^2$, a 7-adic unit, so it has good reduction at 7. Its reduction is $y^2 = x^3 - x$. The complete affine count is

x	0	1	2	3	4	5	6
$x^3 - x \pmod 7$	0	0	6	3	4	1	0
$\#\{y : y^2 = x^3 - x\}$	1	1	0	0	2	2	1

and gives seven affine points. Adding the point at infinity yields $\#E_c(\mathbb{F}_7) = 8$, hence $a_7 = 0$. The good-reduction Frobenius theorem identifies the characteristic polynomial on $E_c[3]$ with $T^2 - a_7T + 7$, reduced modulo 3 [71, Chapters V and VII]. This is $T^2 + 1$, which has no root in $\mathbb{F}_3$. A rational 3-isogeny would supply a Galois-stable line in $E_c[3]$, on which Frobenius would have an eigenvalue in $\mathbb{F}_3$. This is impossible. ☐

Proposition 18.1 and Lemma 18.4 now exclude every odd-prime rational isogeny from E_c.

For the quotient at $(0,0)$ of $y^2 = u^3 + Au^2 + Bu$, the usual integral model is

$$Y^2 = X^3 - 2AX^2 + (A^2 - 4B)X. \tag{165}$$

For example the formula $X = u + A + B/u$, $Y = y(1 - B/u^2)$ verifies the equation and extends to the degree-two quotient with kernel $\{O, (0,0)\}$. The discriminant of the remaining quadratic factor on the right of (165) is $16B$. Hence the quotient has extra rational 2-torsion exactly when B is a rational square.

Translate each of the three rational points of order two of E_c to $(0,0)$. The resulting values of B are, respectively,

$$-(c-1), \qquad c, \qquad c(c-1).$$

The first is negative, and the other two are 2 modulo 8. An integer that is a rational square is an integer square, and a square cannot be 2 modulo 8. Thus each of the three quotients has exactly one nonzero rational point of order two. Its only rational degree-two quotient is the dual edge back to E_c, up to rational isomorphism.

Denote these quotient curves by E_0, E_a, E_b, where the last subscript refers to the kernel at $x = -(c-1)$. Their equations are

$$\begin{aligned} E_c :&\quad y^2 = x^3 + (c-2)x^2 - (c-1)x, \\ E_0 :&\quad Y^2 = X^3 + (4-2c)X^2 + c^2X, \\ E_a :&\quad Y^2 = X^3 - 2(c+1)X^2 + (c-1)^2X, \\ E_b :&\quad Y^2 = X^3 + (4c-2)X^2 + X. \end{aligned} \tag{166}$$

Direct computation of their Weierstrass invariants gives

(167)

curve	$c_4/16$	Δ_{disp}
E_c	$c^2 - c + 1$	$16(c-1)^2c^2$
E_0	$c^2 - 16c + 16$	$-256(c-1)c^4$
E_a	$c^2 + 14c + 1$	$256c(c-1)^4$
E_b	$16c^2 - 16c + 1$	$256c(c-1)$

Here the subscript indicates the discriminant of the displayed integral model, without an assertion of minimality at 2.

Theorem 18.5. *Every elliptic curve over $\mathbb{Q}$ isogenous to E_c over $\mathbb{Q}$, by an isogeny of any degree, is $\mathbb{Q}$-isomorphic to one of the four curves in* (166).

Proof. The quotient computation exhausts the connected rational 2-isogeny graph: E_c has precisely the three indicated subgroups of order two, and each quotient has only its dual edge back. Lemma 18.3 excludes every odd-prime isogeny from every vertex of this graph.

Any rationally isogenous curve is connected to E_c by a prime-degree path, by Lemma 18.2. Every initial 2-edge stays among the four displayed vertices. A first odd-prime edge is impossible by the preceding paragraph. Thus the entire path stays in the list. Possible isomorphisms between displayed vertices only shorten the list and do not affect the conclusion. ☐

The theorem has no non-CM or conjectural Galois-surjectivity hypothesis. Its external inputs are the stated cyclic-isogeny classification, the basic quotient and dual theory, and the good-reduction Frobenius theorem.

18.3. **Matching bounds throughout the class.** Let $\mathcal{I}_c$ be the nonempty finite set of rational isomorphism classes in Theorem 18.5, and define

$$D_{\max}(c) = \max_{F\in\mathcal{I}_c} |\Delta_{\min}(F)|, \qquad J_{\min,\infty}(c) = \min_{F\in\mathcal{I}_c} |j(F)|.$$

Here $\Delta_{\min}$ is the global minimal discriminant over $\mathbb{Q}$. The absolute value in $J_{\min,\infty}$ is the single complex absolute value of the rational number $j(F)$; it is not its global Weil height.

Theorem 18.6. *For every $c = c_n$ in (164),*

$$\frac{c^5}{32} \le D_{\max}(c) \le 256c^5, \qquad 2c \le J_{\min,\infty}(c) \le 32c. \tag{168}$$

Proof. Every absolute displayed discriminant in (167) is at most $256c^5$. Passing to a minimal integral model cannot increase a prime valuation of the discriminant. Theorem 18.5 therefore gives the upper bound for $D_{\max}$ over the entire class.

For its lower bound use E_0. At an odd prime $q \mid c$, its c_4 invariant is congruent to 256 modulo q. At an odd prime $q \mid c-1$, it is congruent to 16 modulo q. Thus c_4 is a unit at every odd prime dividing the displayed discriminant, and the integral model is minimal there. We use here the usual minimality criterion for integral Weierstrass models [71, Chapter VII]: lowering the discriminant by q^{12} would require $q^4 \mid c_4$. Since $v_2(c) = 1$ and $c-1$ is odd, the odd part of the displayed discriminant is retained, giving

$$|\Delta_{\min}(E_0)| \ge (c-1)(c/2)^4 = \frac{(c-1)c^4}{16} \ge \frac{c^5}{32}. \tag{169}$$

No calculation of the minimal model at 2 is needed.

For the j-bounds, $c \ge 1794 > 32$ implies that each polynomial in the $c_4/16$ column of (167) is at least $c^2/2$. In particular,

$$c^2 - 16c + 16 - \frac{c^2}{2} = \frac{c(c-32)}{2} + 16 \ge 0;$$

the other three inequalities follow directly for $c \ge 2$. Consequently every displayed curve has $|c_4| \ge 8c^2$ and

$$|j| = \frac{|c_4|^3}{|\Delta_{\mathrm{disp}}|} \ge \frac{512c^6}{256c^5} = 2c.$$

This ratio is unchanged when one passes to a minimal model. Finally $0 < c^2 - 16c + 16 \le c^2$ for our values of c, so E_0 gives

$$|j(E_0)| = \frac{16(c^2-16c+16)^3}{(c-1)c^4} \le \frac{16c^2}{c-1} \le 32c.$$

This proves both pairs of inequalities. □

With $\log^+ x = \max\{0, \log x\}$, the bounds imply

$$\lim_{n\to\infty} \frac{\log D_{\max}(c_n)}{\log c_n} = 5, \qquad \lim_{n\to\infty} \frac{\min_{F\in\mathcal{I}_{c_n}} \log^+ |j(F)|}{\log c_n} = 1. \tag{170}$$

The second limit concerns complex j-values, not a height of a rational point on an auxiliary elliptic curve.

Corollary 18.7. *For every real $\theta > 5$ and constant $C > 0$, there are positive primitive abc triples for which no elliptic curve $F/\mathbb{Q}$ rationally isogenous to the actual Frey curve satisfies*

$$c^\theta \le C|\Delta_{\min}(F)|.$$

Proof. Choose n so large that $c_n^{\theta-5} > 256C$, and apply (168). □

Thus a uniform $c^{6-\delta}$ recovery for $0 < \delta < 1$, or a $c^{6-o(1)}$ recovery along this family, fails even when the representative can depend arbitrarily on the triple. This refutes the specified discriminant-replacement step. It does not refute a Szpiro conjecture or abc: no sufficiently small radical has been established for these triples. Bounds retaining the archimedean contribution, or using curves outside this rational isogeny class, remain separate possibilities.

Remark 18.8 (Formalization boundary). The Lean arithmetic companion verifies the actual primitive family, the actual elliptic-curve point count over $\mathbb{F}_7$, the absence of a root of $T^2 + 1$ over $\mathbb{F}_3$, the four Weierstrass models and their invariants, and the displayed discriminant and rational absolute j-bounds. It does not formalize the cyclic-degree classification, the Frobenius interpretation of the point count, the entire isogeny-class identification, or minimal-discriminant theory. These are the external mathematical steps explicitly used above.

19. Exact Weil heights within the rational isogeny class

The complex absolute value in Theorem 18.6 is a single local quantity. We now determine the absolute Weil height of the same four j-invariants, including every cancellation in their rational coordinates. The minimum over the entire rational isogeny class is $3\log(c^2 - 16c + 16)$. In particular its leading term remains $6\log c$, although the logarithm of the minimum complex absolute value has leading term $\log c$.

19.1. **Reduced rational coordinates.** Throughout this section put

$$n \in \mathbb{Z}_{\geq 1}, \qquad c = 1792n + 2, \qquad u = c/2 = 896n + 1. \tag{171}$$

Thus $c \geq 1794 > 32$, and $u, c-1$ are positive odd integers. Let E_c, E_0, E_a, E_b be the four actual curves of Theorem 18.5, and write

$$\begin{aligned} P &= c^2 - c + 1, & Q &= c^2 - 16c + 16, \\ R &= c^2 + 14c + 1, & S &= 16c^2 - 16c + 1. \end{aligned} \tag{172}$$

For a rational number $z = N/D$ in lowest terms, with $D > 0$, we use the absolute logarithmic and multiplicative Weil heights

$$\mathrm{h}(z) = \log \max\{|N|, D\}, \qquad \mathrm{H}(z) = \max\{|N|, D\}. \tag{173}$$

Over $\mathbb{Q}$ the relative and absolute normalizations agree.

Proposition 19.1. *The following are the signed reduced rational coordinates of the actual j-invariants. Every denominator is positive.*

curve	*numerator N*	*denominator D*
E_c	$64P^3$	$u^2(c-1)^2$
E_0	$-Q^3$	$(c-1)u^4$
E_a	$8R^3$	$u(c-1)^4$
E_b	$8S^3$	$u(c-1)$

Proof. The previously computed invariants give

$$j(E_c) = \frac{256P^3}{c^2(c-1)^2}, \quad j(E_0) = -\frac{16Q^3}{(c-1)c^4},$$
$$j(E_a) = \frac{16R^3}{c(c-1)^4}, \quad j(E_b) = \frac{16S^3}{c(c-1)}.$$

Substitution of $c = 2u$ gives the displayed coordinates. The sign of $j(E_0)$ follows from its negative displayed discriminant. This computation uses $j = c_4^3/\Delta$ and does not require the displayed models to be minimal at 2.

For coprimality, the exact residue identities give the table

	$\bmod u$	$\bmod (c-1)$
P	1	1
Q	16	1
R	1	16
S	1	1

For example, $Q = 16+u(4u-32) = 1+(c-1)(c-15)$ and $R = 1+u(4u+28) = 16+(c-1)(c+15)$. Because $u, c-1$ are odd, 16 is coprime to either modulus. Consequently each polynomial is coprime to both factors of every denominator. Taking powers and multiplying numerator factors 64 or 8 preserves this coprimality.

For completeness the exact 2-parts are also visible directly: P, R, S are odd, whereas

$$Q = 4(u^2 - 8u + 4), \qquad v_2(Q) = 2.$$

The 2-adic valuations of the four absolute numerators are respectively $6, 6, 3, 3$, and every denominator is odd. Thus no odd prime or power of 2 can cancel in any of the four displayed fractions. $\square$

19.2. **The two exact minima.** Let $\mathcal{I}_c$ denote the rational isogeny class of E_c, modulo rational isomorphism, and define

$$\mathrm{h}_{\min}(c) = \min_{F\in\mathcal{I}_c} \mathrm{h}(j(F)), \qquad J_{\min,\infty}(c) = \min_{F\in\mathcal{I}_c} |j(F)|_\infty.$$

Theorem 18.5 supplies the entire-class identification used in these minima; the following arithmetic calculations do not provide a replacement proof of that classification.

Lemma 19.2. *The unique minimizing rational isomorphism class for the complex absolute value is represented by E_0, and*

$$J_{\min,\infty}(c) = \frac{Q^3}{(c-1)u^4} = \frac{16Q^3}{(c-1)c^4} > 2c > 1. \tag{174}$$

In particular all four absolute numerators in Proposition 19.1 exceed their denominators.

Proof. For $c \geq 32$ one has

$$\begin{aligned} Q - \frac{c^2}{2} &= \frac{c(c-32)}{2} + 16 > 0, \quad Q < c^2,\\ P - Q &= 15(c-1) > 0,\\ R - Q &= 15(2c-1) > 0,\\ S - Q &= 15(c^2-1) > 0. \end{aligned} \tag{175}$$

Thus all four polynomials are positive. Since $Q > c^2/2$ and $c - 1 < c$, the absolute value of $j(E_0)$ is greater than $2c$. The exact ratios of the other three absolute values to this one are

$$\begin{aligned} \frac{|j(E_c)|}{|j(E_0)|} &= \frac{16c^2}{c-1}\left(\frac{P}{Q}\right)^3 > 1,\\ \frac{|j(E_a)|}{|j(E_0)|} &= \left(\frac{cR}{(c-1)Q}\right)^3 > 1,\\ \frac{|j(E_b)|}{|j(E_0)|} &= \left(\frac{cS}{Q}\right)^3 > 1. \end{aligned} \tag{176}$$

These inequalities follow from (175) and $c > 1$. The entire-class theorem now gives the assertion, including uniqueness. Strictness separates E_0 from all other displayed models, regardless of any possible coincidence among those others. $\square$

Theorem 19.3. *The absolute logarithmic Weil heights of the actual four j-invariants are*

$$\begin{aligned} \mathrm{h}(j(E_c)) &= \log 64 + 3\log P, \\ \mathrm{h}(j(E_0)) &= 3\log Q, \\ \mathrm{h}(j(E_a)) &= \log 8 + 3\log R, \\ \mathrm{h}(j(E_b)) &= \log 8 + 3\log S. \end{aligned} \tag{177}$$

The unique minimizer in the entire rational isogeny class is represented by E_0, with

$$\mathrm{h}_{\min}(c) = 3\log(c^2 - 16c + 16), \qquad \min_{F\in\mathcal{I}_c} \mathrm{H}(j(F)) = Q^3. \tag{178}$$

Moreover,

$$\begin{aligned} 6\log c - 3\log 2 < \mathrm{h}_{\min}(c) < 6\log c, \\ c^6/8 < \min_{F\in\mathcal{I}_c} \mathrm{H}(j(F)) < c^6. \end{aligned} \tag{179}$$

Proof. The fractions in Proposition 19.1 are reduced, and Lemma 19.2 shows that their absolute numerators exceed their denominators. Therefore (173) gives exactly (177). The inequalities $P, R, S > Q > 0$, together with $64, 8 > 1$, show that each height other than that of E_0 is strictly greater than $3\log Q$. The entire-class theorem extends this minimum and uniqueness to every rationally isogenous curve. Finally $c^2/2 < Q < c^2$ gives (179), first by cubing and then by taking logarithms. □

19.3. **Finite-place contribution and the bounded gain.** The minimizers above are the same, but the two minimum values measure different quantities. If N/D is reduced, each prime dividing D contributes exactly $v_q(D)\log q$ to the finite-place sum in the logarithmic Weil height; primes not dividing D contribute zero. The entire finite-place contribution is therefore $\log D$. For E_0, whose complex absolute value exceeds one, this proves the exact identity

$$\begin{aligned} \mathrm{h}_{\min}(c) - \log J_{\min,\infty}(c) &= \log((c-1)u^4) \\ &= 5\log c - \log 16 + \log(1 - 1/c). \end{aligned} \tag{180}$$

Equivalently, the exact formulas yield

$$\begin{aligned} \mathrm{h}_{\min}(c) - 6\log c = 3\log(1 - 16/c + 16/c^2) \longrightarrow 0, \\ \frac{J_{\min,\infty}(c)}{c} = 16\,\frac{(1 - 16/c + 16/c^2)^3}{1 - 1/c} \longrightarrow 16. \end{aligned} \tag{181}$$

All limits here and below are along the unbounded sequence (171). Continuity of log at positive arguments proves these limits. In particular

$$\frac{\mathrm{h}_{\min}(c)}{\log c} \longrightarrow 6, \qquad \frac{\log J_{\min,\infty}(c)}{\log c} \longrightarrow 1.$$

The denominator accounts for the missing term of size $5\log c$; it cannot be discarded when using the absolute Weil height.

The gain achieved by choosing the best representative in the entire class is also exact:

$$\mathrm{h}(j(E_c)) - \mathrm{h}_{\min}(c) = \log 64 + 3\log(P/Q) \in (6\log 2, 9\log 2). \tag{182}$$

Indeed $P > Q$ by (175), while $P < c^2$ and $Q > c^2/2$ imply $P/Q < 2$. These strict inequalities prove the stated interval, using $\log 64 = 6\log 2$. Since $P/Q \to 1$, the gain tends to $6\log 2$. Thus optimization over the entire class saves only a bounded additive amount in this family.

Corollary 19.4. *For every $\delta > 0$ and $C \in \mathbb{R}$, there exists an $n \geq 1$ such that every elliptic curve $F/\mathbb{Q}$ rationally isogenous to E_{c_n} satisfies*

$$\mathrm{h}(j(F)) > (6 - \delta)\log c_n + C.$$

Equivalently, for every $\theta < 6$ and $C_0 > 0$, some $n \geq 1$ satisfies

$$\mathrm{H}(j(F)) > C_0 c_n^{\theta} \qquad \textit{for every } F \in \mathcal{I}_{c_n}.$$

Proof. Choose n with $\delta \log c_n > C + 3\log 2$ and apply the lower bound in (179). Such n exist because $c_n \to \infty$. For the multiplicative formulation, choose $c_n^{6-\theta} > 8C_0$ and use the corresponding lower bound. □

This corollary excludes the specific proposed step of lowering the leading sixth-power j-height size by selecting a different rationally isogenous representative. It is not a counterexample to a conductor–height estimate, modified Szpiro, or abc. No upper bound for the radicals of these triples is asserted, and their unresolved height–radical comparison is unaffected by this calculation.

19.4. **Formalization boundary.** The Lean height companion reuses the four actual Weierstrass curves from the preceding section. It proves the signed rational equalities, integer coprimality, and equality with the canonical `Rat.num` and `Rat.den` coordinates. It then uses the actual multiplicative and logarithmic height functions in mathlib and proves compatibility with the existing normalized-height interface over $\mathbb{Q}$. This gives the exact height tables, attained minima over the actual model enumeration, uniqueness, and the logarithmic bounds (179). No artificially defined height is substituted for the library functions.

The enumeration is not asserted in Lean to exhaust a rational isogeny class. That passage uses the separately proved Theorem 18.5, including its stated external isogeny and Frobenius inputs. The strict comparison of the complex minimum, the explicit finite-place identity, the asymptotic limits, the bounded-gain formula, and Corollary 19.4 are paper proofs in this section; they are not added to the companion's formal claims. The module adds no axioms or conjectural hypotheses.

20. Standard-type theta points in a declared additive carrier

We now restrict the local source to numerical representatives of standard-type theta values, with their finite root-of-unity ambiguity. For each independent choice of these representatives, one actual full-Galois Kummer operator attains every point-hull bound below. The operator acts on a declared native additive carrier. This does not identify its action there with an action on the original multiplicative theta Kummer class, or establish membership in the global output family of [62, Corollary 3.12].

Page references to the theta and IUT sources in this section use the archived author versions: December 2008 for [79], May 2020 for [63, 62], and December 2020 for [64].

20.1. **The normalized evaluation and its finite ambiguity.** Retain the field and roots of (114). Thus $\ell \geq 7$ and p are primes, $p \equiv -1 \pmod{30\ell}$, $v_p(b_0) = 2$, and

$$\begin{gathered} e = 15\ell, \quad d = 2e, \quad h = (\ell-1)/2, \quad K_0 = \mathbb{Q}_p(\mu_{30\ell}), \\ \varpi^e = b_0, \quad E = K_0(\varpi), \quad q = b_0^2, \quad r_0 = \varpi^{15} = q^{1/(2\ell)}, \\ \beta = \varpi^{(e+1)/2}/p, \quad \kappa = 1 - 1/e, \quad I = \beta^{1-e}\mathcal{O}_E. \end{gathered} \tag{183}$$

Here ϖ is the element denoted π in (114), and is not a uniformizer. Lemma 14.1 gives $I = p^{-1}\log \mathcal{O}_E^\times$, $\operatorname{Tr} I = \mathbb{Z}_p$, and $\mu_{2\ell} \subset K_0$. In particular, $e \leq p-2$, $p \nmid d$, and $E/\mathbb{Q}_p$ is Galois with residue degree two.

Choose $i^2 = -1$ in an algebraic closure. The series of [79, Proposition 1.4, pp. 20–21] is

$$\vartheta(U) = \sum_{n\in\mathbb{Z}} (-1)^n q^{n(n+1)/2} U^{2n+1}, \qquad f(U) = \vartheta(U)/\vartheta(i). \tag{184}$$

The standard-type theta function is the reciprocal of an ℓ-th root of this normalized function [63, Example 3.2(ii),(iv)].

Lemma 20.1 (The normalized theta coset)**.** *For every integer a and every nonzero U in a finite extension of E,*

$$\vartheta(q^{a/2}U) = (-1)^a q^{-a^2/2} U^{-2a} \vartheta(U), \qquad f(\pm i q^{a/2}) = \pm q^{-a^2/2}.$$

The reciprocals of the ℓ-th roots of these two evaluations form exactly the coset

$$\mu_{2\ell} r_0^{a^2}. \tag{185}$$

Proof. At the indicated points the term valuations grow quadratically as $|n| \to \infty$, so reindexing is legitimate. Substitution $n \mapsto n + a$ gives the functional equation. Also $\vartheta(-U) = -\vartheta(U)$. At $U = i$ the terms $n = 0, -1$ sum to $2i$, and all other terms have positive valuation. Thus $\vartheta(i)$ is a unit. Since $i^{-2a} = (-1)^a$, the two normalized evaluations follow. Their reciprocal roots have ℓ-th powers $\pm q^{a^2/2}$, whose complete root set, for odd ℓ, is (185). □

In the standard cyclotomic and Kummer marking, [64, Remark 2.5.1(i), p. 72] identifies the unperfected evaluation set at label j with the Kummer image of $\mu_{2\ell} r_0^{j^2}$, where the exponent uses the integer representative $0 \le j \le h$. The ambiguities at different labels are independent [64, Remark 2.5.1(iii), p. 73]. They are not arbitrary elements of $\mathcal{O}_E^\times$: standard-type normalization removes that larger constant-unit ambiguity [64, Remark 1.12.2(i),(ii)].

20.2. **One arrow for independent torsion representatives.** For each $1 \le j \le h$, choose $\zeta_j \in \mu_{2\ell}$ independently and put

$$\begin{gathered} x_j = \zeta_j r_0^{j^2}, \qquad r_j = v_p(x_j) = 2j^2/\ell, \\ n_j = \lfloor r_j \rfloor, \qquad k_j = n_j + 1, \qquad T_j = p^{-k_j} x_j, \qquad V = I/pI, \quad V_0 = \beta I/pI. \end{gathered} \tag{186}$$

The quotient V/V_0 has dimension two over $\mathbf{F}_p$. Bars will denote reduction modulo pI.

Lemma 20.2 (Content, trace and projective inertia)**.** *For every choice of the ζ_j,*

$$\begin{gathered} k_j = \lfloor r_j + \kappa \rfloor, \qquad T_j \in \beta I \setminus pI, \qquad \mathrm{Tr}_{E/\mathbb{Q}_p}(T_j) = 0, \\ 1 \in \beta I \setminus pI, \qquad \mathrm{Tr}_{E/\mathbb{Q}_p}(1) = d \in \mathbb{Z}_p^\times. \end{gathered} \tag{187}$$

If $\sigma(\beta) = \epsilon\beta$ generates $\mathrm{Gal}(E/K_0)$*, with ϵ of order e, then*

$$\sigma(T_j) = \epsilon^{30j^2} T_j. \tag{188}$$

The projective orbit of $\overline{T_j}$ over $\mathbf{F}_p$ has exactly ℓ elements.

Proof. Write $2j^2 = \ell n_j + \delta_j$, where $1 \le \delta_j \le \ell - 1$. Then $0 < \delta_j/\ell - 1/e < 1$, proving the formula for k_j. Moreover

$$e\, v_p(T_j) = 15\delta_j - e, \qquad 2 - e < 15 - e \le 15\delta_j - e \le -15 < 1.$$

The exponents $2 - e$ and 1 are the thresholds for βI and pI. This proves the two membership assertions, including the analogous assertion for 1, whose exponent is zero.

With $\gamma = p^2/b_0 \in \mathbb{Z}_p^\times$, (115) gives $r_0 = \gamma^{15}\beta^{30}$. Both γ and ζ_j lie in K_0 and are fixed by σ, proving (188). Its character has order $e/\gcd(e, 30j^2) = \ell$. Summing its inertia conjugates gives zero trace over K_0, and then zero absolute trace.

These roots reduce injectively and $\mu_\ell \cap \mathbf{F}_p^\times = \{1\}$, since $p \equiv -1 \pmod{\ell}$. If two inertia images of $\overline{T_j}$ are proportional over $\mathbf{F}_p$, their character residues are therefore equal. Indeed, a nonzero difference of residues is a unit of K_0, and multiplying $T_j \notin pI$ by such a unit cannot put it in pI. Thus proportionality forces the inertia exponents to agree modulo ℓ, while agreement gives equality already in E. This proves the projective orbit assertion. □

Let Γ_E be the actual integral Kummer operators of Section 11, extended $\mathbb{Q}_p$-linearly to E. They preserve I; we do not identify this group of operators with every integral linear automorphism.

Theorem 20.3 (Common attainment for the finite theta source)**.** *For every tuple* $(\zeta_1, \ldots, \zeta_h) \in \mu_{2\ell}^h$*, there is one $F \in \Gamma_E$ such that*

$$v_p(F(1)) = v_p(F(T_j)) = -\kappa \qquad (1 \le j \le h). \tag{189}$$

The chosen F may depend on the tuple.

Proof. The hypotheses of Lemma 11.3 hold: d is even, $2 \le e \le p-2$, and $\operatorname{Tr} I = \mathbb{Z}_p$. Use its basis

$$I = \mathbb{Z}_p a_{\mathrm{JW}} \oplus \mathbb{Z}_p b_{\mathrm{JW}} \oplus W, \qquad I \cap \ker \operatorname{Tr} = \mathbb{Z}_p a_{\mathrm{JW}} \oplus W,$$

and let B be the b_{JW}-coordinate. The trace of b_{JW} is a unit, so $B = \operatorname{Tr} / \operatorname{Tr}(b_{\mathrm{JW}})$.

For each j, at most one of the first ℓ inertia powers can place $\overline{T_j}$ on $\mathbf{F}_p \bar{a}_{\mathrm{JW}}$. Since $h < \ell$, one power avoids this line for every label. It fixes 1 and preserves V_0 and trace. We now verify every remaining hypothesis of Lemma 11.8, with $k = \mathbf{F}_p$, $L = V_0$, $U = \overline{1}$ and the transformed vectors $\overline{T_j}$. All lie in L by Lemma 20.2; $B(U) \neq 0$, $B(T_j) = 0$, and $T_j \notin \mathbf{F}_p a_{\mathrm{JW}}$. The trace kernel maps onto V/V_0: for $y \in I$, subtract $\operatorname{Tr}(y)/d \in \mathcal{O}_E \subset \beta I$. Finally its cardinality condition is $h+1 < p$.

That lemma consequently gives one parabolic composition sending all these vectors outside V_0. Its integer-coordinate crosses and symplectic operations have actual full-group lifts by Proposition 11.2 and Lemma 11.4, as used in the arithmetic conclusion of Lemma 11.8. Compose them with the chosen field inertia automorphism. The resulting canonical action M sends every input into $I \setminus \beta I$, of valuation $-\kappa$.

Proposition 11.5 gives Kummer action $c_\alpha M_\alpha^{-1}$, with $c_\alpha \in \mathbb{Z}_p^\times$. Using the inverse of the constructed group automorphism therefore gives a unit multiple of M, proving (189). ☐

This proves $\forall(\zeta_j)\,\exists F\,\forall j$, not the stronger assertion that one F works for every tuple. It does not commute multiplication by ζ_j with F, which need not be K_0-linear.

20.3. **The additive point hull and the logarithmic bridge.** For $m = m_j = j+1$, let

$$\mathcal{T}_m = E^{\otimes_{\mathbb{Q}_p} m}, \qquad A_m = \mathcal{O}_E^{\otimes_{\mathbb{Z}_p} m}, \qquad B_m = \prod_{\mathrm{Gal}(E/\mathbb{Q}_p)^{m-1}} \mathcal{O}_E$$

under the decomposition (90). Thus B_m is the integral closure of A_m. The multiplication and literal unit 1 here belong to the fixed native field E. The normalization $I = p^{-1} \log \mathcal{O}_E^\times$ specifies a lattice in that field; it does not replace its multiplication by a structure transported through ρ. In the declared additive tensor carrier put

$$Z_{j,\zeta_j} = 1 \otimes \cdots \otimes x_j, \qquad H_{j,\zeta_j} = \overline{\operatorname{span}_{B_m}\{F^{\otimes m}(Z_{j,\zeta_j}) : F \in \Gamma_E\}}. \tag{190}$$

The literal ones agree with the single-slot embedding of [62, Proposition 3.1(ii), p. 93]. Choosing this embedding does not itself prove global source membership.

Theorem 20.4 (An attained point-source hull)**.** *For every label and every finite torsion representative,*

$$H_{j,\zeta_j} = P_j = \beta^{ek_j - (e-1)m_j} B_{m_j}. \tag{191}$$

Here the uniformizer exponent is taken in every field component. For each tuple of representatives, one F generates these ideals simultaneously for all labels. With $\mu(B_m) = 1$ and $D_m = d^m$,

$$D_{m_j}^{-1} \log \mu(H_{j,\zeta_j}) = (m_j \kappa - k_j) \log p. \tag{192}$$

Proof. Every F preserves I and $p^{k_j} I$. Since $x_j \in p^{k_j} I$ and $1 \in I$, each tensor component has valuation at least $k_j - m_j \kappa$. The component embeddings preserve valuation, giving containment in P_j. The common F from Theorem 20.3 gives $v_p(Fx_j) = k_j - \kappa$ and $v_p(F1) = -\kappa$. Its tensor has the required valuation in every component, so it is a generator of P_j times an element of $B_{m_j}^\times$. This proves the reverse inclusion before closure; the fractional ideal is closed. It is also the direct specialization $u = 1$, $t = T_j$ of Proposition 11.10.

There are d^{m-1} field components, each with residue degree two. Hence $\log \mu(\beta^\nu B_m) = -2d^{m-1} \nu \log p$. Division by d^m proves (192). No further division by m_j is made. ☐

The entire finite orbit $\mu_{2\ell} r_0^{j^2}$ gives the same hull. The source in this proof contains no arbitrarily chosen $\mathcal{O}_E^\times$-multiple and need not contain $r_0^{j^2} B_m$ or a product of fractional ideals.

Proposition 20.5 (A comparison for each arrow)**.** *For $0 \neq x \in \mathcal{O}_E$, set $\lambda(x) = p^{-1}\log(1+px)$, $r = v_p(x)$ and $k = \lfloor r + \kappa \rfloor$. For every $F \in \Gamma_E$,*

$$\begin{aligned} v_p(\lambda(x) - x) \geq 1 + 2r \geq k + 1 - \kappa, \qquad \lambda(x) - x \in p^{k+1}I, \\ v_p(F(\lambda(x))) = v_p(F(x)). \end{aligned} \tag{193}$$

Thus, for $t_j = \lambda(x_j)$ and $u = \lambda(1)$, the point tensors with factors $(x_j, 1, \ldots, 1)$ and $(t_j, u, \ldots, u)$ generate the same principal B_{m_j}-ideal after each individual $F^{\otimes m_j}$. Both orbit hulls are P_j, with the same common attainer.

Proof. For $n \geq 2$, the n-th term of $\lambda(x) - x$ has valuation $(n-1) + nr - v_p(n)$. For odd p, $v_p(n) \leq n - 2$, so this is at least $1 + 2r$. Convergence and ultrametricity give the first inequality; the second uses $k \leq r + \kappa$ and $r \geq 0$. By the definition of k, $x \in p^k I \setminus p^{k+1} I$. Since F is an automorphism of this filtered lattice,

$$v_p(Fx) < k + 1 - \kappa \leq v_p(F(\lambda(x) - x)).$$

Strict ultrametricity proves the valuation equality. Also $u \in \mathbb{Z}_p^\times$, so $\mathbb{Q}_p$-linearity gives $F(u) = uF(1)$. All component valuations of the two tensors therefore agree, proving the ideal assertions. Finally $(t_j - x_j)/p^{k_j} \in pI$, so the common attainer for T_j also works for t_j/p^{k_j}. □

Here $\lambda(x)$ is the coefficient of the separately defined class $\mathrm{Kum}_p(1+px)$ in $\rho = p^{-1}\log^{\mathrm{std}}_{\mathrm{BK}}$. It is not the theta evaluation class. In particular, exact zero trace was proved for T_j, not for t_j. If all m_j coordinates of this unit-cohomology packet are changed from ρ to $\log^{\mathrm{std}}_{\mathrm{BK}}$, the tensor is multiplied by p^{m_j}. Consequently

$$\begin{aligned} P_j^{\mathrm{std}} = p^{m_j} P_j = \beta^{e k_j + m_j} B_{m_j}, \\ D_{m_j}^{-1} \log \mu(P_j^{\mathrm{std}}) = -(k_j + m_j/e)\log p. \end{aligned} \tag{194}$$

The change is $-m_j \log p$. This rescales every coordinate of the specified cohomological source; it does not give the original theta function an additional factor p^{m_j}.

20.4. **Synchronized source conditions.** The constructions distinguish $\mathrm{Kum}(x_j)$, the numerical scalar x_j, and $\mathrm{Kum}_p(1+px_j)$. Their valuations already separate the first and third: $v_p(x_j) = 2j^2/\ell > 0$, whereas $v_p(1+px_j) = 0$. Under normalized logarithm the scalar x_j instead comes from the unit $\exp(px_j)$. Equal additive hulls therefore do not identify the multiplicative inputs.

The LGP construction uses the prescribed log-link pullback before the single-slot insertion [62, Propositions 3.4(ii) and 3.5, pp. 102–106]. A tautological local logarithmic marking can give this pullback the numerical coordinate x_j. It neither makes the log-link a ring homomorphism from the predecessor's original field nor permits independent choices at all labels. The theater log-link comes from one isomorphism and retains its synchronization [62, Proposition 1.3(i) and Remark 1.3.1, pp. 42–43].

Theorem 20.4 alone does not establish membership in the possible-image family of [62, Corollary 3.12, pp. 173–175]. Section 21 supplies one globally synchronized copy representative and proves the canonical local inclusion in its fixed base branch. Compatibility of the complete family of log-Kummer maps and reference data with the unindetermined q-pilot, including the required Ind3 comparison, remains separate. The full local Galois constructions used here are mathematical inputs from Section 11; their local class field theory and source identifications are not newly formalized in Lean by this section. No global abc conclusion is asserted.

21. A synchronized base branch of the theta-pilot images

We now place the local points of Section 20 in a specified base branch of the possible-image construction in [62, Corollary 3.12]. The construction uses one concrete global theta-pilot, two tagged copies of its Hodge theater, and the log-link representative induced by their single global copy-identity. The resulting local inclusion concerns the raw union of possible images in one vertically coric column. It does not identify the complete global family with the one-prime arithmetic bundles of Section 22.

We use the reconstruction results of the cited sources, with their original objects and variance conventions. We do not reprove all those results, or assume the global multiradial and horizontal compatibility assertions of [62, Theorem 3.11] in order to establish the inclusion below. References to [63, 62], [64], and [4] use respectively the archived May 2020, December 2020, and November 2015 author versions.

21.1. **Fixed initial data and the standard log-field.** Fix a member of Corollary 15.6. Its global initial theta data, including its core, auxiliary cover, distinguished quotient element and place section, have already been constructed. At its distinguished rational prime p, retain (183): in particular,

$$e = 15\ell, \qquad d = 2e, \qquad h = (\ell-1)/2, \qquad I = p^{-1}\log\mathcal{O}_E^\times = \beta^{1-e}\mathcal{O}_E, \qquad r_0^{2\ell} = q = b_0^2.$$

Here $v_p(b_0) = 2$, $E/\mathbb{Q}_p$ is Galois with residue degree two, $e \le p-2$, and $\mu_{2\ell} \subset K_0 = \mathbb{Q}_p(\mu_{30\ell})$. For $1 \le j \le h$, put

$$\begin{aligned} m_j = j+1, \qquad a_j = r_0^{j^2}, \qquad x_j = \zeta_j a_j, \quad \zeta_j \in \mu_{2\ell}, \\ \mathcal{T}_{m_j} = E^{\otimes_{\mathbb{Q}_p} m_j}, \qquad Z_{j,\zeta_j} = 1^{\otimes j} \otimes x_j. \end{aligned} \tag{195}$$

The distinguished last slot is the label-j slot in the source procession; it is not an extra factor.

The field of moduli is $F_{\mathrm{mod}} = \mathbb{Q}$. The source's selected place set V maps bijectively to the places of F_{mod}, so it has exactly one selected v above p. This is not a claim that the global torsion field has only one prime above p. It means that the finite local packet at this rational prime has no other selected direct summands: its marked carrier is $\mathcal{T}_{m_j}$.

Let E^- denote the original predecessor field. Definition 1.1(i) of [62, pp. 23–25] equips the perfection of its multiplicative unit group with a *new field structure*, denoted here by L^-. In the concrete model, standard logarithm gives a unital field isomorphism

$$\mathcal{L} : L^- \xrightarrow{\sim} E. \tag{196}$$

It transports both field operations from E; the new multiplication is not the predecessor's old multiplication of units.

For $x \in E$, choose N sufficiently large and write

$$y(x) = p^{-N}[\exp(p^N x)] \in L^-. \tag{197}$$

The bracket is written additively in the old unit perfection. The equality $\exp(p^{N+1}x) = \exp(p^N x)^p$ proves independence of sufficiently large N, and $\mathcal{L}(y(x)) = x$. For $x \in I$, one may take $N = 1$, since $pI = \mathfrak{m}_E$ and $1/e > 1/(p-1)$.

In the chart (196), the pre-log-shell has coordinates $\log\mathcal{O}_E^\times$; the canonical shell is its multiple $p^{-1}\log\mathcal{O}_E^\times = I$. This division by p does not change the field multiplication or its unit. The new field unit is $y(1)$, of coordinate 1, whereas the old multiplicative unit $1 \in (E^-)^\times$ has coordinate zero after logarithm. Thus the background entries of (195) are the new field units.

For comparison, changing the *entire field chart* to $u = x/p$ would instead give

$$\begin{aligned} u \star v = p\,uv, \qquad 1_\rho = 1/p, \qquad \mathbb{Q}_p \ni t \longmapsto t/p, \\ v_\rho(u) = v_E(pu), \qquad \mathcal{O}_\rho = \mathcal{O}_E/p, \qquad I_\rho = I/p. \end{aligned} \tag{198}$$

The equation $a^n = b$ would become $p^{n-1}a_\rho^n = b_\rho$. In an m-fold packet, both the tensor coordinates and the reference integer module would be transported by p^{-m}. This is different from changing only the outputs of a fixed unit-cohomology source from $\rho = p^{-1}\log_{\mathrm{BK}}^{\mathrm{std}}$ to $\log_{\mathrm{BK}}^{\mathrm{std}}$, which gives the factor p^m in (194). Neither rescaling is made in (196).

21.2. **A global identity and three distinct cyclotomes.** Take distinct tagged copies $\mathcal{H}_{n,r}$, $n, r \in \mathbb{Z}$, of the fixed concrete Hodge theater. Retain the full vertical and horizontal poly-arrows required by [62, Definition 3.8(iii), pp. 113–114]. Inside the vertical arrow $\mathcal{H}_{0,-1} \to \mathcal{H}_{0,0}$, select the representative induced by the global copy-identity

$$\Xi : \mathcal{H}_{0,-1}^D \xrightarrow{\sim} \mathcal{H}_{0,0}^D. \tag{199}$$

Selecting a representative does not replace a full poly-arrow by a singleton.

Proposition 21.1 (Synchronized standard-log representatives)**.** *The single isomorphism* (199) *gives, at every finite place and every labeled prime-strip occurrence, the log-field map*

$$\lambda_v = \mathrm{copy}_v \circ \mathcal{L}_v : L_{0,-1,v} \xrightarrow{\sim} E_{0,0,v}. \tag{200}$$

These are simultaneously allowed representatives of the theater log-link. At the distinguished place, the pullbacks of x_j and of each background field unit have standard-log coordinates x_j and 1*.*

Proof. Definition 1.1(i) of [62] constructs the natural isomorphism from the tautological log-Frobenioid to the original Frobenioid. The group acts compatibly on all the monoids in that construction. Remark 1.1.2(i), pp. 29–30, identifies its underlying local fundamental groups by the tautological identity. Composing this natural isomorphism with the copy map therefore covers the D-prime-strip isomorphism induced by Ξ.

The map from F-prime-strip isomorphisms to their D-prime-strip isomorphisms is bijective by [63, Corollary 5.3(ii), p. 144]. Consequently the displayed local map is the unique lift used in [62, Proposition 1.3(i), p. 42]. That proposition applies these lifts to the one Ξ across every prime-strip of both bridges; Remark 1.3.1, p. 43, makes the synchronization requirement explicit. The construction at the archimedean constituents uses the corresponding standard tautological isomorphism.

The concrete marked theater has the synchronized constant-monoid identifications of [64, Corollary 4.6(iii), p. 138]. Its copy-identity neither permutes the labels nor changes these identifications. In those fixed coordinates (200) is the identity field map. The asserted pullbacks follow. ☐

The proposition is about this Ξ, not every theater isomorphism. An arbitrary such isomorphism may change label markings. A unital local field isomorphism fixing $q \in \mathbb{Q}_p$ preserves the coset defined by $X^{2\ell} = q^{j^2}$, but that observation alone does not provide a synchronized global representative.

It is essential to distinguish the three coefficient objects

$$\mu_{\mathrm{prev}}, \qquad \mu_{\log}, \qquad \mu_{\mathrm{current}},$$

arising respectively from multiplication in the old field E^-, the new log-field L^-, and the current field E^+. The old *linear unit-Kummer coordinate* is

$$\kappa^-_{\mathrm{lin}}(x) = p^{-N}\,\mathrm{Kum}_{\mu_{\mathrm{prev}}}(\exp(p^N x)) \in H^1_f(G_E, \mathbb{Q}_p(1)_{\mathrm{prev}}). \tag{201}$$

It is the inverse standard Bloch–Kato logarithm on the unit subspace. On I it takes values in $p^{-1}H^1_f(G_E, \mathbb{Z}_p(1)_{\mathrm{prev}})$.

There is separately the new multiplicative Kummer map

$$(L^-)^{\times,\log} \longrightarrow H^1(G_{L^-}, \mathbb{Q}_p(1)_{\log}).$$

The field isomorphism λ of (200) gives the usual field-theoretic Kummer comparison

$$\mathrm{Kum}_{\mu_{\log}}(y(x)) \longmapsto \mathrm{Kum}_{\mu_{\mathrm{current}}}(\lambda(y(x))). \tag{202}$$

Its group and coefficient maps are those induced by that field isomorphism. The operation $y \mapsto \mathrm{Kum}_{\mu_{\log}}(y)$ is not linear in the old unit-H1 coordinate (201). The new multiplication used here is already part of [62, Definition 1.1(i), p. 24], and its monoid, units and action are used explicitly in [62, Proposition 3.4(i), p. 102].

Indeed, a same-group integral-coefficient H1 arrow cannot identify the two Kummer maps just displayed. If $v_E(x) > 0$, the raw class $\mathrm{Kum}_{\mu_{\mathrm{current}}}(x)$ has nonzero valuation component, while $\kappa^-_{\mathrm{lin}}(x)$ has valuation component zero. The unit subspace is the annihilator of unramified characters under local Tate duality, so a Galois-induced isomorphism with compatible integral Tate coefficients preserves it. The comparison (202) concerns the new multiplicative structure and does not assert this impossible old-H1 identification.

No global exponential is being used. The element $\exp(p^N x)$ in (197) need only belong to the completion E. It is not a global rational-function torsor generator. The source global localization is pulled back along the globally synchronized local isomorphisms [62, Proposition 3.3].

The same distinction is explicit in [4, Definition 5.4(ii), pp. 125–126]: the global log construction re-embeds the global object through its natural local isomorphisms and keeps the underlying global Galois theater. Its Remark 5.4.1, p. 129, warns that the local logarithm diagrams do not extend as a logarithm on the global number field.

21.3. **The canonical core coordinate and the preceding layer.** Let $\mathcal{C}_E$ denote the finite Galois-invariant part of the canonical module $k^{\sim}(G_E)$ in [4, Proposition 5.8(ii), p. 139]. Use the local reciprocity convention of that unit reconstruction, $\mathrm{rec}_E : E^\times \to G_E^{\mathrm{ab}}$. The concrete model of $\mathcal{C}_E$ is the rationalized old unit group with its finite torsion removed.

Proposition 21.2 (The specified unit comparison)**.** *There is a standard comparison*

$$\sigma_E : E \xrightarrow{\sim} \mathcal{C}_E, \qquad \sigma_E(x) = p^{-N}[\mathrm{rec}_E(\exp(p^N x))], \tag{203}$$

and the source shell is $\sigma_E(I)$. For the unit Kummer comparison

$$\eta_E : H^1_f(G_E, \mathbb{Q}_p(1)_{\mathrm{prev}}) \xrightarrow{\sim} \mathcal{C}_E, \qquad \eta_E(\mathrm{Kum}(u)) = [\mathrm{rec}_E(u)], \tag{204}$$

one has

$$\sigma_E = \eta_E \kappa^-_{\mathrm{lin}}. \tag{205}$$

Proof. The brackets in (203) refer to the unit-perfection module. Independence of sufficiently large N follows from the exponential identity used for (197). Standard logarithm on a sufficiently small unit subgroup proves additivity, $\mathbb{Q}_p$-linearity and bijectivity. The actual unit image is the pre-log-shell, and its multiple p^{-1} is the shell. This gives $\sigma_E(I)$, with no further factor p^{-1} in (203).

The Kummer sequence and Hilbert 90 identify the rationalized torsion-free unit group with H^1_f, so (204) is well defined. Evaluating it on $p^{-N}\,\mathrm{Kum}(\exp(p^N x))$ proves (205).

This is the particular comparison specified by the reconstruction source. Parts (c) and (d) of [4, Corollary 5.10(iv), p. 148] give the natural comparison from the field-theoretic shell to the canonical mono-analytic shell, compatible with the shells and volumes. Its proof on p. 149 compares the base field inside abelianizations with its Kummer reconstruction by the fundamental-class and cyclotomic isomorphisms of Corollary 1.10(a),(c). The proof identifies the result as the reciprocity map of local class field theory. On a concrete unit u, this is (204). Thus the comparison used here is not an arbitrary linear isomorphism chosen after the point. □

Write $k^\times_-, k^\times_+$ for the original constant-monoid reconstruction maps of [64, Corollary 4.6(iii)], from the two copied theaters to the common core. In the fixed concrete model they reconstruct the same labeled field. Before forgetting multiplication, the relevant diagram is

$$\begin{array}{ccc} L^- & \xrightarrow{\lambda} & E^+ \\ {\scriptstyle \log(k^\times_-)} \downarrow & & \downarrow {\scriptstyle k^\times_+} \\ L^{\mathrm{c}} & \xrightarrow{\lambda_{\mathrm{c}}} & E^{\mathrm{c}}. \end{array} \tag{206}$$

The lower map is the corresponding tautological standard-log representative. Both paths send $y(x)$ to the constant-field element of coordinate x. To check this directly, first take a sufficiently small old unit: its constant-monoid reconstruction is its actual reconstructed constant, and standard logarithm commutes term by term with this copy-identity field reconstruction. Division by p^N extends the equality to every $y(x)$. The left arrow is formed from the old-unit map and then the new log-field structure; its value in the core additive carrier is $\sigma_E(x)$. The right arrow is a current multiplicative constant-monoid map. These are not two versions of the same H1 arrow.

Let

$$\mathcal{C}_{j,p} = I^{\mathbb{Q}}(S^{\pm}_{j+1};{}^{0,\circ}D^{\perp}_p), \qquad \iota_j = \sigma_E^{\otimes m_j} : \mathcal{T}_{m_j} \xrightarrow{\sim} \mathcal{C}_{j,p}.$$

The superscript $\perp$ here denotes the source's mono-analytic prime-strip, not an orthogonal complement. Its identification with the tensor of $\mathcal{C}_E$ is the natural one in [62, Proposition 3.2(i),(ii), pp. 97–99]. Denote the label-j components of the split LGP monoids at level 0 and at the core by $\mathcal{S}_j^+$ and $\mathcal{S}_j^{\mathrm{c}}$. Applying the distinguished-slot recipe to (206) gives

$$\begin{array}{ccc} \mathcal{S}_j^+ & \hookrightarrow & \mathcal{T}_{m_j} \\ k_{\mathrm{LGP},0}\downarrow & & \downarrow \iota_j \\ \mathcal{S}_j^{\mathrm{c}} & \hookrightarrow & \mathcal{C}_{j,p}. \end{array} \tag{207}$$

The upper right is written in the standard-log coordinates of the *preceding* field L^-. Before choosing coordinates the right-hand map is $\iota_j \mathcal{L}^{\otimes m_j}$.

Here the upper inclusion is [62, Proposition 3.4(ii), pp. 102–103]: pull the current Gaussian monoid back to the preceding log-field, then tensor with the new units in the other slots. The lower inclusion is the same recipe on the reconstructed constant monoids. Proposition 3.5(i), pp. 103–104, explicitly uses the comparisons at *both* indices $r' = r, r-1$. The commutativity of (206), followed by this tensor operation, verifies (207) in the selected concrete base representative. It is not obtained by replacing the preceding carrier with the current H1 space.

The same diagram at all places and the copy-identity of the global field give the corresponding base fractional-ideal comparison. Both recipes use the same local ideals and the same global rational-function monoid acting on them. This is the construction recorded in [62, Proposition 3.10(i), pp. 147–148], again with both adjacent indices. We use this fixed representative, not the later assertion of compatibility for every vertical iterate.

Proposition 21.3 (The variance of the actual core action)**.** *For a continuous automorphism α of G_E, let M_α be its action on the unit abelianization, expressed in logarithmic coordinates:*

$$M_\alpha(\log u) = \log\Big(\mathrm{rec}_E^{-1}\,\alpha^{\mathrm{ab}}(\mathrm{rec}_E(u))\Big). \tag{208}$$

Extend this map $\mathbb{Q}_p$-linearly to E. The actual canonical core map Φ_α satisfies

$$\Phi_\alpha\sigma_E = \sigma_E M_\alpha^{-1}, \qquad \Phi_\alpha(\sigma_E(I)) = \sigma_E(I). \tag{209}$$

Proof. The unit image in the abelianization is preserved by the reconstruction algorithm. Proposition 5.8(i) of [4, p. 139] specifies its *contravariant* functoriality by transfer. For a group isomorphism this transfer is the inverse of the induced abelianization map; there is no index factor. Thus

$$\Phi_\alpha(\sigma_E(x)) = p^{-N}[\alpha^{-1,\mathrm{ab}}(\mathrm{rec}_E(\exp(p^N x)))] = \sigma_E(M_\alpha^{-1}x).$$

The unit image and its perfection are preserved. Dividing the unit image by p proves the shell assertion. ☐

For comparison, a Kummer action with a chosen compatible integral Tate-module isomorphism δ is

$$F_{\alpha,\delta} = c_{\alpha,\delta}M_\alpha^{-1}, \qquad c_{\alpha,\delta}\in\mathbb{Z}_p^\times,$$

by Proposition 11.5. We do not silently identify this with (209). All pointwise statements below use the actual canonical core operator $F = M_\alpha^{-1}$. In particular, if a full Galois lift α_0 has canonical matrix M, then α_0^{-1} gives core matrix $M_{\alpha_0^{-1}}^{-1} = M$. No additional scalar choice is required.

21.4. **Local ideal elements of one global pilot.**

Lemma 21.4 (Points belonging to one pilot object)**.** *In the synchronized representative above, there is one global theta-pilot $\mathcal{P}$, in the local fractional-ideal realization of* [62, Definition 3.8(i)]*, whose label-j ideal at the distinguished place contains Z_{j,ζ_j} for every $\zeta_j \in \mu_{2\ell}$.*

Proof. Choose one standard normalized theta-evaluation representative. Its label-j value is a $\mu_{2\ell}$-multiple of a_j, by Lemma 20.1 and [64, Remark 2.5.1(i), p. 72]. The Gaussian splitting monoid is generated, up to torsion, by this value-profile [64, Corollary 3.6(ii),(iii), pp. 99–100]; Corollary 4.6(iv), pp. 138–139, supplies it in the chosen theater.

Proposition 21.1 and [62, Proposition 3.4(ii)] pull that generator to the preceding log-field with the same standard coordinate. The single-slot insertion of [62, Proposition 3.1(ii), p. 93] tensors it with the new field units. The local fractional ideal generated by this element contains its generator times the unit. Multiplication by a root of unity does not change that ideal, since these are units of the new log-field. This proves the local claim for every ζ_j.

At the other bad places choose the prescribed splitting generators of the same theater, and use its standard good-place constituents. The finite collection of bad-place local ideals defines a global object by [62, Proposition 3.7(v), pp. 111–112]. Definition 3.8(i), p. 112, is precisely the theta-pilot designation for that object. Thus all the local ideal elements in the statement belong to one $\mathcal{P}$. The construction does not require the local generators to be a single element of the global number field. □

The lemma uses elements of the actual local ideals. It need not assert that every independent tuple (ζ_j) arises from one strictly chosen global theta-evaluation representative. The ideals, and hence the global pilot object, are unchanged by those torsion choices.

We also record exactly which local Galois operations are permitted. The Ind1 object in [62, Theorem 3.11(i), pp. 153–154] is the output mono-analytic procession of [63, Proposition 6.9(ii)]. At the selected nonarchimedean place, its constituent category is the category of finite transitive continuous G_E-sets. Pullback along any continuous $\alpha \in \mathrm{Aut}(G_E)$ is a self-equivalence, with inverse the pullback along α^{-1}. Use this equivalence at every repeated occurrence above p and the identity elsewhere, keeping the place and capsule indices fixed.

Definition 4.1(iii),(iv), p. 96, of [63] makes this a constituent prime-strip isomorphism. Its Section 0's capsule-full poly-isomorphisms contain the chosen collection with the identity index map. Definition 4.10, pp. 119–120, makes these precisely the representative data permitted at the stages of the procession. The same choice at repeated occurrences is allowed, and conjugation leaves the full connecting collections unchanged. Thus the collection occurs inside an allowed procession poly-automorphism, as in the representative argument of Section 7. No extension of α to the earlier curve covering category or global Hodge theater is imposed by this output object.

Theorem 21.5 (Membership in a specified base branch)**.** *Let $\mathcal{P}$ be the fixed global pilot of Lemma 21.4, and let ${}^{0,\circ}U^{\mathrm{raw}}_{j,p}$ denote the raw possible-image union defined in* [62, Corollary 3.12, p. 174]*, before its holomorphic hull is taken. For every continuous $\alpha \in \mathrm{Aut}(G_E)$, put $F = M_\alpha^{-1}$. Then, simultaneously for all labels,*

$$\iota_j\big(F^{\otimes m_j} Z_{j,\zeta_j}\big) \in {}^{0,\circ}U^{\mathrm{raw}}_{j,p} \qquad (1 \le j \le h). \tag{210}$$

These images use the fixed adjacent-layer branch (207)*, the same local Ind1 representative at every occurrence above p, the identity elsewhere, and identity Ind2. The assertion is membership for this branch, not compatibility of all branches.*

Proof. Lemma 21.4 puts Z_{j,ζ_j} in the local ideal of $\mathcal{P}$. The concrete base construction (207), including its fractional-ideal realization, therefore places $\iota_j Z_{j,\zeta_j}$ in the core realization of that same pilot. The permitted representative described above acts on this local packet by $\Phi_\alpha^{\otimes m_j}$. Proposition 21.3 gives

$$\Phi_\alpha^{\otimes m_j}(\iota_j Z_{j,\zeta_j}) = \iota_j F^{\otimes m_j} Z_{j,\zeta_j}.$$

This is thus an element of the image of an actual local ideal of the pilot under an allowed representative.

The raw set on p. 174 is the union of such possible images. The basic Kummer image is the branch specified in [62, Proposition 3.5(ii)(a)(1), pp. 104–105]; here it has not been precomposed with a positive log-iterate. Including the further vertical branches in that union does not remove the chosen image. This proves (210). In particular, we have not treated Ind3 as a group with an identity element or used a claimed commutativity of every vertical Kummer map. □

Remark 21.6 (Transport of an ideal and its test module). The source gives the splitting monoid both as a subset of the packet and as a multiplicative action [62, Theorem 3.11(i)(b), pp. 153–154]. If its local ideal is presented as $J = m_z(R)$, with $1 \in R$, then any additive carrier isomorphism Φ satisfies

$$\Phi J = (\Phi m_z \Phi^{-1})(\Phi R), \quad (\Phi m_z \Phi^{-1})(\Phi 1) = \Phi z. \tag{211}$$

Both identities follow by applying the maps to an element of R. They require no multiplicativity of Φ. Using the conjugated operator on a fixed native 1 or a fixed native integer module would be a different operation. The proof of Theorem 21.5 uses ΦJ, with the test module and vector transported as in (211). Its embedded global field is also transported; it is not required to be stabilized in its old native coordinates.

For general initial data with more than one selected place above p, the ambient carrier is $\bigotimes_{\text{slots}}(\bigoplus_{v|p} E_v)$. The point $1^{\otimes j} \otimes x_j$ in a specified all-v block is then the projection of the source's actual single-slot point. The corresponding statement is membership in the projection of the full possible-image set, with the same global pilot image as preimage. Zero extension of that block does not follow. The condition $F_{\text{mod}} = \mathbb{Q}$ above is what permits the literal membership (210) in the whole local packet.

21.5. **An attained local lower inclusion and its boundary.** Let

$$\Gamma_E^{\text{can}} = \{M_\alpha^{-1} : \alpha \in \text{Aut}_{\text{cont}}(G_E)\}.$$

These are the canonical core operators, expressed in the fixed standard-log chart. Retain $k_j = \lfloor 2j^2/\ell \rfloor + 1$, $\kappa = 1 - 1/e$, and the normalized tensor integer ring B_{m_j} of Section 20.

Corollary 21.7 (An attained lower ideal in the possible images)**.** *For every tuple* $(\zeta_1, \ldots, \zeta_h) \in \mu_{2\ell}^h$*, there is one* $F \in \Gamma_E^{\text{can}}$ *such that*

$$v_E(F(1)) = -\kappa, \quad v_E(F(x_j)) = k_j - \kappa \qquad (1 \le j \le h). \tag{212}$$

For every label and fixed torsion representative,

$$\overline{\text{span}_{B_{m_j}}\{F^{\otimes m_j} Z_{j,\zeta_j} : F \in \Gamma_E^{\text{can}}\}} = P_j, \quad P_j = \beta^{ek_j-(e-1)m_j} B_{m_j}. \tag{213}$$

Consequently, in this fixed local chart,

$$P_j \subseteq \overline{\text{span}_{B_{m_j}}\left(\iota_j^{-1}({}^{0,\circ}U_{j,p}^{\text{raw}})\right)}. \tag{214}$$

Proof. The proof of Theorem 20.3 first constructs a canonical matrix M from a genuine full-Galois lift, by tame inertia and the explicit parabolic operations. Before making the final Tate-coefficient comparison, this M already sends 1 and every x_j/p^{k_j} to $I \setminus \beta I$. If its lift is α_0, Proposition 21.3 shows that α_0^{-1} has canonical core action M. This proves (212) without a coefficient rescaling.

Every canonical core operator preserves I, hence each $p^k I$. Since $1 \in I$ and $x_j \in p^{k_j} I$, each field component of its tensor image has valuation at least $k_j - m_j\kappa$. The component embeddings preserve the native valuation, so every such image belongs to P_j. The common operator just constructed gives this exact valuation in every component. Its tensor is therefore a generator of P_j times a unit of B_{m_j}. This gives the reverse inclusion before closure; the fractional ideal is closed. Finally Theorem 21.5 places all these orbit points in the inverse image of the raw union, proving (214). ☐

The quantifier order remains $\forall(\zeta_j)\, \exists F\, \forall j$. Nothing requires one operator to work for all torsion tuples. Equation (214) does not identify the whole possible-image hull with P_j. Other allowed representatives and vertical branches may enlarge it.

The difference between the pure log-field point and a one-plus-unit Kummer input can also be detected exactly.

Lemma 21.8 (Trace distinguishes two points with the same ideal behavior)**.** *For $s = j^2$, $\zeta \in \mu_{2\ell}$, put*

$$a = r_0^s, \quad k = \lfloor 2s/\ell + \kappa \rfloor, \quad t = p^{-1}\log(1+p\zeta a), \quad \varepsilon = \zeta^\ell \in \{1,-1\}.$$

Then

$$\begin{aligned} &\mathrm{Tr}_{E/\mathbb{Q}_p}(\zeta a/p^k) = 0, \\ &\mathrm{Tr}_{E/\mathbb{Q}_p}(t/p^k) = \frac{30}{p^{k+1}} \log(1+\varepsilon p^\ell b_0^s) \neq 0. \end{aligned} \tag{215}$$

The valuation of the second line is $\ell - 1 + 2s - k \geq \ell$.

Proof. The tame character of r_0^s has order ℓ, because $\ell \nmid j^2$. Its ℓ conjugates sum to zero, each occurring fifteen times in E/K_0; ζ is fixed by this inertia. This proves the first trace equality. Multiplication of the same conjugates gives

$$N_{E/K_0}(1+p\zeta r_0^s) = (1+\zeta^\ell p^\ell b_0^s)^{15}.$$

The sign is positive because ℓ is odd. The right side lies in $\mathbb{Q}_p$, so its norm from K_0 squares it. Taking logarithms on principal units and dividing by p^{k+1} proves the second line of (215). Since $p \nmid 30$, its first nonzero logarithmic term gives valuation $\ell + 2s - k - 1$. Finally $k = \lfloor 2s/\ell \rfloor + 1 \leq s$ for $s \geq 1$ and $\ell \geq 7$. The lower bound follows. □

Allowed integral-coefficient Galois/Kummer maps preserve the trace-zero kernel by Proposition 7.5. Canonical core maps differ from such maps by a unit normalization, so they preserve it as well. The two points in (215) cannot literally lie in the same such orbit. This is consistent with the per-operator principal-ideal equality of Proposition 20.5. The membership theorem used the pure points from the actual log-field pullback, not a substitution of the one-plus-log points for them.

The remaining boundary is global. The initial curve, prime ℓ, places and q were fixed at the start. Before Ind1, the copy-identity preserves the root equation and its marked value-group generator. After Ind1, however, one cannot conclude that F fixes the native scalar q, the native ring unit or the old global field embedding. The transported multiplicative data retain their abstract value-group generator; its valuation in the old native chart is a different datum.

The q-pilot in [62, Corollary 3.12, p. 174] is explicitly left outside Ind1, Ind2 and Ind3. Keeping that original object in the notation is not a proof of the full horizontal compatibility assertion. Theorem 21.5 concerns a fixed base branch in one column, not a transport of that possible-image set through a horizontal theta-link.

One still has to match the complete globally synchronized pilot family, including the other places and all required vertical branches, with the precise holomorphic reference, global weights and region used in the subsequent upper bound. The local inclusion does not identify the one-prime bundle of Theorem 22.1 with that full pilot. Nor does it establish the restriction to prime-strip-interpretable regions made later in the proof on p. 175 of [62]. The reconstruction comparisons, actual full-Galois action and source membership in this section are mathematical arguments using the specified external reconstruction results; they are not new Lean formalizations. No global *abc* conclusion is inferred.

22. Arithmetic bundles for the specified native hulls

The local equalities already proved in Section 14 admit an explicit realization by arithmetic vector bundles on the actual level field. We construct these bundles, compute their normalized degrees, and descend a sufficiently powered weighted determinant to the field of moduli $\mathbb{Q}$. This realizes ordinary arithmetic objects with the required numerical normalization. It does not establish their membership in the full possible-output family of IUT III, Corollary 3.12.

The relevant source conventions are the relative weighted determinant and sufficiently divisible positive tensor powers in [62, Remark 3.9.5(vii)(Ob3)], the label average in [62, Proposition 3.9(i)–(iii)], and the normalized local weights in [65, Proposition 1.4(i), Remark 1.7.1]. The additional Ind3 comparisons and the particular global prime-strip identification in [62, Theorem 3.11(ii), Corollary 3.12] are separate requirements; none is assumed in the arithmetic arguments below.

22.1. **The fixed native input and its reference.** Fix a member of the family in Theorem 16.2, with primes $\ell \equiv 43 \pmod{60}$ and $p \equiv -1 \pmod{30\ell}$, and write

$$D_A: \ y^2 = x(x - A^2)(x + A^2 + 1), \qquad K = \mathbb{Q}(i, D_A[30\ell]).$$

The field K is Galois over $\mathbb{Q}$. At the specified place over p, its completion is the field of (114):

$$E = \mathbb{Q}_p(\mu_{30\ell}, \varpi), \qquad \varpi^{15\ell} = b_0, \qquad b_0 \in \mathbb{Q}_p, \qquad v_p(b_0) = 2.$$

Put

$$\begin{gathered} e = 15\ell, \quad f = 2, \quad d = ef, \quad h = (\ell - 1)/2, \quad \kappa = (e-1)/e, \\ \beta = \varpi^{(e+1)/2}/p, \qquad r_0 = \varpi^{15}, \\ m_j = j + 1, \quad N_j = d^{m_j - 1}, \quad D_j = d^{m_j}, \\ a_j = r_0^{j^2}, \qquad k_j = \left\lfloor \frac{2j^2}{\ell} \right\rfloor + 1, \\ \eta_j = e k_j - (e-1) m_j \quad (1 \le j \le h). \end{gathered} \tag{216}$$

The element β is a uniformizer by Lemma 14.1; ϖ has valuation $2/e$ and is not one. All roots are fixed before the permitted arrows are chosen.

For each label define

$$\begin{gathered} T_j = E^{\otimes_{\mathbb{Q}_p} m_j}, \qquad A_j = \mathcal{O}_E^{\otimes_{\mathbb{Z}_p} m_j}, \\ B_j = \prod_{\mathrm{Gal}(E/\mathbb{Q}_p)^{m_j - 1}} \mathcal{O}_E \subset T_j. \end{gathered}$$

The product decomposition uses the first factor as coefficient field and has N_j copies of E. In general $A_j \neq B_j$. The native lattice is $I = p^{-1} \log \mathcal{O}_E^\times = \beta^{1-e}\mathcal{O}_E$. Let Γ_E be the already constructed family of actual native integral Kummer arrows on I, using the same arrow in repeated slots. We retain precisely the marked point inputs

$$\begin{gathered} \rho = p^{-1} \log^{\mathrm{std}}_{\mathrm{BK}}, \\ u = \rho(\mathrm{Kum}(1+p)) = p^{-1}\log(1+p), \\ \tau_j = \rho(\mathrm{Kum}(1 + p a_j)) = p^{-1} \log(1 + p a_j). \end{gathered} \tag{217}$$

Their closed B_j-span after the permitted arrows is

$$P_j^\rho = \overline{\mathrm{span}_{B_j}\{F(\tau_j) \otimes (Fu)^{\otimes j} : F \in \Gamma_E\}}.$$

Corollaries 14.5 and 14.7, with the per-arrow bridge of Proposition 12.1, give

$$M_j^\rho := \mathcal{M}_{\Gamma_E}(a_j) = P_j^\rho = \beta^{\eta_j} B_j. \tag{218}$$

The first term is the pre-transport principal-ideal hull already defined in (102). Equality here identifies these specified native hulls, not their input points or the full published theta-pilot family. No additional choice of a torsion representative, vertical level, or Ind3 enlargement is included in (217)–(218).

Let μ_{B_j} be ordinary additive Haar measure on T_j with $\mu_{B_j}(B_j) = 1$, and write

$$V_j^B(H) = D_j^{-1} \log \mu_{B_j}(H). \tag{219}$$

The positive root $\mu_{B_j}^{1/D_j}$ is not another additive Haar measure. For a general field of moduli F_0, the normalized weight of an ordered word $(v_0, \ldots, v_j)$ over a rational prime is

$$\frac{\prod_{i=0}^{j} [(F_0)_{v_i} : \mathbb{Q}_p]}{[F_0 : \mathbb{Q}]^{m_j}}.$$

Indeed the sum of the numerator over all words is the m_j-th power of $\sum_{v|p}[(F_0)_v : \mathbb{Q}_p] = [F_0 : \mathbb{Q}]$. In the rational branch there is one selected-place word and this weight is 1. The remaining procession normalization is $h^{-1}\sum_{j=1}^{h}$, with no further division by m_j.

22.2. **Actual fractional ideals and normalized degree.** At a finite place w of K, put $q_w = \#k(w)$ and $|x|_w = q_w^{-\operatorname{ord}_w(x)}$. This is the local norm absolute value. At an infinite place σ, use ordinary real or complex modulus and multiplicity $n_\sigma = 1$ or 2, respectively. For a metrized line $\overline{L}$, declare its finite local lattices to be the unit balls. For a nonzero rational section $s \in L \otimes_{\mathcal{O}_K} K$, define

$$\widehat{\deg}_K(\overline{L}) = -\frac{1}{[K:\mathbb{Q}]}\left(\sum_{w\ \text{finite}} \log\|s\|_w + \sum_{\sigma|\infty} n_\sigma \log\|s\|_\sigma\right). \tag{220}$$

For a fractional ideal with the metrics below, the finite sum is finite. Replacing s by xs adds zero: the finite norm sum for $x \in K^\times$ is $-\log|N_{K/\mathbb{Q}}(x)|$, whereas the infinite sum is $\log|N_{K/\mathbb{Q}}(x)|$. This proves independence of the rational section with the indicated multiplicities.

For a fractional ideal $\mathfrak{a} \subset K$ with standard ambient infinite metric $\|x\|_\sigma = |\sigma(x)|$, the section 1 gives

$$\widehat{\deg}_K(\overline{\mathfrak{a}}) = -[K:\mathbb{Q}]^{-1}\log N(\mathfrak{a}). \tag{221}$$

In fact if $\mathfrak{a}_w = \pi_w^b\mathcal{O}_{K_w}$, then $\|1\|_w = q_w^b$. Thus a positive finite ideal exponent has negative degree in this lattice convention. Degree of a vector bundle means degree of its determinant, with the induced determinant metric.

Theorem 22.1 (Arithmetic realization of the native hulls)**.** *Set*

$$\mathfrak{a}_j = \prod_{w|p}\mathfrak{p}_w^{\eta_j}, \qquad \mathcal{E}_j = \mathfrak{a}_j^{\oplus N_j} \subset K^{N_j}, \tag{222}$$

and give $\mathcal{E}_j$ the standard ambient Hermitian metric at every infinite place. These are projective $\mathcal{O}_K$-modules of rank N_j. At each $w \mid p$ their local lattices identify with the conjugate of M_j^ρ over its first coefficient field. At all other finite places they are the standard lattices. Moreover

$$\frac{\widehat{\deg}_K(\overline{\mathcal{E}_j})}{N_j} = -\frac{\eta_j}{e}\log p = V_j^B(M_j^\rho). \tag{223}$$

Proof. A fractional ideal of the Dedekind domain $\mathcal{O}_K$ is invertible, hence projective of rank one. This applies also to negative η_j. Since $K/\mathbb{Q}$ is Galois, all completions above p have the same ramification index e, residue degree f, and local degree d. The product coordinates of T_j identify its first-factor lattice with N_j copies of $\mathcal{O}_{K_w}$. The depth η_j of (218) in every component is precisely the depth specified by (222). Uniformizer units do not change the lattice. Assigning the same exponent to every place over p also makes $\mathfrak{a}_j$ Galois-stable.

If g places lie above p, then $[K:\mathbb{Q}] = gef$ and $N(\mathfrak{p}_w) = p^f$. Since $\det\mathcal{E}_j = \mathfrak{a}_j^{N_j}$, (221) gives

$$\widehat{\deg}_K(\overline{\mathcal{E}_j}) = -\frac{gfN_j\eta_j}{gef}\log p = -\frac{N_j\eta_j}{e}\log p.$$

The rational section used here is the standard coordinate wedge, of norm 1 at each infinite place. Thus the infinite terms are exactly zero for the stated metrics, not omitted terms.

The local index calculation, valid for either sign of η_j, gives $\mu_{B_j}(M_j^\rho) = p^{-fN_j\eta_j}$. Dividing its logarithm by $D_j = efN_j$ proves (223). $\square$

Equivalently the contribution per rank is

$$\sum_{w|p}\frac{[K_w:\mathbb{Q}_p]}{[K:\mathbb{Q}]}\left(-\frac{\eta_j}{e}\log p\right) = -\frac{\eta_j}{e}\log p.$$

Thus passing from one selected completion to all conjugate places introduces no further degree or tensor-length factor.

22.3. **Weighted determinants, common twists, and descent.**

Proposition 22.2. *Let C be any positive integer divisible by every hN_j. The metrized line*

$$\overline{\mathcal{L}} = \bigotimes_{j=1}^{h} (\det \overline{\mathcal{E}_j})^{\otimes C/(hN_j)} \tag{224}$$

is defined using positive integral tensor powers and satisfies

$$\frac{\widehat{\deg}_K(\overline{\mathcal{L}})}{C} = \frac{1}{h} \sum_{j=1}^{h} V_j^B(M_j^\rho). \tag{225}$$

Replacing every $\overline{\mathcal{E}_j}$ by $\overline{\mathcal{E}_j} \otimes \overline{Q}$, for one metrized line $\overline{Q}$, tensors (224) by $\overline{Q}^{\otimes C}$.

Proof. The divisibility hypothesis makes all powers integral and positive. Additivity of degree and (223) prove (225). The canonical determinant isometry

$$\det(\overline{\mathcal{E}_j} \otimes \overline{Q}) = \det \overline{\mathcal{E}_j} \otimes \overline{Q}^{\otimes N_j}$$

shows that the total exponent of $\overline{Q}$ is $\sum_j N_j C/(hN_j) = C$. □

The reference determinant is present in this computation: $\det \mathcal{O}_K^{N_j}$ has the same standard metric and degree zero. Each determinant in (224) can equivalently be replaced by $\det \mathcal{E}_j \otimes (\det \mathcal{O}_K^{N_j})^{-1}$. If covolumes are instead computed after restriction of scalars to $\mathbb{Q}$, the field-discriminant and reference covolume factors appear in both equal-rank terms and cancel in this relative determinant.

Corollary 22.3 (Descent to the field of moduli)**.** *If C is divisible by every ehN_j, put*

$$t = \frac{C}{eh} \sum_{j=1}^{h} \eta_j \in \mathbb{Z}. \tag{226}$$

Then $\overline{\mathcal{L}}$ is the isometric pullback from $\mathbb{Q}$ of the actual metrized line $p^t\mathbb{Z} \subset \mathbb{Q}$ with its ordinary ambient absolute-value metric. In rational tensor powers, $\overline{\mathfrak{a}_j}$ is the pullback of the arithmetic line with finite exponent η_j/e at p and the standard infinite metric.

Proof. Unique ideal factorization and the common ramification index give

$$p\mathcal{O}_K = \prod_{w|p} \mathfrak{p}_w^e, \qquad \mathfrak{a}_j^e = p^{\eta_j}\mathcal{O}_K. \tag{227}$$

Multiplication identifies the tensor metric on the left of the second equality with the ambient metric on the right, since absolute values multiply at every embedding. This proves the rational-tensor-power statement.

In (224), the exponent at each $w \mid p$ is

$$\sum_{j=1}^{h} \eta_j N_j \frac{C}{hN_j} = \frac{C}{h} \sum_{j=1}^{h} \eta_j = et.$$

There is no finite support elsewhere. Hence the actual lattice is $p^t\mathcal{O}_K$, and the multiplication identification preserves its standard infinite metric. This is the stated pullback. In particular its normalized degree is $-t \log p$. □

An explicit simultaneous choice is $C = ehd^h$, since $N_j = d^j$ and $C/(hN_j) = e\,d^{h-j} \in \mathbb{Z}_{>0}$. Thus the descent concerns an actual powered determinant object, not merely an equality of degrees. Galois stability by itself would not have supplied unpowered integral descent; (227) is the additional step.

22.4. **The exact procession average.**

Proposition 22.4. *Let*

$$R_\ell = \sum_{j=1}^{h} (2j^2 \bmod \ell),$$

where residues are taken in $\{1, \ldots, \ell-1\}$*. Then*

$$\frac{\widehat{\deg}_K(\overline{\mathcal{L}})}{C \log p} = \frac{\ell+1}{12} - \frac{\ell+5}{60\ell} + \frac{2R_\ell}{\ell(\ell-1)}. \tag{228}$$

Each summand $V_j^B(M_j^\rho)$ *in* (225) *is strictly positive.*

Proof. Since ℓ is odd prime and $1 \le j \le h < \ell$, the indicated residue is nonzero. The fractional part of $2j^2/\ell$ is therefore at least $1/\ell > 1/e$, which also verifies $k_j = \lfloor 2j^2/\ell + \kappa \rfloor$. Elementary summation gives

$$\sum_{j=1}^{h} m_j = \frac{h(\ell+5)}{4}, \qquad \sum_{j=1}^{h} \left\lfloor \frac{2j^2}{\ell} \right\rfloor = \frac{\ell^2-1}{12} - \frac{R_\ell}{\ell}.$$

For the second identity, sum $2j^2/\ell$ and subtract the fractional parts. Substitution into the average

$$h^{-1} \sum_j (\kappa m_j - k_j)$$

gives (228). Moreover $2j^2/\ell < j$, so $k_j \le j$ and

$$\kappa m_j - k_j \ge 1 - \frac{j+1}{e} > 0.$$

□

As a numerical check at $\ell = 43$, direct integer summation gives $R_{43} = 473$, $\sum_j k_j = 164$, and $\sum_j \eta_j = -56508$. The right side of (225) evaluates to $18836 \log p/4515$. No distributional assertion about quadratic residues is used here. In particular this positive degree is a property of (222), not a sign assertion about the complete theta-pilot quantity in the source.

22.5. **The tensor-order correction.**

Proposition 22.5. *Let* μ_{A_j} *be additive Haar measure with* $\mu_{A_j}(A_j) = 1$*. For every measurable region* $H \subset T_j$ *of finite positive volume,*

$$D_j^{-1} \log \mu_{A_j}(H) = D_j^{-1} \log \mu_{B_j}(H) + \frac{(m_j-1)\kappa}{2} \log p. \tag{229}$$

Proof. Tameness gives the discriminant exponent $f(e-1)$ for $\mathcal{O}_E/\mathbb{Z}_p$. On a tensor integral basis of A_j, the Gram matrix of the absolute algebra trace is the Kronecker product of the factor trace Gram matrices. Its determinant therefore has valuation

$$v_p \operatorname{disc}(A_j) = m_j d^{m_j-1} f(e-1).$$

The product order B_j has $N_j = d^{m_j-1}$ copies of $\mathcal{O}_E$, so

$$v_p \operatorname{disc}(B_j) = d^{m_j-1} f(e-1).$$

Under a finite-index change of lattice, the trace determinant changes by the square of the index. Consequently

$$\log[B_j : A_j] = \frac{m_j-1}{2} d^{m_j-1} f(e-1) \log p.$$

Finally $\mu_{A_j} = [B_j : A_j]\mu_{B_j}$. Dividing by $D_j = efd^{m_j-1}$ proves the formula. □

The factor-weight prescription in [56, (9.10.3.1), (9.10.5.1)], for this rational branch, uses exponent $1/d$ on each factor. On tensor lattices this agrees with $\mu_{A_j}^{1/D_j}$, since tensoring a lattice change in one factor raises its determinant to the power D_j/d. It therefore has the correction (229) relative to the maximal-order reference. Agreeing on the factor weights alone does not remove the A_j/B_j index.

22.6. All-slot scaling and transport of the infinite metrics.

Proposition 22.6. *Write $\mathcal{E}_j^\rho = \mathcal{E}_j$ for the realization (222). Replace all m_j entries of the tuple $(\tau_j, u, \ldots, u)$ specified in (217) by their standard Bloch–Kato logarithms, retaining the same cohomology classes, arrows, and ambient reference B_j. Then*

$$\begin{aligned} M_j^{\mathrm{std}} &= p^{m_j} M_j^\rho = \beta^{\eta_j + em_j} B_j = \beta^{ek_j + m_j} B_j, \\ V_j^B(M_j^{\mathrm{std}}) &= V_j^B(M_j^\rho) - m_j \log p. \end{aligned} \tag{230}$$

For the global realization, making this change only at the specified rational prime p, while keeping the ambient infinite metrics fixed, gives

$$\begin{aligned} \mathcal{E}_j^{\mathrm{std}} &= p^{m_j} \mathcal{E}_j^\rho, \\ \frac{\widehat{\deg}_K(\overline{\mathcal{E}_j^{\mathrm{std}}}) - \widehat{\deg}_K(\overline{\mathcal{E}_j^\rho})}{N_j} &= -m_j \log p. \end{aligned} \tag{231}$$

The label-averaged change is $-(\ell+5)\log p/4$. If instead the infinite metrics are transported along multiplication by p^{m_j}, the map is an isometry of metrized bundles and their degrees are equal.

Proof. The two coordinates differ by the rational scalar p at every entry, including all backgrounds. The permitted arrows are $\mathbb{Q}_p$-linear, and the tensor map is multilinear. Thus every image tensor is multiplied by p^{m_j}. Closure and B_j-span commute with this invertible scalar. Its determinant on the D_j-dimensional space has absolute value $p^{-m_j D_j}$, which proves (230).

At finite places away from p the scalar is a unit, and at every place above p its uniformizer exponent is em_j. The degree calculation in Theorem 22.1 proves (231). The average of $m_j = j+1$ is $(\ell+5)/4$.

For the transported metric, the new infinite norm on a vector y is $p^{-m_j}\|y\|_\sigma$. The determinant norm is therefore multiplied by $p^{-m_j N_j}$. Since $\sum_{\sigma|\infty} n_\sigma = [K:\mathbb{Q}]$, its normalized infinite degree contribution increases by $m_j N_j \log p$, cancelling the finite change. This also follows from the resulting isometry. □

Thus scaling a finite coordinate with fixed references differs from transporting a whole metrized global object. Neither operation allows the active root alone to change scale while the other source entries are silently left unchanged.

22.7. An almost-everywhere reference test.

Proposition 22.7. *Fix one global member and one label j. Outside a finite set of rational primes p', take the declared background class $\mathrm{Kum}(1+p')$ at the selected completion, its m_j-fold tuple, and the native integral Kummer arrows. Relative to the maximal order $B_{j,p'}$, its native hull is $B_{j,p'}$, whereas its standard-coordinate hull is $(p')^{m_j} B_{j,p'}$. With the reference unchanged, the sum of standard-coordinate normalized local log-volumes is $-\infty$, and the volume product is zero. This is not a finite-support fractional-ideal modification of a fixed arithmetic bundle.*

Proof. Exclude 2, the primes ramified in K, the bad primes of the curve, and the other finitely many distinguished primes. The remaining completion $E_{p'}/\mathbb{Q}_{p'}$ is unramified. The logarithm and exponential on principal units give

$$\log \mathcal{O}_{E_{p'}}^\times = p' \mathcal{O}_{E_{p'}}, \qquad I_{p'} = \mathcal{O}_{E_{p'}}, \qquad u_{p'} = (p')^{-1} \log(1+p') \in \mathbb{Z}_{p'}^\times.$$

The residue-root-of-unity part has logarithm zero. The tensor product of the unramified integral orders is finite etale over $\mathbb{Z}_{p'}$ and is already its maximal product order.

Every integral automorphism of $I_{p'}$ preserves nonzero classes modulo $p'I_{p'}$. Since the field is unramified, this is precisely the unit condition. Each permitted image of $u_{p'}$ is therefore a unit, and its repeated tensor has unit value in every field component. The identity arrow is present. The $B_{j,p'}$-span of the native images is consequently exactly $B_{j,p'}$. The all-slot argument of Proposition 22.6 gives the standard hull $(p')^{m_j}B_{j,p'}$, with normalized log-volume $-m_j \log p'$, independently of the local degree.

Infinitely many rational primes remain, and $\sum_{p'} m_j \log p' = +\infty$, since every summand is at least $m_j \log 2 > 0$. This proves the assertions about the sum and product. A lattice in a fixed finite-dimensional K-space agrees with a fixed integral reference at all but finitely many places. Nonzero depth m_j at infinitely many distinct rational primes violates this condition. A finite contribution from fixed metrics at infinity cannot cancel the divergence. □

Equivalently, the local scalars $(p')_{p'}$ do not form an idele: their inverses are nonintegral at infinitely many primes. One can replace the reference by $(p')^{m_j}B_{j,p'}$ at the same time, obtaining local volume 1 again, but that specifies a different restricted product and requires an explicit comparison with the old reference.

Remark 22.8 (The remaining identification)*.* The almost-everywhere test concerns the declared native branch, not the complete Ind3 family. Likewise, the lattices assigned away from p in (222) and their standard infinite metrics are part of our construction. They have not been identified with the other components of the full theta-pilot output. Theorems about these actual arithmetic bundles therefore supply neither the particular membership in IUT III, Corollary 3.12, nor the required comparison across Ind3. Establishing that membership must retain the source classes, allowed arrows, reference structures, weights, and the link to the same global q-pilot. Equality of degrees of separately constructed objects does not provide it.

23. Three-prime support and mixed-full Campana points

This section records two complementary descriptions of the first locus not covered by the two-prime theorem. The first is an exact support decomposition. The second keeps multiplicities rather than support size and isolates the log-general-type Campana locus studied most recently by Browning–Verzobio [8].

23.1. **The support-three gate.** Let $a + b = c$ be a positive primitive triple with $\omega(abc) \leq 3$. Pairwise coprimality gives

$$\omega(abc) = \omega(a) + \omega(b) + \omega(c).$$

If $a, b > 1$, every term on the right is positive. Hence equality holds in the support bound, and there are distinct primes p, q, r and positive integers x, y, z such that

$$a = p^x, \qquad b = q^y, \qquad c = r^z. \tag{232}$$

If one input is one, the support split between the two consecutive nonunits is exactly $(1, 1)$, $(1, 2)$, or $(2, 1)$. Thus, after exchanging the inputs, the remaining equations are

$$1 + p^x = q^y, \qquad 1 + p^x = q^y r^z, \qquad 1 + p^x q^y = r^z, \tag{233}$$

with distinct displayed prime bases. This proves an exhaustion rather than a density reduction: the two semiprime branches in (233) remain moving Diophantine problems.

The all-nonunit branch has the following useful rigidity.

Theorem 23.1 (Common-exponent rigidity)**.** *Suppose that distinct primes satisfy $p^x + q^y = r^z$ and $d = \gcd(x, y, z) > 1$. Then, up to exchanging the inputs,*

$$2^4 + 3^2 = 5^2.$$

Proof. Set $X = p^{x/d}$, $Y = q^{y/d}$, and $Z = r^{z/d}$. Then $X^d + Y^d = Z^d$. Fermat's Last Theorem [84, 81] forces $d = 2$. In the resulting primitive Pythagorean triple the even leg is a power of two, say $X = 2^A$. Euclid's parametrization gives

$$X = 2mn, \qquad Y = m^2 - n^2, \qquad Z = m^2 + n^2,$$

where m, n are coprime and of opposite parity. Since mn is a power of two, one has $n = 1$ and $m = 2^k$. Now

$$Y = (2^k - 1)(2^k + 1)$$

is a prime power, while the two odd factors are coprime. The smaller factor must be one, so $k = 1$ and $(X, Y, Z) = (4, 3, 5)$. Squaring proves the assertion. □

For $x, y, z \geq 2$, sorting the exponents shows that $1/x + 1/y + 1/z \geq 1$ occurs only for

$$(2, 2, n),\ (2, 3, 3),\ (2, 3, 4),\ (2, 3, 5),\ (3, 3, 3),\ (2, 4, 4),\ (2, 3, 6).$$

Elementary difference-of-powers arguments exclude the full signatures $(2, 3, 3)$ and $(2, 3, 6)$ in every output orientation. Theorem 23.1 excludes $(3, 3, 3)$ and $(2, 4, 4)$. For $(2, 2, n)$, the odd-exponent orientation reduces to $S^2 + 4 = R^n$; the theorem of Arif–Abu Muriefah [3] then yields the second of exactly two solutions

$$2^4 + 3^2 = 5^2, \qquad 2^2 + 11^2 = 5^3.$$

The other orientations follow by factoring a difference of squares; even n follows from Theorem 23.1.

Consequently, if $c > \operatorname{rad}(abc)$ in an all-nonunit support-three triple, then the exponent gcd is one and the all-nonlinear remainder consists only of $(2, 2, 3)$, $(2, 3, 4)$, $(2, 3, 5)$, or signatures with reciprocal sum below one. The first case is the genuine triple $2^2 + 11^2 = 5^3$, for which $125 > 110$. Linear exponents are essential:

$$7 + 181^2 = 2^{15}, \qquad \frac{2^{15}}{2 \cdot 7 \cdot 181} > 12.$$

This example refutes a putative extension $c \leq \operatorname{rad}(abc)$ to support three; it is not an abc counterexample. The unresolved branches are the two semiprime unit equations, linear exponents, the prime-base signatures $(2, 3, 4)$ and $(2, 3, 5)$, and the union over varying hyperbolic signatures. Darmon–Granville [32] gives finiteness for each fixed hyperbolic signature, not uniformly over that union.

23.2. **Different fullness exponents.** An integer $n > 0$ is m-full when every prime divisor occurs with valuation at least m. Let $P = (a, b, c)$ be a primitive abc point and suppose that a, b, c are respectively p-, q-, and r-full, where $p, q, r > 0$. Put

$$\sigma_{p,q,r} = \frac{1}{p} + \frac{1}{q} + \frac{1}{r}.$$

Theorem 23.2 (Mixed-full radical compression)**.** *One has*

$$\operatorname{rad}(abc)^{pqr} \leq c^{qr+pr+pq}, \tag{234}$$

$$\log \operatorname{rad}(abc) \leq \sigma_{p,q,r} \log c. \tag{235}$$

Moreover,

$$\sigma_{p,q,r} < 1 \quad \iff \quad qr + pr + pq < pqr.$$

Proof. Fullness gives $\operatorname{rad}(a)^p \mid a$, and similarly for b, c. Raise these three divisibilities to qr, pr, pq, multiply them, and use radical multiplicativity across a primitive triple. Since $a, b < c$, this gives

$$\operatorname{rad}(abc)^{pqr} \leq a^{qr} b^{pr} c^{pq} \leq c^{qr+pr+pq},$$

which proves (234). Taking the three logarithmic inequalities separately gives

$$\log \operatorname{rad}(abc) \leq \frac{\log a}{p} + \frac{\log b}{q} + \frac{\log c}{r} \leq \sigma_{p,q,r} \log c,$$

proving (235). Clearing the positive denominator pqr proves the final equivalence. □

Corollary 23.3 (A strict Campana disproof gate)**.** *Fix $p, q, r > 0$ with $\sigma_{p,q,r} < 1$. The abc conjecture implies a uniform height bound for every primitive abc point with the three prescribed fullness conditions. Conversely, an unbounded family of such points implies the negation of the standard abc conjecture.*

Proof. Choose $\varepsilon > 0$ so that $(1+\varepsilon)\sigma_{p,q,r} < 1$. If abc holds with constant C_ε, Theorem 23.2 gives

$$\big(1-(1+\varepsilon)\sigma_{p,q,r}\big)\log c \le C_\varepsilon.$$

The coefficient is positive, so c is bounded. This proves the first claim. The same displayed inequality contradicts unbounded height and proves the second. □

Browning–Verzobio [8, Theorems 1.1 and 1.2] study exactly these points. If $p = r+u$, $q = r+v$, with fixed $u \ge v \ge 0$, they prove, for all sufficiently large r,

$$N(B) \ll B^{1/p+1/q-\eta_{u,v}(r)+\varepsilon}, \qquad \eta_{u,v}(r) = r^{-2} + O_{u,v}(r^{-5/2}) > 0.$$

They also obtain a power saving for every $r \ge 2$ in the special family $(p,q,r) = (r+2, r+1, r)$. This is a genuine improvement over the trivial $B^{1/p+1/q}$ count and applies precisely to the low-radical locus in Corollary 23.3. Its exponent remains positive, so it proves neither the predicted boundedness nor the existence of an unbounded family. The determinant-method and function-field inputs of that paper are cited mathematical results and are not declared as Lean axioms.

24. A coefficient-below-six obstruction and finite selected-place hulls

We next close two proposed shortcuts. On the Frey route, optimization over an entire rational isogeny class cannot lower the coefficient below six. On the local IUT route, projected membership does suffice after the actual finite product-order hull, but that finite statement does not determine a global Arakelov degree.

24.1. An exact conductor and an isogeny-class obstruction.

For $n \ge 1$, put $c = 1792n+2$ and

$$E_c : y^2 = x(x-1)(x+c-1) = x^3 + (c-2)x^2 + (1-c)x.$$

The preceding sections determine the four vertices of its rational isogeny class and prove that the minimum absolute logarithmic Weil height in that class is

$$3\log(c^2 - 16c + 16). \tag{236}$$

Proposition 24.1 (Exact local and global conductor)**.** *The displayed equation is minimal at every prime. At 2 it has Kodaira symbol III and conductor exponent five. Every odd prime $s \mid c(c-1)$ is multiplicative of type $I_{2v_s(c(c-1))}$ and has conductor exponent one. Hence every elliptic curve $F/\mathbb{Q}$ rationally isogenous to E_c has*

$$N(F) = 16\,\mathrm{rad}(c(c-1)). \tag{237}$$

If $D = (c-1)(c/2)^4$ is the reduced denominator of the height-minimizing j-invariant, then

$$N(F) = 32\,\mathrm{rad}(D). \tag{238}$$

Proof. Write $A = c-2$ and $B = 1-c$. Then

$$v_2(A) \ge 8, \qquad v_2(B) = 0, \qquad B \equiv -1 \pmod 4.$$

The corresponding row of the complete two-torsion model table in Mulholland [50, Theorem 2.1 and Table 2.1], which specializes Papadopoulos's classification, gives type III, conductor exponent five, and minimality at two. The invariants are

$$c_4 = 16(c^2 - c + 1), \qquad \Delta = 16(c-1)^2c^2.$$

If an odd prime s divides c or $c-1$, then $c^2-c+1 \equiv 1 \pmod s$. Thus c_4 is a unit and the model is minimal multiplicative, with the asserted type and conductor exponent. All other odd primes are good. This proves (237) for E_c. Rational isogenies identify the rational Tate modules and hence preserve every local Artin conductor, so the same formula holds throughout the class.

Writing $u = c/2$, one has $D = (2u-1)u^4$ and $\gcd(u, 2u-1) = 1$. Therefore

$$\mathrm{rad}(c(c-1)) = \mathrm{rad}(2u(2u-1)) = 2\,\mathrm{rad}(D),$$

which gives (238). □

The congruence modulus permits an infinite power subfamily. Put

$$c_k = 2 \cdot 897^k, \qquad n_k = \frac{897^k - 1}{896} = 1 + 897 + \cdots + 897^{k-1}.$$

Then $c_k = 1792n_k + 2$ and, since $897 = 3 \cdot 13 \cdot 23$,

$$N(E_{c_k}) = 28704\,\mathrm{rad}(2 \cdot 897^k - 1) < 28704c_k. \tag{239}$$

Theorem 24.2 (Uniform failure of every coefficient below six)**.** *For every* $0 \le \theta < 6$ *and* $C \in \mathbb{R}$, *all sufficiently large* k *satisfy*

$$h(j(F)) > \theta \log N(F) + C$$

simultaneously for every $F/\mathbb{Q}$ *rationally isogenous to* E_{c_k}.

Proof. For $c_k > 32$, (236) gives

$$h(j(F)) > 6 \log c_k - 3 \log 2.$$

Equation (239) and $\theta \ge 0$ give

$$\theta \log N(F) < \theta \log c_k + \theta \log 28704.$$

Their difference is greater than

$$(6 - \theta) \log c_k - 3 \log 2 - \theta \log 28704,$$

which tends to infinity because $c_k = 2 \cdot 897^k$ and $\theta < 6$. □

This rules out every subcritical coefficient obtained merely by changing the representative inside the rational isogeny class. It leaves the coefficient-six and all-epsilon estimate $(6 + \varepsilon) \log N + C_\varepsilon$ untouched; in particular, it does not prove that six is an attainable or optimal coefficient. On this family the remaining input is a lower bound for $\mathrm{rad}(2 \cdot 897^k - 1)$; the elementary upper bound in (239) cannot supply it.

24.2. **Finite products of selected-place components.** Fix a rational prime and one label. After decomposing the finite-etale tensor packet into field factors, write

$$T = \prod_{w \in W} T_w, \qquad B = \prod_{w \in W} B_w,$$

where W is the finite set of ordered selected-place words and B_w is the product of the integer rings in the field factors of T_w.

Proposition 24.3 (Exact product-order span)**.** *For every subset* $S \subset T$,

$$\mathrm{span}_B(S) = \prod_{w \in W} \mathrm{span}_{B_w}(\pi_w S). \tag{240}$$

The same identity holds after topological closure.

Proof. Projection of a B-linear combination gives the forward inclusion. For the reverse inclusion, let $e_w \in B$ be the central idempotent supported on the w-component. If $y_w = \sum_i b_i \pi_w(s_i)$, then $\sum_i (e_w b_i) s_i$ belongs to $\mathrm{span}_B(S)$, has component y_w at w, and has every other component zero. Sum this construction over the finite set W. Closure commutes with a finite product. □

Suppose now that S satisfies the first-branch hypotheses of IUT III, Remark 3.9.5(i): it is relatively compact and contains a relatively compact finite-log-volume subset. The source holomorphic hull is the smallest λB containing S, with every field component of λ nonzero. Its definition is componentwise, and therefore

$$\mathrm{Hull}_B(S) = \prod_{w \in W} \mathrm{Hull}_{B_w}(\pi_w S). \tag{241}$$

The compactness hypothesis is essential: in the other branch of the source definition, the hull is the entire ambient space.

If full-rank product fractional ideals P_w satisfy

$$P_w \subset \overline{\mathrm{span}_{B_w}(\pi_w S)} \qquad (w \in W),$$

then (240) gives

$$\prod_w P_w \subset \mathrm{Hull}_B(S).$$

For an ordered word $w = (v_0, \dots, v_j)$, set

$$h_v = [(F_{\mathrm{mod}})_v : \mathbb{Q}_p], \qquad \omega_w = \frac{\prod_i h_{v_i}}{[F_{\mathrm{mod}} : \mathbb{Q}]^{j+1}}.$$

IUT IV, Remark 1.7.1 gives precisely these normalized weights. They sum to one because

$$\sum_w \prod_i h_{v_i} = \left(\sum_{v|p} h_v\right)^{j+1} = [F_{\mathrm{mod}} : \mathbb{Q}]^{j+1}.$$

Monotonicity of Haar measure now yields the finite weighted lower bound

$$\sum_w \omega_w V_w(\pi_w \,\mathrm{Hull}(S)) \geq \sum_w \omega_w V_w(P_w), \tag{242}$$

where $V_w = D_w^{-1} \log \mu_w$ and $D_w = \dim_{\mathbb{Q}_p} T_w$. No single raw member must realize all word projections: the central idempotents combine witnesses only after taking the B-span.

Equation (242) is a finite-place result. It does not specify the archimedean metric or prove that the finite lattices localize one global projective module. Indeed, keep every finite lattice of $L = \mathbb{Z} \subset \mathbb{Q}$ fixed and set $\|x\|_{\infty,t} = e^{-t}|x|$. All finite projections, hulls, and Haar volumes are independent of t, while $\widehat{\deg}(\overline{L}_t) = t$. Thus finite projected membership alone cannot determine a global Arakelov inequality. The remaining source problem is to construct the same globally metrized marked family and compare it, through the actual Ind3 and horizontal links, with the fixed q-pilot.

25. Subcritical uniformity and a Pell counterexample gate

This section runs the positive and counterexample programs against the same unchanged target. The positive result identifies the exact pointwise uniformity statement equivalent to *abc*. The counterexample analysis shows why fixed residual coefficients and polynomial identities cannot supply an unbounded strict mixed-full family, while retaining one explicit conditional Pell route. No condition introduced below is asserted to hold without proof.

25.1. **The exact positive gate.** For an *abc* point P, retain

$$H(P) = \log c, \qquad R(P) = \log \mathrm{rad}(abc).$$

Consider the pointwise uniformity statement

$$\forall \mu \in [0,1) \quad \exists B_\mu \in \mathbb{R} \quad \forall P, \qquad R(P) \leq \mu H(P) \Longrightarrow H(P) \leq B_\mu. \tag{243}$$

Theorem 25.1 (Subcritical-locus equivalence)**.** *The standard logarithmic abc conjecture is equivalent to* (243)*.*

Proof. Assume *abc*, fix $0 \leq \mu < 1$, and set $\eta = (1-\mu)/2$. Then

$$d = 1 - (1+\eta)\mu = \frac{(1-\mu)(2-\mu)}{2} > 0.$$

If C_η is the *abc* constant and $R(P) \leq \mu H(P)$, then

$$dH(P) \leq C_\eta,$$

so $B_\mu = C_\eta / d$ works.

Conversely, assume (243). Given $\varepsilon > 0$, put $\mu = (1+\varepsilon)^{-1}$ and let B_μ be its bound. If $R(P) \leq \mu H(P)$, then

$$H(P) \leq B_\mu \leq (1+\varepsilon)R(P) + \max\{B_\mu, 0\}.$$

If the displayed premise fails, then $H(P) < (1+\varepsilon)R(P)$. Thus the single constant $C_\varepsilon = \max\{B_\mu, 0\}$ works for every P. $\square$

This is a proved equivalence, rather than a replacement definition. It shows that density zero, an exceptional-set estimate, or a positive power saving cannot by itself prove *abc*: every point on every fixed subcritical slope must be bounded.

25.2. **What counting would have to accomplish.** The logical gap persists even under very rapid separation. Put

$$X_n = 2^{2^n}.$$

Then $X_{n+1} = X_n^2$, but the sequence is infinite. If $A(X) = \#\{n : X_n \leq X\}$, then for every integer $X \geq 2$,

$$A(X)^2 \leq X. \tag{244}$$

Indeed, two base-two logarithms give $A(X) \leq 1 + \lfloor \log_2 \log_2 X \rfloor$. The elementary induction $(N+1)^2 \leq 2^{2^N}$ completes (244). Thus an actual exponent-1/2 count and an exact square-gap law are compatible with infinitely many objects. This is a counterexample to that proposed counting inference, not to *abc*.

Two upgrades do reach pointwise boundedness. If the number of objects in the k-th shell is at most $C2^{-\delta k}$ with $\delta > 0$, then the integer shell cardinality is eventually strictly below one and hence zero. There is also a useful amplification criterion.

Proposition 25.2 (Single-source amplification)**.** *Suppose $C > 0$, $\theta, \kappa, \beta \geq 0$, and a target class satisfies $|T(Y)| \leq CY^\theta$ for $Y \geq 1$. If every bad source of height $X \geq 1$ produces a set of distinct targets F_X with*

$$F_X \subseteq T(X^\kappa), \qquad |F_X| \geq X^\beta, \qquad \beta > \kappa\theta,$$

then every such source satisfies

$$X \leq C^{1/(\beta - \kappa\theta)}.$$

Proof. For each individual source,

$$X^\beta \leq |F_X| \leq |T(X^\kappa)| \leq CX^{\kappa\theta}.$$

Division and the positive exponent $\beta - \kappa\theta$ give the result. No overlap estimate between fibres of different sources is used. ☐

For the Browning–Verzobio bound in Section 23, the resulting design inequality is

$$\beta > \kappa\left(\frac{1}{p} + \frac{1}{q} - \eta_{u,v}(r) + \varepsilon\right).$$

Their present positive exponent reaches neither an empty-shell bound nor this amplification inequality without a new source-to-target construction.

25.3. **Residual-kernel escape and polynomial rigidity.** For $m \geq 1$ and $n > 0$, define

$$\kappa_m(n) = \prod_{\ell \mid n} \ell^{v_\ell(n) \bmod m}, \qquad \rho_m(n) = \prod_{\ell \mid n} \ell^{\lfloor v_\ell(n)/m \rfloor}.$$

Primewise Euclidean division gives the unique decomposition

$$n = \kappa_m(n)\rho_m(n)^m, \tag{245}$$

with $\kappa_m(n)$ m-th-power-free.

Proposition 25.3 (Kernel escape)**.** *Fix $p, q, r \geq 2$ with $1/p + 1/q + 1/r < 1$. In every infinite family of distinct primitive points whose coordinates are respectively p-, q-, and r-full, the triples*

$$\big(\kappa_p(a), \kappa_q(b), \kappa_r(c)\big)$$

take infinitely many values.

Proof. For a fixed residual triple (A, B, C), (245) converts $a + b = c$ into

$$Ax^p + By^q = Cz^r.$$

Primitivity of (a, b, c) implies $\gcd(x, y, z) = 1$. Darmon and Granville [32, Theorem 2] prove that this fixed equation has only finitely many proper integer solutions in the strict reciprocal range. A finite residual packet would therefore give a finite union of finite sets. $\square$

The proposition rules out a finite list of fixed generalized-Fermat coefficient triples. It does not rule out a fixed Pell or elliptic source curve whose coordinate values induce infinitely many residual kernels; the Pell gate below has exactly this form.

There is a separate obstruction to putting all multiplicities into one polynomial identity.

Proposition 25.4 (Mason–Stothers obstruction)**.** *Let nonzero, pairwise coprime polynomials A, B, C over a characteristic-zero field satisfy $A+B = C$. If they are respectively polynomial-p-, q-, and r-full and at least one is nonconstant, then*

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1.$$

Proof. Put $D = \max\{\deg A, \deg B, \deg C\} > 0$. The polynomial *abc* theorem of Stothers [33] gives

$$D + 1 \leq \deg \operatorname{rad}(ABC).$$

Fullness and pairwise coprimality give

$$\deg \operatorname{rad}(ABC) \leq \frac{\deg A}{p} + \frac{\deg B}{q} + \frac{\deg C}{r} \leq D \left(\frac{1}{p} + \frac{1}{q} + \frac{1}{r} \right).$$

The reciprocal sum cannot be at most one. $\square$

This excludes polynomial-full identities. It does not exclude a sparse set of specializations at which polynomials with simple algebraic factors take full integer values.

25.4. **A strict conditional Pell route.** The classical consecutive-powerful-number Pell orbit studied by Walker [35]

$$x_0 = 1, \quad y_0 = 0, \qquad x_{n+1} = 3x_n + 8y_n, \quad y_{n+1} = x_n + 3y_n$$

satisfies

$$1 + 8y_n^2 = x_n^2. \tag{246}$$

Its positive coordinates are strictly increasing. Hence it gives an unbounded primitive family $(1, 8y_n^2, x_n^2)$ in which both nonunit coordinates are always squarefull.

If y is itself squarefull, then $8y^2$ is cubefull: the prime 2 occurs to exponent at least three, and every odd prime occurs in y^2 to exponent at least four. The unit is 7-full and x^2 is 2-full, so

$$(1, 8y^2, x^2)$$

has the strict signature $(7, 3, 2)$, with

$$\frac{1}{7} + \frac{1}{3} + \frac{1}{2} = \frac{41}{42} < 1.$$

Theorem 25.5 (Pell-squarefull disproof gate)**.** *If an unbounded subsequence of the positive Pell roots y_n is squarefull, then the standard abc conjecture is false.*

Proof. The corresponding points from (246) are positive and primitive, have the fixed strict mixed-full signature $(7, 3, 2)$, and have unbounded height. The strict mixed-full disproof gate from Section 23 applies to the unchanged standard target. $\square$

The actual radical slope is stronger than 41/42. Squarefullness gives $\mathrm{rad}(y)^2 \le y$; also $\mathrm{rad}(x) \le x$ and $y < x$. Therefore

$$\mathrm{rad}(8y^2x^2)^2 \le 4\,\mathrm{rad}(y)^2\,\mathrm{rad}(x)^2 \le 4yx^2 < 4x^3.$$

With $H = \log(x^2)$ and $R = \log \mathrm{rad}(8y^2x^2)$ this yields

$$R \le \frac{3}{4}H + \log 2. \tag{247}$$

The unresolved premise is squarefullness of infinitely many y_n. Here $\alpha = 3+\sqrt{8}$ and $\beta = 3-\sqrt{8}$ are algebraic integers with $\alpha+\beta = 6$, $\alpha\beta = 1$, and coprime nonzero integral sum and product; moreover $\alpha/\beta = \alpha^2 > 1$ is not a root of unity and $y_n = (\alpha^n - \beta^n)/(\alpha - \beta)$. Thus this is a nondegenerate Lucas pair in the sense required by Bilu–Hanrot–Voutier. Their Theorem 1.4 says that every Lucas or Lehmer term of index greater than 30 has a primitive divisor [34]. Their theorem does not assert that this divisor has valuation one, so it alone does not negate the premise. The first twelve terms were factored only to check the recurrence and formulas; only $y_1 = 1$ is squarefull among them, and no asymptotic conclusion is drawn.

The Lean companion proves the Pell norm recurrence and unboundedness, the squarefull-to-cubefull transfer, the exact $(7,3,2)$ slope, and the conditional implication to the original `ABCConjecture`. It does not assume or prove an unbounded squarefull subsequence. The Darmon–Granville, Mason–Stothers, three-quarter radical, and Lucas–Wieferich arguments remain paper proofs.

26. Balanced persistence: affine amplification and explicit counterexample gates

This section records a further simultaneous attack on the positive and negative directions. A route is kept active whenever its missing statement is merely difficult or unknown. We mark a construction mechanism as closed only when a theorem or an explicit counterexample contradicts the statement that mechanism actually needs. A local failure is never promoted to a failure of the surrounding route.

The standard conjecture remains the target throughout. No existence premise for a squarefull Pell subsequence, a squarefull Hall remainder, or a large low-radical affine subfibre is assumed.

26.1. **A complete two-parameter primitive shear.** Fix a primitive positive seed

$$a + b = c, \qquad \gcd(a,b) = 1,$$

and put

$$P = abc, \qquad R = \mathrm{rad}(P), \qquad \rho = \frac{\log P}{\log c}, \qquad \sigma = \frac{\log R}{\log c}.$$

After interchanging a and b, assume $b \ge c/2$. The elementary bounds

$$c(c-1) \le P \le \frac{c^3}{4} \tag{248}$$

will be used repeatedly.

For integers $h, k \ge 1$ define

$$\begin{aligned} U &= 1 + Ph, \\ V &= 1 + P(h + ck), \\ W &= 1 + P(h + bk). \end{aligned} \tag{249}$$

Theorem 26.1 (Primitive affine shear)**.** *If* $\gcd(U,k) = 1$*, then*

$$(aU, bV, cW)$$

is a positive primitive abc point. The ordered output determines the parameter pair (h,k)*.*

Proof. The three identities

$$V - U = cPk, \qquad W - U = bPk, \qquad V - W = aPk$$

and $a + b = c$ give

$$aU + bV = cW.$$

All three cofactors are congruent to 1 modulo P, so they avoid every prime of the seed. A common divisor of U and V divides cPk and is coprime to cP; it must therefore divide k, contrary to $\gcd(U, k) = 1$. The same argument gives $\gcd(U, W) = 1$.

A common divisor of V and W divides aPk. It avoids aP, while $V \equiv U \pmod{k}$ and hence is coprime to k. Thus U, V, W are pairwise coprime. The seed coordinates are pairwise coprime, and all cross terms are coprime because the cofactors avoid P. This proves primitivity. Finally $U = (aU)/a$ recovers h, and $V - U = cPk$ then recovers k. □

The construction is not a thin parametrization.

Lemma 26.2 (Uniform admissible count)**.** *For all sufficiently large M,*

$$\#\{1 \le h, k \le M : \gcd(1 + Ph, k) = 1\} \ge \frac{M^2}{8}.$$

Inside the upper square $\lceil M/2 \rceil \le h, k \le M$, the count is at least $M^2/32$.

Proof. A bad pair has a prime $\ell \le M$ dividing both k and $1 + Ph$; such a prime does not divide P. For each ℓ, there are at most M/ℓ choices of k and at most $M/\ell + 1$ choices of h. Hence

$$\#\{\text{bad pairs}\} \le M^2 \sum_{\substack{\ell \le M \\ \ell \text{ prime}}} \ell^{-2} + M \sum_{2 \le n \le M} n^{-1} \le \frac{3M^2}{4} + M \log M.$$

Here $\sum_{n \ge 2} n^{-2} \le 3/4$. This proves the first assertion for large M.

For the upper square let L denote its side length. The contribution of ℓ is at most $(L/\ell + 1)^2$, so the total number of bad pairs is at most

$$\frac{3L^2}{4} + 2L \log M + \pi(M).$$

For large M the final two terms are at most $L^2/8$. Since $L \ge M/2$, at least $L^2/8 \ge M^2/32$ pairs remain. □

For fixed $K > 5$, set

$$M_K = \left\lfloor \frac{c^{K-2}}{4P} \right\rfloor.$$

Equation (248), Theorem 26.1, and Lemma 26.2 give

$$\#\{\text{distinct outputs of height at most } c^K\} \gg M_K^2 \gg c^{2K-10}. \tag{250}$$

Indeed,

$$cW \le c + cPM_K(1 + b) < c^K.$$

At $K = 8$ this produces at least $c^6/32$ outputs after increasing the fixed lower threshold for c.

26.2. **The actual-radical gate.** The raw count does not inherit the seed's radical saving automatically. For $n \ge 1$ put

$$E(n) = \frac{n}{\operatorname{rad}(n)}.$$

Because the cofactors in (249) are pairwise coprime and avoid P, radical multiplicativity gives the exact identity

$$\operatorname{rad}(aUbVcW) = R \operatorname{rad}(UVW) = \frac{RUVW}{E(U)E(V)E(W)}. \tag{251}$$

Writing $H = cW$, the condition

$$\operatorname{rad}(aUbVcW) < H^{\mu}$$

is therefore equivalent to

$$E(U)E(V)E(W) > \frac{R}{c}UVH^{1-\mu}. \tag{252}$$

This equivalence is the positive route's exact remaining arithmetic object.

At $K = 8$, $\mu = 3/4$, and on the upper parameter square, every exceptional output has

$$E(U)E(V)E(W) > \frac{R}{8192}c^{14}. \tag{253}$$

To see the direction of the height inequality explicitly, the lower bounds in the upper square give $UVW \geq c^{20}/8192$, while $H < c^8$, and

$$\frac{R}{c}UVH^{1/4} = \frac{RUVW}{H^{3/4}}.$$

Define the square-root part

$$Q(n) = \prod_p p^{\lfloor v_p(n)/2 \rfloor}.$$

Primewise, $Q(n)^2 \mid n$ and $Q(n)^2 \geq E(n)$. Thus (253) places every desired output in the tail

$$Q(UVW) \gg R^{1/2}c^7, \tag{254}$$

whereas the parameter box has side $M \ll c^4$. This explains why an ordinary small-modulus squarefree sieve does not settle either direction.

There is nevertheless a uniform unconditional upper budget. Let $\mathcal{E}_c$ be the $K = 8$, $\mu = 3/4$ exceptional subfibre. Then

Theorem 26.3 (Three-way exceptional budget)**.** *For every fixed subcritical source locus and every $\delta > 0$,*

$$|\mathcal{E}_c| \ll_\delta c^{\min\{12-2\rho,\, 6-\sigma,\, 8-\rho-\sigma/3\}+\delta}, \tag{255}$$

uniformly in the seed.

Proof. The first term is the size of the full two-parameter fibre. The second is the actual-support S-unit estimate from Section 4, applied with target radical below c^6 and inherited radical R.

For the third term, (251) implies

$$D := \operatorname{rad}(UVW) < \frac{c^6}{R}.$$

Since the three cofactors are pairwise coprime, one of their radicals is below

$$Z = (c^6/R)^{1/3} = c^{2-\sigma/3}.$$

The de Bruijn radical count used by Bernert–Browning–Lichtman–Teräväinen [7] bounds the number of possible values of that cofactor by $O_\delta(c^{2-\sigma/3+\delta})$. The exponent is uniform: for a fixed source slope $\sigma \in [0, \lambda]$, the relevant de Bruijn exponent ranges over a compact interval, so a finite mesh and monotonicity reduce the claim to finitely many fixed estimates.

For fixed U, the parameter h is unique and there are at most M choices of k. Fixed V or W has no larger representation multiplicity. Since $M \ll c^{6-\rho}$, the third exponent follows. □

Theorem 26.4 (Seed-sensitive affine upper gate)**.** *Fix a positive primitive seed $a + b = c$, and put $R = \operatorname{rad}(abc)$. Let $X \geq 2$ and $0 < \mu < 1$. For an arbitrary subset of the admissible positive integral parameters, let $\mathcal{E}(X)$ be its set of distinct affine-shear outputs satisfying $H = cW \leq X$ and $\operatorname{rad}(aUbVcW) < H^\mu$. Then, for every $\varepsilon > 0$,*

$$\#\mathcal{E}(X) \ll_{\mu,\varepsilon} R^{-2/3}X^{2\mu/3+\varepsilon}. \tag{256}$$

The implied constant is independent of the seed and of the chosen parameter subset.

Proof. Positivity in $aU + bV = cW = H$ gives $U, V, W \le H \le X$. Exact support avoidance and cofactor coprimality give

$$\mathrm{rad}(aUbVcW) = R\,\mathrm{rad}(U)\,\mathrm{rad}(V)\,\mathrm{rad}(W),$$

so every exceptional output satisfies

$$\mathrm{rad}(U)\,\mathrm{rad}(V)\,\mathrm{rad}(W) < X^\mu/R.$$

All three pair projections (U, V), (U, W), and (V, W) are injective. The two-coordinate consequence of the de Bruijn radical count is

$$\#\{(m, n) : m, n \le X,\ \mathrm{rad}(m)\,\mathrm{rad}(n) < Y\} \ll_\varepsilon YX^\varepsilon.$$

Writing r_1, r_2, r_3 for the three cofactor radicals, the identity $(r_1r_2)(r_1r_3)(r_2r_3) = (r_1r_2r_3)^2$ shows that one pair product is below $(X^\mu/R)^{2/3}$. Apply the pair count through the corresponding injective projection and sum over the three choices. This proves (256). □

The relaxed region $\rho \le 3$, $\sigma < 1$ leaves every exponent in (255) above 4; the infimum of the third term is $14/3$. This is a lower envelope of the available upper-budget exponents, not a pointwise upper bound $|\mathcal{E}_c| \ll c^{14/3}$. For a fixed source slope $\lambda < 1$ its corresponding relaxed envelope is $5 - \lambda/3$.

At $\mu = 3/4$, Proposition 1.1 of Bernert–Browning–Lichtman–Teräväinen gives exponent $2\mu/3 = 1/2$ in target height [7]. Their Theorem 1.3 gives the different uniform exponent $3/5$ for fixed $\mu < 1$; at $\mu = 3/4$ it is the weaker of the two applicable estimates. Hence the following statement would close the positive route through the subcritical-locus equivalence.

Corollary 26.5 (Exact conditional closing step)**.** *Fix $\lambda < 1$. Suppose every sufficiently large seed satisfying $R < c^\lambda$ has at least $c^{17/4}$ members of $\mathcal{E}_c$. Then the height of that source locus is bounded.*

Proof. Every member is a distinct primitive point of height below c^8. For every $\delta > 0$, the cited exceptional-set theorem gives

$$|\mathcal{E}_c| \ll_\delta c^{4+8\delta}.$$

Take $\delta = 1/64$. Then $4 + 8\delta = 33/8 < 17/4$, so comparison with the hypothesized lower bound bounds c. □

The fixed exponent $17/4$ is sufficient but no longer optimal. By (256), the sharp conditional comparison at $X = c^8$ is

$$|\mathcal{E}_c| \ge R^{-2/3}c^{4+\eta} \qquad (\eta > 0).$$

For a seed with $R = c^{\sigma+o(1)}$, the required exponent is therefore $4-2\sigma/3+\eta$. This is a conditional lower target, not a consequence of the upper bound.

There is also a precise local warning. Conditional on local admissibility at $p \nmid P$, each of $p^2 \mid U, V, W$ has probability $1/(p(p + 1))$, but the three events are pairwise disjoint. Thus the model asserting independence of the three same-prime events is refuted. Finite local data at distinct primes remain exactly independent by the Chinese remainder theorem. This local counterexample closes only that probabilistic model; it does not refute the affine lower-bound route.

No theorem presently supplies the lower bound in Corollary 26.5. The upper budget does not refute it. The affine route therefore remains active.

26.3. **Square-root descent on the balancing Pell orbit.** Let

$$u_0 = 0, \quad u_1 = 1, \quad u_{n+2} = 6u_{n+1} - u_n,$$

and define x_n by

$$x_n^2 - 8u_n^2 = 1.$$

Equivalently,

$$(3 + \sqrt{8})^n = x_n + u_n\sqrt{8}.$$

Introduce the companion coordinates

$$(1+\sqrt{2})^n = A_n + B_n\sqrt{2}.$$

Theorem 26.6 (Coprime square-root factorization)**.** *For every $n \geq 1$,*

$$x_n = A_n^2 + 2B_n^2, \qquad u_n = A_nB_n, \qquad A_n^2 - 2B_n^2 = (-1)^n, \qquad \gcd(A_n, B_n) = 1. \tag{257}$$

Consequently u_n is squarefull if and only if both A_n and B_n are squarefull.

Proof. Squaring $(1+\sqrt{2})^n$ and using $(1+\sqrt{2})^2 = 3+\sqrt{8}$ gives the first two identities. Taking norms gives the third. A common divisor of A_n and B_n divides the norm, hence is one. The prime supports of the two factors are therefore disjoint, which proves the squarefull equivalence. □

Adding and subtracting the norm identity gives

$$\left(\frac{x_n-1}{2}, \frac{x_n+1}{2}\right) = \begin{cases} (2B_n^2, A_n^2), & n \text{ even}, \\ (A_n^2, 2B_n^2), & n \text{ odd}. \end{cases} \tag{258}$$

Thus the associated primitive point Q_n is obtained by inserting the unit between the two entries in (258).

Theorem 26.7 (One-half radical slope)**.** *If u_n is squarefull and c_n is the larger nonunit entry of Q_n, then*

$$\log \operatorname{rad}(Q_n) < \frac{1}{2}\log c_n + \log 2. \tag{259}$$

Hence an unbounded squarefull subsequence of (u_n) would disprove the standard abc conjecture.

Proof. Theorem 26.6 makes both factors squarefull, so

$$\operatorname{rad}(Q_n) = \operatorname{rad}(2A_n^2B_n^2) \leq 2\operatorname{rad}(A_n)\operatorname{rad}(B_n) \leq 2\sqrt{A_nB_n}.$$

For even n, $c_n = A_n^2$ and $B_n < A_n$; for odd n, $c_n = 2B_n^2$ and $A_n < 2B_n$. Thus $A_nB_n < c_n$ in both cases, proving (259). Applying standard *abc* with, for example, $\varepsilon = 1/2$ would bound the heights, contradicting an unbounded subsequence. □

On an even squarefull index, both nonunit entries are 4-full, so the fixed signature is $(3, 4, 4)$ with reciprocal sum $5/6$. This is an additional conversion of the same unresolved squarefull premise, rather than a new source of indices.

26.4. **Valuation saturation and prime-index descent.** For a prime p, let $z(p)$ be its rank of apparition in (u_n). Strong divisibility gives $p \mid u_n$ if and only if $z(p) \mid n$. For odd p, the order calculation in the quadratic residue algebra gives

$$z(p) \mid p - \left(\frac{2}{p}\right). \tag{260}$$

Sanna's exact valuation theorem [36] specializes to

$$v_p(u_n) = \begin{cases} v_p(u_{z(p)}) + v_p(n), & z(p) \mid n, \\ 0, & z(p) \nmid n, \end{cases} \qquad (p \text{ odd}), \tag{261}$$

and

$$v_2(u_n) = 0 \quad (n \text{ odd}), \qquad v_2(u_n) = v_2(n) \quad (n \text{ even}).$$

Equations (260)–(261) yield an exact saturation rule. If $p \parallel u_{z(p)}$, $z(p) \mid N$, and u_N is squarefull, then $p \mid N/z(p)$. Iterating only the factorizations

$$\begin{array}{r|l} 2 & 2\cdot 3 \\ 3 & 5\cdot 7 \\ 4 & 2^2\cdot 3\cdot 17 \\ 5 & 29\cdot 41 \\ 6 & 2\cdot 3^2\cdot 5\cdot 7\cdot 11 \\ 7 & 13^2\cdot 239 \end{array}$$

proves the finite obstruction

$$u_N \text{ squarefull and } 2 \mid N \quad \Longrightarrow \quad 22{,}318{,}790{,}340 \mid N. \tag{262}$$

The factor 13^2 at rank seven supplies no outgoing exponent-one edge. This is a real obstruction to the naive use of primitive-divisor theorems:

$$u_7 = 13^2 \cdot 239. \tag{263}$$

Carmichael's theorem in Yabuta's formulation [40] provides a primitive divisor at every relevant index, but (263) proves that "primitive" does not mean "valuation one."

The strongest unconditional descent obtained here reduces the full squarefull question to prime indices.

Theorem 26.8 (Largest-prime descent)**.** *Suppose $N > 1$ and u_N is squarefull. If ℓ is the largest prime divisor of N, then ℓ is odd and u_ℓ is squarefull.*

Proof. If N were a power of two, then $z(3) = 2$ and (261) would give $v_3(u_N) = 1$, a contradiction. Thus ℓ is odd.

Let $q \mid u_\ell$. Since u_ℓ is odd and ℓ is prime, $z(q) = \ell$. Equation (260) gives $\ell \mid q \pm 1$. An odd prime with this property is larger than ℓ: the first positive representatives $\ell - 1$ and $\ell + 1$ are even. Hence $q \nmid N$ by the maximality of ℓ. Since $\ell \mid N$, valuation lifting gives

$$v_q(u_N) = v_q(u_\ell) + v_q(N) = v_q(u_\ell).$$

The left side is at least two. This holds for every prime divisor of u_ℓ, proving squarefullness. □

An odd prime p for which $p^2 \mid u_{z(p)}$ is a balancing-Wieferich prime. Write $(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}$. At an odd prime index ℓ, $u_\ell = A_\ell B_\ell$ and $\gcd(A_\ell, B_\ell) = 1$. The perfect-power classifications of Cohn and Ljunggren [45, 46, 47] sharpen the prime-index alternative as follows.

Theorem 26.9 (Prime-index four-prime packet)**.** *For every odd prime ℓ, either some prime satisfies $q \parallel u_\ell$, or $\ell \neq 7$ and u_ℓ is squarefull with at least four distinct support primes of rank exactly ℓ: at least two divide A_ℓ and at least two divide B_ℓ. In this second alternative every such prime has first valuation at least two, and each channel contains a prime whose first valuation is at least three.*

Proof. If no support valuation equals one, the term is squarefull. Coprimality makes both channels squarefull. Cohn's associated-Pell theorem says that $A_\ell > 1$ is not a perfect power. The standard Pell classifications say the same for B_ℓ, except at $\ell = 7$; there $u_7 = 13^2 \cdot 239$ and $239 \parallel u_7$. A squarefull integer which is not a perfect power has at least two support primes and an odd valuation at least three. Applying this separately to the two channels proves the packet. The rank and valuation claims follow from (260)–(261). □

The channel orders also force

$$q \mid A_\ell \Longrightarrow q \equiv 1 \text{ or } 2\ell + 1 \pmod{4\ell}, \qquad q \mid B_\ell \Longrightarrow q \equiv 1 \text{ or } 2\ell - 1 \pmod{4\ell}.$$

Combining Theorem 26.9 with the largest-prime descent shows that every hypothetical nonunit squarefull balancing term forces such a packet at an odd prime rank. The adjacent Pell point consequently has the sharpened bound

$$\operatorname{rad}(2A_\ell^2 B_\ell^2) \leq 2\sqrt{\frac{u_\ell}{4\ell^2 - 1}}. \tag{264}$$

No theorem currently rules out all such packets or constructs infinitely many of them. The Pell route remains active.

26.5. **Elliptic and Hall alternatives.** The Mordell curve

$$E : y^2 = x^3 - 2, \qquad P = (3, 5)$$

supplies an independent critical family. Write

$$nP = \left(\frac{\mathsf{X}_n}{\mathsf{D}_n^2}, \frac{\mathsf{Y}_n}{\mathsf{D}_n^3}\right)$$

in lowest terms. Then

$$\mathsf{Y}_n^2 + 2\mathsf{D}_n^6 = \mathsf{X}_n^3. \tag{265}$$

The three entries are pairwise coprime. Since

$$2P = \left(\frac{129}{10^2}, \frac{-383}{10^3}\right),$$

the denominator elliptic divisibility sequence satisfies $\mathsf{D}_2 \mid \mathsf{D}_{2k}$, so D_n is even at every even index [49]. Equation (265) then has critical signature $(2, 6, 3)$. It becomes strict if Y_n or X_n is squarefull on an unbounded subsequence, or if the odd part of D_n is squarefull. Elliptic primitive-divisor theorems control new support, but not its first valuation, so none of these upgrade gates is currently refuted.

There is also a size-sensitive Hall gate.

Proposition 26.10 (Squarefull Hall gate)**.** *Suppose positive primitive data satisfy*

$$X^3 + K = Y^2, \qquad K^2 \le X,$$

and K is squarefull. Then

$$\operatorname{rad}(X^3 K Y^2)^{12} < (Y^2)^{11}, \tag{266}$$

so an unbounded family of such data would disprove standard abc.

Proof. Put $D = \operatorname{rad}(X^3 K Y^2)$. Radical submultiplicativity gives $D \le X \operatorname{rad}(K) Y$. Squarefullness and the size condition give

$$\operatorname{rad}(K)^{12} \le K^6 \le X^3.$$

Therefore $D^{12} \le X^{15} Y^{12}$. Since $K > 0$, $X^3 < Y^2$, whence $X^{15} < Y^{10}$. This proves (266). Its logarithmic slope is strictly below $11/12$. ☐

Danilov's identity, as recorded by Dujella [41, 42], is

$$(z^2 + 6z + 4)^3 - (z^2 + 1)(z^2 + 9z + 19)^2 = -27(2z + 11). \tag{267}$$

Choose positive solutions

$$z^2 - 5w^2 = -1, \qquad z \equiv 57 \pmod{125}.$$

There are infinitely many: one is $(682, 305)$, and multiplication by a fixed positive power of the norm-one unit $9 + 4\sqrt{5}$ preserves the congruence and gives an unbounded orbit. Put

$$A = z^2 + 6z + 4, \qquad B = z^2 + 9z + 19, \qquad L = 2z + 11,$$
$$X = A/5, \qquad Y = wB/5, \qquad K = 27L/125.$$

The congruence gives $125 \mid L$, $A \equiv 20 \pmod{25}$, and $5 \mid w$. Equation (267) becomes

$$X^3 + K = Y^2. \tag{268}$$

Moreover $\gcd(A, B) = 1$ and $\gcd(A, wB) = 5$, so $\gcd(X, Y) = 1$. Thus this is an unbounded primitive Hall family with

$$X \sim \frac{z^2}{5}, \qquad K \sim \frac{54z}{125} \sim \frac{54\sqrt{5}}{125}\sqrt{X}, \qquad \frac{54\sqrt{5}}{125} < 1.$$

Consequently $K^2 < X$ on a sufficiently large tail of the orbit. The first point is the exact identity

$$93844^3 + 297 = 28748141^2.$$

Here $297 = 3^3 \cdot 11$ is not squarefull. This refutes only the universal claim that every remainder in the congruence orbit is squarefull; it does not refute an unbounded squarefull subsequence. Such a subsequence would satisfy Proposition 26.10 after discarding its finite initial segment and would disprove *abc*.

For each fixed $m \geq 2$, forcing the moving factor $(2z + 11)/125$ to be one exact mth power leads to the hyperelliptic equation

$$(2w)^2 = 3125t^{2m} - 550t^m + 25.$$

The polynomial on the right is squarefree of degree $2m$, so its smooth completion has genus $m-1 \geq 1$. Siegel's theorem yields only finitely many integral points for that fixed m. This closes the fixed exact-power mechanism. It does not close the squarefull route because the square and cube kernels may both move.

Certified finite computation. A reproducible exact audit in the companion computation archive checked both open squarefull routes without extrapolating from finite data. For the balancing recurrence, every one of the 999 terms u_n with $2 \leq n \leq 1000$ has an explicit prime p and residue q such that

$$u_n \equiv pq \pmod{p^2}, \qquad 1 \leq q < p.$$

Thus each of those terms is rigorously non-squarefull. The extended certified run through $n = 2000$ now settles 1997 of the 1999 nonunit terms and leaves precisely

$$\{1873, 1951\}$$

unresolved. These two indices are neither squarefull hits nor certified non-hits. The seven new exclusions use local recursive Pocklington proofs and two independent recurrences modulo p^2.

An exhaustive scan of all 183,071 odd primes $q \leq 2{,}500{,}000$ tested $u_{q-(2/q)}$ modulo q^3. The only balancing-Wieferich hits are $13, 31, 1546463$, all of exact depth two; no depth-three hit occurs in this finite range. This is pressure against the four-prime packet, not an infinite nonexistence theorem.

On the Danilov orbit obtained by iterating $(9+4\sqrt{5})^{10}$ from $(682, 305)$, the audit verifies the Pell equation, integrality, primitivity, $X^3 + K = Y^2$, and $K^2 \leq X$ for $0 \leq t \leq 80$; every one of the 81 remainders also has a certified prime with $v_p(K) = 1$. The five-small-prime periodic sieve already fails to decide $t = 326$. A subsequent symbolic prime-square lift eliminates the finite-search boundary: if K_t is squarefull, then

$$\begin{aligned} t &\equiv 1221369550325650259678094491108403475378 27 \\ &\qquad \pmod{183205432548847538951714173666260521306741}. \end{aligned} \tag{269}$$

Hence every nonnegative index below the displayed representative is ruled out as a squarefull index. The surviving index residue class is nonempty, but the theorem asserts no squarefull value in it and does not exclude all such values. The Danilov route remains active.

The Cohn–Nitaj constructions provide genuine unbounded critical $(3, 3, 3)$ families [43, 44]. Walsh's Theorem 1.1 supplies another infinite family for an odd prime p under the explicit hypothesis that $Y^2 = X^3 - 432p^2$ has positive rank [48]; without a proved rank statement for the selected p, that branch remains conditional. In each case the reciprocal sum equals one. A powerful-coordinate subsequence would make a strict signature, and no cited theorem either constructs or refutes that subsequence. These routes also remain active.

26.6. **Formalization boundary and route ledger.** Seven Lean modules formalize the elementary deterministic core after the paper proofs were written:

Module	Kernel-checked content
`PellAdjacentFactor Counterexample20260831`	Half-factor existence, primitive adjacent point, radical square bound, one-half conductor inequality, fourth-full transfer, and the conditional implication to ¬`ABCConjecture`.
`PellSquareRoot Descent20260831`	The $(1+\sqrt{2})^n$ recurrence, norm, coprimality, exact square shapes, fullness splitting, and equivalence with the Pell coordinate.
`PellPrimeIndex Dichotomy20260831`	Odd squarefull exponents have depth at least three, the complete index-seven obstruction $239 \parallel u_7$, and the two-channel numerical product bound.
`HallSquarefull Counterexample20260831`	The primitive Hall point, the twelfth-power radical inequality, the 11/12 conductor inequality, and the unbounded-family disproof gate.
`AffineShear Amplification20260831`	The shear equation, avoidance of the seed support, all cofactor and endpoint coprimality statements, the actual `ABCPoint`, and injectivity of the parameter map.
`AffineExcess UpperBound20260831`	Injectivity of all three pair projections, same-prime double-divisibility exclusion, and the integer geometric-mean lemma.
`DanilovGlobalIndex Sieve20260831`	The 11- and 89-adic stages, ten prime-square lift packets, exact CRT, and the global progression and lower-index exclusion in (269).

The analytic BBLT and S-unit estimates; the Cohn–Ljunggren perfect-power classifications; Sanna's valuation theorem; Carmichael–Yabuta and Siegel; elliptic-divisibility and primitive-divisor results; the Danilov parametrization, primitivity, and orbit unboundedness; and the Cohn–Nitaj and conditional Walsh existence results all remain outside this Lean increment. They are proved on paper or cited, rather than inserted as project axioms.

The resulting route ledger is deliberately narrow:

- the affine repeated-prime lower bound, simultaneous squarefull Pell coordinates, exclusion or construction of the forced four-prime balancing-Wieferich packets, the surviving Danilov residue class, and powerful elliptic-coordinate subsequences are *active*;
- a fixed strict generalized-Fermat equation, a finite residual-kernel packet, a primitive polynomial-full identity, and the fixed exact-power Danilov specialization are closed only as the stated construction mechanisms;
- the example $K = 297$ refutes "every Danilov remainder is squarefull" but leaves all unbounded-subsequence variants open; the same-prime independent affine model is also refuted, without closing the affine route.

No unconditional closed term of type `ABCConjecture`, and no rigorous unconditional disproof, is obtained in this increment.

27. Radical-step fibres, global prime packets, and the same-pilot type boundary

This section continues the positive and negative searches simultaneously. Failure of a finite search, absence of a known theorem, and present technical difficulty do not retire a route. A counterexample below is used only against the fully quantified statement that it satisfies; the surrounding route is kept whenever a stronger or differently typed mechanism survives.

27.1. **Replacing the product step by the seed radical.** Let $a + b = c$ be primitive, put $P = abc$ and $R = \mathrm{rad}(P)$, and let Q be any positive multiple of R. For $h, k \geq 1$ define

$$U = 1 + Qh, \qquad V = 1 + Q(h + ck), \qquad W = 1 + Q(h + bk). \tag{270}$$

Theorem 27.1 (Radical-step primitive shear)**.** *If* $\gcd(U,k)=1$*, then* (aU,bV,cW) *is a primitive abc triple. Each of the three pair projections* (U,V)*,* (U,W)*, and* (V,W) *determines* (h,k)*. If* $K\geq 2$ *and*

$$M_Q=\left\lfloor \frac{c^{K-2}}{4Q}\right\rfloor,$$

then every parameter pair $1\leq h,k\leq M_Q$ *has height below* c^K*. For sufficiently large* M_Q*, at least* $M_Q^2/8$ *of these pairs are admissible.*

Proof. Every prime dividing a seed coordinate divides Q, while all three cofactors are congruent to one modulo Q. Thus every cofactor avoids the entire seed support. The identities

$$V-U=cQk,\qquad W-U=bQk,\qquad V-W=aQk$$

show that a common divisor of U and V, or of U and W, must divide k after the factors Q and the appropriate seed coordinate have been removed. The hypothesis $\gcd(U,k)=1$ excludes both possibilities. A common divisor of V and W divides aQk; it avoids aQ, and $V\equiv U \pmod{k}$, so it also avoids k. Hence U,V,W are pairwise coprime. Combining this with support avoidance and the pairwise coprimality of a,b,c proves primitivity. The identity $aU+bV=cW$ follows by expansion.

Any one of the three displayed differences recovers k from the indicated pair. If the pair contains U, then $U=1+Qh$ recovers h; from (V,W) one instead uses $h=(V-1)/Q-ck$. This proves the three injectivity statements. Since $b+1\leq c$,

$$W\leq 1+QcM_Q\leq 1+\frac{c^{K-1}}{4},$$

and therefore $cW<c^K$ for $c\geq 2$ and $K\geq 2$. Finally, the elementary sieve used in Lemma 26.2 applies verbatim with Q in place of P: the bad pairs are covered by primes dividing both k and $1+Qh$. It leaves at least $M_Q^2/8$ pairs once M_Q is sufficiently large. □

The minimal step $Q=R$ is therefore optimal among these multiples for the raw box. At $K=8$ its elementary lower count has scale

$$\frac{c^{12}}{R^2},\tag{271}$$

instead of c^{12}/P^2. This is a genuine improvement when the seed has large prime powers. It does not itself count low-radical outputs. Indeed, the pair-projection upper argument is intrinsic to the output factors and remains

$$\#\mathcal{E}_Q(c^8)\ll_\varepsilon R^{-2/3}c^{4+\varepsilon}\tag{272}$$

on the $\mu=3/4$ locus.

There is no gain from averaging the same construction over multiples of R. If $Q=sR$, then the Q-point at (h,k) is exactly the R-point at (sh,sk). Moreover $1+Qh\equiv 1 \pmod{s}$, so admissibility at (h,k) implies admissibility at (sh,sk). Thus the larger-step fibres are subfibres of the minimal one after rescaling their boxes.

The often stated "matching lower bound" should also be interpreted with care. Assume the uniform upper estimate (272). For fixed $\lambda<1$ and $\eta>0$, the assertion that there are constants $A>0,c_0$ such that every seed with $R<c^\lambda$ and $c\geq c_0$ satisfies

$$\#\mathcal{E}_R(c^8)\geq AR^{-2/3}c^{4+\eta}\tag{273}$$

is equivalent to boundedness of the heights of the seeds in that subcritical locus. The forward implication compares (273) with (272) at $\varepsilon=\eta/2$. Conversely, after a height bound has already been proved, one may choose c_0 above it and the lower assertion is vacuous. Consequently (273) is an exact closing gate, but it is not by itself an explanatory distribution theorem.

A fixed congruence template cannot provide that explanation. Suppose d_1,d_2,d_3 are pairwise coprime, $D=d_1d_2d_3$, and $\gcd(D,Qabc)=1$. The conditions

$$d_1\mid U,\qquad d_2\mid V,\qquad d_3\mid W$$

have exactly D solutions (h,k) modulo D, hence density D^{-1}. The deterministic lower certificate for repeated-prime excess supplied solely by these divisibilities is

$$\Delta = \prod_{i=1}^{3} \frac{d_i}{\operatorname{rad}(d_i)} \le D. \tag{274}$$

On the upper half of the $K = 8$ radical-step box, forcing the target radical inequality requires $\Delta \gg Rc^{14}$. It therefore also requires $D \gg Rc^{14}$, while the periodic main term is at most

$$\frac{M_R^2}{D} \ll \frac{c^{12}/R^2}{Rc^{14}} = \frac{1}{c^2R^3}.$$

This rules out obtaining the matching lower bound from the periodic main term of a single template alone. It leaves large unions of templates, boundary terms, and accidental high valuations open.

There are also exact finite warnings against replacing excess by a visual "square cofactor" condition. For the subcritical seed $(1,8,9)$, $R = 6$, the choices

$$(Q,h,k) = (6,840,60) \quad \text{and} \quad (72,70,5)$$

both give

$$(U,V,W) = (5041,8281,7921) = (71^2,91^2,89^2).$$

Nevertheless the output height is 71289, its radical is 3450174, and the desired fourth-power versus third-power radical inequality fails. A second example is

$$(Q,h,k) = (72,1160,798), \qquad (U,V,W) = (289^2,775^2,737^2),$$

and it fails the same target. These are counterexamples to the assertion that three square cofactors automatically give an exceptional output. They do not refute the eventual arithmetic lower gate.

27.2. **Exact order stagnation in the balancing orbit.** Let

$$u_0 = 0, \quad u_1 = 1, \quad u_{n+2} = 6u_{n+1} - u_n,$$

and put $\alpha = 3 + 2\sqrt{2}$ and $\gamma = \alpha^2 = 17 + 12\sqrt{2}$. Then

$$u_n = \alpha^{1-n}\frac{\gamma^n - 1}{\gamma - 1}. \tag{275}$$

For an odd rational prime q, let $z(q)$ be its rank of apparition and put $e(q) = v_q(u_{z(q)})$.

Theorem 27.2 (Order-stagnation tower)**.** *For every odd prime q and $k \ge 1$,*

$$z(q^k) = z(q)q^{\max(0,k-e(q))}. \tag{276}$$

In particular, depth two is exactly one failure of order growth and depth three is exactly two consecutive failures.

Proof. In $K = \mathbb{Q}(\sqrt{2})$, every odd q is unramified and $N(\gamma - 1) = -32$. At every prime ideal $\mathfrak{q} \mid q$, (275) identifies $z(q)$ with the residual order of γ and $e(q)$ with $v_{\mathfrak{q}}(\gamma^{z(q)} - 1)$. If y is a local unit with $v(y-1) = e \ge 1$, the odd-prime binomial expansion gives

$$v(y^m - 1) = e + v_q(m).$$

Thus the least multiplier needed to raise the congruence from level one to level k is $q^{\max(0,k-e)}$. Substitution gives (276). $\square$

At an odd prime index ℓ, write $(1+\sqrt{2})^\ell = A_\ell + B_\ell\sqrt{2}$. The previous packet theorem says that a squarefull $u_\ell = A_\ell B_\ell$ forces four distinct rank-ℓ repeated primes, two in each coprime channel, with a depth-three prime in each channel. There is also a first-order relation among all packet primes. If

$$A_\ell = 1 + 2\ell a, \qquad \left(\frac{2}{\ell}\right) B_\ell = 1 + 2\ell b,$$

then the Pell norm gives the exact identity

$$a - 2b + \ell(a^2 - 2b^2) = 0. \tag{277}$$

Expanding the prime factorizations modulo $4\ell^2$ turns its reduction modulo ℓ into a coupled congruence between the quotient coordinates in the two channels. This is stronger than independent residue conditions, although the free quotient variables prevent a contradiction at present.

The latest number-field input yields a useful global alternative. Fellini and Murty state that finiteness of base-x super-Wieferich prime ideals forces infinitely many non-Wieferich prime ideals [39]. An adversarial audit of their Section 8 architecture gives a sharper prime-order consequence, but it also requires two local proof repairs and one textual correction. At a selected prime index ℓ, a prime ideal $\mathfrak{p} \mid \Phi_\ell(x)$ has residual order dividing ℓ. Order one would imply both $x \equiv 1$ and $\ell \equiv 0$ modulo $\mathfrak{p}$; the defining exclusion of primes above ℓ which divide $(x-1)$ rules this out. Thus every such factor has order exactly ℓ, and its exponent in $\Phi_\ell(x)$ is its first-occurrence valuation. Excluding the finitely many super-Wieferich orders leaves exponents one or two. If only finitely many prime-order exponent-one ideals existed, the Section 8 localization, finite square-class decomposition, and Siegel argument would again place infinitely many distinct points on a finite family of genus-one curves.

This streamlined prime-index proof deliberately avoids the printed second case of Lemma 6.4. The displayed equality there is generally false: with $x = 2, p = 3, f = 2, i = 1$, it identifies $\Phi_6(2) = 3$ with $(2^6 - 1)/(2^2 - 1) = 21$. Its binomial proof also treats $\binom{p}{p} = 1$ as if it were divisible by p; separating the last term repairs the valuation estimate. A contradictory sentence on printed p. 226 is avoided by allowing the localized square generator to be a unit. These counterexamples invalidate intermediate printed steps. They do not invalidate the repaired valuation conclusion or the resulting Siegel argument.

Specializing this proof extraction to $x = \gamma$ and descending prime ideals to rational primes gives the unconditional alternative. After discarding the finitely many prime ideals above 2, the local dictionary applies to odd rational primes:

$$\begin{aligned} &\text{either infinitely many odd primes } q \text{ satisfy } e(q) \geq 3, \\ &\text{or infinitely many odd prime indices } \ell \text{ admit } q \parallel u_\ell \text{ with } z(q) = \ell. \end{aligned} \tag{278}$$

This is a repaired refinement of the cited proof architecture, not a quotation of its theorem statement. In the first branch it gives neither a common rank nor opposite channels; in the second it gives infinitely many nonsquarefull prime-index terms but not all prime indices. It therefore does not settle the forced four-prime packet.

Three natural shortcuts have exact counterexamples. At index seven,

$$B_7 = 13^2, \qquad 13^2 \parallel u_7, \qquad z(13) = 7, \qquad 13 = 2 \cdot 7 - 1.$$

Moreover

$$F_7(X) = X^6 - 5X^4 + 6X^2 - 1, \quad F_7(6) = 40391 = 13^2 \cdot 239, \quad F_7'(6) \equiv 2 \pmod{13}.$$

Thus a primitive divisor need not be simple, equality in the minimal channel bound does not force simplicity, and a nonsingular polynomial root modulo q does not force the fixed integer value to avoid q^2. The opposite channel has the further exact example

$$1546463 = 2 \cdot 773231 + 1, \quad z(1546463) = 773231, \quad 1546463^2 \parallel A_{773231},$$

with both displayed rank parameters prime. These examples close only the three stated implications.

Two independent exact programs exhaust all 5,761,454 odd primes $q \leq 10^8$. The only repeated primes are $13, 31, 1546463$, all of exact depth two; there is no depth-three hit. Hence both depth-three primes in a hypothetical squarefull prime-index packet exceed 10^8. This is a finite certificate and is never used as an infinitude conclusion.

27.3. **Recursive prime-square lifts on the Danilov progression.** On the surviving Danilov progression, write $t = T + Qr$ and maintain

$$3T + 1 = hQ, \qquad h \in \{1, 2\}. \tag{279}$$

The quadratic-unit orbit gives the exact Fibonacci factorization

$$L_T = 5F_{10hQ}F_{10hQ-5}. \tag{280}$$

If $p \parallel F_{10Q}$, $p \nmid 3375Q$, then the packet modulo p^2 is nondegenerate. Squarefullness forces the unique residue

$$h + 3r \equiv 0 \pmod{p}. \tag{281}$$

For its representative $0 \leq \rho < p$, divisibility and the inequalities $0 < h + 3\rho < 3p$ give a unique $h' \in \{1, 2\}$ with $h + 3\rho = h'p$. The update

$$T' = T + Q\rho, \qquad Q' = Qp$$

then preserves $3T' + 1 = h'Q'$.

Carmichael's theorem in Yabuta's form supplies a primitive divisor of F_{10Q} outside its explicit small exceptions [40]. Rank of apparition makes such a divisor fresh relative to Q. The theorem does not control its first valuation. This distinction yields the exact conditional closure.

Theorem 27.3 (Conditional simple-primitive closure)**.** *Assume that at every adaptively generated state, F_{10Q} has a primitive prime divisor p with $p \parallel F_{10Q}$. Then no index in the initial Danilov survivor progression has squarefull remainder.*

Proof. The simple divisor supplies a nondegenerate fresh packet and (281) nests the next progression while preserving (279). Repetition constructs arbitrarily many distinct fresh primes. If one fixed index t survived as squarefull, every chosen prime would divide the one fixed nonzero integer L_t. A finite integer has only finitely many distinct prime divisors, a contradiction. $\square$

The valuation-one hypothesis is essential to this proof and is not a consequence of the cited primitive-divisor theorem. Nor does one abstract packet automatically generate another. In $\mathbb{Z}[\sqrt{5}]$, take

$$\eta = (9 + 4\sqrt{5})^8, \qquad \alpha_0 = 19 + \sqrt{5}, \qquad L_r = K_r = 2\operatorname{Re}(\alpha_0\eta^r) + 11.$$

Modulo 49, $\eta = 1 + 35\sqrt{5}$ and

$$L_r \equiv 7r \pmod{49}. \tag{282}$$

At $(T, Q, p) = (0, 1, 7)$ the slope is nonzero, the forced root is $r = 0$, and $K_0 = 49$ is squarefull. The updated modulus is 7, while every prime divisor of $L_0 = 49$ already divides it. Hence no fresh successor exists. This satisfies the full abstract local-packet mechanism, but its initial norm is not the Danilov norm. It refutes automatic abstract continuation and does not refute the Danilov-specific simple-primitive route.

An exact adaptive computation constructs 626 distinct packets. They occur in thirteen nonempty batches. The final state has a 4398-digit modulus and exactly 638 distinct prime factors; no next eligible packet occurs below 10^8. Every selected prime has exact valuation one, and every CRT update is independently replayed. This certifies the finite recursion, while the endpoint remains open above its bound.

27.4. **The simple-primitive gate and Wall–Sun–Sun support.** The missing valuation-one premise can be characterized exactly for the Fibonacci indices arising above. If $n > 5$, $5 \mid n$, and p is a primitive divisor of F_n, then the rank of apparition satisfies

$$z(p) = n \mid p - \left(\frac{5}{p}\right).$$

Reduction modulo five and quadratic reciprocity exclude the inert sign, so

$$p \equiv 1 \pmod{n}. \tag{283}$$

Sanna's exact valuation formula then gives

$$v_p(F_n) = v_p(F_{p-1}). \tag{284}$$

Thus a primitive divisor is repeated precisely when it is a Fibonacci–Wieferich, or Wall–Sun–Sun, prime [36, 37]. No such prime is presently known; that empirical fact is not used as a nonexistence theorem.

Let Q_* be the final 4398-digit squarefree modulus in the certified recursive chain and put $n_* = 10Q_*$. Its complete factorization consists of exactly 638 primes, each below 10^8, and $\gcd(Q_*, 30) = 1$. Let C_n denote the Fibonacci cyclotomic factor. For $n > 2$ divisible by 10, Yabuta's correction analysis says that C_n has at most one nonprimitive prime factor; if present, it is the greatest prime factor of n and occurs once. At n_* this correction cannot occur. Indeed, squarefreeness would give $n_* = qz(q)$, whence $n_* \leq q(q+1) \leq 10^8(10^8+1)$, contrary to the number of digits.

Theorem 27.4 (Exact surviving simple-primitive failure)**.** *If F_{n_*} has no simple primitive divisor, then C_{n_*} is powerful and every prime $p \mid C_{n_*}$ is Wall–Sun–Sun and satisfies $p \equiv 1 \pmod{n_*}$. At least one such prime satisfies*

$$p \geq 41n_* + 1.$$

Proof. The correction just excluded shows that every factor of C_{n_*} is primitive with its full F_{n_*} multiplicity. Equations (283) and (284) therefore give the powerful and Wall–Sun–Sun assertions. Hong's explicit large primitive-divisor theorem with $\kappa = 40$ applies because $\log n_* > 4398 \log 10 > 10036$ [38]. It produces a primitive prime outside $jn_* \pm 1$ for $1 \leq j \leq 40$. The split congruence leaves only the plus sign, so its coefficient is at least 41. ☐

This theorem sharply locates the surviving bad case but does not contradict it. A broad real-Lucas shortcut is false under its full standard hypotheses: for

$$U_{m+2} = 2U_{m+1} + 3U_m, \qquad U_{10} = 14762 = 2 \cdot 11^2 \cdot 61,$$

the prime 11 is the unique primitive divisor at index ten and occurs exactly twice. The roots are $3, -1$, so the sequence is real, nondegenerate and has coprime parameters. This counterexample closes only the sequence-uniform transfer; it does not satisfy the Fibonacci-specific hypothesis. An exact search over the 252 squarefree $Q \leq 1000$ with $\gcd(Q, 30) = 1$ finds simple primitive certificates for 207 cases below $5 \cdot 10^7$. The remaining 45 cases are unresolved by that finite bound, not counterexamples.

27.5. **The current LANA specification and the pointed same-pilot gate.** The public Project LANA Lean repository was audited at the pinned commit `ddaddc2` [60]. Its declared Corollary 3.12 variant shares initial theta data between a q-pilot scalar and an independently supplied right-hand-side container. At that commit the right-hand-side record has a prior consistency defect.

Theorem 27.5 (Empty-set obstruction in the pinned low-resolution signature)**.** *For every initial theta datum D, the pinned type `RHSData D` is uninhabited.*

Proof. The admissible prime satisfies $\ell \geq 5$, and the required standard procession therefore has positive length. Choose one capsule index and the rational prime $p = 2$. The record asserts that the finite sum of the weights over the p-fibre is one, so that fibre cannot be empty. A constant function from the capsule labels to a point of the fibre supplies a component c.

For this component the record further asserts, for every set U, that

$$\mathrm{vol}((x \mapsto px)^{-1}U) = \mathrm{vol}(U) + \log p. \tag{285}$$

Take $U = \varnothing$. Its inverse image is empty, so cancellation in (285) gives $\log 2 = 0$, contrary to $\log 2 > 0$. ☐

Thus the assembled variant-data type, which contains an `RHSData` field, is also empty. A universal theorem over that type would be vacuous, and there can be no counterexample satisfying all its fields. This result rules out the pinned low-resolution signature as a satisfiable specification. It says nothing against a corrected Haar-volume interface, the published IUT argument, or *abc*. A direct repair restricts (285) to nonempty measurable regions of finite nonzero volume and proves stability of that domain under preimages; alternatively one may use an extended-real logarithmic codomain with explicit zero-volume semantics.

After this repair, the main typing gap remains. The two sides currently share the initial datum but have no field realizing the q-pilot as a region in the right-hand container, no identity for that region's volume, no bridge between the two weight systems, no multiradial output map, no IPL/SHE/APT or Ind3 data, no determinant normalization, and no volume monotonicity theorem. Sharing a parameter supplies no order relation between two independently chosen real outputs.

The July 2026 LANA report isolates a stronger target: for a suitable output choice S, prove equality of maps

$$\eta^q = \eta_S^{\mathrm{anab}} : \mathbb{R}^{\mathrm{val}} \longrightarrow \mathbb{R}^{\mathrm{ss}}.$$

The report states that this equality is not currently proved and does not declare it manifestly false [59]. The numerical conclusion needs less.

Proposition 27.6 (Minimal pointed same-pilot certificate)**.** *Let p_q be the canonical q-pilot point, let ν be the correctly normalized volume coordinate, and put $q_* = -|\log q|$, $T = -|\log \Theta|$. Suppose a globally synchronized, source-permitted output S satisfies*

$$\nu(\eta^q(p_q)) = q_*, \qquad \eta^q(p_q) = \eta_S^{\mathrm{anab}}(p_q), \qquad \nu(\eta_S^{\mathrm{anab}}(p_q)) \leq T.$$

Then $q_* \leq T$.

Proof. Apply ν to the pointed equality and substitute the two inequalities. □

The proposition is elementary after its hypotheses are available. Its content is the construction of the middle equality before taking the volume quotient: the diagram must start with the fixed q-pilot arithmetic line, follow the theta link, IPL/SHE/APT construction, the required indeterminacies, log-Kummer correction and determinant normalization, and return to the same right-hand arithmetic holomorphic structure with compatible local and global metrics. Defining the quotient equality by equality of the desired scalar values would be circular. A source-defined witness in $\Phi(P_q)$ also suffices, but membership in that class already contains the lower volume inequality and must be constructed independently.

The pinned specification failure is therefore a narrow no-go theorem, while the pointed same-pilot construction remains an active positive route. The public report's full map equality is a stronger active target. None of these results yields an unconditional term of type `ABCConjecture` or its negation.

28. Corrected volume holonomy, affine density, and deep-prime escape

This section records the next continuation of the four active routes. The results are deliberately asymmetric. On the positive side we construct an inhabited valuation-ball model for the local volume law, prove an affine squarefree-bulk theorem, and amplify the arithmetic consequences of a hypothetical Pell or Danilov survivor. On the negative side we give complete counterexamples to two scalar reconstruction mechanisms. Neither counterexample satisfies the stronger object-level hypotheses of the corresponding IUT or arithmetic route.

28.1. **A corrected local volume model and its holonomy ledger.** Fix a rational prime p, normalize $|p|_p = p^{-1}$, and for $k \in \mathbb{Z}$ put

$$B_p(k) = \{x \in \mathbb{Q}_p : |x|_p \leq p^{-k}\}.$$

These compact-open balls avoid the empty-set obstruction in Theorem 27.5 while retaining the intended scaling law.

Proposition 28.1 (Valuation-ball transport)**.** *For every $k \in \mathbb{Z}$,*

$$(x \mapsto px)^{-1} B_p(k) = B_p(k-1).$$

Moreover, $B_p(k) \subseteq B_p(m)$ if and only if $m \leq k$. If $\operatorname{vol}(B_p(k)) = p^{-k}$, then

$$\log \operatorname{vol}((x \mapsto px)^{-1} B_p(k)) = \log \operatorname{vol}(B_p(k)) + \log p.$$

The same additive shift survives every normalized finite weighted packet and every positive-length procession average.

Proof. The first assertion is the chain

$$|px|_p \leq p^{-k} \iff p^{-1}|x|_p \leq p^{-k} \iff |x|_p \leq p^{-(k-1)}.$$

The inclusion criterion follows because $p^{-k} \leq p^{-m}$ precisely when $m \leq k$; necessity may also be tested at p^k. The volume formula is $\log p^{-(k-1)} = \log p^{-k} + \log p$. Finally, a common additive shift passes through a weighted sum whose weights total one, and through division by a positive finite procession length. ☐

Let a logarithmic transport from a pointed object x to a pointed object y with coordinate L and shift δ mean

$$L(y) = L(x) + \delta.$$

Composition adds shifts and reversal negates them.

Theorem 28.2 (Zero logarithmic holonomy)**.** *Every transport returning to the same pointed object has total shift zero. Consequently an uncorrected loop whose edge shifts are nonnegative and whose total shift is positive cannot be a same-pilot loop. A correcting edge in a closed loop must contribute the negative of the accumulated shift.*

Proof. If $y = x$, then $L(x) = L(x) + \delta$, and cancellation gives $\delta = 0$. The remaining assertions are immediate applications to a sum of edge shifts. ☐

For rational coefficients the scalar test can be sharpened place by place.

Theorem 28.3 (Prime-local rational ledger)**.** *Let $p_1, \ldots, p_r$ be distinct rational primes and $c_i \in \mathbb{Q}$. If*

$$\sum_{i=1}^{r} c_i \log p_i = 0,$$

then $c_i = 0$ for every i. Hence a closed rational-prime logarithmic transport must cancel separately at every rational place.

Proof. Clear a common denominator and write the resulting coefficients as $m_i \in \mathbb{Z}$. Exponentiation gives $\prod_i p_i^{m_i} = 1$. Moving negative powers to the other side produces an equality of positive integers, and unique factorization forces every m_i to vanish. ☐

The coefficient field and the amount of retained local data are essential. Arbitrary real coefficients are dependent, for example $(\log 3)\log 2 - (\log 2)\log 3 = 0$. Positivity and normalization do not repair the loss of labelled information.

Proposition 28.4 (Positive normalized scalar collision)**.** *There are two different triples of strictly positive real weights, each having sum one, whose weighted averages of* $\log 2, \log 3, \log 5$ *are equal.*

Proof. Put $a = \log 2 < b = \log 3 < c = \log 5$ and define

$$W = \left(\frac{c-b}{2(c-a)}, \frac{1}{2}, \frac{b-a}{2(c-a)} \right), \qquad V = \left(\frac{c-b}{3(c-a)}, \frac{2}{3}, \frac{b-a}{3(c-a)} \right).$$

All six entries are positive, and each row sums to one. More generally, for $n = 2, 3$,

$$\frac{c-b}{n(c-a)} a + \left(1 - \frac{1}{n}\right) b + \frac{b-a}{n(c-a)} c = b,$$

because $(c-b)a+(b-a)c = b(c-a)$. The middle coordinates 1/2 and 2/3 show that the triples differ. $\square$

Proposition 28.4 is a full counterexample to recovering labelled weights from one positive normalized scalar. It does not refute a construction which transports the labelled local objects, nor the stronger same-pilot map equality discussed after Proposition 27.6.

28.2. **Squarefree bulk and the coupled long-arm affine gate.** Return to the minimal radical-step shear in Theorem 27.1. Thus $a+b=c$ is primitive, $R = \operatorname{rad}(abc)$, and

$$U = 1 + Rh, \qquad V = 1 + R(h+ck), \qquad W = 1 + R(h+bk).$$

Proposition 28.5 (Support-closed gap recovery)**.** *For $h,k>0$,*

$$\gcd(V-U, W-U) = Rk, \qquad (a,b,c) = \frac{1}{Rk}(V-W, W-U, V-U).$$

Conversely, let $1 < U < W < V$, put $g = \gcd(V-U, W-U)$, and divide the three consecutive gaps by g to obtain (a,b,c). If $\operatorname{rad}(abc) \mid g$ and $\operatorname{rad}(abc) \mid U-1$, then the triple is exactly a minimal positive-parameter radical-step shear with $k = g/R$ and $h = (U-1)/R$.

Proof. The forward identities follow from $(V-W, W-U, V-U) = Rk(a,b,c)$ and $\gcd(b,c)=1$. Conversely the definition of g makes the normalized gaps integral and primitive, while addition of the first two gives the third. The two divisibilities define integral h,k; the strict inequality $U>1$ gives $h>0$, and substitution recovers the three displayed affine forms. The weaker condition $U \ge 1$ is false: $(U,W,V) = (1,3,5)$ gives $(a,b,c) = (1,1,2)$ and $R = g = 2$, but $h = 0$. $\square$

Write $E(n) = n/\operatorname{rad}(n)$. The inherited three-quarter exceptional inequality in the upper-half canonical box is

$$E(U)E(V)E(W) > \frac{Rc^{14}}{8192},$$

while $U \le c^6$ and $V, W \le c^7$. It follows pointwise that

$$8192E(V) > Rc, \qquad 8192E(W) > Rc. \tag{286}$$

Thus both long arms must have large repeated-prime excess.

The same box nevertheless contains a uniformly positive squarefree bulk. Put

$$M = \left\lfloor \frac{c^6}{4R} \right\rfloor, \qquad I_M = \{\lfloor M/2 \rfloor + 1, \dots, M\}, \qquad N = |I_M|.$$

Theorem 28.6 (Uniform squarefree-admissible bulk)**.** *Fix $\theta < 5/2$. For all sufficiently large primitive seeds satisfying $R < c^\theta$, at least*

$$\frac{1}{2}\left(5 - \frac{\pi^2}{2}\right) N^2$$

pairs $(h,k) \in I_M^2$ obey $\gcd(U,k) = 1$ and have U,V,W all squarefree. Every resulting abc output has radical at least its height and is therefore nonexceptional for every fixed exponent below one.

Proof. For a prime $p \nmid R$, failure of $\gcd(U,k) = 1$ prescribes one residue class for both h and k modulo p, and hence occurs for at most $(N/p+1)^2$ pairs. For a fixed $F \in \{U,V,W\}$, the condition $p^2 \mid F$ prescribes one class of h modulo p^2 after k is fixed, and hence occurs for at most $N^2/p^2 + N$ pairs. Primes dividing R cause neither event. Since $2 \mid R$, the union bound and

$$\sum_{j \ge 1} \frac{1}{(2j+1)^2} = \frac{\pi^2}{8} - 1$$

leave at least

$$\left(5 - \frac{\pi^2}{2}\right) N^2 - 2N(1 + \log M) - M - 3Nc^{7/2}$$

good pairs. Here every possible square divisor is covered because $U, V, W \leq c^7$. The hypothesis $R < c^\theta$ gives $N \gg c^{6-\theta}$, so the three error terms are smaller than half the positive main term when $\theta < 5/2$.

For a good pair, pairwise coprimality and avoidance of the seed support give $\mathrm{rad}(aUbVcW) = RUVW$. Also $U \geq c$ for all sufficiently large seeds in this range, and therefore $RUVW \geq cW$, the output height. □

Theorem 28.6 and (286) point in opposite directions without contradiction. The desired exceptional lower count occupies a vanishing fraction of the raw box, so a positive-density generic bulk cannot be complemented into that lower bound. A proof must create simultaneous high-excess tails in both long arms or exploit the support-closed reverse description in Proposition 28.5.

28.3. **Prime-rank depth and second-order Pell coupling.** Let $u_0 = 0, u_1 = 1, u_{n+2} = 6u_{n+1} - u_n$ and write $(1+\sqrt{2})^n = A_n + B_n\sqrt{2}$. At an odd prime index ℓ, $u_\ell = A_\ell B_\ell$, and the inherited valuation dictionary identifies every odd divisor of either channel with rank ℓ.

Theorem 28.7 (Pointwise channel escape)**.** *For every odd prime ℓ, the A channel has either a prime $q \parallel A_\ell$ or an odd-exponent divisor $p^{e(p)} \mid A_\ell$ with rank ℓ and $e(p) \geq 3$. The same holds for the B channel except for $B_7 = 13^2$. Consequently each channel separately has simple divisors at all but finitely many prime indices, or has infinitely many distinct prime-rank depth-three primes. If the global set of depth-three primes is finite, then both simple divisors occur simultaneously at every sufficiently large prime index.*

Proof. An integer which is not a perfect power either has an exponent-one prime or has an odd prime exponent at least three. Apply this observation to the perfect-power classifications of the two Pell coordinates. Different prime indices give different ranks, hence different primes in the second branch. If only finitely many depth-three primes exist, delete their finitely many ranks and the exceptional index seven; both channels must then take the simple branch at every remaining prime index [45, 46, 47]. □

Factoring the channels and expanding every signed local factor through its quadratic term yields a necessary congruence one digit deeper than the first-order ledger. With the notation K_A, K_B, C_A, C_B for the linear and quadratic factor coefficients, respectively,

$$K_A - 2K_B + \ell(K_A^2 - 2K_B^2) + 2\ell(C_A - 2C_B) \equiv 0 \pmod{4\ell^2}. \tag{287}$$

Quadratic reciprocity also gives the all-pairs identity

$$\prod_{q \mid A_\ell} \prod_{r \mid B_\ell} \left(\frac{q}{r}\right)^{v_q(A_\ell)v_r(B_\ell)} = \left(\frac{2}{\ell}\right).$$

These ledgers constrain a full four-prime packet but do not yet force an inconsistency.

Applying the repaired Fellini–Murty prime-order argument separately to $1+\sqrt{2}$ and its negative produces an alternative: either infinitely many rational balancing primes have depth at least three, or infinitely many prime indices have a simple A-divisor and infinitely many prime indices have a simple B-divisor. The latter two index sets are not asserted to intersect; the synchronized conclusion in Theorem 28.7 uses the stronger finite-depth-three hypothesis instead [39].

An exhaustive certificate checks every one of the 50,847,533 odd primes $q \leq 10^9$. Independent segmented-sieve/Lucas-doubling and dense-sieve/quotient-ring implementations agree: the only depth-two primes are

$$13, \quad 31, \quad 1546463,$$

and each has exact depth two; no depth-three prime occurs. Arbitrary-precision replay verifies the three residues modulo q^3. Thus each depth-three prime required by an actual opposite-channel packet exceeds 10^9. This is a finite lower bound, not a nonexistence result.

28.4. **Divisor-pair amplification on the Danilov survivor.** Let F_n be the Fibonacci sequence. The Danilov factorization has the form

$$K_t = \frac{27}{25} F_N F_{N-5}, \qquad N = 10(3t+1),$$

with $\gcd(F_N, F_{N-5}) = 5$. The final recursive certificate supplies a squarefree, 638-prime, 4398-digit modulus Q_*, coprime to 30, such that $Q_* \mid 3t+1$ throughout the surviving progression.

Theorem 28.8 (Wall–Sun–Sun divisor-pair amplification)**.** *Let $Q > 1$ be squarefree and coprime to 10, let R be a positive multiple of Q, and put $N = 10R$. If $(27/25)F_N F_{N-5}$ is squarefull and Q has s distinct prime factors, then there are at least 2^{s-1} distinct Wall–Sun–Sun primes. They may be indexed by the divisors $d \mid Q$ satisfying $d < Q/d$, and the indexed prime p_d obeys*

$$z(p_d) = N/d, \qquad p_d \equiv 1 \pmod{N/d}, \qquad p_d > N/\sqrt{Q}.$$

For the final Danilov progression this supplies a distinguished subfamily of at least 2^{637} such primes, each exceeding 10^{2199}.

Proof. The involution $d \mapsto Q/d$ pairs the 2^s divisors of nonsquare Q, so exactly 2^{s-1} satisfy $d < Q/d$. Put $m_d = N/d$. Carmichael–Yabuta gives a primitive divisor p_d of F_{m_d} [40]. Since $10 \mid m_d$, its rank is m_d and the split-rank congruence gives $p_d \equiv 1 \pmod{m_d}$. The inequality $d < \sqrt{Q}$ implies $m_d > d$, whence p_d divides neither m_d nor d and therefore does not divide $N = m_d d$.

Now $p_d \mid F_N$ and, because $p_d > 5$, it does not divide F_{N-5} or the rational prefactor. Squarefullness forces $v_{p_d}(F_N) \geq 2$. The exact Lucas valuation formula and $p_d \nmid N$ transfer this valuation to F_{m_d}, making p_d a Wall–Sun–Sun prime [36]. Equality of two selected primes would force equality of their ranks and hence equality of the corresponding divisors, so they are distinct. Finally $p_d \geq N/d + 1 > N/\sqrt{Q}$.

For Q_* take $s = 638$. Since Q_* has 4398 decimal digits and $R \geq Q_*$, the size bound gives $p_d > 10\sqrt{Q_*} > 10^{2199}$. □

The certified upper bound for the prime factors of Q_* permits almost all divisors, rather than only one representative from each complementary pair.

Theorem 28.9 (Factor-bound amplification)**.** *Under the hypotheses of Theorem 28.8, suppose every prime factor of Q is at most B, and put*

$$E_B(Q) = \#\{e : e \mid Q,\ 10e \leq B\}.$$

Then squarefullness forces at least $2^s - E_B(Q)$ distinct Wall–Sun–Sun primes. At the verified endpoint $B = 99{,}966{,}059$ and exact truncated subset-product enumeration gives $E_B(Q_*) = 622$. Thus a survivor forces at least $2^{638} - 622$ distinct Wall–Sun–Sun primes. At least one forced prime is at least $41N + 1 > 10^{4399}$.

Proof. Write $R = kQ$. For every divisor $e \mid Q$ with $10e > B$, put $d = Q/e$ and $m_e = N/d = 10ke$. A primitive divisor p_e of F_{m_e} is congruent to one modulo m_e, so $p_e > m_e > B$. It divides neither Q nor m_e, and hence does not divide $N = m_e d$. The squarefullness and valuation transfer argument from Theorem 28.8 makes it a Wall–Sun–Sun prime. Different e give different ranks and hence different primes. Exactly $E_B(Q)$ of the 2^s divisors were omitted.

For Q_* the certified factor list has maximum $B = 99{,}966{,}059$; exact enumeration below $B/10$ returns 622 divisors. The final large-prime claim uses Hong's primitive-divisor theorem with $\kappa = 40$: since $\log N > 10036$, it supplies a primitive divisor of F_N outside $jN \pm 1$ for $1 \leq j \leq 40$. The split congruence leaves the plus sign and coefficient at least 41, and squarefullness transfers repeatedness as before [38]. □

No known theorem excludes $2^{638} - 622$ Wall–Sun–Sun primes at these sizes, so Theorems 28.8 and 28.9 are strong, explicitly quantified conditional obstructions, not a contradiction. The cyclotomic-derivative shortcut also fails: the complete example

$$\Phi_{10}(-3) = 121 = 11^2, \qquad \Phi_{10}'(-3) = -142, \qquad \gcd(121, 142) = 1$$

shows that a squared value need not come from a multiple root. Hensel uniqueness does not prevent a fixed algebraic unit from landing on the unique lift. This counterexample closes only that derivative mechanism; the Fibonacci and Danilov routes remain active.

28.5. **Formal and logical boundary.** The elementary parts of Propositions 28.1 and 28.4, Theorems 28.2 and 28.3, the affine gap and excess identities, the Pell second-order ledger, and the abstract Danilov divisor-pair transfer have independent Lean implementations. Across the four companion modules these comprise 65 theorem or lemma declarations, 18 definitions, four abbreviations, and five structures, for 92 declarations; 58 embedded `#print axioms` queries report only `propext`, `Classical.choice`, and `Quot.sound`. The squarefree-bulk estimate, the perfect-power classifications and Fellini–Murty argument, the primitive-divisor and valuation theorems, Hong's theorem, and the large finite certificates remain explicit paper, literature, or computation inputs and are not introduced as Lean axioms. The computation packages freeze source, output, byte count, and SHA-256 manifests.

At this stage the four live endpoints are: construct an object-level same-pilot transport satisfying the prime-local zero-holonomy ledger; produce or exclude the simultaneous affine high-excess tails; rule out or construct opposite-channel prime-rank depth-three Pell packets; and exclude or realize the enormous Wall–Sun–Sun population forced by a Danilov survivor. None is closed by difficulty or by a finite no-hit, and no unconditional term of type `ABCConjecture` or its negation is claimed.

29. A literal distinct-prime-factor countermodel and the retained radical target

Carella's current preprint [14], Theorem 5.1(iii), p. 12, prints the hypothesis

$$\#\{n \le x : \omega(n) > w\} = o(x^{3/5}), \qquad w \le 2\log\log x. \tag{288}$$

The prose following the display refers instead to smooth integers in the short interval $[x, x+x^{3/5}]$. The two sets are different. We first audit the displayed global statement literally.

Let $p_1 = 2 < p_2 < \cdots$ be the primes and put $Q_r = \prod_{j=1}^{r} p_j$.

Theorem 29.1 (Primorial-multiple counterfamily)**.** *For a real number $w \ge 0$, let $r = \lfloor w \rfloor + 1$. Then, for $x \ge Q_r$,*

$$\#\{n \le x : \omega(n) > w\} \ge \left\lfloor \frac{x}{Q_r} \right\rfloor. \tag{289}$$

If $w = w(x) \le 2\log\log x$ for all sufficiently large x, then the left side is $x^{1-o(1)}$ and is therefore not $o(x^{3/5})$.

Proof. For every $1 \le m \le \lfloor x/Q_r \rfloor$, the integer mQ_r is at most x and is divisible by the r distinct primes $p_1, \ldots, p_r$. Thus $\omega(mQ_r) \ge r > w$, and multiplication by the positive integer Q_r is injective. This proves (289).

Bertrand's postulate gives $p_{j+1} < 2p_j$, hence $p_j \le 2^j$ and

$$\log Q_r \le \frac{r(r+1)}{2} \log 2.$$

Under $w(x) \le 2\log\log x$, this is $O((\log\log x)^2) = o(\log x)$. Consequently $Q_r = x^{o(1)}$, and the lower bound in (289) is $x^{1-o(1)}$. Dividing it by $x^{3/5}$ gives $x^{2/5-o(1)} \to \infty$. $\square$

Thus (288) is false with all of its displayed conditions. Moreover, Theorem 5.1 is conditional on that statement, while the subsequent proof of the preprint's Theorem 1.1 treats the conditional result as available for every large x. The claimed unconditional conclusion does not follow.

This literal counterfamily is separate from the earlier local audit in Section 2. If the display is repaired by restricting to B-smooth integers in $[x, x+x^{3/5}]$, the theorem above no longer applies literally. Theorem 2.3 then shows that almost every smooth integer in the relevant Younis regime has much more than $2\log\log x$ distinct prime divisors, contradicting the proposed majority and moment mechanism. Neither result excludes one exceptionally sparse low-radical neighbour in infinitely many prime-power-centred intervals.

The final radical calculation of the proposed construction is in fact valid under such an existence premise. The following formulation keeps the larger route without requiring smoothness or a bound on ω.

Theorem 29.2 (Three-fifths radical-neighbour gate)**.** *Fix $k \geq 1$, $\sigma \geq 0$, and suppose*

$$\sigma < \frac{2}{5} - \frac{1}{k}. \tag{290}$$

If there are infinitely many primes p and integers c, at unbounded $X = p^k$, for which

$$X < c \leq X + X^{3/5}, \qquad p \nmid c, \qquad \mathrm{rad}(c) \leq X^{\sigma+o(1)}, \tag{291}$$

then the standard abc conjecture is false.

Proof. Set $a = c - p^k$ and $b = p^k$. Since $p \nmid c$, the three entries are pairwise coprime. Radical submultiplicativity gives

$$\mathrm{rad}(abc) \leq a\, p\, \mathrm{rad}(c) \leq X^{3/5+1/k+\sigma+o(1)}.$$

Condition (290) makes the fixed exponent $q = 3/5 + 1/k + \sigma$ strictly smaller than one. Choose $\varepsilon > 0$ with $(1+\varepsilon)q < 1$. After absorbing the $o(1)$ term into half of the positive gap,

$$\frac{c}{\mathrm{rad}(abc)^{1+\varepsilon}} \geq X^{(1-(1+\varepsilon)q)/2} \longrightarrow \infty.$$

This contradicts the existence of the uniform *abc* constant for that fixed ε. $\square$

The conclusion shows that the condition $\mathrm{rad}(c) = X^{o(1)}$ is stronger than needed. At $k = 4$, for example, every fixed $\sigma < 3/20$ suffices; the allowed upper exponent tends to $2/5$ as k grows.

There is an exact extra restriction if $k \geq 3$ and one asks for candidates at a positive proportion of prime centres. The Rankin radical count proved in the underlying research report gives

$$\#\{n \leq X : \mathrm{rad}(n) \leq X^{\sigma}\} \ll_\eta X^{\sigma+\eta}.$$

Disjointness of the $X^{3/5}$ intervals and the prime number theorem then force $\sigma \geq 1/k$.

Proposition 29.3 (Exact density-compatible exponent window)**.** *There exists a real σ such that*

$$\frac{1}{k} \leq \sigma < \frac{2}{5} - \frac{1}{k} \tag{292}$$

if and only if $k > 5$. For every integer $k \geq 6$, the uniform choice $\sigma = 1/5$ is feasible and gives

$$\frac{3}{5} + \frac{1}{k} + \frac{1}{5} \leq \frac{29}{30} < 1.$$

Proof. The interval in (292) is nonempty precisely when $1/k < 2/5 - 1/k$, equivalently $2/k < 2/5$, or $k > 5$. If $k \geq 6$, then $1/k \leq 1/6 < 1/5$ and the displayed total-slope bound follows. $\square$

For $k = 1, 2$, the target (290) is already empty because $\sigma \geq 0$. The positive-density versions with $3 \leq k \leq 5$ are rejected by the counting contradiction. A zero-density unbounded subsequence is not rejected for $3 \leq k \leq 5$. For $k \geq 6$, Proposition 29.3 provides a concrete positive target that current density estimates do not forbid. What remains is an arithmetic theorem producing the neighbours, not another manipulation of the final radical budget.

30. A faithful prime-unit-label vector bridge

The scalar holonomy ledger in Section 28 cannot reconstruct a pilot: two different positive normalized labelled packets may have the same one-dimensional weighted value. We therefore retain the rational prime, complete unit part, and fixed label until after the object comparison. IUT III [62] distinguishes unit/coric data and mono-analytic log-shells before the final log-volume operation. The pinned Project LANA container [59, 60] likewise retains the rational place fibre and procession labels in a packet component. These sources motivate the coordinate type; they do not prove that the actual multiradial transport preserves the coordinates below.

Fix a rational prime p. For $x \in \mathbb{Q}^\times$, put

$$e_p(x) = v_p(x), \qquad u_p(x) = \frac{x}{p^{e_p(x)}}, \qquad C_p(x) = (e_p(x), u_p(x)). \tag{293}$$

Negative powers have their ordinary field meaning.

Theorem 30.1 (Exact prime-unit reconstruction)**.** *For every $x \in \mathbb{Q}^\times$,*

$$v_p(u_p(x)) = 0, \qquad p^{e_p(x)}u_p(x) = x. \tag{294}$$

Consequently C_p is injective. The identical assertions hold on the actual complete local field $\mathbb{Q}_p^\times$ with its normalized additive valuation.

Proof. Valuation additivity and $v_p(p) = 1$ give

$$v_p(u_p(x)) = v_p(x) - e_p(x)v_p(p) = 0.$$

The second equality in (294) is cancellation of the same nonzero field power used in the definition. Equal coordinate pairs therefore have equal reconstruction expressions. This proves injectivity. The proof uses only the corresponding field and valuation identities in $\mathbb{Q}_p$ and is unchanged there. $\square$

There is also a valuation-free field-summand template. If K is any field, $\pi \in K^\times$, and $e : K^\times \to \mathbb{Z}$ is any function, then

$$u_{\pi,e}(x) = \frac{x}{\pi^{e(x)}} \neq 0, \qquad \pi^{e(x)}u_{\pi,e}(x) = x. \tag{295}$$

Thus $x \mapsto (e(x), u_{\pi,e}(x))$ is injective. When e is a normalized discrete valuation and π a uniformizer, the first calculation in Theorem 30.1 additionally proves that the second coordinate is a genuine unit. Hence the reconstruction mechanism applies separately to every actual field summand once that summand and its transport have been constructed.

Let L be a label set and let a packet be a map $P : L \to \mathbb{Q}^\times$. Define its complete labelled signature by

$$\Sigma_p(P) = \big(C_p(P(i))\big)_{i\in L}. \tag{296}$$

Theorem 30.2 (Labelled and region reconstruction)**.** *The map Σ_p is injective. More generally, let X be a carrier with a family of exponent and unit coordinates whose joint value is faithful at each point, and let Σ be the corresponding labelled signature. For packet regions $A, B \subseteq (L \to X)$,*

$$\Sigma(A) \subseteq \Sigma(B) \implies A \subseteq B. \tag{297}$$

If μ is a monotone finite real-valued region functional, then $\mu(B) \le T$ and (297) imply $\mu(A) \le T$.

Proof. Equality of two signatures gives equality of both local coordinates at every fixed label. Theorem 30.1, or the assumed pointwise faithfulness, gives equality of the packet entries and then of the packets.

For the region statement, take $P \in A$. Image containment supplies $Q \in B$ with $\Sigma(Q) = \Sigma(P)$. Injectivity gives $Q = P$, hence $P \in B$. Monotonicity of μ and transitivity of the real order give the last assertion. $\square$

This is a non-circular route from vector data to a scalar bound. It does not assume equality of the regions or equality of their volumes. It requires an independently proved containment of complete signature images, followed by a monotonicity theorem on the correct finite-positive-volume domain.

The unit coordinate also completes the holonomy ledger. Suppose a labelled transport $P \to Q$ has exponent shifts $\delta_i \in \mathbb{Z}$ and unit twists $\tau_i \in \mathbb{Q}$ satisfying

$$e_p(Q(i)) = e_p(P(i)) + \delta_i, \qquad u_p(Q(i)) = u_p(P(i))\tau_i. \tag{298}$$

Proposition 30.3 (Pointed vector holonomy)**.** *Every twist in* (298) *is nonzero and has p-adic valuation zero. Transports compose labelwise by*

$$(\delta, \tau) \circ (\epsilon, \upsilon) = (\delta + \epsilon, \tau\upsilon).$$

If the labelled transport is a pointed loop, $Q = P$, then $\delta_i = 0$ and $\tau_i = 1$ for every fixed label i.

Proof. Both unit parts in the second equation of (298) are nonzero, so $\tau_i \neq 0$. Applying v_p and Theorem 30.1 gives $0 = 0 + v_p(\tau_i)$. Substitution of two consecutive transport equations gives the composition law. If $Q = P$, cancellation in the exponent equation gives $\delta_i = 0$, and cancellation of the nonzero unit part gives $\tau_i = 1$. □

Each of the three retained layers is necessary for this particular reconstruction theorem.

Proposition 30.4 (Complete counterexamples to three weakened interfaces)**.** *The following data satisfy all the indicated weakened hypotheses.*

(1) *At $p = 5$, the one-label packets $P = 1$, $Q = 2$, with weight one, have the same exponent zero, the same fixed label, and the same positive normalized weight, but are unequal.*
(2) *At $p = 5$, the one-label packets $P = 1$, $Q = 6$, with weight one, have equal exponent zero and equal first unit residue modulo* 5*, but are unequal.*
(3) *At $p = 5$, take $P = (1, 2)$, $Q = (2, 1)$ and weights* $(1/2, 1/2)$*. Their multisets of complete coordinates agree after transposition. At the fixed labels the exponent shifts vanish and the unit twists are* $(2, 1/2)$*, whose product is one, but the labelled packets differ and neither twist is one.*

Proof. The integers $1, 2, 6$ are 5-adic units, and $1 \equiv 6 \pmod{5}$. The stated weights are positive and sum to one. In the two-label example, transposition exchanges the two complete coordinates, while at fixed labels the quotient unit twists are $2/1$ and $1/2$. Their product is one and each is nontrivial. Every asserted inequality is immediate. □

The first example closes only exponent-only reconstruction; the second closes only first-residue truncation; the third closes only unordered reconstruction and aggregate unit-holonomy closure. The third example does not refute a transport that records and returns an explicit label permutation.

The active IUT target is therefore a source-level theorem on the actual common carrier: through the theta link, log-Kummer correction, determinant normalization, and every required Ind1–Ind3 branch, the globally synchronized output must preserve the faithful prime-unit coordinate at every required place and every fixed label, or at least contain the input signature image. Theorem 30.2 would then transfer the output bound back to the native pilot before taking scalar log-volume. No such global preservation theorem is proved here, and none of the finite counterexamples disproves it. The IUT route remains active.

31. Refined-factor zero-aware packet signatures

The prime-unit signature of Section 30 must be corrected at two source-level seams. A place-tuple tensor algebra need not be a field: IUT III, Proposition 3.1, splits it into a direct sum of field factors [62]. Moreover its integral packet includes zero, so a coordinate defined only on the multiplicative group is not a faithful coordinate on the source carrier.

Let K be a field, choose $\pi \in K^\times$, and allow an arbitrary map $e : K^\times \to \mathbb{Z}$. Define

$$\mathcal{C}_{\pi,e}(x) = \begin{cases} \mathsf{none}, & x = 0, \\ \mathsf{some}\left(e(x), x/\pi^{e(x)}\right), & x \neq 0. \end{cases}$$

Theorem 31.1 (Zero-aware reconstruction)**.** *The map $\mathcal{C}_{\pi,e}$ is injective. Its reconstruction map is*

$$\mathsf{none} \longmapsto 0, \qquad \mathsf{some}(n, u) \longmapsto \pi^n u.$$

No valuation property of e is needed for injectivity.

Proof. The zero tag separates zero from every nonzero input. For $x \neq 0$, $\pi^{e(x)}(x/\pi^{e(x)}) = x$. Applying reconstruction to equal coordinates therefore recovers equal inputs. □

If e is a normalized discrete valuation and π a uniformizer, the second component is the complete unit part. The signature is functorial only after the necessary multiplicative data have been supplied.

Theorem 31.2 (Factorwise covariance)**.** *Let $\phi : K \simeq K'$ be a field equivalence. Suppose the target scale is $\phi(\pi)$ and*

$$e'(\phi(x)) = e(x) \qquad (x \neq 0).$$

Then

$$\mathcal{C}_{\phi(\pi),e'}(\phi(x)) = \widehat{\phi}\big(\mathcal{C}_{\pi,e}(x)\big),$$

where $\widehat{\phi}$ fixes the exponent and applies ϕ to the second coordinate.

Proof. The zero case is immediate. For $x \neq 0$, preservation of division and integer powers gives

$$\frac{\phi(x)}{\phi(\pi)^{e(x)}} = \phi\left(\frac{x}{\pi^{e(x)}}\right).$$

□

Fix one finite-dimensional stage of the source inductive system. Let C be its place-tuple index and let A_c be the associated finite étale algebra. The correct refined index at this stage is

$$D = \coprod_{c \in C} \operatorname{MaxSpec}(A_c), \qquad L_{(c,\mathfrak{m})} = A_c/\mathfrak{m}.$$

The semisimple decomposition gives an algebra equivalence

$$\Psi : \prod_{c \in C} A_c \simeq \prod_{d \in D} L_d.$$

Choose nonzero scales and exponent maps on every L_d.

Corollary 31.3 (Faithful refined packet signature)**.** *The map*

$$x \longmapsto (\mathcal{C}_{\pi_d, e_d}(\Psi(x)_d))_{d \in D}$$

is injective. It is covariant under a reindexing $D \simeq D'$ and a family of field equivalences satisfying the scale and exponent hypotheses of Theorem 31.2.

Proof. Each factor coordinate is injective by Theorem 31.1; equality in the dependent product is componentwise. Compose with the injective algebra equivalence Ψ. Covariance is the factorwise theorem after reindexing. □

If the target uniformizer is chosen independently, a unit twist must be recorded. With $\eta_d = \phi_d(\pi_d)/\pi'_{\sigma(d)}$, the unit coordinate transforms as

$$u'_{\sigma(d)}(\phi_d(x)) = \phi_d(u_d(x))\eta_d^{e_d(x)}.$$

Raw equality of units is therefore too strong unless $\eta_d = 1$.

The source currently transports less information. IUT III, Theorem 3.11(ii)(a), supplies chosen-branch poly-isomorphisms and compatible log-volumes; Proposition 3.9 supplies the relevant permutation invariance. The indeterminacy (Ind3) is described by inclusions or surjections, so its faithful abstraction is a reachability relation followed by image containment. These statements do not by themselves produce the componentwise field equivalences and valuation covariance required above. The zero-labelled singleton core from IUT I, Corollary 6.10(iii), survives as an object-level remnant [63], but it does not select a complete unit coordinate.

At the integral level, let R be a discrete valuation ring with fraction field F, let each K_i/F be finite separable, and set $T = \bigotimes_R \mathcal{O}_{K_i}$ and $B = \prod_m \mathcal{O}_{L_m}$ inside $\bigotimes_F K_i \simeq \prod_m L_m$, then

$$T \subseteq B, \qquad B/T \text{ is finitely generated torsion over } R,$$

Equality can fail in a ramified quadratic example; the resulting Fitting ideal records an index correction. Relating that ideal to a different or conductor requires an additional theorem.

The counterexample audit fixes the scope of the route. The decomposition $E \otimes_F E \simeq E \times E$ refutes one-field-per-tuple indexing. A determinant-one shear preserves a lattice and Haar volume while changing a component valuation, so volume compatibility alone does not imply signature covariance. Equal determinants do not reconstruct rank-two lattices, and a proper sublattice with surjective coordinate projections need not be a product of independent exponent enlargements. These examples reject only the weakened interfaces just named. The current LANA strand remains a specification rather than a construction of the multiradial output [59, 60]; no full-premise counterexample to the source-level refined covariance target is known, so the IUT route remains active.

32. Audit of the current concrete IUT implication

After the preceding packet work was completed, the public LANA IUT repository added a conditional implication strand. We audited commit `6e963070`. Its main declaration has the schematic type

$$\begin{aligned}
&\mathsf{P}(T) \to (\forall K, d,\ \mathsf{C}_{22}(T,K,d)) \to \\
&\quad (\forall K, d,\ \mathsf{E}(K,d)) \to \mathsf{Cheb} \to \mathsf{PNT} \to \\
&\quad (\forall D, LT, TL, QI,\ \mathsf{C}_{312}(D, LT, TL, QI)) \to \\
&\quad \mathtt{ABC}(T),
\end{aligned} \tag{299}$$

Here $\mathsf{P}(T)$ abbreviates $\mathtt{T.ProofPackage}$, $\mathsf{C}_{22}(T,K,d)$ abbreviates $\mathtt{Corollary22Inputs}(T,K,d)$, and $\mathsf{Cheb}, \mathsf{PNT}$ denote the two stated prime bounds. The remaining abbreviations are

$$\begin{aligned}
\mathsf{E}(K,d) &:= \mathtt{ConcreteThetaDataExistence}(I(K,d)),\\
\mathsf{C}_{312}(D, LT, TL, QI) &:= \mathtt{Corollary312Variant}\big(\\
&\qquad\qquad \mathtt{concreteVariantData}(D, LT, TL, QI)\big).
\end{aligned}$$

Here $LT : \mathtt{LocalTheory}(D.K_t)$ and $TL : \mathtt{ThetaLocalData}(D, LT)$. Moreover,

$$\mathtt{ABC}(T) := T.\mathtt{StatementI}.$$

Thus (299) is a proved conditional implication, not a construction of its inputs and not a closed proof of the integer statement (1).

The proof graph is nevertheless useful. Concrete theta-data existence is converted to theta-data existence restricted by a predicate; its invariants, certificate and local estimate enter Theorem 1.10; the assumed Corollary 3.12 variant and the prime bounds yield Corollary 2.2; finite exceptional sets are absorbed to prove $T.\mathtt{StatementII}$; finally the `genl` proof package proves

$$T.\mathtt{StatementII} \Longrightarrow T.\mathtt{StatementI}.$$

Every source assumption remains visible in the type.

The new concrete interface has, however, a complete elementary counterexample. Its local log-volume field requires, for every finite label type ι, every prime p, every place tuple c, and every set U in the tensor packet,

$$V\big((x \mapsto px)^{-1}U\big) = V(U) + \log p, \qquad V : \mathcal{P}(A) \longrightarrow \mathbb{R}. \tag{300}$$

Theorem 32.1 (Empty-set obstruction to the current `LocalTheory`)**.** *At the audited commit,* `LocalTheory`(K) *is empty for every number field K.*

Proof. Take the empty finite label type, its unique place tuple, $p = 2$, and $U = \varnothing$. The preimage of the empty set under every function is empty, so (300) becomes

$$V(\varnothing) = V(\varnothing) + \log 2.$$

As $V(\varnothing)$ is a finite real number, subtraction gives $\log 2 = 0$, contradicting $\log 2 > 0$. □

This counterexample satisfies every premise of the displayed volume law. It therefore rejects that exact total-real-valued specification. It does not reject the Haar scaling identity on nonempty measurable sets of finite positive measure. The upstream change that adds nonemptiness to the packet-product combination law does not repair (300), where U remains unrestricted. Each nonvacuous witness requested by `ConcreteThetaDataExistence` contains a term of `LocalTheory`; consequently the advertised concrete branch cannot currently be instantiated. The definition is implication-valued and may still hold vacuously in a degenerate model with no qualifying input point, which is not the intended theta-data construction.

There is a second independent boundary. The repository target `ABCConjecture` is the explicit integer inequality (1), whereas T.`StatementI` is an abstract bounded-discrepancy inequality for hyperbolic curves. No checked declaration in the audited upstream snapshot proves

$$T.\texttt{StatementI} \Longrightarrow \texttt{ABCConjecture}. \tag{301}$$

A proof of (301) must construct the intended height theory, send every primitive positive triple to a degree-one tripod point, compare the canonical height with $\log \max(a, b, c)$, compare log-different plus log-conductor with $\log \operatorname{rad}(abc)$, and unpack bounded discrepancy into one uniform additive constant.

We therefore retain the reproducible pin `ddaddc274281adb5674d647e24fa478745ac6d40`. A direct update would also change packet summands from fields to commutative rings, integral subrings to sets, and nonzero hull scales to units. The refined-factor signature then needs an explicit equivalence of each tensor packet with a finite product of fields. A successful compilation after an immediate pin change would not cure the empty-set contradiction.

The constructive continuation is precise:

(1) restrict logarithmic volume to a proof-carrying domain of measurable finite positive-volume regions, and prove closure under the scaling preimages;
(2) construct an inhabited local theory on that corrected domain;
(3) provide the finite-product-of-fields decomposition needed by the refined zero-aware coordinates;
(4) construct the compatible 2ℓ-th Tate-parameter roots and the remaining concrete theta data;
(5) prove the actual Corollary 3.12 statement; and
(6) prove the integer bridge (301).

Difficulty of these steps does not eliminate the corrected IUT route.

The companion Lean audit proves the universal empty-set theorem for any real-valued set-volume with the unrestricted prime-preimage law, specializes the already pinned public no-go, and introduces the honest proposition

$$\texttt{StatementIToIntegerABCBridge}(T) := T.\texttt{StatementI} \to \texttt{ABCConjecture}.$$

It proves only that a supplied term of this bridge consumes a supplied term of T.`StatementI`. No bridge or IUT premise is populated by a hidden axiom.

33. An admissible log-volume model and an explicit integer bridge

The empty-set obstruction in Section 32 identifies a defect in one total real-valued interface, rather than in Haar scaling. We give an explicit extensional repair and then state the exact comparison data that transfer the abstract general-position inequality to the repository's integer target.

Fix $s : X \to X$ and $\delta \in \mathbb{R}$. An admissible preimage log-volume consists of an inhabited region type $\mathcal{R}$, an injective carrier map $C : \mathcal{R} \to \mathcal{P}(X)$ with $C(U) \neq \varnothing$, a map $P : \mathcal{R} \to \mathcal{R}$, and $V : \mathcal{R} \to \mathbb{R}$ such that

$$C(PU) = s^{-1}C(U), \qquad V(PU) = V(U) + \delta. \tag{302}$$

Injectivity prevents assigning several volumes to the same underlying set; the typed operation P supplies the required closure proof.

Theorem 33.1 (Valuation-ball model)**.** *For every rational prime p, the interface* (302) *with $X = \mathbb{Q}_p$, $s(x) = px$, and $\delta = \log p$ is inhabited.*

Proof. For $k \in \mathbb{Z}$, let

$$B_k = \{x \in \mathbb{Q}_p : |x|_p \leq p^{-k}\}, \qquad P(k) = k - 1, \qquad V(k) = -k \log p.$$

Every ball contains zero. Since $|p|_p = p^{-1}$, $px \in B_k$ if and only if $x \in B_{k-1}$, proving carrier closure. The inclusions among these balls recover the integer index, so the carrier map is injective. Finally $V(k-1) = -k\log p + \log p = V(k) + \log p$. □

The shift law also imposes an exact consistency condition. Induction gives

$$V(P^n U) = V(U) + n\delta. \tag{303}$$

If $\delta \neq 0$, no positive iterate can fix U; if $P^m U = P^n U$, applying V and cancelling δ gives $m = n$. Every orbit is therefore infinite. This supplies full-premise contradictions to proposed repairs with a fixed region, a positive-period cycle, or only finitely many regions. The valuation lattice passes all three tests. It is a model only of this local scaling layer, not a construction of LANA's tensor packets or of IUT.

We next make the abstract-to-integer bridge non-tautological. Let T be a height theory, let P_{abc} be the set of positive primitive triples, and put

$$H(a,b,c) = \log \max(a,b,c), \qquad R(a,b,c) = \log \operatorname{rad}(abc).$$

A uniform tripod comparison consists of an encoding $\iota : P_{abc} \to T.\mathrm{Pt}(T.\mathrm{tripod})$, membership of every $\iota(t)$ in the degree-at-most-one set, and constants $A_H, A_R \in \mathbb{R}$ such that

$$H(t) \leq h_{\mathrm{can}}(\iota(t)) + A_H, \tag{304}$$

$$\log \operatorname{Diff}(\iota(t)) + \log \operatorname{Cond}(\iota(t)) \leq R(t) + A_R. \tag{305}$$

Theorem 33.2 (Bounded-discrepancy transfer)**.** *If T.StatementI and a uniform tripod comparison are supplied, then the repository's exact `ABCConjecture` holds.*

Proof. Fix $\epsilon > 0$. Apply T.StatementI to the tripod, degree bound one, and ϵ. It supplies C_ϵ such that every encoded point satisfies

$$h_{\mathrm{can}}(\iota(t)) \leq (1+\epsilon)(\log \operatorname{Diff}(\iota(t)) + \log \operatorname{Cond}(\iota(t))) + C_\epsilon.$$

Since $1 + \epsilon > 0$, the two comparisons give

$$H(t) \leq (1+\epsilon)R(t) + C_\epsilon + A_H + (1+\epsilon)A_R.$$

The final three terms form one constant independent of t, which is exactly the required quantifier order. □

Each field is logically necessary at this level. For example, on $\mathbb{N}$, $h = k = R = 0$ and $H(n) = n$ satisfy the source inequality and radical comparison but refute every uniform target without the height comparison. Taking $h = k = H = n$ and $R = 0$ does the same without the radical comparison. Restricting the source inequality to the empty set shows why degree membership is necessary, and replacing A_H by the point-dependent error $A_H(n) = n$ destroys uniformity. These are full-premise countermodels within this bare sequence-transfer schema to the corresponding weakened bridges. They are not abc counterexamples and do not retire the IUT/same-pilot research route. Constructing the intended degree-one encoding and proving (304)– (305) remain open arithmetic work.

34. The exact rational-tripod shadow and its semantic boundary

The comparison in Section 33 has an exact arithmetic realization on the open rational tripod

$$U(\mathbb{Q}) = \{x \in \mathbb{Q} : 0 < x < 1\}.$$

For a positive primitive triple $t = (a, b, c)$, put $\lambda(t) = a/c$. Conversely, if $x = m/n$ is reduced in $U(\mathbb{Q})$, set $t(x) = (m, n - m, n)$. These maps are inverse. With H_{tr} the rational Weil height and N_{tr} the truncated counting function for the three marked points $0, 1, \infty$, one has

$$H_{\mathrm{tr}}(\lambda(t)) = \log c = \log \max(a, b, c), \qquad N_{\mathrm{tr}}(\lambda(t)) = \log \operatorname{rad}(abc). \tag{306}$$

The first identity follows because a/c is reduced and $0 < a < c$; the second follows because the supports of a/c and $1 - a/c = b/c$ have union exactly the prime support of abc.

Define a one-curve shadow height theory T_{rat}. Its point set is $U(\mathbb{Q})$; its degree-at-most-zero locus is empty and every point has exact degree one; and

$$h_{\mathrm{can}} = H_{\mathrm{tr}}, \qquad \log \mathrm{Diff} = 0, \qquad \log \mathrm{Cond} = N_{\mathrm{tr}}.$$

The remaining structural fields have elementary witnesses. The identity map on $U(\mathbb{Q})$ therefore gives a degree-one comparison, while (306) gives both normalization errors equal to zero.

Theorem 34.1 (Exact shadow boundary)**.** *Statement I for T_{rat} is equivalent to the standard integer abc conjecture:*

$$T_{\mathrm{rat}}.\mathrm{StatementI} \iff \mathrm{ABCConjecture}.$$

Proof. The forward implication is the bounded-discrepancy transfer of Section 33, with both errors zero. For the reverse implication, integer *abc* transfers through the inverse reduced triple $t(x)$ to the uniform rational-tripod inequality. Since the shadow has one curve, the degree-zero case is empty and every positive degree bound contains all of $U(\mathbb{Q})$, that inequality is exactly its Statement I. □

For an arbitrary height theory T, an open-rational tripod comparison is a map from $U(\mathbb{Q})$ to the degree-at-most-one points of the distinguished tripod, together with uniform height and radical errors. Composition with $t \mapsto a/c$ turns this into the integer comparison package. Composition with $x \mapsto t(x)$ gives the converse, with the same constants. Thus the rational and integer semantic comparison packages have exactly the same logical strength.

Three full-premise models delimit what the current bare structural fields can prove. First, a one-point theory with every degree locus empty satisfies Statement I vacuously but admits no comparison from the nonempty set $U(\mathbb{Q})$. Second, on the actual rational points take

$$h_{\mathrm{can}} = 0, \quad \log \mathrm{Diff} = 0, \quad \log \mathrm{Cond} = N_{\mathrm{tr}}.$$

Statement I, degree membership, and the radical identity all hold, but no uniform height error exists because $H_{\mathrm{tr}}(1/(n+1)) = \log(n+1)$ is unbounded. Third, take

$$h_{\mathrm{can}} = H_{\mathrm{tr}}, \quad \log \mathrm{Diff} = e^{2H_{\mathrm{tr}}}, \quad \log \mathrm{Cond} = N_{\mathrm{tr}}.$$

Statement I follows from $e^{2H} \geq 1 + 2H \geq H$ for $H \geq 0$, and the height comparison is exact, but a uniform radical error would bound $e^{2H_{\mathrm{tr}}}$, again impossible.

These are counterexamples to automatic degree realization and to automatic uniform normalization from the present abstract fields. They are not counterexamples to the intended arithmetic height theory, IUT, Statement I for that theory, or *abc*. The positive route remains the source-level construction of the rational point, its degree-one membership, and the two triple-independent normalization constants in the intended objects. The Lean companion constructs the shadow, proves Theorem 34.1, and verifies all three pressure models without placeholders or new axioms.

35. A source-faithful rational degree-one tripod realization

The shadow construction of Section 34 can be compared directly with the source definitions in Mochizuki's *Arithmetic Elliptic Curves in General Position*. Put

$$P = \mathbb{P}^1_{\mathbb{Q}}, \qquad C = \{0, 1, \infty\}, \qquad U = P \setminus C, \qquad U_{(0,1)}(\mathbb{Q}) = \{x \in \mathbb{Q} : 0 < x < 1\}.$$

The map $j(x) = [x : 1]$ is an injective map to $U(\mathbb{Q})$. Its points have minimal field of definition $\mathbb{Q}$, and hence exact degree one. The recurrence in the abstract LANA height theory already forces

$$\mathrm{ptLE}(X, 1) = \mathrm{ptEQ}(X, 1), \tag{307}$$

because the degree-at-most-zero locus is empty.

Let h be the normalized logarithmic Weil height and let n be the truncated count for the three marked points. For the genuine arithmetic objects, one has

$$\mathrm{ht}_{\omega_P(C)} \circ j \approx h, \qquad \log\text{-diff}_P \circ j = 0, \qquad \log\text{-cond}_C \circ j \approx n. \tag{308}$$

Indeed, $\omega_P(C) \cong \mathcal{O}_{\mathbb{P}^1}(1)$, and invariance of the height BD class under isomorphism gives the first comparison. The different ideal of $\mathbb{Q}/\mathbb{Q}$ is the unit ideal, giving the middle equality. Finally, if $x = m/r$ is reduced with $0 < m < r$, pulling $0, 1, \infty$ back to $\operatorname{Spec} \mathbb{Z}$ gives the divisors of $m, r - m, r$. Reduction of the pullback divisor therefore gives

$$\log\text{-cond}_C(j(x)) = \sum_{p \mid m(r-m)r} \log p = n(x)$$

for the standard model. Changing the proper flat model changes this function only by bounded discrepancy.

For an abstract height theory T, call a map $j : U_{(0,1)}(\mathbb{Q}) \to T.\mathrm{Pt}(T.\mathrm{tripod})$ a source-faithful degree-one realization when it has exact-degree-one image and satisfies the three relations in (308) after replacing the arithmetic functions by the corresponding fields of T. Write H_T, D_T, N_T for the three pulled-back functions. Define the restricted Statement I by

$$\forall \epsilon > 0 \; \exists C_\epsilon \; \forall x, \qquad H_T(x) \le (1 + \epsilon)(D_T(x) + N_T(x)) + C_\epsilon. \tag{309}$$

We denote this proposition by $\mathrm{RStmtI}(T, j)$.

Theorem 35.1 (Source-faithful degree-one transport)**.** *For every source-faithful degree-one realization,*

$$\mathrm{RStmtI}(T, j) \iff \mathrm{ABCConjecture}.$$

Consequently, full T.StatementI *together with such a realization implies the integer abc conjecture.*

Proof. Choose uniform constants A_+, A_-, B_+, B_- such that

$$H_T \le h + A_+, \quad h \le H_T + A_-, \qquad N_T \le n + B_+, \quad n \le N_T + B_-.$$

If (309) holds, then, using $D_T = 0$,

$$h(x) \le (1 + \epsilon)n(x) + C_\epsilon + A_- + (1 + \epsilon)B_+.$$

This is the uniform rational-tripod formulation of integer *abc*. Conversely, that formulation gives

$$H_T(x) \le (1 + \epsilon)N_T(x) + C_\epsilon + A_+ + (1 + \epsilon)B_-,$$

which is (309). Finally, full Statement I restricts along j because (307) puts every image point in its degree-one domain. $\square$

This theorem closes the elementary source normalization but not the IUT argument. At the pinned versions, `Genl.HeightTheory` is only an abstract record, no genuine arithmetic instance is constructed, and the pinned IUT repository has no connection to Statement I. A later public IUT snapshot has only a conditional connection that still takes the height theory, its proof package, and the IUT and analytic inputs as premises.

The distinction cannot be erased by adding the abstract proof package. The degree-empty, height-zero, and different-inflated pressure models all admit full `ProofPackage` instances. They show respectively that this package does not force rational degree-one points, the canonical height normalization, or the rational different. A separate conductor-inflated model preserves exact points, degree, height, and zero different while destroying every uniform upper conductor comparison. These are full-premise countermodels to their precisely stated weakened extraction claims; they do not contradict the genuine computations above, IUT, or *abc*. For clarity, the first three models retain the abstract proof package. The conductor-inflated model retains the height theory, Statement I, and the stated point, degree, height, and different data, but no proof-package instance is claimed for it.

The Lean companion proves the recurrence (307), formalizes the abstract transport theorem, constructs the rational-shadow consistency witness, and checks the pressure models without placeholders or new axioms. Constructing the genuine arithmetic height-theory instance and proving its Statement I remain active upstream tasks.

36. Admissible scaling at the current IUT interface and an exact order index

We now compare the preceding source-faithful bridge with Project LANA commit `c65b28c` [61]. This snapshot contains a genuine positive advance: `Iut/Concrete/Existence.lean` constructs the previously abstract `ConcreteThetaDataExistence` datum from an elliptic curve and its arithmetic, Tate, representation-theoretic, and anabelian inputs. The new theorem `CurveInputs.concreteThetaDataExistence` therefore removes one assembly gap. The endpoint `cor312Variant_implies_abc_curves`, however, remains conditional. Its hypotheses still include

- the height-theory proof package `Genl.HeightTheory.ProofPackage` and the curve inputs;
- anabelian existence, together with providers for local theory, theta-local data, and tower arithmetic; and
- the analytic prime bounds and the concrete Corollary 3.12 comparison.

It is therefore not an unconditional proof of Statement I or of integer *abc*.

The same snapshot retains a logically impossible quantifier in its local log-volume interface. Abstract the issue as follows. Let $s : X \to X$, let $V : \mathcal{P}(X) \to \mathbb{R}$, and fix $\delta \in \mathbb{R}$.

Theorem 36.1 (Total-set and nonempty-set obstructions)**.** *If*

$$V(s^{-1}U) = V(U) + \delta \qquad (U \subseteq X), \tag{310}$$

then $\delta = 0$. If X is nonempty, the same conclusion holds when (310) is required only for nonempty U. More generally, the conclusion holds on any asserted domain containing a set U with $s^{-1}U = U$.

Proof. For the total law take $U = \varnothing$; its preimage is again empty and additive cancellation gives $\delta = 0$. Under the nonempty restriction take $U = X$, whose preimage is X. The last assertion is the same cancellation with the stated fixed set. ☐

In the current `LocalTheory.componentVol_prime_preimage` field one has $\delta = \log p$ for a prime p. Since $p \geq 2$, this shift is positive, so no inhabitant of the literal all-set structure can exist. Merely appending `U.Nonempty` would move the contradiction from the empty set to the whole set and would not repair it.

The source-level repair used in our audit makes the domain explicit. It adds a predicate $\mathcal{A}$ of admissible finite, nonzero-volume regions and requires

$$U \in \mathcal{A} \Longrightarrow U \neq \varnothing, \qquad U \in \mathcal{A} \Longrightarrow s^{-1}U \in \mathcal{A}, \qquad U \in \mathcal{A} \Longrightarrow V(s^{-1}U) = V(U) + \log p. \tag{311}$$

The predicate is transported from `LocalTheory` to `LogVolumeData` and through `Concrete.Container.vol`. No numeric equality is imposed on sets outside its domain. This is the minimal quantifier correction suggested by the finite, nonzero-volume regime of IUT III, Proposition 3.9 and Remark 3.9.5 [62].

Proposition 36.2 (Admissible orbit law)**.** *Assume* (311)*. For $U \in \mathcal{A}$ and $r \geq 0$,*

$$V(s^{-r}U) = V(U) + r \log p.$$

If $\log p \neq 0$, the admissible preimage orbit of U is injective and in particular is not periodic.

Proof. Induction on r proves the formula: closure under preimage supplies the admissibility hypothesis needed at each successor step. If $s^{-r}U = s^{-t}U$, the formula gives $(r-t)\log p = 0$, whence $r = t$. $\square$

The patched versions of the three affected source files compile together with both upstream targets `Iut` and `Iut4Sec1`; the build finishes all 8767 tasks. This verifies that the repaired dependency shape is compatible with the current source tree. It does not construct the local Haar volume or prove the comparison theorem.

There is a second accounting seam at the passage from an integral tensor order to the product of maximal factor orders. For $n \in \mathbb{N}$ put

$$\mathcal{O}_n = \{(a,b) \in \mathbb{Z}^2 : n \mid a-b\}, \qquad \delta_n(a,b) = a - b \pmod{n}. \tag{312}$$

Theorem 36.3 (Exact congruence-order quotient)**.** *The map $\delta_n : \mathbb{Z}^2 \to \mathbb{Z}/n\mathbb{Z}$ is a surjective additive homomorphism with kernel $\mathcal{O}_n$. Consequently*

$$\mathbb{Z}^2/\mathcal{O}_n \simeq \mathbb{Z}/n\mathbb{Z}, \qquad [\mathbb{Z}^2 : \mathcal{O}_n] = n.$$

For $n > 1$, $\mathcal{O}_n$ is a proper finite-index subring of $\mathbb{Z}^2$, although both coordinate projections $\mathcal{O}_n \to \mathbb{Z}$ are surjective.

Proof. Additivity is immediate, and every residue class is the image of $(a, 0)$. Moreover $\delta_n(a,b) = 0$ exactly when $n \mid a - b$, so the first isomorphism theorem gives the quotient. The kernel is a subring: it contains $(0,0)$ and $(1,1)$, is additively closed, and for two of its elements

$$ac - bd = a(c-d) + d(a-b),$$

so it is closed under componentwise multiplication. Its quotient cardinality is n. If $n > 1$, the point $(1,0)$ is absent, proving properness; every (z,z) is present, proving both projection statements. $\square$

The theorem includes $n = 0$ under the `Nat.card` index convention: both the quotient and $\mathbb{Z}/0\mathbb{Z} \cong \mathbb{Z}$ are infinite and their recorded natural cardinality is zero. For positive m, n, exactness also gives

$$[\mathbb{Z}^2 : \mathcal{O}_{mn}] = [\mathbb{Z}^2 : \mathcal{O}_m][\mathbb{Z}^2 : \mathcal{O}_n], \qquad \log[\mathbb{Z}^2 : \mathcal{O}_{mn}] = \log[\mathbb{Z}^2 : \mathcal{O}_m] + \log[\mathbb{Z}^2 : \mathcal{O}_n]. \tag{313}$$

Thus a finite integral-order defect contributes an additive logarithmic term. Identifying the actual IUT tensor-order quotient with a local different or conductor exponent is a further arithmetic theorem, not a consequence of this model.

The empty and whole sets are full-premise counterexamples to the two stated overbroad volume laws. The orders $\mathcal{O}_n$, $n > 1$, are full-premise counterexamples to the assertion that surjective component projections force the product order. These witnesses retire precisely those shortcuts. They do not satisfy the premises of an admissible, index-aware same-pilot theorem, so that corrected route remains active. Its unresolved obligations are to construct the actual admissible Haar regions, calculate the tensor-order indices at every factor and place, prove covariance through the horizontal link and Ind3, and establish the concrete Corollary 3.12 comparison.

The Lean companion proves the two no-go theorems, identifies the exact kernel and quotient in (312), proves the index and its logarithmic multiplication law, and verifies properness together with both surjective projections. The source ledger separately freezes the upstream commit, exact patch replay, and complete build log. No same-pilot theorem, Statement I, or *abc* consequence is introduced as a formal assumption.

37. Actual Haar admissibility and packet normalization

The preceding source audit leaves a concrete local question: can the repaired admissible-domain interface be realized by an actual measure, with the sign and coefficient required by the source? Proposition 3.9(i) of [62] takes the nonarchimedean log-volume domain to be the nonempty compact open subsets, normalizes the integral structure to log-volume zero, and uses packet normalization so that multiplication by the local rational prime subtracts its logarithm. We now construct the underlying Haar model and separate raw local volume from that packet normalization.

Let K be a nontrivially normed field with proper metric, let μ be a regular additive Haar measure, and let $a \in K^\times$. The local fields used below have this properness property. Write $m_a(x) = ax$ and let $\Delta_\mu : K^\times \to \mathbb{R}_{>0}$ be the distributive Haar character, so that

$$\mu(aU) = \Delta_\mu(a)\mu(U).$$

Define the finite-positive domain

$$\mathcal{F}_\mu = \{U : U \text{ is measurable and } 0 < \mu(U) < \infty\}, \qquad L_\mu(U) = \log \mu(U).$$

The conversion from an extended nonnegative measure to a real logarithm is made only after both inequalities in the definition of $\mathcal{F}_\mu$ have been proved.

Theorem 37.1 (Actual scalar-preimage law)**.** *For every $U \in \mathcal{F}_\mu$,*

$$m_a^{-1}(U) = a^{-1}U \in \mathcal{F}_\mu, \qquad L_\mu(m_a^{-1}U) = L_\mu(U) + \log \Delta_\mu(a^{-1}). \tag{37.1.1}$$

If U is also compact and open, then $m_a^{-1}(U)$ is compact and open.

Proof. The carrier identity follows in both directions from $ax \in U \iff x = a^{-1}(ax) \in a^{-1}U$. Multiplication by a is a homeomorphism, so its inverse image preserves measurability, openness, and compactness. The Haar change-of-variables identity gives

$$\mu(m_a^{-1}U) = \mu(a^{-1}U) = \Delta_\mu(a^{-1})\mu(U).$$

The character is finite and strictly positive. The right side is therefore finite and positive, and taking real logarithms proves the formula. □

Suppose now that the valuation ring $\mathcal{O}_K$ is a discrete valuation ring, the residue field has cardinality q, and $\mu(\mathcal{O}_K) = 1$. Let π be a uniformizer.

Theorem 37.2 (Uniformizer orbit and its obstruction)**.** *Put $B_n = \pi^{-n}\mathcal{O}_K$ for $n \geq 0$. Every B_n is a nonempty compact open member of $\mathcal{F}_\mu$, and*

$$m_\pi^{-1}(B_n) = B_{n+1}, \qquad L_\mu(B_n) = n \log q. \tag{37.2.1}$$

Consequently $n \mapsto B_n$ is injective, and no single $V \in \mathcal{F}_\mu$ contains every B_n.

Proof. The q additive cosets of $\pi\mathcal{O}_K$ partition $\mathcal{O}_K$. Translation invariance and $\mu(\mathcal{O}_K) = 1$ imply $\mu(\pi\mathcal{O}_K) = q^{-1}$, hence $\Delta_\mu(\pi) = q^{-1}$. Theorem 37.1 therefore adds $\log q$ under a uniformizer preimage. The unit ball is a nonempty open ball; properness makes it compact, and scalar multiplication preserves these properties. Induction gives the displayed identities. Since $q > 1$, the log-volumes are strictly increasing, proving injectivity. If every B_n lay in one finite-positive V, monotonicity would give $n \log q \leq L_\mu(V)$ for every n, contrary to the Archimedean property. □

The last conclusion rules out only a bounded, finite, or periodic implementation of the complete preimage orbit. It does not conflict with the least-hull construction for one fixed relatively compact region in Remark 3.9.5(i) of [62], and it does not rule out the Haar/IUT route.

The rational-prime coefficient requires one further normalization. Let $K/\mathbb{Q}_p$ have ramification index e and residue degree f. Choose a unit u with

$$p = u\pi^e, \qquad q = p^f.$$

Multiplication by u preserves Haar measure. Thus raw Haar log-volume changes under rational-prime preimage by

$$\log \Delta_\mu(p^{-1}) = e \log q = ef \log p. \tag{37.2.2}$$

Theorem 37.3 (Local-degree and packet normalization)**.** *Define*

$$L_\mu^{\mathrm{pkt}}(U) = \frac{L_\mu(U)}{ef}.$$

Then

$$L_\mu^{\mathrm{pkt}}(p^{-1}U) = L_\mu^{\mathrm{pkt}}(U) + \log p. \tag{37.3.1}$$

For finitely many components, if $\sum_i w_i = 1$*, the weighted packet volume* $\sum_i w_i L_{\mu_i}^{\mathrm{pkt}}(U_i)$ *also changes by exactly* $\log p$ *under simultaneous rational-prime preimage.*

Proof. Divide the raw identity $ef \log p$ by the positive local degree ef. Every normalized component then has the same increment $\log p$, so the weighted increment is $(\sum_i w_i) \log p = \log p$. □

This two-stage calculation explains which coefficient has to be realized in the actual tensor components. Ordinary weight normalization cannot perform the first stage.

Proposition 37.4 (Full-premise raw-volume counterexample)**.** *The condition* $\sum_i w_i = 1$ *alone does not turn raw Haar log-volume into a* $\log p$ *rational-prime shift.*

Proof. Let $K/\mathbb{Q}_p$ be the unramified quadratic extension, take the one-element packet, and give it weight one. Then $e = 1$, $f = 2$, $q = p^2$, and all local field, Haar, positivity, and weight premises hold. The raw shift is $\log(p^2) = 2 \log p \neq \log p$. Division by $ef = 2$ restores the conclusion of Theorem 37.3. □

Proposition 37.4 retires exactly the weaker raw-volume claim. The corrected route remains active. It must still construct $p = u\pi^e$ and $q = p^f$ uniformly in every actual tensor component, identify the component functional with $L_\mu/(ef)$, transport the compact-open admissible class through the direct-sum and least-hull operations, and then prove the horizontal same-pilot and Ind3 comparison. None of these global requirements follows from a local change-of-variables formula.

38. Refined tensor Haar volume and the pointed same-pilot gate

Author: ChatGPT. This section isolates an unconditional local theorem that is useful at the present Project-LANA/IUT interface. It does not assert IUT III, Corollary 3.12, or the *abc* conjecture. In particular, the finite-etale factorization, Haar-positivity, Ind3-openness, and pointed-horizontal identification hypotheses below are logically distinct.

Primary-source boundary. The terminology follows Mochizuki's May 2020 author version of *Inter-universal Teichmuller Theory III.* The exact locators are Proposition 3.9(i), Remark 3.9.5(i), Theorem 3.11(i)–(ii), and Corollary 3.12, especially proof Steps (xi-b)–(xi-f). The identification still missing after the three indeterminacies is also recorded in Sections 6–7 of the July 2026 LANA formalization report. The frozen Project LANA source is commit

`c65b28c9f9631635e742294c3a5df15759e7c74c`.

No source claim is imported as a premise unless it is displayed below.

38.1. Primitive factors and the prime Jacobian.

Fix a rational prime p. For a place tuple c, let

$$A_c = \bigotimes_{j \in S, \mathbb{Q}_p} K_{c_j}.$$

Since the local extensions are finite and separable, A_c is a finite-etale $\mathbb{Q}_p$-algebra. Thus

$$A_c \simeq \prod_{d \in D_c} L_{c,d}, \qquad N_c := \dim_{\mathbb{Q}_p} A_c = \sum_{d \in D_c} [L_{c,d} : \mathbb{Q}_p]. \tag{314}$$

For a factor $L = L_{c,d}$, write $e = e(L/\mathbb{Q}_p)$ and $f = f(L/\mathbb{Q}_p)$.

Primitive-factor theorem. For every uniformizer $\pi \in \mathcal{O}_L$, there is $u \in \mathcal{O}_L^\times$ such that

$$p = u\pi^e, \qquad |k_L| = p^f, \qquad [L : \mathbb{Q}_p] = ef. \tag{315}$$

Indeed, unique factorization in the discrete valuation ring gives $p = u\pi^m$. Applying the integer-valued valuation normalized by $v_L(\pi) = 1$ yields $m = v_L(p) = e$. The residue field is an f-dimensional vector space over $\mathbb{F}_p$, and the fundamental equality for finite extensions of complete discretely valued fields gives $[L : \mathbb{Q}_p] = ef$. Consequently,

$$N_c = \sum_{d \in D_c} e_{c,d} f_{c,d} > 0. \tag{316}$$

The last identity follows by taking dimensions in (314); it is not an assumption about an unsplit tensor product.

Normalize additive Haar measure $\mu_{c,d}$ by $\mu_{c,d}(\mathcal{O}_{c,d}) = 1$. For nonempty compact-open $U_{c,d} \subseteq L_{c,d}$, set

$$U_c = \prod_d U_{c,d}, \qquad \mu_c = \prod_d \mu_{c,d}, \qquad \lambda_c(U_c) = \frac{1}{N_c} \log \mu_c(U_c).$$

All measures in this display are finite and strictly positive. The additive Haar change-of-variables formula and (315) give

$$\mu_{c,d}(p^{-1}U_{c,d}) = p^{e_{c,d}f_{c,d}} \mu_{c,d}(U_{c,d}).$$

Taking products and using (316) proves

$$\lambda_c(p^{-1}U_c) = \lambda_c(U_c) + \log p. \tag{317}$$

Moreover, with $n_{c,d} = [L_{c,d} : \mathbb{Q}_p]$ and $\rho_{c,d} = n_{c,d}/N_c$, one has

$$\sum_d \rho_{c,d} = 1, \qquad \lambda_c(U_c) = \sum_d \rho_{c,d} \frac{\log \mu_{c,d}(U_{c,d})}{n_{c,d}}. \tag{318}$$

Thus the normalization is by total local degree, rather than by the number of primitive factors.

If a tuple has weight W_c, its primitive factors must receive

$$\widetilde{W}_{c,d} = W_c \rho_{c,d}. \tag{319}$$

Then $\sum_d \widetilde{W}_{c,d} = W_c$, so normalized tuple weights remain normalized after primitive refinement.

38.2. **Finite-positive orbit regions.** Let X be a finite product of locally compact primitive fields with product Haar measure μ. Let $U \subseteq X$ be nonempty and open, and let $\{\phi_a : X \simeq X\}_{a \in A}$ be homeomorphisms, one of which is the identity. Suppose that every $\phi_a(U)$ lies in a common set E with compact closure. Then

$$\Theta(U) := \bigcup_{a \in A} \phi_a(U) \tag{320}$$

is nonempty and open. It therefore has positive Haar measure. Its closure is a closed subset of the compact set $\overline{E}$, so it is compact; Haar measure is finite on that closure. Hence

$$0 < \mu(\Theta(U)) < \infty. \tag{321}$$

When A is finite and U is compact-open, $\Theta(U)$ is itself compact-open. For an infinite family the stated hypotheses yield an open region with compact closure; they do not imply that the orbit union is closed.

The same argument applies to an inductively generated Ind1–Ind3 system only under the following separate premises: ordinary branches are open and one is nonempty; Ind1 and Ind2 preserve openness; every relational Ind3 target of an open reachable region is open; and all constructors preserve the common compact-envelope containment. Structural induction then proves openness and envelope containment for every reachable region, after which (321) follows. The finite-etale theorem above does not supply any of these source-level Ind hypotheses.

38.3. **The pointed same-pilot implication.** Let P be the actual native q-pilot region, Θ the actual union of reachable theta outputs, and H a finite-positive hull. If the object-level construction proves the pointed inclusions

$$P \subseteq \Theta \subseteq H, \tag{322}$$

then Haar monotonicity and monotonicity of the real logarithm give

$$\log \mu(P) \leq \log \mu(\Theta) \leq \log \mu(H). \tag{323}$$

Thus any independent estimate $\log \mu(H) \leq T$ implies $\log \mu(P) \leq T$. No volume-preservation assertion for the Ind maps is needed for this conditional implication. However, naming some generated branch "native" does not establish (322). The open source-level obligation is to transport the original pointed pilot through the theta link, IPL/SHE/APT, log-Kummer correction, determinant normalization, and Ind3.

38.4. **Exact counterexamples and route status.** Each counterexample below satisfies all premises of the deliberately weaker claim that it refutes.

(1) A one-factor quadratic local field has factor count 1 and total degree 2. Its raw prime-preimage shift is $2 \log p$, so division by factor count does not give (317). This refutes only factor-count normalization.
(2) For an unramified quadratic Galois extension $K/\mathbb{Q}_p$, $K \otimes_{\mathbb{Q}_p} K \simeq K \times K$. Copying a parent weight 1 to both factors gives total weight 2; the relative-degree weights $(1/2, 1/2)$ satisfy (319). This refutes only copied parent weights.
(3) On $\mathbb{R}$ with Lebesgue Haar measure, an Ind3 relation may send $(-1, 1)$ to $\{0\}$ while both lie in the compact envelope $[-1, 1]$. The target has measure zero. This refutes the claim that envelope preservation alone yields finite-positive Ind3 outputs; it does not refute a concrete Ind3 theorem that proves openness or positivity.
(4) In the discrete three-point group with counting Haar measure, take $P = \{0, 1\}$ and $\Theta = H = \{0\}$. All regions are compact-open and finite-positive and $\Theta \subseteq H$, but $\log \mu(P) = \log 2 > 0 = \log \mu(H)$. This refutes only an unpointed inference.

The finite-etale factor identities and total-degree Haar normalization are proved. The relative-degree weights, bounded-open orbit lemma, and conditional pointed sandwich are also proved. The identification of the concrete IUT Ind1 and Ind2 maps, openness or positive measure of the required Ind3 targets, and the pointed IPL, SHE, and APT horizontal transport remain open gaps. No counterexample satisfying all premises of those source statements is presently supplied; therefore those routes remain active. IUT III, Corollary 3.12, IUT, and the standard *abc* conjecture remain open in this work.

38.5. **Formalization boundary.** The companion Lean development derives the identities in (315) and (316) on the actual primitive quotients of a `TupleFiniteEtalePacket`. It also formalizes the product-Haar formulas, the refined weights, the orbit and Ind interface lemmas and the pointed sandwich. It checks coefficient-level versions of the factor-count and copied-weight obstructions and set-level versions of the envelope and unpointed obstructions. The quadratic local-field and tensor decomposition witnesses for the first two remain paper proofs. Its axiom audit reports only `propext`, `Classical.choice`, and `Quot.sound`; there are no project axioms or admitted proofs.

The following source-specific work remains on paper: identifying each IUT capsule's completed tensor algebra with the finite-etale packet; comparing the tensor product of integral orders with the product of maximal factor orders, including finite-index/different corrections; instantiating the abstract Ind1/Ind2 homeomorphisms and common envelope; proving the required Ind3 openness or positivity; and constructing the pointed horizontal map. Finite computation audits the rational normalization formulas only and gives no evidence for these source-specific obligations or for *abc*.

39. A two-arm CRT packet for the minimal affine shear

Let $a+b=c$ be primitive, put $R=\operatorname{rad}(abc)$, and consider the minimal-step affine shear

$$U=1+Rh, \qquad V=1+R(h+ck), \qquad W=1+R(h+bk). \tag{324}$$

Write $E(n)=n/\operatorname{rad}(n)$. The density analysis in Section 28 shows that every three-quarter exception in the upper-half canonical box must satisfy

$$Rc<8192E(V), \qquad Rc<8192E(W). \tag{325}$$

The two inequalities carry more joint information than two unrelated numerical lower bounds.

Theorem 39.1 (Coprime joint carrier)**.** *If V,W,T,K are positive integers, $(V,W)=1$, and $T<KE(V)$ and $T<KE(W)$, then*

$$(E(V),E(W))=1, \qquad E(V)E(W)\mid VW, \qquad T^2<K^2E(V)E(W).$$

Proof. Each powerful part divides its parent integer. Divisor monotonicity of coprimality gives $(E(V),E(W))=1$, and multiplication gives the divisibility. Multiplying the two strict positive inequalities proves the last assertion. For (325), take $T=Rc$ and $K=8192$. □

The diagonal $h=k$ is automatically admissible because $(1+Rk,k)=1$, and its long arms are

$$V=1+R(c+1)k, \qquad W=1+R(b+1)k.$$

Theorem 39.2 (Diagonal two-arm CRT packet)**.** *Let D,F be positive coprime squarefree integers such that*

$$(D,R(c+1))=1, \qquad (F,R(b+1))=1.$$

There is a unique residue class $k_0 \pmod{D^2F^2}$ such that

$$k\equiv k_0 \pmod{D^2F^2} \iff D^2\mid V \ \text{ and } \ F^2\mid W.$$

Every member is admissible and satisfies $D\mid E(V)$ and $F\mid E(W)$.

Proof. The two coefficients $R(c+1)$ and $R(b+1)$ are units modulo D^2 and F^2, respectively. Each affine congruence therefore has one root, and the Chinese remainder theorem combines them into one class modulo D^2F^2. If a squarefree D satisfies $D^2\mid n$, then each prime of D occurs in n with valuation at least two and hence occurs in $E(n)$; thus $D\mid E(n)$. Apply this observation to both arms. Admissibility was noted above. □

The theorem supplies simultaneous repeated-prime mass, but the following exact calculation shows that the two marginal gates do not contain enough information to force exceptionality.

Proposition 39.3 (A counted packet and a complete insufficiency witness)**.** *For the seed $(a,b,c)=(1,242,243)$ one has $R=66$ and canonical scale*

$$M=\left\lfloor\frac{243^6}{4\cdot 66}\right\rfloor=779890651873.$$

The diagonal progression

$$h=k=356+1225t$$

has exactly 318322715 distinct points with $M/2<k\le M$. Every point satisfies $5\mid E(V)$, $7\mid E(W)$, and both inequalities in (325). At its first point,

$$h=k=389945326231, \qquad (E(U),E(V),E(W))=(1,5,7),$$

the primitive output is

$$(A,B,C)=(25736391531247,1519682447137014050,1519708183528545297), \qquad A+B=C,$$

but $\operatorname{rad}(ABC)^4>C^3$. Hence the point is not a three-quarter exception.

Proof. On the progression,

$$V = 25(229321 + 789096t), \qquad W = 49(116521 + 400950t).$$

The canonical inequalities are equivalent to $318322715 \le t \le 636645429$, an interval of exactly 318322715 integers; the affine function of t is injective. The square divisibilities and the squarefree-carrier lemma give the two excess divisibilities, while $66 \cdot 243 < 8192 \cdot 5$ and $66 \cdot 243 < 8192 \cdot 7$ give the gates.

For the first point, exact factorization gives

$$\begin{aligned} U &= 17 \cdot 1513905384191, \\ V &= 5^2 \cdot 23 \cdot 37 \cdot 139267 \cdot 2119433, \\ W &= 7^2 \cdot 17431 \cdot 7322098141. \end{aligned}$$

All residual factors displayed here are prime. A shorter certificate for nonexceptionality is the squarefree divisor

$$d = 139267 \cdot 2119433 \cdot 17431 = 5145057294975341$$

of ABC: it divides $\operatorname{rad}(ABC)$ and direct integer comparison gives $d^4 > C^3$. □

Proposition 39.3 refutes only the assertion that the two necessary long-arm gates are sufficient. It does not refute their necessity or the affine route. The remaining positive theorem must produce, uniformly over every fixed subcritical seed range $R < c^\lambda$, enough canonical admissible parameters satisfying the full pointwise inequality

$$E(U)E(V)E(W) > \frac{RUVW}{(cW)^{3/4}} = \frac{R}{c} UV(cW)^{1/4}. \tag{326}$$

Within the present density architecture, one seeks constants $\eta, \kappa > 0$ and at least $\kappa R^{-2/3} c^{4+\eta}$ such parameters for every sufficiently large seed. Growing squarefree carriers and a correlated powerful-value theorem for all three arms remain distinct active ways to reach (326).

40. Large-modulus separation and affine certificate entropy

The periodic estimate for a fixed divisibility template in the minimal affine shear has an error term too large at the modulus forced by the full three-arm excess condition. This section removes that boundary error by a separation argument. The result controls one prescribed certificate, and then quantifies how many such certificates a matching lower bound would need. It does not bound a pointwise adaptive or otherwise correlated family of certificates.

40.1. **A three-form separation lemma.** Let $0 < B < C$ and let $d_U, d_V, d_W, L, X_U, X_V, X_W$ be positive integers. Put $D = d_U d_V d_W$, and assume that the three d's are pairwise coprime. For two pairs $(h,k), (h',k') \in \mathbb{Z}^2$, set

$$x = h - h', \qquad y = k - k', \qquad H = \max\{|x|, |y|\}.$$

Suppose

$$d_U \mid x, \qquad d_V \mid x + Cy, \qquad d_W \mid x + By, \tag{327}$$

and impose the six cancellation conditions

$$\begin{gathered} (d_V, C) = (d_W, B) = 1, \qquad (d_U, C) = (d_W, C - B) = 1, \\ (d_U, B) = (d_V, C - B) = 1. \end{gathered} \tag{328}$$

Theorem 40.1 (Cubic-product separation). *Let $T > 0$. In addition to (327)–(328), assume*

$$\begin{gathered} d_U \le X_U, \qquad d_V \le X_V, \qquad d_W \le X_W, \\ D > T > (C+1)^2 L^3, \qquad LX_U < T, \quad LX_V < T, \quad LX_W < T. \end{gathered} \tag{329}$$

If $(h,k) \ne (h',k')$, then $H > L$.

Proof. Suppose first that x, $x+Cy$, and $x+By$ are all nonzero. Pairwise coprimality and (327) give

$$D \mid x(x+Cy)(x+By).$$

Since $B<C$, a nonzero multiple of D therefore satisfies

$$D \le |x|\,|x+Cy|\,|x+By| \le (C+1)^2H^3. \tag{330}$$

The assumption $H\le L$ would contradict the middle inequality in (329).

The zero factors require the individual modulus caps. If $x=0$, then (328) permits cancellation of C and B from the last two divisibilities. Pairwise coprimality gives $d_Vd_W \mid y$. Distinctness gives $y\neq 0$, so $d_Vd_W \le |y| \le H$. Under $H\le L$ this would imply

$$D = d_Ud_Vd_W \le X_UL < T,$$

contrary to $D>T$. If $x+Cy=0$, cancellation from $d_U \mid Cy$ and $d_W \mid (C-B)y$ gives $d_Ud_W \mid y$; hence $H\le L$ would give $D\le X_VL<T$. If $x+By=0$, the same argument gives $d_Ud_V \mid y$ and then $D\le X_WL<T$. These four cases exhaust the possibilities. ☐

Proposition 40.2 (Product-cell packing)**.** *Let $S\subseteq\{1,\dots,M\}^2$ and suppose distinct members of S have sup distance greater than L. Then*

$$|S| \le \left\lceil \frac{M}{L+1}\right\rceil^2 \le \left(\frac{M}{L+1}+1\right)^2. \tag{331}$$

Proof. Divide each coordinate interval into consecutive cells of length $L+1$. Two points in one product cell have both coordinate differences at most L, so every product cell contains at most one member of S. ☐

40.2. **The canonical affine box.** Let $a+b=c$ be a positive primitive seed, orient it so that $b\ge c/2$, and write $\mathcal{R}=\mathrm{rad}(abc)$. With the radical step, the three cofactor forms are

$$U = 1+\mathcal{R}h, \qquad V = 1+\mathcal{R}(h+ck), \qquad W = 1+\mathcal{R}(h+bk). \tag{332}$$

Take

$$M = \left\lfloor \frac{c^6}{4\mathcal{R}}\right\rfloor, \qquad L = \left\lfloor \frac{c^4}{13}\right\rfloor. \tag{333}$$

Assume $c\ge 6$ and $\mathcal{R}<c$. For $1\le h,k\le M$ one has

$$U<c^6, \qquad V<c^7, \qquad W<c^7. \tag{334}$$

Indeed $\mathcal{R}M \le c^6/4$, and $\mathcal{R}(1+c)M \le (1+c)c^6/4$; the claimed strict inequalities follow for $c\ge 6$, while $b<c$ makes the W estimate no larger.

The seed conditions also imply $\mathcal{R}\ge 6$. If c is odd, then $c\ge 7$, the product abc is even, and an odd prime divides c. If c is even, coprimality makes a,b odd, and $c\ge 6$ makes one of them at least three. In both cases distinct primes 2 and $q\ge 3$ divide abc, whence $\mathcal{R}\ge 2q\ge 6$.

For a positive integer n, write $E(n)=n/\mathrm{rad}(n)$. If $d\mid n$, then $d/\mathrm{rad}(d)\le E(n)$, as follows prime by prime from $v_p(d)-1\le v_p(n)-1$ on the support of d. Fix positive moduli d_U,d_V,d_W, and let S be the parameter pairs in the box satisfying $\gcd(U,k)=1$, $d_U\mid U$, $d_V\mid V$, and $d_W\mid W$. Put

$$\Delta = \prod_{Z\in\{U,V,W\}} \frac{d_Z}{\mathrm{rad}(d_Z)}. \tag{335}$$

Thus $\Delta > \mathcal{R}c^{14}/8192$ is a sufficient fixed certificate for the full necessary excess inequality

$$E(U)E(V)E(W) > \frac{\mathcal{R}c^{14}}{8192} \tag{336}$$

from the upper-half three-quarter analysis.

For every point of S, the identities

$$V-U=\mathcal{R}ck, \qquad W-U=\mathcal{R}bk, \qquad V-W=\mathcal{R}ak$$

and $\operatorname{rad}(abc) = \mathcal{R}$ show that U, V, W are pairwise coprime and each is coprime to $\mathcal{R}abc$. If S is nonempty, one point of S then shows that the fixed divisors are pairwise coprime and satisfy every cancellation condition in (328) with $(B, C) = (b, c)$. For two points of S, subtracting the corresponding template divisibilities gives divisibility of $\mathcal{R}x$, $\mathcal{R}(x + cy)$, and $\mathcal{R}(x + by)$. Each d_Z is coprime to $\mathcal{R}$, so cancellation gives (327).

Theorem 40.3 (One full-strength affine template). *If S is nonempty and $\Delta > \mathcal{R}c^{14}/8192$, then*

$$|S| < \frac{12c^4}{\mathcal{R}^2}. \tag{337}$$

Proof. Set $T = \mathcal{R}c^{14}/8192$. Since $D = d_U d_V d_W \geq \Delta > T$ and $6(c+1) \leq 7c$, equations (333) and (334) give

$$(c+1)^2 L^3 \leq \frac{49c^{14}}{36 \cdot 13^3} < T.$$

Here the strict comparison is $49 \cdot 8192 = 401408 < 474552 = 36 \cdot 6 \cdot 13^3 \leq 36\mathcal{R}13^3$. They also give $Ld_U < T$ and $Ld_V, Ld_W < T$. The weakest of these comparisons is

$$\frac{c^4}{13}c^7 < \frac{\mathcal{R}c^{14}}{8192},$$

which reduces to $8192 < 13\mathcal{R}c^3$ and holds already at the conservative endpoint $\mathcal{R} = c = 6$, where the right side is 16848. Theorem 40.1 now applies to every two points of S.

Because $L + 1 > c^4/13$ and $M \leq c^6/(4\mathcal{R})$,

$$\frac{M}{L+1} < \frac{13c^2}{4\mathcal{R}}.$$

The last quantity exceeds one when $\mathcal{R} < c$. Also $\mathcal{R} \leq c - 1$, and the elementary factorization of $5c^2 - 36c + 36$ at its roots $6/5$ and 6 gives

$$\frac{\mathcal{R}}{c^2} \leq \frac{c-1}{c^2} \leq \frac{5}{36}, \qquad \left(\frac{13}{4} + \frac{5}{36}\right)^2 = \left(\frac{61}{18}\right)^2 = \frac{3721}{324} < 12.$$

Proposition 40.2 therefore gives

$$|S| < \left(\frac{13c^2}{4\mathcal{R}} + 1\right)^2 < \frac{12c^4}{\mathcal{R}^2}.$$

□

Corollary 40.4 (Certificate entropy). *Let $\kappa, \eta > 0$. Suppose a union of N fixed templates satisfying Theorem 40.3 contains at least $\kappa\mathcal{R}^{-2/3}c^{4+\eta}$ parameter pairs. Then*

$$N > \frac{\kappa}{12}\mathcal{R}^{4/3}c^{\eta}. \tag{338}$$

Proof. The union bound and (337) give $|\bigcup_{i=1}^N S_i| < N(12c^4/\mathcal{R}^2)$. Comparing this with the assumed lower bound and cancelling positive factors proves (338). □

The corollary closes the large-modulus boundary for a single full-strength template and rules out every fixed finite list as a source of the matching lower bound. It leaves active an unbounded correlated family, pointwise certificate selection, an algebraic parametrization, and excess contributed by unprescribed residual factors.

40.3. **Why the individual caps cannot be removed.** The determinant inequality alone does not imply separation. Take

$$B = 1, \quad C = 2, \quad L = 1, \quad T = 10, \quad d_U = 31, \quad d_V = d_W = 1, \quad X_U = X_V = X_W = 1$$

and the two points $(30, 1)$ and $(30, 2)$. The moduli are pairwise coprime, all six conditions in (328) hold, and the three divisibilities (327) hold because $x = 0$. Moreover

$$D = 31 > T = 10 > (C+1)^2 L^3 = 9, \qquad LX_U = LX_V = LX_W = 1 < T,$$

but the sup distance is $1 = L$. Both points have $U = 1 + h = 31$ when the step is one. This is a complete counterexample to the strengthening obtained by deleting $d_Z \le X_Z$: it fails exactly $d_U \le X_U$. It does not refute Theorem 40.1 or any broader affine route.

The exact constants and this counterexample are independently replayed in the companion affine template-entropy computation directory. That finite replay also stress-tests 6244 bounded abstract parameter cases and 494515 canonical pairs $(c, \mathcal{R})$. These no-hit checks are diagnostic; the general conclusions above come from the proofs, and no route is retired by finite computation.

41. Determinant layers and canonical-kernel entropy

We refine the fixed-template estimate of Section 40 by using all three affine congruences simultaneously. Let $S \subseteq [1, M]^2$, put $N = M - 1$, and suppose every difference vector (x, y) in one fixed template satisfies

$$d_U \mid x, \qquad d_V \mid x + Cy, \qquad d_W \mid x + By,$$

where d_U, d_V, d_W are pairwise coprime. Set $D = d_U d_V d_W$.

Proposition 41.1 (Three-form determinant divisor)**.** *For any two template difference vectors $v = (x, y)$ and $w = (x', y')$,*

$$D \mid \det(v, w).$$

Proof. The modulus d_U divides $xy' - yx'$. The identities

$$\det(v, w) = (x + Cy)y' - y(x' + Cy') = (x + By)y' - y(x' + By')$$

give divisibility by d_V and d_W. Pairwise coprimality multiplies the three divisibilities. $\square$

Choose a shortest nonzero difference vector v and put $\delta = \|v\|_\infty$. The determinant map $p \mapsto \det(v, p - p_0)$ has values in $D\mathbb{Z}$, range length at most $N(|v_1| + |v_2|)$, and level sets parallel to v. Projection to a dominant coordinate gives the following capacity.

Theorem 41.2 (Determinant-layer capacity)**.** *If $|S| \ge 2$, then*

$$|S| \le \left(\left\lfloor \frac{N}{\delta} \right\rfloor + 1 \right) \left(\left\lfloor \frac{N(|v_1| + |v_2|)}{D} \right\rfloor + 1 \right) \le \frac{4N^2}{D} + \frac{N}{\delta} + 1.$$

If $D > 2N^2$, all points of S are collinear. If the complete three-form separation theorem also yields $\delta > L$, then

$$|S| \le \left\lfloor \frac{M-1}{L+1} \right\rfloor + 1.$$

Proof. An interval of the displayed determinant range contains at most the second number of multiples of D; a level contains at most the first number of points. Expanding and using $\delta \le N$ gives the coarse inequality. When $D > 2N^2$, every determinant has absolute value below D and hence is zero. The final estimate is one-dimensional interval packing. $\square$

Return to a positive primitive seed $a + b = c$, oriented by $b \ge c/2$, with $R = \operatorname{rad}(abc) < c$ and

$$M = \left\lfloor \frac{c^6}{4R} \right\rfloor, \qquad T_0 = \frac{Rc^{14}}{8192}.$$

The cases $c = 6, 7, 8$ are incompatible with $R < c$, so $c \ge 9$; also $R \ge 6$. Consequently $T_0 > 2(M - 1)^2$. Every complete certificate with repeated-loss product above T_0 is therefore in the zero determinant layer.

Theorem 41.3 (Canonical one-dimensional capacities)**.** *At the fixed scale $L_{12} = \lfloor c^4/12 \rfloor$ every canonical template satisfies*

$$|S| < \frac{31c^2}{10R}.$$

More generally, if $s \ge 1$ and $s^3 \le R$, then at $L_s = \lfloor sc^4/22 \rfloor$ one has

$$|S| < \frac{57c^2}{10Rs}.$$

Proof. The cubic separation inequality at scale 12 follows from

$$8192 \cdot 100 < 81 \cdot 12^3 \cdot 6,$$

and the weakest long-arm cap from $8192 < 12 \cdot 6 \cdot 9^3$. At scale $s/22$, use $s^3 \le R$ and $8192 \cdot 100 < 81 \cdot 22^3$, together with $8192 < 22 \cdot 9^3$. One-dimensional packing gives respectively $|S| < 3c^2/R + 1$ and $|S| < 11c^2/(2Rs) + 1$; the inequalities $10R < c^2$ and $5Rs < c^2$ absorb the final terms. □

For $n > 0$ define its repeated kernel by

$$K(n) = \prod_{p^e \| n,\ e \ge 2} p^e.$$

Primewise comparison proves

$$K(n) \mid n, \qquad \frac{K(n)}{\operatorname{rad} K(n)} = \frac{n}{\operatorname{rad} n},$$

and $K(n)$ divides every divisor of n preserving this exact loss. Thus partitioning exceptional affine points by $(K(U), K(V), K(W))$ is the unique coarsest exact-lossless template partition.

Corollary 41.4 (Forced canonical-kernel entropy)**.** *Let $\kappa, \eta > 0$. If the exceptional packet has size at least $\kappa R^{-2/3} c^{4+\eta}$ and N_{ker} canonical-kernel triples are realized, then, for any integer $s \ge 1$ with $s^3 \le R$,*

$$N_{\mathrm{ker}} > \max\left\{ \frac{10\kappa}{31} R^{1/3} c^{2+\eta}, \frac{10\kappa}{57} s R^{1/3} c^{2+\eta} \right\}.$$

Taking $s = \lfloor R^{1/3} \rfloor$ yields

$$N_{\mathrm{ker}} > \frac{5\kappa}{57} R^{2/3} c^{2+\eta}.$$

Every realized triple additionally satisfies

$$\operatorname{rad}\big(K(U)K(V)K(W)\big) < \frac{8192 c^6}{R}.$$

Proof. The canonical-kernel triples partition the exceptional packet into disjoint fixed-template classes. Each class retains the complete loss threshold $\Delta > T_0$, so Theorem 41.3 applies to it at both scales. Summing the two strict class bounds gives

$$|\mathcal{E}| < N_{\mathrm{ker}} \frac{31c^2}{10R}, \qquad |\mathcal{E}| < N_{\mathrm{ker}} \frac{57c^2}{10Rs}.$$

Combining these inequalities with the assumed lower bound and dividing by the positive capacities gives the displayed maximum.

For $s = \lfloor R^{1/3} \rfloor$, the standing bound $R \ge 6$ implies $s \ge 1$, $s^3 \le R$, and $s \ge R^{1/3}/2$. Substitution in the second term of the maximum yields $N_{\mathrm{ker}} > (5\kappa/57) R^{2/3} c^{2+\eta}$.

Finally put $P = K(U)K(V)K(W)$. The repeated-kernel identities, together with the pairwise coprimality of the primitive arms, give $P/\operatorname{rad} P = \Delta > T_0$. Also $K(n) \mid n$ and the canonical arm bounds give $P \le UVW < c^{20}$. Hence

$$\operatorname{rad} P = \frac{P}{P/\operatorname{rad} P} < \frac{c^{20}}{T_0} = \frac{8192 c^6}{R},$$

as required. □

The positivity of κ is essential to this exact formulation. If it is omitted, $\kappa = 0$, an empty exceptional packet, and $N_{\mathrm{ker}} = 0$ satisfy the remaining premises but contradict the strict conclusion. This full-premise counterexample retires only the zero-density strengthening.

For a fractional cover in which each template meets $\mathcal{E}$ in at most K_s points and each point receives total nonnegative weight at least one, the finite inequality $|\mathcal{E}| \le K_s \sum_\tau w_\tau$ handles correlated overlaps exactly. It does not produce the required adaptive weight bound. Two complete integer witnesses mark precise boundaries: one certified template has $4M^2 < D$, disproving an area-only estimate, while two adjacent points carry different complete certificates, disproving the assertion that a union of pointwise selected templates inherits fixed-template separation. The latter witness lies outside the canonical box. Neither counterexample removes the canonical adaptive-kernel route; the unresolved task is to produce or exclude the required $R^{2/3}c^{2+\eta}$ family under the actual exception inequality.

42. Adaptive three-point kernels in the affine shear

The fixed-template argument of Section 41 does not allow the three arm certificates to vary from point to point. We now isolate the part of the determinant mechanism that survives completely adaptive choices.

Retain the minimal-radical shear

$$U(h,k) = 1 + Rh, \qquad V(h,k) = 1 + R(h+ck), \qquad W(h,k) = 1 + R(h+bk),$$

and let p, q, r be three admissible parameters in $[1,M]^2$. At each point s choose arbitrary divisors

$$d_U(s) \mid U(s), \qquad d_V(s) \mid V(s), \qquad d_W(s) \mid W(s).$$

Define the triplewise common moduli

$$g_Z = \gcd(d_Z(p), d_Z(q), d_Z(r)) \quad (Z = U, V, W), \qquad G = g_U g_V g_W.$$

Lemma 42.1 (Sharp square-triangle determinant)**.** *If three points lie in an axis-parallel square of side N, then*

$$|\det(q-p, r-p)| \le N^2.$$

The constant is sharp.

Proof. Translate the square to $[0,N]^2$ and order the three first coordinates as x_1, x_1+s, x_1+s+t, where $s, t \ge 0$ and $s+t \le N$. If the corresponding second coordinates are Y_1, Y_2, Y_3, translation gives

$$D = tY_1 - (s+t)Y_2 + sY_3.$$

The positive contribution and the absolute value of the negative contribution are each at most $(s+t)N \le N^2$. A permutation changes only the sign. Three corners attain equality. □

At any one admissible point the three arms are pairwise coprime and each is coprime to R. For a difference vector (x,y) between two of the points, cancellation of R therefore gives

$$g_U \mid x, \qquad g_V \mid x+cy, \qquad g_W \mid x+by.$$

Applying this to both $q-p$ and $r-p$ and using

$$\det((x,y),(x',y')) = (x+cy)y' - y(x'+cy') = (x+by)y' - y(x'+by')$$

shows that every g_Z divides the determinant. The three moduli are pairwise coprime, hence their product divides it.

Theorem 42.2 (Adaptive common-kernel determinant bound)**.** *If $p, q, r \in [1,M]^2$ are noncollinear, then*

$$G \le (M-1)^2.$$

Proof. The preceding congruences give $G \mid \det(q-p, r-p)$. Noncollinearity makes the determinant nonzero, so G is at most its absolute value. Apply Lemma 42.1 to the square of side $M-1$. □

This theorem allows unrelated certificates at all three points. What it does not provide is a density theorem forcing a triple whose three triplewise gcds have large product. That is now the precise positive selection problem.

There are also useful canonical constant improvements. Put

$$M = \left\lfloor \frac{c^6}{4R} \right\rfloor, \qquad c \ge 9, \qquad E(n) = \frac{n}{\operatorname{rad}(n)}.$$

For every $(h,k) \in [1,M]^2$,

$$3U \le c^6, \qquad 3V \le c^7, \qquad 3W \le c^7. \tag{339}$$

Indeed, $4RM \le c^6$. The short-arm estimate follows from $12 + 12Rh \le 12 + 3c^6 \le 4c^6$. For a long arm,

$$4(3V) \le 12 + 3c^6(c+1) \le 4c^7,$$

where the last inequality is $12 \le (c-3)c^6$; $b \le c$ gives the same bound for W.

Suppose the affine point meets the necessary exceptional threshold

$$Rc^{14} < 8192E(U)E(V)E(W). \tag{340}$$

Since $3E(U) \le c^6$, cancellation gives

$$3Rc^8 < 8192E(V)E(W).$$

Thus at least one long arm $Z \in \{V, W\}$ satisfies

$$3Rc^8 < 8192E(Z)^2. \tag{341}$$

Combining $Z = \mathrm{rad}(Z)E(Z)$, $3Z \le c^7$, and (341) yields

$$27R\,\mathrm{rad}(Z)^2 < 8192c^6. \tag{342}$$

For the full repeated kernel

$$K(n) = \prod_{p^e \| n,\ e \ge 2} p^e, \qquad K = K(U)K(V)K(W),$$

pairwise coprimality gives

$$K = \mathrm{rad}(K)E(U)E(V)E(W).$$

Moreover, (339) gives $27K \le 27UVW \le c^{20}$. Multiplying (340) by $27\,\mathrm{rad}(K)$ and cancelling c^{14} proves the improved support bound

$$27R\,\mathrm{rad}(K) < 8192c^6. \tag{343}$$

The geometric alternatives have been tested on actual parameters. For the primitive seed $(a, b, c) = (1, 8, 9)$ one has $R = 6$ and $M = 22143$. The admissible points $(20, 1)$ and $(20, 2)$ have arm triples

$$(121, 175, 169), \qquad (121, 229, 217).$$

They share $K(U) = 121$, have distance one and vertical difference, while

$$121 > (b+1)(c+1) = 90, \qquad 121 > (c+1)^2 = 100.$$

Taking divisor certificates $(121, 25, 169)$ and $(121, 1, 1)$ gives common gcds $(121, 1, 1)$ and product 121. Thus this is a full-premise in-box counterexample to the precise local implication that deletes the zero-first-coordinate branch and concludes $G \le (c+1)^2\|p - q\|_\infty^3 = 100$.

Likewise $(160, 1), (160, 2), (160, 3)$ are admissible, collinear and share $U = K(U) = 961$. Their diameter is two and

$$961 > (c+1)^2 2^3 = 800.$$

Choosing certificates $(961, 1, 1)$ at every point gives triplewise gcds $(961, 1, 1)$ and satisfies all six difference congruences. Hence the precise full-premise local cubic implication to noncollinearity is false. This is not a counterexample to Theorem 42.2, nor to its numerical box-wide conclusion: $961 < (22143 - 1)^2$. It refutes only the local diameter-scale strengthening and the invalid step from divisibility to a size bound when the determinant may be zero.

Lean verifies the sharp determinant lemma for every point order, the derivation of the adaptive congruence interface from actual pointwise arm divisors and triplewise gcds, all three bounds in (339), the long-arm dichotomy, (342), the arithmetic seam of (343), and the negations of both exact local implications. No adaptive common-kernel density input is assumed. The affine route therefore remains active at the triple-selection gate.

43. Incidence budgets at the affine triple-selection gate

The adaptive determinant theorem leaves one precise selection problem: an exceptional point may choose its own arm divisors, while the determinant bound needs three points with a large *common* divisor product. We record the finite geometry and the exact incidence budget available for this step.

Let $S \subseteq [1, M]^2 \cap \mathbb{Z}^2$, put $N = M - 1$, and retain the affine arms U, V, W of Section 42. For a fixed actual arm-divisor label

$$\lambda = (d_U, d_V, d_W), \qquad D(\lambda) = d_U d_V d_W,$$

define

$$F_\lambda = \{x \in S : d_U \mid U(x),\ d_V \mid V(x),\ d_W \mid W(x)\}.$$

Lemma 43.1 (Integer-line cap)**.** *If every triple in S is collinear, then $|S| \le M$. Equivalently, more than M points in the square contain a noncollinear triple.*

Proof. Choose distinct $p, q \in S$ when $|S| > 1$. If $p_1 = q_1$, collinearity forces every point of S to have that first coordinate, so projection to the second coordinate is injective and has at most M values. If $p_1 \ne q_1$, and $r, s \in S$ have $r_1 = s_1$, subtracting $\det(q-p, r-p) = \det(q-p, s-p) = 0$ gives $(q_1 - p_1)(r_2 - s_2) = 0$. Hence $r = s$, so projection to the first coordinate is injective and again $|S| \le M$. □

Theorem 43.2 (Large-label fibre and incidence bounds)**.** *Assume the three arms are pairwise coprime at every point of S. If $D(\lambda) > N^2$, then $|F_\lambda| \le M$. Consequently, for every finite family $\mathcal{L}$ of such large labels,*

$$\sum_{\lambda \in \mathcal{L}} |F_\lambda| \le M|\mathcal{L}|.$$

If every x in a set $E \subseteq S$ belongs to at least μ of these fibres, then

$$\mu|E| \le M|\mathcal{L}|. \tag{344}$$

Proof. If $|F_\lambda| > M$, Lemma 43.1 produces a noncollinear triple in the fibre. Use the same label at all three points. Actual arm divisibility yields the six difference congruences, and pairwise coprimality of the arms makes d_U, d_V, d_W pairwise coprime. The adaptive determinant theorem then gives $D(\lambda) \le N^2$, a contradiction. Summing the fibre bound proves the first displayed incidence inequality. Double-counting $\{(x, \lambda) : x \in E \cap F_\lambda\}$ proves (344). □

The associated energy identity shows exactly what an analytic refinement must estimate. For positive labels $d_Z(x)$, set

$$\mathcal{E}_3 = \sum_{x,y,z \in S} \prod_{Z \in \{U,V,W\}} \gcd(d_Z(x), d_Z(y), d_Z(z))$$

and

$$n(e) = \#\{x \in S : e_U \mid d_U(x),\ e_V \mid d_V(x),\ e_W \mid d_W(x)\}.$$

Then the divisor identity $m = \sum_{e|m} \varphi(e)$ gives, by expanding three finite sums,

$$\mathcal{E}_3 = \sum_{e_U, e_V, e_W} \varphi(e_U)\varphi(e_V)\varphi(e_W) n(e)^3. \tag{345}$$

Thus total energy alone is insufficient: one must place enough of it on $e_U e_V e_W > N^2$ and control the contribution of collinear triples.

That collinear qualification cannot be removed. For the canonical seed $(a, b, c) = (1, 8, 9)$, $R = 6$ and $M = 22143$. At

$$(21480, 282), \quad (21480, 6211), \quad (21480, 12140)$$

use the common square label

$$(d_U, d_V, d_W) = (128881, 49, 121) = (359^2, 7^2, 11^2).$$

The corresponding arm rows are

	U	V	W
1	128881	144109	142417
2	128881	464275	427009
3	128881	784441	711601.

The label divides every row, all canonical admissibility and coprimality conditions hold, and the vertical increment $5929 = 49 \cdot 121$ proves all six congruences. Yet

$$128881 \cdot 49 \cdot 121 = 764135449 > 22142^2 \quad \text{and} \quad \det(q-p, r-p) = 0.$$

This is a full-premise counterexample to the extra implication "common product above the box square implies noncollinearity." It does not meet the noncollinearity premise of the valid determinant theorem and therefore does not retire the affine route.

The remaining positive gate is now quantitative. One may close this branch by constructing a finite family $\mathcal{L}$ and a multiplicity μ with $\mu|E| > M|\mathcal{L}|$, or by proving that the large-product part of (345) exceeds every possible collinear contribution. No full-premise counterexample to either conditional criterion is known.

44. Direction periods and collinear affine energy

The common-kernel selection in Section 43 leaves a collinear concentration problem. We now extract an exact arithmetic period along each nonnegative ray. Write

$$U(h,k) = 1 + Rh, \quad V(h,k) = 1 + R(h+Ck), \quad W(h,k) = 1 + R(h+Bk)$$

and parametrize the ray by $x(n) = (h_0 + ns, k_0 + nt)$. For a positive pairwise-coprime label $\lambda = (d_U, d_V, d_W)$ define

$$A_U = s, \qquad A_V = s + Ct, \qquad A_W = s + Bt,$$
$$r_Z = \frac{d_Z}{\gcd(d_Z, A_Z)}, \qquad T_\lambda = r_U r_V r_W,$$
$$C_\lambda = \gcd(d_U, A_U)\gcd(d_V, A_V)\gcd(d_W, A_W).$$

Theorem 44.1 (Exact ray period and capacity)**.** *If the same label divides the corresponding three arms at $x(n)$ and $x(m)$, then*

$$T_\lambda \mid |n-m|, \qquad T_\lambda C_\lambda = d_U d_V d_W.$$

Consequently, if all occurrences have indices in $[0,H]$, then

$$n_\lambda \le \left\lfloor \frac{H}{T_\lambda} \right\rfloor + 1.$$

Proof. Each d_Z divides a number congruent to one modulo R, hence is coprime to R. Subtracting the two arm values and cancelling R gives $d_Z \mid |n-m|A_Z$. Dividing d_Z and A_Z by their gcd shows $r_Z \mid |n-m|$. The r_Z remain pairwise coprime because they divide the pairwise-coprime components d_Z, so their product divides the gap. The factorization follows from $(d_Z/\gcd(d_Z,A_Z))\gcd(d_Z,A_Z) = d_Z$. All indices lie in one residue class modulo the positive integer T_λ, which gives the capacity. □

Assume $s > 0$ and $N > 0$, put $L = \max(s,t)$ and $K = (B+1)(C+1)$, and suppose $HL \le N$. Then every direction coefficient is positive and

$$C_\lambda \le s(s+Ct)(s+Bt) \le KL^3.$$

Theorem 44.2 (Nonconstant-ray cube-root cap)**.** *If $N^2 < d_U d_V d_W$, then*

$$(n_\lambda - 1)^3 < KN.$$

For any finite catalogue $\mathcal{L}$ of such labels and nonnegative integer weights w_λ,

$$\sum_{\lambda \in \mathcal{L}} w_\lambda n_\lambda^3 \le 4(B+1)(C+1)N \sum_{\lambda \in \mathcal{L}} w_\lambda.$$

Proof. Put $a = n_\lambda - 1$ and $T = T_\lambda$. The period capacity gives $aT \le H$, hence $aTL \le N$. Since

$$N^2 < TC_\lambda \le TKL^3,$$

squaring the first inequality and cancelling positive factors gives $a^2T < KL$. Thus $a^2 < KL$ and $aL \le N$; multiplication yields $a^3 < KN$. For positive integers n, X, the implication $(n-1)^3 < X \Rightarrow n^3 \le 4X$ follows from

$$4(a^3+1) - (a+1)^3 = 3(a-1)^2(a+1) \ge 0.$$

Apply this with $X = KN$, multiply by w_λ, and sum. The constant four is optimal at this integer level: $n = X = 2$ defeats the constant three. □

Ambient positivity is indispensable. The pure ledger

$$(n, H, T, C, D, N, K, L) = (1, 0, 1, 1, 1, 0, 1, 1)$$

satisfies every other period, factorization, capture, and strict large-label premise, while the proposed conclusion becomes $0 < 0$. This refutes exactly the $N > 0$-omitted strengthening.

Taking $w_\lambda = \varphi(d_U)\varphi(d_V)\varphi(d_W)$ converts the collinear part of the third-gcd identity to a total totient-weight problem. The remaining cost is the weight of the actual large-label catalogue and the number of supporting rays, rather than the raw length of a line.

The hypothesis $s > 0$ is essential. On a vertical ray the U component is constant and may be absorbed in full. If $d_U \le X_U$, $B, C, t > 0$, and $Ht \le N$, then

$$C_\lambda \le X_U BC\, t^2, \qquad (n_\lambda - 1)^2 < X_U BC.$$

Signed analogues for the two negative-slope arm-level directions, together with their individual component caps, remain part of the active route.

There is a full actual pressure test. For the primitive seed $(1, 8, 9)$ take

$$R = 6, \quad N = 22142, \quad x(n) = (21480, 282 + n), \quad \lambda = (128881, 49, 121)$$

and $S = \{0, 5929, 11858, 17787\}$. All four points lie in the canonical box, satisfy the admissibility conditions, and carry the displayed pairwise- coprime label, whose product exceeds N^2. Here

$$T_\lambda = 5929, \qquad C_\lambda = 128881, \qquad |S| = 17787/5929 + 1 = 4,$$

whereas $\lfloor 17787/(d_U d_V d_W)\rfloor + 1 = 1$. This refutes the stronger claim that the whole label product is itself a ray period and also refutes deleting the nonconstant-direction premise. It attains the correct period capacity and does not contradict the cube-root or vertical theorems.

The formal companion proves all period, capacity, cube-root, weighted-energy, and vertical statements above and checks this actual counterexample. No full-premise counterexample is known to the remaining catalogue-weight and signed arm-level transfer problems, so those routes remain active.

45. Actual affine catalogue weight and cubic aggregation

The ray-period argument reduces the remaining non-arm-level collinear term to the Euler-totient weight of the divisor labels that actually arise from canonical repeated kernels. We now compute that weight and aggregate its cubic energy without assuming that different kernel catalogues are disjoint.

For a positive kernel triple $d = (d_U, d_V, d_W)$ put $D(d) = d_U d_V d_W$ and

$$\mathcal{D}(d) = \{(e_U, e_V, e_W) : e_Z \mid d_Z\}, \qquad w(e) = \varphi(e_U)\varphi(e_V)\varphi(e_W).$$

For a threshold T, define

$$L_T(d) = \sum_{\substack{e \in \mathcal{D}(d) \\ e_U e_V e_W > T}} w(e).$$

Theorem 45.1 (Actual catalogue weight and finite tail)**.** *For every positive triple d,*

$$\sum_{e\in\mathcal{D}(d)} w(e) = D(d)$$

and

$$D(d) \le L_T(d) + T\tau(d_U)\tau(d_V)\tau(d_W).$$

Moreover, if $T < D(d)$, then the top label gives

$$\varphi(d_U)\varphi(d_V)\varphi(d_W) \le L_T(d).$$

Proof. The three sums separate, and $\sum_{e|n}\varphi(e) = n$. On the complementary range $e_Ue_Ve_W \le T$, one has $w(e) \le e_Ue_Ve_W \le T$, and there are at most $\tau(d_U)\tau(d_V)\tau(d_W)$ labels. Finally, $e = d$ occurs in the large range whenever $T < D(d)$. □

At a canonical exceptional affine point let $d = (K(U), K(V), K(W))$, where K retains the exact prime powers of exponent at least two. With

$$M = \left\lfloor \frac{c^6}{4R} \right\rfloor, \qquad N = M - 1,$$

the established arm and excess inequalities give

$$27D(d) \le c^{20}, \qquad D(d) > \frac{Rc^{14}}{8192}, \qquad N^2 < \frac{c^{12}}{16R^2}.$$

The elementary bound $\tau(n) \ll_\epsilon n^\epsilon$, applied with $\epsilon = 1/40$ to the three components, therefore yields uniformly

$$\frac{D(d) - L_{N^2}(d)}{D(d)} \ll c^{-3/2}.$$

Thus asymptotically all of the actual totient weight lies above the sharp box-square determinant threshold.

Let $\mathcal{K}$ be the realized canonical kernel triples, and let m_κ be the size of the corresponding class in the exceptional-set partition. If $n(e)$ is the number of exceptional points whose canonical kernel is divisible componentwise by e, then

$$n(e) = \sum_{\kappa : e|\kappa} m_\kappa.$$

Theorem 45.2 (Monotone catalogue aggregation)**.** *For the determinant-large part of the third-gcd energy,*

$$\sum_{\kappa\in\mathcal{K}} m_\kappa^3 L_T(\kappa) \le \sum_{\substack{e \\ e_Ue_Ve_W > T}} w(e)n(e)^3.$$

Consequently, at $T = N^2$ the right side is at least

$$\bigl(1 - O(c^{-3/2})\bigr) \sum_{\kappa\in\mathcal{K}} D(\kappa)m_\kappa^3.$$

Proof. For each fixed label,

$$\sum_{\kappa : e|\kappa} m_\kappa^3 \le \left(\sum_{\kappa : e|\kappa} m_\kappa\right)^3 = n(e)^3.$$

Multiply by $w(e)$, sum over e, and interchange the two finite sums. The asymptotic consequence follows from Theorem 45.1. □

This theorem controls energy, including its diagonal part. It does not by itself force two kernel classes to share a large label. If every point introduces a new catalogue, all relevant occupancies may still be one. A bound for deduplicated catalogue weight or singleton novelty remains necessary for a genuine reuse conclusion.

The form best matched to ray capacities is shifted. For nonnegative weights and integer occupancies put

$$W = \sum_i w_i, \qquad I = \sum_i w_i n_i, \qquad J = \sum_i w_i(n_i - 1).$$

Weighted Hölder gives

$$J^3 \le W^2 \sum_i w_i (n_i - 1)^3.$$

Since every occurring label has $n_i \ge 1$, the termwise identity $n_i = 1 + (n_i - 1)$ gives $I = W + J$. Thus the same estimate is the explicit overlap inequality

$$(I - W)^3 \le W^2 \sum_i w_i (n_i - 1)^3.$$

Hence individual caps $(n_i - 1)^3 \le X_i$ imply

$$J^3 \le W^2 \sum_i w_i X_i.$$

Labels may use different supporting lines and different X_i; no raw number-of-rays factor occurs. This also avoids the factor four used when the earlier paper converted shifted cubes to n_i^3.

The finite tail cannot be strengthened by simply subtracting T. The positive pairwise-coprime powerful triple $(d_U, d_V, d_W) = (9, 25, 1)$ at $T = 5$ has total weight 225, small-label weight $1 + 2 + 4 = 7$, and therefore

$$L_5(d) = 218 < 220 = 225 - 5.$$

This is a complete counterexample only to $L_T(d) \ge D(d) - T$. It leaves the correct divisor-count tail and the affine route intact.

The Lean companion checks the exact catalogue identities, tail bounds, top-label baseline, cubic aggregation, polynomial and shifted weighted Hölder inequalities, and this precise counterexample. The active next steps are a signed period/capture theorem for every non-arm-level supporting line, the two negative-slope arm-level caps, and an arithmetic bound on deduplicated catalogue novelty. No full-premise counterexample to those corrected targets is known.

46. Signed ray capture and exact canonical arm ceilings

We close the signed-direction and constant-arm losses left by Sections 44 and 45. Retain the three affine arms

$$U(h,k) = 1 + Rh, \qquad V(h,k) = 1 + R(h + Ck), \qquad W(h,k) = 1 + R(h + Bk), \qquad 0 < B < C.$$

A line through at least two lattice points has a primitive signed direction (s,t) and a parametrization $(h_0 + js, k_0 + jt)$. Put

$$A_U = s, \qquad A_V = s + Ct, \qquad A_W = s + Bt, \qquad L = \max(|s|, |t|). \tag{346}$$

If the index interval has length H inside a square of side difference N, then $HL \le N$.

For a positive pairwise-coprime divisor label $\lambda = (d_U, d_V, d_W)$, set

$$r_Z = \frac{d_Z}{\gcd(d_Z, |A_Z|)}, \qquad T_\lambda = r_U r_V r_W, \qquad C_\lambda = \prod_Z \gcd(d_Z, |A_Z|), \qquad D_\lambda = d_U d_V d_W. \tag{347}$$

Theorem 46.1 (Signed period and strict capture squeeze)**.** *Suppose λ occurs n_λ times on the segment, and put $a_\lambda = n_\lambda - 1$. Then*

$$T_\lambda \mid |i - j| \quad \textit{for two occurrence indices}, \qquad T_\lambda C_\lambda = D_\lambda, \qquad a_\lambda T_\lambda L \le N. \tag{348}$$

If $N > 0$ and $D_\lambda > N^2$, then

$$a_\lambda N L < C_\lambda, \tag{349}$$

$$a_\lambda^2 T_\lambda L^2 < C_\lambda, \qquad a_\lambda^3 T_\lambda^2 L^3 < C_\lambda N, \tag{350}$$

$$(a_\lambda T_\lambda L)^3 < D_\lambda N. \tag{351}$$

Proof. Subtract the two arm divisibilities. Actual occurrence makes every d_Z coprime to R, so cancellation gives $d_Z \mid (i-j)A_Z$. Dividing by the gcd in (347) gives $r_Z \mid |i-j|$. The r_Z remain pairwise coprime, hence their product divides the gap; the factorization of D_λ is immediate. All occurrence indices lie in one residue class modulo T_λ, so $a_\lambda T_\lambda \le H$ and then $a_\lambda T_\lambda L \le N$.

Now

$$a_\lambda T_\lambda LN \le N^2 < T_\lambda C_\lambda.$$

Cancel T_λ to obtain (349). Squaring the period capacity and using the same strict inequality gives $a_\lambda^2 T_\lambda L^2 < C_\lambda$; multiplication by the capacity proves the cubic inequality. Multiplying once more by T_λ gives (351). ☐

This theorem includes all signs and all zero coefficients. For a non-arm direction none of A_U, A_V, A_W vanishes. With

$$P(s,t) = |s|\,|s+Ct|\,|s+Bt|, \qquad K = (B+1)(C+1), \tag{352}$$

one has

$$C_\lambda \le P(s,t) \le KL^3. \tag{353}$$

The constant is sharp at $s = t = 1$. Consequently every repeated large non-arm label obeys

$$a_\lambda NL < P(s,t), \qquad a_\lambda^3 T_\lambda^2 < KN, \qquad KL^2 > N. \tag{354}$$

The last condition follows because $KL^2 \le N$ would contradict the first inequality as soon as $a_\lambda \ge 1$. Thus non-arm repetitions are confined to large primitive directions and retain an inverse-square period saving.

The canonical relation is $C = B+1$. There are then three zero-coefficient primitive directions, up to sign.

Theorem 46.2 (Exact canonical arm capture)**.** *For the constant U, V, W directions, respectively, the absolute coefficient triples, scales, and captures are*

constant arm	$(\lvert A_U\rvert, \lvert A_V\rvert, \lvert A_W\rvert)$	L	C_λ
U	$(0, C, B)$	1	d_U
V	$(C, 0, 1)$	C	d_V
W	$(B, 1, 0)$	B	d_W.

Hence a large occurring label satisfies the three respective linear caps

$$a_\lambda N < d_U, \qquad a_\lambda NC < d_V, \qquad a_\lambda NB < d_W. \tag{355}$$

Proof. Each component of an actual label divides an arm value congruent to one modulo $R = \operatorname{rad}(BC)$, and is therefore coprime to B and C. For the U direction,

$$C_\lambda = \gcd(d_U, 0)\gcd(d_V, C)\gcd(d_W, B) = d_U.$$

For the V direction the coefficients are $(C, 0, C-B) = (C, 0, 1)$, giving $C_\lambda = d_V$. The W direction is $(B, -1, 0)$ and gives $C_\lambda = d_W$. Substitution into (349) proves (355). ☐

With the preceding canonical point bounds $3d_U \le C^6$ and $3d_V, 3d_W \le C^7$, this gives

$$3a_\lambda N < C^6, \qquad 3a_\lambda NC < C^7, \qquad 3a_\lambda NB < C^7.$$

These are linear in the relevant label component and improve the former square-root treatment of the vertical direction.

We next pass from a shifted occupancy to the full cubic energy. For every $n \in \mathbb{N}$, with truncated $a = n-1$,

$$n^3 \le 1 + 7a^3. \tag{356}$$

For $n \ge 2$, the difference of the right side and $(a+1)^3$ is $3a(a-1)(2a+1) \ge 0$; the cases $n = 0, 1$ are immediate. Equality at $n = 2$ shows that seven is optimal.

Let $\mathcal{K}$ be the realized selected canonical kernel triples $\kappa = (k_U, k_V, k_W)$, with $D_\kappa = k_U k_V k_W$. Define $\mathcal{L}$ as the deduplicated union of their downward divisor catalogues, then retain only labels whose coordinate product exceeds N^2. Let n_λ be the number of selected canonical points realizing the label λ. For

$$w_\lambda = \varphi(d_U)\varphi(d_V)\varphi(d_W), \qquad W_0 = \sum_{\lambda\in\mathcal{L}} w_\lambda, \qquad E = \sum_{\lambda\in\mathcal{L}} w_\lambda n_\lambda^3, \qquad E_{\mathrm{sh}} = \sum_{\lambda\in\mathcal{L}} w_\lambda (n_\lambda - 1)^3,$$

equation (356) gives

$$E \le W_0 + 7E_{\mathrm{sh}}. \tag{357}$$

This charges a singleton once and introduces no number-of-lines factor.

Choose one owner $o(\lambda) \in \mathcal{K}$ for every deduplicated label. Euler's divisor identity gives, for each coordinate Z,

$$\sum_{\lambda\in\mathcal{L}} w_\lambda d_Z(\lambda)^3 \le \sum_{\kappa\in\mathcal{K}} D_\kappa k_Z(\kappa)^3, \qquad W_0 \le \sum_{\kappa\in\mathcal{K}} D_\kappa. \tag{358}$$

Indeed, labels owned by κ form a subset of its full downward catalogue, whose total totient weight is D_κ, and $d_Z^3 \le k_Z^3$.

Partition repeated labels into non-arm labels and the three arm directions. Let

$$\mathcal{S}_{\mathrm{non}} = \sum_{\substack{\lambda \text{ repeated} \\ \lambda \text{ non-arm}}} \frac{w_\lambda}{T_\lambda^2}.$$

Theorem 46.3 (Global signed affine energy ceiling)**.** *In the canonical setup,*

$$\begin{aligned} E \le{}& \sum_{\kappa\in\mathcal{K}} D_\kappa + 7KN\mathcal{S}_{\mathrm{non}} \\ &+ \frac{7}{N^3}\sum_{\kappa\in\mathcal{K}} D_\kappa\left(k_U^3 + \frac{k_V^3}{C^3} + \frac{k_W^3}{B^3}\right). \end{aligned} \tag{359}$$

Every term in $\mathcal{S}_{\mathrm{non}}$ is supported on $KL^2 > N$.

Proof. The non-arm cubic cap in (354) gives $E_{\mathrm{non}} \le KN\mathcal{S}_{\mathrm{non}}$. Cubing (355), multiplying by w_λ, and summing gives the three coordinate moments divided by N^3, N^3C^3, N^3B^3. Enlarge each arm subset to all labels and apply (358). Add the terms and use (357). ☐

The arm-level contribution is therefore explicit. The remaining affine upper-bound task is the non-arm inverse-period sum $\mathcal{S}_{\mathrm{non}}$, together with the comparison of the singleton baseline against catalogue-class multiplicities. These are active gates.

Several complete pressure tests determine the exact boundary. Replacing $N^2 < D_\lambda$ by $N^2 \le D_\lambda$ fails at $(n, H, T, C_\lambda, D, N, L) = (2, 1, 1, 1, 1, 1, 1)$. The tuple

$$(n, H, T, C_\lambda, D, N, K, L) = (2, 3, 3, 6, 18, 3, 6, 1)$$

satisfies the period, span, factorization, strict-threshold, and capture premises, but refutes adding one more factor of T to either higher capture inequality. Scaling a primitive arm direction by five gives full-premise examples showing why primitivity is required, and four separate primitive examples show why each coefficient-coprimality premise is required. At occupancy two, coefficient six in (356) fails. These witnesses retire only the stated strengthenings; actual canonical labels retain the strictness, primitivity, and coprimality premises.

Catalogue membership is also essential. In the actual box $(B, C, M, N, R) = (1, 2, 10, 9, 2)$, with 82 admissible points, the totient weight of all arm-divisor labels above N^2 is 972,496, whereas the owner mass of the twelve distinct powerful-kernel triples is only 1,072. Thus deleting membership in the selected downward-catalogue union would assert the false inequality $972{,}496 \le 1{,}072$. This full-premise counterexample retires precisely that enlarged owner statement, while leaving the selected-catalogue theorem unchanged.

An independent exact-integer audit checks 15,840 signed directions, 1,776,807 cubic ledgers, 2,390,018 normalized quadratic ledgers, 43,403 exact arm-capture cases, and four actual canonical boxes. Those boxes contain 4,307 distinct local large labels and 259 repeated labels, testing every ray-level identity. The selected powerful-kernel subcatalogues for $M = 10$ contain 110 large labels and 12 repeated labels. A separate $(B, C, M) = (4, 5, 30)$ stress box contains 113 selected large labels, nine repeated labels, and one repeated non-arm label. All owner and global energy checks pass. These finite audits are pressure tests and supply no asymptotic inference.

The Lean companion kernel-checks the signed period/capture algebra, all three canonical arm identities, the sharp constant K, the optimal coefficient seven, the finite weighted owner bounds, the global shifted-energy interface, and the full-premise boundary countermodels. It assumes no exceptional-point lower bound, no catalogue novelty theorem, and no *abc* statement.

47. Affine incidence cancellation and inverse-period catalogues

We refine the selected powerful-kernel catalogue without enlarging it to all arm divisors. Let $\mathcal{K}$ be the finite set of kernel classes, let m_κ be the number of selected points in class κ, and let

$$\mathcal{L}_N(\kappa) = \{d = (d_U, d_V, d_W) : d_Z \mid k_Z(\kappa),\ d_U d_V d_W > N^2\}.$$

Give d the weight $w_d = \varphi(d_U)\varphi(d_V)\varphi(d_W)$ and put

$$L_\kappa = \sum_{d \in \mathcal{L}_N(\kappa)} w_d.$$

On the deduplicated union $\mathcal{L}$, write n_d for the number of selected points supporting d, and define

$$I = \sum_{d\in\mathcal{L}} w_d n_d, \qquad E = \sum_{d\in\mathcal{L}} w_d n_d^3, \qquad E_{\mathrm{sh}} = \sum_{d\in\mathcal{L}} w_d (n_d - 1)^3.$$

Proposition 47.1 (Exact point-label incidence)**.** *One has*

$$I = \sum_{\kappa\in\mathcal{K}} m_\kappa L_\kappa. \tag{47.1.1}$$

No disjointness assumption on the class tails is required.

Proof. For every label, $n_d = \sum_{\kappa : d \in \mathcal{L}_N(\kappa)} m_\kappa$. Substitute this identity in I and interchange the two finite sums. The inner sum for a fixed class is exactly L_κ. □

The preceding monotone-overlap theorem gives $\sum_\kappa m_\kappa^3 L_\kappa \le E$. The correct conversion of E uses the incidence I, rather than paying the deduplicated union weight as an unrelated error.

Lemma 47.2 (Optimal incidence shift)**.** *For every $n \in \mathbb{N}$,*

$$n^3 \le n + 6(n-1)^3, \tag{47.2.1}$$

where subtraction is in $\mathbb{N}$. The coefficient six is optimal.

Proof. The cases $n = 0, 1$ are equalities. For $n \ge 2$, write $a = n - 1$; then

$$n + 6a^3 - n^3 = a(a-1)(5a+2) \ge 0.$$

At $n = 2$ equality reads $8 = 2 + 6$, so no smaller coefficient works. □

Theorem 47.3 (Singleton cancellation)**.** *With*

$$\Delta_{\mathrm{mult}} = \sum_{\kappa\in\mathcal{K}} (m_\kappa^3 - m_\kappa) L_\kappa,$$

one has

$$\Delta_{\mathrm{mult}} \le 6E_{\mathrm{sh}}. \tag{47.3.1}$$

Every class of multiplicity one cancels exactly from the left side. For occupancy at least three, the coefficient six may be replaced by three; the two coefficients are separately sharp at occupancies two and three.

Proof. Multiply Lemma 47.2 by w_d and sum to obtain $E \le I + 6E_{\mathrm{sh}}$. Combine this with the monotone-overlap inequality and Proposition 47.1, then subtract the finite common incidence. For $n \ge 3$, the refinement follows from $n^3 \le n+3(n-1)^3$; equality at $n = 3$ proves sharpness. □

There are two distinct sources of shifted incidence. Put

$$A_0 = \sum_\kappa L_\kappa, \qquad A_1 = \sum_\kappa (m_\kappa - 1)L_\kappa, \qquad W = \sum_{d\in\mathcal{L}} w_d, \qquad \Omega = A_0 - W.$$

If r_d is the number of supporting kernel classes, then $\Omega = \sum_d w_d(r_d - 1)$. Double counting gives

$$J := \sum_d w_d(n_d - 1) = A_1 + \Omega. \tag{47.3.2}$$

Weighted Hölder therefore yields the exact normalized pressure inequality

$$(A_1 + \Omega)^3 \le (A_0 - \Omega)^2 E_{\mathrm{sh}}. \tag{47.3.3}$$

The term A_1 records multiplicity inside one class; Ω records overlap of distinct class catalogues. Neither can be deleted.

We next retain the inverse period. For two selected points write their difference as $q(s,t)$ with (s,t) primitive, set

$$A_U = s, \qquad A_V = s + Ct, \qquad A_W = s + Bt, \qquad L = \max(|s|,|t|), \quad qL \le N,$$

and assume the direction is non-arm. Let

$$g_Z = \gcd(k_Z(x), k_Z(y)), \quad c_Z = \gcd(g_Z, |A_Z|), \quad G = \prod_Z g_Z, \quad C_g = \prod_Z c_Z, \quad T_g = G/C_g.$$

Pairwise coprimality of the three arms and cancellation of $R = \mathrm{rad}(BC)$ give

$$T_g \mid q, \qquad G = T_g C_g \le q C_g. \tag{47.3.4}$$

For $k \ge 1$ and $a \ge 0$, define

$$F(k,a) = \sum_{d\mid k} \varphi(d) \frac{\gcd(d,a)^2}{d^2}.$$

If $p^e \parallel k$ and $r = \min(e, v_p(a))$, with $r = e$ for $a = 0$, a direct prime-power summation gives

$$F(p^e, a) = \begin{cases} p^e, & r = e, \\ p^r + p^{r-1}(1 - p^{-(e-r)}), & r < e. \end{cases} \tag{47.3.5}$$

Thus F has an exact Euler product and $F(k,a) \le \gcd(k,a)\prod_{p\mid k}(1 + p^{-1})$.

Let $Q(x,y)$ be the sum of w_d/T_d^2 over all common labels with $d_U d_V d_W > N^2$, where $T_d = \prod_Z d_Z/\gcd(d_Z,|A_Z|)$. Also put

$$H(k) = \sum_{d\mid k} \frac{\varphi(d)}{d} = \prod_{p^e\parallel k} \left(1 + e(1 - p^{-1})\right).$$

Theorem 47.4 (Hybrid pair-catalogue bound)**.** *Every non-arm pair satisfies*

$$Q(x,y) \le \min\left\{ \prod_Z F(g_Z, |A_Z|), \frac{C_g^2 G}{N^4}, \frac{C_g^2 H(G)}{N^2} \right\}. \tag{47.4.1}$$

If q_0 and $P_0 = \prod_Z |A_Z|^{(R)}$ denote the R-free parts, then the first term is at most $P_0 q_0/\varphi(q_0)$ and $G \mid q_0 P_0$.

Proof. Dropping the large-product restriction makes the three divisor sums independent and gives the Euler-product term. For a retained label, $D_d > N^2$ and $C_d \le C_g$, hence $T_d^{-2} < C_g^2/N^4$; summing all totient weights gives G. Alternatively,

$$\frac{w_d}{T_d^2} = \frac{w_d}{D_d}\frac{C_d^2}{D_d} < \frac{C_g^2}{N^2}\frac{w_d}{D_d},$$

and the enlarged divisor sum is $H(G)$. The R-free conclusions follow from $T_g \mid q_0$, $C_g \mid P_0$, and the local Euler factors. □

Because G is powerful, the same pair also obeys a useful direction filter. Writing $\mathfrak{E}(n) = n/\operatorname{rad}(n)$, one has

$$T_g\mathfrak{E}(P_0) > N, \qquad \mathfrak{E}(P_0) > L. \tag{47.4.2}$$

Indeed $G > N^2$ and powerfulness give $\mathfrak{E}(G) > N$, while $\mathfrak{E}(G) \le T_g\mathfrak{E}(C_g) \le T_g\mathfrak{E}(P_0)$; then use $T_g \mid q$ and $qL \le N$. In the canonical branch $C = B+1$, the three R-free coefficients are pairwise coprime. If one of them has powerful excess greater than $L^{1/3}$, then the corresponding original linear coefficient has at least that excess. The original coefficient, rather than the operation of removing R-factors, is injective along each side of the direction square. The elementary estimate

$$\#\{n \le X : \mathfrak{E}(n) > Y\} = O(X/\sqrt{Y})$$

follows by putting $m = \prod_p p^{\lfloor v_p(n)/2\rfloor}$: then $m^2 \mid n$ and $m^2 \ge \mathfrak{E}(n) > Y$, after which one sums X/m^2 over $m > \sqrt{Y}$. On the shell $\max(|s|,|t|) = L$, this counts $O(CL^{5/6})$ eligible values on the nonconstant sides. The exceptional vertical-side values of s are summed by the same square-divisor estimate in dyadic blocks. Summing the shells reduces the number of eligible primitive directions of scale at most N from $O(N^2)$ to $O(CN^{11/6})$. This is a real saving, although it does not yet count the maximal powerful intersection tops on each direction.

Finally, let $S_{\mathrm{non}} = \sum w_d/T_d^2$ over repeated non-arm labels. A label of occupancy n_d occurs in exactly $\binom{n_d}{2}$ pair catalogues, so

$$\sum_{\{x,y\}} Q(x,y) = \sum_{d\ \mathit{mrepeated,nonarm}} \binom{n_d}{2}\frac{w_d}{T_d^2}.$$

Choosing one loop for every class of multiplicity at least two and one edge for every relevant pair of singleton classes gives a smaller support skeleton $\mathcal{E}$ with

$$S_{\mathrm{non}} \le \sum_{e\in\mathcal{E}} Q_e. \tag{47.4.3}$$

The inverse period cannot supply a uniform saving by itself. In the actual canonical box

$$(B, C, R, M, N) = (8, 9, 6, 22143, 22142)$$

the two admissible points $(3128, 10183)$ and $(21897, 11423)$ lie in distinct singleton kernel classes and share the sole large label

$$d = (137^2, 173^2, 1), \qquad D_d = 561737401 > N^2.$$

Their primitive direction has coefficients $(18769, 29929, 28689)$, so $C_d = D_d$ and $T_d = 1$. Hence

$$Q_{\{x,y\}} = S_{\mathrm{non}} = \varphi(137^2)\varphi(173^2) = 554413792. \tag{47.4.4}$$

This full-premise example has $R < C$ and refutes any strict fractional saving deduced only from the subcritical branch, singleton support, or a positive period lower bound. It does not refute the pair-skeleton theorem.

The active normalized contradiction target is

$$(A_1+\Omega)^3 > (A_0-\Omega)^2\left[(B+1)(C+1)N\sum_{e\in\mathcal{E}} Q_e + \frac{1}{N^3}\sum_{\kappa} L_\kappa\left(k_U^3 + \frac{k_V^3}{C^3} + \frac{k_W^3}{B^3}\right)\right]. \tag{47.4.5}$$

Proving this requires a global aggregation theorem for the divisibility-maximal powerful intersection tops. No counterexample to that complete statement is known, so the affine route remains active.

48. Maximal-intersection ownership aggregation

We continue the affine catalogue notation of Section 47. This section proves an unconditional compression theorem for the non-arm catalogue. It does not prove the *abc* conjecture: the final strict comparison isolated below remains open.

Let X be the selected canonical parameter points in a square of side difference N. Write $\kappa(x) = (k_U(x), k_V(x), k_W(x))$ for the powerful-kernel triple of x, and order triples by coordinatewise divisibility. For distinct $x, y \in X$, set

$$\gamma(x, y) = \gcd_{\mathrm{coord}}(\kappa(x), \kappa(y)).$$

Let Γ be the distinct values $\gamma(x, y)$ having coordinate product greater than N^2 and a non-arm supporting line, and let $\mathcal{M}$ be the divisibility-maximal elements of Γ. Put

$$\mathrm{Supp}(\mu) = \{x \in X : \mu \preceq \kappa(x)\}, \qquad r_\mu = |\,\mathrm{Supp}(\mu)|.$$

Proposition 48.1 (Maximal supports and the reduced skeleton)**.** *Every repeated non-arm large label divides a maximal top on the same unique non-arm line. For $\mu \in \mathcal{M}$, one has $r_\mu \geq 2$, and every pair of distinct points in* $\mathrm{Supp}(\mu)$ *has exact top μ. Consequently distinct maximal supports meet in at most one point and*

$$\sum_{\mu \in \mathcal{M}} \binom{r_\mu}{2} \leq \binom{|X|}{2}. \tag{48.1.1}$$

Moreover, the reduced class skeleton consisting of one loop for every kernel class of multiplicity at least two and one edge for every pair of singleton classes is cofinal in Γ, and hence has exactly the same maximal top values.

Proof. A repeated label is contained in the exact gcd top of any two of its support points. Finiteness supplies a maximal top above it. The top is itself a large label, so large-label line uniqueness keeps every comparable top on the original non-arm line. If two points support a maximal top μ, their exact top lies above μ and on that line; maximality forces equality. Two common points of distinct maximal supports would therefore have two different exact tops, proving the intersection and pair count assertions.

For cofinality, consider an exact pair top omitted by the reduced skeleton. At least one endpoint belongs to a class κ of multiplicity at least two. Its retained loop has top κ, which dominates the omitted gcd. Every point in this class supports the omitted large label, so the loop is on the same unique non-arm line and is eligible. A cofinal subset of a finite poset has the same maximal elements: dominate a whole-poset maximum by a subset element, and dominate any whole-poset extension of a subset maximum once more inside the subset; antisymmetry then gives equality. ☐

Fix a total order on $\mathcal{M}$, and assign every repeated non-arm label λ to the first maximal top $o(\lambda)$ divisible by it. Define

$$\mathcal{O}_\mu = \{\lambda : o(\lambda) = \mu\}, \qquad S_\mu = \sum_{\lambda \in \mathcal{O}_\mu} \frac{w_\lambda}{T_\lambda^2}, \qquad E_\mu = \sum_{\lambda \in \mathcal{O}_\mu} w_\lambda (n_\lambda - 1)^3.$$

These are exact partitions of S_{non} and E_{non}. If $\lambda \in \mathcal{O}_\mu$, then

$$n_\lambda \geq r_\mu, \qquad T_\lambda \mid T_\mu. \tag{48.1.2}$$

The second relation is in this direction: for $d \mid m$, $d/\gcd(d, a) \mid m/\gcd(m, a)$ coordinatewise. It does not permit T_λ^{-2} to be bounded above by T_μ^{-2}.

On the line of μ, let

$$Q(\mu) = \sum_{\substack{d \preceq \mu \\ D_d > N^2}} \frac{w_d}{T_d^2}, \qquad B_\mu \geq S_\mu, \qquad \mathcal{H}_3 = \sum_{\mu \in \mathcal{M}} \frac{B_\mu}{(r_\mu - 1)^3}.$$

One valid owner-independent choice has

$$B_\mu = \min\left\{F(\mu, A_\mu), \frac{C_\mu^2 G_\mu}{N^4}, \frac{C_\mu^2 H(G_\mu)}{N^2}\right\}, \qquad S_\mu \le Q(\mu) \le B_\mu, \tag{48.1.3}$$

with the notation of Theorem 47.4.

Theorem 48.2 (Ownership Cauchy inequality and strict ray closure)**.** *One has*

$$S_{\mathrm{non}}^2 \le E_{\mathrm{non}}\mathcal{H}_3. \tag{48.2.1}$$

Writing $K = (B+1)(C+1)$, the nonempty non-arm domain satisfies the strict bounds

$$S_{\mathrm{non}} < KN\mathcal{H}_3, \qquad E_{\mathrm{non}} < (KN)^2\mathcal{H}_3. \tag{48.2.2}$$

For an empty domain the corresponding weak inequalities hold with both sides zero.

Proof. For each maximal top, catalogue coverage and support containment give

$$S_\mu \le B_\mu, \qquad S_\mu \le \sum_{\lambda \in \mathcal{O}_\mu} w_\lambda \le \frac{E_\mu}{(r_\mu - 1)^3}.$$

Multiplication and Cauchy–Schwarz over the finite index set $\mathcal{M}$, rather than over labels or point pairs, prove the first claim. The signed-ray theorem gives termwise $(n_\lambda - 1)^3 T_\lambda^2 < KN$. Positivity of w_λ / T_λ^2 makes the summed inequality $E_{\mathrm{non}} < KNS_{\mathrm{non}}$ strict whenever the exact summation domain is nonempty. In that case $S_{\mathrm{non}} > 0$, and the owner cap of the nonempty owner set also gives $\mathcal{H}_3 > 0$. Combining the two inequalities and dividing by S_{non} proves the strict conclusions. □

Proposition 48.3 (Full-catalogue multiplicity is paid by energy)**.** *The maximal-top catalogues satisfy*

$$\sum_{\mu \in \mathcal{M}} Q(\mu) \le E_{\mathrm{non}}. \tag{48.3.1}$$

Thus the exact choice $B_\mu = Q(\mu)$ gives $\mathcal{H}_3 \le \sum_\mu Q(\mu) \le E_{\mathrm{non}}$.

Proof. For a repeated non-arm label λ, let $m_\lambda = \#\{\mu \in \mathcal{M} : \lambda \preceq \mu\}$. Every divisor term of $Q(\mu)$ is an actual repeated label on the same unique line, so finite summation gives the exact identity

$$\sum_{\mu \in \mathcal{M}} Q(\mu) = \sum_{\lambda \text{ repeated,nonarm}} \frac{w_\lambda}{T_\lambda^2} m_\lambda.$$

Choose one realizing pair for every top counted by m_λ. These pairs lie in the n_λ-point support and are distinct, because one pair has one exact gcd top. Hence $m_\lambda \le \binom{n_\lambda}{2}$. For $n_\lambda \ge 2$ and $T_\lambda \ge 1$,

$$\frac{1}{T_\lambda^2}\binom{n_\lambda}{2} \le (n_\lambda - 1)^3.$$

Multiplying by w_λ and summing proves the claim. Finally $r_\mu \ge 2$, so the cubic support denominator only decreases $Q(\mu)$. □

Let $\mathcal{A}_{\mathrm{arm}}$ denote the three proved arm caps and let $\widehat{Q}_{\mathrm{dir}}$ be the sum of the pairwise-coprime direction-lcm envelopes. Exact label containment in those envelopes and the preceding theorem give

$$E_{\mathrm{sh}} \le \min\{KN\widehat{Q}_{\mathrm{dir}}, (KN)^2\mathcal{H}_3\} + \mathcal{A}_{\mathrm{arm}}. \tag{48.2.3}$$

Since $J = A_1 + \Omega$, $W = A_0 - \Omega$, and $J^3 \le W^2 E_{\mathrm{sh}}$, every selected configuration with $W > 0$ obeys

$$\left(\frac{A_1 + \Omega}{A_0 - \Omega}\right)^3 \le \min\left\{KN\frac{\widehat{Q}_{\mathrm{dir}}}{A_0 - \Omega}, (KN)^2\frac{\mathcal{H}_3}{A_0 - \Omega}\right\} + \frac{\mathcal{A}_{\mathrm{arm}}}{A_0 - \Omega}. \tag{48.2.4}$$

This is the ownership-preserving normalized gate.

For completeness, in the canonical consecutive-coefficient setting $1 \le B$, $C = B + 1$, the arithmetic divisor enumeration gives, for every $\varepsilon > 0$,

$$\begin{aligned} |\mathcal{M}| &\ll_\varepsilon (CN)^\varepsilon \min\{N^2, CN^{11/6}\}, \\ \sum_{\mu \in \mathcal{M}} B_\mu &\ll_\varepsilon K(CN)^\varepsilon \min\{N^5, CN^{29/6}\}, \end{aligned} \tag{48.2.5}$$

provided $S_\mu \le B_\mu \le F(\mu, A_\mu)$. On $r_\mu - 1 \ge R_0 \ge 1$, the pressure has the further factor R_0^{-4}. Indeed a closest support pair has step q_μ satisfying $q_\mu(r_\mu - 1)L \le N$; the fixed-direction candidates divide the three integers $q_\mu|A_Z|$; and the original, before R-free reduction, coefficient is finite-to-one on each side of the direction square. Here the three direction coefficients are pairwise coprime after removing the primes of $\mathrm{rad}(BC)$: their pairwise gcds divide B, C, and $C - B = 1$. The square-divisor excess estimate then gives $O(CL^{5/6})$ ordinary directions on shell L, and summation gives the displayed exponents.
Certified boundaries. The following statements are refuted only with the premises stated here. All examples are exact finite certificates.

(1) The implication $T_\lambda \ge T_\mu$ for every owned label is false under canonical admissibility, non-arm maximal ownership, and full catalogue containment. At $B = 7, C = 8, M = 400, N = 399$, the unique top $\mu = (9, 121, 529)$ has $T_\mu = 3$, while its proper owned label $(3, 121, 529)$ has period one; the exact catalogue has two terms and $Q(\mu) = 445280/3$.
(2) The assertion that one kernel class belongs to at most one maximal top is false under canonical admissibility and global divisibility maximality. At $B = 5, C = 6, M = 254, N = 253$, the singleton class $(1, 841, 43681)$ lies in the two incomparable maximal supports with tops $(1, 841, 121)$ and $(1, 841, 361)$, whose intersection is exactly that one point.
(3) Raw pair enumeration is not ownership preserving. At $B = 4, C = 5, M = 390, N = 389$, all three pairs of three canonical points have the same top $(49, 361, 9)$. Its support has size three, its period-one catalogue has mass 86184, and equality holds in the local cubic Cauchy bound after the single top is deduplicated.
(4) Neither $Q(\mu) \le w_\mu$ nor $Q(\mu) \le 2w_\mu$ follows from canonical admissibility, non-arm maximal ownership, and full catalogue coverage. At $B = 55123, C = 55124, M = 3, N = 2$, the two-point top $\mu = (1, 1, 55125)$ has 34 period-one terms,

$$Q(\mu) = 55122, \qquad w_\mu = 25200, \qquad \frac{Q(\mu)}{w_\mu} = \frac{9187}{4200} > 2.$$

The explicit three-term cap equals 55125, with inflation $35/16 > 2$.
(5) Maximality, ownership, cap coverage, $r_\mu \ge 2$, and linear support alone imply neither a strict Cauchy saving nor a linear number of tops. The abstract four-vertex construction assigns an edge-specific prime square to each pair, realizes all six pair tops, and has $S_{\mathrm{non}} = E_{\mathrm{non}} = \mathcal{H}_3$. This item is an abstract ledger certificate and does not claim canonical affine realizability.

The first four examples are canonical admissible local selections, but they are not asserted to satisfy the exceptional-point inequality. They refute the listed local implications and no stronger statement that adds exceptional-point geometry. The fifth omits the affine divisibility and powerful-excess filters. None of these certificates closes the affine route.
Open gate. The unresolved task is to prove the strict reverse of

$$(A_1 + \Omega)^3 \le (A_0 - \Omega)^2 \left[\min\{KN\widehat{Q}_{\mathrm{dir}}, (KN)^2\mathcal{H}_3\} + \mathcal{A}_{\mathrm{arm}}\right] \tag{48.2.6}$$

for every hypothetical exceptional selection, or to obtain an actual standard *abc* counterexample. This requires a canonical low-support supersaturation or catalogue-sparsity theorem coupling the normalized incidence, $\mathcal{H}_3$, and the arm term. The finite scan found no same-line or same-direction multiplicity in its tested range, but these are explicitly finite no-hit observations and are not used as theorems.

Formalization boundary. The companion Lean module proves 24 finite or algebraic theorems and prints the axiom dependencies of all 24. It formalizes finite-poset cofinality, abstract maximal-support and codegree arguments, owner sums, Cauchy aggregation, strict-ray closure, the polynomial normalized gate, abstract tree subtraction, and the numerical cores of the boundary certificates. It does not formalize the affine realization of the maximal cover, the general period-divisibility lemma, the exact catalogue double count and its support-pair injection, the direction Euler products and tails, the spanning-tree realization, the closest-pair shell counts, the arm and weighted Hölder instantiations, or the complete canonical-kernel verification of the finite examples. The analytic bounds above remain paper proofs; the complete finite canonical premises are independently replayed by the accompanying Python certificate.

49. Residue-parity localization in the balancing-Pell packet

Write

$$(1+\sqrt{2})^n = A_n + B_n\sqrt{2}.$$

For odd n these coprime coordinates satisfy $A_n^2 - 2B_n^2 = -1$. The preceding four-prime analysis proves that, at an odd prime index ℓ, the odd-exponent prime sets O_A, O_B obey

$$\prod_{q\in O_A}\prod_{r\in O_B}\left(\frac{q}{r}\right) = \left(\frac{2}{\ell}\right). \tag{360}$$

We now localize the depth-three primes by congruence class.

Multiplication by $1+\sqrt{2}$ gives

$$A_{n+1} = A_n + 2B_n, \qquad B_{n+1} = A_n + B_n.$$

Since

$$(1+\sqrt{2})^8 = 577 + 408\sqrt{2} \equiv 1 \pmod 8,$$

the coordinate pair is periodic modulo eight. Direct evaluation gives

$$\begin{array}{c|cccc} n \pmod 8 & 1 & 3 & 5 & 7 \\ \hline A_n \pmod 8 & 1 & 7 & 1 & 7 \\ B_n \pmod 8 & 1 & 5 & 5 & 1. \end{array} \tag{361}$$

If an odd prime q divides A_n, the norm equation gives $2B_n^2 \equiv 1 \pmod q$, whence $q \equiv 1, 7 \pmod 8$. If an odd prime r divides B_n, then $A_n^2 \equiv -1 \pmod r$, whence $r \equiv 1, 5 \pmod 8$.

We use the following finite-product observation. If $\rho^2 = 1$ and $\rho \neq 1$ in a commutative monoid, every x_i is 1 or ρ, and

$$\prod_i x_i^{e_i} = \rho,$$

then some $x_i = \rho$ has odd exponent. Otherwise each nontrivial factor would occur to even exponent and the product would be one. If every $e_i \geq 2$, the selected exponent is at least three.

Theorem 49.1 (Forced-residue depth-three localization)**.** *Let ℓ be an odd prime and suppose $A_\ell B_\ell$ is squarefull. Then:*

1. *if $\ell \equiv 3, 7 \pmod 8$, some prime $q_A \mid A_\ell$ occurs to an odd exponent at least three and satisfies $q_A \equiv 7 \pmod 8$;*
2. *if $\ell \equiv 3, 5 \pmod 8$, some prime $q_B \mid B_\ell$ occurs to an odd exponent at least three and satisfies $q_B \equiv 5 \pmod 8$.*

Proof. The residue table shows that both coordinates are odd. Reduce the two prime-power factorizations modulo eight. The allowed local classes are $\{1, 7\}$ and $\{1, 5\}$, respectively. In the stated index classes, (361) makes the total product the nontrivial involution 7 or 5. The finite-product observation selects an odd exponent. Since $\gcd(A_\ell, B_\ell) = 1$, squarefullness of $A_\ell B_\ell$ makes every positive exponent in either channel at least two, so the selected odd exponent is at least three. $\square$

For $\ell \equiv 3 \pmod 8$, both conclusions hold simultaneously. Thus the two previously unspecified deep primes can be required to occupy the 7-class in the A channel and the 5-class in the B channel.

The Lean statement uses an explicit factor-list packet interface. Each entry stores a prime and its exponent; the packet records the allowed local classes and the squarefull lower bound. If the A residue product is 7 or the B residue product is 5, the corresponding list contains the prime and odd exponent asserted above. Thus the formal theorem combines the residue-product hypothesis and the depth conclusion in one statement, while passage from an actual Pell factorization to that interface is the elementary argument in this proof.

Theorem 49.2 (A forced cross-channel nonresidue)**.** *If $\ell \equiv 3, 5 \pmod 8$ and $A_\ell B_\ell$ is squarefull, then there are odd-exponent primes $q \mid A_\ell$ and $r \mid B_\ell$, each of exponent at least three, such that*

$$\left(\frac{q}{r}\right) = -1.$$

Proof. Here $(2/\ell) = -1$. Equation (360) is a finite product of nonzero signs with value -1: coprimality of the two channels prevents a zero Legendre symbol. Hence at least one factor is negative. Membership in $O_A \times O_B$ gives odd exponents; squarefullness makes them at least three. ☐

Here too the Lean statement is combined rather than left as two unrelated lemmas: a flattened factor-list packet stores every pair in $O_A \times O_B$, its two odd squarefull exponents, and its actual Legendre symbol. A negative product yields a stored pair with symbol -1 and both exponents at least three.

The existential quantifier is sharp under the ordinary actual prime-index premises. At $\ell = 11$ one has

$$A_{11} = 8119 = 23 \cdot 353, \qquad B_{11} = 5741,$$

with all three factors prime, while

$$\left(\frac{23}{5741}\right) = -1, \qquad \left(\frac{353}{5741}\right) = 1.$$

Indeed, $252^2 \equiv 353 \pmod{5741}$, whereas quadratic reciprocity reduces the first symbol to $\left(\frac{14}{23}\right) = -1$. The product is still $-1 = \left(\frac{2}{11}\right)$ because $11 \equiv 3 \pmod 8$. The factorization shows that $23, 353 \in O_A$ and $5741 \in O_B$. This is a full actual counterexample to the strengthening that every cross pair is a nonresidue. It has three exponent-one factors, so it does not satisfy the squarefull premise and does not close the Pell route.

The companion Lean module checks the integral recurrence and norm, the eight-step formula, all residue classes in (361), the abstract odd-exponent and depth-three extraction, the two explicit factor-list packet theorems, and one combined index-eleven certificate containing the prime-index premise, global character, coordinates, factorization, exponent-one witnesses and Legendre values. The open point remains an actual opposite-channel depth-three exclusion; the new residue and character conditions narrow that packet without yet making it inconsistent.

50. Odd kernels and a third-order balancing-Pell packet

Continue with $(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}$ and an odd prime index ℓ. Suppose, for pressure testing, that both A_ℓ and B_ℓ are squarefull. A finite squarefull factorization has the canonical form

$$N = D^3 C^2, \qquad D = \prod_{v_p(N) \text{ odd}} p.$$

Indeed an even exponent is $2 + 2c$ and an odd exponent is $3 + 2c$. The inherited perfect-power classifications force each exponent vector in a hypothetical packet to have gcd one, so both odd kernels are nontrivial:

$$A_\ell = D_A^3 X^2, \qquad B_\ell = D_B^3 Y^2, \qquad D_A, D_B > 1. \tag{362}$$

Every kernel prime consequently occurs at depth at least three in its assigned channel. The negative Pell equation becomes

$$D_A^6 X^4 - 2D_B^6 Y^4 = -1.$$

Let $s_\ell = (2/\ell)$. The rank congruences, sign product, and residue orbit from the preceding Pell sections give

$$D_A \equiv 1 \pmod{2\ell}, \qquad D_B \equiv s_\ell \pmod{2\ell}, \tag{363}$$

$$X^2 \equiv 1 \pmod{2\ell}, \qquad Y^2 \equiv 1 \pmod{2\ell}. \tag{364}$$

Because ℓ is prime and the square cores are odd, each of X, Y is congruent to one of ± 1 modulo 2ℓ. The complete mod-eight data are

$\ell \bmod 8$	1	3	5	7
$D_A \bmod 8$	1	7	1	7
$D_B \bmod 8$	1	5	5	1.

Writing $d_A = (D_A - 1)/(2\ell)$ and $d_B = (D_B - s_\ell)/(2\ell)$ further gives

$\ell \bmod 8$	1	3	5	7
$d_A \bmod 4$	0	1	0	1
$d_B \bmod 4$	0	1	3	0.

The two kernels also form a three-edge Jacobi-character triangle with the index:

$$\left(\frac{D_A}{D_B}\right) = s_\ell, \qquad \left(\frac{\ell}{D_A}\right) = \left(\frac{-1}{\ell}\right), \qquad \left(\frac{\ell}{D_B}\right) = \left(\frac{s_\ell}{\ell}\right).$$

Thus the three signs, in the order displayed, are

$\ell \bmod 8$	1	3	5	7
(D_A/D_B)	+1	−1	−1	+1
(ℓ/D_A)	+1	−1	+1	−1
(ℓ/D_B)	+1	−1	+1	+1.

In the 3 (mod 8) class all three aggregate edges are negative. This does not assert that every individual cross pair is negative; an actual index- eleven factorization already refutes that stronger statement.

One additional quotient digit can be retained. For a finite list T, let $E_j(T)$ be its jth elementary symmetric sum. The exact expansion

$$\prod_i (1 + xt_i) \equiv 1 + xE_1(T) + x^2 E_2(T) + x^3 E_3(T) \pmod{x^4}$$

follows by induction. Repeat each A-channel quotient according to its valuation to form T_A, and repeat each signed B-channel quotient to form T_B. Write $(K_A, C_A, H_A) = (E_1, E_2, E_3)(T_A)$ and similarly for B. For

$$a = \frac{A_\ell - 1}{2\ell}, \qquad b = \frac{s_\ell B_\ell - 1}{2\ell},$$

cancellation of 2ℓ gives

$$a \equiv K_A + 2\ell C_A + 4\ell^2 H_A, \qquad b \equiv K_B + 2\ell C_B + 4\ell^2 H_B \pmod{8\ell^3}.$$

Substitution into the exact identity $a - 2b + \ell(a^2 - 2b^2) = 0$ yields the necessary third-order ledger

$$\begin{aligned} 0 \equiv K_A - 2K_B + \ell(K_A^2 - 2K_B^2) + 2\ell(C_A - 2C_B) \\ + 4\ell^2(H_A - 2H_B + K_A C_A - 2K_B C_B) \\ + 4\ell^3(C_A^2 - 2C_B^2) \pmod{8\ell^3}. \end{aligned}$$

Its reduction modulo $4\ell^2$ recovers the preceding second-order ledger. The new congruence is a necessary condition and is not asserted to be a contradiction.

A deterministic two-direction search examined all 668 odd prime indices $3 \le \ell \le 5000$ and all primes $q \le 2{,}000{,}000$ in the necessary classes $q \equiv \pm 1 \pmod{2\ell}$. It certified 481 indices with an actual exponent-one divisor of A_ℓ or B_ℓ; 187 bounded-search rows remain open. An independent verifier recomputed every witness modulo q^2, checked primality, and, for every odd prime $\ell \le 191$, verified a simple divisor from the exact coordinates. The largest such witness,

$$1355877461004671178070 1 \parallel B_{59},$$

is supported by a complete Pocklington certificate. Hence

$$\boxed{A_\ell B_\ell \text{ is not squarefull for every odd prime } \ell \le 191.}$$

This is a finite theorem. It neither proves eventual simplicity nor treats an unresolved row as a squarefull packet.

The Lean companion checks the cubic-square factor identity, residue transport, Jacobi reciprocity steps, arbitrary-list third-order expansion, quotient cancellation, and the coupled ledger. The remaining positive gate is an actual opposite-channel depth-three exclusion satisfying all kernel, residue, character, and quotient constraints. No full-premise packet counterexample was found, so this route remains active.

51. All-order Pell–Lucas staircases and a paired companion splitter

The two balancing-Pell channels admit a useful norm-one splice. Put

$$(1+\sqrt{2})^n = A_n + B_n\sqrt{2}, \qquad \alpha = 3+2\sqrt{2}, \quad \beta = 3-2\sqrt{2},$$

and let

$$u_0 = 0, \quad u_1 = 1, \quad u_{n+2} = 6u_{n+1} - u_n,$$

with companion $v_0 = 2, v_1 = 6, v_{n+2} = 6v_{n+1} - v_n$. Binet's formulas give

$$u_n = \frac{\alpha^n - \beta^n}{\alpha - \beta} = \frac{B_{2n}}{2} = A_n B_n, \qquad v_n = 2A_{2n}, \qquad v_n^2 - 32u_n^2 = 4. \tag{365}$$

Thus the norm-one Lucas sequence sees the coprime channel product rather than either channel alone.

We use Zhang's all-orders multiplication formula [88, Proposition 5.1]. This source is an arXiv v1 preprint, so we record the specialization and the additional argument in full. Fix an odd prime ℓ, set

$$A = A_\ell, \qquad B = B_\ell, \qquad U = u_\ell = AB, \qquad \theta = \frac{\ell - 1}{2},$$

and define

$$c_r(\ell) = \frac{\ell \prod_{j=1}^{r}(\ell^2 - (2j-1)^2)}{4^r (2r+1)!}, \qquad a_r = 32^r c_r(\ell) \qquad (0 \le r \le \theta). \tag{366}$$

The source proves $c_r(\ell) \in \mathbf{Z}_{>0}$ and the exact identity

$$Q_\ell := \frac{u_{\ell^2}}{u_\ell} = \sum_{r=0}^{\theta} a_r U^{2r}. \tag{367}$$

The prime-index rank congruences from the preceding Pell sections imply that every prime $p \mid U$ satisfies

$$p \equiv \left(\frac{2}{p}\right) \pmod{2\ell}, \qquad p \ge 2\ell - 1. \tag{368}$$

This lower bound makes every coefficient in (367) a unit on the actual support.

Lemma 51.1 (All multiplication coefficients are support units)**.** *For* $0 \le r \le \theta$*,*

$$\gcd(a_r, U) = 1.$$

Moreover $c_\theta(\ell) = 1$.

Proof. Let $p \mid U$. For $1 \le j \le r$, equation (368) gives

$$0 < \ell - (2j-1) < p, \qquad 0 < \ell + (2j-1) \le 2\ell - 2 < p.$$

Consequently p divides none of the numerator factors in (366). Since the numerator equals the denominator times the integer $c_r(\ell)$, the prime cannot divide $c_r(\ell)$. It also cannot divide 32. This proves the coprimality.

For the final coefficient write $\ell = 2\theta + 1$. Then

$$\prod_{j=1}^{\theta} (\ell^2 - (2j-1)^2) = 4^{\theta}(2\theta)!,$$

so the numerator and denominator of c_θ coincide. □

For $0 \le r \le \theta$, subtract the first r terms and put

$$D_r = Q_\ell - \sum_{i<r} a_i U^{2i}, \qquad E_r = \sum_{i=r}^{\theta} a_i U^{2(i-r)}.$$

Theorem 51.2 (All-order support-unit staircase)**.** *For every* $0 \le r \le \theta$*,*

$$D_r = U^{2r} E_r, \qquad E_r \equiv a_r \pmod{U^2}, \qquad \gcd(E_r, U) = 1. \tag{369}$$

Hence, for $p \mid U$ *and* $e_p = v_p(U)$*,*

$$v_p(D_r) = 2re_p. \tag{370}$$

At the last step,

$$D_\theta = 32^{\theta} U^{\ell-1}.$$

Proof. Factoring U^{2r} from the exact tail proves the first two statements in (369). The leading coefficient is coprime to U by Lemma 51.1; adding a multiple of U^2 preserves that coprimality. This proves the exact valuation. The final formula uses $c_\theta = 1$. □

For a hypothetical squarefull packet, every $e_p \ge 2$, and at an odd-kernel prime $e_p \ge 3$. Thus the rth deviation has exact depth at least $4r$, respectively $6r$, at those primes. This is an all-order necessary condition, not a contradiction.

The first tail also couples to the companion. Define

$$W = \frac{u_{\ell^2}/u_\ell - \ell}{U^2}, \qquad S = \frac{v_{\ell^2} - v_\ell}{U^2}.$$

The fourth-order formulas in [88, Corollary 5.2] yield

$$W \equiv \frac{4\ell(\ell^2-1)}{3} \pmod{U^2}, \tag{371}$$

$$S \equiv 4v_\ell(\ell^2-1) \pmod{U^2}. \tag{372}$$

In particular W is a support unit.

Theorem 51.3 (Paired companion splitter)**.** *One has*

$$\ell S \equiv 3v_\ell W \pmod{U^2}, \tag{373}$$

and therefore

$$\ell S \equiv 6W \pmod{A^2}, \qquad \ell S \equiv -6W \pmod{B^2}. \tag{374}$$

The factor $6W$ *is invertible modulo* U^2*. The unique* $Z \pmod{U^2}$ *satisfying*

$$6WZ \equiv \ell S \pmod{U^2}$$

obeys

$$Z \equiv A_{2\ell} \pmod{U^2}, \qquad Z \equiv 1 \pmod{A^2}, \qquad Z \equiv -1 \pmod{B^2}, \qquad Z^2 \equiv 1 \pmod{U^2}.$$

Proof. Multiply (372) by ℓ and (371) by $3v_\ell$; their right sides coincide. Next,

$$\frac{v_\ell}{2} = A_{2\ell} = 2A^2 + 1 = 4B^2 - 1,$$

which gives the opposite channel signs in (374). Every prime divisor of U exceeds $\ell + 1$ and is therefore coprime to $6W$. Cancellation shows that the splitter residue is $A_{2\ell}$. Since $A^2 - 2B^2 = -1$, one has $\gcd(A, B) = 1$; hence its two squared signs combine modulo $A^2B^2 = U^2$ and give the last congruence. □

There is also a precise negative result. A single normalized correction is not rigid. Consider the full-premise claim that every positive odd $k \equiv 1 \pmod{u_3^2}$ satisfies

$$\frac{u_{3k}/u_3 - 1}{u_3^2} \equiv 0 \pmod 5.$$

It is false. Since $u_3 = 35$, take

$$k = 2451 = 1 + 2 \cdot 35^2.$$

Two independent recurrence computations, one linear and one by binary powering in $\mathbf{Z}[T]/(T^2 - 6T + 1)$, give

$$u_{7353} \equiv 85785 = 35 \cdot 2451 \pmod{214375 = 35(5 \cdot 35^2)}.$$

After the justified divisions, the normalized correction is 2 modulo five. This is an explicit instance of Zhang's local-surjectivity theorem [88, Theorem 5.6]. It retires only the exact fixed-zero subclaim. The all-order staircase and paired companion route remain active because that theorem does not independently prescribe both corrections.

The Lean companion kernel-checks the even-power tail algebra, coefficient support argument, paired congruence, channel signs, modular cancellation, square-root reconstruction, and the local-surjectivity logical boundary. It also checks the displayed recurrence residue by kernel reduction, without `native_decide`. The remaining sufficient target is a correlated opposite-channel depth-three exclusion satisfying this complete staircase, the earlier kernel character triangle, and the third-order factorization ledger. No full-premise packet counterexample was found, so that route is not retired.

52. Correlated Pell–Lucas staircases and opposite-channel incidence

We continue the norm-one splice of Section 51. For a norm-one Lucas pair write

$$u_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}, \qquad v_n = \alpha^n + \beta^n, \qquad \alpha\beta = 1,$$

and let $\delta = (\alpha - \beta)^2$. For a positive odd integer $k = 2\theta + 1$, define

$$d_j(k) = \binom{\theta + j}{2j}, \qquad \Psi_k(W) = \sum_{j=0}^{\theta} d_j(k) W^j. \tag{375}$$

Proposition 52.1 (Exact companion multiplication polynomial)**.** *For every $n \geq 1$,*

$$v_{nk} = v_n \Psi_k(\delta u_n^2). \tag{376}$$

Thus the notation v_{nk}/v_n below denotes an exact integer identity and does not assume division in a residue ring.

Proof. In a splitting ring put $x = \alpha^n$ and $W = (x - x^{-1})^2 = \delta u_n^2$. No quotient in the splitting ring is needed. Put $S_m = x^{2m+1} + x^{-(2m+1)}$. The exact Laurent-polynomial recurrence is $S_{m+1} = (W + 2)S_m - S_{m-1}$, with $S_0 = x + x^{-1}$ and $S_1 = (x + x^{-1})(1 + W)$. The polynomials $P_m(W) = \sum_{j=0}^{m} \binom{m+j}{2j} W^j$ have the same recurrence and initial values, by Pascal's identity. Induction therefore gives the exact factorization $S_m = (x + x^{-1})P_m(W)$ without cancelling a possibly nonunit factor. Taking $m = \theta$ is (376); being symmetric in x, x^{-1}, it descends to the integer identity. □

Let $\Phi_k(W) = \sum c_j(k) W^j$ be the first-sequence multiplication polynomial from [88, Proposition 5.1], where

$$c_j(k) = \frac{k \prod_{h=1}^{j} (k^2 - (2h-1)^2)}{4^j (2j+1)!}.$$

Theorem 52.2 (Coefficientwise companion correlation)**.** *For $0 \le j \le \theta$,*

$$(2j+1)c_j(k) = k d_j(k). \tag{377}$$

Consequently

$$\Phi_k(W) + 2W\Phi_k'(W) = k\Psi_k(W). \tag{378}$$

Proof. Since $k = 2\theta + 1$,

$$\frac{1}{4^j} \prod_{h=1}^{j} (k^2 - (2h-1)^2) = \frac{(\theta+j)!}{(\theta-j)!}.$$

Substitution gives $c_j(k) = \frac{k}{2j+1}\binom{\theta+j}{2j}$, proving (377). Coefficient comparison then proves (378). $\square$

Specialize to the balancing-Pell pair at an odd prime index ℓ:

$$A = A_\ell, \qquad B = B_\ell, \qquad U = AB = u_\ell, \qquad v_\ell = 2A_{2\ell}, \qquad \delta = 32.$$

Put $\theta = (\ell-1)/2$, $a_j = 32^j c_j(\ell)$, $b_j = 32^j d_j(\ell)$ and, for $0 \le s \le (\ell-1)/2$, define

$$\mathcal{E}_s(X) = \sum_{j=s}^{\theta} a_j X^{j-s}, \qquad \mathcal{F}_s(X) = \sum_{j=s}^{\theta} b_j X^{j-s}, \qquad E_s = \mathcal{E}_s(U^2), \quad F_s = \mathcal{F}_s(U^2).$$

Theorem 52.3 (Paired all-order support-unit staircase)**.** *Removing the first s terms from either multiplication polynomial leaves respectively $U^{2s}E_s$ and $U^{2s}F_s$. Moreover*

$$\gcd(E_s, U) = \gcd(F_s, U) = 1, \qquad (2s+1)\mathcal{E}_s(X) + 2X\mathcal{E}_s'(X) = \ell\mathcal{F}_s(X), \tag{379}$$

and in particular

$$(2s+1)E_s \equiv \ell F_s \pmod{U^2}. \tag{380}$$

If $p \mid U$ and $e_p = v_p(U)$, both unnormalized tails have exact valuation $2se_p$ at p.

Proof. The tail factorizations are exact by construction. Every support prime $p \mid U$ satisfies $p \ge 2\ell - 1 > \ell$. All factorial arguments in $d_s(\ell) = \binom{(\ell-1)/2+s}{2s}$ are therefore below p, so d_s is a p-adic unit. The first-sequence support-unit argument from the preceding section makes c_s a p-adic unit, and $p \nmid 32$. Hence the leading terms a_s, b_s, and thus E_s, F_s modulo U^2, are units at every support prime.

Apply (377) term by term. The coefficient of X^{j-s} on the left of the differential identity is $(2s+1+2(j-s))a_j = (2j+1)a_j = \ell b_j$. This proves (379) and its reduction (380) after setting $X = U^2$. The exact valuations follow from the unit leading tails. $\square$

Set $T_s = v_\ell F_s$ and $Z_0 = v_\ell/2 = A_{2\ell}$.

Theorem 52.4 (Coherent every-order channel splitters)**.** *For every s, the congruence*

$$2(2s+1)E_s Z_s \equiv \ell T_s \pmod{U^2} \tag{381}$$

has the unique solution $Z_s \equiv Z_0 \pmod{U^2}$. Consequently

$$Z_s \equiv 1 \pmod{A^2}, \qquad Z_s \equiv -1 \pmod{B^2}, \qquad Z_s^2 \equiv 1 \pmod{U^2}. \tag{382}$$

For two levels s, t one has

$$(2t+1)E_t T_s \equiv (2s+1)E_s T_t \pmod{U^2}. \tag{383}$$

If $q^3 \mid A$ and $r^3 \mid B$, every Z_s satisfies

$$Z_s \equiv 1 \pmod{q^6}, \qquad Z_s \equiv -1 \pmod{r^6}. \tag{384}$$

Proof. Multiplying (380) by v_ℓ gives

$$\ell T_s \equiv (2s+1)v_\ell E_s = 2(2s+1)E_s Z_0 \pmod{U^2}.$$

Every prime divisor of U exceeds ℓ, so $2(2s+1)E_s$ is invertible modulo U^2. This proves uniqueness and identifies Z_s. The Pell identities $A_{2\ell} = 2A^2+1 = 4B^2-1$ give the two signs; coprimality of A, B combines them modulo U^2. Cross-multiplication at levels s, t and cancellation of ℓ, which is coprime to U, proves (383). Finally $q^3 \mid A$ and $r^3 \mid B$ imply $q^6 \mid A^2$ and $r^6 \mid B^2$. ☐

The splitters therefore form one coherent projective line; higher orders are not independent tests. The character information can nevertheless be resolved vertex by vertex. In a hypothetical squarefull packet write

$$A = D_A^3 X^2, \qquad B = D_B^3 Y^2,$$

with D_A, D_B squarefree, and let O_A, O_B be their prime supports.

Theorem 52.5 (Vertexwise character incidence)**.** *For $r \in O_B$ and $q \in O_A$ one has*

$$\prod_{q\in O_A}\left(\frac{q}{r}\right) = \left(\frac{2}{r}\right), \qquad \prod_{r\in O_B}\left(\frac{q}{r}\right) = \left(\frac{B}{q}\right). \tag{385}$$

Both total products equal $(2/\ell)$. If $q \equiv 1 \pmod 8$, the second endpoint is the quartic-two sign $2^{(q-1)/4} \pmod q$. For $\ell \equiv 3$ or $5 \pmod 8$, there exist $q \in O_A, r \in O_B$ such that

$$v_q(A) \geq 3, \quad v_r(B) \geq 3, \quad \left(\frac{q}{r}\right) = -1, \quad 2\ell \mid q+r, \tag{386}$$

and the every-order signs (384) hold at this same pair.

Proof. For $r \mid B$, the Pell equation gives $A^2 \equiv -1 \pmod r$. Every B-channel prime is $1 \pmod 4$, so Euler's criterion yields $(A/r) = (2/r)$. Removing the square core from $A = D_A^3 X^2$ proves the first product. Removing the square core from B and using quadratic reciprocity, whose sign is trivial because $r \equiv 1 \pmod 4$, proves the second. Multiplying all rows or all columns gives the same edge product, and the inherited kernel ledger evaluates it as $(2/\ell)$.

If $q \equiv 1 \pmod 8$, the congruence $2B^2 \equiv 1 \pmod q$ implies $(B/q) = 2^{(q-1)/4}$ because this quarter-power is a sign. Finally, in the two stated index classes the inherited table gives $D_B \equiv 5 \pmod 8$. Thus some $r \in O_B$ is $5 \pmod 8$ and its row product is negative; one edge (q/r) is negative. The rank congruences are $q \equiv 1 \pmod{2\ell}$ and $r \equiv -1 \pmod{2\ell}$, proving the last divisibility. Squarefull odd exponents are at least three, and Theorem 52.4 supplies the depth-six signs. ☐

This necessary packet is highly constrained but has not been contradicted. An independently replayed computation proves a separate finite theorem: for every one of the 57 odd prime indices $3 \leq \ell \leq 271$, at least one of A_ℓ, B_ℓ has a certified prime divisor of exponent exactly one. Hence $A_\ell B_\ell$ is not squarefull throughout that range. The verifier checks each divisor, its primality, and the coordinate modulo its square; the large witness at $\ell = 59$ uses a complete Pocklington certificate. It also checks 138,675 coefficient pairs, 228 recurrence evaluations, and the incidence graphs at all 13 prime indices through 43.

A bounded search of 527,352 candidate primes up to $2 \cdot 10^6$ finds $13^2 \parallel B_7$ but no depth-three opposite pair. This is only a finite no-hit and retires no assertion. At index 11, the row at $r = 5741$ contains both $(23/r) = -1$ and $(353/r) = +1$, a full-premise counterexample to the stronger claim that every edge of a negative row is negative. It is not a squarefull packet and therefore does not refute the active exclusion route.

Any surviving packet must simultaneously satisfy the earlier third-order congruence modulo $8\ell^3$, both support-unit staircases, every cross-order determinant, the depth-six signs, and the vertexwise incidence laws. The missing theorem must couple the prime-factor quotients in the third-order ledger to this all-order projective line. Its current absence is a difficult open gate, not evidence against the route.

The Lean companion checks the rational coefficient identity, exact weighted list identity, normalized-tail congruence, every-order recovery, cross-order determinant, sixth-power propagation, negative-row extraction, and the quartic-two power algebra. The Lucas multiplication theorem, rank theorem, finite factor certificates, and the open uniform exclusion are kept at their stated mathematical or computational boundaries and are not added as axioms.

53. Factor-quotient jets and Pell projective curvature

Write

$$(1+\sqrt{2})^n = A_n + B_n\sqrt{2}, \qquad U = A_\ell B_\ell, \qquad v = 2A_{2\ell}, \qquad x = 2\ell,$$

where ℓ is an odd prime. Put

$$a = \frac{A_\ell - 1}{x}, \qquad b = \frac{s_\ell B_\ell - 1}{x}, \qquad s_\ell = \left(\frac{2}{\ell}\right).$$

For a prime factor $q \mid A_\ell$, write $q = 1 + xk_q$ and repeat k_q according to its valuation. For $r \mid B_\ell$, write $r = s_r + xh_r = s_r(1 + xs_r h_r)$ and repeat $s_r h_r$. If the first three elementary symmetric functions of the two resulting lists are (K_A, C_A, H_A) and (K_B, C_B, H_B), respectively, exact finite-product expansion gives

$$a \equiv K_A + xC_A + x^2 H_A \pmod{x^3}, \qquad b \equiv K_B + xC_B + x^2 H_B \pmod{x^3}. \tag{53.1}$$

Three copies of one quotient t have the exact fingerprint

$$\begin{aligned} E_1(t,t,t,R) &= 3t + K_0,\\ E_2(t,t,t,R) &= 3t^2 + 3tK_0 + C_0,\\ E_3(t,t,t,R) &= t^3 + 3t^2K_0 + 3tC_0 + H_0, \end{aligned} \tag{53.2}$$

where (K_0, C_0, H_0) is the residual triple. A depth-three carrier is therefore visible in the third-order ledger even when its actual depth is larger.

The factor quotients also determine a jet of the Lucas companion.

Theorem 53.1 (Two-channel companion jet)**.** *Define*

$$\begin{aligned} V_A(K,C,H) &= 6 + 8xK + x^2(8C + 4K^2) + x^3(8H + 8KC),\\ V_B(K,C,H) &= 6 + 16xK + x^2(16C + 8K^2) + x^3(16H + 16KC). \end{aligned}$$

Then

$$v \equiv V_A(K_A, C_A, H_A) \pmod{x^4}, \qquad v \equiv V_B(K_B, C_B, H_B) \pmod{x^4}. \tag{53.1.1}$$

Proof. The negative Pell identity gives $v = 4A_\ell^2 + 2 = 8B_\ell^2 - 2$. Substitute $A_\ell = 1 + xa$ in the first expression and $s_\ell B_\ell = 1 + xb$ in the second. Modulo x^4, the square of $K + xC + x^2H$ contributes only $K^2 + 2xKC$ after multiplication by x^2. The displayed polynomials follow by collecting coefficients. ☐

Let $\theta = (\ell-1)/2$ and let α_j, β_j be the correlated Lucas coefficients from the preceding section, satisfying

$$(2j+1)\alpha_j = \ell\beta_j, \qquad \alpha_\theta = \beta_\theta = 32^\theta.$$

For the tail polynomials E_j, F_j, put $T_j = vF_j(U^2)$ and define

$$\Delta_{\text{top}} = T_{\theta-1}\ell E_\theta(U^2) - T_\theta(\ell-2)E_{\theta-1}(U^2).$$

Theorem 53.2 (Exact endpoint curvature)**.** *For every odd prime index,*

$$\Delta_{\text{top}} = 2v\,32^{\ell-1}U^2. \tag{53.2.1}$$

Moreover

$$\gcd(2v32^{\ell-1}, U) = 1, \qquad v_p(\Delta_{\text{top}}) = 2v_p(U) \quad (p \mid U). \tag{53.2.2}$$

Thus U^2 always divides the determinant and U^3 never does.

Proof. The two top tails are

$$E_{\theta-1} = \alpha_{\theta-1} + \alpha_\theta U^2, \quad F_{\theta-1} = \beta_{\theta-1} + \alpha_\theta U^2, \quad E_\theta = F_\theta = \alpha_\theta.$$

After substituting $T_j = vF_j$, the constant part of the determinant is

$$v\alpha_\theta\{\ell\beta_{\theta-1} - (\ell-2)\alpha_{\theta-1}\} = 0$$

by coefficient correlation. Its U^2 coefficient is $2v\alpha_\theta^2 = 2v32^{\ell-1}$, proving the first identity. Odd-index Pell coordinates make U odd, while

$$v^2 - 32U^2 = 4.$$

Every common odd divisor of v and U would divide four. The quotient in the first identity is therefore a unit at every prime of U, proving the valuation statement and sharpness. ☐

With $\kappa_\ell = \Delta_{\mathrm{top}}/U^2 = 2v32^{\ell-1}$, Theorems 53.1 and 53.2 give the direct coupling

$$\kappa_\ell \equiv 2\,32^{\ell-1}V_A(K_A, C_A, H_A) \equiv 2\,32^{\ell-1}V_B(K_B, C_B, H_B) \pmod{16\ell^4}. \tag{53.2.3}$$

Thus the third-order factor-quotient ledger determines the 2ℓ-adic jet of the exact unit measuring failure of integral projective collinearity. Since the rank theorem gives $\gcd(2\ell, U) = 1$, the jet modulus and U^2 are coprime; the Chinese remainder theorem combines them without contradiction. A closing argument must relate their values through reciprocity, height, or distribution.

Two full-premise counterexamples delimit stronger local claims. First, the local-only assertion that the forced depth-three quotient data and complete third-order ledger are inconsistent is false. Take

$$\ell = 3, \qquad q = 7 = 1 + 6, \qquad r = 797 = -1 + 6 \cdot 133,$$

and quotient lists $[1, 1, 1]$ and $[-133, -133, -133]$. They satisfy the required congruence and character table, and the full ledger is divisible by $8\ell^3$. The synthetic values are squarefull of depth three. They do not satisfy the global Pell equation:

$$q^6 - 2r^6 + 1 = -512601560592751008 \neq 0.$$

This retires only the local inconsistency claim.

Second, the genuine Pell index $\ell = 7$ has

$$A_7 = 239, \quad B_7 = 169, \quad U = 40391, \quad v = 228486,$$

and

$$\Delta_{\mathrm{top}} = 8004950881858947309895 68, \qquad \Delta_{\mathrm{top}} \bmod U^3 = 24354030047568 \neq 0.$$

It is a complete counterexample to the proposed U^3 strengthening and an instance of the exact U^2 theorem. Since $239 \parallel A_7$, it is not a counterexample to a hypothetical squarefull Pell packet.

Independent certificates exhibit a prime divisor of exponent one in $A_\ell B_\ell$ for every one of the 57 odd prime indices $3 \le \ell \le 271$. This proves finite nonsquarefullness in that range and nothing beyond it. The active route must still rule out or construct an unbounded family satisfying simultaneously squarefullness, the opposite depth-three nonresidue pair, both factor-quotient jets, the complete third-order ledger, all correlated Lucas tails, exact endpoint curvature, and vertexwise character incidence. No full-premise counterexample to that global statement is known.

54. Transverse Hensel specialization in the Pell–Lucas packet

Let

$$F_0(T) = 0, \quad F_1(T) = 1, \quad F_{n+2}(T) = TF_{n+1}(T) + F_n(T)$$

and

$$L_0(T) = 2, \quad L_1(T) = T, \quad L_{n+2}(T) = TL_{n+1}(T) + L_n(T).$$

For $(1+\sqrt{2})^n = A_n + B_n\sqrt{2}$ one has $F_n(2) = B_n$ and $L_n(2) = 2A_n$. Recent work on powerful Fibonacci and Horadam polynomials clarifies polynomial multiplicities [10], while primitive-divisor theory over polynomial rings supplies a complementary global context [11]. Polynomial squarefreeness does not, by itself, control the valuation of a specialized integer. The results below make that distinction exact.

Put $\Delta = T^2 + 4$. The Binet formulas in the quadratic extension of $\mathbb{Q}(T)$ give

$$L_n(T)^2 - \Delta F_n(T)^2 = 4(-1)^n, \qquad L_n'(T) = nF_n(T), \tag{387}$$

and

$$\Delta F_n'(T) = nL_n(T) - TF_n(T). \tag{388}$$

Indeed, for $\alpha = (T + \sqrt{\Delta})/2$ and $\beta = (T - \sqrt{\Delta})/2$, one has $\alpha\beta = -1$, $F_n = (\alpha^n - \beta^n)/(\alpha - \beta)$, and $L_n = \alpha^n + \beta^n$; subtraction and differentiation prove the three identities. At $T = 2$ they become

$$L_n'(2) = nB_n, \qquad 4F_n'(2) = nA_n - B_n.$$

Consequently, if ℓ is an odd prime and an odd support prime $p \nmid \ell$ divides B_ℓ, then $F_\ell'(2) \not\equiv 0 \pmod{p}$. Likewise, if $q \nmid \ell$ divides A_ℓ, then $L_\ell'(2) \not\equiv 0 \pmod{q}$. Coprimality of A_ℓ and B_ℓ is the only additional input. Thus every actual repeated support factor is a transverse specialization coincidence rather than a multiple root of the polynomial.

Lemma 54.1 (One-digit Hensel displacement). *Let $f \in \mathbb{Z}[T]$, let p be prime, and suppose $p^e \mid f(t)$ with $e \geq 1$. Then*

$$\frac{f(t + p^e h)}{p^e} \equiv \frac{f(t)}{p^e} + hf'(t) \pmod{p}.$$

If $p \nmid f'(t)$, there is a unique $h \pmod{p}$ for which $p^{e+1} \mid f(t + p^e h)$, namely

$$h \equiv -\frac{f(t)}{p^e} f'(t)^{-1} \pmod{p}.$$

The old representative t persists precisely when $p^{e+1} \mid f(t)$.

Proof. The integral Taylor expansion has the form $f(t + z) = f(t) + zf'(t) + z^2 G(t, z)$. Substituting $z = p^e h$ and using $2e \geq e + 1$ gives the congruence after division by p^e. Its right-hand side is affine with nonzero slope modulo p, proving existence and uniqueness of the displayed digit. Taking $h = 0$ proves the final assertion. □

For a simple support root write

$$\lambda_{f,p,e}(t) = -\frac{f(t)}{p^e} f'(t)^{-1} \pmod{p}.$$

The product $A_\ell B_\ell$ is squarefull exactly when every level-one displacement at the fixed parameter $T = 2$ vanishes in both channels. A support factor of exponent at least three also has zero level-two displacement. This is an exact reformulation, not a proof of the required global exclusion: the remaining problem is to prevent one fixed integer $T = 2$ from agreeing simultaneously with all of the relevant lifted roots.

Local Hensel conditions cannot supply that global prohibition when the parameter is allowed to move. For distinct primes p_i and simple roots $p_i \mid f_i(t_0)$, Lemma 54.1 gives one class modulo p_i^2 for each lifted root, and the Chinese remainder theorem gives a unique simultaneous class modulo $\prod_i p_i^2$. At index three this construction is completely explicit:

$$F_3(T) = T^2 + 1, \qquad L_3(T) = T^3 + 3T.$$

The lifts of the F-channel prime 5 and the L-channel prime 7 from $T = 2$ meet at $T = 282 \pmod{1225}$. At the least positive representative,

$$F_3(282) = 79525 = 5^2 \cdot 3181,$$
$$\frac{L_3(282)}{2} = 11213307 = 3^2 \cdot 7^2 \cdot 47 \cdot 541,$$

and $D = (282^2 + 4)/4 = 19882 = 2 \cdot 9941$ is squarefree. Specializing (387) yields the genuine global point

$$11213307^2 - 19882\, 79525^2 = -1.$$

This is a full counterexample to the precise claim that simple opposite channel roots cannot both become repeated while the parameter moves and the Pell coefficient remains squarefree. It has coefficient 19882, and both coordinates retain exponent-one factors, so it is neither a counterexample to the fixed coefficient-two packet nor to the abc conjecture.

There is also an actual fixed-parameter collision:

$$F_7(T) = T^6 + 5T^4 + 6T^2 + 1,$$
$$F_7(2) = 169 = 13^2, \qquad F_7'(2) = 376 \equiv -1 \pmod{13}.$$

The next Hensel digit is one and

$$13^3 \mid F_7(171) = 25006385400373, \qquad 13^4 \nmid F_7(171).$$

Thus a simple polynomial root can specialize to an integer with exponent two. This refutes exactly the stronger specialization claim; it does not give a squarefull Pell packet because $A_7 = 239$ occurs only to the first power.

Finally, an exhaustive external search over positive even $T = 2s$ with $s \le 10^7$ uses the unique representation $N = x^2y^3$ of a positive squarefull integer with y squarefree. It checks 43,355,470 canonical representations. The sole squarefull value in the full F_3 channel is

$$F_3(682) = 465125 = 5^3 \cdot 61^2,$$

whereas

$$\frac{L_3(682)}{2} = 158608307 = 11 \cdot 13 \cdot 31 \cdot 37 \cdot 967$$

is not squarefull. Independent Python and C++ implementations agree. Lean checks the singleton identities, the nonsquarefull witness, the Pell norm, and squarefreeness of $116282 = 2 \cdot 53 \cdot 1097$; the entire 43-million-case enumeration is not yet replayed in the kernel and is therefore reported as an external finite computation.

The original formal companion proves the integer Hensel divisibility core, supplied derivative readouts, index-three Taylor and CRT certificates, and the displayed numerical points. At that stage the all-index identities, general Taylor and simultaneous-CRT theorems, all-support squarefull equivalence, and one bundled moving witness were paper-level obligations. The supplemental Section 55 now proves those exact items in Lean under their displayed hypotheses. It does not supply the fixed-$T = 2$ all-support transversality or the global exclusion described below.

The fixed $D = 2$ route remains open. Its smallest surviving target is to prove that every odd prime index has a support prime with nonzero level-one displacement at $T = 2$, or more narrowly to exclude the opposite-channel level-one/level-two packet after imposing the previously proved rank, character, factor-quotient, all-order-tail, and endpoint-curvature constraints. No counterexample meeting those complete premises is known, so difficulty does not retire the route.

55. All-index Pell polynomials and complete Hensel steering

Author: ChatGPT. This section closes the algebraic formalization gaps left in Section 54. It does not prove the fixed coefficient-two squarefull exclusion and does not prove or disprove the *abc* conjecture. Its purpose is to put the all-index identities, the full polynomial Hensel law, the all-support equivalence, finite simultaneous steering, and the complete moving counterexample under one exact interface.

Let $F_n, L_n \in \mathbb{Z}[T]$ be defined by

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+2} = TF_{n+1} + F_n,$$
$$L_0 = 2, \quad L_1 = T, \quad L_{n+2} = TL_{n+1} + L_n.$$

Theorem 55.1 (All-index polynomial identities)**.** *For every $n \geq 0$,*

$$L_n^2 - (T^2+4)F_n^2 = 4(-1)^n, \tag{389}$$

$$L_n' = nF_n, \tag{390}$$

$$(T^2+4)F_n' = nL_n - TF_n. \tag{391}$$

Evaluation at any $t \in \mathbb{Z}$ preserves the two recurrences and all three identities.

Proof. Two-step induction gives

$$L_n = 2F_{n+1} - TF_n, \qquad L_{n+1} = TF_{n+1} + 2F_n. \tag{392}$$

Ordinary induction in the F recurrence gives Cassini's identity

$$F_{n+1}^2 - TF_{n+1}F_n - F_n^2 = (-1)^n.$$

Substitution of the first equality in (392) proves (389). Differentiating the L recurrence and using the second equality in (392) proves (390) by two-step induction. For the remaining formula, differentiate the F recurrence and use its two preceding instances together with

$$(T^2+4)F_{n+1} = TL_{n+1} + 2L_n,$$

which follows from (392). Polynomial evaluation is a ring homomorphism, so it transports every displayed equality. □

Theorem 55.2 (Polynomial Taylor–Hensel law)**.** *Let $f \in \mathbb{Z}[T]$, let $p, t, c, h \in \mathbb{Z}$, and let $e \geq 1$. For every fixed $z \in \mathbb{Z}$ there is an integer $G = G(f, t, z)$ such that*

$$f(t+z) = f(t) + zf'(t) + z^2G.$$

In particular,

$$f(t+p^eh) \equiv f(t) + p^ehf'(t) \pmod{p^{e+1}}.$$

If $p \neq 0$ and $f(t) = p^ec$, then

$$p^{e+1} \mid f(t+p^eh) \iff p \mid c + hf'(t). \tag{393}$$

If moreover $\gcd(f'(t), p) = 1$, a digit satisfying (393) exists and is unique modulo p.

Proof. The integral Taylor expansion gives the first formula. Substitution of $z = p^eh$ and the inequality $e+1 \leq 2e$ give the congruence. After writing $f(t) = p^ec$, cancellation of the nonzero factor p^e gives (393). Bézout's identity supplies an admissible digit, and subtracting two admissible affine congruences proves uniqueness. □

Call $N \in \mathbb{N}$ squarefull when every prime $p \mid N$ satisfies $p^2 \mid N$. For a polynomial f, an integer t, and a support prime p, define zero first displacement to mean

$$p \mid f(t), \qquad \gcd(f'(t), p) = 1, \qquad p^2 \mid f(t). \tag{394}$$

By Theorem 55.2, the final condition is equivalent to the unique next Hensel digit being zero.

Proposition 55.3 (All-support two-channel equivalence)**.** *Let $A, B \in \mathbb{N}$ be coprime, let $f_A(t) = u_AA$ and $f_B(t) = u_BB$, and suppose that each scale u_A, u_B is coprime to every prime in the corresponding support. Assume also that all support roots of f_A, f_B at t are simple. Then AB is squarefull if and only if (394) holds at every support prime in both channels.*

In particular, this applies formally to

$$f_A = L_\ell, \quad u_A = 2, \quad f_B = F_\ell, \quad u_B = 1, \quad t = 2,$$

once the odd-support scale condition and both all-support transversality conditions have been proved.

Proof. For coprime A, B, Euclid's lemma gives

$$AB \text{ squarefull} \iff A \text{ and } B \text{ squarefull}.$$

In each channel, a support prime square divides u_AN if and only if it divides N, since it is coprime to the scale. The asserted equivalence now follows directly from (394). □

Theorem 55.4 (Finite simultaneous Hensel steering)**.** *Let $f_1, \dots, f_m \in \mathbb{Z}[T]$, let $p_1, \dots, p_m$ be distinct primes, and suppose*

$$p_i \mid f_i(t_0), \qquad p_i \nmid f_i'(t_0) \quad (1 \le i \le m).$$

There is a unique residue class

$$t \pmod{\prod_{i=1}^{m} p_i^2}$$

such that

$$t \equiv t_0 \pmod{p_i}, \qquad p_i^2 \mid f_i(t) \quad (1 \le i \le m).$$

Proof. Theorem 55.2 gives one and only one local class modulo p_i^2. The square moduli are pairwise coprime, so the integer Chinese remainder theorem gives a joint class and uniqueness modulo their product. Polynomial evaluation respects congruence. Conversely, any integer satisfying the displayed base and divisibility conditions lies in each unique local class, hence in the joint CRT class. □

At index three the recurrence gives

$$F_3(T) = T^2 + 1, \qquad L_3(T) = T^3 + 3T.$$

The selected roots at $t_0 = 2$ are

$$7 \mid L_3(2), \quad 7 \nmid L_3'(2), \qquad 5 \mid F_3(2), \quad 5 \nmid F_3'(2).$$

Their lifted classes are $t \equiv 37 \pmod{49}$ and $t \equiv 7 \pmod{25}$; the explicit joint representative used below is 282.

Proposition 55.5 (Complete moving witness)**.** *Put*

$$t = 282, \quad A = 11213307, \quad B = 79525, \quad D = 19882.$$

Then

$$2A = L_3(t), \qquad B = F_3(t), \qquad 4D = t^2 + 4,$$
$$A = 3^2\, 7^2\, 47\, 541, \qquad B = 5^2\, 3181, \qquad D = 2\, 9941,$$

all displayed factors are prime, D is positive and squarefree, and

$$A^2 - DB^2 = -1.$$

The selected factors satisfy $7^2 \mid A$ and $5^2 \mid B$, while

$$47^2 \nmid A, \qquad 541^2 \nmid A, \qquad 3181^2 \nmid B.$$

Thus this tuple satisfies every premise of the moving-parameter exclusion and refutes that exact strengthening. It is not a squarefull two-channel packet and has $D \neq 2$.

Proof. All evaluations and factorizations are direct integer identities. Trial division proves the primality assertions; in particular 9941 is prime, so $D = 2 \cdot 9941$ is squarefree. The global norm is the specialization of (389), divided by four. The three failed square divisibilities certify the stated non-squarefull scope. □

The companion Lean module checks each result above. Its structure `HGlobalMoveWitness` contains the prime-index, base-root, simple-derivative, lifted-residue, repeated-divisibility, factorization, primality, positivity, squarefree-coefficient, global-norm, and non-squarefull fields in one object. The associated axiom audit reports only the standard Mathlib principles `propext`, `Classical.choice`, and `Quot.sound`.

The arithmetic route remains open at a precise point: one must discharge the all-support transversality premises for the actual fixed Pell packet and then exclude simultaneous zero displacements at $T = 2$. No counterexample meeting those full fixed-parameter premises has been found, so the route is retained.

56. A matrix Frobenius anchor at the fixed Pell parameter

Write
$$(1+\sqrt{2})^n = A_n + B_n\sqrt{2}, \qquad F_n(2) = B_n, \qquad L_n(2) = 2A_n.$$
The preceding Hensel section left derivative transversality at all actual support primes as an explicit premise. It can be discharged without using a rank-of-apparition theorem.

Set
$$M = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \qquad K = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}.$$
Then $M = I + K$, $K^2 = 2I$, and the first column of M^n is $(A_n, B_n)^t$. If $\ell = 2m+1$ is prime, the intermediate binomial coefficients are divisible by ℓ, so for an integral matrix R,
$$M^\ell = I + K^\ell + \ell K R = I + 2^m K + \ell K R.$$
Taking the first column proves the following statement.

Theorem 56.1 (Prime-index Frobenius anchor)**.** *For every odd prime $\ell = 2m+1$ there are integers r_A, r_B such that*
$$A_\ell = 1 + \ell r_A, \qquad B_\ell = 2^m + \ell r_B.$$
In particular $\ell \nmid A_\ell B_\ell$.

Proof. Only the last assertion remains. Divisibility of the first coordinate by ℓ would give $\ell \mid 1$. Divisibility of the second would give $\ell \mid 2^m$, hence $\ell = 2$, contrary to oddness. □

The all-index differential identities specialize to
$$L_\ell'(2) = \ell B_\ell, \qquad 4F_\ell'(2) + B_\ell = \ell A_\ell.$$

Theorem 56.2 (All-support transversality at $T=2$)**.** *If $q \mid A_\ell$ is prime, then* $\gcd(L_\ell'(2), q) = 1$. *If $p \mid B_\ell$ is prime, then* $\gcd(F_\ell'(2), p) = 1$.

Proof. The coordinates are coprime. In the first channel, q divides neither ℓ nor B_ℓ, so the first differential identity proves the claim. In the second, p divides neither ℓ nor A_ℓ and is odd. Reducing the second identity modulo p gives $4F_\ell'(2) \equiv \ell A_\ell \not\equiv 0 \pmod{p}$. □

For a simple support root let $Z_1(f,t,p)$ mean that $p \mid f(t)$, $gcd(f'(t),p) = 1$, and $p^2 \mid f(t)$. The one-digit Hensel law identifies this with zero first displacement.

Theorem 56.3 (Exact fixed-parameter gate)**.** *For every odd prime ℓ,*
$$A_\ell B_\ell \textit{ is squarefull} \iff \begin{cases} Z_1(L_\ell, 2, q) & (q \mid A_\ell), \\ Z_1(F_\ell, 2, p) & (p \mid B_\ell), \end{cases}$$
where both quantifiers range over all rational support primes.

Proof. The preceding theorem supplies simplicity at every support prime. The factor 2 in $L_\ell(2) = 2A_\ell$ is a unit at every such prime, and the two coordinates are coprime. Thus the square divisibilities in Z_1 are exactly the defining square divisibilities for squarefullness of the two coordinates. □

This theorem instantiates the earlier conditional all-support statement; it does not prove either side. The universal exclusion of simultaneous zero displacements is exactly the still-open prime-index squarefull exclusion.

The stronger assertion that no individual support displacement vanishes is false. At $\ell = 7$,
$$A_7 = 239, \qquad B_7 = 169 = 13^2, \qquad F_7'(2) = 376 \equiv -1 \pmod{13}.$$
Hence $Z_1(F_7, 2, 13)$ holds with every premise. This does not refute the all-support exclusion because 239 has exponent one.

An independent bounded replay examined all odd prime $\ell \leq 20000$ and all primes $q \leq 10^7$ in the necessary classes $q = \pm 1 \pmod{2\ell}$. Its 3,091,963 candidate tests found only the same valuation-two hit $(7, 13, B)$, no valuation-three hit, and no opposite-channel repeated pair. The

789 indices without an in-bound simple certificate remain unresolved, as do all larger support primes. The finite no-hit is not an asymptotic theorem.

Lean proves Theorems 56.1– 56.3. It also proves that the surviving zero-displacement exclusion is equivalent to squarefull exclusion, together with the complete index-seven refutation of the stronger local assertion. The standard abc conjecture and the fixed-Pell squarefull exclusion both remain open.

57. Signed trace projectors for a prime-index Pell squarefull gate

Author: ChatGPT.
Summary. Let $(1+\sqrt{2})^n = A_n + B_n\sqrt{2}$. We give an exact signed-trace encoding of the still open assertion that $A_\ell B_\ell$ is not squarefull at an odd prime index. At every odd index,

$$A_{2n} - 1 = 2A_n^2, \qquad A_{2n} + 1 = 4B_n^2.$$

Thus squared support in either coordinate is equivalent to fourth-power support in the corresponding adjacent trace factor. We also identify a canonical Chinese-remainder projector $E_n = (A_{2n}+1)/2 = 2B_n^2$ whose exact idempotent defect is $2(A_nB_n)^2$, and show that one explicit Newton correction raises its precision from $(A_nB_n)^2$ to $(A_nB_n)^4$.

An exhaustive computation over odd prime $\ell \le 800000$ and support prime $q \le 2000000$, independently replayed by matrix powering, finds exactly two depth-two support collisions, one in each channel. Both give signed-trace depth exactly four and therefore disprove several tempting strengthenings. The index-seven row is nonsquarefull because its opposite coordinate is the exponent-one prime 239; the global squarefull status of the large row is unresolved. Neither row is a counterexample to the *abc* conjecture. All algebraic claims and the small counterexample are kernel checked in Lean.

57.1. The Pell coordinates and the unresolved gate.

Define nonnegative integers A_n, B_n by

$$(1+\sqrt{2})^n = A_n + B_n\sqrt{2}.$$

Equivalently, $(A_0, B_0) = (1, 0)$ and

$$(A_{n+1}, B_{n+1}) = (A_n + 2B_n, A_n + B_n).$$

The norm identity is

$$A_n^2 - 2B_n^2 = (-1)^n. \tag{395}$$

For an integer $N > 0$, *squarefull* means that $p \mid N$ implies $p^2 \mid N$ for every prime p.

The present work begins with the following fixed-parameter result from the companion transversality analysis.

Proposition 57.1 (fixed-parameter gate)**.** *Let ℓ be an odd prime. Every support prime of A_ℓ and B_ℓ is transverse at the Pell polynomial parameter $T = 2$. Furthermore, all first Hensel displacements vanish if and only if $A_\ell B_\ell$ is squarefull.*

Proposition 57.1 is recalled as an input, rather than silently being folded into a computation. Classical rank-of-apparition theory for Lucas sequences supplies the necessary candidate classes $q \equiv \pm 1 \pmod{2\ell}$ used below; primitive-divisor results such as [34] provide the broader historical setting. The exclusion of a squarefull product at every odd prime index remains open.

57.2. Depth doubling in adjacent signed trace factors.

Lemma 57.2 (addition law)**.** *For all $m, n \ge 0$,*

$$A_{m+n} = A_mA_n + 2B_mB_n,$$
$$B_{m+n} = A_mB_n + B_mA_n.$$

Proof. Multiply $(A_m + B_m\sqrt{2})(A_n + B_n\sqrt{2})$ and compare the rational and $\sqrt{2}$ coefficients. This is also an induction using the defining recurrence. ∎

Put $Z_n = A_{2n}$. Taking $m = n$ in Lemma 57.2 gives

$$Z_n = A_n^2 + 2B_n^2, \qquad B_{2n} = 2A_nB_n.$$

Theorem 57.3 (signed trace-square identities)**.** *If n is odd, then*

$$Z_n - 1 = 2A_n^2, \qquad Z_n + 1 = 4B_n^2.$$

Proof. For odd n, equation (395) gives $2B_n^2 = A_n^2 + 1$. Substitute this identity into $Z_n = A_n^2 + 2B_n^2$. □

Theorem 57.4 (exact depth dictionary)**.** *Let n be odd.*

(1) *If q is an odd prime dividing A_n, then*

$$q^4 \mid Z_n - 1 \iff q^2 \mid A_n.$$

(2) *If q is an odd prime dividing B_n, then*

$$q^4 \mid Z_n + 1 \iff q^2 \mid B_n.$$

Moreover, coordinate depth exactly two gives signed-trace depth exactly four.

Proof. Use Theorem 57.3. The factors 2 and 4 are coprime to an odd prime. Unique factorization gives both fourth-power equivalences; the same comparison at powers two, three, four, and five gives the exact depth-two-to-four assertion. □

For odd integers A, B and a trace value Z satisfying the two identities of Theorem 57.3, define

$$\mathrm{ST}_4(A, B, Z) \iff \begin{cases} q^4 \mid Z - 1 & \text{for every prime } q \mid A, \\ q^4 \mid Z + 1 & \text{for every prime } q \mid B. \end{cases}$$

Corollary 57.5 (exact signed-packet gate)**.** *For every odd prime ℓ,*

$$\begin{aligned} &\textit{all fixed-}T = 2 \textit{ first Hensel displacements vanish} \\ &\iff A_\ell B_\ell \textit{ is squarefull} \\ &\iff \mathrm{ST}_4(A_\ell, B_\ell, A_{2\ell}). \end{aligned}$$

Proof. The first equivalence is Proposition 57.1. The coordinates are coprime, so their product is squarefull exactly when both coordinates are squarefull. Apply Theorem 57.4 to every support prime. □

Corollary 57.6 (radical compression)**.** *If both coordinates at an odd index are squarefull, then*

$$\mathrm{rad}(A_n)^4 \mid Z_n - 1, \qquad \mathrm{rad}(B_n)^4 \mid Z_n + 1,$$

and

$$\mathrm{rad}(A_nB_n)^4 \mid (Z_n - 1)(Z_n + 1) = 8(A_nB_n)^2.$$

Proof. Squarefullness gives $\mathrm{rad}(A_n)^2 \mid A_n$ and $\mathrm{rad}(B_n)^2 \mid B_n$. Square these divisibilities and use Theorem 57.3. Coprimality combines the radicals, while direct multiplication gives the displayed product identity. □

The last divisibility is compatible with squarefullness and is not a contradiction. Its content is the placement of the two complete support packets in adjacent factors of one trace coordinate.

57.3. **The exact projector and its Newton continuation.** Fix an odd index and abbreviate $A = A_n$, $B = B_n$, $U = AB$, and $Z = Z_n$. Define

$$E = \frac{Z+1}{2} = 2B^2 = A^2 + 1.$$

Theorem 57.7 (channel projector)**.** *The integer E satisfies*

$$E \equiv 1 \pmod{A^2}, \qquad E \equiv 0 \pmod{B^2},$$

and has exact defect

$$E^2 - E = 2U^2. \tag{396}$$

Thus E is idempotent modulo U^2. If $U \geq 3$, it is not idempotent modulo U^3.

Proof. The congruences follow from the two expressions for E. Their product gives

$$E(E-1) = 2B^2A^2 = 2U^2.$$

If U^3 divided the last expression, then U would divide 2, which is impossible for $U \geq 3$. $\square$

The sharpness of the raw projector does not terminate the construction. For integers e, d, set

$$\mathcal{N}(e,d) = e - (2e-1)d.$$

Theorem 57.8 (one Newton step)**.** *If $e^2 - e = d$, then*

$$\mathcal{N}(e,d)^2 - \mathcal{N}(e,d) = d^2(4d-3).$$

For $d = 2U^2$, the corrected projector is congruent to e modulo U^2 and is idempotent modulo U^4.

Proof. Write $z = 2e - 1$. Since $z^2 = 4d + 1$, expansion yields

$$\begin{aligned}(e-zd)^2 - (e-zd) &= d - z^2d + z^2d^2 \\ &= d^2(4d-3).\end{aligned}$$

The correction zd is divisible by U^2, and the new defect contains $d^2 = 4U^4$. $\square$

57.4. **Complete-premise finite counterexamples.** The search enumerates every odd prime $\ell \leq 800000$ and every prime $q \leq 2000000$ in the necessary classes $q \equiv \pm 1 \pmod{2\ell}$. It computes (A_ℓ, B_ℓ) modulo q^2, and recomputes every repeated hit modulo q^5. The producer uses binary powering in $\mathbb{Z}[\sqrt{2}]$. A verifier independently powers

$$\begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}^\ell.$$

A separate certificate checks primality by complete trial division and recomputes each collision by a third, linear-recurrence algorithm.

Proposition 57.9 (replayed finite search)**.** *In the stated rectangle there are* 63950 *prime indices,* 764366 *candidate-prime tests, and* 12356 *actual support hits. Exactly two support hits have depth at least two. They are the rows in Table 1; both have coordinate depth exactly two and signed trace depth exactly four. No depth-three hit and no index with repeated support in both channels occurs in the rectangle.*

Table 1. The complete repeated-support rows. The two quotient columns are nonzero residues modulo q; C_ℓ denotes the indicated channel and $S_\ell = A_{2\ell} \mp 1$ its signed trace shift.

ℓ	q	ch.	C_ℓ/q^2	$C'_\ell(2)$	S_ℓ/q^4
7	13	B	1	12	4
773231	1546463	A	1090979	326969	1407445

Proof. The exhaustive producer and independent matrix replay agree on every count and on both complete rows. For the larger row, complete trial division through 879 and 1243 proves the primality of ℓ and q, respectively. Three orbit algorithms agree modulo $q^5 = 8844996565598309452666138088543$. The nonzero residues in Table 1 prove $q^2 \parallel C_\ell$, transversality of the fixed parameter, and $q^4 \parallel A_{2\ell} \mp 1$. $\square$

The first row can also be read without software:

$$B_7 = 169 = 13^2, \qquad A_{14} + 1 = 114244 = 4 \cdot 13^4.$$

It follows that $13^5 \nmid A_{14} + 1$.

Corollary 57.10 (refuted strengthenings)**.** *Each of the following universal statements is false:*

(1) *no support prime repeats at an odd prime index;*

(2) *repetition is confined to the A channel;*
(3) *repetition is confined to the B channel;*
(4) *the minimal representative $q = 2\ell \pm 1$ forces simple support;*
(5) *coordinate depth two always promotes to signed-trace depth five.*

Proof. The two rows of Table 1 meet every premise of the relevant claims. The first supplies the B-channel counterexample, the second the A-channel counterexample, and both use the minimal residue representative. Exact trace depth four refutes the last assertion. ☐

Remark 57.11 (logical boundary). The two rows are local collisions. Neither row says that both coordinates at its index are squarefull. Consequently neither refutes the signed-packet exclusion in Corollary 57.5, and neither is an *abc* counterexample. Bounded absence of a simultaneous row is recorded only as a finite observation.

57.5. **Formal and computational record.** The Lean module `PellSignedTraceProjector20260903.lean` formalizes the addition and doubling laws, Theorems 57.3–57.8, the radical consequences, and the complete index-seven refutation. It compiles with `-DwarningAsError=true`. Its companion audit prints the axioms of every declaration. The union consists only of the standard imported library axioms

$$\texttt{propext}, \qquad \texttt{Classical.choice}, \qquad \texttt{Quot.sound}.$$

The reproducibility bundle contains the producer, matrix verifier, complete-premise collision certifier, JSON outputs, exit codes, compiler logs, environment metadata, and SHA-256 manifests. No finite no-hit is promoted to a global theorem.

57.6. **Remaining problem.** The prime-index squarefull gate is now the existence problem for two fourth-power support packets in the adjacent factors $A_{2\ell} - 1$ and $A_{2\ell} + 1$, together with their exact product and one explicit Newton correction step. The depth-two collisions prove that neither a single channel nor one additional trace digit can close the problem. A successful continuation must couple all support primes across both adjacent factors, or construct a genuine simultaneous packet. Since no complete-premise counterexample to that global exclusion is presently known, the route remains open.

58. Prime-layer radical bounds and the corrected Mersenne order ledger

Put $M_m = 2^m - 1$ and

$$W_m = \frac{M_m}{\operatorname{rad}(M_m)}.$$

For every odd prime q, let $d_q = \operatorname{ord}_q(2)$ and $w_q = v_q(2^{d_q} - 1)$. Define the base excess of the exact-order block d by

$$E_d = \prod_{\substack{q \text{ prime} \\ d_q = d}} q^{w_q - 1}.$$

The index-lifting contribution must be kept separate:

$$B_m = \prod_{d \mid m} E_d, \qquad L_m = \prod_{\substack{q \text{ prime} \\ q \mid M_m}} q^{v_q(m/d_q)}.$$

Theorem 58.1 (Correct exact-order decomposition)**.** *For every $m \geq 1$,*

$$W_m = L_m B_m = L_m \prod_{d \mid m} E_d. \tag{397}$$

Moreover $L_m \mid m$, so $\log L_m \leq \log m = o(m)$.

Proof. If $q \mid M_m$, then $d_q \mid m$, and odd-prime LTE gives

$$v_q(M_m) = w_q + v_q(m/d_q).$$

After one copy of q is removed into the radical, the first summand supplies q^{w_q-1} in the unique block E_{d_q} and the second supplies its factor in L_m. Multiplying over the prime support proves (397). Since $d_q \mid q-1$, one has $q \nmid d_q$ and therefore $v_q(m/d_q) = v_q(m)$. Thus L_m takes from m only the full powers of a subset of its prime divisors, proving $L_m \mid m$. $\square$

The factor L_m cannot be deleted. At $m = 6$,

$$M_6 = 63 = 3^2 \cdot 7, \qquad \mathrm{rad}(M_6) = 21, \qquad W_6 = 3.$$

Here $d_3 = 2$, $w_3 = 1$, $d_7 = 3$, and $w_7 = 1$, so the base product is one, whereas $L_6 = 3$. This is a complete counterexample to the formula $W_m = \prod_{d|m} E_d$ without an index-lifting term. It supports the corrected decomposition rather than refuting the order-block route.

At a prime index the lifting factor disappears, and the radical admits an elementary uniform lower bound.

Theorem 58.2 (Composite prime-layer square carrier)**.** *Let ℓ be an odd prime. If M_ℓ is composite, then it has at least two distinct prime divisors, every prime divisor q satisfies*

$$\mathrm{ord}_q(2) = \ell, \qquad 2\ell \mid q-1, \qquad q \geq 2\ell + 1,$$

and consequently

$$(2\ell+1)^2 \leq \mathrm{rad}(M_\ell), \qquad (2\ell+1)^2 E_\ell \leq \Phi_\ell(2). \tag{398}$$

If one prime factor is at least H, the square in (398) may be replaced by $H\,(2\ell+1)$.

Proof. First, $2^\ell - 1$ is not a nontrivial prime power. Indeed, if $2^\ell - 1 = q^v$, then an even v contradicts reduction modulo eight. If $v \geq 3$ is odd, the factorization

$$q^v + 1 = (q+1)(q^{v-1} - q^{v-2} + \cdots - q + 1)$$

has an odd second factor strictly greater than one, impossible for a power of two. Hence a composite value has two distinct prime factors.

For a prime divisor q, its order divides the prime ℓ and is not one, so it equals ℓ. Lagrange's theorem and oddness of q give $2\ell \mid q-1$. Two distinct factors therefore contribute at least $(2\ell+1)^2$ to the radical. Since $L_\ell = 1$ and $\mathrm{rad}(M_\ell)E_\ell = M_\ell = \Phi_\ell(2)$, the excess form follows. If one factor is at least H, combine it with the distinct factor at least $2\ell + 1$. $\square$

Writing $P(N)$ for the greatest prime factor of N, Erdős and Shorey prove $P(2^\ell - 1) \gg \ell \log \ell$ for prime ℓ [15]; the later account of Ford, Luca, and Shparlinski records the same uniform input [16]. The asymmetric part of Theorem 58.2 therefore yields

$$\frac{E_\ell}{\Phi_\ell(2)} \ll \frac{1}{\ell^2 \log \ell} \tag{399}$$

for every sufficiently large prime ℓ, including the prime M_ℓ case trivially. This is only a polynomial saving. It leaves

$$\log E_\ell \leq \ell \log 2 - 2\log \ell - \log\log \ell + O(1),$$

whereas the Mersenne endpoint needs $\log E_\ell = o(\ell)$.

The corrected global ledger identifies a clean sufficient target.

Proposition 58.3 (Base-block totient criterion)**.** *If*

$$\log E_d = o(\varphi(d)),$$

then $\log W_m = o(m)$ as $m \to \infty$.

Proof. Put $e_d = \log E_d \geq 0$. Given $\varepsilon > 0$, choose D so that $e_d \leq \varepsilon\varphi(d)$ for $d \geq D$, and put $C_D = \sum_{d<D} e_d$. The divisor identity $\sum_{d|m} \varphi(d) = m$ gives

$$\log B_m = \sum_{d|m} e_d \leq C_D + \varepsilon m.$$

Theorem 58.1 now gives $\log W_m \leq \log m + C_D + \varepsilon m$. Divide by m, let m tend to infinity, and then let ε tend to zero. $\square$

Proposition 58.4 (Cyclotomic reformulation)**.** *For every $d > 1$,*

$$E_d = \frac{\Phi_d(2)}{\mathrm{rad}(\Phi_d(2))}, \qquad \log \Phi_d(2) = \varphi(d) \log 2 + O(1), \tag{400}$$

where the absolute value of the $O(1)$ term is at most two. If V_d is the powerful part of $\Phi_d(2)$, then

$$E_d \le V_d \le E_d^2.$$

Hence the hypothesis of Proposition 58.3 is equivalent both to

$$\log \mathrm{rad}(\Phi_d(2)) = \varphi(d) \log 2 + o(\varphi(d))$$

and to $\log V_d = o(\varphi(d))$.

Proof. Murty and Wong record the unconditional cyclotomic classification that a prime divisor of $\Phi_d(a, b)$ is 1 (mod d), except possibly for the largest prime factor of d, and that the exceptional divisor occurs to at most the first power [17]. Pomerance gives the same classification directly for $\Phi_d(2)$ and records the sharper bounds $2^{\varphi(d)-1} \le \Phi_d(2) < 2^{\varphi(d)+1}$ [19]. For $(a, b) = (2, 1)$ the first class consists of exact-order-d primes; the exceptional class disappears on division by the radical. This proves the first identity in (400). The cyclotomic product formula gives

$$\log \Phi_d(2) = \varphi(d) \log 2 + \sum_{e|d} \mu(d/e) \log(1 - 2^{-e}).$$

Since $-\log(1-x) \le 2x$ for $0 \le x \le 1/2$, the last sum has absolute value at most $\sum_{e\ge1} 2^{1-e} = 2$. A prime power q^a, $a \ge 2$, contributes q^{a-1} to E_d and q^a to V_d; now $a-1 \le a \le 2(a-1)$. Taking logarithms proves the equivalences. □

The reformulation permits an unconditional removal of all sufficiently small repeated-prime support. Put

$$\mathcal{W}_d = \{q : \mathrm{ord}_q(2) = d,\ v_q(2^d - 1) \ge 2\},$$
$$T_d = \prod_{q \in \mathcal{W}_d} q, \qquad D_d = \prod_{\substack{\mathrm{ord}_q(2)=d \\ v_q(2^d-1)\ge 3}} q^{v_q(2^d-1)-2}.$$

Thus $E_d = T_d D_d$.

Proposition 58.5 (Near-quadratic tail reduction)**.** *Set*

$$Y_d = \frac{\varphi(d)^2}{\log \log(3d)}, \qquad T_d^{\le Y} = \prod_{\substack{q \in \mathcal{W}_d \\ q \le Y_d}} q.$$

Then

$$\log T_d^{\le Y} = o(\varphi(d)). \tag{401}$$

If $\log E_d = o(\varphi(d))$ fails, then there are some $\epsilon > 0$ and a sequence $d_j \to \infty$ such that, for each fixed $\delta > 0$, there is a subsequence on which at least one of the following persists:

(1) $\log D_{d_j} \ge \epsilon\varphi(d_j)/4$;

(2) *the interval $Y_{d_j} < q \le d_j^{2+\delta}$ contains at least*

$$\frac{\epsilon\varphi(d_j)}{4(2 + \delta) \log d_j}$$

distinct base-two Wieferich primes, all of exact order d_j;

(3)
$$\sum_{\substack{\mathrm{ord}_q(2)=d_j,\ v_q(2^{d_j}-1)\ge2 \\ q>d_j^{2+\delta}}} \log q \ge \epsilon\varphi(d_j)/4.$$

Every prime in the third arm has $\mathrm{ord}_q(2) < q^{1/(2+\delta)}$ *and hence belongs to the zero-density exceptional set in the Erdős–Murty almost-all order theorem* [18]*. That density theorem does not control the weighted mass on a sparse sequence.*

Proof. Exact order d implies $q \equiv 1 \pmod{d}$. Fix $\theta \in (1/2, 1)$. The uniform bound $\varphi(d) \gg d/\log\log(3d)$ gives $d < Y_d^{\theta}$ for all sufficiently large d. The weighted Brun–Titchmarsh inequality of Murty–Séguin [20, Theorem 2.4] then gives

$$\log T_d^{\le Y} \le \sum_{\substack{q \le Y_d \\ q \equiv 1 \pmod{d}}} \log q \le \frac{2Y_d \log Y_d}{\varphi(d)\log(Y_d/d)} = o(\varphi(d)),$$

which proves (401). If the target fails, choose d_j with $\log E_{d_j} \ge \epsilon\varphi(d_j)$. Split $E_d = T_d D_d$ into the small factor, the two displayed support ranges, and D_d. For large j the small logarithm is at most $\epsilon\varphi(d_j)/4$; one of the other three nonnegative logarithms is therefore at least $\epsilon\varphi(d_j)/4$. Infinite pigeonhole fixes an arm. In the transition range every summand is at most $(2+\delta)\log d_j$, giving the asserted cardinality. In the extreme range $q > d_j^{2+\delta}$ and exact order give $d_j < q^{1/(2+\delta)}$. The final literature comparison follows from the almost-all lower order bound of Erdős and Murty. □

Corollary 58.6 (A fixed fraction of the square budget)**.** *Let* $s_d = |\mathcal{W}_d|$ *and put*

$$B_d = \frac{(\varphi(d)+1)\log 2}{2\log(d+1)}.$$

Then $s_d \le B_d$*. On the transition alternative of Proposition 58.5,*

$$\frac{|\{q \in \mathcal{W}_d : Y_d < q \le d^{2+\delta}\}|}{B_d} \ge \frac{\epsilon}{2(2+\delta)\log 2}\,\frac{\varphi(d)}{\varphi(d)+1}\,\frac{\log(d+1)}{\log d}.$$

Thus that alternative consumes a fixed positive proportion of the maximum support allowed by square divisibility alone.

Proof. Every $q \in \mathcal{W}_d$ is at least $d+1$, while $T_d^2 \mid \Phi_d(2)$. Pomerance's bound $\Phi_d(2) < 2^{\varphi(d)+1}$ [19] therefore gives

$$s_d \log(d+1) \le \log T_d \le \tfrac{1}{2}\log \Phi_d(2) < \tfrac{1}{2}(\varphi(d)+1)\log 2.$$

Divide the transition-cardinality bound in the preceding proposition by B_d. □

This is a necessary structure, not a contradiction. One very deep lift can still carry the first arm, and no unconditional theorem excludes the fixed-proportion cluster in the second arm.

Because $q \nmid (q-1)/d_q$, LTE also gives

$$v_q(2^{q-1} - 1) = v_q(2^{d_q} - 1) = w_q.$$

Thus T_d is exactly the same-order Wieferich support and D_d is its super-Wieferich excess. A prime enters only one canonical block. Finiteness of base-two super-Wieferich primes would remove the first arm eventually but would leave the two support tails. Current fixed-base results, including Fellini–Murty's finite-super implication, do not provide a weighted estimate inside one exact-order block [39]. Under uniformly bounded canonical valuations—in particular, under finiteness of base-two super-Wieferich primes—Murty–Séguin's bound for the largest prime is only at a polynomial scale [20]. Pomerance's unconditional results establish many composite primitive parts; his stronger distinct-factor theorem is abc-conditional [19]. Neither conclusion yields exponential radical saturation on the $\varphi(d)$ scale.

Proposition 58.7 (A nontrivial canonical block)**.** *The canonical exact-order block at* $d = 364$ *is nontrivial; more precisely,*

$$1093 \mid E_{364}.$$

Proof. Direct integer arithmetic gives

$$1093^2 \mid 2^{364} - 1, \qquad 1093^3 \nmid 2^{364} - 1,$$

and

$$2^{182} \equiv 1092, \qquad 2^{52} \equiv 27, \qquad 2^{28} \equiv 121 \pmod{1093}.$$

The number 1093 is prime and $364 = 2^2 \cdot 7 \cdot 13$. The prime-divisor criterion for multiplicative order therefore gives $\mathrm{ord}_{1093}(2) = 364$: exponent 364 kills 2, while division by any prime divisor of 364 does not. The first two divisibilities show that the valuation at 1093 is exactly two. Thus the defining product for E_{364} contains 1093^{2-1}. □

Proposition 58.7 is a complete counterexample to the strengthening $E_d = 1$ for every d. It does not contradict $\log E_d = o(\varphi(d))$, because a single fixed block has no effect on the limit at infinity.

Two further exact witnesses delimit tempting universal strengthenings:

$$2^{37} - 1 = 223 \cdot 616318177$$

has exactly two prime factors, refuting the claim that every composite prime layer with $\ell \geq 37$ has at least three; and

$$2^{11} - 1 = 23 \cdot 89, \qquad \mathrm{rad}(M_{11}) = 2047 < (2 \cdot 11 + 1)^3$$

refutes a cubic radical replacement valid at every composite odd-prime layer. These witnesses do not refute an eventual three-factor estimate, a weighted base-block theorem, or Proposition 58.3. A deterministic scan of all prime indices $3 \leq \ell \leq 61$ found no repeated factor; that is recorded only as a finite no-hit.

Lean verifies the elementary prime-layer theorems, the three prime-layer witnesses, and the finite totient-tail inequality. Companion modules additionally verify exact-order LTE, $L_m \mid m$, the finite product $W_m = L_m \prod_{d|m} E_d$, equality of relative and canonical blocks on divisors, the exact logarithmic divisor-sum identity, and the conditional asymptotic conclusion of Proposition 58.3, as well as the nontrivial canonical-block witness $1093 \mid E_{364}$. The finite core of the powerful-part comparison, the deep/transition/extreme mass trichotomy, and both the realized-support and exact ambient square-budget ratio implications is checked in a fifth module. That module takes the cyclotomic identification, the Brun–Titchmarsh small-arm estimate, and the prime-order distribution theorem as explicit paper inputs; it does not assert them as Lean theorems. The cited largest-prime-factor theorem remains an external published input, and $\log E_d = o(\varphi(d))$ remains an open hypothesis. The lifting remainder and the passage from that hypothesis are now kernel-checked.

59. The weighted small-order tail

This section sharpens the third alternative in Proposition 58.5. It also records a logical limitation of the almost-all order theorem: density in the prime-size variable does not control logarithmic mass in a moving exact-order fibre. None of the results in this section closes the Mersenne route unconditionally.

For an odd prime p, write

$$f(p) = \mathrm{ord}_p(2), \qquad \alpha_p = v_p(2^{f(p)} - 1).$$

The valuation formula of Murty–Séguin and the intrinsic-prime classification give, for $d > 1$,

$$\log E_d = \sum_{f(p)=d} (\alpha_p - 1) \log p = \log T_d + \log D_d. \tag{402}$$

Here T_d contains one copy of every p with $f(p) = d$ and $\alpha_p \geq 2$, while D_d contains the remaining depth. In particular,

$$T_d^2 \mid \Phi_d(2), \qquad \log T_d < \frac{\varphi(d) + 1}{2} \log 2. \tag{403}$$

Proposition 59.1 (Global order-index budget)**.** *For $Y \geq 2$,*

$$\sum_{\substack{p \text{ odd prime} \\ f(p) \leq Y}} \alpha_p \log p \leq \sum_{d \leq Y} \log \Phi_d(2) = O(Y^2).$$

Consequently

$$\#\{p \text{ odd prime} : f(p) \leq Y\} = O(Y^2/\log Y).$$

Proof. If $f(p) = d$, the full power p^{α_p} belongs to the exact-order part of $\Phi_d(2)$. The order is unique, so contributions belonging to different d do not overlap. Sum the logarithms and use $\Phi_d(2) < 2^{\varphi(d)+1}$ from [19]. For the counting assertion, split the primes at Y^2; every prime above the split contributes at least $2 \log Y$. □

This is a quadratic cumulative estimate. Restricting it to a single order cannot produce the required $o(\varphi(d))$ bound.

Proposition 59.2 (Weighted consequence of the almost-all order theorem)**.** *Fix $\delta > 0$ and set*

$$\mathcal{B}_\delta = \{p : f(p) \leq p^{1/(2+\delta)}\}.$$

Then

$$\#\{p \leq x : p \in \mathcal{B}_\delta\} = o(x/\log x), \qquad \sum_{\substack{p \leq x \\ p \in \mathcal{B}_\delta}} \log p = o(x).$$

Proof. Erdős–Murty prove that, for every positive $\eta(p) \to 0$, outside $o(\pi(x))$ primes one has $f(p) > p^{1/(2+\eta(p))}$ [18]. Take $\eta(p) < \delta$ eventually. If $A_\delta(x)$ denotes the exceptional count, Abel summation gives

$$\sum_{\substack{p \leq x \\ p \in \mathcal{B}_\delta}} \log p = A_\delta(x) \log x - \int_2^x \frac{A_\delta(t)}{t}\, dt = o(x).$$

□

At the beginning of the extreme arm, $x = d^{2+\delta}$, the last estimate need not be $o(\varphi(d))$. The weighted Brun–Titchmarsh theorem [20, Theorem 2.4] gives a little-oh precisely in the range

$$x = o(\varphi(d)^2), \qquad d < x^\theta, \qquad \frac{\log x}{\log(x/d)} = O(1)$$

for a fixed $\theta < 1$. It gives only the natural $O(\varphi(d))$ scale when $x \asymp \varphi(d)^2$. Thus neither result covers the extreme range.

In a dyadic shell $(X, 2X]$, the available inputs reduce to

$$\sum_{\substack{p \in \mathcal{W}_d \\ X < p \leq 2X}} \log p \leq \min \left\{ o(X), O\left(\frac{X \log X}{\varphi(d) \log(X/d)}\right), O(\varphi(d)) \right\}. \tag{404}$$

The final term is (403); at every relevant scale it is the only generally effective one, and it has no strict saving. Across larger X, the available theorems furnish no uniform $o(\varphi(d))$ saving, so the known shell estimates do not create the missing little-oh when summed.

Proposition 59.3 (Coarse information does not imply a fibrewise bound)**.** *There exist primes p_j and integers $d_j \to \infty$ such that*

$$p_j \equiv 1 \pmod{d_j}, \qquad d_j < p_j^{1/(2+\delta)}, \qquad \#\{j : p_j \leq x\} = O(\log \log x),$$

for every fixed $\delta > 0$, while positive numbers ϕ_j satisfy

$$\log p_j \asymp \phi_j, \qquad 2 \log p_j < (\phi_j + 1) \log 2.$$

Hence global zero density, the congruence forced by exact order, the small-order scale, and the numerical square budget do not by themselves imply $o(\phi_j)$ mass in a moving labelled fibre.

Proof. Take $d_j = 2^j$, $\phi_j = \varphi(d_j) = 2^{j-1}$, fix $0 < c < \log 2/2$, and put $X_j = \exp(c\phi_j)$. Since $d_j = O(\log X_j)$, the classical Siegel–Walfisz theorem supplies, for large j, a prime

$$p_j \in [X_j, 2X_j], \qquad p_j \equiv 1 \pmod{d_j}.$$

The logarithms grow geometrically, proving the count. Moreover,

$$\sum_{p_j \le x} \log p_j = O(\log x \log\log x) = o(x),$$

so even the global logarithmic mass is sparse. Exponential growth gives the small-order inequality, and $\log p_j = c\phi_j + O(1)$ gives the stated linear fibre mass. Finally, $\Phi_{2^j}(2) = 2^{\phi_j} + 1$, and eventually

$$2\log p_j < \phi_j \log 2 < \log \Phi_{d_j}(2),$$

which proves the asserted numerical square budget. □

In Proposition 59.3, assigning abstract labels $\widetilde{f}(p_j) = d_j$ and $\widetilde{\alpha}_{p_j} = 2$ makes every fibre a singleton of linear logarithmic mass. These labels are not asserted to equal the actual multiplicative order and valuation. The proposition therefore refutes only an inference from the four coarse estimates; it is not an arithmetic counterexample to the tail and does not close the route.

Proposition 59.4 (Summable-profile shell localization)**.** *Suppose a finite nonnegative mass satisfies $H \le M \le \sum_{k\in I} M_k$, where $H > 0$. If $a_k \ge 0$ and $\sum_{k\in I} a_k \le 1$, then some $k \in I$ satisfies*

$$M_k \ge H a_k.$$

Proof. Otherwise $\sum_k M_k < H \sum_k a_k \le H$, a contradiction. □

For

$$L_d = (2+\delta)\log d, \qquad \mathcal{S}_{d,k} = \{p \in \mathcal{W}_d : 2^k L_d < \log p \le 2^{k+1} L_d\},$$

take $a_k = 6/(\pi^2(k+1)^2)$. An extreme mass at least $\epsilon\varphi(d)/4$ forces, for some k,

$$\sum_{p\in\mathcal{S}_{d,k}} \log p \ge \frac{3\epsilon}{2\pi^2(k+1)^2}\varphi(d)$$

and consequently

$$|\mathcal{S}_{d,k}| \ge \frac{3\epsilon\varphi(d)}{2\pi^2(k+1)^2\, 2^{k+1}(2+\delta)\log d}. \tag{405}$$

A summable-envelope upper bound of the reverse form, with a common coefficient tending to zero, would close the extreme arm.

There is also a useful concentration split. At any logarithmic height $h > 0$, a mass at least H either has mass at least $H/2$ above h, or contains at least $H/(2h)$ points below h. The square budget bounds the number of points above h by $O(\varphi(d)/h)$. Thus a failed tail has either a macroscopic component of rank $O(\varphi(d)/h)$ or an increasingly diffuse submacroscopic component. The first rank is bounded when h is chosen comparable to $\varphi(d)$.

The pointwise block estimate is stronger than the endpoint actually requires. Put

$$B_m = \prod_{d|m} E_d, \qquad A(m) = \log B_m = \sum_{d|m} \log E_d.$$

The exact lifting identity and $1 \le L_m \le m$ give

$$W_m = L_m B_m, \qquad \log W_m = A(m) + \log L_m, \qquad \log L_m = o(m).$$

Theorem 59.5 (Exact divisor-average endpoint)**.** *One has*

$$\log W_m = o(m) \quad\Longleftrightarrow\quad \sum_{d|m} \log E_d = o(m).$$

Proof. From left to right, subtract the actual LTE term $\log L_m = o(m)$ from $\log W_m = \log B_m + \log L_m$. From right to left, add that same term. Both steps use the exact product identity. □

The quantifier in both little-oh statements is $\forall\epsilon > 0\,\exists M\,\forall m \geq M$. The identity holds for every positive m, and $L_m \mid m$ is the LTE lifting input that proves $0 \leq \log L_m \leq \log m = o(m)$. Thus both directions include the complete lifting remainder. In particular, the reverse implication is not inferred from a product inequality: it uses $W_m = L_m B_m$ and the negligible lifting factor.

Let s_d be the support mass below $Y_d = \varphi(d)^2/\log\log(3d)$, and write

$$\log E_d = s_d + r_d,$$

where

$$r_d = \sum_{\substack{f(p)=d,\ \alpha_p\geq 2\\ p>Y_d}} \log p + \sum_{f(p)=d} (\alpha_p - 2)_+ \log p. \tag{406}$$

The proved small-arm estimate is $s_d = o(\varphi(d))$.

Corollary 59.6 (Divisor-average remainder criterion)**.** *The Mersenne endpoint follows from*

$$\forall\epsilon > 0\ \exists M\ \forall m \geq M, \qquad \sum_{d\mid m} r_d \leq \epsilon m.$$

Proof. The identity

$$\sum_{d\mid m} \varphi(d) = m$$

converts the small-arm little-oh into

$$\sum_{d\mid m} s_d = o(m).$$

Adding the displayed hypothesis and applying Theorem 59.5 completes the proof. ☐

This criterion is strictly weaker than $r_d = o(\varphi(d))$. To see strictness, select a rapidly increasing divisibility chain of squarefree primorials $n_1 \mid n_2 \mid \cdots$ such that

$$\sum_{i<k} n_i = o(n_k), \qquad \varphi(n_k)/n_k \longrightarrow 0,$$

and define $r_{n_k} = \varphi(n_k)$ and $r_d = 0$ otherwise. Pointwise $o(\varphi(d))$ fails along n_k. If n_k is the largest selected primorial dividing m, then

$$\frac{1}{m}\sum_{d\mid m} r_d \leq \frac{\varphi(n_k) + \sum_{i<k} n_i}{n_k} = o(1).$$

When the largest such k remains bounded, the numerator is fixed and the ratio again tends to zero. The example is abstract and asserts nothing about the actual Mersenne remainders. The divisor-average criterion opens a second analytic attack on the tail, through divisor lattices rather than uniform control of every fibre.

For a fixed $\delta > 0$, a pointwise proof still reduces to the three nonnegative gates

$$\sum_{f(p)=d} (\alpha_p - 2)_+ \log p = o(\varphi(d)),$$

$$\#\{p \in \mathcal{W}_d : Y_d < p \leq d^{2+\delta}\} = o(\varphi(d)/\log d),$$

and

$$\sum_{\substack{p\in\mathcal{W}_d\\ p>d^{2+\delta}}} \log p = o(\varphi(d)).$$

The current literature supplies none of these three estimates in the required uniform form. The exact divisor-average criterion and the summable-profile localization are formalized in `MersenneWeightedOrderTail20260901`; all analytic inputs remain external to the Lean kernel.

60. Exact-order blocks

Let

$$M_m = 2^m - 1, \qquad W_m = \frac{M_m}{\operatorname{rad}(M_m)}.$$

For an odd prime q, put

$$d_q = \operatorname{ord}_q(2), \qquad w_q = v_q(2^{d_q} - 1). \tag{407}$$

Since $d_q \mid q-1$ and $q \nmid (q-1)/d_q$, the odd-prime lifting-the- exponent formula gives

$$v_q(2^{q-1} - 1) = w_q. \tag{408}$$

Thus $w_q \geq 2$ is precisely the base-two Wieferich condition, while $w_q \geq 3$ is the base-two super-Wieferich condition.

For each $d \geq 1$, define the canonical exact-order block and its two depth components by

$$E_d = \prod_{d_q = d} q^{w_q - 1}, \qquad T_d = \prod_{\substack{d_q = d \\ w_q \geq 2}} q, \tag{409}$$

$$S_d^{(3)} = \prod_{\substack{d_q = d \\ w_q \geq 3}} q, \qquad D_d = \prod_{\substack{d_q = d \\ w_q \geq 3}} q^{w_q - 2}. \tag{410}$$

All these products are finite: every prime in the order-d fibre divides $2^d - 1$. Comparing the exponent of each prime gives the exact identity

$$E_d = T_d D_d. \tag{411}$$

The factor T_d records one copy of every repeated exact-order prime; D_d records only the multiplicity beyond the second copy.

For completeness, the order blocks recover the whole Mersenne power loss after an elementary lifting factor is retained. Set

$$L_m = \prod_{q \mid M_m} q^{v_q(m/d_q)}. \tag{412}$$

Here $d_q \mid m$ whenever $q \mid M_m$.

Proposition 60.1 (Exact Mersenne ledger)**.** *For every $m \geq 1$,*

$$W_m = L_m \prod_{d \mid m} E_d, \qquad L_m \mid m. \tag{413}$$

Consequently $0 \leq \log L_m \leq \log m = o(m)$.

Proof. If $q \mid M_m$, then $d_q \mid m$, and odd-prime LTE gives

$$v_q(M_m) = w_q + v_q(m/d_q).$$

After the radical removes one copy of q, the first term contributes $q^{w_q - 1}$ to the unique block E_{d_q}, and the second contributes its factor to L_m. Multiplication over the prime support proves the first identity in (413). Moreover, $d_q \mid q-1$ implies $q \nmid d_q$, hence $v_q(m/d_q) = v_q(m)$. The product L_m therefore takes the full q-primary component of m for only a subset of the prime divisors of m, and so $L_m \mid m$. □

The lemmas of Murty and Séguin classify prime powers in cyclotomic values [20, Lemmas 4.2–4.3]. At the unique exact-order index they correspond to Wieferich and super-Wieferich primes. Their results do not give a weighted upper bound within a moving exact-order fibre. Fellini and Murty obtain quantitative lower bounds for non-Wieferich prime ideals under a number-field *abc* conjecture or under finiteness of the relevant super-Wieferich primes [39, Theorems 1.2–1.4]. Neither hypothesis supplies an unconditional estimate for the fixed-base depth masses studied below.

61. The exact depth layer cake

For $j \geq 3$, define the logarithmic depth layer

$$\Theta_{d,j} = \sum_{\substack{d_q=d \\ w_q \geq j}} \log q. \tag{414}$$

Proposition 61.1 (Finite layer-cake identity)**.** *For every d,*

$$\log D_d = \sum_{j \geq 3} \Theta_{d,j}. \tag{415}$$

The sum is finite. More precisely, if $M \geq w_q$ for every prime in the order-d super-Wieferich support, then the right-hand side may be truncated at $j = M$.

Proof. A fixed prime q contributes $(w_q - 2)\log q$ to $\log D_d$. It occurs once in every layer with $3 \leq j \leq w_q$, and therefore contributes

$$\sum_{j=3}^{w_q} \log q = (w_q - 2)\log q.$$

The prime support is finite, and every valuation is finite. Interchanging the resulting two finite sums and then summing the displayed identity over q proves (415). The same calculation proves the stated finite truncation. □

The identity distinguishes the first super-Wieferich layer $\Theta_{d,3} = \log S_d^{(3)}$ from the accumulated mass at higher depths. A small first layer alone does not control the full sum unless the high-depth layers are controlled as well.

62. Moving-depth truncation

For an integer $K \geq 3$, put

$$R_d(K) = \sum_{\substack{d_q=d \\ w_q \geq 3}} (w_q - K)_+ \log q, \qquad (x)_+ = \max\{x, 0\}. \tag{416}$$

Proposition 62.1 (Exact truncation and support bound)**.** *For every $K \geq 3$,*

$$\log D_d = \sum_{\substack{d_q=d \\ w_q \geq 3}} \min\{w_q - 2, K - 2\} \log q + R_d(K), \tag{417}$$

$$\log D_d \leq (K-2)\log S_d^{(3)} + R_d(K). \tag{418}$$

Moreover,

$$R_d(K) = \sum_{j \geq K+1} \Theta_{d,j}. \tag{419}$$

Proof. For integers $w \geq 0$ and $K \geq 2$, truncated subtraction gives

$$w - 2 = \min\{w-2, K-2\} + (w-K)_+ \tag{420}$$

whenever $w \geq 2$. Multiplying (420) by the nonnegative number $\log q$ and summing over the super-Wieferich support proves (417). The first coefficient is at most $K - 2$; summing that pointwise inequality yields (418). Finally, $w - K$ is the number of integers j with $K+1 \leq j \leq w$. The finite sum interchange used in Proposition 61.1 therefore proves (419). □

Corollary 62.2 (Frequency–depth dichotomy)**.** *If $K > 2$ and $\log D_d \geq A$, then*

$$\frac{A}{2(K-2)} \leq \log S_d^{(3)} \qquad \text{or} \qquad \frac{A}{2} \leq R_d(K). \tag{421}$$

Proof. If the first alternative fails, then $(K-2)\log S_d^{(3)} < A/2$. Combining this strict inequality with (418) and $\log D_d \geq A$ forces $R_d(K) > A/2$. □

Thus a large deep factor cannot be explained merely by the existence of some super-Wieferich primes. At every chosen threshold it forces either large one-copy support mass or large mass above that threshold.

63. Sufficient asymptotic gates

All little-oh statements in this section are taken along positive integers tending to infinity.

Theorem 63.1 (Moving-threshold gate). *Let $K_d \geq 3$ be an integer-valued threshold. If*

$$(K_d - 2)\log S_d^{(3)} = o(\varphi(d)), \tag{422}$$

$$R_d(K_d) = o(\varphi(d)), \tag{423}$$

then

$$\log D_d = o(\varphi(d)). \tag{424}$$

If, in addition, $\log T_d = o(\varphi(d))$, then

$$\log \frac{2^m - 1}{\operatorname{rad}(2^m - 1)} = o(m). \tag{425}$$

Proof. Apply (418) with $K = K_d$. Its two nonnegative right-hand terms are little-oh of $\varphi(d)$ by (422)–(423), proving (424). The exact identity (411) then gives $\log E_d = o(\varphi(d))$ under the additional hypothesis.

For clarity, the last implication uses only the standard divisor identity

$$\sum_{d|m} \varphi(d) = m.$$

Given $\varepsilon > 0$, choose D so that $\log E_d \leq \varepsilon\varphi(d)$ for $d \geq D$, and put $C_D = \sum_{d<D} \log E_d$. Then

$$\sum_{d|m} \log E_d \leq C_D + \varepsilon m.$$

Proposition 60.1 and $\log L_m = o(m)$ now imply (425). $\square$

The two hypotheses in Theorem 63.1 are sufficient, not necessary. In particular, a uniform depth bound makes (423) vanish for a fixed sufficiently large threshold but does not by itself prove the support estimate (422). Conversely, sparsity of $S_d^{(3)}$ does not control an unbounded high-depth tail.

The actual Mersenne endpoint admits a weaker divisor-average interface.

Proposition 63.2 (Exact divisor-average endpoint). *One has*

$$\log W_m = o(m) \quad \Longleftrightarrow \quad \sum_{d|m} \log E_d = o(m). \tag{426}$$

Proof. Taking logarithms in (413) gives

$$\log W_m = \log L_m + \sum_{d|m} \log E_d.$$

All terms are nonnegative and $\log L_m = o(m)$. Adding or subtracting this negligible lifting term proves the two implications in (426). $\square$

Let s_d be a controlled part of the one-copy repeated support and let U_d be the remaining part, so that

$$\log T_d = s_d + U_d, \qquad s_d, U_d \geq 0. \tag{427}$$

In the Mersenne order-block route, s_d is the Brun–Titchmarsh small arm and U_d collects the surviving one-copy support tails.

Theorem 63.3 (Divisor-average threshold gate)**.** *Assume*

$$\sum_{d|m} s_d = o(m). \tag{428}$$

Let $K_d \geq 3$*. If*

$$\sum_{d|m} \Big(U_d + (K_d - 2) \log S_d^{(3)} + R_d(K_d) \Big) = o(m), \tag{429}$$

then (425) *holds.*

Proof. Equations (411) and (427), together with (418), give the termwise nonnegative bound

$$\log E_d \leq s_d + U_d + (K_d - 2) \log S_d^{(3)} + R_d(K_d).$$

Sum over $d \mid m$ and apply (428)–(429). The right-hand side is $o(m)$, so Proposition 63.2 yields (425). ☐

If $s_d = o(\varphi(d))$, then (428) follows from $\sum_{d|m} \varphi(d) = m$. Likewise, separate pointwise $o(\varphi(d))$ estimates for the three terms in (429) imply that divisor-average condition. No converse is asserted: a divisor average can tolerate large values at sparse orders. Theorem 63.3 is therefore the weaker interface and is often the more appropriate target.

64. An exact odd-order counterexample

Factorizations involving $2^{d/2} \pm 1$ may suggest restricting repeated exact-order primes to even d. The second classical base-two Wieferich prime rules out that restriction.

Proposition 64.1 (The prime 3511)**.** *The integer* 3511 *is prime and*

$$\operatorname{ord}_{3511}(2) = 1755, \qquad v_{3511}(2^{1755} - 1) = 2. \tag{430}$$

In particular, the exact order is odd.

Proof. First, $3511 - 1 = 3510 = 2 \cdot 3^3 \cdot 5 \cdot 13$. The congruences

$$7^{3510} \equiv 1 \pmod{3511}$$

and

r	2	3	5	13
$7^{3510/r} \bmod 3511$	3510	2754	1800	494
$\gcd(7^{3510/r} - 1, 3511)$	1	1	1	1

give a Pocklington certificate for the primality of 3511: the known factor is $F = 3510 > \sqrt{3511}$.

Direct modular exponentiation gives

$$2^{1755} \equiv 1 \pmod{3511^2}, \tag{431}$$

$$2^{1755} \equiv 21954602502 \not\equiv 1 \pmod{3511^3}, \tag{432}$$

$$2^{585} \equiv 756 \pmod{3511}, \tag{433}$$

$$2^{351} \equiv 1578 \pmod{3511}, \tag{434}$$

$$2^{135} \equiv 88 \pmod{3511}. \tag{435}$$

The first congruence shows that the order divides $1755 = 3^3 \cdot 5 \cdot 13$. The last three congruences exclude division of the order by each prime divisor 3, 5, and 13; hence the order is exactly 1755. Finally, (431)–(432) show that its canonical valuation is exactly two. ☐

Proposition 64.1 refutes the precise universal statement

$$q^2 \mid 2^{\operatorname{ord}_q(2)} - 1 \implies 2 \mid \operatorname{ord}_q(2). \tag{436}$$

The scope is narrow. Since the valuation at 3511 is two, this prime is absent from $S_{1755}^{(3)}$ and contributes nothing to D_{1755}. It is not a super-Wieferich counterexample and does not contradict any condition in Theorem 63.1 or Theorem 63.3.

65. Finite computational audit

A finite scan enumerated every prime at most 10^7, tested $2^{q-1} \equiv 1 \pmod{q^2}$, computed the exact order for every hit, and then increased the prime-power modulus until the congruence first failed. The generator used a full Eratosthenes sieve. An independent verifier used a segmented sieve and recomputed every Fermat congruence, order certificate, and depth certificate.

The two enumerations agree on all 664,579 primes in the interval. Their complete list of base-two Wieferich hits is

q	$\mathrm{ord}_q(2)$	w_q	order parity
1093	364	2	even
3511	1755	2	odd

Thus the interval contains no base-two super-Wieferich hit.

This is strictly a finite audit. The absence of a depth-three hit below 10^7 proves neither finiteness nor density zero and supplies no asymptotic estimate for $S_d^{(3)}$, $R_d(K)$, or D_d. It is used only to check the explicit certificate in Section 64 and the implementation of the depth definitions.

66. Remaining analytic gate

The exact identities reduce the deep obstruction without estimating it. A weaker decomposed sufficient target is to choose moving thresholds $K_d \geq 3$ and prove

$$\sum_{d|m} \Big(U_d + (K_d - 2) \log S_d^{(3)} + R_d(K_d) \Big) = o(m), \tag{437}$$

while retaining the already controlled divisor-average small arm. A stronger pointwise route would instead establish

$$\log T_d = o(\varphi(d)), \qquad (K_d - 2) \log S_d^{(3)} = o(\varphi(d)), \qquad R_d(K_d) = o(\varphi(d)).$$

Neither Murty–Séguin nor Fellini–Murty supplies these fixed-base, weighted exact-order estimates unconditionally. The truncation bound is a one-way sufficient mechanism, and no necessity claim is made. The super-Wieferich depth route therefore remains open.

67. Totient-weighted concentration at the Mersenne endpoint

The exact order-block identity of Theorem 59.5 changes the remaining Mersenne problem from a pointwise estimate into a divisor-average problem. This section identifies the natural probability measure on that divisor lattice and localizes the unresolved arithmetic mass to divisors extremely close to the ambient index.

For an integer $m \geq 1$ and a nonnegative sequence a_d put

$$A(m) = \sum_{d|m} a_d, \qquad \mu_m(d) = \frac{\varphi(d)}{m} \quad (d \mid m).$$

The identity $\sum_{d|m} \varphi(d) = m$ makes μ_m a probability measure. Assume throughout the boundedness condition

$$0 \leq a_d \leq C\varphi(d) \tag{438}$$

with one constant $C \geq 0$, and define

$$B_\epsilon(m) = \sum_{\substack{d|m \\ a_d > \epsilon\varphi(d)}} \varphi(d).$$

Theorem 67.1 (Exact totient exceptional-mass criterion)**.** *Under* (438)*,*

$$A(m) = o(m) \quad \Longleftrightarrow \quad B_\epsilon(m) = o(m) \quad \text{for every fixed } \epsilon > 0.$$

Proof. For every positive ϵ one has the finite inequalities

$$\epsilon B_\epsilon(m) \le A(m) \le \epsilon m + C B_\epsilon(m).$$

The first implication follows from the lower inequality. For the reverse implication, given $\eta > 0$, apply the hypothesis with $\epsilon = \eta/2$ and require $B_\epsilon(m) \le \eta m/(2(C+1))$. □

The divisor probability has an exact logarithmic moment. If $m = \prod_{p^a \| m} p^a$ and D is distributed according to μ_m, then

$$\mathbb{E}_{\mu_m} \log \frac{m}{D} = \sum_{p^a \| m} \frac{1-p^{-a}}{p-1} \log p. \tag{439}$$

Indeed, the exponent $J_p = v_p(D)$ has probability $\varphi(p^j)/p^a$, and the tail-sum identity gives $\mathbb{E}(a - J_p) = \sum_{r=1}^{a} p^{-r}$.

Theorem 67.2 (Moving near-diagonal concentration)**.** *There is an absolute C_0 such that*

$$\mathbb{E}_{\mu_m} \log \frac{m}{D} \le C_0 \log\log(3m).$$

Consequently, if $H(m) > 0$ eventually and $H(m)/\log\log(3m) \to \infty$, then

$$\frac{1}{m} \sum_{\substack{d \mid m \\ d \le m \exp(-H(m))}} \varphi(d) \longrightarrow 0.$$

Proof. Bound (439) by $\sum_{p|m} \log p/(p-1)$. Enlarge the absolute constant to absorb the finitely many small m, and assume $z = \log m > 1$. Split at z. Chebyshev's estimate $\vartheta(x) = O(x)$ bounds every dyadic shell below z by an absolute constant, giving $O(\log z)$ in total. Above z the sum is at most $\log \operatorname{rad}(m)/(z-1) = O(1)$. Markov's inequality at height $H(m)$ proves the assertion. □

The growth condition on H cannot uniformly be replaced by $H = C \log\log m$ with fixed C. For the primorial $m_y = \prod_{p \le y} p$, the complementary divisor m_y/D contains each $p \le y$ independently with probability $1/p$. Choose an integer $r > C+1$ and disjoint intervals $0 < u_1 < v_1 < \cdots < u_r < v_r < 1$ with $\sum_i u_i > C+1$. For each i, let $A_i(y)$ be the event that m_y/D contains a prime in $(y^{u_i}, y^{v_i}]$. Mertens's prime-product theorem [21] gives

$$\Pr(A_i(y)^c) = \prod_{y^{u_i} < p \le y^{v_i}} (1 - 1/p) \longrightarrow u_i/v_i.$$

The events are independent because their intervals are disjoint, hence $\liminf_y \Pr(\bigcap_i A_i(y)) = \prod_i (1 - u_i/v_i) > 0$. On this intersection $m_y/D > y^{C+1}$; Chebyshev's two-sided bounds for $\vartheta(y) = \log m_y$ imply $y^{C+1} > (\log m_y)^C$ eventually. Therefore, for every fixed $C > 0$,

$$\liminf_{y \to \infty} \mu_{m_y} \left\{ D \le \frac{m_y}{(\log m_y)^C} \right\} > 0.$$

This is a full-premise counterexample to the proposed uniform strengthening for arbitrary bounded masses (take $a_d = \varphi(d)$). It is not a counterexample to the actual Mersenne blocks.

For those blocks let $a_d = \log E_d$. If a prime p occurs in E_d, then $\operatorname{ord}_p(2) = d$, so $d \mid p-1$ and $p \nmid d$. In the identity $\prod_{e|d} \Phi_e(2) = 2^d - 1$, a divisibility $p \mid \Phi_e(2)$ with $e \mid d$ would, since $p \nmid e$, make the reduction of 2 a primitive eth root; hence $e = d$. Thus the full p-adic multiplicity of $2^d - 1$ is carried by $\Phi_d(2)$. The exponent used in E_d is one smaller, and the distinct prime-power factors are pairwise coprime, proving $E_d \mid \Phi_d(2)$. The root product then gives

$$0 \le \log E_d \le \varphi(d) \log 3. \tag{440}$$

The relevant valuation identities and current distribution results are consistent with [20, 19, 39]; none supplies the following exceptional-weight estimate. The primitive-divisor refinement [22] and the recent generalized-Wieferich congruence criteria [23] likewise control local phenomena rather than the required moving totient mass.

Corollary 67.3 (Localized exact Mersenne gate)**.** *Fix $\delta > 0$. Then*

$$\log W_m = o(m) \quad \Longleftrightarrow \quad \sum_{\substack{d\mid m \\ d>m^{1-\delta} \\ \log E_d>\epsilon\varphi(d)}} \varphi(d) = o(m) \quad \text{for every } \epsilon > 0.$$

The same statement remains valid when the near-top range is narrowed to

$$d > m\exp\bigl(-\{\log\log(3m)\}^2\bigr) = m^{1-o(1)}.$$

Proof. Apply Theorem 67.1 using (440). Theorem 67.2 makes the complementary divisor range negligible. Finally use the exact equivalence $\log W_m = o(m) \Longleftrightarrow \sum_{d\mid m} \log E_d = o(m)$. □

The endpoint still forces $a_m = o(m)$, so pointwise control remains necessary on primes and fixed prime powers. What is gained is permission for large normalized blocks on orders with small $\varphi(d)/d$, provided their near-diagonal totient mass is negligible. No checked source, including the recent powerful-number count [8], proves this final gate, and no full-premise counterexample to it is known.

The Lean companion checks the exact moment, its dyadic $O(\log\log(3m))$ estimate, the general moving-threshold Markov theorem and its squared-loglog specialization. It also proves the exact-order prime-power divisibility in $\Phi_d(2)$ and hence (440). Combining these results gives a kernel-checked equivalence between the Mersenne endpoint and the remaining near-diagonal exceptional-totient estimate; that estimate itself is neither assumed nor proved.

68. Fixed-polylogarithmic co-divisors at the Mersenne endpoint

The squared-loglog localization in Corollary 67.3 can be sharpened without assuming a stronger concentration theorem. The sharpening proved here is a family of fixed windows with a quantified outer limit. The present moment argument does not show that the complement of one fixed window is negligible.

Retain $a_d = \log E_d$ and put

$$B_\epsilon(m) = \sum_{\substack{d\mid m \\ a_d>\epsilon\varphi(d)}} \varphi(d), \qquad \Lambda_m = \log\log(3m).$$

For a fixed positive integer k, define

$$B^{\mathrm{poly}}_{\epsilon,k}(m) = \sum_{\substack{d\mid m \\ \log(m/d)<k\Lambda_m \\ a_d>\epsilon\varphi(d)}} \varphi(d).$$

Writing $q = m/d$, its support is the genuinely polylogarithmic range

$$q < \exp(k\Lambda_m) = \{\log(3m)\}^k.$$

Let

$$\Delta(m) = \frac{1}{m}\sum_{d\mid m} \varphi(d)\log\frac{m}{d}.$$

The exact moment formula in Section 67, together with the elementary Chebyshev bound, supplies constants $C_0 > 0$ and m_0 such that $\Delta(m) \le C_0\Lambda_m$ for $m \ge m_0$. Finite weighted Markov therefore gives, for every fixed $k \ge 1$,

$$\sum_{\substack{d\mid m \\ k\Lambda_m\le\log(m/d)}} \varphi(d) \le \frac{C_0}{k}m. \tag{441}$$

Consequently

$$0 \le B_\epsilon(m) - B^{\mathrm{poly}}_{\epsilon,k}(m) \le \frac{C_0}{k}m. \tag{442}$$

The right side is uniform in m, but it is not $o(m)$ for one fixed k.

Theorem 68.1 (Fixed-polylogarithmic co-divisor gate)**.** *For the actual canonical Mersenne blocks,*

$$\log W_m = o(m) \quad \iff \quad \text{for every } \epsilon > 0 \text{ and every fixed } k \geq 1, \quad B^{\mathrm{poly}}_{\epsilon,k}(m) = o(m).$$

The same equivalence holds when k ranges over all fixed positive real numbers.

Proof. The cyclotomic bound $0 \leq a_d \leq (\log 3)\varphi(d)$ and the exceptional-mass criterion give

$$\log W_m = o(m) \quad \iff \quad B_\epsilon(m) = o(m) \quad \text{for every } \epsilon > 0.$$

The forward implication follows by restricting a nonnegative sum. For the reverse implication, fix $\epsilon > 0$ and a requested normalized error $\eta > 0$. First choose one integer $k > 2C_0/\eta$. This k is now fixed, so the hypothesis eventually gives $B^{\mathrm{poly}}_{\epsilon,k}(m) < \eta m/2$. Equation (442) then gives $B_\epsilon(m) < \eta m$. Letting η vary proves the required little-oh statement. Integer windows imply the real-window formulation, while the converse follows by specialization. □

The order of quantifiers in the proof is essential:

$$\forall \eta > 0\ \exists k = k(\eta)\ \forall m \geq m_0(\eta, k).$$

An abstract primorial divisor model shows that, for general totient-bounded masses, the far tail for a single fixed k need not be $o(m)$. Thus the Markov and moment estimates alone cannot replace the theorem by a one-window assertion. This model is not the actual Mersenne mass $\log E_d$ and does not refute a future Mersenne-specific one-window strengthening.

There is a useful dichotomy in the negation. If the endpoint fails, then for some fixed $\epsilon, \eta, k > 0$ and an unbounded sequence m_j,

$$\sum_{\substack{q \mid m_j \\ q < \{\log(3m_j)\}^k \\ \log E_{m_j/q} > \epsilon\varphi(m_j/q)}} \frac{\varphi(m_j/q)}{m_j} \geq \eta. \tag{443}$$

Let $\mathcal{Q}_j$ be the set of co-divisors counted in (443), and let M_j be its largest summand. After passage to a subsequence, exactly one of the following pressure regimes remains.

(1) If $M_j \geq \tau > 0$, then $\varphi(m_j/q)/m_j \leq 1/q$ forces $q \leq 1/\tau$; a further subsequence fixes $q = q_0$. With $d_j = m_j/q_0$ one obtains

$$\frac{\varphi(d_j)}{d_j} \geq q_0\tau, \qquad \frac{\log E_{d_j}}{d_j} > \epsilon q_0 \tau.$$

This is the atomic, fixed-co-divisor obstruction and still contains the prime-index problem.

(2) If $M_j \to 0$, the packet contains at least $\eta/M_j \to \infty$ exceptional co-divisors, and

$$\sum_{q \in \mathcal{Q}_j} \log E_{m_j/q} > \epsilon \eta m_j.$$

This is the diffuse multi-fibre obstruction.

The transition, extreme-order, and deep-lift decomposition of Section 69 applies inside the same fixed-polylogarithmic packet.

The prime-index test confirms that no core difficulty has been hidden: for $m = \ell$ prime, $q = 1$ lies in every window with weight $(\ell - 1)/\ell$, so the theorem still requires $\log E_\ell = o(\ell)$. The certified occurrences $1093 \mid E_{364}$ and $3511 \mid E_{1755}$ refute small universal pointwise constants but are finite and do not refute the eventual gate. Conversely, the sparse primorial model from the earlier totient audit satisfies every fixed-k gate while its pointwise normalized blocks need not converge to zero. These tests verify both directions of the quantifier boundary and produce no full-premise counterexample to the actual Mersenne endpoint.

The Lean companion proves the exact exceptional-set partition, the finite C_0/k estimate, a general uniform-in-fixed-index transfer theorem, and the displayed equivalence for the repository's actual `mersennePowerLoss`. All moment and cyclotomic estimates are reused from previously checked theorems; no distribution hypothesis enters the kernel.

69. Global near-diagonal triage of the Mersenne excess

The localized gate of Corollary 67.3 identifies the moving divisor range on which an obstruction can survive. We now combine that localization with the exact-order Wieferich decomposition. The result does not estimate the remaining mass, but it proves that a failure cannot be distributed among unspecified cyclotomic effects: one of three explicit mechanisms must carry a positive linear logarithmic mass.

For an odd prime q, put

$$d_q = \mathrm{ord}_q(2), \qquad w_q = v_q(2^{d_q} - 1),$$

and, for $d \geq 1$, let

$$\mathcal{W}_d = \{q : d_q = d,\ w_q \geq 2\}, \qquad D_d = \prod_{\substack{d_q = d \\ w_q \geq 3}} q^{w_q - 2}.$$

Fix $\delta > 0$ and set

$$Y_d = \frac{\varphi(d)^2}{\log\log(3d)}.$$

The logarithm of the canonical excess block has the exact four-layer decomposition

$$\log E_d = S_d + T_{d,\delta} + X_{d,\delta} + G_d, \tag{444}$$

where

$$S_d = \sum_{\substack{q \in \mathcal{W}_d \\ q \leq Y_d}} \log q, \qquad T_{d,\delta} = \sum_{\substack{q \in \mathcal{W}_d \\ Y_d < q \leq d^{2+\delta}}} \log q,$$

$$X_{d,\delta} = \sum_{\substack{q \in \mathcal{W}_d \\ q > d^{2+\delta}}} \log q, \qquad G_d = \log D_d.$$

Indeed, the exponent of q in E_d is $w_q - 1 = 1 + (w_q - 2)$. Moreover, LTE gives

$$v_q(2^{q-1} - 1) = v_q(2^{d_q} - 1) = w_q,$$

so $\mathcal{W}_d$ consists precisely of base-two Wieferich primes of exact order d.

Put

$$H(m) = \{\log\log(3m)\}^2, \qquad \mathcal{N}_m = \{d \mid m : d > me^{-H(m)}\}.$$

The cyclotomic cap $E_d \leq 3^{\varphi(d)}$ and the exact logarithmic-deficit moment prove

$$\sum_{\substack{d \mid m \\ d \notin \mathcal{N}_m}} \log E_d = o(m). \tag{445}$$

The uniform Brun–Titchmarsh argument for primes $q \equiv 1 \pmod{d}$ below Y_d gives $S_d = o(\varphi(d))$. Since the least member of $\mathcal{N}_m$ tends to infinity and $\sum_{d \mid m} \varphi(d) = m$, this pointwise uniform estimate sums to

$$\sum_{d \in \mathcal{N}_m} S_d = o(m). \tag{446}$$

Theorem 69.1 (Global three-arm obstruction)**.** *Suppose that, for some $\epsilon > 0$ and an unbounded sequence m_j,*

$$\sum_{d \mid m_j} \log E_d \geq \epsilon m_j.$$

After passage to a subsequence, one fixed alternative holds for every sufficiently large j:

(1) $\sum_{d \in \mathcal{N}_{m_j}} G_d \geq \epsilon m_j / 6$*;*

(2) *there are at least*
$$\frac{\epsilon m_j}{6(2+\delta)\log m_j}$$
distinct base-two Wieferich primes q such that
$$d_q \mid m_j, \quad d_q > m_j e^{-H(m_j)}, \quad Y_{d_q} < q \le d_q^{2+\delta};$$

(3) $\sum_{d\in\mathcal{N}_{m_j}} X_{d,\delta} \ge \epsilon m_j/6$. *Every prime counted in the last alternative satisfies* $\operatorname{ord}_q(2) < q^{1/(2+\delta)}$.

Proof. By (445) and (446), the far mass and the near-small mass are each at most $\epsilon m_j/4$ after deleting finitely many indices. Summing (444) therefore gives
$$\sum_{d\in\mathcal{N}_{m_j}} (G_d + T_{d,\delta} + X_{d,\delta}) \ge \frac{\epsilon m_j}{2}.$$
One of the three sums is at least $\epsilon m_j/6$; infinite pigeonhole fixes the same arm on a subsequence. In the transition case each summand is at most $(2+\delta)\log m_j$. Exact-order prime sets for different d are disjoint, so division by this cap proves the asserted cardinality. The extreme inequality $q > d_q^{2+\delta}$ gives the final order bound. $\square$

Corollary 69.2 (Three sufficient aggregate gates)**.** *For one fixed $\delta > 0$, assume*
$$\sum_{d\in\mathcal{N}_m} G_d = o(m), \qquad \#\{\text{transition primes at } m\} = o(m/\log m), \qquad \sum_{d\in\mathcal{N}_m} X_{d,\delta} = o(m).$$
Then $\log W_m = o(m)$.

Proof. The transition sum is at most $(2+\delta)\log m$ times its support size. Adding the three hypotheses to (445) and (446) yields $\sum_{d|m}\log E_d = o(m)$. The exact factorization $W_m = L_m\prod_{d|m} E_d$, with $L_m \mid m$, finishes the proof. $\square$

None of the three disjuncts can be deleted by positivity alone. On a one-point near set, take target mass one, far and small masses zero, and respectively
$$(G,T,X) = (1,0,0), \quad (0,1,0), \quad (0,0,1).$$
These are complete counterexamples to each corresponding two-disjunct ordered-ring strengthening. They are abstract mass ledgers rather than Mersenne blocks, so they do not eliminate an arithmetic arm.

The companion Lean module checks the finite near/far subtraction, the 1/6 trichotomy, the transition-cardinality lower bound, the five-arm closure inequality, and all three full-premise counterexamples. Its analytic inputs remain the proved and cited statements above; no distribution assertion is inserted as a formal axiom. Present results on primitive cyclotomic factors and generalized Wieferich congruences [22, 39, 23] do not bound any one of the three weighted moving arms. All three routes therefore remain active.

70. A threshold-free fixed-polylogarithmic Mersenne gate

Let E_d be the canonical exact-order excess block and put $a_d = \log E_d$. Previous sections prove

$$0 \le a_d \le (\log 3)\varphi(d), \qquad \log W_m = o(m) \iff A(m) := \sum_{d|m} a_d = o(m). \tag{447}$$

For $L_m = \log\log(3m)$ and a fixed positive integer k, define
$$P_k(m) = \sum_{\substack{d|m\\ \log(m/d)<kL_m}} a_d, \qquad F_k(m) = \sum_{\substack{d|m\\ kL_m\le\log(m/d)}} a_d.$$
The strict and closed inequalities give the exact partition $A(m) = P_k(m) + F_k(m)$.

Theorem 70.1 (Actual fixed-window equivalence)**.** *One has*

$$\log W_m = o(m) \quad \Longleftrightarrow \quad \text{for every fixed } k \geq 1, \quad P_k(m) = o(m).$$

Proof. The forward implication follows from $0 \leq P_k \leq A$. Conversely, the exact logarithmic-deficit moment and weighted Markov inequality provide an absolute C_0 such that, eventually and uniformly in $k \geq 1$,

$$\sum_{\substack{d \mid m \\ kL_m \leq \log(m/d)}} \varphi(d) \leq \frac{C_0}{k} m.$$

By (447), $F_k(m) \leq (\log 3) C_0 m/k$. Given $\eta > 0$, first fix k so large that this upper bound is below $\eta m/2$, and only then use the fixed-k hypothesis to make $P_k(m) < \eta m/2$. Hence $A(m) = o(m)$, and (447) finishes the proof. □

Inside this window, write the exact-order Wieferich decomposition

$$a_d = S_d + T_{d,\delta} + X_{d,\delta} + G_d,$$

where S_d contains one copy of primes $p \leq Y_d = \varphi(d)^2/\log\log(3d)$, $T_{d,\delta}$ contains one copy for $Y_d < p \leq d^{2+\delta}$, $X_{d,\delta}$ contains one copy for $p > d^{2+\delta}$, and

$$G_d = \sum_{\substack{\mathrm{ord}_p(2)=d \\ v_p(2^d-1)\geq 3}} (v_p(2^d-1)-2)\log p.$$

The proved bound $S_d = o(\varphi(d))$ and the fact that all window indices tend to infinity combine with

$$\sum_{d \mid m} \varphi(d) = m$$

to show that the localized S-mass is $o(m)$.

Theorem 70.2 (Sharp fixed-window three-arm obstruction)**.** *Fix $\delta > 0$. If $\log W_m \neq o(m)$, then there are fixed $k \geq 1$ and $\epsilon > 0$, and an unbounded sequence m_j, for which $P_k(m_j) \geq \epsilon m_j$. For every fixed $0 < \gamma < 1/3$, after passage to a subsequence one same alternative holds eventually:*

(1) *the localized sum of G_d is at least $\gamma\epsilon m_j$;*
(2) *at least*

$$\frac{\gamma \epsilon m_j}{(2+\delta)\log m_j}$$

distinct base-two Wieferich primes lie in the localized transition arm;
(3) *the localized sum of $X_{d,\delta}$ is at least $\gamma\epsilon m_j$; each such prime satisfies $\mathrm{ord}_p(2) < p^{1/(2+\delta)}$.*

In particular, one may take $\gamma = 1/4$.

Proof. The contrapositive of Theorem 70.1 fixes k and a linear lower bound along a subsequence. After subtracting the localized $o(m_j)$ small arm, the three remaining nonnegative masses total $(\epsilon - o(1))m_j$. Because $3\gamma < 1$, one carries at least $\gamma\epsilon m_j$ for all large j; infinite pigeonhole fixes the arm. In the transition range, $p \leq d^{2+\delta} \leq m_j^{2+\delta}$, so each summand is at most $(2+\delta)\log m_j$. Exact-order prime sets are disjoint, yielding the cardinality bound. The extreme-order inequality is the rearrangement of $p > d^{2+\delta}$. □

The constants are sharp at the stated algebraic level. The full quarter-budget mass ledger

$$(R, S, G, T, X) = (4, 1, 1, 1, 1)$$

refutes every uniform coefficient greater than $1/4$. The ledger $(3, 0, 1, 1, 1)$ refutes every coefficient greater than $1/3$ when the small arm is zero. These ledgers are not asserted to arise from Mersenne blocks.

Actual finite pressure comes from the sealed exhaustive scan through 10^7, independently rechecked by a segmented sieve. Its only base-two Wieferich hits are

$$(p, \mathrm{ord}_p(2), v_p(2^{\mathrm{ord}_p(2)}-1)) = (1093, 364, 2), \quad (3511, 1755, 2).$$

They refute the finite strengthenings that every localized block vanishes, or that every localized repeated prime must already be deep, transition, or extreme. Two isolated indices do not refute any little-oh assertion, and the absence of depth at most 10^7 has no asymptotic consequence.

For any fixed $\delta > 0$, closure now reduces to proving, for every fixed k, that the localized deep mass and extreme mass are $o(m)$ and that the transition support is $o(m/\log m)$. No full-premise counterexample to these three arithmetic gates is known, so all three remain active.

71. Multiplier-index compression of the localized Mersenne loss

We retain the notation of Section 70. For an odd prime p put

$$d_p = \operatorname{ord}_p(2), \qquad w_p = v_p(2^{d_p} - 1), \qquad r_p = \frac{p-1}{d_p}.$$

Lagrange's theorem gives $d_p \mid p - 1$, so r_p is a positive integer and $p = 1 + d_p r_p$. In particular, for fixed d, the multiplier r_p determines p injectively inside the exact-order fibre $d_p = d$. This elementary observation removes one entire localized mass arm.

Set

$$L_m = \log\log(3m), \qquad \Lambda_m = \log(3m) L_m^2.$$

For $d \mid m$, split the one-copy repeated-prime support as

$$U_d(m) = \sum_{\substack{d_p = d,\ w_p \geq 2 \\ p \leq d^2/\Lambda_m}} \log p,$$

$$B_d(m) = \sum_{\substack{d_p = d,\ w_p \geq 2 \\ p > d^2/\Lambda_m}} \log p,$$

and retain the deep excess

$$G_d = \sum_{\substack{d_p = d \\ w_p \geq 3}} (w_p - 2) \log p.$$

Prime by prime, the canonical loss is exactly

$$a_d = U_d(m) + B_d(m) + G_d. \tag{448}$$

Proposition 71.1 (Uniform low-multiplier estimate)**.** *For every fixed positive integer k, and all sufficiently large m,*

$$\sum_{\substack{d \mid m \\ \log(m/d) < kL_m}} U_d(m) \leq \frac{2m \log m}{\Lambda_m}(1 + kL_m) = o(m).$$

Proof. If p contributes to $U_d(m)$, then

$$dr_p = p - 1 < p \leq d^2/\Lambda_m, \qquad r_p < d/\Lambda_m.$$

The positive integral multipliers are distinct, so this fibre has at most d/Λ_m elements. Since $p \leq d^2 \leq m^2$ eventually, $U_d(m) \leq 2d \log m/\Lambda_m$. Writing $q = m/d$ and enlarging the divisor sum to all positive integers $q < (\log(3m))^k$ gives

$$\sum U_d(m) \leq \frac{2m \log m}{\Lambda_m} \sum_{q < (\log(3m))^k} \frac{1}{q} \leq \frac{2m \log m}{\Lambda_m}(1 + kL_m).$$

After division by m, the final expression is at most $2(1 + kL_m)/L_m^2$, which tends to zero. $\square$

Theorem 71.2 (Two-arm near-square-root obstruction)**.** *Suppose that the Mersenne endpoint $\log W_m = o(m)$ fails. Then there are a fixed $k \geq 1$, an $\epsilon > 0$, and an unbounded sequence m_j such that, for every fixed $0 < \gamma < 1/2$, after passage to a subsequence one same alternative holds for all large j:*

(1) *the localized deep mass satisfies*

$$\sum_{\substack{d \mid m_j \\ \log(m_j/d) < kL_{m_j}}} G_d \ge \gamma\epsilon m_j;$$

(2) *the localized B-mass satisfies*

$$\sum_{\substack{d \mid m_j \\ \log(m_j/d) < kL_{m_j}}} B_d(m_j) \ge \gamma\epsilon m_j,$$

and every prime in this sum obeys

$$\operatorname{ord}_p(2) < \sqrt{p\Lambda_{m_j}}.$$

Proof. The fixed-window equivalence supplies k, ϵ and m_j for which the localized sum of a_d is at least ϵm_j. Sum (448) and apply Proposition 71.1. The two unresolved arms then total at least $(\epsilon - o(1))m_j$. If both were smaller than $\gamma\epsilon m_j$, their sum would be smaller than $2\gamma\epsilon m_j$, contradicting $2\gamma < 1$ for large j. Infinite pigeonhole fixes an arm on a subsequence. Finally, $p > d^2/\Lambda_{m_j}$ implies $d < \sqrt{p\Lambda_{m_j}}$. ☐

Thus this route closes if, for every fixed k, both localized sums of G_d and $B_d(m)$ are $o(m)$. The coefficient $1/2$ is sharp at the exact nonnegative-ledger level: $(R, U, G, B) = (2, 0, 1, 1)$ satisfies every displayed mass premise and defeats each coefficient greater than $1/2$. This tuple is an abstract full-premise counterexample to the stronger ledger conclusion; it is not a Mersenne block.

The exact-order lower bounds of Erdős–Murty [18] are global and unweighted, and leave a genuine $p^{1/2-o(1)}$ to $p^{1/2+o(1)}$ band. The valuation dictionary of Murty–Séguin [20] and the cyclotomic description in Pomerance [19] justify the block language, but do not give the localized Wieferich-weighted estimate needed here. The conditional alternatives in Fellini–Murty [39] likewise do not supply either remaining unconditional little-oh statement. Hence both arms remain active.

The formal companion proves the integral multiplier representation and injectivity, the finite cardinality and logarithmic-mass bounds, the exact two-arm subtraction, its fixed-window finite-sum form, and the sharp ledger counterexample. The harmonic asymptotic and the realization of the three mass functions remain explicit paper mathematics; no distribution or squarefreeness hypothesis is installed as a Lean axiom.

72. Balanced Mersenne multiplier localization

We refine Section 71 in two directions. The one-copy multiplier cutoff can be moved closer to its critical harmonic scale, and a published pointwise p-adic logarithm bound removes a growing low-multiplier interval from the deep-valuation arm. No density conjecture is used.

For an odd prime p, retain

$$d_p = \operatorname{ord}_p(2), \qquad w_p = v_p(2^{d_p} - 1), \qquad r_p = \frac{p-1}{d_p},$$

and let $a_d = \sum_{d_p = d}(w_p - 1)\log p$. Fix a positive integer k and a real number $\eta > 0$. Put

$$L_m = \log\log(3m), \qquad F_{m,\eta} = \log(3m)L_m^{1+\eta}, \qquad H_{m,\eta} = \left\lfloor \sqrt{\frac{\log(3m)}{L_m^{1+\eta}}} \right\rfloor. \tag{449}$$

Split the one-copy repeated support into

$$U_d^{(\eta)}(m) = \sum_{\substack{d_p=d,\ w_p\geq 2 \\ p\leq d^2/F_{m,\eta}}} \log p,$$

$$B_d^{(\eta)}(m) = \sum_{\substack{d_p=d,\ w_p\geq 2 \\ p>d^2/F_{m,\eta}}} \log p,$$

and split the deep excess by multiplier:

$$V_d^{(\eta)}(m) = \sum_{\substack{d_p=d,\ w_p\geq 3 \\ r_p<H_{m,\eta}}} (w_p-2)\log p,$$

$$G_d^{(\eta)}(m) = \sum_{\substack{d_p=d,\ w_p\geq 3 \\ r_p\geq H_{m,\eta}}} (w_p-2)\log p.$$

The four terms give the exact identity

$$a_d = U_d^{(\eta)}(m) + B_d^{(\eta)}(m) + V_d^{(\eta)}(m) + G_d^{(\eta)}(m). \tag{450}$$

Proposition 72.1 (Adaptive one-copy cutoff)**.** *For fixed $k\geq 1$ and $\eta>0$, eventually in m,*

$$\sum_{\substack{d\mid m \\ \log(m/d)<kL_m}} U_d^{(\eta)}(m) \leq \frac{2m\log m}{F_{m,\eta}}(1+kL_m) = O_{k,\eta}\left(\frac{m}{L_m^{\eta}}\right) = o(m).$$

Proof. Write $q=m/d$. If p occurs in $U_d^{(\eta)}(m)$, multiplier injectivity and $p=1+dr_p$ give

$$r_p < \frac{p}{d} \leq \frac{d}{F_{m,\eta}}.$$

There are therefore at most $d/F_{m,\eta}$ such primes. Eventually $F_{m,\eta}\geq 1$, so $p\leq d^2\leq m^2$ and

$$U_d^{(\eta)}(m) \leq \frac{2d\log m}{F_{m,\eta}}.$$

The window is $q<(\log(3m))^k$. Hence

$$\sum U_d^{(\eta)}(m) \leq \frac{2m\log m}{F_{m,\eta}} \sum_{\substack{q\mid m \\ q<(\log(3m))^k}} \frac{1}{q} \leq \frac{2m\log m}{F_{m,\eta}}(1+kL_m).$$

Dividing by m gives $O_k((1+L_m)/L_m^{1+\eta}) = O_k(L_m^{-\eta})$. ☐

Thus the exponent two on L_m in the previous cutoff can be replaced by $1+\eta$ for every fixed positive η. The endpoint $\eta=0$ is not claimed: the same calculation then gives only a bounded normalized error. Relative to the old cutoff this is strictly sharper for $0<\eta<1$, equal at $\eta=1$, and a valid but weaker localization for $\eta>1$. That limitation is not an arithmetic counterexample, so the critical cutoff remains an active route.

The deep estimate uses Yamada's unconditional theorem [85]. With

$$C_Y = 283\log 3\log 6,$$

his Theorem 1.2, equation (7), states for every prime p that

$$v_p(2^{p-1}-1) \leq \left\lfloor C_Y\frac{p-1}{(\log p)^2}\right\rfloor + 4. \tag{451}$$

Lemma 72.2 (Exact-order Yamada envelope)**.** *For an odd prime p, with $d=d_p$, $r=r_p$, and $w=w_p\geq 2$,*

$$(w-2)\log p \leq C_Y\frac{dr}{\log p} + 2\log p.$$

If $d > 1$ and $r < H$, then

$$(w-2)\log p \le C_Y \frac{dr}{\log d} + 2\log(1+dH). \tag{452}$$

Proof. Since $p-1 = dr$ and $0 < r < p$, one has $p \nmid r$. LTE gives

$$v_p(2^{p-1}-1) = v_p((2^d)^r - 1) = v_p(2^d-1) + v_p(r) = w.$$

Substitute this equality into (451), discard the floor, subtract two, and multiply by $\log p$. The final assertion uses $p > d$ and $p = 1 + dr < 1 + dH$. □

Lemma 72.3 (Triangular multiplier energy)**.** *If positive integral labels $r(x) < H$ are injective on a finite set S, then*

$$|S| \le H-1, \qquad \sum_{x\in S} r(x) \le \frac{H(H-1)}{2}.$$

Consequently, for $d > 1$ and $H \ge 2$,

$$\sum_{\substack{d_p=d,\ w_p\ge 3\\ r_p<H}} (w_p-2)\log p \le \frac{C_Y d}{\log d}\frac{H(H-1)}{2} + 2(H-1)\log(1+dH). \tag{453}$$

Proof. The image of the labels lies in $\{1,\dots,H-1\}$, proving both finite bounds. Apply (452) prime by prime and use injectivity of $p \mapsto r_p$ in the exact-order fibre. □

Theorem 72.4 (Low-multiplier deep mass is negligible)**.** *For every fixed $k \ge 1$ and $\eta > 0$,*

$$\sum_{\substack{d|m\\ \log(m/d)<kL_m}} V_d^{(\eta)}(m) = o(m).$$

More precisely, if $Q_m = (\log(3m))^k$ and $H = H_{m,\eta}$, then eventually

$$\begin{aligned}\sum V_d^{(\eta)}(m) \le{}& \frac{C_Y m H(H-1)}{2\log(m/Q_m)}(1+kL_m)\\ &+ 2HQ_m\log(1+mH).\end{aligned} \tag{454}$$

Proof. In the window $q = m/d < Q_m$, hence $d > m/Q_m$. Sum (453). The coefficient of the triangular term is bounded using

$$\sum_{\substack{q|m\\ q<Q_m}} d = m\sum_{\substack{q|m\\ q<Q_m}} \frac{1}{q} \le m(1+kL_m),$$

and $1/\log d \le 1/\log(m/Q_m)$. There are fewer than Q_m integral co-divisors, giving the second line of (454). Finally

$$H^2 \le \frac{\log(3m)}{L_m^{1+\eta}}, \qquad \log(m/Q_m) = \log m - kL_m \sim \log m.$$

The first line divided by m is $O_k(L_m^{-\eta})$, and the second line is polylogarithmic in m, hence $o(m)$. □

The same slack L_m^η therefore balances the normalized losses in Proposition 72.1 and Theorem 72.4.

Theorem 72.5 (Balanced two-arm obstruction)**.** *Suppose the Mersenne endpoint $\log W_m = o(m)$ fails, and fix $\eta > 0$. Then there are fixed $k \ge 1$, $\epsilon > 0$, and an unbounded sequence m_j such that for every fixed $0 < \gamma < 1/2$, after passage to a subsequence one same alternative holds for all large j:*

(1)
$$\sum_{\substack{d\mid m_j\\ \log(m_j/d)<kL_{m_j}}} G_d^{(\eta)}(m_j) \geq \gamma\epsilon m_j,$$
and every counted prime has $r_p \geq H_{m_j,\eta}$ *and* $p \geq 1 + d_p H_{m_j,\eta}$*;*

(2)
$$\sum_{\substack{d\mid m_j\\ \log(m_j/d)<kL_{m_j}}} B_d^{(\eta)}(m_j) \geq \gamma\epsilon m_j,$$
and every counted prime has
$$d_p < \sqrt{p\log(3m_j)L_{m_j}^{1+\eta}}.$$

Proof. The fixed-window equivalence supplies a localized a_d-mass at least ϵm_j. Sum (450) and use the two little-oh estimates above. The $G^{(\eta)}$ and $B^{(\eta)}$ sums total $(\epsilon - o(1))m_j$, so one carries every fixed share below one half on a subsequence. The displayed prime restrictions are respectively the definition of $G^{(\eta)}$ and the rearrangement $p > d_p^2/F_{m_j,\eta}$. □

Thus it now suffices to prove that the two surviving localized sums are $o(m)$ for one fixed $\eta > 0$ and every fixed k. The first concerns deep base-two lifts only after their order multiplier tends to infinity; the second lies within a factor $\sqrt{\log(3m)L_m^{1+\eta}}$ of the square-root order threshold. Neither estimate follows from the global Erdős–Murty theorem [18] or the Murty–Séguin valuation dictionary [20].

The exact scope of newer literature is also important. Li–Zhao [86] fix one prime ideal and prove eventual stabilization of successive prime-power order growth; their threshold depends on that prime ideal and therefore does not bound initial depths uniformly as p varies. Large-sieve results for Fermat quotients such as Shparlinski [87] average over intervals of the base variable and do not specialize to fixed base two with the simultaneous exact-order, divisor-window, and square-divisibility restrictions. The conditional non-Wieferich conclusions of Fellini–Murty [39] likewise do not close either surviving arm.

There are two exact sharpness boundaries. For every integer $H \geq 1$, the complete positive packet $S_H = \{1, \dots, H-1\}$ satisfies
$$\sum_{r\in S_H} r = \frac{H(H-1)}{2}.$$
For any fixed C, choosing an integer $H > 2C + 1$ gives $\sum_{r\in S_H} r > CH$; hence injectivity plus a pointwise linear envelope does not imply a uniform $O(H)$ energy bound. The member $H = 4$ already refutes the coefficient-one inequality. At the two-arm level, $(R, U, V, G, B) = (2, 0, 0, 1, 1)$ defeats every coefficient greater than one half. Both are complete abstract counterexamples to the respective stronger finite inferences. In actual arithmetic,
$$\mathrm{ord}_{3511}(2) = 1755, \qquad v_{3511}(2^{1755} - 1) = 2, \qquad \frac{3511 - 1}{1755} = 2,$$
so the universal assertion that every repeated exact-order prime has multiplier at least three is also false with all premises present. This finite depth-two row does not refute either asymptotic surviving arm.

The companion Lean module formalizes triangular multiplier energy, the positive-label capacity, the exact LTE transport from $p-1$ to the canonical exact-order valuation, the affine weighted envelope and its realization on the actual exact-order fibre, the two-controlled/two-surviving finite ledger, the complete triangular sharpness family and the two-arm sharpness example, and the full 3511 order, square-divisibility, and multiplier certificate. Yamada's theorem and the real asymptotic summation above remain cited paper mathematics; no valuation, density, or abc assertion is introduced as a Lean axiom.

73. Critical slow-slack compression in the Mersenne route

The balanced multiplier argument of Section 72 can be pushed below every fixed positive power of $\log\log m$. For an odd prime p, retain

$$d_p = \mathrm{ord}_p(2), \qquad w_p = v_p(2^{d_p}-1), \qquad r_p = \frac{p-1}{d_p}, \qquad L_m = \log\log(3m).$$

Call a positive function σ *slow-admissible* when

$$\sigma(m) \longrightarrow \infty, \qquad \frac{L_m\sigma(m)}{\log(3m)} \longrightarrow 0. \tag{455}$$

Set

$$F_{m,\sigma} = \log(3m)L_m\sigma(m), \qquad H_{m,\sigma} = \left\lfloor \sqrt{\frac{\log(3m)}{L_m\sigma(m)}} \right\rfloor. \tag{456}$$

The latter tends to infinity. For $d \mid m$, split the repeated one-copy mass and the deep excess as

$$U_d^{(\sigma)}(m) = \sum_{\substack{d_p=d,\ w_p\ge 2\\ p\le d^2/F_{m,\sigma}}} \log p, \qquad B_d^{(\sigma)}(m) = \sum_{\substack{d_p=d,\ w_p\ge 2\\ p> d^2/F_{m,\sigma}}} \log p,$$
$$V_d^{(\sigma)}(m) = \sum_{\substack{d_p=d,\ w_p\ge 3\\ r_p< H_{m,\sigma}}} (w_p-2)\log p, \qquad G_d^{(\sigma)}(m) = \sum_{\substack{d_p=d,\ w_p\ge 3\\ r_p\ge H_{m,\sigma}}} (w_p-2)\log p.$$

These four nonnegative terms still sum exactly to the block mass a_d.

Theorem 73.1 (Critical slow-slack removal)**.** *Let σ satisfy (455). For every fixed positive integer k,*

$$\sum_{\substack{d\mid m\\ \log(m/d)<kL_m}} U_d^{(\sigma)}(m) = o(m), \qquad \sum_{\substack{d\mid m\\ \log(m/d)<kL_m}} V_d^{(\sigma)}(m) = o(m). \tag{457}$$

If also $\sigma(m) = o(L_m^{\eta})$ for every fixed $\eta>0$, both surviving cutoff regions are eventually contained in the corresponding fixed-η regions of Section 72.

Proof. Write $q = m/d$ and $Q_m = (\log(3m))^k$. In the displayed divisor window, $q < Q_m$. If p occurs in $U_d^{(\sigma)}$, then

$$0 < r_p = \frac{p-1}{d} < \frac{p}{d} \le \frac{d}{F_{m,\sigma}}.$$

The integral multipliers are distinct on a fixed exact-order fibre. Hence there are at most $d/F_{m,\sigma}$ such primes, each at most $d^2 \le m^2$, and

$$\sum U_d^{(\sigma)}(m) \le \frac{2m\log m}{F_{m,\sigma}} \sum_{q<Q_m} \frac{1}{q} \le \frac{2m\log m}{F_{m,\sigma}}(1+kL_m) = o(m). \tag{458}$$

The last step uses only $\sigma(m)\to\infty$.

For the deep term, Yamada's pointwise theorem [85] and exact-order LTE give, for $d>1$ and $H\ge 2$,

$$\sum_{\substack{d_p=d,\ w_p\ge 3\\ r_p<H}} (w_p-2)\log p \le \frac{C_Y d}{\log d}\frac{H(H-1)}{2} + 2(H-1)\log(1+dH), \qquad C_Y = 283\log 3\log 6.$$

Summing with $H = H_{m,\sigma}$ and $d > m/Q_m$ gives

$$\begin{aligned}\sum V_d^{(\sigma)}(m) \le{}& \frac{C_Y mH(H-1)}{2\log(m/Q_m)}(1+kL_m)\\ &+ 2HQ_m\log(1+mH).\end{aligned}$$

Since $H^2 \le \log(3m)/(L_m\sigma(m))$, the first line divided by m is $O_k(\sigma(m)^{-1})$; the second is polylogarithmic and hence $o(m)$.

Finally,

$$\frac{F_{m,\sigma}}{F_{m,\eta}} = \frac{\sigma(m)}{L_m^\eta} \longrightarrow 0, \qquad \frac{\sqrt{\log(3m)/(L_m\sigma(m))}}{\sqrt{\log(3m)/L_m^{1+\eta}}} = \sqrt{\frac{L_m^\eta}{\sigma(m)}} \longrightarrow \infty.$$

Thus the near-square survivor threshold rises and the high-multiplier threshold rises in the required directions. Taking floors preserves the second strict inequality eventually because both raw scales diverge. □

The explicit choice

$$\sigma_*(m) = \log(3 + L_m) \tag{459}$$

satisfies every hypothesis. It replaces every fixed L_m^η loss by one further iterated logarithm. The cutoff regions are strictly nested as real intervals; for a particular finite m the actual prime sets may coincide if the removed band happens to contain no prime.

Theorem 73.2 (Critical two-arm alternative)**.** *If the Mersenne endpoint*

$$\log \frac{2^m - 1}{\mathrm{rad}(2^m - 1)} = o(m)$$

fails, then for any slow-admissible σ there are fixed $k \ge 1$, $\epsilon > 0$, and an unbounded sequence m_j such that, after passage to a subsequence, one of the following carries at least $\gamma\epsilon m_j$ for every fixed $0 < \gamma < 1/2$:

(1) *the deep excess with $r_p \ge H_{m_j,\sigma}$;*
(2) *the repeated one-copy mass with*

$$d_p < \sqrt{p \log(3m_j) L_{m_j}\sigma(m_j)}.$$

The same alternative occurs for all sufficiently large indices of the chosen subsequence.

Proof. The fixed-window equivalence supplies k, ϵ and a localized total block mass at least ϵm_j. Subtract the two $o(m_j)$ terms in (457). The G and B masses then total $(\epsilon - o(1))m_j$. If both were below $\gamma\epsilon m_j$, their sum would be below $2\gamma\epsilon m_j$, a contradiction for large j. Infinite pigeonhole fixes one arm on a subsequence. The order bound is the rearrangement $p > d_p^2/F_{m_j,\sigma}$. □

Exact order also gives a useful filter that is independent of the asymptotic argument.

Proposition 73.3 (Euler character of the exact-order multiplier)**.** *For an odd prime p, if $d = \mathrm{ord}_p(2)$ and $p - 1 = dr$, then*

$$\left(\frac{2}{p}\right) = (-1)^r.$$

The allowed residue classes modulo eight are

$d \bmod 8$	$r \bmod 8$
0	$0, 2, 4, 6$
1	$0, 6$
2	$0, 1, 4, 5$
3	$0, 2$
4	$0, 1, 2, 3, 4, 5, 6, 7$
5	$0, 6$
6	$0, 3, 4, 7$
7	$0, 2$

Proof. If d is odd, the even number $p - 1 = dr$ forces r even, and Euler's half-power is one. If d is even, exactness implies $2^{d/2} \not\equiv 1 \pmod{p}$, while its square is one; hence it is -1 and $2^{(p-1)/2} \equiv (-1)^r$. Euler's criterion gives the identity. Combining it with the supplementary law for $(2/p)$ and $p \equiv 1 + dr \pmod{8}$ yields the table by the 64 residue checks. □

The exact arithmetic row

$$p = 1093, \qquad d_p = 364, \qquad w_p = 2, \qquad r_p = 3$$

has $1093^2 \mid 2^{364} - 1$ and $1093^3 \nmid 2^{364} - 1$. It is therefore a full-premise counterexample to the proposed assertion that every repeated base-two exact-order prime has even multiplier. That assertion is retired. It lies on the unrestricted $d \equiv 4 \pmod 8$ row and does not refute the character theorem or either slow-slack estimate. A deterministic scan of all primes through 10^5 recovers only the known depth-two rows 1093 and 3511; this finite no-hit statement has no asymptotic consequence.

There is a separate abstract endpoint boundary. Labels $r = 1, \ldots, H-1$ with weights $x_r = r$ satisfy positivity, injectivity, the cutoff $r < H$, and the pointwise affine envelope, but have total $H(H-1)/2$. With co-divisors $q \le Q$, weights $x_{q,r} = r/q$ attain one half of $H(H-1)\sum_{q \le Q} 1/q$. This refutes the inference of a subquadratic energy bound from those abstract premises alone. It is not an arithmetic prime packet, so the critical arithmetic target $\sigma \equiv 1$ remains active. Current order and Wieferich literature, including the exact valuation dictionary of Murty–Séguin [20], supplies no localized weighted estimate that closes it.

The Lean companion checks support monotonicity, nonnegative survivor-mass monotonicity, the exact cutoff balance, the finite two-arm ledger, the Euler half-power identity on actual exact-order fibres, and the full 1093 certificate. The real asymptotic estimates and Yamada's theorem remain paper mathematics with explicit citations; no density statement or *abc* consequence is introduced as a Lean axiom.

74. Exact-order coupling at the Mersenne endpoint

We return to the fixed polylogarithmic co-divisor window. Put

$$A_m = \log(3m), \qquad L_m = \log A_m, \qquad Q_m = A_m^k,$$

where $k > 0$ is fixed, and let

$$F_m = A_m L_m, \qquad H_m = \left\lfloor \sqrt{A_m/L_m} \right\rfloor.$$

For a prime p of exact base-two order d_p, write

$$w_p = v_p(2^{d_p} - 1), \qquad r_p = \frac{p-1}{d_p}.$$

For $d \mid m$ with $q = m/d < Q_m$, split the repeated-prime mass as

$$\begin{aligned} U_d &= \sum_{\substack{d_p = d,\ w_p \ge 2 \\ p \le d^2/F_m}} \log p, & B_d &= \sum_{\substack{d_p = d,\ w_p \ge 2 \\ p > d^2/F_m}} \log p, \\ V_d &= \sum_{\substack{d_p = d,\ w_p \ge 3 \\ r_p < H_m}} (w_p - 2)\log p, & G_d &= \sum_{\substack{d_p = d,\ w_p \ge 3 \\ r_p \ge H_m}} (w_p - 2)\log p. \end{aligned} \tag{74.1}$$

Their sum is exactly the logarithmic repeated-prime mass of the canonical dth exact-order block

$$E_d = \prod_{\mathrm{ord}_p(2) = d} p^{v_p(2^d - 1) - 1}.$$

It need not equal the full radical loss of $2^d - 1$, because that integer can also contain primes whose exact order properly divides d. The earlier independent multiplier estimate reached only $O(m)$ when the slack parameter was set to one. Summing primes with their congruence weights removes that endpoint loss.

Theorem 74.1 (The low one-copy arm at $\sigma = 1$)**.** *For every fixed $k > 0$,*

$$\sum_{\substack{d \mid m \\ m/d < Q_m}} U_d = o(m).$$

Proof. For one divisor d, put $X = d^2/F_m$. Every prime in U_d is $1 \pmod{d}$, hence

$$U_d \le \vartheta(X; d, 1) = \sum_{\substack{p \le X \\ p \equiv 1 \pmod{d}}} \log p.$$

The window gives $d > m/Q_m$. Uniformly there, $d/F_m \to \infty$ and $F_m^2 < d$ for sufficiently large m; the latter is equivalent to $d < X^{2/3}$. The weighted Brun–Titchmarsh estimate [20, Theorem 2.4], with the fixed exponent $2/3$, gives

$$\vartheta(X; d, 1) \le \frac{2X \log X}{\varphi(d) \log(X/d)}.$$

Moreover $\log X \le 2 \log d$ and, uniformly in the window, $\log(X/d) = \log(d/F_m) \ge \frac{1}{2} \log d$. Consequently

$$U_d \ll \frac{d^2}{F_m \varphi(d)}.$$

The uniform estimate $d/\varphi(d) \ll \log\log(3d) \ll L_m$ yields $U_d \ll d/A_m$. On writing $d = m/q$ and enlarging the divisor sum to all positive integers $q < Q_m$, we obtain

$$\sum_{\substack{d \mid m \\ m/d < Q_m}} U_d \ll \frac{m}{A_m} \sum_{q < Q_m} \frac{1}{q} \ll_k m \frac{L_m}{A_m} = o(m).$$

□

The remaining low-multiplier deep arm must retain the common-index correlation. Define $\mathcal{S}_k(m)$ to consist of triples (q, r, p) such that

$$q \mid m, \quad q < Q_m, \quad d = m/q = \operatorname{ord}_p(2), \quad w_p \ge 3, \quad p = 1 + dr, \quad 1 \le r < H_m,$$

and put

$$E_k(m) = \sum_{(q,r,p) \in \mathcal{S}_k(m)} \frac{r}{q}, \qquad N_k(m) = |\mathcal{S}_k(m)|.$$

Proposition 74.2 (Cross-fibre injection and first moment)**.** *Distinct rows of $\mathcal{S}_k(m)$ have distinct rational values r/q, and*

(74.2.1) $$E_k(m) = \frac{1}{m} \sum_{(q,r,p) \in \mathcal{S}_k(m)} (p - 1).$$

Proof. If $r_1/q_1 = r_2/q_2$, then $d_1 r_1 = (m/q_1) r_1 = (m/q_2) r_2 = d_2 r_2$. Thus $p_1 = p_2$. The order of two modulo a fixed prime is unique, so $d_1 = d_2$, whence $q_1 = q_2$ and $r_1 = r_2$. For each row, $r/q = dr/(dq) = (p-1)/m$; summation proves the identity. □

Using Yamada's published pointwise valuation estimate in the form

(74.2.2) $$(w_p - 2) \log p \le C_Y \frac{d_p r_p}{\log p} + 2 \log p,$$

one obtains the following correlation-preserving transfer.

Theorem 74.3 (Farey-energy reduction)**.** *For fixed k and all sufficiently large m,*

(74.3.1) $$\sum_{\substack{d \mid m \\ m/d < Q_m}} V_d \le \frac{C_Y m}{\log(m/Q_m)} E_k(m) + 2N_k(m) \log(1 + mH_m),$$

where $N_k(m) \le H_m Q_m$ and the second term is $o(m)$. Hence

(74.3.2) $$E_k(m) = o(\log m)$$

is sufficient for the entire low-multiplier deep arm to be $o(m)$. Equivalently, the missing condition is

$$\sum_{(q,r,p) \in \mathcal{S}_k(m)} (p - 1) = o(m \log m).$$

Proof. On a row, $d_p r_p = m(r_p/q)$, $\log p \ge \log d_p \ge \log(m/Q_m)$, and $p = 1 + d_p r_p \le 1 + mH_m$. Substitution into the pointwise estimate and summation gives the displayed bound. There are fewer than Q_m possible q and fewer than H_m possible r, and the preceding injection forbids duplicate rows. Therefore

$$N_k(m) \le H_m Q_m, \qquad H_m Q_m \log(1 + mH_m) = O_k(A_m^{k+3/2} L_m^{-1/2}) = o(m).$$

Finally $\log(m/Q_m) \sim \log m$, which proves the implication. □

The condition in Theorem 74.3 is not inferred from distinct slopes alone. Indeed the reduced Farey box $\{(q,r) : q, r \le N, (q,r) = 1\}$ has distinct slopes but total energy asymptotic to $(3/\pi^2)N^2 \log N$. This pressure example omits the common divisibility, primality, exact-order, and depth-three premises, so it refutes only that abstract inference and leaves the arithmetic route active.

There is also an exact prime-power interpretation. If $d = \mathrm{ord}_p(2)$ and $w = v_p(2^d - 1)$, then for every $j \ge 1$,

$$\mathrm{ord}_{p^j}(2) = d \iff j \le w. \tag{74.3.3}$$

One direction follows from $p^j \mid 2^d - 1$; in the other, the order modulo p^j divides d, while reduction modulo p makes d divide it. Thus $B + G$ is exactly the logarithmic mass of the selected stable exact-order prime-power layers: layer two above the size cutoff, together with layers $j \ge 3$ above the multiplier cutoff. The elementary implication

$$dH_m \le d^2/F_m, \quad p = 1 + dr > d^2/F_m \quad \Longrightarrow \quad r \ge H_m$$

holds for every sufficiently large m in the fixed window. Hence both surviving arms eventually lie on a common high-multiplier support.

Two complete arithmetic witnesses delimit stronger pointwise claims. For $p = 1093$, $d = 364$, $q = 10^{12}$, $m = dq$, and $k = 8$, one has

$$\Phi_{364}(2) = 1093^2 \cdot 4733 \cdot 8861085190774909 \cdot 556338525912325157.$$

All four factors have exact order 364, only 1093 has depth two, and the endpoint inequalities put its mass in B_{364} with $U_{364} = V_{364} = G_{364} = 0$. This refutes $B = 0$, every universal fibrewise inequality $B \le CG$, and the assertion that every repeated high-multiplier row is deep. For $p = 3511$, $d = 1755$, $q = 10^{71}$, and $k = 32$, the row lies in B but has $r = 2 < H_m = 5$. It refutes a finite pointwise implication $B \Rightarrow r \ge H_m$, while the sufficiently-large window theorem remains intact.

An exhaustive sieve of all 50,847,534 primes through 10^9 finds only the two repeated-order carriers 1093 and 3511, both of depth two. This is a finite certificate and has no asymptotic consequence. The exact active targets are still

$$E_k(m) = o(\log m), \qquad \sum_{(d,p,j,q) \in \mathcal{C}_k(m)} \log p = o(m),$$

where $\mathcal{C}_k(m)$ is the stable high-multiplier layer packet described above. Neither target has a counterexample or a proof at this checkpoint.

75. Denominator entropy at the Mersenne Farey endpoint

We continue with the exact-order endpoint support

$$\mathcal{S}_k(m) = \left\{ (q,r,p) : \begin{array}{c} q \mid m, \quad q < A_m^k, \quad d = m/q, \\ \mathrm{ord}_p(2) = d, \quad p = 1 + dr, \\ 1 \le r < H_m, \quad v_p(2^d - 1) \ge 3 \end{array} \right\},$$

where

$$A_m = \log(3m), \qquad L_m = \log A_m, \qquad H_m = \left\lfloor \sqrt{A_m/L_m} \right\rfloor.$$

The surviving low-multiplier energy is

$$E_k(m) = \sum_{(q,r,p)\in\mathcal{S}_k(m)} \frac{r}{q}.$$

The preceding reduction needs $E_k(m) = o(\log m)$. We do not prove that estimate here. We instead show that any failure forces a polynomial swarm of distinct base-two super-Wieferich primes, improving the direct cardinality reduction by a fixed power of $\log m$.

Define

$$W_3(x) = \#\{p \le x : p \text{ prime and } p^3 \mid 2^{p-1} - 1\}.$$

For $1 \le T < A_m^k$, let

$$\Delta_3(m;T) = \#\left\{p : \ p \text{ prime}, \quad \frac{m}{A_m^k} < p < 1 + \frac{mH_m}{T}, \quad p^3 \mid 2^{p-1} - 1\right\}.$$

Proposition 75.1 (Depth transport)**.** *Every row of $\mathcal{S}_k(m)$ is a base-two super-Wieferich prime, and the map from rows to their primes is injective. Moreover $p \le 1 + mH_m$.*

Proof. For a row, $p - 1 = dr$ and $p^3 \mid 2^d - 1$. Hence

$$2^d - 1 \mid 2^{dr} - 1 = 2^{p-1} - 1.$$

If two rows carry the same prime, uniqueness of the exact order gives the same d, after which $q = m/d$ and $r = (p-1)/d$ agree. Finally $p = 1 + (m/q)r \le 1 + mH_m$. □

For an integer $1 \le T < A_m^k$, let

$$N_{>T}(m) = \#\{(q, r, p) \in \mathcal{S}_k(m) : q > T\}, \qquad \mathcal{H}_T = \sum_{q\le T} \frac{1}{q}.$$

Theorem 75.2 (Denominator-entropy inequality)**.** *One has*

$$E_k(m) \le \frac{H_m(H_m - 1)}{2}\mathcal{H}_T + \frac{H_m}{T}N_{>T}(m).$$

Proof. For fixed $q \le T$, the pair (q, r) determines $p = 1 + (m/q)r$, so the fibre contains each $1 \le r < H_m$ at most once. Its energy is at most $H_m(H_m - 1)/(2q)$. If $q > T$, each row has $r/q < H_m/T$. Summing the two estimates proves the assertion. □

Theorem 75.3 (Super-Wieferich swarm forced by endpoint failure)**.** *Fix $k > 0$. If, for some $\varepsilon > 0$ and infinitely many m,*

$$E_k(m) \ge \varepsilon A_m,$$

then, with

$$\eta = \min\{\varepsilon/4, k/2\} > 0,$$

one has along that sequence

$$\Delta_3(m; \lfloor A_m^\eta \rfloor) \gg_\varepsilon A_m^{1/2+\eta} L_m^{1/2},$$

and therefore

$$W_3(1 + mH_m) \gg_\varepsilon A_m^{1/2+\eta} L_m^{1/2}.$$

Proof. Set $T = \lfloor A_m^\eta \rfloor$. The prefix in Theorem 75.2 is at most

$$\frac{A_m}{2L_m}(1 + \log T) = \frac{\eta}{2}A_m + o(A_m) \le \frac{\varepsilon}{8}A_m + o(A_m).$$

It is therefore at most $\varepsilon A_m/4$ for all sufficiently large m. The tail must contribute at least $3\varepsilon A_m/4$, whence

$$N_{>T}(m) \ge \frac{3\varepsilon}{4}\frac{A_m T}{H_m} \gg_\varepsilon A_m^{1/2+\eta} L_m^{1/2}.$$

For every tail row, $q > A_m^\eta$ and $q < A_m^k$ give

$$\frac{m}{A_m^k} < p < 1 + \frac{mH_m}{A_m^\eta};$$

depth transport and injectivity therefore give the localized assertion and then the global one. □

Corollary 75.4 (Critical counting exponent one half)**.** *If*

$$\limsup_{x\to\infty} \frac{\log \max\{1, W_3(x)\}}{\log\log x} \le \frac{1}{2},$$

then $E_k(m) = o(\log m)$ *for every fixed* $k > 0$*.*

Proof. A failure would give Theorem 75.3 for a fixed $\eta > 0$. Since

$$\log\log(1 + mH_m) = L_m + o(1),$$

the resulting lower bound makes the displayed limsup at least $1/2 + \eta$, a contradiction. □

The hypothesis of Corollary 75.4 is open. It is weaker than finiteness of super-Wieferich primes and also weaker than the direct sufficient bound $W_3(x) = o(\sqrt{\log x \log\log x})$. Current multiplicative-order, Brun–Titchmarsh, and Fermat-quotient large-sieve results [18, 20, 85, 87] do not provide it for the fixed base 2. The recent conditional results [39, 86] likewise do not upper-bound this packet unconditionally.

There is a sharp combinatorial warning. Let $m_n = \mathrm{lcm}(1, \dots, n)$ and take all reduced pairs

$$q \le n/2, \qquad 1 \le r < H_{m_n}, \qquad (q, r) = 1.$$

Then $q \mid m_n$, the integers $1 + (m_n/q)r$ exist, and the slopes are distinct. Standard totient summation gives

$$\sum \frac{r}{q} = \left(\frac{3}{\pi^2} + o(1)\right) A_{m_n}.$$

Thus common divisibility and Farey injectivity alone do not give a little-oh. These abstract labels need not be prime, have the prescribed exact order, or have depth three, so this is not a counterexample to the Mersenne route.

The exact computation bundle constructs these common-index models through $n = 10000$ and obtains energy ratio $0.2961553070\ldots$, while checking the denominator inequality with rational arithmetic. It also replays the exhaustive scan of all 50,847,534 primes through 10^9: the only base-two Wieferich primes are 1093 and 3511, and both have depth exactly two. The published Dorais–Klyve search gives the corresponding ordinary Wieferich exclusion through 6.7×10^{15}. These are finite boundaries, not asymptotic evidence inserted into the proof.

Lean formalizes the finite prefix and tail inequalities, the resulting lower bound for the number of tail rows, depth transport, and the two exact depth-two witnesses. Its theorem-level dependency union is only `Classical.choice`, `Quot.sound`, and `propext`. The critical counting estimate and every analytic limit remain outside the kernel.

76. A quantitative kernel for the Mersenne Farey swarm

This section turns the finite denominator split of Section 75 into a kernel-checked quantitative and subsequential statement. It also records a sharp counterexample boundary. No estimate for the distribution of base-two super-Wieferich primes is inserted.

Retain the notation

$$\mathcal{H}_T = \sum_{q=1}^{T} \frac{1}{q}, \qquad C_H = \sum_{r=1}^{H-1} r,$$

and write $E_{\le T}$ and $E_{>T}$ for the two pieces of the Farey energy.

Proposition 76.1 (Harmonic and square prefix bounds)**.** *For every* $T \ge 1$*,*

$$\mathcal{H}_T \le 1 + \log T.$$

If every denominator fibre is contained in $\{1, \dots, H-1\}$*, then*

$$E_{\le T} \le H^2(1 + \log T).$$

Proof. The first sum is the classical harmonic number. Isolating its first term and comparing the remaining decreasing summand with the integral of $1/x$ gives the first inequality. The exact denominator split gives $E_{\le T} \le C_H \mathcal{H}_T$, while $C_H = H(H-1)/2 \le H^2$. □

The common-index scale itself also admits an unconditional formal bracket. For $M_n = \operatorname{lcm}(1, \dots, n)$,

$$n \log 2 - \log(n+1) \le \log M_n \le (\log 4 + 4)n.$$

Indeed, Mathlib identifies $\log M_n$ with Chebyshev's function $\psi(n)$. The lower bound follows from $2^n \le (n+1)M_n$, and the upper bound is the explicit elementary Chebyshev estimate for ψ. This proves the correct linear scale without the prime number theorem; the sharper ratio $\log M_n/n \to 1$ is not claimed.

Theorem 76.2 (Cleared quantitative swarm)**.** *Let $T > 0$. Suppose the prefix fibres lie in $\{1, \dots, H-1\}$, all tail numerators are at most H, and*

$$\varepsilon A \le E \le E_{\le T} + E_{>T}, \qquad H^2(1 + \log T) \le \kappa A.$$

Then

$$T(\varepsilon - \kappa)A \le N_{>T} H.$$

Proof. Proposition 76.1 bounds the prefix by κA. The tail must therefore carry at least $(\varepsilon - \kappa)A$. Since every tail row contributes at most H/T, multiplication by T proves the assertion. □

For $X \ge 0$, put

$$\mathcal{W}_3(X) = \{p \le X : p \text{ prime and } p^3 \mid 2^{p-1} - 1\}.$$

Proposition 76.3 (Finite exact-order row transport)**.** *Fix a common index m and a finite set S of actual exact-order endpoint rows. If every $x \in S$ satisfies*

$$p_x \text{ prime}, \qquad p_x^3 \mid 2^{d_x} - 1, \qquad p_x \le X,$$

then

$$|S| \le |\mathcal{W}_3(X)|.$$

Proof. Exact order gives $d_x = \operatorname{ord}_{p_x}(2)$, so the depth transport theorem yields $p_x^3 \mid 2^{p_x - 1} - 1$. Hence the prime-coordinate map lands in $\mathcal{W}_3(X)$. The same exact-order identity makes that coordinate map injective on endpoint rows. Restricting the injection to S proves the finite cardinality inequality. □

This statement closes the finite row-to-prime bookkeeping only for rows whose primality, exact-order and depth premises have been proved. It neither turns arbitrary Farey rows into actual endpoint rows nor estimates the size of $\mathcal{W}_3(X)$.

For a predicate P_m, write $\operatorname{Freq}(P_m)$ when $\forall N\, \exists m \ge N : P_m$.

Proposition 76.4 (Exact failure witness for little-oh)**.** *If $f, g : \mathbb{N} \to \mathbb{R}_{\ge 0}$, then*

$$f \ne o(g) \quad \Longleftrightarrow \quad \exists \varepsilon > 0 : \operatorname{Freq}(\varepsilon g(m) < f(m)).$$

Proof. The definition of $f = o(g)$ says that for every positive ε the inequality $|f(m)| \le \varepsilon |g(m)|$ holds eventually. Nonnegativity removes the absolute values. Negating the two quantifiers gives precisely the displayed frequent strict reverse inequality. □

Corollary 76.5 (Frequent quantitative swarm)**.** *Suppose $E_m, A_m \ge 0$, $E \ne o(A)$, and the finite packet conditions hold eventually. If for every $\delta > 0$ one eventually has*

$$H_m^2(1 + \log T_m) \le \frac{\delta}{2} A_m,$$

then for some fixed $\varepsilon > 0$,

$$\operatorname{Freq}\left(T_m \frac{\varepsilon}{2} A_m \le N_{>T_m}(m) H_m\right).$$

Proof. Proposition 76.4 supplies one fixed $\varepsilon > 0$ and a frequent set on which $\varepsilon A_m < E_m$. Intersect it with the eventual structural set and apply Theorem 76.2 with $\kappa = \varepsilon/2$. □

The coefficient of the exact prefix estimate cannot be uniformly reduced. Indeed, take the full fibres $R_q = \{1, \dots, H-1\}$. Then

$$E_{\le T} = C_H \mathcal{H}_T.$$

For $T = 1$ and $H = 2$, all three quantities equal one. Hence any proposed constant $c < 1$ satisfying $E_{\le T} \le cC_H\mathcal{H}_T$ for every admissible packet would force $1 \le c$, a contradiction. This full-premise example deletes only that strict universal improvement. It leaves the arithmetic route intact: future gain must use primality, exact order, depth, or cross-fibre correlation.

The Lean module `MersenneFareyQuantitativeSwarm20260903` proves all statements in this section at their displayed quantifiers. Its theorem-level dependency union is only `propext`, `Classical.choice`, and `Quot.sound`. The critical upper bound

$$\limsup_{x\to\infty} \frac{\log \max\{1, W_3(x)\}}{\log\log x} \le \frac{1}{2}$$

remains an open arithmetic input, so neither the Mersenne endpoint nor the abc conjecture is proved here.

77. Alternative quality metrics and the packing boundary

Sankaran's 2026 preprint introduces doubly geometric mean (DGM) and power-mean qualities and proves unconditional divergence for an alternative quality using Chen's theorem [26]. For the standard and DGM qualities it identifies the exact packing factor

$$\eta = \frac{G}{A}, \qquad q_{\mathrm{std}} = \eta q_{\mathrm{DGM}}, \tag{AQ.1}$$

where G and A are the geometric and arithmetic means of the logarithms of the distinct primes dividing *abc*.

Proposition 77.1 (Exact packing-coordinate transfer)**.** *If $q_s = \eta q_D$ and $\eta > 0$, then, for every real B,*

$$q_s \le B \iff q_D \le B/\eta.$$

If also $0 \le \eta \le 1$ and $q_D \ge 0$, then $0 \le q_s \le q_D$.

Proof. The equivalence follows by dividing $\eta q_D \le B$ by the positive factor η. Nonnegativity gives the second lower bound, and multiplication of $\eta \le 1$ by $q_D \ge 0$ gives the upper bound. ☐

Consequently, the inequality $q_{\mathrm{DGM}} \le (1+\varepsilon)/\eta$ is an exact rewriting of the standard abc quality bound. It becomes a proof tool only after an independent estimate for the reciprocal packing efficiency is supplied. Alternative-quality divergence alone has no such force.

Proposition 77.2 (Clustered-logarithm lower bound)**.** *If $L > 0$, $C \ge 0$, and $L \le G \le A \le L + C$, then*

$$\frac{G}{A} \ge \frac{L}{L+C}.$$

Proof. The factor $L/(L+C)$ is nonnegative, so $A \le L + C$ gives $LA/(L+C) \le L \le G$. Divide by $A > 0$. ☐

For distinct primes in $[X, e^C X]$ with fixed $C > 0$, this lower bound tends to one as $X \to \infty$, regardless of the number of selected primes. This identifies a scope mismatch in the extension following Theorem 4.13 of the source: the decay part of its Lemma 4.12 fixes both the number of primes and the complementary product while one prime grows, whereas the later application allows the prime count and complementary product to vary. The claimed decay does not follow from the displayed largest-prime condition alone. Moreover, $\log P_n = O(\log c_n)$ is only a power upper bound and is not equivalent to $P_n \sim c_n^{\kappa}$. The critical-boundary proof of Theorem 4.10 also derives an upper estimate before calling its limiting constant exact; the limsup use in Theorem 4.13 needs only the upper estimate. These issues do not affect the exact packing identity or the Chen-based alternative-quality divergence result, and a cluster of prime coordinates is not asserted to be realized by additive *abc* triples.

Proposition 77.3 (Exact abstract no-go for divergence transfer)**.** *There are positive sequences* $\eta_n, q_{D,n}, q_{s,n}$ *with* $0 < \eta_n \le 1$ *and* $q_{s,n} = \eta_n q_{D,n}$ *such that* q_D *is unbounded above while* $q_{s,n} = 1$ *for every* n*.*

Proof. Take $q_{D,n} = n + 1$, $\eta_n = (n + 1)^{-1}$ and $q_{s,n} = 1$. The product identity and all positivity conditions are exact. The Archimedean property shows that q_D is unbounded, whereas q_s is constant. □

This witness satisfies every premise of the metric-only implication it refutes. It is not a sequence of integer *abc* triples, so it neither refutes the standard conjecture nor invalidates the arithmetic search. It locates the surviving gate: on actual triples one must control the correlation between growth of q_{DGM} and decay of η.

The source PDF and verification capsule are archived at

`research/sources/alternative_quality_metrics_2026_09_03/`.

The Lean companion module

`AlternativeQualityPackingBridge20260903`

checks the displayed algebra and the abstract countermodel.

78. Synchronized divisor packets and orientation rigidity

Author: ChatGPT. This section extracts a finite incidence structure directly from the equation $a + b = c$. Its points are triples of divisors whose squares collide modulo the opposite arms. The construction records exponent caps, the allocation of primes among the three arms, and local square-root signs. It is therefore different from a scalar reweighting of the usual radical. We prove the finite algebraic statements first and state the Lean boundary only after the traditional proofs. No form of the *abc* conjecture is assumed or proved.

78.1. **The packet spectrum.** Let

$$a, b, c \in \mathbf{Z}_{>0}, \qquad a + b = c, \qquad (a, b) = 1, \qquad a > 1, \quad b > 1. \tag{460}$$

For positive integers r, s, write

$$\Delta(r, s) = |r^2 - s^2|.$$

Definition 78.1 (Synchronized divisor packet)**.** A synchronized divisor packet for (a, b, c) is a triple $Q = (x, y, z)$ satisfying

$$\begin{gathered} x, y, z > 1, \qquad x \mid a, \quad y \mid b, \quad z \mid c, \\ a \mid \Delta(y, z), \qquad b \mid \Delta(x, z), \qquad c \mid \Delta(x, y). \end{gathered} \tag{461}$$

The finite set of all packets is denoted by $\mathcal{S}(a, b, c)$. A packet is proper if it differs from (a, b, c). We attach to a packet the quantities

$$D(Q) = \Delta(y, z)\Delta(x, z)\Delta(x, y), \tag{462}$$

$$B(Q) = \max(y, z)^2 \max(x, z)^2 \max(x, y)^2, \tag{463}$$

$$T(Q) = \max(x, y, z). \tag{464}$$

The restriction $a, b > 1$ removes unit-arm degeneracies from this section. It is a stated scope restriction and does not remove unit-arm triples from the *abc* conjecture.

Proposition 78.2 (Nonempty finite spectrum)**.** *The coordinates of every packet are pairwise coprime and pairwise distinct. Moreover,* $(a, b, c) \in \mathcal{S}(a, b, c)$*, so the spectrum is finite and nonempty.*

Proof. The equation and $(a, b) = 1$ give

$$(a, c) = (a, a + b) = 1, \qquad (b, c) = (b, a + b) = 1.$$

Common divisors of packet coordinates divide the corresponding pairs of arms, so $(x, y) = (x, z) = (y, z) = 1$. The coordinates exceed one and are therefore distinct.

For the full packet, the congruences

$$c \equiv b \pmod{a}, \qquad c \equiv a \pmod{b}, \qquad a \equiv -b \pmod{c}$$

give the three square congruences in (461). Finally, each arm has finitely many divisors. ☐

Theorem 78.3 (Product divisibility and height envelope)**.** *Every $Q \in \mathcal{S}(a,b,c)$ satisfies*

$$abc \mid D(Q), \qquad abc \le D(Q) \le B(Q) \le T(Q)^6. \tag{465}$$

Proof. The packet congruences give positive integers α, β, γ with

$$\Delta(y,z) = a\alpha, \qquad \Delta(x,z) = b\beta, \qquad \Delta(x,y) = c\gamma.$$

Their product proves $abc \mid D(Q)$. The coordinate gaps are nonzero by Proposition 78.2, so a positive multiple of abc is at least abc. For every pair r, s,

$$\Delta(r,s) = \max(r,s)^2 - \min(r,s)^2 \le \max(r,s)^2.$$

Multiplication yields $D(Q) \le B(Q)$, and each pair maximum is at most $T(Q)$. ☐

The pair-max term in (465) is the exact output of the argument. Passing immediately to T^6 loses anisotropic information.

78.2. **Index one and an infinite exact-gap stratum.** The divisibility in Theorem 78.3 defines the positive integer

$$\kappa(Q) = \frac{D(Q)}{abc}. \tag{466}$$

Proposition 78.4 (Index-one gap rigidity)**.** *One has $\kappa(Q) = 1$ if and only if*

$$\Delta(y,z) = a, \qquad \Delta(x,z) = b, \qquad \Delta(x,y) = c. \tag{467}$$

In that case z lies strictly between x and y.

Proof. Using the positive quotients α, β, γ from the preceding proof, we have $\kappa(Q) = \alpha\beta\gamma$. A product of positive integers is one exactly when every factor is one. Under (467), the equation $c = a + b$ becomes

$$|x^2 - y^2| = |y^2 - z^2| + |x^2 - z^2|.$$

Equality in the triangle inequality on the real line places z^2 between x^2 and y^2. Positivity then places z between x and y. ☐

Proposition 78.5 (Parametric exact-gap packets)**.** *For every integer $t \ge 2$, put*

$$\begin{aligned} a_t &= 2t+1, \quad b_t = t(3t+2), \quad c_t = (t+1)(3t+1), \\ x_t &= 2t+1, \quad y_t = t, \qquad\qquad z_t = t+1. \end{aligned} \tag{468}$$

Then (a_t, b_t, c_t) satisfies (460), and (x_t, y_t, z_t) is a proper packet of index one.

Proof. Expansion gives

$$a_t + b_t = 3t^2 + 4t + 1 = (t+1)(3t+1) = c_t.$$

We have $(2t+1, t) = 1$, while

$$2(3t+2) - 3(2t+1) = 1;$$

hence $(a_t, b_t) = 1$. The three coordinate divisibilities in (461) are visible in (468). Finally,

$$(t+1)^2 - t^2 = 2t+1 = a_t,$$
$$(2t+1)^2 - (t+1)^2 = t(3t+2) = b_t,$$
$$(2t+1)^2 - t^2 = (t+1)(3t+1) = c_t.$$

Thus the gaps are exact and Proposition 78.4 applies. The packet is proper because $b_t > y_t$ and $c_t > z_t$. ☐

In this family $a_tb_tc_t \asymp 18t^5$ and $T(Q_t) = 2t+1 \asymp 2t$. Consequently no exponent strictly smaller than five can replace six in a universal estimate with constant one. The proposition is a square-difference identity and overlaps familiar quadratic constructions. Its role here is to exhibit an exact stratum of the finite packet spectrum; it is not presented as a renamed independent discovery.

78.3. **Prime-power orientations and a rigid global sign.** For an odd prime power $p^e \| a$, the first square congruence factors as

$$p^e \mid (y-z)(y+z).$$

The prime p cannot divide both factors: otherwise it divides $2y$ and $2z$, and oddness together with $(y,z)=1$ gives a contradiction. The whole prime power therefore occupies exactly one of the channels

$$p^e \mid y-z \qquad \text{or} \qquad p^e \mid y+z. \tag{469}$$

The same statement holds cyclically on the other arms. Different prime powers on one composite arm may choose different signs. At the unique possible even arm the factors need not be coprime, so the 2-primary branch must be retained separately.

Definition 78.6 (Canonical orientation)**.** A packet is canonically oriented when

$$y \equiv z \pmod a, \qquad x \equiv z \pmod b, \qquad x \equiv -y \pmod c. \tag{470}$$

These are the three whole-arm signs used by the full packet.

Theorem 78.7 (Canonical-orientation rigidity)**.** *A canonically oriented packet is the full packet* (a,b,c)*.*

Proof. The divisor conditions give $x \le a$, $y \le b$, and $z \le c$. The final congruence in (470) says $c \mid x+y$. Because $x+y>0$,

$$c \le x+y \le a+b = c.$$

It follows that $x+y=a+b$; the two coordinate inequalities then force $x=a$ and $y=b$.

We claim that $c < ab$. Since $a,b>1$, the product $(a-1)(b-1)$ is positive. If it were one, then $a=b=2$, contradicting $(a,b)=1$. Hence $(a-1)(b-1)\ge 2$, which is equivalent to $a+b<ab$.

After substituting $x=a$ and $y=b$, the first two congruences say that z and $c=a+b$ have the same residue modulo both a and b. The coprime Chinese remainder theorem yields $z \equiv c \pmod{ab}$. But $0<z\le c<ab$, so $z=c$. $\square$

Every proper packet therefore has an actual orientation defect. On odd prime powers it changes or fragments at least one of the canonical signs; the possible even arm carries the separate 2-adic ambiguity.

78.4. **Connection with radical, height, and quality.** Let

$$\mathcal{R} = \operatorname{rad}(abc), \qquad q_s(a,b,c) = \frac{\log c}{\log \mathcal{R}}.$$

The spectrum is finite and nonempty, so one may define

$$\mathcal{E}_{\mathrm{sync}}(a,b,c) = \min_{Q\in\mathcal{S}(a,b,c)} \frac{\log B(Q)}{\log \mathcal{R}}. \tag{471}$$

Corollary 78.8 (Packet-energy majorant)**.** *For every triple in the present scope,*

$$q_s(a,b,c) \le \mathcal{E}_{\mathrm{sync}}(a,b,c).$$

Proof. Theorem 78.3 gives $c \le abc \le B(Q)$ for every packet. Take logarithms, divide by the positive number $\log\mathcal{R}$, and then take the minimum. $\square$

The original candidate global statement is now visible. If, for each $\varepsilon > 0$ and all but finitely many primitive nonunit triples, one could produce a packet with

$$B(Q) \le \mathcal{R}^{1+\varepsilon}, \tag{472}$$

then Corollary 78.8 would imply the usual abc inequality. The infinite primitive family established in the companion radical-excess note refutes (472) at $\varepsilon = 1/3$. Only this uncompensated gate is retired. The corrected live gate inserts the factor ab:

$$B(Q) \le ab\,\mathcal{R}^{1+\varepsilon}. \tag{473}$$

Together with $abc \le B(Q)$, this would imply $c \le \mathcal{R}^{1+\varepsilon}$. No eventual estimate of the compensated form is established or refuted here.

For comparison, Sankaran's alternative-quality construction [26] assigns to the distinct primes $p_1, \ldots, p_\omega$ dividing abc the arithmetic and geometric means

$$A = \frac{1}{\omega}\sum_i \log p_i, \qquad G = \left(\prod_i \log p_i\right)^{1/\omega}, \qquad \eta = G/A,$$

and proves the exact factorization

$$q_s = \eta q_{\mathrm{DGM}}, \qquad q_{\mathrm{DGM}} = \frac{\log c}{\omega G}.$$

The packet spectrum is not this scalar factorization under another name. It depends on exponent caps, arm assignment, simultaneous modular incidence, and orientation channels. For example, $(2,3,5)$ and $(3,5,8)$ have the same radical 30 and hence the same η, but the first has one packet and the second has the additional proper packet $(3,5,2)$.

78.5. **Counterexample and finite-search audit.** The following complete-premise examples eliminate tempting stronger claims:

(a,b,c)	packet (x,y,z)	eliminated claim
$(3,5,8)$	$(3,5,2)$	uniqueness of the full packet
$(5,7,12)$	$(5,7,2)$	$abc \le T^3$
$(5,16,21)$	$(5,2,3)$	$abc \le T^4$ and $abc \le (xyz)^2$
$(385,527,912)$	$(7,31,24)$	$abc \le T^5$

In the last two rows the three gaps equal (a,b,c) exactly. In particular,

$$385 \cdot 527 \cdot 912 = 185040240 > 31^5 = 28629151.$$

These counterexamples do not touch the pair-max/sixth-power theorem.

A deterministic integer search scanned all 3,795,230 normalized primitive triples with $c \le 5000$. It enumerated every packet for $c \le 1000$, for the top 200 standard-quality triples through $c = 5000$, and for every triple in that range with $q_s \ge 1$. Across 151,244 enumerated triples it found 151,711 packets, including 467 proper packets and 105 index-one packets. Exact assertions checked the three packet congruences, the product divisibility, and both bounds in (465). The family in Proposition 78.5 was separately checked for $2 \le t \le 100000$. The selection $q_s \ge 1$ used the equivalent exact integer condition $c \ge \mathrm{rad}(abc)$; floating-point logarithms were used for ranking and reported ratios.

The twenty highest-quality triples in the finite range had only their full packet. This is adverse finite evidence, but a finite range neither proves eventual packet rigidity nor decides the compensated all-but-finitely-many assertion.

78.6. **Lean boundary and audit.** The independent Lean companion formalizes the exact finite core: the primitive data, synchronized divisor packets, inherited coprimality, product divisibility, pair-max and sixth-power bounds, canonical-orientation rigidity, and generic channel lemmas. It also defines the actual squarefree radical, standard quality, packet energy, and the minimum over the finite nonempty packet spectrum, and proves the quality majorant in Corollary 78.8. Each counterexample in the preceding table is an explicit Lean datum and packet satisfying all packet premises. The deterministic computation and both global radical estimates are absent from this original formalization; in particular no upper bound on the minimum packet energy is assumed. The companion radical-excess module formalizes the counterfamily and the rational-exponent compensated implication. A separate axiom audit lists all dependencies.

Thus the surviving output of this route is a finite structural invariant and the compensated radical-control problem. The original uncompensated problem has a full-premise counterfamily. Neither result proves or disproves the *abc* conjecture.

79. Radical-excess obstructions to packet compression

Author: ChatGPT. We continue with the synchronized divisor packets of Section 78. The purpose of this section is twofold. First, we identify the exact order-statistic geometry hidden in the pair-max envelope. Second, we correct the proposed radical-scale packet gate by giving an infinite complete-premise counterfamily. The counterfamily disproves the packet gate, not the *abc* conjecture.

Let

$$\mathcal{R} = \operatorname{rad}(abc), \qquad E = \frac{abc}{\mathcal{R}}.$$

The integer E will be called the radical excess. Recall that every synchronized packet $Q = (x, y, z)$ satisfies $abc \le B(Q)$.

79.1. The shape identity.

Put

$$m(Q) = \min(x, y, z), \qquad T(Q) = \max(x, y, z),$$

and

$$L(Q) = \max(y, z) \max(x, z) \max(x, y).$$

Thus $B(Q) = L(Q)^2$.

Theorem 79.1 (Symmetric packet-shape identity)**.** *For every synchronized packet,*

$$m(Q)L(Q) = xyzT(Q), \qquad m(Q)^2 B(Q) = \big(xyzT(Q)\big)^2.$$

Proof. Sort the coordinates as $r \le s \le t$. Their pair maxima are s, t, t, so $m = r$, $T = t$, and $L = st^2$. Consequently $mL = rst^2 = (rst)t = xyzT$. Squaring proves the second identity. Since both sides are symmetric, the computation is independent of the sorting permutation. □

Proposition 79.2 (Support and full-envelope monotonicity)**.** *Every synchronized packet satisfies*

$$xyz \mid abc, \qquad \operatorname{rad}(xyz) \mid \mathcal{R}, \qquad B(Q) \le B(a, b, c).$$

Proof. Multiply the three coordinate divisibilities. Radical divisibility is monotone under divisibility for nonzero natural numbers. Finally, $x \le a$, $y \le b$, and $z \le c$; hence every pair maximum for Q is no larger than its full-packet counterpart. Square and multiply. □

Equality of coordinate support with full radical support is false. The complete-premise packet $(5, 7, 2)$ over $(5, 7, 12)$ has coordinate radical 70, whereas $\operatorname{rad}(5 \cdot 7 \cdot 12) = 210$.

79.2. The radical-excess obstruction.

Theorem 79.3 (Necessary condition for rational packet compression)**.** *Let $m, n \ge 0$. If*

$$B(Q)^m \le \mathcal{R}^{m+n},$$

then

$$E^m \le \mathcal{R}^n.$$

Equivalently, $\mathcal{R}^n < E^m$ implies $\mathcal{R}^{m+n} < B(Q)^m$ for every synchronized packet Q over the same primitive triple.

Proof. Since $\mathcal{R} \mid abc$, one has $\mathcal{R}E = abc$. Therefore

$$\mathcal{R}^m E^m = (abc)^m \le B(Q)^m \le \mathcal{R}^{m+n} = \mathcal{R}^m \mathcal{R}^n.$$

The radical is positive, so cancellation of $\mathcal{R}^m$ gives the assertion. □

This condition depends only on the underlying triple. In particular, no choice inside the packet spectrum can repair its failure.

Theorem 79.4 (Infinite four-thirds obstruction)**.** *For $k \geq 0$, set*

$$A_k = 2^{k+4}, \qquad P_k = (A_k, 3, A_k + 3).$$

These are pairwise distinct primitive nonunit triples. If $\mathcal{R}_k = \mathrm{rad}(A_k \cdot 3 \cdot (A_k + 3))$, then every synchronized packet over P_k satisfies

$$\mathcal{R}_k^4 < B(Q)^3.$$

Consequently the locus on which every packet violates $B^3 \leq \mathcal{R}^4$ is infinite.

Proof. Put $h = 2^{k+3}$, so $A_k = 2h$ and $h \geq 8$. The triple is primitive because A_k is a power of 2 and is coprime to 3. Radical submultiplicativity gives

$$\mathcal{R}_k \leq \mathrm{rad}(A_k)\,\mathrm{rad}(3)\,\mathrm{rad}(A_k + 3) \leq 2 \cdot 3 \cdot (A_k + 3) = 6(2h + 3).$$

Writing $E_k = A_k 3(A_k + 3)/\mathcal{R}_k$, the equality $\mathcal{R}_k E_k = 6h(A_k + 3)$ and the preceding upper bound imply $h \leq E_k$. Moreover,

$$\mathcal{R}_k \leq 6(2h + 3) = 12h + 18 < 64h \leq h^3 \leq E_k^3.$$

Theorem 79.3, with $(m, n) = (3, 1)$, now gives $\mathcal{R}_k^4 < B(Q)^3$ for every packet. The map $k \mapsto P_k$ is injective because its first coordinate is strictly increasing. □

For positive real $\mathcal{R}$ and B, an estimate $B \leq \mathcal{R}^{4/3}$ implies $B^3 \leq \mathcal{R}^4$. Theorem 79.4 therefore refutes the earlier all-but-finitely-many packet gate $B(Q) \leq \mathcal{R}^{1+\varepsilon}$ at the exact value $\varepsilon = 1/3$. It does not produce triples violating $c \leq \mathcal{R}^{1+\varepsilon}$, and hence gives no counterexample to *abc*.

79.3. **A correctly compensated forward gate.**

Theorem 79.5 (Compensated packet implication)**.** *Let $m, n \geq 0$. If*

$$B(Q)^m \leq (ab)^m \mathcal{R}^{m+n},$$

then

$$c^m \leq \mathcal{R}^{m+n}.$$

Proof. The synchronized product bound gives

$$(ab)^m c^m = (abc)^m \leq B(Q)^m \leq (ab)^m \mathcal{R}^{m+n}.$$

Cancel the positive factor $(ab)^m$. □

Thus the factor ab is the exact elementary compensator that changes a packet bound for the full product into the desired bound for the sum arm. No eventual compensated estimate is proved here. Its pointwise four-thirds form already fails at $(2, 3, 5)$, whose packet spectrum consists only of the full packet. The all-but-finitely-many compensated direction remains open because this section supplies no infinite counterfamily to it.

79.4. **Formal and computational audit.** The companion Lean module formalizes every identity and implication above, the infinite dyadic obstruction, and the finite counterexamples. Its axiom audit contains only `propext`, `Classical.choice`, and `Quot.sound`.

An exact integer search exhausts every normalized primitive nonunit triple with $c \leq 3000$ and every divisor triple satisfying all packet premises. It checks 1,365,095 primitive triples, 1,366,531 packets, and 1,436 proper packets. The same program separately factors and exhausts the first twenty-one dyadic rows. Null finite searches are recorded only as evidence and are not promoted to universal statements.

80. Canonical gain surfaces and defect flags

Müller and Taktikos propose separating the quality of certain *abc* equations into an approximation gain and a power gain [27]. The factorization itself is exact and useful, but the proposed fixed bounds on the two factors do not supply the coefficient-one quantifier in the standard *abc* conjecture. We record the precise correlation that is needed and then introduce a finite-dimensional defect flag on which different arithmetic mechanisms can act.

The source requires two cautions. Definition 2.6 prints $\log(q_n^s k)$ in the numerator when $d_n > 0$, whereas the proof of Theorem 2.7 uses $\log(p_n^3)$ for the same sign. These are distinct when $p_n^3 = kq_n^3 + d_n$ and $d_n > 0$. Moreover, the printed proof of Theorem 2.8 obtains a limiting equality from a one-sided asymptotic estimate and refers to a finite verification below an unspecified cutoff. We therefore use no continued-fraction assertion from that theorem. The frozen source and its audit extraction are recorded under `research/sources/abc_gain_surface_2026_09_03/`.

80.1. **The gain surface.** Let $H, N, \mathcal{R} > 1$ and write

$$h = \log H, \qquad n = \log N, \qquad \rho = \log \mathcal{R}.$$

Define

$$q(H, \mathcal{R}) = \frac{h}{\rho}, \qquad A(H, N) = \frac{h}{n}, \qquad P(N, \mathcal{R}) = \frac{n}{\rho}.$$

Proposition 80.1 (Exact gain surface)**.** *For every intermediate scale $N > 1$,*

$$q(H, \mathcal{R}) = A(H, N)P(N, \mathcal{R}).$$

Proof. Both n and ρ are positive, and $(h/n)(n/\rho) = h/\rho$. □

The level set $q = 1 + \varepsilon$ is the hyperbola $AP = 1 + \varepsilon$. Separate bounds $A \le A_0$ and $P \le P_0$ give only $q \le A_0 P_0$. Thus the proposed constants $3/2$ and 3 give a weak exponent $9/2$, rather than the standard $1 + \varepsilon$ for every positive ε.

This failure is not an issue of endpoints. In logarithmic coordinates take $(\rho, h, n) = (2, 4, 5)$ and exponentiate. Then $1 < \mathcal{R} < H < N$ while

$$A = \frac{4}{5} < \frac{3}{2}, \qquad P = \frac{5}{2} < 3, \qquad q = 2 > \frac{3}{2}.$$

This is a full-premise countermodel to the abstract implication from the two independent bounds. It is not an *abc* triple.

80.2. **The canonical arithmetic corridor.** Let $a + b = c$ be a primitive positive triple with $a, b > 1$, and set

$$M = abc, \qquad \mathcal{R} = \operatorname{rad}(abc), \qquad A_{\text{can}} = \frac{\log c}{\log M}, \qquad P_{\text{can}} = \frac{\log M}{\log \mathcal{R}}.$$

Theorem 80.2 (Canonical gain corridor)**.** *Every such triple satisfies*

$$c^2 < M < c^3, \qquad \frac{1}{3} < A_{\text{can}} < \frac{1}{2}, \qquad q(a, b, c) = A_{\text{can}} P_{\text{can}}.$$

Consequently

$$\frac{1}{3} P_{\text{can}} < q(a, b, c) < \frac{1}{2} P_{\text{can}}.$$

Proof. Because $a, b > 1$ and $(a, b) = 1$, the pair cannot be $(2, 2)$, so $(a-1)(b-1) > 1$. Expansion and $a + b = c$ give $ab > c$, whence $abc > c^2$. Also $a < c$ and $b < c$, so $abc < c^3$. Strict monotonicity of the logarithm gives

$$2 \log c < \log M < 3 \log c,$$

which is equivalent to the claimed corridor. The factorization is Proposition 80.1; multiplying the corridor by the positive number P_{can} proves the last display. □

Corollary 80.3 (Necessary canonical power concentration)**.** *If $q(a,b,c) \geq 1+\varepsilon$, then*

$$P_{\mathrm{can}} > 2(1+\varepsilon).$$

Conversely, $P_{\mathrm{can}} \leq 2(1+\varepsilon)$ implies $q(a,b,c) < 1+\varepsilon$.

Proof. Use $q < P_{\mathrm{can}}/2$ from Theorem 80.2. □

The coefficient two is the same sharp threshold found by the symmetric product transfer, but the factorized form exposes the cancellation that the coefficient statement hides.

80.3. **The affine defect plane.** Put $h = \log c$, $m = \log M$, and $\rho = \log \mathcal{R}$, and define

$$X = \frac{m-\rho}{\rho}, \qquad Y = \frac{m-h}{\rho}.$$

Here X is the normalized multiplicity discarded by the radical and Y is the normalized product-to-height slack.

Theorem 80.4 (Exact cancellation)**.** *One has*

$$\boxed{q(a,b,c) = 1 + X - Y.}$$

Consequently the effective logarithmic inequality

$$h \leq (1+\varepsilon)\rho + C$$

is equivalent to

$$X - Y \leq \varepsilon + \frac{C}{\rho}.$$

Proof. Cancellation gives

$$1 + \frac{m-\rho}{\rho} - \frac{m-h}{\rho} = \frac{h}{\rho}.$$

The second assertion follows by subtracting one and dividing by $\rho > 0$. □

Thus a large power excess can be harmless when the approximation slack is comparably large. Independent constant bounds erase this relation.

80.4. **A scale groupoid and higher-dimensional flags.** On real scale coordinates define

$$\delta(x,y) = y - x.$$

This is an additive one-cocycle on the pair groupoid:

$$\delta(x,x) = 0, \qquad \delta(x,z) = \delta(x,y) + \delta(y,z).$$

On nonzero coordinates, $g(x,y) = y/x$ is the corresponding multiplicative cocycle. Both identities are immediate by cancellation.

A *defect flag* is a finite path

$$\rho = s_0, s_1, \ldots, s_k = m, \qquad d_i = s_i - s_{i-1}.$$

Its total is path-independent:

$$\sum_{i=1}^{k} d_i = m - \rho.$$

In particular every closed scale flag has trivial additive and multiplicative holonomy. Refining a factorization cannot manufacture a saving; arithmetic must restrict the admissible paths or bound their costs.

Theorem 80.5 (Defect-budget half-space)**.** *Suppose*

$$\sum_i d_i \le \sum_i B_i$$

and

$$\sum_i B_i \le (m-h) + \varepsilon\rho + C.$$

Then

$$h \le (1+\varepsilon)\rho + C.$$

Proof. The cocycle law telescopes, so $m - \rho = \sum_i d_i \le \sum_i B_i$. Combining this with the second budget and cancelling m gives the result. □

For a fixed total allowance and no sign constraints, the vectors $(B_1, \dots, B_k)$ lie in an affine half-space. Its intersection with the nonnegative orthant is a simplex when the allowance is nonnegative. Packet compression, Pell transversality, IUT log-volume estimates, and analytic radical bounds may therefore be assigned to different coordinates, but their total must remain below the approximation slack plus the ε-allowance.

80.5. **An exact counterexample boundary.** The tempting universal strengthening $P_{\mathrm{can}} \le 3$ is false. For the primitive hit

$$3 + 125 = 128$$

one has

$$M = 48000 = 2^7 \cdot 3 \cdot 5^3, \qquad \mathcal{R} = 30,$$

and $48000 > 30^3 = 27000$. Hence

$$P_{\mathrm{can}} = \frac{\log 48000}{\log 30} > 3.$$

This retires only the universal power-three cap. It does not refute the correlated defect barrier or the *abc* conjecture.

An exhaustive computation of the 5,465,583 unordered primitive nonunit triples with $c \le 6000$ found no violation of the canonical approximation corridor, 65 triples of quality above one, and 14 triples with canonical power gain above three. These data are finite evidence only.

The exact open gate is, for every $\varepsilon > 0$, to prove

$$\sum_i d_i \le (m-h) + \varepsilon\rho + C_\varepsilon.$$

This is equivalent to standard *abc*, so the coordinate construction alone does not close the conjecture. It supplies a common, noncircular accounting surface on which the different routes can be combined.

81. A labelled valuation-incidence complex

This section develops a higher-dimensional object directly from the prime factorisation of an *abc* triple. It is deliberately restricted to *primitive nonunit triples*

$$a, b, c \in \mathbb{N}_{>0}, \qquad a > 1, \quad b > 1, \quad a + b = c, \quad \gcd(a,b) = 1.$$

The restriction is part of every statement below. Unit-arm triples require a separate extension. The construction is finite and exact. The next two sections refute its first one-face quantitative selectors and preserve the remaining weighted, multi-face, three-arm, and homological design problem.

Put $x_A = a$, $x_B = b$, and $x_C = c$. For $i \in \{A, B, C\}$ let

$$S_i = \{p : p \text{ prime},\ p \mid x_i\}, \qquad e_i(p) = v_p(x_i).$$

Primitivity and $a + b = c$ make the three coordinates pairwise coprime. Thus their prime supports are disjoint, although we retain the arm label because it also records a different local congruence.

Definition 81.1 (Valuation-incidence complex)**.** A vertex is a labelled pair (i, p) with $p \in S_i$, carrying the datum $(i, p, e_i(p))$. A face is a triple

$$F = (F_A, F_B, F_C), \qquad F_i \subseteq S_i.$$

We write $F \preceq G$ if $F_i \subseteq G_i$ on every arm. Union and intersection are taken armwise. The empty face is $\varnothing_P = (\varnothing, \varnothing, \varnothing)$ and the top face is $\top_P = (S_A, S_B, S_C)$.

The face relation is the Boolean face lattice on $S_A \sqcup S_B \sqcup S_C$. In particular, it is reflexive, transitive and antisymmetric; the empty and top faces are respectively least and greatest; and armwise union is the least common upper bound. These assertions follow coordinatewise from the corresponding elementary facts for finite subsets. If

$$\nu(F) = |F_A| + |F_B| + |F_C|, \qquad \dim_{\mathbb{N}}(F) = \max\{\nu(F) - 1, 0\},$$

then a nonempty top face with N vertices has dimension $N - 1$. There is no a priori bound on N.

For a face F and an arm i, define

$$R_i(F) = \prod_{p \in F_i} p, \qquad D_i(F) = \prod_{p \in F_i} p^{e_i(p)-1},$$
$$M_i(F) = \prod_{p \in F_i} p^{e_i(p)}, \quad d_i(F) = \sum_{p \in F_i} (e_i(p) - 1).$$

Thus R_i records selected radical mass, D_i selected multiplicity beyond the first layer, M_i the corresponding prime-power modulus, and d_i its additive defect degree. The exact six-coordinate realisation is

$$T(F) = ((R_A(F), D_A(F)), (R_B(F), D_B(F)), (R_C(F), D_C(F))).$$

Applying logarithms produces an additive display in $\mathbb{R}^6$, but logarithms play no role in face membership or in the filtration below.

Proposition 81.2 (Exact face factorisation)**.** *For every face and every arm,*

$$R_i(F)D_i(F) = M_i(F), \qquad M_i(F) \mid x_i.$$

At the top face,

$$M_i(\top_P) = x_i, \qquad R_i(\top_P) = \mathrm{rad}(x_i), \qquad \mathrm{rad}(x_i)D_i(\top_P) = x_i.$$

If F and G are armwise disjoint, each of R_i, D_i, M_i is multiplicative under $F \cup G$, while $d_i(F \cup G) = d_i(F) + d_i(G)$.

Proof. For every selected prime, positivity of $e_i(p)$ gives $p\,p^{e_i(p)-1} = p^{e_i(p)}$. Multiplication over F_i proves the first identity. The selected product is a subproduct of the canonical factorisation of x_i, so it divides x_i; on the top face it is the full factorisation. Retaining one copy of each support prime gives the usual squarefree radical. The disjoint-union formulas are the standard product and sum formulas over disjoint finite sets. □

For a budget $u = (u_A, u_B, u_C) \in \mathbb{N}^3$, define the filtered subcomplex

$$\mathcal{K}_u(P) = \{F : d_i(F) \le u_i \text{ for } i = A, B, C\}.$$

Proposition 81.3 (Defect filtration)**.** *If $G \preceq F$ and $F \in \mathcal{K}_u(P)$, then $G \in \mathcal{K}_u(P)$. The empty face lies in every filtration level. If F, G are armwise disjoint, $F \in \mathcal{K}_u(P)$ and $G \in \mathcal{K}_v(P)$, then $F \cup G \in \mathcal{K}_{u+v}(P)$. Moreover,*

$$F \in \mathcal{K}_{(0,0,0)}(P) \iff e_i(p) = 1 \textit{ for every selected } (i, p).$$

Proof. Every summand $e_i(p) - 1$ is nonnegative. Restricting to a subset can only decrease its sum, and the empty sums vanish. The graded-union assertion uses the additive formula in Proposition 81.2. At budget zero, a finite sum of nonnegative integers is zero precisely when each summand is zero; positivity of $e_i(p)$ then gives the final equivalence. □

The arm colours also retain the additive equation at selected prime-power resolution.

Proposition 81.4 (Local incidence and CRT rigidity)**.** *Every face satisfies*

$$c \equiv b \pmod{M_A(F)}, \qquad c \equiv a \pmod{M_B(F)}, \qquad a+b \equiv 0 \pmod{M_C(F)}.$$

*The three moduli are pairwise coprime. Call F AB-*reconstructing *when $c < M_A(F)M_B(F)$. Every top face is AB-reconstructing. More generally, if F is AB-reconstructing and*

$$0 \le x < M_A(F)M_B(F), \quad x \equiv b \pmod{M_A(F)}, \quad x \equiv a \pmod{M_B(F)},$$

then $x = c$.

Proof. Proposition 81.2 gives $M_A(F) \mid a$, $M_B(F) \mid b$, and $M_C(F) \mid c$; subtracting within $a+b=c$ proves the three congruences. Since each selected modulus divides one member of a pairwise coprime triple, the moduli are pairwise coprime. At the top face, $M_A = a$ and $M_B = b$. For nonunits one has $a+b \le ab$, with equality only at $a=b=2$; primitivity excludes that case, so $a+b < ab$. For a general reconstructing face, the displayed residues show that x and c are congruent modulo both coprime moduli, hence modulo their product. Both lie in the same half-open interval of that length, so they are equal. □

The top point recovers the ordinary conductor exactly:

$$R_A(\top_P)R_B(\top_P)R_C(\top_P) = \operatorname{rad}(abc).$$

It also gives a precise half-space form of a rational-power abc bound.

Theorem 81.5 (Exact top-face half-space)**.** *Write $R_i = R_i(\top_P)$ and $D_C = D_C(\top_P)$. For all $m, n \in \mathbb{N}$,*

$$c^m \le \operatorname{rad}(abc)^{m+n} \iff D_C^m \le (R_AR_B)^{m+n}R_C^n.$$

Furthermore, for any face F, the stronger selected-face inequality

$$D_C^m \le \big(R_A(F)R_B(F)\big)^{m+n}R_C^n$$

implies $c^m \le \operatorname{rad}(abc)^{m+n}$.

Proof. The top-face identities give $c = R_CD_C$ and $\operatorname{rad}(abc) = R_AR_BR_C$. Substitution turns the left inequality into

$$R_C^mD_C^m \le R_C^m\big((R_AR_B)^{m+n}R_C^n\big).$$

Since $R_C > 0$, cancellation proves the equivalence. For the second claim, $F_i \subseteq S_i$ gives $R_A(F) \le R_A$ and $R_B(F) \le R_B$; enlarge the right side of the selected-face inequality and apply the reverse implication of the equivalence. □

Theorem 81.5 is an exact change of coordinates. It does not prove that a useful selected face exists. The following fully realised example also prevents two tempting degeneracy assumptions.

Proposition 81.6 (A five-dimensional complete-premise boundary)**.** *The primitive nonunit triple*

$$12 + 833 = 845, \qquad 12 = 2^2 \cdot 3, \quad 833 = 7^2 \cdot 17, \quad 845 = 5 \cdot 13^2$$

has a six-vertex top face, hence natural dimension five, and

$$T(\top_P) = ((6,2),(119,7),(65,13)), \qquad (d_A, d_B, d_C) = (1,1,1).$$

Every arm has nontrivial defect, and on every arm the two valuation exponents are different. The face selecting 3*,* 17*, and* 5 *lies at zero budget.*

Proof. The displayed factorizations list the six vertices and give all coordinates by their definitions. They also show directly that one exponent on each arm is 2 and the other is 1. Thus the defects are $2, 7, 13$, all nontrivial, and every defect degree is one. Selecting the three exponent-one vertices gives defect degree zero on every arm. □

This example refutes exactly the pointwise universal statements that some arm must have defect one and that the valuations must be constant on each arm. It does not refute eventual or density variants and does not refute any estimate that permits all three arms to be defective.

The first falsifiable gate proposed for this higher-dimensional route used an absolute budget. For fixed $m, n \geq 1$, it asked whether a number $t = t(m, n)$ exists such that all but finitely many primitive nonunit triples possess a face F satisfying

$$\text{(VIC-ABS-1)}\quad d_A(F), d_B(F) \leq t, \qquad c < M_A(F)M_B(F), \qquad D_C^m \leq \big(R_A(F)R_B(F)\big)^{m+n} R_C^n.$$

Theorem 81.5 proves that VIC-ABS-1 would imply the corresponding rational-exponent height bound on this domain. The next section constructs an infinite complete-premise family that refutes this absolute budget formulation. That obstruction does not refute the complex or its weighted, multi-face, three-arm, and homological replacements.

The companion Lean module `ABCValuationIncidenceComplex20260903` constructs the actual faces with `Nat.primeFactors` and the labels with `Nat.factorization`. It kernel-checks the finite-poset, factorisation, filtration, congruence, CRT, half-space, selected-face, and witness statements above. No selector is inserted as an axiom or an inhabited interface.

82. The fixed-budget incidence obstruction

We now test VIC-ABS-1 against every one of its necessary reconstruction premises. The resulting obstruction is asymptotic and uses actual primitive nonunit triples; it therefore retires the exact all-but-finitely-many absolute-budget statement.

For $k \geq 0$, define

$$P_k = (2^{k+4}, 3, 2^{k+4} + 3).$$

Every P_k is primitive and nonunit, and the family is injective through its first coordinate.

Theorem 82.1 (No fixed-budget reconstruction on the dyadic tail)**.** *Let $t, k \in \mathbb{N}$ with $t < k+3$. No face of P_k satisfies both*

$$d_A(F) \leq t \qquad \textit{and} \qquad c < M_A(F)M_B(F).$$

Consequently, for every fixed t, infinitely many primitive nonunit triples have no face that both reconstructs through the two summand arms and has A-defect at most t. In particular, VIC-ABS-1 is false for every pair of rational-power parameters (m, n).

Proof. The A-coordinate of P_k is 2^{k+4}. Its support consists only of 2, with valuation $k+4$. If a face selected this vertex, then its contribution to $d_A(F)$ would be

$$v_2(2^{k+4}) - 1 = k + 3 > t,$$

contrary to the budget. Hence F_A is empty and $M_A(F) = 1$. The selected B-modulus divides the coordinate 3, so $M_B(F) \leq 3$. It follows that

$$M_A(F)M_B(F) \leq 3 < 2^{k+4} + 3 = c,$$

which rules out reconstruction. For a fixed t, every $k \geq t$ satisfies $t < k + 3$; the injective tail therefore supplies infinitely many failures. Every full VIC-ABS-1 face would satisfy the two conditions just excluded, so its additional quantitative inequality cannot repair the selector. □

The theorem retires the fixed absolute defect budget and nothing more. In particular, it leaves intact all face-poset, local-incidence, CRT, and half-space results of Section 81. It also suggests where the missing scale entered: selecting one high-multiplicity arm can cost logarithmically in the height.

The first attempted correction was therefore scale-sensitive. Put

$$\ell_2(c) = \min\{s \in \mathbb{N} : c \leq 2^s\}.$$

For fixed $m, n \geq 1$, VIC-1R asks whether a slack $t = t(m, n)$ exists such that all but finitely many primitive nonunit triples possess a face satisfying

$$\begin{aligned} d_A(F) + d_B(F) &\leq \ell_2(c) + t, \\ c &< M_A(F)M_B(F), \\ D_C^m &\leq \big(R_A(F)R_B(F)\big)^{m+n} R_C^n. \end{aligned} \tag{VIC-1R}$$

The dyadic obstruction does not violate the scale budget merely by selecting the prime 2. Nevertheless, the next companion section gives a different balanced two-prime family that refutes VIC-1R itself. The surviving parent route must therefore use weighted budgets, several faces with combined congruence data, all three labelled arms, or another invariant absent from this one-face selector. Each replacement remains subject to simultaneous positive proof construction and complete-premise counterexample search.

The Lean module `ABCValuationIncidenceFixedBudgetObstruction20260903` defines the full pointwise selector and its all-but-finitely-many form. It kernel-checks the dyadic valuation, support exclusion, modulus bounds, infinite failure set, and the negation of VIC-ABS-1. It does not formalise or assume VIC-1R; that separate statement is handled by the scale-budget obstruction module.

83. The coefficient-one scale-budget obstruction

We now test the first scale-sensitive repair of the fixed-budget incidence selector. Write

$$\ell_2(c) = \min\{s \in \mathbb{N} : c \leq 2^s\}.$$

For fixed $m, n \geq 1$, VIC-1R asks for a fixed additive slack $t = t(m, n)$ such that all but finitely many primitive nonunit triples admit a face F with

$$\begin{aligned} d_A(F) + d_B(F) &\leq \ell_2(c) + t, \\ c &< M_A(F)M_B(F), \\ D_C^m &\leq \big(R_A(F)R_B(F)\big)^{m+n} R_C^n. \end{aligned} \tag{VIC-1R}$$

The following obstruction satisfies all ambient arithmetic premises and fails the first two necessary conditions together on an infinite tail.

For $r \geq 1$, put

$$Q_r = (2^{2r}, 3^r, 2^{2r} + 3^r).$$

The two summands are coprime, so Q_r is primitive; all three coordinates exceed one. The family is injective through its first coordinate.

Theorem 83.1 (No coefficient-one scale-budget reconstruction)**.** *Let $r \geq 1$ and let F be an AB-reconstructing face of Q_r. Then*

$$d_A(F) + d_B(F) = 3r - 2 \qquad \textit{and} \qquad \ell_2(2^{2r} + 3^r) \leq 2r + 1.$$

Consequently, for every fixed $t \in \mathbb{N}$, every $r > t + 3$ fails

$$d_A(F) + d_B(F) \leq \ell_2(c) + t.$$

In particular, VIC-1R is false for every pair $m, n \geq 1$.

Proof. The A-support is the singleton $\{2\}$ and the B-support is the singleton $\{3\}$. If 2 were omitted, then $M_A(F) = 1$ and $M_B(F) \mid 3^r$, whence

$$M_A(F)M_B(F) \leq 3^r < 2^{2r} + 3^r = c,$$

contrary to reconstruction. If 3 were omitted, the symmetric estimate $M_A(F)M_B(F) \leq 2^{2r} < c$ gives the same contradiction. Thus both vertices are selected, and their excess valuations give

$$d_A(F) + d_B(F) = (2r - 1) + (r - 1) = 3r - 2.$$

Since $3^r < 4^r = 2^{2r}$ for $r \geq 1$,

$$c = 2^{2r} + 3^r < 2^{2r+1},$$

so the defining minimum satisfies $\ell_2(c) \leq 2r + 1$. If $r > t + 3$, then

$$3r - 2 > 2r + 1 + t \geq \ell_2(c) + t.$$

Hence no reconstructing face meets the budget. Infinitely many distinct Q_r occur for every fixed t. The third line of VIC-1R cannot create a face after the first two necessary conditions have become incompatible, so the all-but-finitely-many selector fails for every m, n. □

This theorem retires the coefficient-one binary-scale one-face selector and nothing broader. It does not refute the valuation-incidence complex, its exact half-space theorem, or standard *abc*. Surviving constructions may use logarithmically weighted defect, a larger scale coefficient coupled to a new quantitative restriction, several faces with joint CRT data, three-arm reconstruction, or a homological invariant of the filtered complex. None of these replacements is assumed here.

The ordinary proof above was fixed before formalization. The Lean module `ABCValuationIncidenceScaleBudgetObstruction20260903` reindexes by $r = k + 1$, defines the binary scale by a least-witness construction, proves the exact defect $3k+1$ and scale bound $2k+3$, constructs an infinite failure set for every slack, and negates the full pointwise and all-but-finitely-many VIC-1R predicates. Its one-for-one audit covers all 34 public declarations. The dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`; there are no custom axioms, placeholders, or native decision oracles.

84. Three-arm incidence covers and complement transport

The preceding obstructions rule out two one-face selectors, but they do not rule out the labelled incidence complex itself. We now allow a face to use prime-power vertices on all three arms and charge every selected excess layer by its actual logarithmic weight.

Let $P = (a, b, c)$ be a positive primitive triple with $a + b = c$. For $i \in \{A, B, C\}$ put $x_A = a$, $x_B = b$, and $x_C = c$. A three-arm face F selects a subset F_i of the prime support of each x_i. Define

$$R_i(F) = \prod_{p \in F_i} p, \qquad \overline{R}_i(F) = \prod_{\substack{p \mid x_i \\ p \notin F_i}} p,$$

$$D_i(F) = \prod_{p \in F_i} p^{v_p(x_i)-1}, \quad M_i(F) = \prod_{p \in F_i} p^{v_p(x_i)}.$$

Write $R(F)$, $\overline{R}(F)$, $D(F)$, and $M(F)$ for the products over the three arms, and write $R(P) = \operatorname{rad}(abc)$. We say that F covers the endpoint when $c \leq M(F)$.

Proposition 84.1 (Exact three-arm products and CRT rigidity)**.** *Every three-arm face satisfies*

$$R(F)D(F) = M(F), \qquad R(F)\overline{R}(F) = R(P) \qquad \textit{(3C-1).}$$

If $c < M(F)$, $0 \leq x < M(F)$, *and*

$$x \equiv b \pmod{M_A(F)}, \qquad x \equiv a \pmod{M_B(F)}, \qquad x \equiv 0 \pmod{M_C(F)},$$

then $x = c$.

Proof. For a selected prime, the identity $p\,p^{v_p(x_i)-1} = p^{v_p(x_i)}$ gives the first product formula. Selected and unselected primes partition the support on each arm, so their radicals multiply to $\operatorname{rad}(x_i)$. Pairwise coprimality of the coordinates then gives the second formula.

Each selected arm modulus divides its coordinate. The equation $a + b = c$ therefore gives the three displayed congruences for c. The three moduli are pairwise coprime, so the Chinese remainder theorem gives $x \equiv c \pmod{M(F)}$. Both numbers lie in $[0, M(F))$, hence $x = c$. □

The first weighted proposal asks, for every $\varepsilon > 0$, for a constant C_ε such that every P has a covering face satisfying

$$\log D(F) \leq \varepsilon \log R(P) + C_\varepsilon \qquad \text{(RAW-3C).}$$

Coverage and (3C-1) would then give

$$\log c \leq \log R(P) + \log D(F) \leq (1+\varepsilon)\log R(P) + C_\varepsilon,$$

so RAW-3C would imply standard *abc*. The following complete-premise family shows that this proposal is still too strong.

For $t \geq 1$, put

$$x_t = 2t+1, \qquad y_t = 2t(t+1), \qquad z_t = 2t^2+2t+1,$$

and set $P_t = (x_t^2, y_t^2, z_t^2)$. Euclid's identity gives $x_t^2 + y_t^2 = z_t^2$, while the consecutive-parameter gcd calculation gives $\gcd(x_t, y_t) = 1$. Thus the P_t are distinct positive primitive triples.

Theorem 84.2 (Infinite obstruction to RAW-3C)**.** *RAW-3C is false, already at* $\varepsilon = 1/4$.

Proof. Every selected valuation on P_t is at least two. Consequently $R(F) \leq D(F)$ and

$$M(F) = R(F)D(F) \leq D(F)^2.$$

If F covers, then $z_t^2 \leq M(F)$ and hence $z_t \leq D(F)$. On the other hand, $\mathrm{rad}(u^2) = \mathrm{rad}(u) \leq u$ for every positive integer u, and $x_t, y_t \leq z_t$. Therefore

$$R(P_t) \leq x_t y_t z_t \leq z_t^3.$$

If RAW-3C held at $1/4$, every t would admit a covering face with

$$\log z_t \leq \log D(F) \leq \frac{1}{4}\log R(P_t) + C \leq \frac{3}{4}\log z_t + C.$$

This would bound $\frac{1}{4}\log z_t$ by the fixed constant C, whereas z_t is unbounded. □

The failure identifies the missing resource: the radical vertices omitted from a covering face. For every selected prime p and every layer above its first occurrence, create a source token of weight $\log p$ and key p. For every unselected prime q, create a sink token of capacity $\log q$ and key q. A complement transport is a nonnegative fractional flow from sources to sinks, subject to row and column capacities, in which positive flow from p to q is allowed only when $p \leq q$. Its unmatched source mass is

$$U(F,f) = \sum_s \left(w(s) - \sum_q f(s,q) \right).$$

Proposition 84.3 (Transport accounting)**.** *For every complement transport,*

$$\sum_s w(s) = \log D(F), \qquad \sum_q w(q) = \log \overline{R}(F),$$

and

$$\log D(F) \leq \log \overline{R}(F) + U(F,f) \qquad \text{(3C-2)}.$$

For every prime threshold T one also has the necessary Hall inequality

$$\sum_{p(s)>T} w(s) - \sum_{q>T} w(q) \leq U(F,f) \qquad \text{(3C-3)}.$$

Proof. A selected vertex p contributes $v_p(x_i) - 1$ source tokens, so summing their weights gives $\log D(F)$. The sink identity is the logarithm of the complementary squarefree product. If W is the carried mass, then $U = \log D(F) - W$, while the column capacities give $W \leq \log \overline{R}(F)$; this proves (3C-2). A source with key above T can send mass only to a sink whose key is also above T. Comparing that source-tail mass with the available sink-tail capacity proves (3C-3). □

The ordered gate CT-3C asks, for every $\varepsilon > 0$, for a constant C_ε such that every positive primitive P has a covering face and a complement transport satisfying

$$U(F,f) \leq \varepsilon \log R(P) + C_\varepsilon \qquad \text{(CT-3C)}.$$

Theorem 84.4 (Complement transport implies standard *abc*)**.** *If CT-3C holds, then the standard abc conjecture holds.*

Proof. Coverage, (3C-1), and (3C-2) give

$$\begin{aligned}\log c &\le \log R(F) + \log D(F)\\ &\le \log R(F) + \log \overline{R}(F) + U(F,f)\\ &= \log R(P) + U(F,f).\end{aligned}$$

Substitution of CT-3C yields $\log c \le (1+\varepsilon)\log R(P) + C_\varepsilon$ with the standard quantifier order. □

Section 86 gives an infinite complete-premise refutation of CT-3C. Before that obstruction was isolated, the exact-support computation enumerated all 218,893 unordered primitive triples with $a \le b$ and $a+b=c \le 1200$, all covering faces of each triple, the balanced family $(2^{2r}, 3^r, 2^{2r}+3^r)$ for $1 \le r \le 12$, and the first eighty members of the Pythagorean-square family. It finds 1,669 triples without a zero-defect covering face. The largest displayed value of $\min U/\log R(P)$ in the exhaustive range is approximately 0.613147, at $1+8=9$. These finite ratios can be absorbed by an additive constant and by themselves neither prove nor refute an asymptotic gate. Integer factorisation, support, coverage, radical, modulus, and defect decisions are exact. Floating-point logarithms enter only the finite flow values and their ranking.

For completeness, the descending greedy calculation is optimal for this finite nested graph. Group equal keys and process groups from largest to smallest, adding the current sink capacity before matching the current source mass. If Δ_r is cumulative source mass minus cumulative sink capacity after r groups and U_r is the greedy unmatched mass, conservation gives

$$U_r = \max(U_{r-1}, \Delta_r), \qquad U_0 = 0.$$

Thus $U_r = \max(0, \Delta_1, \ldots, \Delta_r)$. These cumulative differences are precisely the Hall upper-tail excesses, every one of which is a lower bound for any transport. The greedy value therefore equals the finite min-cut formula. The implementation cross-checks the two calculations and includes the regression instance with sources $(5,1),(2,1)$ and sink $(3,1)$, whose unmatched optimum is 1.

The ordinary arguments above were fixed before their Lean translation. The companion module is

`ABCThreeArmIncidenceSuccessor20260903`.

It formalises the exact products, local signatures, three-modulus CRT rigidity, both implications to `ABCConjecture`, the infinite RAW-3C refutation, the source and sink mass identities, the Hall obstruction, and the exact bound $\mathrm{height}(P) \le \mathrm{conductor}(P) + U(F,f)$. Its one-for-one audit covers all 65 public declarations. The dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`; no custom axiom or placeholder is used. The converse finite Hall theorem remains a useful exact description of this ordered graph, but the prime-square obstruction shows that no bound on its optimized tail excess can establish CT-3C. Subsequent work must change the transport geometry while retaining the parent incidence data.

85. Signed endpoint prime-token transport

We next attach a finite transport problem to every positive primitive *abc* triple; unit arms are allowed in this section. Set

$$K_c = \frac{c}{\mathrm{rad}(c)}, \qquad E_{ab} = \mathrm{rad}(a)\,\mathrm{rad}(b), \qquad R = \mathrm{rad}(abc).$$

Proposition 85.1 (Endpoint-core balance)**.** *Every positive primitive triple satisfies*

$$E_{ab} = \mathrm{rad}(ab), \qquad R = E_{ab}\,\mathrm{rad}(c), \qquad E_{ab}c = RK_c.$$

Consequently

$$\delta_c(a,b,c) := \log K_c - \log E_{ab} = \log c - \log R.$$

Proof. The three coordinates are pairwise coprime, so the squarefree radical is multiplicative across their supports. Since $c = \mathrm{rad}(c)K_c$, multiplication gives $E_{ab}c = RK_c$. Every factor is positive; taking logarithms and rearranging proves the last identity. □

For $p \mid c$, write $e_p = v_p(c)$. The source tokens and their masses are

$$\mathcal{S}_c = \{(p,j) : p \mid c,\ 1 \le j < e_p\}, \qquad w(p,j) = \log p.$$

Thus one source token is retained for each prime layer above the first copy. The sink tokens and capacities are

$$\mathcal{T}_{ab} = \{q : q \mid ab\}, \qquad v(q) = \log q,$$

with one sink for every distinct external prime. Canonical factorisation immediately gives

$$\text{(EPT-1)} \qquad \sum_{s \in \mathcal{S}_c} w(s) = \log K_c, \qquad \sum_{q \in \mathcal{T}_{ab}} v(q) = \log E_{ab}.$$

Definition 85.2 (Fractional monotone endpoint transport)**.** A transport is a family $F(s,q) \in \mathbb{R}_{\ge 0}$ such that

$$\sum_q F(s,q) \le w(s), \qquad \sum_s F(s,q) \le v(q),$$

and $F((p,j),q) > 0$ only when $p \le q$. Write

$$U(F) = \sum_s \left(w(s) - \sum_q F(s,q) \right), \quad V(F) = \sum_q \left(v(q) - \sum_s F(s,q) \right)$$

for unmatched source mass and unused sink capacity.

The zero transport always exists. The content of the construction is the size of $U(F)$, together with the order restriction on its edges.

Theorem 85.3 (Exact accounting and height control)**.** *Every fractional monotone endpoint transport satisfies*

$$U(F) = \delta_c(a,b,c) + V(F).$$

In particular,

$$\delta_c(a,b,c) \le U(F), \qquad \log c \le \log R + U(F).$$

Proof. Let $C(F) = \sum_{s,q} F(s,q)$. Finite interchange of summation gives

$$U(F) = \sum_s w(s) - C(F), \qquad V(F) = \sum_q v(q) - C(F).$$

Subtracting and using the two token-mass identities gives $U(F) - V(F) = \delta_c(a,b,c)$. The column-capacity inequalities make $V(F) \ge 0$, and Proposition 85.1 converts the first resulting inequality into the height bound. ☐

The monotone edge condition carries information that the scalar identity does not see. For $t \in \mathbb{R}$, put

$$S_{>t} = \sum_{(p,j) \in \mathcal{S}_c,\ p>t} \log p, \qquad T_{>t} = \sum_{q \in \mathcal{T}_{ab},\ q>t} \log q.$$

Theorem 85.4 (Weighted Hall tail obstruction)**.** *Every fractional monotone endpoint transport satisfies*

$$U(F) \ge S_{>t} - T_{>t} \qquad (t \in \mathbb{R}).$$

Proof. Mass emitted by a source over $p > t$ can enter only a sink $q \ge p > t$. Consequently the total carried mass from that source tail is at most the total capacity $T_{>t}$ of the sink tail. Its unmatched mass is therefore at least $S_{>t} - T_{>t}$, and it is at most the total unmatched mass because every row deficit is nonnegative. ☐

We do not use a converse max-flow formula. The proved tail inequalities are necessary tests for any proposed transport construction.

Definition 85.5 (Uniform endpoint-flow gate)**.** For every $\varepsilon > 0$, require a constant C_ε such that every positive primitive triple admits a monotone fractional transport satisfying

$$\text{(EPF)} \qquad U(F) \le \varepsilon \log R + C_\varepsilon.$$

Theorem 85.6 (EPF implies standard *abc*)**.** *If the uniform endpoint-flow gate holds, then the standard abc conjecture holds.*

Proof. For the flow supplied by EPF, Theorem 85.3 gives

$$\log c \le \log R + U(F) \le (1+\varepsilon)\log R + C_\varepsilon.$$

Exponentiating yields the usual constant-times $\operatorname{rad}(abc)^{1+\varepsilon}$ inequality, with the same quantifier order. □

The gate EPF is refuted by the infinite prime-square family in Section 86. Two smaller complete-premise examples separately show why an integral replacement is too rigid. A *full integral matching* assigns every source token injectively to a sink token whose prime is at least the source prime.

Proposition 85.7 (Integral matching boundaries)**.** *For* $3+13=16$*, the source multiset is* $\{2,2,2\}$ *and the sink set is* $\{3,13\}$*, so no full integral matching exists. For* $9+16=25$*, the source has one token over* 5 *and the sinks are* $\{2,3\}$*, so no monotone edge exists, although*

$$K_c = 5 \le 6 = E_{ab}.$$

Proof. The first claim is the pigeonhole principle applied to three distinct layer tokens and two sinks. In the second example every available sink prime is strictly below 5. The displayed aggregate inequality follows from $\operatorname{rad}(25)=5$, $\operatorname{rad}(9)=3$, and $\operatorname{rad}(16)=2$. □

These witnesses refute universal full integral matching and the inference from $K_c \le E_{ab}$ to such a matching. On their own they do not settle the fractional epsilon gate; the later infinite family does.

There is also an infinite obstruction to removing the epsilon allowance from the scalar endpoint-core estimate.

Proposition 85.8 (No fixed zero-epsilon multiplier)**.** *Let* $X_k = 4^{3^k}$*. Then*

$$3^{k+1} \mid X_k - 1, \qquad 3^k \operatorname{rad}(X_k-1) \le X_k - 1,$$

and the primitive triple $(1, X_k-1, X_k)$ *satisfies*

$$\frac{K_{X_k}}{\operatorname{rad}(1)\operatorname{rad}(X_k-1)} > \frac{3^k}{2}.$$

Consequently, for every $K \in \mathbb{N}$ *there is a positive primitive triple with* $K_c > K \operatorname{rad}(a)\operatorname{rad}(b)$*.*

Proof. The divisibility is by induction. It is true at $k=0$ because $X_0 - 1 = 3$. If $3^{k+1} \mid X_k - 1$, then $X_k \equiv 1 \pmod 3$ and hence $3 \mid X_k^2 + X_k + 1$. The factorisation

$$X_{k+1} - 1 = X_k^3 - 1 = (X_k-1)(X_k^2+X_k+1)$$

supplies the next power of 3. Write $X_k - 1 = 3^{k+1}u$. Radical submultiplicativity and $\operatorname{rad}(u) \le u$ give

$$\operatorname{rad}(X_k-1) \le 3\operatorname{rad}(u) \le 3u,$$

which is the second inequality. Since X_k is a power of 2, $K_{X_k} = X_k/2$; therefore

$$\frac{K_{X_k}}{\operatorname{rad}(X_k-1)} \ge \frac{(X_k/2)3^k}{X_k-1} > \frac{3^k}{2}.$$

Choosing, for example, $k = 2K+1$ makes $3^k > 2K$ and proves the final claim. □

This is an infinite, full-quantifier counterexample to a fixed natural multiplier at epsilon zero. This particular family does not refute *abc* or the epsilon gate: the displayed certified logarithmic lower bound grows only linearly in k, while $\log X_k = 3^k \log 4$. The logically separate prime-square family in the next section does refute the ordered epsilon gate.

The companion module `ABCSignedEndpointPrimeTokenTransport20260903` formalises the exact core identity, finite weighted-flow accounting, Hall tail obstruction, the quantified implication EPF $\Rightarrow$ `ABCConjecture`, both finite integral counterexamples, and the elementary radical-budget mechanism used by the zero-epsilon family. The module does not postulate EPF. Its quantified negation and the common CT-3C obstruction are checked in the companion module of Section 86.

86. A common obstruction to ordered prime transport

The three-arm and endpoint constructions use the same hard orientation: positive mass may move from a source prime r to a sink prime q only when $r \leq q$. The next family tests that orientation with every quantifier and every arithmetic premise present.

For an odd prime p, put

$$P_p = (1, p^2 - 1, p^2).$$

These are positive primitive *abc* triples. Indeed, the additive relation is immediate, the unit arm is coprime to both other arms, and $\gcd(p^2 - 1, p^2) = 1$.

Lemma 86.1 (Prime separation). *If p is an odd prime and q is a prime divisor of $p^2 - 1$, then $q < p$.*

Proof. Since $p^2 - 1 = (p-1)(p+1)$, Euclid's lemma gives $q \mid p-1$ or $q \mid p+1$. The first alternative gives $q < p$. In the second, $p+1$ is an even composite integer greater than 2, so its prime divisor q is a proper divisor and hence $q \leq (p+1)/2 < p$. □

Theorem 86.2 (The ordered endpoint gate is false). *The uniform endpoint-flow gate* (EPF) *is false. More precisely, every monotone endpoint transport on P_p has*

$$U(F) = \log p.$$

Proof. The endpoint p^2 has one excess layer, at the prime p, so the source consists of one token of mass $\log p$. Every sink prime divides $p^2 - 1$ and is therefore smaller than p by Lemma 86.1. The order condition forbids every edge from the source token. Thus no source mass is carried and $U(F) = \log p$.

Moreover, radical multiplicativity and the elementary bound $\operatorname{rad}(n) \leq n$ give

$$R(P_p) = p \operatorname{rad}(p^2 - 1) \leq p(p^2 - 1) < p^3. \tag{OPS-1}$$

If EPF held at $\varepsilon = 1/4$ with constant C, then every odd prime p would satisfy

$$\log p = U(F) \leq \frac{1}{4} \log R(P_p) + C < \frac{3}{4} \log p + C.$$

This bounds $\frac{1}{4} \log p$ by C, contrary to the existence of arbitrarily large odd primes. □

Theorem 86.3 (The ordered three-arm gate is false). *The uniform three-arm complement-transport gate* (CT-3C) *is false. For every covering face of P_p and every one of its ordered complement transports,*

$$U(F, f) \geq \log p.$$

Proof. Suppose first that a face does not select the prime p on the C-arm. The A-arm has no prime vertex, the selected C-arm modulus is then 1, and the selected B-arm modulus divides $p^2 - 1$. Hence $M(F) \leq p^2 - 1 < p^2$, contradicting coverage. Every covering face must therefore select p on the C-arm. Its valuation is two, so this vertex creates a source token of mass $\log p$.

The A-arm has no complementary prime. The only C-arm prime has already been selected. Every remaining complementary prime is a divisor of $p^2 - 1$ and is smaller than p by Lemma 86.1. The distinguished source token can therefore send no mass, regardless of the other choices in the face. This proves $U(F, f) \geq \log p$. Combining the claimed CT-3C estimate at $\varepsilon = 1/4$ with (OPS-1) gives the same impossible bound $\frac{1}{4} \log p < C$ for all odd primes. □

The conditional implications

$$\text{EPF} \Longrightarrow abc, \qquad \text{CT-3C} \Longrightarrow abc$$

remain valid: the counterexample proves that their antecedents are false. It does not disprove standard *abc*, the labelled incidence complex, or a transport with a different geometry. Taking finitely many independent covering faces under the same hard order creates no admissible sink for the distinguished p-source; a repair must change either the edge geometry or the way different faces interact.

There is a useful exact description of the Hall obstruction. For a three-arm face and a threshold T, let $H_{>T}(F)$ be the logarithm of the selected full prime-power modulus contributed by selected primes above T, and let $R_{>T}(P)$ be the logarithm of the full radical contributed by all arm-labelled primes above T. Separating selected from unselected radical vertices gives

$$S_{>T}(F) - T_{>T}(F) = H_{>T}(F) - R_{>T}(P). \tag{OPS-2}$$

Indeed, a selected prime of valuation e contributes $(e-1)\log p$ to the source side, an unselected prime contributes $-\log p$, and adding and subtracting the selected $\log p$ gives the right-hand side. At $T = p-1$ for P_p, the right-hand side is $2\log p - \log p = \log p$. Thus the obstruction is an exact high-prime valuation surplus, not a failure of the max-flow algorithm.

Simply deleting the order condition is not by itself a new arithmetic advance. For a complete bipartite fractional flow with nonnegative source weights of total mass A and sink capacities of total mass B, the minimum unmatched mass is

$$\min U = (A-B)_+. \tag{OPS-3}$$

The capacity inequalities give the lower bound. If $0 < A \le B$, the product coupling $f_{ij} = a_i b_j / B$ fills every source; if $A > B > 0$, the coupling $f_{ij} = a_i b_j / A$ fills every sink and leaves $A - B$ unmatched. The zero cases are immediate. At the endpoint, (OPS-3) becomes

$$\min U = \big(\log c - \log R(P)\big)_+.$$

After harmless enlargement of the additive constant to be nonnegative, a uniform epsilon-bound for this quantity is equivalent to the usual scalar *abc* inequality. Unordered transport therefore supplies no independent mechanism.

A nontrivial successor must allow the downward edges exposed by P_p while retaining information beyond total mass. One candidate assigns a nonnegative relative displacement cost

$$\kappa(r,q) = \frac{(\log r - \log q)_+}{\log r}$$

to mass sent from r to q, and studies $U + \sum f(r,q)\kappa(r,q)$ on one or several faces. A uniform epsilon-small bound for that sum still implies *abc*, because the added cost is nonnegative. The family P_p no longer causes the formal "no admissible edge" failure. No uniform cost bound is asserted here. The next section performs the required adversarial test and refutes this exact relative-drop calibration on a different infinite family. Weighted multi-face and homological incidence constructions remain active only as precisely stated future interfaces.

The ordinary arguments in this section were fixed before formalization. The companion obstruction module is

`ABCThreeArmComplementTransportObstruction20260903`.

It checks the complete premises, both quantified negations, the exact tail identity (OPS-2), and their shared prime-square mechanism without assuming either transport gate or `ABCConjecture`. The costed bidirectional successor and its conditional implication are isolated in

`ABCBidirectionalPrimeTransportSuccessor20260903`.

The two modules expose respectively 37 and 44 public declarations, with one axiom query for each declaration. Their dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`; no custom axiom or placeholder is used. No uniform instance of the successor is postulated; its exact BEP instance is refuted in Section 87.

87. A square-root obstruction to relative-drop energy

The bidirectional endpoint model removes the false hard order and assigns an edge from p to q the relative drop

$$d(p,q) = \max\left\{0, \frac{\log p - \log q}{\log p}\right\}.$$

For a feasible flow f, its proposed energy is $E(f) = U(f) + \sum_{s,q} f(s,q)d(p_s,q)$. The prime-square family of Section 86 no longer fails because its downward edges are absent: all capacity-compatible edges now exist. A different infinite family shows that charging every downward edge at this scale is still too rigid.

First observe the exact finite optimization problem. Put

$$\rho(p,q) = 1 - d(p,q) = \min\left\{1, \frac{\log q}{\log p}\right\}.$$

If S is total source mass, then

$$E(f) = S - \sum_{s,q} \rho(p_s,q) f(s,q). \tag{BRE-1}$$

Thus minimizing energy is a maximum-reward transportation problem. Its dual minimizes

$$\sum_s w(s)u_s + \sum_q v(q)v_q \quad \text{subject to} \quad u_s + v_q \geq \rho(p_s,q), \qquad u_s, v_q \geq 0.$$

For one source, the optimum fills sink capacities in nonincreasing prime order until either the source or all sinks are exhausted. This follows by moving any mass on a lower-reward sink to an unused higher-reward sink. The description is exact and supplies finite primal–dual certificates; it does not imply an asymptotic estimate.

Choose an arbitrarily large prime $p \equiv 1 \pmod 4$. Fermat's two-square theorem gives

$$p = m^2 + n^2, \qquad m > n > 0.$$

Primality makes m and n coprime, and their parities are opposite. Put

$$X = m^2 - n^2, \qquad Y = 2mn.$$

Then $X^2 + Y^2 = p^2$ and $\gcd(X,Y) = 1$, so

$$Q_p = (X^2, Y^2, p^2) \tag{BRE-2}$$

is a positive primitive *abc* triple.

Lemma 87.1 (Square-root support). *Every prime q dividing XY satisfies*

$$q \leq m + n, \qquad q^2 \leq 2p.$$

Proof. The factorizations $X = (m-n)(m+n)$ and $Y = 2mn$ show that q divides one of $m-n, m+n, 2, m, n$. Each is at most $m+n$. Finally,

$$(m+n)^2 = m^2 + n^2 + 2mn \leq 2(m^2+n^2) = 2p.$$

□

Theorem 87.2 (Bidirectional relative-drop obstruction). *The gate* (BEP) *is false. It is refuted at $\varepsilon = 1/12$ by the family* (BRE-2)*.*

Proof. The endpoint p^2 has exactly one excess token, of capacity $L = \log p$. Every sink prime divides XY, and the lemma gives

$$\log q \leq \frac{L + \log 2}{2}, \qquad d(p,q) \geq \delta_p := \frac{1}{2} - \frac{\log 2}{2L}.$$

For $p \geq 5$, one has $0 < \delta_p < 1$. If an arbitrary flow carries total mass W, its single row gives $0 \leq W \leq L$, while

$$U = L - W, \qquad D \geq \delta_p W.$$

Consequently

$$E \geq L - (1-\delta_p)W \geq \delta_p L = \frac{L - \log 2}{2}. \tag{BRE-3}$$

The radical of (BRE-2) satisfies

$$\operatorname{rad}(X^2Y^2p^2) \leq XYp \leq p^3,$$

because $X, Y \leq p$. Hence its conductor is at most $3L$. If BEP held at $\varepsilon = 1/12$ with constant C, (BRE-3) would imply

$$\frac{L - \log 2}{2} \leq E \leq \frac{1}{12}(3L) + C,$$

or $L/4 \leq C + (\log 2)/2$. Primes congruent to one modulo four are unbounded, so this is impossible. □

The proof bounds every admissible flow and uses an infinite family with all positivity, addition, and coprimality premises present. It therefore retires the exact relative-drop energy gate. It does not refute standard *abc*, bidirectional transport as a parent construction, or a differently calibrated packet or homological cost.

The mechanism suggests the correction. Prime-hypotenuse Pythagorean squares naturally place leg factors at square-root scale. A candidate that assigns a fixed positive proportional charge to every edge down to that scale will fail for the same reason. The next successor must aggregate several sinks into a packet, record an algebraic degree, or permit cancellation across an incidence boundary. A quadratic calibration such as

$$\max\left\{0, \frac{\log p - 2\log q}{\log p}\right\}$$

passes this particular scale test, but its uniform estimate is not asserted and must be subjected to the same complete-premise audit.

The ordinary proof above precedes its Lean realization in

`ABCBidirectionalEnergyPythagoreanObstruction20260903`.

The companion audit queries every public declaration. The module proves the unbounded prime-hypotenuse construction, the support and energy bounds, and the quantified negation of BEP without assuming `ABCConjecture`. The ordinary estimate above is the sharper square-root bound and gives $\varepsilon = 1/12$; the Lean proof uses the simpler integral consequence $q^4 \leq p^3$, obtains energy at least $(\log p)/4$, and closes the same negation at $\varepsilon = 1/24$. The module exposes 40 public declarations and its audit has exactly 40 corresponding queries; the dependency union is only `propext`, `Classical.choice`, and `Quot.sound`.

88. Prime-packet boundary transport

The two transport obstructions distinguish individual edge geometry from aggregate radical capacity. We next aggregate all excess layers at one endpoint prime into one source and assign each external radical prime, as an indivisible sink, to at most one source packet. Several small primes may therefore act together, and no cost is attached to their individual distance from the source.

Let finite sources have weights $x_i \geq 0$ and finite sinks have weights $y_j \geq 0$. A packet assignment is a map

$$o : J \longrightarrow I \sqcup \{\varnothing\}.$$

Put

$$M_i(o) = \sum_{o(j)=i} y_j, \qquad B_i(o) = (x_i - M_i(o))_+, \qquad B(o) = \sum_i B_i(o).$$

Proposition 88.1 (Packet boundary inequality)**.** *Every packet assignment satisfies*

$$\sum_i x_i - \sum_j y_j \leq B(o). \tag{PBT-1}$$

Proof. The packets are disjoint, so $\sum_i M_i(o)$ is at most the total sink weight. For each source, $x_i - M_i(o) \leq (x_i - M_i(o))_+$. Summing these inequalities gives

$$\sum_i x_i - \sum_j y_j \leq \sum_i (x_i - M_i(o)) \leq \sum_i B_i(o).$$

□

For an abc point $P = (a, b, c)$, use one source for each prime $p \mid c$, with aggregate weight

$$x_p = (v_p(c) - 1) \log p,$$

and one indivisible sink for each prime $q \mid ab$, with weight $\log q$. Canonical factorization gives

$$\sum_{p|c} x_p = \log \frac{c}{\operatorname{rad}(c)}, \qquad \sum_{q|ab} \log q = \log \operatorname{rad}(ab).$$

Together with (PBT-1), the exact endpoint balance yields

$$\text{(PBT-2)} \qquad \log c \le \log \operatorname{rad}(abc) + B(o).$$

The packet-boundary gate PBT asks that, for every $\varepsilon > 0$, one constant C_ε work for all positive primitive P, with a packet assignment such that

$$\text{(PBT)} \qquad B(o) \le \varepsilon \log \operatorname{rad}(abc) + C_\varepsilon.$$

Theorem 88.2 (Packet boundary implies standard abc)**.** *PBT implies the standard abc conjecture.*

Proof. Substitute (PBT) into (PBT-2). Since $a + b = c$ and all coordinates are positive, c is their maximum. The resulting inequality has exactly the standard epsilon–constant quantifier order. □

PBT is not the divisible unordered relaxation pointwise. Two sources of weight one and a single indivisible sink of weight two have equal total source and sink mass. Divisible flow has zero residual, whereas a packet assignment can cover at most one source; its optimal packet boundary is one. Thus the packet objective cannot be recovered from the two total masses alone. We do not claim that this proves logical independence of its uniform arithmetic gate from abc.

The two preceding counterexample mechanisms do not automatically refute PBT. Whenever $c = p^2$, there is one aggregate source of weight $\log p$. Assigning all external primes to its packet gives exactly

$$B = (\log p - \log \operatorname{rad}(ab))_+ .$$

For both $(1, p^2 - 1, p^2)$ and the prime-hypotenuse Pythagorean-square family, the individual inequalities $q < p$ or $q \le \sqrt{2p}$ create neither a forbidden edge nor a per-edge charge. This observation does not prove PBT: an infinite family with small aggregate external radical, or competition among several powerful endpoint primes, may still refute it.

The next positive problem is therefore an arithmetic logarithmic bin-covering theorem: partition the distinct primes of ab among the powerful primes of c while leaving only epsilon-small uncovered aggregate weight. The next counterexample search must optimize the full indivisible assignment on an unbounded family; a finite bad value is absorbed by C_ε and retires nothing. If the bare packet gate fails, congruence labels or a genuine signed multi-face boundary must be added without discarding the parent construction.

The ordinary proof was completed before the Lean module

`ABCPrimePacketBoundaryTransportSuccessor20260903`.

That module checks the finite boundary inequality, endpoint specialization, conditional implication, a two-source noncollapse certificate, and the one-source formula. PBT itself is a definition and has no asserted inhabitant in this construction. The next section gives a complete counterexample to that inhabitance claim. The module exposes 39 public declarations and its audit has exactly 39 corresponding queries; the dependency union is only `propext`, `Classical.choice`, and `Quot.sound`.

89. A Linnik obstruction to exclusive prime packets

The packet-boundary gate of the preceding section has a second, genuinely different obstruction. Its exclusive ownership rule prevents one external prime from recording simultaneous congruences at several powerful endpoint primes. Linnik's least-prime theorem turns this defect into an unconditional infinite counterfamily.

For a finite packet problem, let the source and sink weights be $x_i \geq 0$ and $y_j \geq 0$, and put $X = \sum_i x_i$, $Y = \sum_j y_j$. If M_i is the mass assigned to source i, then

$$\sum_i (x_i - M_i)_+ = X - \sum_i \min\{x_i, M_i\}. \tag{LPO-1}$$

This is the clipped-reward identity. In particular, if there are k selected sources with weights $z_1 \geq \cdots \geq z_k$ and only t positive sinks, disjoint ownership lets at most t of those sources receive positive mass. Every assignment therefore has residual at least

$$\sum_{i=t+1}^{k} z_i. \tag{LPO-2}$$

For one sink of weight y, the exact optimum is

$$R_* = X - \max_i \min\{x_i, y\}. \tag{LPO-3}$$

These formulas isolate a cardinality loss which is absent from divisible transport.

We use Linnik's unconditional theorem in the following uniform form: there are absolute constants $C \geq 1$ and $L \geq 1$ such that the least prime in any reduced residue class $a \pmod{Q}$ is at most CQ^L. This is precisely the uniform polynomial estimate proved by Linnik and sharpened, with an explicit admissible exponent, by Xylouris [83, Theorem 1.1]. No optimal value of L is needed.

Let

$$p_1 < p_2 < \cdots < p_k, \qquad M_k = \prod_{i=1}^{k} p_i, \qquad Q_k = M_k^2,$$

where the p_i are the first k primes. Apply Linnik's theorem to the reduced class $-1 \pmod{Q_k}$ and choose a prime ℓ_k with

$$\ell_k \equiv -1 \pmod{M_k^2}, \qquad \ell_k \leq CM_k^{2L}. \tag{LPO-4}$$

Then

$$P_k = (1, \ell_k, \ell_k + 1) \tag{LPO-5}$$

is a positive primitive *abc* point. The family is infinite because $\ell_k + 1 \geq M_k^2$.

Lemma 89.1 (One sink and many compulsory sources)**.** *Every packet assignment at P_k has residual*

$$R(P_k) \geq \log M_k - \log p_k. \tag{LPO-6}$$

Proof. The external product is the prime ℓ_k, so there is exactly one sink. For every $i \leq k$, (LPO-4) gives $p_i^2 \mid \ell_k + 1$; hence the aggregate source at p_i has weight

$$(v_{p_i}(\ell_k + 1) - 1) \log p_i \geq \log p_i.$$

The unique sink can belong to at most one source packet. Apply (LPO-2) and discard at most the largest of the k guaranteed weights. □

The conductor remains comparable with $\log M_k$. Indeed,

$$\begin{aligned} r(P_k) &\leq \log\bigl(\ell_k(\ell_k + 1)\bigr) \\ &\leq 2\log \ell_k + \log 2 \leq 4L \log M_k + B, \end{aligned} \tag{LPO-7}$$

where $B = 2\log C + \log 2$. At the same time,

$$\frac{\log p_k}{\log M_k} \longrightarrow 0. \tag{LPO-8}$$

For completeness, Bertrand's postulate gives $p_k < 2^k$ for large k, while $p_i \geq i + 1$ gives $M_k \geq (k+1)!$; the last half of the factorial factors already proves (LPO-8).

Theorem 89.2 (Exclusive packet obstruction)**.** *The gate* PBT *is false.*

Proof. Set $\varepsilon_0 = 1/(8L)$. Combining (LPO-6) and (LPO-7) gives

$$R_*(P_k) - \varepsilon_0 r(P_k) \geq \frac{1}{2}\log M_k - \log p_k - \frac{B}{8L}. \tag{LPO-9}$$

By (LPO-8), the right side tends to infinity. Thus every proposed additive constant fails at some P_k, even after minimizing over all assignments. This has exactly the negated epsilon–constant–point–assignment quantifier order of PBT. □

This is not an *abc* counterexample. Since M_k is even, $\operatorname{rad}(\ell_k+1) \geq 2$, and therefore

$$\operatorname{rad}\big(\ell_k(\ell_k+1)\big) = \ell_k \operatorname{rad}(\ell_k+1) \geq 2\ell_k \geq \ell_k + 1. \tag{LPO-10}$$

Hence every point in the family satisfies $h(P_k) \leq r(P_k)$ and has zero positive scalar defect. PBT fails solely through its indivisibility gap. Indeed, if $\Delta(P) = (h(P) - r(P))_+$ and $G(P) = R_*(P) - \Delta(P)$, then the exact identity $R_* = \Delta + G$ shows that PBT is equivalent to standard logarithmic *abc* together with a uniform sublinear bound on G. The Linnik family refutes the second conjunct while satisfying the first with constant zero.

The correct structural lesson is that one external prime may encode many CRT incidences at once:

$$\ell_k \equiv -1 \pmod{p_i^2} \qquad (1 \leq i \leq k).$$

A successor must permit this shared incidence while charging its logarithmic capacity only once. Allowing arbitrary divisible sharing without incidence data returns to the scalar defect, so neither repair alone is sufficient. The next positive target is a shared CRT-incidence boundary with a pointwise height bridge and an audit showing zero artificial loss on (LPO-5).

The ordinary proof above uses the published Linnik theorem. The current Mathlib tree does not contain that analytic theorem. The companion Lean module therefore checks the finite packet identities, the prime-neighbour arithmetic core, and the implication from a transparent Linnik-style escape hypothesis to the negation of PBT; it does not insert Linnik's theorem as a project axiom or mislabel the resulting source boundary as kernel-closed.

The accompanying exact computation independently enumerates all 1,368,094 normalized primitive triples with $c \leq 3000$. It finds 572 strict packet fragmentation gaps, 567 of them with zero scalar defect, and replays all optima with a second state-set algorithm. Five finite square-primorial prime-predecessor rows exhibit the same one-sink mechanism. These data are regression and discovery evidence only; Theorem 89.2 uses the infinite Linnik family and does not extrapolate from the scan.

90. Shared CRT incidence after exclusive packets

The Linnik counterfamily to exclusive prime packets identifies an arithmetic loss rather than a shortage of total radical mass. One prime $\ell \equiv -1 \pmod{M^2}$ certifies all congruences modulo the prime-power factors of M^2 simultaneously. Its capacity $\log \ell$ must therefore be shareable, but it may be charged only once. This section gives a first capacity-correct shared-incidence successor, proves its height bridge, and separates it from the scalar relaxation. Its uniform estimate remains open.

Let $P = (a, b, c)$ be a positive primitive point:

$$a, b, c \in \mathbb{Z}_{>0}, \qquad a + b = c, qquad \gcd(a,b) = \gcd(b,c) = \gcd(c,a) = 1.$$

Put

$$\operatorname{core}(n) = \frac{n}{\operatorname{rad}(n)}, \qquad h(P) = \log c, \qquad r(P) = \log \operatorname{rad}(abc),$$

and define the endpoint sources and external sinks by

$$I = I(P) = \{p : p \mid c,\ v_p(c) \geq 2\}, \qquad J = J(P) = \{q : q \mid ab\}.$$

For $e_p = v_p(c)$, set

$$x_p = (e_p - 1)\log p, \qquad y_q = \log q,$$

and write

$$X = \sum_{p\in I} x_p = \log \operatorname{core}(c), \qquad Y = \sum_{q\in J} y_q = \log \operatorname{rad}(ab). \tag{SCRT-1}$$

For $S \subseteq I$ and $T \subseteq J$, let

$$\begin{gathered} m(S) = \prod_{p\in S} p^{e_p}, \\ a_T = \prod_{\substack{q\in T \\ q\mid a}} q^{v_q(a)}, \qquad b_T = \prod_{\substack{q\in T \\ q\mid b}} q^{v_q(b)}, \\ x(S) = \sum_{p\in S} x_p, \qquad y(T) = \sum_{q\in T} y_q. \end{gathered} \tag{SCRT-2}$$

The empty products are one. Thus a_T, b_T are unitary divisors carrying the complete valuation of every selected external prime.

Definition 90.1 (Shared CRT block)**.** A pair (S, T) with $\varnothing \neq S \subseteq I$ and $\varnothing \neq T \subseteq J$ is *CRT compatible* when

$$m(S) \mid a_T + b_T. \tag{SCRT-3}$$

It is *saturated* when also

$$x(S) \leq y(T). \tag{SCRT-4}$$

A saturated compatible pair is a *shared CRT block*.

Compatibility is arithmetic, rather than merely an incidence convention: the truncated relation $a_T + b_T$ simultaneously certifies every prime-power modulus indexed by S. When I and J are nonempty, the full pair is always compatible because $a_J = a$, $b_J = b$, and $m(I) \mid c$.

Definition 90.2 (SCRT-0 configuration)**.** An SCRT-0 configuration consists of a finite family $\mathcal{H}$ of saturated blocks (S_ν, T_ν) whose source sets are pairwise disjoint and whose sink sets are pairwise disjoint. Put

$$I_0 = I \setminus \bigcup_\nu S_\nu, \qquad J_0 = J \setminus \bigcup_\nu T_\nu.$$

The remaining data are an exclusive assignment

$$o : J_0 \longrightarrow I_0 \sqcup \{\varnothing\}.$$

Thus a residual sink is unused or belongs to one residual source, while several sinks may belong to the same source. For $p \in I_0$, put

$$M_p(o) = \sum_{\substack{q\in J_0 \\ o(q)=p}} y_q, \qquad R_p(o) = \max\{x_p - M_p(o), 0\},$$

and define

$$B(\mathcal{H}, o) = \sum_{p\in I_0} R_p(o), \qquad B_{\mathrm{SCRT}}(P) = \min_{(\mathcal{H},o)} B(\mathcal{H}, o). \tag{SCRT-5}$$

The minimum exists because the underlying sets are finite; the empty block family with every sink unused is an admissible configuration. Sources in a saturated block have zero boundary. Choosing a block consumes all of $y(T_\nu)$, including any surplus over $x(S_\nu)$.

The sink union T_ν occurs in exactly one block, and its total credit is at most $y(T_\nu)$. Every residual sink occurs in at most one exclusive packet. Thus no logarithmic capacity is counted twice.

Proposition 90.3 (Once-charged height bridge)**.** *Every SCRT-0 configuration satisfies*

$$X \le Y + B(\mathcal{H}, o) \tag{SCRT-6}$$

and consequently

$$h(P) \le r(P) + B(\mathcal{H}, o). \tag{SCRT-7}$$

Proof. Saturation gives $x(S_\nu) \le y(T_\nu)$ block by block. At every remaining source, $x_p \le M_p(o) + R_p(o)$. Disjoint blocks and exclusive ownership give

$$X \le \sum_\nu y(T_\nu) + \sum_{p \in I_0} M_p(o) + B(\mathcal{H}, o) \le Y + B(\mathcal{H}, o),$$

which is (SCRT-6). Prime factorisation gives

$$h(P) - r(P) = \log \operatorname{core}(c) - \log \operatorname{rad}(ab) = X - Y,$$

which proves (SCRT-7). □

Let $B_{\mathrm{PBT}}(P)$ denote the optimum of the earlier exclusive packet model and put $\Delta(P) = \max\{X - Y, 0\}$.

Proposition 90.4 (Sandwich and easy strata)**.** *For every positive primitive point,*

$$\Delta(P) \le B_{\mathrm{SCRT}}(P) \le B_{\mathrm{PBT}}(P). \tag{SCRT-8}$$

Moreover, $X \le Y$ implies $B_{\mathrm{SCRT}}(P) = 0 = \Delta(P)$, and $|I| \le 1$ implies $B_{\mathrm{SCRT}}(P) = \Delta(P)$.

Proof. Nonnegativity and (SCRT-6) give the lower bound. Taking no shared blocks recovers exactly the PBT assignment problem, giving the upper bound. If $I = \varnothing$, then $X = 0$ and the empty configuration has zero boundary. If $I \ne \varnothing$ and $X \le Y$, then $J \ne \varnothing$ and the full pair (I, J) is a valid saturated block, again giving zero boundary. Finally, if I has one member, assign every sink to that source; the resulting residual is $\max\{X - Y, 0\}$ and attains the lower bound. □

Thus an additional shared-incidence gap can occur only when $X > Y$ and at least two endpoint primes carry positive excess weight.

Definition 90.5 (Uniform shared-CRT gate)**.** SCRT-0 is the statement

$$\forall \varepsilon > 0\ \exists C_\varepsilon\ \forall P\ \exists (\mathcal{H}, o) : \quad B(\mathcal{H}, o) \le \varepsilon r(P) + C_\varepsilon. \tag{SCRT-0}$$

Here P ranges over all positive primitive *abc* points.

Theorem 90.6 (Conditional reduction to *abc*)**.** *If SCRT-0 holds, then the standard logarithmic abc conjecture holds.*

Proof. For the configuration supplied by SCRT-0, Proposition 90.3 gives

$$h(P) \le r(P) + B(\mathcal{H}, o) \le (1 + \varepsilon) r(P) + C_\varepsilon.$$

This has the standard quantifier order. □

No instance of SCRT-0 is assumed or proved here.

Both clauses in the definition are necessary. If the capacity of one shared sink union were repeated once per incident source, the point $(1, 224, 225)$ would be assigned zero boundary: its two source weights are $\log 3, \log 5$ and its external mass is $\log 14$, available through the full relation. Yet

$$h - r = \log(15/14) > 0,$$

so the height bridge would fail. At the other extreme, if the full block may distribute arbitrary partial credits z_p with $\sum z_p \le Y$, its optimal boundary is

$$X - \min(X, Y) = (X - Y)_+. \tag{SCRT-9}$$

The unrestricted model is therefore exactly the scalar defect. In the reverse logical implication from *abc*, the additive constant may first be replaced by its maximum with zero before taking positive parts.

SCRT-0 is strictly different pointwise. At

$$(343, 625, 968) = (7^3, 5^4, 2^3 11^2),$$

the source weights are $\log 4, \log 11$ and the sink weights are $\log 5, \log 7$. The joint block is unsaturated because $\log 44 > \log 35$. The saturated blocks are

$$(\{2\}, \{7\}), \quad (\{2\}, \{5, 7\}), \quad (\{11\}, \{5, 7\}),$$

with resulting best residuals $\log(11/5), \log 11, \log 4$, respectively. The empty block family and split assignment $5 \mapsto 2$, $7 \mapsto 11$ attain $\log(11/7)$, and the finite list of other assignments is worse. Thus

$$B_{\mathrm{SCRT}}(343, 625, 968) = \log(11/7) > \log(44/35) = \Delta. \tag{SCRT-10}$$

The recorded infinite counterfamilies do not reproduce their earlier artificial losses. Both $(1, p^2 - 1, p^2)$ and the prime-hypotenuse Pythagorean-square family have one endpoint source, so $B_{\mathrm{SCRT}} = \Delta$. The same holds for the fixed-prime family $(1, p^N - 1, p^N)$. In the Linnik family $(1, \ell, \ell + 1)$, the single sink ℓ gives a full shared block: since ℓ is odd,

$$\operatorname{core}(\ell + 1) \le (\ell + 1)/2 \le \ell,$$

so that block is saturated and has zero boundary. For the shifted family $(2, \ell, \ell+2)$ with $\ell \equiv -2 \pmod{M^2}$ and M odd, the full sink union $\{2, \ell\}$ likewise closes every forced source once; indeed $\operatorname{core}(\ell+2) \le (\ell+2)/3 \le \ell$. For $(4^r, 3^r, 4^r + 3^r)$, the full block works whenever $\operatorname{core}(4^r + 3^r) \le 6$, and a single repeated endpoint prime again gives $B_{\mathrm{SCRT}} = \Delta$. The earlier balanced-family obstruction does not prove an infinite subsequence with several powerful endpoint primes and a bad SCRT partition, so it does not refute this gate.

A harder child model, SCRT-SAT, deletes the exclusive residual stage and charges every source outside saturated blocks in full. It is false. For

$$n = 284 + 310t, \qquad P_n = (2, 15^n - 2, 15^n),$$

direct modular exponentiation gives $15^{284} \equiv 2 \pmod{31^2}$ and $15^{310} \equiv 1 \pmod{31^2}$. Hence $31^2 \mid 15^n - 2$ and

$$Y = \log\bigl(2 \operatorname{rad}(15^n - 2)\bigr) < \log(2 \cdot 15^n/31) < (n - 1)\log 15 = X. \tag{SCRT-11}$$

Disjoint saturated blocks cannot close both endpoint sources, so SCRT-SAT leaves at least $(n - 1)\log 3$. Since its conductor is less than $n \log 15$, every fixed $0 < \varepsilon < \log 3/\log 15$ refutes its uniform gate. SCRT-0 retains the external prime-log partition and therefore is not refuted by this argument; (SCRT-11) alone supplies only the constant total-capacity gap $X - Y > \log(31/30)$.

The saturated rule currently consumes the whole capacity $y(T)$, including surplus beyond $x(S)$. This is a design choice. A surplus-reuse successor would need a proved fractional-cover or signed-chain identity which keeps each sink charged once without admitting the unrestricted pooling in (SCRT-9).

The Lean module `ABCSharedCRTIncidenceSuccessor20260903` formalises the abstract once-charged certificate and the mass inequality (SCRT-6). It proves the endpoint height bridge and the conditional implication to `ABCConjecture`. It also checks the duplicated-capacity failure and the scalar optimum (SCRT-9). The module has 30 public declarations and a one-for-one 30-query axiom audit whose union is only `propext`, `Classical.choice`, and `Quot.sound`. The arithmetic CRT-block configuration, the optimum (SCRT-10), and the SCRT-SAT counterfamily are ordinary proofs at this checkpoint and are not represented by axioms or opaque gate inhabitants. Proving or refuting the uniform SCRT-0 estimate is the next unresolved step.

91. Primary-literature gate audit through September 2026

We re-audited primary sources available through September 3, 2026 against the exact gates used in this paper. This is a theorem-by-theorem compatibility check, not a survey by title or terminology. No checked source supplies a generally accepted unconditional proof or disproof of standard abc, and no source closes one of the remaining repository gates without an additional uniformity statement.

Laniewski's parity coordinates give the clearest direct match to the canonical defect route [25, Proposition 3.7 and Corollary 3.9]. For $m = \log(abc)$, $r = \log \operatorname{rad}(abc)$, and $h = \log c$, the cited identity is exactly

$$q = 1 + \frac{(m-r)-(m-h)}{r} = 1 + X - Y.$$

The equality follows by expanding the numerator: $(m-r)-(m-h) = h-r$. It therefore confirms the coordinate choice used in the canonical gain section, but it gives no upper bound for $X-Y$. Laniewski's Theorem 4.3 similarly gives an exact parity-class transgression threshold in terms of linear radical defect and the smaller summand. It is a certificate for transgression, not an estimate excluding it. The conditional dichotomy in Theorem 10.5 starts from a hypothetical fixed-epsilon transgressive sequence and hence cannot supply that missing estimate.

The exceptional-set theorems of Bernert–Browning–Lichtman–Teräväinen give, in their notation,

$$N_\lambda(X) \ll_{\lambda,\varepsilon} X^{(23\lambda+3)/40+\varepsilon}$$

and a stronger fixed-$\lambda < 1$ exponent $3/5 + \varepsilon$ [7, Theorems 1.2–1.3]. These are unconditional aggregate counts. They do not imply a pointwise bound on $X - Y$, an eventual compensated packet estimate, or a fixed-index Pell valuation theorem. The distinction between aggregate scarcity and an all-triples uniform inequality is retained throughout our formal interfaces.

Baek and Lee prove Mason–Stothers over arbitrary fields with the necessary alternative that all three derivatives vanish [9, Theorem 1]. Their formal development is the source of the Mathlib theorem `Polynomial.abc` already used by our `MasonSpecializationBarrier` module. In positive characteristic the triple

$$-1, \qquad -x^p, \qquad (1+x)^p$$

has vanishing derivatives, so deleting that branch would be false. The polynomial theorem has no specialization mechanism that turns it into the needed integer radical inequality; the repository's specialization barrier therefore remains.

For the Pell route, Laniewski's Theorem 4.31 provides a positive-density composite-index family in which quality exceeding one is detected by non-squarefreeness of a Pell-coordinate product. Non-squarefree is weaker than squarefull. At its only odd-prime seed $\ell = 7$, the repository's exact factorization is

$$A_7B_7 = 239 \cdot 13^2, \qquad v_{239}(A_7B_7) = 1.$$

Thus this row satisfies the non-squarefree condition but not the simultaneous squarefull gate. Sanna's valuation formulas determine later valuations once the rank-of-apparition valuation is known [36]; Carmichael's primitive-divisor theorem guarantees a primitive divisor [40], but being primitive does not imply exponent one. These results leave exactly the first-rank exponent problem isolated by the signed trace module.

Browning and Verzobio obtain power-saving counts for sums of three powerful numbers in a broad log-general-type range [8, Theorem 1.2]. The squarefull exponent triple $(2, 2, 2)$ relevant to the elementary Pell surface is explicitly outside that range. Their theorem is therefore a valid aggregate input in its stated domain, but not a squarefull-Pell exclusion.

The alternative-quality identity $q_{\mathrm{std}} = \eta q_{\mathrm{DGM}}$ remains usable [26]. Divergence of q_{DGM} alone does not control the product. The claimed later phase-boundary transfer uses a one-sided estimate as an equality and applies a fixed-support lemma in a growing-support regime; we do not import those conclusions. Likewise, the fixed-epsilon exceptional-set claim in Carella depends on replacing an $O(h)$ error by $O(h\rho(u))$ while $\rho(u) \to 0$ [14, Lemma 4.2]. That step does not follow. These findings reject only the dependent inferences, not the broader smooth-neighbour or alternative-metric routes. The exact gain factorization in Müller–Taktikos remains motivational, but the preprint changes the approximation-gain numerator between its definition and subsequent theorem and supplies no standard-abc coefficient-one estimate [27].

Finally, IUT III Corollary 3.12 is stated inside the situation of Theorem 3.11 and the preceding IUT constructions [62]. Zhou's explicit rational inequalities expressly invoke its μ_6 version [31]. Published sources therefore exist, but they do not eliminate the source-faithful obligation carried by our IUT-facing route: all places, the Ind1–Ind3 action, actual hulls and integral orders, and pointed same-pilot transport must inhabit one compatible interface. That chain remains open in this repository; no source-dependent conclusion is silently promoted to an independent Lean theorem.

The detailed version-locked ledger, with proposition and page references, is `research/ABC_PRIMARY_LITERATURE_GATE_AUDIT_2026_09_03.md`. Its classifications are local: a flawed inference retires that inference only, while every unrefuted parent route continues.

92. Steinberg valuation contact surfaces and a Veronese obstruction

Author: ChatGPT. This section constructs a two-dimensional valuation invariant of a primitive abc point, proves its affine and five-term identities, and audits two proposed closure estimates against full-premise counterexamples. No form of the abc conjecture, Vojta's conjecture, or an IUT comparison is assumed.

92.1. The divisor triangle.

Let

$$\mathrm{Div}_{\mathbf{Q}} = \bigoplus_{p \text{ prime}} \mathbf{Z}[p], \qquad d(q) = \sum_p v_p(q)[p].$$

For a positive primitive equation $a + b = c$, put $A = d(a)$, $B = d(b)$, $C = d(c)$, and $x = a/c$. Since $1 - x = b/c$, define the oriented valuation contact surface

$$\Omega(A, B, C) = (A - C) \wedge (B - C) = d(x) \wedge d(1 - x). \tag{474}$$

Proposition 92.1 (Three-leg expansion and affine covariance)**.** *One has*

$$\Omega(A, B, C) = A \wedge B + B \wedge C + C \wedge A.$$

Moreover, for every divisor vector T, integer m, and permutation $\sigma \in S_3$,

$$\Omega(A + T, B + T, C + T) = \Omega(A, B, C),$$
$$\Omega(mA, mB, mC) = m^2 \Omega(A, B, C),$$
$$\Omega(A_{\sigma(1)}, A_{\sigma(2)}, A_{\sigma(3)}) = \mathrm{sgn}(\sigma)\Omega(A, B, C).$$

Proof. Bilinearity and alternation give

$$(A - C) \wedge (B - C) = A \wedge B - A \wedge C - C \wedge B.$$

This is the displayed three-leg sum. Translation cancels in both differences, scaling factors from both exterior arguments, cyclic permutations preserve the three-leg sum, and a transposition reverses it. □

Primitivity makes the prime supports S_a, S_b, S_c pairwise disjoint. If $w_p \geq 0$ and $L_w(A) = \sum_{p \in S_a} v_p(a) w_p$, define

$$\left\| \sum_{p<q} z_{pq}[p] \wedge [q] \right\|_{1,w} = \sum_{p<q} |z_{pq}| w_p w_q.$$

Proposition 92.2 (Exact mixed contact area)**.** *For a primitive abc point,*

$$\|\Omega(A, B, C)\|_{1,w} = L_w(A)L_w(B) + L_w(B)L_w(C) + L_w(C)L_w(A).$$

Proof. The three terms of Proposition 92.1 lie on the disjoint rectangles $S_a \times S_b$, $S_b \times S_c$, and $S_c \times S_a$. On each rectangle the absolute coefficient is a product of its two positive valuation coefficients. The double sum therefore factors, and the three rectangle norms add. □

With $w_p = \log p$, write

$$\Phi = \log a \log b + \log b \log c + \log c \log a. \tag{475}$$

If $r_a = \log \operatorname{rad}(a)$ and similarly for the other legs, the radical skeleton area is

$$\Psi = r_a r_b + r_b r_c + r_c r_a, \qquad 3\Psi \le (r_a + r_b + r_c)^2.$$

Writing $\log a = r_a + \delta_a$ and cyclically gives the exact positive excess

$$\Phi - \Psi = r_a\delta_b + \delta_a r_b + \delta_a\delta_b + r_b\delta_c + \delta_b r_c + \delta_b\delta_c + r_c\delta_a + \delta_c r_a + \delta_c\delta_a.$$

92.2. **Five-term gluing before the Milnor quotient.** For rational x, y in the domain of the following expressions, set

$$z_1 = x, \quad z_2 = y, \quad z_3 = y/x, \quad z_4 = \frac{y(1-x)}{x(1-y)}, \quad z_5 = \frac{1-x}{1-y}.$$

Theorem 92.3 (Five-term contact relation)**.**

$$\Omega(z_1) - \Omega(z_2) + \Omega(z_3) - \Omega(z_4) + \Omega(z_5) = 0.$$

Proof. Put $X = d(x)$, $U = d(1-x)$, $Y = d(y)$, $V = d(1-y)$, and $Z = d(x-y)$. Multiplicativity of divisors and $d(-1) = 0$ give the five pairs

$$(X, U),\ (Y, V),\ (Y - X, Z - X),\ (Y + U - X - V, Z - X - V),\ (U - V, Z - V).$$

After inserting the alternating signs, bilinear expansion cancels every basis wedge. $\square$

The terminology is deliberate. Equation (474) is the image under $\bigwedge^2 d$ of the standard Bloch boundary $\delta([x]) = x \wedge (1-x)$. It is not a new nonzero Milnor K_2 invariant: the Milnor quotient imposes $\{x, 1-x\} = 0$. We work before that quotient. Theorem 92.3 allows one contact cell to be replaced by four auxiliary cells and exposes the new divisor $d(x-y)$.

Proposition 92.4 (Quadratic Veronese peeling)**.** *For $x \in \mathbf{Q} \setminus \{0, 1, -1\}$,*

$$\Omega(x^2) = 2\Omega(x) - 2\Omega\left(\frac{x}{1+x}\right).$$

If $x = X/Z$ with $0 < X < Z$ and $\gcd(X, Z) = 1$, this becomes

$$\Omega(X^2, Z^2 - X^2, Z^2) = 2\Omega(X, Z - X, Z) - 2\Omega(X, Z, X + Z).$$

Proof. Set $y = x^2$ in Theorem 92.3. The five arguments are $x, x^2, x, x/(1+x), 1/(1+x)$. The last is the complement of the fourth, and $\Omega(1-z) = -\Omega(z)$. Substitution and rearrangement give the first identity. The second follows by putting $x = X/Z$ in lowest terms. $\square$

This exact oriented identity replaces a coherent square cell by two cells whose entries are linear in X, Z. It exposes primes of $Z - X$ and $Z + X$ to the filling cost, and explains why a single-cell square obstruction does not rule out every five-term filling policy.

92.3. **A non-circular gate, and why it is false.** Put $h_a = \log a$, $h_b = \log b$, $H = \log c$, and $\rho = r_a + r_b + r_c = \log \operatorname{rad}(abc)$. Define the mixed polarization

$$\mathcal{M} = h_a r_b + r_a h_b + h_b r_c + r_b H + H r_a + r_c h_a.$$

Consider the uniform single-cell assertion

$$\Phi \le \frac{1+\epsilon}{2}\mathcal{M} + C_\epsilon H. \tag{476}$$

Proposition 92.5 (The gate would imply abc)**.** *If (476) holds for every $\epsilon > 0$ with one constant valid for all primitive points, then the logarithmic abc conjecture holds.*

Proof. Every leg height is at most H, so $\mathcal{M} \le 2H\rho$. Also $c \le 2ab$, whence $H - \log 2 \le h_a + h_b$ and $\Phi \ge H(H - \log 2)$. Substitute these two estimates in (476) and cancel $H > 0$ to obtain

$$H \le (1+\epsilon)\rho + C_\epsilon + \log 2.$$

□

This implication is non-circular: the premise compares a quadratic two-vector norm with a one-sided radical polarization. It is nevertheless false.

Theorem 92.6 (Primitive Pythagorean-square obstruction)**.** *Gate* (476) *fails with all of its quantifiers and all primitive abc premises satisfied.*

Proof. For $t \ge 1$, let

$$x = 2t+1, \qquad y = 2t(t+1), \qquad z = 2t^2 + 2t + 1.$$

Expansion gives $x^2 + y^2 = z^2$. The odd number x is coprime to 2, t, and $t+1$, so $\gcd(x, y) = 1$; the equation gives pairwise coprimality of x, y, z. Thus (x^2, y^2, z^2) is a primitive abc point.

For every $u > 0$,

$$\log \operatorname{rad}(u^2) \le \log u = \tfrac{1}{2}\log(u^2).$$

Consequently $\mathcal{M} \le \Phi$. But

$$\frac{\Phi}{H} \ge \log(y^2) = 2\log(2t(t+1)) \longrightarrow \infty.$$

Taking $\epsilon = 1/2$ in (476) would imply $\Phi \le (3/4)\Phi + C_{1/2}H$, or $\Phi \le 4C_{1/2}H$, a contradiction. □

The same family refutes the direct skeleton gate $\Phi \le 3(1+\epsilon)^2\Psi + K_\epsilon^2$ for sufficiently small positive ϵ, since $\Psi \le \Phi/4$ and Φ is unbounded. The finite primitive point $9 + 16 = 25$ separately refutes coefficient-one thinness: its $[3] \wedge [2]$ coefficient has absolute value 8, while the radical coefficient is 1. None of these is an abc counterexample.

92.4. **Veronese correction and open route.** For $n > 1$, put $g(n) = \gcd_{p|n} v_p(n)$ and write $n = u^{g(n)}$. If $h = \log n$, $r = \log \operatorname{rad}(n)$, and $u_h = h/g$, then

$$h - r = (g-1)u_h + (u_h - r). \tag{477}$$

For a unit leg we set $g(1) = 1$ and $h = r = u_h = 0$, so both defect components vanish. Thus the definitions remain total on auxiliary primitive cells. The first summand is coherent Veronese-ray thickness; the second is the residual thickness of a base whose prime exponents have gcd one. Polarizing (477) on all three legs gives

$$2\Phi - \mathcal{M} = \mathcal{V} + \mathcal{R},$$

where, with $u_i = h_i/g_i$,

$$\nu_i = (g_i - 1)u_i, \qquad \sigma_i = u_i - r_i,$$
$$\mathcal{V} = \sum_{i<j}(\nu_i h_j + \nu_j h_i), \qquad \mathcal{R} = \sum_{i<j}(\sigma_i h_j + \sigma_j h_i).$$

These coherent and residual contact losses are nonnegative and total on unit legs. The coherent Veronese term $\mathcal{V}$ alone is already large enough to supply the square-family obstruction. This does not assert that $\mathcal{R} = 0$ and does not refute separate control of $\mathcal{R}$.

For a rational cell, write $z = m/n$ in lowest terms with $n > 0$ and $m(n-m) \ne 0$. We use the signed relation $m + (n-m) = n$ for orientation and the pairwise-coprime absolute legs $|m|, |n-m|, n$ for valuations, heights, radicals, and exponent gcds. Thus every scalar cell has a canonical normalization, even when an auxiliary argument lies outside $(0,1)$. One may instead work in the conservative positive subcomplex $0 < y < x < 1$: all five arguments then lie in $(0,1)$ and have unique reduced positive primitive triples. The quadratic move $y = x^2$ is in this domain for $0 < x < 1$.

For a five-term-equivalent chain $\Gamma = \sum_j n_j[z_j]$, define

$$\mathcal{Q}(\Gamma) = \sum_j |n_j|(\mathcal{M}_j + \mathcal{V}_j), \qquad \mathcal{R}(\Gamma) = \sum_j |n_j|\mathcal{R}_j.$$

Proposition 92.7 (Automatic filling-boundary inequality)**.** *Every finite five-term-equivalent chain satisfies*

$$\Phi(a,b,c) \leq \tfrac{1}{2}\big(\mathcal{Q}(\Gamma)+\mathcal{R}(\Gamma)\big).$$

Proof. Five-term equivalence gives the exact oriented relation $\Omega_0 = \sum_j n_j\Omega_j$. The weighted ℓ^1 triangle inequality and the cellwise identity $2\Phi_j = \mathcal{M}_j + \mathcal{V}_j + \mathcal{R}_j$ give

$$\Phi_0 = \|\Omega_0\|_{1,\log} \leq \sum_j |n_j|\|\Omega_j\|_{1,\log} = \frac{1}{2}\sum_j |n_j|(\mathcal{M}_j + \mathcal{V}_j + \mathcal{R}_j).$$

Only finitely many prime coordinates occur. □

The surviving open target is to construct, uniformly, a chain satisfying

$$\begin{aligned} \mathcal{R}(\Gamma) &\leq \epsilon\mathcal{Q}(\Gamma) + K_\epsilon H, \\ \mathcal{Q}(\Gamma) &\leq 2H\rho + L_\epsilon H. \end{aligned} \tag{478}$$

Together with Proposition 92.7, these estimates give

$$\Phi \leq (1+\epsilon)H\rho + \tfrac{1}{2}\big((1+\epsilon)L_\epsilon + K_\epsilon\big)H.$$

The lower bound $\Phi \geq H(H-\log 2)$ therefore implies abc. Gate (478) is not a renamed height inequality: it requires an actual five-term filling and separate control of its coherent and primitive-residual costs. It remains open. The Pythagorean-square family eliminates the uncalibrated one-cell policy, but it does not eliminate every auxiliary-cell filling.

The exponent triple also defines a generalized Fermat signature with $\chi_{\text{orb}} = 1/g_a + 1/g_b + 1/g_c - 1$. Results for a fixed hyperbolic signature do not provide the uniform estimate over varying signatures that (478) requires.

Primary-source boundary. The relevant primary sources have sharply different arithmetic settings:

- Ru–Wang, *Vojta's abc conjecture for entire curves in toric varieties highly ramified over the boundary*, `https://arxiv.org/abs/2410.19395v2`, is a complex entire-curve theorem.
- Pasten, *On the arithmetic case of Vojta's conjecture with truncated counting functions*, `https://arxiv.org/abs/2205.07841v3`, proves arithmetic approximation inequalities and bounds toward abc.
- Gasbarri–Guo–Wang, *Campana conjecture for coverings of toric varieties over function fields*, `https://arxiv.org/abs/2401.13186v2`, works over function fields.

None is an unconditional integer abc theorem over $\mathbf{Q}$. Cached primary PDFs and SHA-256 metadata are stored under `research/sources/steinberg_contact_surface_2026_09_02/`.

Formal status. The Lean modules `SteinbergValuationContactSurface20260902.lean` and its axiom audit formalize the exterior coefficient identities, five-term relation, scalar area identities, quadratic Veronese peeling, MC⇒abc, the coefficient-one audit, the complete Gate-MC Pythagorean-square obstruction, the abstract scalar Veronese/residual split for supplied data, and the finite-coordinate filling-boundary inequality. The integer exponent-gcd construction, actual finite-support cell norm, and conditional exact-boundary chain instantiation are supplied by the supplemental Section 93. The algebraic and positive five-term-generated relations and their exact boundary bridge are formalized in Section 94. What remains open is a positive filling-existence theorem for an arbitrary target and the two analytic Gate VF cost estimates. The quantified Gate-SC obstruction is formalized at the fixed admissible value $\epsilon = 1/10$. These modules do not inhabit Gate VF or close `ABCConjecture`.

93. Integer exponent gcds and finite-support Steinberg chains

Author: ChatGPT. This section supplies two algebraic bridges needed by the valuation-contact route. First, it constructs the canonical common-exponent decomposition of every positive integer and separates its coherent and primitive residual defects. Second, it realizes contact surfaces in an actual finite-support coordinate module and proves the exact weighted cell norm and a conditional finite-chain estimate. No form of the *abc* conjecture and no analytic filling estimate is assumed.

93.1. **The canonical common-exponent decomposition.** For $n > 1$, write

$$n = \prod_{p \in S(n)} p^{e_p}, \qquad e_p > 0,$$

where $S(n)$ is nonempty, and put

$$g(n) = \gcd_{p \in S(n)} e_p, \qquad u(n) = \prod_{p \in S(n)} p^{e_p/g(n)}. \tag{479}$$

For the unit we adopt the convention

$$g(1) = 1, \qquad u(1) = 1. \tag{480}$$

The second convention does not turn the gcd of an empty exponent family into one; it only defines the decomposition at the unit.

Proposition 93.1 (Integer exponent-gcd decomposition)**.** *For every positive integer n,*

$$g(n) > 0, \qquad n = u(n)^{g(n)}, \qquad u(n) > 0, \qquad \operatorname{rad}(u(n)) = \operatorname{rad}(n).$$

If $n > 1$, then $u(n)$ has the same nonempty prime support as n and

$$\gcd_{p \mid u(n)} v_p(u(n)) = 1.$$

The last assertion is deliberately restricted to $n > 1$.

Proof. The assertions at $n = 1$ follow from (480). Suppose $n > 1$. Its finite prime support is nonempty, so the gcd $g(n)$ of its positive exponents is positive and divides every e_p. Consequently

$$g(n)\frac{e_p}{g(n)} = e_p$$

for every supported prime, and unique factorization gives

$$u(n)^{g(n)} = \prod_{p \in S(n)} p^{g(n)(e_p/g(n))} = \prod_{p \in S(n)} p^{e_p} = n.$$

Every quotient $e_p/g(n)$ is positive. Thus no supported prime disappears and no new prime appears, proving support and radical equality. Dividing a finite nonzero family by its gcd makes the gcd of the quotient family one, which proves the final assertion. $\square$

Set

$$h(n) = \log n, \qquad h_u(n) = \log u(n), \qquad r(n) = \log \operatorname{rad}(n),$$

and define

$$\nu(n) = (g(n) - 1)h_u(n), \qquad \sigma(n) = h_u(n) - r(n). \tag{481}$$

Proposition 93.2 (Coherent–residual defect split)**.** *For every positive integer n,*

$$h(n) = g(n)h_u(n), \qquad h(n) - r(n) = \nu(n) + \sigma(n), \qquad \nu(n), \sigma(n) \geq 0. \tag{482}$$

At $n = 1$, the three heights h, h_u, r and both thicknesses ν, σ vanish (while the convention gives $g(1) = 1$).

Proof. Proposition 93.1 and the logarithm of a positive integral power give $h(n) = g(n)h_u(n)$. Since $g(n) \geq 1$ and $h_u(n) \geq 0$, one has $\nu(n) \geq 0$. The equality of radicals and the elementary inequality $\operatorname{rad}(u(n)) \leq u(n)$ give $\sigma(n) \geq 0$. Finally,

$$g(n)h_u(n) - r(n) = (g(n) - 1)h_u(n) + (h_u(n) - r(n)).$$

The unit statement follows directly from the convention. $\square$

93.2. **An actual finite-support contact module.** Let

$$\mathcal{D} = \mathbb{N} \to_0 \mathbb{Z}, \qquad \mathcal{E} = (\mathbb{N} \times \mathbb{N}) \to_0 \mathbb{Z},$$

where $\to_0$ denotes finitely supported functions. For $X, Y \in \mathcal{D}$ define

$$(X \otimes Y)_{p,q} = X_p Y_q, \qquad X \wedge Y = X \otimes Y - Y \otimes X,$$

and set

$$\Omega(A, B, C) = (A - C) \wedge (B - C) \in \mathcal{E}.$$

This ordered-pair model stores both orientations of each alternating coordinate. For effective divisors $A, B, C : \mathbb{N} \to_0 \mathbb{N}$, a finite set S, and nonnegative weights w_p, put

$$L_{S,w}(A) = \sum_{p \in S} A_p w_p, \qquad \|\Omega\|_{S,w} = \frac{1}{2} \sum_{p,q \in S} |\Omega_{p,q}| w_p w_q.$$

Proposition 93.3 (Exact finite-support mixed area)**.** *If the supports of A, B, C are pairwise disjoint, then*

$$\|\Omega(A, B, C)\|_{S,w} = L_{S,w}(A) L_{S,w}(B) + L_{S,w}(B) L_{S,w}(C) + L_{S,w}(C) L_{S,w}(A). \tag{483}$$

Proof. The three-leg expansion gives

$$\Omega_{p,q} = A_p B_q - A_q B_p + B_p C_q - B_q C_p + C_p A_q - C_q A_p.$$

At any coordinate at most one of A_p, B_p, C_p is nonzero. Applying this observation at p and at q gives the pointwise identity

$$\begin{aligned} |\Omega_{p,q}| = A_p B_q + B_p A_q + B_p C_q + C_p B_q \\ + C_p A_q + A_p C_q. \end{aligned}$$

After multiplication by $w_p w_q$ and finite summation, each directed rectangle factors. For example,

$$\sum_{p,q \in S} A_p B_q w_p w_q = L_{S,w}(A) L_{S,w}(B).$$

Every cross-leg rectangle occurs in both orientations; the prefactor $1/2$ removes this duplication and yields (483). ☐

A canonical positive rational cell is a triple $P = (a, b, c)$ of positive integers such that

$$a + b = c, \qquad \gcd(a, b) = 1.$$

It represents $a/c \in (0, 1)$. The sum relation also gives $\gcd(a, c) = \gcd(b, c) = 1$, so the three valuation divisors have pairwise disjoint prime supports. Let $S(P)$ be their finite union and take $w_p = \log p$. Unique factorization and Proposition 93.3 give

$$\Phi(P) := \|\Omega(P)\|_{S(P),\log} = \log a \log b + \log b \log c + \log c \log a. \tag{484}$$

Let $M(P)$ be the one-sided radical polarization of the three leg heights. Let $V(P)$ be the bilinear contact loss formed from the three coherent thicknesses ν, and let $R(P)$ be the corresponding loss formed from the three residual thicknesses σ. Polarizing Proposition 93.2 in (484) proves the exact identity

$$2\Phi(P) = M(P) + V(P) + R(P). \tag{485}$$

No estimate is used in (485).

93.3. **Finite chains with an exact boundary.** Let I be finite, let P_0 and P_j be canonical positive rational cells, and let $n_j \in \mathbb{Z}$. Assume the exact equality in the integral finite-support module

$$\Omega(P_0) = \sum_{j\in I} n_j \Omega(P_j). \tag{486}$$

Theorem 93.4 (Concrete exact-boundary filling estimate)**.** *Under* (486)*,*

$$\Phi(P_0) \le \frac{1}{2}\Bigg(\sum_{j\in I} |n_j|\big(M(P_j) + V(P_j)\big) + \sum_{j\in I} |n_j| R(P_j)\Bigg). \tag{487}$$

Proof. Take the finite union S of every prime support occurring in the target and the chain. Evaluate (486) at each $(p, q) \in S \times S$, cast to the reals, and multiply by $\frac{1}{2}\log p \log q$. The triangle inequality gives

$$\Phi(P_0) \le \sum_{j\in I} |n_j| \Phi(P_j).$$

Substitute (485) for each cell and separate the two sums. This is exactly (487). ☐

Theorem 93.4 is conditional only on the displayed finite-support equality. The inductively generated equivalence relation from the permitted rational five-term moves remains *open and unformalized*: this section neither defines all generators and closure operations nor proves that every generated chain satisfies (486). It also proves neither of the two analytic bounds required by Gate VF.

93.4. **Lean realization and formal boundary.** The companion Lean module listed in Section 96 formalizes Propositions 93.1 and 93.2, the actual finite-support types `FiniteDivisor` and `FiniteExteriorSurface`, Proposition 93.3, the canonical rational-cell specialization (484), the split (485), and Theorem 93.4. Its final namespace theorem instantiates the earlier abstract finite-coordinate triangle inequality. The companion axiom audit uses no project-specific axiom, and direct compilation with warnings treated as errors succeeds.

94. Generated five-term relations and the finite-boundary bridge

Author: ChatGPT. The valuation-contact construction of Section 93 provides a concrete finite-support exterior surface for every positive reduced rational cell and an analytic estimate for every finite chain satisfying an exact surface boundary. What remained was to derive that boundary from the relation generated by five-term moves. We supply that derivation here. We first work in a free divisor-symbol module, then pass to the smaller relation generated only by positively realized rational moves. The argument is the divisor shadow of the standard pre-Bloch complex [28, 29, 30]; no K-theoretic or *abc* estimate is imported from that theory.

94.1. **The free symbol chain and its boundary.** Let

$$\mathcal{D} = \mathbb{N} \to_0 \mathbb{Z}, \qquad \mathcal{E} = (\mathbb{N}\times\mathbb{N}) \to_0 \mathbb{Z},$$

and retain the ordered-coordinate exterior product

$$(A \wedge B)_{p,q} = A_p B_q - A_q B_p.$$

Write $\mathcal{S} = \mathcal{D} \times \mathcal{D}$ for the set of divisor symbols and let

$$\mathcal{C}_0 = \mathbb{Z}^{(\mathcal{S})}$$

be the free integral module on those symbols. Its basis element associated with (A, B) is denoted by $[A, B]$. Define a linear map

$$\partial : \mathcal{C}_0 \longrightarrow \mathcal{E}, \qquad \partial[A, B] = A \wedge B. \tag{488}$$

For $X, U, Y, V, Z \in \mathcal{D}$, define the formal five-term chain

$$\begin{aligned}\mathfrak{r}(X, U, Y, V, Z) &= [X, U] - [Y, V] + [Y - X, Z - X]\\ &\quad - [Y + U - X - V, Z - X - V] + [U - V, Z - V].\end{aligned} \tag{489}$$

Theorem 94.1 (Five-term generators are boundary cycles)**.** *For all $X, U, Y, V, Z \in \mathcal{D}$,*

$$\partial\mathfrak{r}(X, U, Y, V, Z) = 0.$$

Proof. By linearity and (488), the claimed boundary is

$$\begin{aligned} &X \wedge U - Y \wedge V \\ &\qquad + (Y - X) \wedge (Z - X) \\ &\qquad - (Y + U - X - V) \wedge (Z - X - V) \\ &+ (U - V) \wedge (Z - V). \end{aligned}$$

Expand each wedge bilinearly. Every elementary wedge between two members of $\{X, U, Y, V, Z\}$ occurs with opposite coefficients and cancels. This is also the finite-support lift of the earlier coefficientwise five-term identity. ☐

Let

$$\mathcal{R}_5 = \operatorname{span}_{\mathbb{Z}}\{\mathfrak{r}(X, U, Y, V, Z) : X, U, Y, V, Z \in \mathcal{D}\} \subseteq \mathcal{C}_0. \tag{490}$$

Theorem 94.2 (Generated relation-to-boundary bridge)**.** *One has*

$$\mathcal{R}_5 \subseteq \ker \partial.$$

Consequently, if

$$c \sim_5 d \iff c - d \in \mathcal{R}_5,$$

then $\partial c = \partial d$.

Proof. The kernel of ∂ is an integral submodule. By Theorem 94.1, it contains every generator of the span in (490); hence it contains the span. If $c - d \in \mathcal{R}_5$, then linearity gives $\partial c - \partial d = \partial(c - d) = 0$. ☐

This proof includes arbitrary finite sums, negatives, and integer multiples. In particular, the exact boundary is a consequence of a generation certificate and is not included among its premises.

94.2. **The positive rational subrelation.** For a positive reduced cell

$$P = (a, b, c), \qquad a + b = c, \qquad \gcd(a, b) = 1,$$

define its divisor symbol by

$$\sigma(P) = \big(d(a) - d(c), d(b) - d(c)\big). \tag{491}$$

Then

$$\partial[\sigma(P)] = (d(a) - d(c)) \wedge (d(b) - d(c)) = \Omega(P). \tag{492}$$

A positive realization of a five-term generator consists of positive reduced cells $P_1, \ldots, P_5$ and divisor data X, U, Y, V, Z for which

$$\begin{array}{r|c} P_1 & (X, U) \\ P_2 & (Y, V) \\ P_3 & (Y - X, Z - X) \\ P_4 & (Y + U - X - V, Z - X - V) \\ P_5 & (U - V, Z - V) \end{array} \tag{493}$$

are the five individual identities for $\sigma(P_i)$. These local symbol identities contain no signed boundary assertion. Let $\mathcal{R}_5^+$ be the integral span of the signed chains

$$[\sigma(P_1)] - [\sigma(P_2)] + [\sigma(P_3)] - [\sigma(P_4)] + [\sigma(P_5)] \tag{494}$$

over all positive realizations.

Theorem 94.3 (Positive generated boundary). *There are inclusions*

$$\mathcal{R}_5^+ \subseteq \mathcal{R}_5 \subseteq \ker \partial.$$

Thus every equivalence generated by actual positive five-term moves preserves the finite-support surface.

Proof. The five identities in (493) identify each chain (494) with the algebraic generator (489). Therefore each generator of $\mathcal{R}_5^+$ belongs to $\mathcal{R}_5$, and the first inclusion follows by taking spans. The second inclusion is Theorem 94.2. □

Suppose now that a target cell P_0 and cells P_j with coefficients $n_j \in \mathbb{Z}$ satisfy the generation statement

$$[\sigma(P_0)] - \sum_j n_j[\sigma(P_j)] \in \mathcal{R}_5^+. \tag{495}$$

Theorem 94.3 and (492) derive

$$\Omega(P_0) = \sum_j n_j \Omega(P_j). \tag{496}$$

Applying the exact finite-chain theorem from the preceding section gives

$$\Phi(P_0) \leq \frac{1}{2}\left(\sum_j |n_j|\big(M(P_j) + V(P_j)\big) + \sum_j |n_j| R(P_j)\right). \tag{497}$$

Unlike the earlier conditional structure, (495) stores only membership in the relation generated by actual moves. Equation (496) is a theorem.

94.3. **An infinite arithmetic family of positive moves.** Let $0 < b < a < c$ and impose the five reducedness conditions

$$\begin{gathered} \gcd(a, c-a) = \gcd(b, c-b) = \gcd(b, a-b) = 1, \\ \gcd\big(b(c-a), c(a-b)\big) = 1, \\ \gcd(c-a, a-b) = 1. \end{gathered} \tag{498}$$

Consider the five cells

(499)

	left	middle	total
P_1	a	$c-a$	c
P_2	b	$c-b$	c
P_3	b	$a-b$	a
P_4	$b(c-a)$	$c(a-b)$	$a(c-b)$
P_5	$c-a$	$a-b$	$c-b.$

Every entry is positive and the hypotheses (498) make every row reduced. The fourth row is a cell because

$$b(c-a) + c(a-b) = a(c-b).$$

Proposition 94.4 (Common-denominator realization). *The cells* (499) *positively realize the five-term move with*

$$x = \frac{a}{c}, \qquad y = \frac{b}{c},$$

and arguments

$$\frac{a}{c}, \quad \frac{b}{c}, \quad \frac{b}{a}, \quad \frac{b(c-a)}{a(c-b)}, \quad \frac{c-a}{c-b}. \tag{500}$$

Proof. Put

$$\begin{aligned} X &= d(a) - d(c), & U &= d(c-a) - d(c), \\ Y &= d(b) - d(c), & V &= d(c-b) - d(c), \\ Z &= d(a-b) - d(c). \end{aligned}$$

The symbols of the first two cells are (X, U) and (Y, V) by definition. Direct subtraction gives the third and fifth rows of (493). For the fourth row, multiplicativity of divisors gives

$$d\big(b(c-a)\big) - d\big(a(c-b)\big) = Y + U - X - V,$$
$$d\big(c(a-b)\big) - d\big(a(c-b)\big) = Z - X - V.$$

Thus all five symbol identities hold, and Theorem 94.3 supplies the surface relation. □

For every odd $c \geq 3$, the choice $a = 2$, $b = 1$ satisfies (498). Indeed, the only nontrivial checks reduce to

$$\gcd(2, c-2) = 1, \qquad \gcd(c-2, c) = \gcd(c-2, 2) = 1.$$

This gives an infinite parameterized family of actual positive generators.

94.4. **A full-premise counterexample to literal chain cancellation.** The generated quotient is essential. Define the augmentation

$$\varepsilon : \mathcal{C}_0 \longrightarrow \mathbb{Z}, \qquad \varepsilon\left(\sum_s n_s[s]\right) = \sum_s n_s.$$

For every input one has

$$\varepsilon\big(\mathfrak{r}(X, U, Y, V, Z)\big) = 1 - 1 + 1 - 1 + 1 = 1. \tag{501}$$

Proposition 94.5 (A generator is never the zero free chain)**.** *For every $X, U, Y, V, Z \in \mathcal{D}$,*

$$\mathfrak{r}(X, U, Y, V, Z) \neq 0.$$

In particular, the boundary map ∂ is not injective.

Proof. If the free chain were zero, its augmentation would be zero, contradicting (501). Its boundary is nevertheless zero by Theorem 94.1, so ∂ has a nonzero kernel. □

This refutes the exact shortcut asserting that the five signed basis chains cancel before quotienting. It does not refute the five-term route. The correct statement is the kernel inclusion of Theorem 94.2.

The failure persists with all positivity and reducedness premises. Taking

$$x = \frac{1}{2}, \qquad y = \frac{1}{6}$$

gives the five distinct arguments

$$\frac{1}{2}, \quad \frac{1}{6}, \quad \frac{1}{3}, \quad \frac{1}{5}, \quad \frac{3}{5}$$

and reduced cell triples

$$(1, 1, 2), \quad (1, 5, 6), \quad (1, 2, 3), \quad (1, 4, 5), \quad (3, 2, 5).$$

Their signed surface boundary is zero, whereas their signed free chain is nonzero; its coefficient at the $1/2$ symbol is one.

94.5. **Formal boundary and remaining analytic gate.** The Lean companion module is

`SteinbergFiveTermBoundaryBridge20260903`.

It formalizes Theorems 94.1, 94.2, and 94.3, the generated filling implication (497), Proposition 94.4, the odd-denominator family, and Proposition 94.5. It also formalizes the five-distinct positive example. Its axiom audit uses only `propext`, `Classical.choice`, and `Quot.sound`.

The algebraic part of S-I4C is therefore closed: a chain generated by actual positive five-term moves has the required finite exterior boundary. The route remains open at the harder existence and analytic stages. For an arbitrary target cell, one must still construct a finite positive generated filling satisfying useful bounds on chain length, coefficients, prime support, and the two costs on the right of (497). More precisely, put

$$\mathcal{Q}(\Gamma) = \sum_j |n_j|(M(P_j) + V(P_j)), \qquad \mathcal{R}(\Gamma) = \sum_j |n_j| R(P_j).$$

For every $\epsilon > 0$, Gate VF still requires constants K_ϵ, L_ϵ and a generated filling for each primitive target of height $H = \log c$ and radical logarithm $\rho = \log \operatorname{rad}(abc)$ such that

$$\mathcal{R}(\Gamma) \le \epsilon \mathcal{Q}(\Gamma) + K_\epsilon H, \qquad \mathcal{Q}(\Gamma) \le 2H\rho + L_\epsilon H.$$

Neither estimate nor such a uniform construction is proved here, and this section proves neither the *abc* conjecture nor its negation.

95. Quadratic Veronese peeling and valuation-layer flags

Author: ChatGPT. This section isolates the specialization $y = x^2$ of the five-term contact relation, transports it to primitive integer triples, and audits the resulting one-move policy on a full-premise Pythagorean-square family. The fixed policy fails three necessary uniform estimates. The unrestricted multi-move contact-complex route remains open.

For $z \in \mathbf{Q} \setminus \{0, 1\}$, let

$$\Omega(z) = d(z) \wedge d(1 - z), \qquad d : \mathbf{Q}^\times \longrightarrow \bigoplus_p \mathbf{Z}[p].$$

Recall the five-term identity

$$\Omega(x) - \Omega(y) + \Omega(y/x) - \Omega\left(\frac{y(1-x)}{x(1-y)}\right) + \Omega\left(\frac{1-x}{1-y}\right) = 0. \tag{502}$$

Theorem 95.1 (Quadratic Veronese peeling)**.** *The specialization $y = x^2$ is admissible in (502) exactly when $x \notin \{0, 1, -1\}$. On this domain,*

$$\Omega(x^2) = 2\Omega(x) - 2\Omega\left(\frac{x}{1+x}\right). \tag{503}$$

Proof. The five-term domain says that x, x^2 are nonzero, different from one, and different from each other. Since $x^2 - 1 = (x-1)(x+1)$ and $x^2 - x = x(x-1)$, this is equivalent to the three stated exclusions. Direct field algebra then gives

$$\frac{x^2}{x} = x, \quad \frac{x^2(1-x)}{x(1-x^2)} = \frac{x}{1+x}, \quad \frac{1-x}{1-x^2} = \frac{1}{1+x}.$$

The final two arguments are complementary. Alternation gives $\Omega(1-z) = -\Omega(z)$, so substitution in (502) proves (503). No sign condition on x is required, since $d(-1) = 0$ for finite-prime divisors. ☐

Writing $X = d(x)$, $U = d(1-x)$, and $W = d(1+x)$ exposes the coefficientwise identity behind the proof:

$$(2X) \wedge (U + W) = 2X \wedge U - 2(X - W) \wedge (-W). \tag{504}$$

95.1. **Primitive integer transport.** Let (a, b, c) be a positive pairwise-coprime triple with $a + b = c$ and put $x = a/c$.

Proposition 95.2 (Quadratic transform)**.** *The triple*

$$T_2(a, b, c) = \big(a^2, b(a + c), c^2\big)$$

is positive, pairwise coprime, and satisfies $a^2 + b(a + c) = c^2$. Moreover,

$$\Omega\big(a^2, b(a + c), c^2\big) = 2\Omega(a, b, c) - 2\Omega(a, c, a + c). \tag{505}$$

Proof. The sum identity follows from $b = c - a$: $a^2 + (c - a)(a + c) = c^2$. A common prime of a^2 and $b(a + c)$ would divide a and either b or $a + c$, contradicting the original pairwise coprimality. The same argument applies to $b(a + c)$ and c^2, while a^2 and c^2 are coprime directly. Equation (505) is Theorem 95.1 with $x = a/c$. □

For $t \geq 1$, set

$$X = 2t + 1, \qquad Y = 2t(t + 1), \qquad Z = 2t^2 + 2t + 1.$$

Then

$$X^2 + Y^2 = Z^2, \qquad Z - Y = 1, \qquad Y + Z = X^2,$$

and X, Y, Z are pairwise coprime. Taking $x = Y/Z$ in (503) yields the full-premise identity

$$\boxed{\Omega(Y^2, X^2, Z^2) = 2\Omega(Y, 1, Z) - 2\Omega(Y, Z, X^2).} \tag{506}$$

Thus the first auxiliary cell has no nontrivial middle divisor, and only the third leg of the second auxiliary cell retains a visible square.

95.2. **Exact positive-cost redistribution.** Put

$$u = \log X, \quad v = \log Y, \quad w = \log Z,$$

and

$$\beta = \log \operatorname{rad}(X), \quad \alpha = \log \operatorname{rad}(Y), \quad \gamma = \log \operatorname{rad}(Z).$$

Write $\delta_X = u - \beta$, $\delta_Y = v - \alpha$, and $\delta_Z = w - \gamma$. For the declared outer-square decomposition, the full, coherent, and residual costs of the squared cell are

$$\begin{aligned} \Phi_{\text{sq}} &= 4(uv + uw + vw), \\ V_{\text{sq}} &= \Phi_{\text{sq}}, \\ R_{\text{sq}} &= 2\{\delta_Y(u + w) + \delta_X(v + w) + \delta_Z(u + v)\}. \end{aligned} \tag{507}$$

After weighting both right-hand cells in (506) by the absolute coefficient two, their costs are

$$\begin{aligned} \Phi_{\text{peel}} &= 2vw + 2\{vw + 2u(v + w)\}, \\ V_{\text{peel}} &= 2u(v + w), \\ R_{\text{peel}} &= 4\delta_Y(u + w) + 2\delta_X(v + w) + 4\delta_Z(u + v). \end{aligned} \tag{508}$$

Proposition 95.3 (Conservation and redistribution)**.** *One has*

$$\Phi_{\text{peel}} = \Phi_{\text{sq}}, \tag{509}$$

$$V_{\text{peel}} < \tfrac{1}{2} V_{\text{sq}}, \tag{510}$$

$$R_{\text{peel}} - R_{\text{sq}} = 2\delta_Y(u + w) + 2\delta_Z(u + v) \geq 0. \tag{511}$$

In addition, $V_{\text{peel}}/V_{\text{sq}} \to 1/4$ as $t \to \infty$.

Proof. Expansion of (508) proves (509). The difference between half the old coherent cost and the new coherent cost is $2vw > 0$. Direct subtraction proves (511). Finally $u \sim \log t$ and $v, w \sim 2 \log t$, which gives the stated limit. □

For calibrated cost $Q = M + V = 2\Phi - R$, the same computation gives

$$Q_{\text{peel}} - Q_{\text{sq}} = -2\delta_Y(u + w) - 2\delta_Z(u + v). \tag{512}$$

The coherent improvement has therefore been transferred to residual cost; the total $Q + R = 2\Phi$ has not decreased.

95.3. **The primewise valuation-layer flag.** For $k \geq 1$, define

$$R_k(n) = \prod_{p:\, v_p(n) \geq k} p, \qquad \lambda_k(n) = \log R_k(n).$$

Then

$$\log n = \sum_{k \geq 1} \lambda_k(n), \qquad \log \operatorname{rad}(n) = \lambda_1(n), \qquad \log \frac{n}{\operatorname{rad}(n)} = \sum_{k \geq 2} \lambda_k(n). \tag{513}$$

This filtration sees repeated prime mass even when the gcd of all exponents is one; $2s^2$ with s odd is the basic example. Bilinearity gives the exact layer-pair expansion

$$\Phi(a,b,c) = \sum_{k,l \geq 1} \big(\lambda_k(a)\lambda_l(b) + \lambda_k(b)\lambda_l(c) + \lambda_k(c)\lambda_l(a)\big). \tag{514}$$

Squaring duplicates every layer:

$$R_{2j-1}(n^2) = R_{2j}(n^2) = R_j(n). \tag{515}$$

Consequently the square-cell layer-pair area is replicated four times, and the layer ledger recovers the conservation law (509).

The flag also rules out a naive uniform depth cutoff. With $t = 2^m$, $v_2(Y^2) = 2m + 2$, so the prime 2 occupies arbitrarily high layers inside the genuine primitive cells (Y^2, X^2, Z^2). Weighted layer decay and primewise matching across several five-term moves are not refuted by this example.

95.4. **Three fixed-policy no-go theorems.** The first no-go is immediate from (509). If some $\eta > 0$ and K satisfied

$$\Phi_{\text{peel}} \leq (1-\eta)\Phi_{\text{sq}} + K(2w)$$

for all t, then $\eta\Phi_{\text{sq}} \leq 2Kw$. But $Z = Y + 1 \leq 2Y$ and $Y \geq 4$, so $w \leq 3v/2$ and

$$\frac{\Phi_{\text{sq}}}{2w} \geq \frac{4}{3}u \longrightarrow \infty.$$

Thus no positive one-step area contraction exists.

For the calibrated-boundary coefficient, expansion gives

$$Q_{\text{peel}} = 2u(v+w) + 4\alpha(u+w) + 4\gamma(u+v) + 2\beta(v+w), \tag{516}$$

$$Q_{\text{peel}} - 2H\rho = 2u(v+w) + 4\alpha u + 4\gamma(u+v-w) + 2\beta(v-w), \tag{517}$$

where $H = 2w$ and $\rho = \alpha + \beta + \gamma$. Since $Y < Z \leq XY$, $0 \leq \beta \leq u$, and $\alpha, \gamma \geq 0$, the last display implies

$$Q_{\text{peel}} - 2H\rho \geq 4uv. \tag{518}$$

Dividing by H and using $w \leq 3v/2$ shows that the excess is an unbounded multiple of height. Hence the fixed chain cannot satisfy $Q_{\text{peel}} \leq 2H\rho + LH$ for any constant L.

Finally, the policy requiring the following bound for every $\varepsilon > 0$ is false:

$$R_{\text{peel}} \leq \varepsilon Q_{\text{peel}} + K_\varepsilon H$$

with a uniform K_ε. Set $t = 2^m$ and $L_m = \log t$. For $m \geq 2$,

$$L_m \leq u \leq 2L_m, \quad 2L_m \leq v, w \leq 3L_m, \quad \delta_Y \geq L_m.$$

Thus

$$R_{\text{peel}} \geq 12L_m^2, \qquad \Phi_{\text{sq}} \leq 84L_m^2, \qquad H \leq 6L_m.$$

As $Q_{\text{peel}} \leq 2\Phi_{\text{sq}}$, at $\varepsilon = 1/100$ one obtains

$$R_{\text{peel}} - \frac{1}{100}Q_{\text{peel}} \geq \frac{258}{25}L_m^2,$$

which cannot be absorbed into any uniform $K_{1/100}H$. Thus the single admissible value $\varepsilon = 1/100$ already refutes the universal-in- ε policy; no assertion is made here that every positive ε fails. This theorem concerns the declared outer-square split. It does not claim that the same subsequence refutes the maximal common-exponent residual.

All triples used above satisfy positivity, the sum equation, and pairwise coprimality. These counterexamples retire the exact one-move policies only. Gate VF permits arbitrary additional five-term moves and remains open. The abc conjecture also remains open.

The Lean companion `QuadraticVeronesePeeling20260902.lean` formalizes the exact domain, integer transport, valuation-layer identities, and all three quantified fixed-policy no-go theorems. Its separate axiom audit covers all 53 public theorem declarations. A deterministic computation checks 20,000 consecutive parameters and a power-of-two subsequence through 2^{18}.

96. Formal verification and remaining obligations

The companion development uses Lean 4.32.0 and the repository's pinned dependencies. The principal companion modules are listed below. Audit modules only display their types and dependency reports.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`SmoothLowOmegaCounting`	Actual prime-power encoding, finite cardinality bounds, and the finite first moment.
`CarellaGlobalOmega Hypothesis20260901`	Positive-multiple injection and the exact N/Q high-ω lower bound; the $\sigma < 2/5 - \kappa$ disproof gate, the density-compatible exponent-window threshold $k > 5$, and a uniform one-fifth feasible exponent for $k \geq 6$.
`SignedPrimeSupport`	Actual prime-exponent sums and the coupled equivalence with the original target.
`SignedEndpointDyadicObstruction`	The genuine dyadic triples and failure of a separate-endpoint slope below one.
`ReachableTensorLatticeCriterion`	Unit-determinant span certificate, symmetric span and its missing off-diagonal tensor.
`FreyPellEffectiveResidual Corridor20260830`	The integral elliptic identity, integer square-factor descent, explicit scalar implications, and the Frey denominator bound.
`AnalyticAmplification Continuation20260830`	CRT uniqueness and finite counts; actual radical inequalities; conic and height identities.
`DVRReachableHaar20260830`	A minimum determinant from actual columns, the exact quotient cardinality, closed span, and full-rank Haar and log-volume.
`GeometryUniformity Continuation20260830`	Norm and ratio identities, logarithmic absorption, explicit corridor implications, and the primitive Frey–Mordell point.
`IUTReachabilityContinuation20260830`	Tensor units, independent scaled vectors, and integral tensor-span containment in a coefficient-module hull.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`MatveevPellFinitePacket20260830`	The strict Pell approximation, normalized-coefficient algebra, absolute-index absorption, and the final height implication with external bounds stated as hypotheses.
`IUTTargetReset20260830`	Covariance of an all-isomorphism collation with repeated labels, its span, and source reindexing under an explicit compatibility identity.
`AnalyticActualRadical UniformGate20260830`	Actual new-prime radical and integer excess identities, the conic budget, exponent-window algebra, and exact numerical counterexamples.
`GeometryGlobalUniformGate20260830`	Canonical integer cube decomposition, auxiliary point and cubic-root identities, finite matrix determinants, and explicit algebraic height bridges.
`IUTAdmissibleGalois UniformGate20260830`	Kernel and valuation consequences of an explicit trace-transport hypothesis, affine-depth separation, and invariance of module span under unit scalars.
`IUTTracePreserving Transvection20260830`	Actual linear automorphisms achieving common avoidance while preserving a functional.
`IUTAllOpenLatticeRigidity20260830`	Scalar rigidity from all shrinking line neighborhoods; an actual $\mathbb{Z}_p$ specialization, with surjectivity forcing a unit scalar.
`IUTFullGaloisWordLift20260830`	An actual free-group automorphism, its inverse words, fixed boundary, and abelian images.
`IUTThreeLabelMinimumLayer20260830`	Actual linear automorphisms, finite-kernel avoidance, and three-label specialization.
`Frey139Tate210Realization20260830`	Actual Frey Weierstrass invariants and point types over finite fields, exact integer certificates and real logarithmic intervals for two curves.
`SL2TransvectionGeneration20260830`	Elementary matrices in the actual special linear group and the normal-subgroup coprime-index argument.
`Frey43BalancedRealization20260830`	The genuine primorial and all-prime exponent bound, actual curve invariants and finite-field point cardinality, and the real logarithmic window.
`IUTGeneralTameSquareLabels20260830`	Uniqueness of the actual exceptional quadratic congruence, floor and ceiling bounds, finite avoidance budget, and the hull-exponent scale identities.
`TraceDualPreidealHull20260831`	The actual algebra trace and integral dual, exact two-factor product duality, a scaled order-ideal inclusion, and its transported B-module span bound.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`ABCTwoPrimeSupport20260831`	Actual prime-support classification, the sharp radical bound on the full two-prime subclass, and its equality cases.
`TraceCovariantRationalReturn20260831`	Actual algebra traces and dimensions, rational scalar return, the entire scalar-line action forced by one nonzero return, and valuation consequences.
`ABCOddPartFibre20260831`	Actual odd parts of primitive triples, the finite fibre and bound two, fixed-parity uniqueness, and the exact two-point fibre over $(1, 3, 7)$.
`FreyEntireIsogenyArithmetic20260831`	Actual primitive family, four Weierstrass models, finite-field point count, and displayed discriminant and rational absolute j-bounds.
`FreyIsogenyWeilHeight20260831`	Actual reduced numerators and denominators, library Weil heights of the four curves, the exact least height, its unique minimizer, and explicit bounds.
`FreyIsogenyConductorSharpness20260831`	Actual radical of the reduced j denominator, the power subfamily, conductor-proxy bound, and subcritical gap for all four actual models.
`ABCMixedFullCampana20260831`	Denominator-free mixed-full radical compression, exact reciprocal slope, abc-conditional height bound, and the standard-target disproof implication from an unbounded family.
`ABCThreePrimeSignatures20260831`	Elementary factor rigidity and all output placements of the prime-base signatures $(2, 3, 3)$ and $(2, 3, 6)$, without importing external Diophantine classifications.
`IUTFiniteProductProjectionSpan20260831`	Central-idempotent reconstruction of a finite product-ring span from coordinate spans and finite nonnegative weighted-sum monotonicity.
`ABCSubcriticalLocusUniformity20260831`	Exact equivalence between standard *abc* and all fixed subcritical-locus height bounds, integer and pointwise amplification thresholds, and an infinite square-gap counting countermodel.
`PellCampanaCounterexample20260831`	Actual Pell recurrence and unboundedness, fullness transfer to the strict $(7, 3, 2)$ signature, and the conditional implication from an unbounded squarefull-root family to the unchanged negation of standard *abc*.
`PellAdjacentFactorCounterexample20260831`	Exact half-factorization, the primitive adjacent point, one-half conductor slope, fourth-full transfer, and the standard-target conditional disproof gate.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`PellSquareRootDescent20260831`	The $(1+\sqrt{2})^n$ orbit, norm and coprimality, exact adjacent square shapes, and the squarefull product equivalence.
`PellPrimeIndexDichotomy20260831`	Odd squarefull valuation depth, the exact index-seven simple divisor, and the two-channel numerical product bound.
`PellPrimeRank Counterexamples20260901`	The exact index-seven repeated divisor and nonsingular-derivative counterexample, together with arithmetic counterexamples to two defective cyclotomic intermediate claims exposed by the source audit.
`HallSquarefull Counterexample20260831`	Primitive Hall points, the twelfth-power radical inequality, the strict $11/12$ conductor slope, and the unbounded-family disproof gate.
`AffineShearAmplification20260831`	The complete affine equation, support avoidance, cofactor and endpoint coprimality, actual `ABCPoint` construction, and parameter injectivity.
`AffineExcessUpperBound20260831`	All three pair projections, same-prime exclusion, and the integer geometric-mean step.
`AffineRadicalStep20260901`	Support-killing moduli, primitive endpoints, all three pair injections, nesting of multiple-step fibres, the exact matching-lower quantifier gate, and the two square-row radical nonexamples.
`DanilovGlobalIndexSieve20260831`	Uniform modular linearization, ten prime-square packets, exact CRT, and the explicit global index progression.
`DanilovRecursiveLift20260901`	The recursive state invariant, finite fixed-index support obstruction, and the complete abstract one-step continuation countermodel.
`DanilovSimplePrimitiveNoGo20260901`	A full standard real-Lucas counterexample to sequence-uniform simplicity, powerful primitive parts, finite correction support, and the elementary $41n+1$ tail.
`IUTLanaSpecificationNoGo20260901`	Uninhabitability of the pinned upstream `RHSData` and assembled variant records, the total set-volume no-go, normalized-weight nonemptiness, and the pointed same-pilot sufficiency lemma.
`AffineDensityAttack20260901`	Exact gap recovery, strict-positive reverse boundary, the complete $U=1$ counterexample to its weaker form, finite-field fibre cardinality, squarefree excess identities, and the two-long-arm cancellation kernel.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`AffineTwoArmCRTPacket20260901`	Coprime joint excess, the unique diagonal CRT class and squarefree-carrier transfer, the exact 318322715-point canonical image, and a complete counterexample to two-marginal-gate sufficiency.
`MersennePrimeLayerRadical20260901`	Non-prime-power composite Mersenne values, exact prime-index order and radical carriers, finite totient-tail and lifting-mass bridges, and complete $m = 6$, 11, and 37 counterexamples to three stronger formulas.
`MersenneOrderBlockDecomposition20260901`	Actual exact-order divisibility and LTE, $L_m = \gcd(m, 2^m - 1)$, the finite fibre grouping $B_m = \prod_{d\mid m} E_d$, the corrected product for W_m, and the complete index-six ledger.
`MersenneOrderBlockAsymptotic20260901`	Canonical index-independent blocks, relative/canonical agreement, exact logarithmic divisor sums, and the implication $\log E_d = o(\varphi(d)) \Rightarrow \log W_m = o(m)$ with its open antecedent explicit.
`MersenneCanonicalBlockWitness20260901`	Exact order $\mathrm{ord}_{1093}(2) = 364$, exact valuation two, the divisibility $1093 \mid E_{364}$, and the negation of the strengthening that every canonical block equals one.
`MersenneWieferichTailReduction20260901`	Finite powerful-part comparison, the deep/transition/extreme mass trichotomy, and the transition-support and square-budget cardinality bounds, with all analytic inputs kept as explicit premises.
`AffineTemplateEntropy20260901`	Three-form cubic-product separation including every zero branch, affine membership cancellation, the canonical separated-set bound $\lvert S\rvert R^2 < 12c^4$ under explicit box and separation hypotheses, a generic finite-union inequality, and the complete counterexample to deleting an individual modulus cap.
`MersenneWeightedOrderTail20260901`	Summable-profile localization, heavy-or-diffuse mass alternatives, divisor-mass little-oh transfer, and the exact equivalence between Mersenne power loss and the $o(m)$ divisor-average block endpoint.
`MersenneSuperWieferichDepth20260901`	Exact finite depth layer cake, moving-threshold support/tail bounds, the pointwise sufficient implication, and the certified odd-order depth-two witness at 3511.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`PellFourPrimeCoupling20260901`	Arbitrary finite-product quadratic truncation, quotient cancellation, the exact Pell coordinate identity, second-order channel transfer modulo $4\ell^2$, and the index-seven certificate.
`PellResidueParityLocalization20260901`	The exact eight-step Pell recurrence and modulo-eight coordinate table, involution product extraction of odd depth at least three under squarefullness, the finite sign-product step, and the complete index-eleven Legendre counterexample to an all-pairs strengthening.
`DanilovWSSEscape20260901`	Primitive-rank injectivity, multiplicity transfer, the exact 2^{637} and $2^{638} - 622$ witness-count kernels, the $41n + 1$ arithmetic implication, and the derivative no-go counterexample.
`IUTCorrectedVolumeHolonomy20260901`	Actual valuation balls, normalized packet shifts, zero transport holonomy, rational-prime log independence, prime-local cancellation, and the positive normalized scalar collision.
`IUTPrimeUnitLabelVectorBridge20260901`	Rational, actual p-adic, and arbitrary-field scale reconstruction; faithful labelled signature-image transfer, vector holonomy, and three exact counterexamples to weakened coordinate interfaces.
`IUTRefinedFactorZeroAwareSignature20260901`	Zero-aware reconstruction on each primitive field factor, faithful dependent products, generic fixed-stage finite-etale packet composition once its instances are supplied, and covariance under explicit refined field transport.
`IUTLanaCurrentConcreteImplicationAudit20260901`	The universal empty-set obstruction to total real log-volume translation, the pinned public specialization, and the explicit missing bridge from abstract `StatementI` to integer `ABCConjecture`.
`IUTAdmissibleVolumeIntegerBridge20260901`	An extensional nonempty admissible-region interface, the actual $\mathbb{Q}_p$ valuation-ball model, fixed/periodic/finite-region no-go theorems, and the explicit uniform-comparison transfer from `StatementI` to the exact integer `ABCConjecture`.
`AffineDeterminantLayerEntropy20260901`	Product-modulus determinant divisibility, forced collinearity, one-dimensional capacity interfaces, canonical scale constants, fractional covers, and two exact arithmetic counterexamples plus the zero-density logical counterexample.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`AffineAdaptiveCommonKernel20260901`	Sharp square triangle-determinant bounds, the derivation from actual pointwise arm divisors and triplewise gcds to six difference congruences, factor-three arm caps, the long-arm square-excess alternative, factor-27 support bounds, and two exact full-premise local counterexamples.
`AffineCommonKernelTriple Selection20260901`	The lattice line cap, fixed-label packing, large-label fibre and incidence-multiplicity bounds, catalogue selection, the exact third-gcd incidence-energy expansion, and the actual canonical-box collinear counterexample to inferring noncollinearity from common-product size.
`MersenneTotientDivisor Concentration20260901`	Exact exceptional-totient criteria, logarithmic-deficit factorization, $O(\log\log(3m))$ moment and moving-window concentration, unconditional cyclotomic cap for the actual blocks, the resulting near-diagonal Mersenne equivalence, and the finite power-of-two weight witness.
`MersennePolylogCodivisor Gate20260901`	Exact partition into fixed-polylogarithmic co-divisor windows, the uniform C/k tail, uniform-in-fixed-index transfer, and the actual Mersenne endpoint equivalence.
`MersenneNearDiagonalGlobal Triage20260901`	Aggregate near/far subtraction, the deep/transition/extreme 1/6 trichotomy, transition-cardinality control, five-arm closure, and three full-premise one-point counterexamples to deleting a disjunct.
`MersenneFixedPolylogBlockMass Triage20260901`	The exact actual-mass window partition, the threshold-free little-oh equivalence, localized deep/transition/extreme selection for every coefficient below 1/3, transition cardinality, and sharp full-premise mass-ledger counterexamples.
`AffineCollinearPeriodEnergy20260901`	Exact reduced direction periods and capture factorization, actual ray divisibility, residue-class capacity, nonconstant-ray cube-root and weighted cubic-energy bounds, the vertical square-root bound under an arm cap, and two full-premise period counterexamples plus the ambient-positivity boundary model.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`IUTRationalTripodShadow Comparison20260901`	Actual open rational points, exact Weil-height and truncated-count conversions, the zero-error shadow comparison, the equivalence of its Statement I with integer *abc*, and three full-premise semantic pressure models.
`MersenneMultiplierIndex TwoArm20260901`	Actual exact-order multiplier representation and injectivity, finite low-multiplier cardinality and log-mass bounds, two-arm finite-sum subtraction, and the sharp one-half mass-ledger counterexample.
`PellOddKernelThirdOrder Packet20260901`	Cubic-square odd-kernel factorization, congruence transport, Jacobi reciprocity rows, arbitrary finite third-order products, quotient cancellation, and the modulo-$8\ell^3$ two-channel ledger.
`AffineCatalogueWeight Overlap20260901`	Exact three-coordinate Euler-totient catalogue mass, divisor-count tail and top-label baseline, monotone cubic aggregation under coincident sublabels, weighted shifted-incidence bounds, and the complete $(9, 25, 1)$ counterexample to subtracting only the threshold.
`IUTRationalDegreeOneSource Realization20260901`	Exact degree-one recurrence, source-realization transport and equivalence with integer *abc*, the rational-shadow witness, and full-premise pressure models for missing height, different, conductor, and degree semantics.
`MersenneBalancedMultiplierDepth Localization20260901`	Triangular multiplier energy, exact-order LTE transport, finite weighted envelopes, the two-controlled/two-surviving mass ledger, its sharpness family, and the complete multiplier-two certificate at 3511.
`PellLucasAllOrderStaircase20260901`	Even-power tail coprimality, coefficient-support exclusion, paired companion congruences, channel-square reconstruction, local-surjectivity logic, and the kernel recurrence certificate refuting fixed-zero rigidity.
`IUTAdmissibleScalingOrder Index20260901`	Empty- and whole-set obstructions to overbroad volume shifts, exact congruence-order kernel and quotient, additive index, properness, and surjective coordinate projections.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`AffineSignedRayCanonical Caps20260901`	Signed direction period and capture, strict large-label squeeze, all three exact arm captures, optimal shifted cubic bridge, weighted owner-catalogue bounds, and exact countermodels to stronger threshold, period, and premise deletions.
`MersenneCriticalSlowSlack Gate20260901`	Nested survivor supports and masses, exact cutoff balance, two-arm subtraction, Euler half-power character on actual exact-order fibres, and the complete odd-multiplier certificate at 1093.
`PellLucasCorrelatedAllOrder Exclusion20260901`	Rational coefficient correlation, exact weighted tail identity, coherent every-order splitters, cross-order determinant, sixth-power channel signs, negative-row extraction, and quartic-two power algebra.
`IUTActualHaarAdmissible Orbit20260902`	Actual normalized additive Haar preimages on finite-positive and compact-open regions, uniformizer orbit and no-envelope theorem, local-degree packet normalization, the raw-weight coefficient counterexample, and the $\mathbb{Q}_p$ residue cardinality.
`AffineInversePeriodCatalogue Novelty20260902`	Optimal incidence-plus-shift inequalities, exact finite class-incidence exchange, multiplicity/novelty identities, owner-tail and pair-catalogue double counts, inverse-period capture algebra, and complete finite period-one certificates.
`PellLucasFactorQuotientProjective Coupling20260902`	Depth-three quotient fingerprints, two companion jets, exact endpoint curvature and U^2 sharpness, together with the local-ledger and genuine index-seven counterexamples.
`MersenneSigmaOneExactOrder Coupling20260902`	Common-index slope injection, exact Farey first-moment identities, stable prime-power order layers, cutoff and two-arm ledgers, and the finite 1093 and 3511 certificates; no analytic limit is asserted in Lean.
`IUTRefinedTensorHaarTheta SamePilot20260902`	Primitive local-factor identities, the ef coefficient, total-degree Haar normalization, relative factor weights, compact-open orbit control, and a conditional pointed same-pilot sandwich; no source-level IUT comparison is asserted.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`MersenneFareyDenominatorEntropy20260902`	Exact finite Farey prefix–tail denominator splitting, a generic finite tail row-count lower bound, exact-order depth-three implications, injectivity, and finite short-window certificates; the complete swarm theorem and critical asymptotic count are not asserted.
`AffineOwnershipMaximalIntersectionAggregation20260902`	Abstract finite-poset maximal-support, owner, and Cauchy skeleton; algebraic strict-ray consequences; and numerical cores of the boundary examples. The concrete maximal cover, large-label migration, and catalogue-sparsity estimate remain outside Lean.
`PellPolynomialHenselSpecialization20260902`	Exact index-three polynomial identities, one-digit Hensel lifting, Taylor/CRT data, and the displayed $T = 282$ numerical packet; no bundled full-premise witness or fixed-$D = 2$ exclusion is claimed in this module.
`PellPolynomialAllIndexFormalization20260902`	All-index Pell/Lucas polynomial recurrences, norm and derivative identities, arbitrary polynomial Hensel lifting, conditional all-support displacement equivalence, finite CRT steering, and the bundled moving witness.
`SteinbergValuationContactSurface20260902`	Divisor exterior contact, standard five-term boundary cancellation, abstract finite-coordinate ℓ^1 boundary control, and complete Pythagorean counterexamples to two single-cell gates.
`SteinbergIntegerFiniteChain20260902`	Positive-integer exponent-gcd normalization, radical/coherent/residual splitting, actual finite-support divisor cells, the exact mixed-area norm, and the filling inequality conditional on an explicit integral surface-boundary equality.
`QuadraticVeronesePeeling20260902`	Exact quadratic five-term peeling, primitive Pythagorean specialization, layer expansion, and full-premise failures of three fixed one-move policies; arbitrary multi-move fillings remain open.
`MersenneFareyQuantitativeSwarm20260903`	Classical harmonic identification and logarithmic prefix control, cleared and divided finite swarm inequalities, exact frequent-subsequence quantifiers, and the finite injection of actual exact-order depth-three rows into the bounded super-Wieferich set.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`AlternativeQualityPacking Bridge20260903`	Exact packing coordinate equivalence, the AM–GM comparison, a clustered-coordinate lower bound, and an abstract full-premise example in which DGM quality is unbounded while standard quality is constant.
`PellFixedTwoTransversality20260903`	Matrix Frobenius congruences at prime indices, unconditional derivative transversality at every actual fixed-$T = 2$ support prime, the exact squarefull/simultaneous-displacement equivalence, and the individual $(7, 13)$ collision boundary.
`PellSignedTraceProjector20260903`	Exact adjacent signed trace-square identities, support-depth doubling, the all-support fourth-power packet equivalence, radical consequences, a sharp raw channel projector and its explicit Newton lift, and the exact index-seven fifth-power boundary.
`SteinbergFiveTermBoundary Bridge20260903`	Algebraic and positive-realization-generated five-term submodules, their inclusion in the boundary kernel, conversion to calibrated finite chains, realization families, and nonzero-cycle counterexamples to literal cancellation and boundary injectivity.
`ABCSynchronizedDivisor Packets20260903`	Actual primitive nonunit abc packet data, finite nonemptiness, product and sixth-power bounds, positive synchronization index, canonical-orientation rigidity, the real-log minimum-energy majorant, an infinite proper exact-gap family, and actual finite packets refuting the recorded stronger height candidates.
`SynchronizedPacketRadicalExcess Obstruction20260903`	Exact packet shape and radical-excess identities, an infinite primitive counterfamily to uncompensated four-thirds compression, and the surviving ab-compensated implication from packet size to the abc height bound.
`ABCCanonicalGainSurface20260903`	Strict canonical gain corridor, exact gain-product and excess/slack identities, additive and multiplicative scale cocycles, equivalence of the uniform defect-flag budget with `ABCConjecture`, and complete countermodels to two stronger gain bounds.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`ABCValuationIncidenceComplex20260903`	Labelled three-arm prime-support faces, exact radical/defect/modulus coordinates, defect filtration, local incidence signatures, CRT reconstruction, the rational-power top-face half-space, a filtered-face implication, and the complete five-dimensional $12 + 833 = 845$ boundary. No eventual selector is asserted.
`ABCValuationIncidenceFixedBudgetObstruction20260903`	The dyadic family $P_k = (2^{k+4}, 3, 2^{k+4} + 3)$, exact support and valuation certificates, forced omission of the A-vertex below a fixed budget, failure of AB reconstruction on an infinite tail, and the negation of the original all-but-finitely-many VIC-ABS-1 selector.
`ABCValuationIncidenceScaleBudgetObstruction20260903`	The balanced family $Q_r = (2^{2r}, 3^r, 2^{2r} + 3^r)$, forced selection of both singleton summand supports, exact defect $3r - 2$, binary-scale upper bound $2r + 1$, an infinite tail of reconstruction-budget failures, and the negation of the full coefficient-one VIC-1R selector.
`ABCThreeArmIncidenceSuccessor20260903`	Exact selected/complement radical partitions, three-arm local signatures and CRT rigidity, the implication of RAW-3C to `ABCConjecture`, its infinite primitive Pythagorean-square refutation, complement-flow mass identities and Hall tails, and the implication of the ordered CT-3C gate to `ABCConjecture`. CT-3C is not asserted and is refuted by the companion module below.
`ABCSignedEndpointPrimeTokenTransport20260903`	Exact endpoint core/radical balance, finite fractional-flow accounting, weighted Hall tail obstruction, implication of the uniform small-unmatched-mass gate to `ABCConjecture`, integral matching counterexamples, and the zero-epsilon prime-power radical-budget obstruction. The ordered uniform fractional gate is not asserted.
`ABCThreeArmComplementTransportObstruction20260903`	The odd-prime family $(1, p^2 - 1, p^2)$, the strict separation of its endpoint source from every external prime, forced selection of that source in every covering three-arm face, the exact full-modulus-minus-radical tail identity, conductor growth at most $3 \log p$, and the quantified negations of both CT-3C and EPF at $\varepsilon = 1/4$.

Module (within `IUTThreeClosures`)	Checked mathematical scope
`ABCBidirectionalPrimeTransport` `Successor20260903`	An all-edge endpoint flow with a nonnegative relative downward-displacement cost, its exact accounting and conditional implication to `ABCConjecture`, and the distinction between that candidate and the scalar completely unordered optimum. The companion obstruction below refutes its exact uniform BEP gate.
`ABCBidirectionalEnergyPythagorean` `Obstruction20260903`	The unbounded prime-hypotenuse Pythagorean-square family, positivity and primitivity of every point, the square-root bound for every external prime, energy at least $(\log p)/4$ for every flow, conductor at most $3 \log p$, and the quantified negation of BEP at $\varepsilon = 1/24$.
`ABCPrimePacketBoundaryTransport` `Successor20260903`	Aggregate endpoint source packets and indivisible external-prime sinks, the finite packet-boundary inequality, the endpoint height bridge, the conditional implication PBT $\Rightarrow$ `ABCConjecture`, an exact two-source/one-sink noncollapse certificate, and the one-source formula for $c = p^2$. No uniform PBT instance is asserted.
`ABCPrimePacketBoundaryLinnik` `Obstruction20260903`	The clipped-reward identity, unique-sink residual lower bound, actual prime-neighbour point, square-product endpoint sources, conductor and zero-scalar-defect bounds, and the implication from the transparent arithmetic proposition `LinnikPrimeNeighborEscape` to the negation of PBT. The published Linnik theorem supplies that proposition in the ordinary proof; it is not inserted as a Lean axiom.
`ABCSharedCRTIncidence` `Successor20260903`	The abstract once-charged capacity certificate, its source-minus-sink and endpoint height bridges, the conditional implication for an explicit arithmetic admissibility predicate, the exact scalar optimum of unrestricted pooling, and the failure of duplicated capacity. Arithmetic CRT blocks, the strict noncollapse optimum, and the SCRT-SAT counterfamily remain ordinary proofs rather than axioms.

The six modules added in the dual-route continuation contain 94 public theorems and 15 definitions or structures. A declaration-level audit checks all 109 items: five have no kernel axioms, and every other dependency is a subset of `propext`, `Classical.choice`, and `Quot.sound`.

There is no `sorryAx`, declaration-style axiom, or unsafe proof. The full 9142-job build succeeds; its 265 warning lines are exactly the frozen previous multiset, so these modules add no warning. The validation record lists the exact commands, results, hashes, and kernel dependency reports. The pre-existing signed-layer module required an explicit import, closure of an elementary algebra branch, and the correct order of unfolding before rewriting; these were repaired without changing its statements.

The four balanced-persistence modules add 62 public theorems and 15 definitions or structures, for 77 audited declarations. Their dedicated audit reports no `sorryAx`; every dependency is a subset of `propext`, `Classical.choice`, and `Quot.sound`, and the axiom-free declarations are recorded separately in the validation artifact. Direct compilation of the four modules and their audit succeeds without warnings. The audit target builds 8767 jobs with 19 warnings from historical dependencies and none from these files. Both the library and repository-default builds complete 9147 jobs with 265 warnings; the warning multisets agree exactly with the frozen dual-route baseline.

The September 1 continuation adds three more modules with 57 theorems and 30 definitions or structures, for 87 further declarations. Direct compilation succeeds, as does the aggregate 9151-job build. In particular, the Danilov finite certificates use kernel reduction rather than native evaluation; declaration-level `#print axioms` reports only `propext`, `Classical.choice`, and `Quot.sound`, with no native-evaluation axiom and no `sorryAx`.

The global-packet and same-pilot continuation adds five modules with 122 theorem or lemma declarations and 42 definitions, structures, or abbreviations, for 164 declarations in their source. Each module compiles directly, and the aggregate target completes 9194 jobs. A token-boundary source scan finds no axiom declaration or proof placeholder and no use of native evaluation, opaque declarations, or unsafe definitions in these files. The embedded kernel-dependency reports contain only `propext`, `Classical.choice`, and `Quot.sound` where dependencies occur. The permanent validation bundle records the commands, complete logs, source hashes, and computation-manifest replays.

The present holonomy-depth continuation adds four modules with 65 theorem or lemma declarations, 18 definitions, four abbreviations, and five structures, for 92 declarations. The files contain 58 embedded `#print axioms` queries; the union of their reported kernel dependencies is exactly `propext`, `Classical.choice`, and `Quot.sound`. A code-only source scan finds no declaration-style axiom, proof placeholder, native evaluation, opaque declaration, or unsafe definition. All four direct compilations are warning-free, and the aggregate target completes 9198 jobs. The analytic squarefree-bulk estimate, the published Pell and Fibonacci inputs, and the large finite certificates retain the explicit paper-and-computation boundary stated in Section 28.

The literal Carella audit adds one module with eleven theorem declarations and two definitions. Direct compilation is warning-free. Its eleven embedded `#print axioms` reports are all either axiom-free or have dependency set contained in `propext`, `Classical.choice`, and `Quot.sound`; there is no proof placeholder or custom analytic axiom. Lean checks the finite positive-multiple count and exact exponent arithmetic. Bertrand's postulate, the Rankin radical count, the prime number theorem, and the passage to the moving asymptotic threshold remain paper inputs rather than new Lean axioms.

The prime-unit-label continuation adds one module with 42 theorem declarations, 19 definitions, four structures, and three abbreviations. Its 43 embedded axiom reports, one of which also audits a definition, have union exactly `propext`, `Classical.choice`, and `Quot.sound`; direct compilation is warning-free. Lean checks reconstruction on both rational multiplicative inputs and Mathlib's actual complete p-adic field, the arbitrary-field scale template, labelled region transfer, vector transport, and the three finite counterexamples. It does not identify those models with the complete IUT packet carrier or assert the missing global coordinate-preservation theorem.

The affine two-arm continuation adds 26 theorem declarations and six definitions. Every theorem has an embedded kernel report. Their dependency union consists of `propext`, `Classical.choice`, and `Quot.sound`. Lean checks the general CRT equivalence, squarefree-carrier transfer, both directions of canonical interval membership, injectivity and exact image cardinality, and every hypothesis of the insufficiency counterexample. The independent replay recomputes the complete factorizations and strict integer inequalities; it is not used to infer an asymptotic statement.

The five Mersenne modules add 97 theorem declarations, ten definitions, and 97 dependency reports, again with union exactly the same three standard kernel principles. They contain no native evaluation or proof placeholder. Lean checks the elementary prime-layer carrier, the finite totient-tail and abstract total-mass inequalities, the exact $m = 6$, 11, and 37 witnesses, the actual exact-order LTE grouping, $L_m \mid m$, the corrected finite product, the limiting implication from the canonical block hypothesis, the nontrivial canonical block $1093 \mid E_{364}$, the finite powerful-part comparison, the three-arm mass dichotomy, and the transition-support cardinality, realized-support, and ambient square-budget ratio implications. The stronger complete identity $E_d = \Phi_d(2)/\operatorname{rad}(\Phi_d(2))$ and the analytic cyclotomic distribution estimates, Brun–Titchmarsh and totient estimates, the Erdős–Murty order theorem, the Erdős–Shorey theorem, and the hypothesis $\log E_d = o(\varphi(d))$ retain the external/open boundary stated in Section 58.

Together, the eight modules in this continuation contain 176 theorem declarations, 37 definitions, three abbreviations, and four structures, for 220 top-level declarations. Their 177 embedded `#print axioms` queries have dependency union exactly `propext`, `Classical.choice`, and `Quot.sound`. All eight direct compilations and the aggregate 9206-job target succeed.

The preceding affine-entropy and Mersenne-depth checkpoint adds three modules with 74 theorem declarations and 21 definitions, for 95 further top-level declarations. Every theorem has an embedded `#print axioms` query. Lean checks the complete finite separation algebra and canonical packing constant, the exact divisor-average endpoint, the finite depth layer cake and threshold bounds, and all premises of the two narrow auxiliary counterexamples. The analytic fixed-base distribution estimates and the final affine exceptional-template lower bound remain paper obligations rather than Lean axioms. Together with the preceding eight-module checkpoint, these eleven modules contain 250 theorems, 58 definitions, three abbreviations, and four structures, for 315 counted top-level declarations and 251 embedded dependency reports. The direct module checks and the aggregate 9209-job target succeed; the dependency union remains exactly `propext`, `Classical.choice`, and `Quot.sound`.

The current determinant, totient-concentration, and refined-factor checkpoint adds three modules with 105 theorem declarations, 20 lemma declarations, and 26 definitions, for 151 counted declarations. A generated same-scope audit covers all 125 proofs one for one, including the 13 private affine proofs that cannot be named after import. Its dependency union is again exactly `propext`, `Classical.choice`, and `Quot.sound`. All three direct compilations and the generated audit are warning-free, and the aggregate 9212-job target succeeds. The sealed replay freezes 469 local Lean and Lake inputs and explicitly records that these checks do not close the standard *abc* conjecture.

The present adaptive-kernel, fixed-polylogarithmic, near-diagonal, and current-IUT audit adds four modules with 52 theorem declarations, four lemma declarations, and sixteen definitions, for 72 declarations in all; three of the affine theorems are private implementation lemmas. A generated same-scope audit issues one `#print axioms` query for each of the 56 proof declarations, including the private proofs; their union is exactly `propext`, `Classical.choice`, and `Quot.sound`. All four direct compilations are warning-free. The aggregate target completes 9216 jobs with no errors; its warning diagnostics are inherited from historical imports and none arises in these modules. A code-only scan finds no proof placeholder, declaration-style axiom, native evaluation, unsafe definition, or opaque declaration, and `git diff --check` succeeds. This aggregate build retains the pinned LANA commit `ddaddc2`. Separately, against a detached checkout of commit `6e963070`, the public `Iut` target builds and an additional audit theorem proves `LocalTheory`$(K) \to$ `False`. Successful construction of the library therefore does not provide an inhabitant of that structure.

The selection, admissible-volume, Pell-residue, and fixed-block-mass checkpoint adds four modules with 92 theorem declarations, two lemma declarations, 22 definitions, and eight structures, for 124 counted declarations. The generated same-scope audit issues exactly 94 `#print axioms` queries, including all eight private affine proofs. Its dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`. The four direct compilations and the audit are warning-free, and the aggregate target completes 9220 jobs. The associated finite Mersenne replay independently checks all 664579 primes through 10^7, recovering precisely the depth-two witnesses 1093 and 3511 and no deeper hit. These computations refute only the explicitly stated finite strengthenings; they do not convert any surviving asymptotic gate into an assumption or close the standard *abc* statement.

The current direction-period, rational-shadow, multiplier-index, and odd-kernel checkpoint adds four modules with 83 theorem declarations, 40 definitions, two structures, and one instance, for 126 counted declarations. A generated same-scope audit covers all 83 proofs, including three private affine helpers. Its dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`; all four direct compilations and the audit are warning-free. The aggregate target completes 9224 jobs. The independent Pell replay checks 648189 bounded repeated- factor candidates, 42 exact simple-divisor certificates covering every odd prime index through 191, and one complete Pocklington certificate. It records 187 unresolved bounded-search indices as unresolved and makes no asymptotic inference. These checks certify the stated formal reductions and narrow counterexamples, not the standard *abc* conjecture or its negation.

The source-realization, balanced-multiplier, catalogue-overlap, and all-order Lucas continuation adds four modules with 67 theorem declarations, 18 definitions, and one structure, for 86 counted declarations. A generated same-scope audit covers all 67 proofs one for one. Its dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`; all four direct compilations and the audit are warning-free, and no target source contains a proof placeholder, declaration- style axiom, native evaluation, unsafe definition, or opaque declaration. The aggregate target completes 9229 jobs. The independent Lucas replay checks all five local residues modulo five, including the fixed-zero counterexample, and checks the full coefficient staircase and companion splitter at $\ell = 3, 5, 7, 11$. Its finite no-hit results are pressure tests, not an asymptotic exclusion or a proof of *abc*.

The signed-ray, critical-slow-slack, correlated-Lucas, and admissible-index checkpoint adds four modules with 90 theorem declarations and 16 definitions, for 106 counted declarations. A generated same-scope audit issues one `#print axioms` query for every theorem. The dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`; all four direct warning-as-error compilations and the generated audit pass. The aggregate target completes 9233 jobs. Exact replays cover 1,776,807 affine cubic ledgers, 2,390,018 quadratic ledgers, 15,840 signed directions, 43,403 arm captures, the complete Mersenne character packet, all 57 Pell prime indices through 271, 138,675 product-derived coefficient pairs, and 527,352 bounded depth candidates. The current-source IUT patch replays byte for byte and the patched upstream `Iut` and `Iut4Sec1` targets build all 8767 jobs. The owner-energy theorem is audited only on the selected powerful- kernel catalogue union; the recorded $972{,}496 > 1{,}072$ witness prevents an invalid all-divisor enlargement. No bounded no-hit, source-interface repair, or local formal theorem is promoted to an unconditional proof or disproof of *abc*.

The actual-Haar, sigma-one, catalogue-novelty, and endpoint-curvature checkpoint adds four modules with 102 theorem declarations, 18 definitions, two structures, and one abbreviation, for 123 counted declarations. Same-scope audits issue exactly 102 `#print axioms` queries, one for every theorem; their dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`. All four direct warning-as-error compilations and both separate audit modules pass, and the aggregate target completes 9239 jobs. The finite replays check the actual local-degree normalization witness, all 50,847,534 primes through 10^9, six affine catalogue boxes and every recorded full-premise witness, and exponent-one certificates at all 57 odd prime Pell indices through 271. The weighted Brun–Titchmarsh and Yamada-type analytic inputs, the tensor/same-pilot realization, the final affine intersection aggregation, and the global squarefull Pell exclusion remain outside Lean and are not introduced as axioms.

The refined-Haar, Farey-entropy, ownership, polynomial-Hensel, contact, and quadratic-peeling checkpoint adds eight modules with 297 theorem declarations, 113 definitions, five abbreviations, four structures, and two named instances, for 421 counted declarations. Its generated cross-audit issues one `#print axioms` query for every counted declaration, not only for proof declarations. All 421 reports are present, and their dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`. Each module and the generated audit passes direct warning-as-error compilation, and the aggregate target completes 9255 jobs. Deterministic replays cover the refined tensor-weight identities, the finite Mersenne denominator certificates, two independent affine checks, the Pell producer/verifier and bounded moving-parameter scan, and the quadratic peeling producer/verifier. At the date of that checkpoint, the fixed-$D = 2$ Pell transversality and rank gate, critical Mersenne count, affine catalogue sparsity, source-level IUT transport, five-term generated-move boundary theorem, and two Gate VF cost estimates remain open and are not inserted as axioms. Thus this checkpoint supplies neither a term of `ABCConjecture` nor a term of its negation.

The September 3 quantitative-transversality and generated-packet checkpoint adds five modules with 169 theorem declarations, 79 definitions, one abbreviation, seven structures, and two named instances, for 258 counted declarations. The generated audit issues one `#print axioms` query for every declaration. Its dependency union is contained in `propext`, `Classical.choice`, and `Quot.sound`; every module, hand-written audit, and generated audit passes direct warning-as-error compilation, and the umbrella target completes 9265 jobs. The fixed-$T = 2$ Pell support-transversality premise and the algebraic five-term boundary bridge are now closed. The Mersenne harmonic and quantifier seams, the synchronized-packet finite energy majorant, and all recorded full-premise local counterexamples are also kernel checked. The global Pell squarefull exclusion, actual Mersenne row supply and critical count, and arbitrary-target positive five-term filling with its two costs remain open. The synchronized radical-scale compression proposed at that checkpoint is refuted below at exponent 4/3 and replaced by the compensated form of Theorem 79.5. None of the open gates is inserted as an axiom, and the abstract alternative-quality countermodel is not an integer *abc* family. Thus this repository contains no unconditional proof or disproof of the standard conjecture.

The gain/packet/trace follow-up adds three principal modules with 114 one-for-one axiom queries. Direct warning-as-error compilation succeeds for each module and audit, and the dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`. Exact finite replays cover 5,465,583 primitive triples for the canonical gain surface, 1,365,095 primitive triples and 1,366,531 synchronized packets for the radical-excess audit, and 63,950 Pell prime indices with 764,366 candidate-prime tests. The two Pell collision rows are also checked by complete trial division and three independent orbit algorithms. These finite records do not decide any unbounded gate.

The incidence/endpoint continuation adds eleven principal modules with 496 public declarations and 496 one-for-one axiom queries. The dependency union is exactly `propext`, `Classical.choice`, and `Quot.sound`. The first two one-face selectors, the raw weighted three-arm gate, CT-3C, the ordered fractional endpoint gate, the first relative-drop bidirectional gate, the exclusive-ownership PBT gate, and the saturated-only SCRT-SAT child are refuted by infinite complete-premise families. The once-charged SCRT-0 successor has a proved pointwise bridge and conditional implication but no asserted uniform estimate. The parent incidence complex remains active through SCRT-0, surplus-reuse, weighted multi-face, and homological successors. The endpoint replay enumerates 3,795,230 primitive nonunit triples, and the three-arm replay enumerates 218,893 unordered primitive positive triples. The exact packet optimizer adds 1,368,094 normalized primitive triples and 1,038 structured rows; an independent implementation replays the full scope. All three searches keep their finite scope explicit and do not infer an asymptotic claim from a bounded scan.

The formal scope is deliberately narrower than the full paper. Theorem 2.3 and Corollary 2.4 use Younis and classical smooth-number asymptotics in their mathematical proofs; those external analytic theorems are not formalized here. The full-rank case of Theorem 6.1, including its actual index and Haar measure, is now checked in its normalized-probability formulation on the compact integral lattice. Identifying this with the restriction of ambient local-field Haar measure and the rank-deficient measure-zero case remain paper proofs. The ellipse count and asymptotic amplification estimates, source-specific Kummer and local-field identifications, general independent-action theorem, Matveev's theorem, (35), the small-generator approximation inputs, and the endomorphism-algebra argument retain their stated paper-proof boundary. In particular, the complete effective finiteness theorem is mathematically unconditional by the cited published theorems, but it has not been formalized as an unconditional Lean theorem. A Lean implication with an explicit approximation or height hypothesis is not a Lean proof of that hypothesis. The new radical count uses the cited S-unit theorem and an analytic divisor sum outside Lean. The actual Ism classification, local class field theory, Galois cohomology, root-logarithm calculation, and categorical representative reconstruction are also paper proofs. The shrinking-line rigidity theorem proves their algebraic core without assuming LCFT as an axiom. The global cubic construction does not formalize Pasten's theorem, number-field orders, Siegel's local approximation, or Nagell–Lutz merely by checking their coordinate identities. These distinctions apply equally to the new module-hull and fixed-support balance statements. Likewise, the free-group word proof does not formalize the Jannsen–Wingberg presentation of a local Galois group, and the elementary matrix development does not formalize an elliptic Tate representation. The integral local logarithm, trace-dual lattices, finite-etale covers, the initial-data construction, Linnik's theorem and the analytic power-free count retain their mathematical proof boundary. The algebraic trace-dual core is checked with the actual library trace and dual submodules, without formalizing the local inverse different or the Galois witness by that fact alone. Exact finite arithmetic is checked against actual library objects, without inserting these external theorems as axioms. The complete two-prime inequality and arithmetic odd-part fibre bound are unconditional Lean theorems about the unchanged canonical objects. The scalar-return module uses the actual algebra trace, with trace covariance as an explicit hypothesis; it does not formalize the Galois, Kummer, or principal-unit logarithm inputs that establish the hypothesis in the mathematical application. The entire rational isogeny-class argument uses the cited cyclic-degree classification and Frobenius and minimal-model theory outside Lean. The companion module checks its arithmetic models and bounds without inserting those inputs as axioms. The height companion additionally checks the reduced fractions and the least absolute logarithmic Weil height on those four actual curves, using the library height and its normalized-height interface over $\mathbb{Q}$. It does not formalize the classification of the entire isogeny class merely by proving the four-model minimum. The new conductor-obstruction companion deliberately calls $16\,\mathrm{rad}(c(c-1))$ a *conductor proxy*: the identification with the Néron conductor, the 2-adic reduction type, and isogeny invariance are the paper inputs of Proposition 24.1. The mixed-full module consumes the unchanged standard `ABCConjecture`; it neither assumes the Browning–Verzobio count nor asserts the existence of an unbounded Campana family. The finite-product module formalizes the heterogeneous finite product of its coordinate rings. Source compactness, finite-etale field decomposition, Haar normalization, and the identification with the IUT hull remain the explicitly cited paper layer. The support-three classifications that use Fermat's Last Theorem and the Arif–Abu Muriefah theorem are not inserted into Lean as axioms. The arithmetic bundles, their metrics and descent also remain mathematical proofs; equality of their degrees alone is not a formal or mathematical proof of source-family membership. The finite theta point-hull theorem uses the previously proved full Galois constructions and original theta evaluation formulas outside Lean. Its common arrow is allowed to depend on the tuple of torsion choices. The per-arrow logarithmic bridge equates principal ideals in the declared carrier, not the original multiplicative Kummer inputs. The source-defined canonical local membership in Section 21 uses the cited reconstruction theorems and local reciprocity outside Lean. The single base branch and its lower ideal are not represented as a formal proof of the complete global comparison.

For the global-packet continuation, Lean checks the exact finite Pell counterexamples, the affine support-modulus algebra and matching-lower quantifier logic, the Danilov recursion invariant and abstract continuation countermodel, the sequence-uniform and parity-free Lucas counterexamples, the elementary powerful-packet consequences, and the literal pinned LANA record contradiction. The repaired Fellini–Murty prime-order/Siegel argument and its rank-tower descent remain paper proofs. The Fibonacci cyclotomic correction, rank-of-apparition valuation transfer, Carmichael–Yabuta primitive-divisor theorem, and Hong's large-prime theorem are cited mathematical inputs rather than Lean axioms. The pinned LANA no-go proves only that the pinned low-resolution record is uninhabited; no corrected Haar-volume interface or object-level pointed same-pilot diagram has been constructed. Thus none of these formal modules turns the surviving Pell, Fibonacci–Danilov, affine, or IUT obligations into an unconditional abc conclusion.

The surviving proof target is a uniform estimate for the coupled expression (25), or a source-derived theorem strong enough to imply it. The IUT route still needs the comparison of the actual output family with the region in the global upper estimate, under compatible normalizations. The analytic disproof route still needs a height-unbounded family satisfying one fixed radical exponent gap; the finite-support bound neither constructs nor forbids such a sparse family. The specified Pell–Chebyshev residual is now effectively finite; the fundamental-unit cases and the absence of a global abc reduction remain outside that conclusion. The actual-radical amplifier still requires a lower count in the surviving window; the cubic route must control the growing omitted endpoint or denominators uniformly. On the local IUT route, explicit full-Galois arrows and exact pre-ideal hulls have now been obtained. The original word weights are explicit on the rational branch, and the computed local hulls admit global determinant objects over the field of moduli. The remaining task is to establish membership and comparison for the complete globally synchronized family with its markings, including the Ind3 and cross-Frobenius comparison for the same measured set and reference. The canonical local membership of one fixed base branch is now supplied by Theorem 21.5; it does not settle these remaining comparisons. The newest public conditional implication does not bypass this gate: its current local-volume interface is empty by Theorem 32.1. The valuation-ball model of Theorem 33.1 proves that an explicit extensional admissible-region replacement is consistent, and Theorem 33.2 removes the logical ambiguity from the final integer bridge. The rational shadow now realizes that bridge with zero error, but Theorem 34.1 shows exactly why this is not a proof: Statement I for the shadow is equivalent to *abc*. The actual degree-one source computation now identifies the rational points, canonical height, different, and conductor with their standard arithmetic counterparts, and Theorem 35.1 proves the exact two-way transport for any source-faithful realization. What remains is to construct the intended arithmetic `HeightTheory` instance in Lean and prove its upstream Statement I from the actual IUT data, including the disputed Corollary 3.12 comparison. The full-premise pressure models show that the current abstract record and proof package cannot manufacture those semantic identifications.

The public source at commit `c65b28c9f9631635e742294c3a5df15759e7c74c` now constructs the old concrete theta-data existence interface from curve data and explicit providers. This closes one assembly seam but retains the anabelian, local-theory, tower, analytic, height-package, and Corollary 3.12 premises. The all-set log-volume law makes a literal local-theory provider impossible; Theorem 36.1 gives the exact counterexample, and the patched admissible interface compiles throughout the current upstream tree. Theorem 36.3 also shows that componentwise surjectivity cannot erase an integral-order index. The active IUT gate is therefore an admissible, index-aware construction of the actual local volumes followed by the full horizontal and Ind3 covariance and the concrete comparison theorem.

On the affine route, determinant divisibility now survives point-adaptive certificates. Theorem 43.2 converts the selection problem into a catalogue-multiplicity inequality; alternatively, one may lower-bound the large-product part of (345) beyond every collinear contribution. The exact ray-period theorem now controls the previously unbounded nonconstant collinear occupancy by a weighted cubic-energy ceiling. The actual downward catalogue now has total totient weight exactly equal to the canonical kernel product, asymptotically all above the determinant threshold, and coincident sublabels only increase the cubic energy. The signed-direction capture squeeze now handles all primitive rays and gives exact linear caps for the three constant-arm directions. The global ceiling (359) leaves one non-arm inverse-square-period sum, supported only on large directions, and the comparison of the singleton baseline with catalogue multiplicities. These are the active gates. The canonical vertical example proves that the label product itself cannot be used as a period, while the $(9, 25, 1)$ catalogue and the new exact pressure tuples disprove only their stated stronger inferences.

On the Mersenne route, multiplier injectivity removes the low one-copy arm in every fixed-polylogarithmic co-divisor packet. The remaining gates are the high-multiplier deep-lift estimate and the localized Wieferich-weighted near-square-root order estimate. The balanced cutoff works with every fixed $\eta > 0$, and the critical slow-slack theorem now works for any divergent σ with $L_m\sigma = o(\log m)$; the explicit $\sigma = \log(3+L_m)$ narrows both surviving regions beyond every fixed positive log-log power. The Euler-character identity filters the exact multiplier residues, but its $d \equiv 4 \pmod 8$ row is unrestricted. No finite computation or current global density theorem proves either surviving little-oh statement at the arithmetic endpoint $\sigma = 1$. The exact primes 3511 and 1093 refute only the stated multiplier-at-least-three and even-multiplier strengthenings.

The Pell packet now carries nontrivial odd kernels, a complete aggregate Jacobi triangle, the third-order quotient ledger, and an all-order norm-one Lucas staircase. The paired companion correction recovers opposite signs on the two channel squares. The companion staircase is now explicit and coefficientwise correlated with the first; every order recovers the same splitter, and all cross-order determinants vanish modulo U^2. For a hypothetical squarefull packet, vertexwise character laws force a nonresidue opposite pair with depth at least three when $\ell \equiv 3, 5 \pmod 8$. Certified simple divisors now exclude squarefull packets at every odd prime index through 271; the finite boundary is kept literal. The $k = 2451$ recurrence witness and the mixed-sign row at index 11 refute only fixed-zero one-channel rigidity and the assertion that all edges of a negative row are negative. No known theorem excludes or constructs an unbounded packet whose two channels satisfy all depth, residue, character, staircase, cross-order, and third-order quotient constraints.

Constructing initial data, descending a determinant, localizing a divisor average, or eliminating Ind2 from a fixed B-module hull does not by itself supply the missing global comparison or distribution estimate. No unconditional closed term of type `ABCConjecture`, or of its negation, is produced by this work.

97. Current boundary and continuing programme

The developments above do not prove or disprove the standard *abc* conjecture. They replace several informal bottlenecks by exact interfaces: an index-aware pointed Haar comparison, a critical exact-order counting problem, an ownership-preserving affine energy inequality, a fixed-parameter Pell displacement problem, a calibrated multi-cell contact filling, and the SCRT-0 shared-incidence uniform estimate or a weighted multi-face or homological successor in the labelled valuation-incidence complex. The two formerly listed ordered transport estimates, the first relative-drop bidirectional estimate, and the exclusive prime-packet estimate are now refuted. In each retained route the local algebra or finite combinatorics is proved before a remaining uniform statement is named.

The counterexamples have a narrower and useful role. They exclude raw or factor-count Haar normalization, reduced Farey reasoning without arithmetic premises, owner-free affine estimates, moving-parameter Hensel exclusions, single-cell contact truncations, fixed one-step quadratic fillings, the proposed $A < 3/2$, $P < 3$ gain inference, uncorrected packet compression, and automatic Pell depth-five promotion. They also exclude universal trivial arm-defect and constant-per-arm valuation claims, both the fixed and coefficient-one binary-scale one-face incidence selectors, the raw weighted three-arm defect gate, both ordered CT-3C and EPF, the relative-drop BEP gate, the exclusive-ownership PBT gate, the saturated-only SCRT-SAT child, full integral endpoint matching, and a fixed natural multiplier at epsilon zero. Each counterexample retires only the exact statement whose complete premises it satisfies. No bounded search is used to infer an asymptotic assertion. We therefore retain every surviving parent route and continue proof construction and counterexample search in parallel.

The next decisive advances would be any one of the following: construct the source-faithful pointed IUT comparison with its integral-order correction; prove the critical one-half counting bound for the relevant depth-three exact-order primes; establish the canonical affine catalogue-sparsity bound; control the canonical power-excess/approximation-slack difference; prove an eventual ab-compensated packet bound; exclude or construct simultaneous fixed-$T=2$ signed fourth-trace Pell packets; or construct a five-term filling satisfying both calibrated cost estimates; construct a weighted, multi-face, or homological incidence successor with a non-tautological sufficient inequality; or prove or refute the exact SCRT-0 uniform estimate, including the composite-base prime-log partition and a capacity-safe surplus-reuse rule. A full-premise counterexample to any one of these exact statements would retire that statement alone and trigger a corrected formulation, while a proof of a sufficient gate would be transported through the already checked reduction.

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